



Business Unit MR

Component Test

Specification (MR)

Scanning

Approver (User): Component Engineer
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1. Introduction

1.1. Purpose

The Component Test Specification describes the method, set-up, test cases and instructions for test execution (e.g., test, evaluation, inspection, etc.) or prescriptions for the recording of the execution. All component test specification must be traceable to tagged items in the component specification.

1.2. Scope

The test cases described in this document formally cover the specifications in Component Specification[1].

1.3. References

Reference	Document Title	Document ID	Revision
[1]	CS Scanning	D002063557	B

1.4. Terms, Definitions and Abbreviations

Terminology & Abbreviations	Description/Definition
RS	Requirement Specification
RTM	Recon Test Machine
STM	Software Test Machine
TB	Test Bay
TR	Test Report
TS	Test Specification
WS	Workstation

2. General Description

2.1. Test approach

The following criteria will be used to decide whether the outcome of the tests have passed or failed:

- If the actual result of a test step matches the expected result the test step has passed.
- If all test steps for a specific test case have passed overall execution result of the test case is passed.
- If the actual result of a test step does not match the expected result the test step has failed.
- If one or more test steps for a specific test case have failed overall execution of the test case is failed.
- If a test step has failed a defect must be entered and a reference to the defect shall be linked to the specific test step.

In case execution of a test step is conform the expected result but unexpected behavior has been observed not related to the specification(s) being tested the test step has passed. A defect must be entered but a reference to the defect shall not be linked to the specific test step.

2.2. Pass/fail criteria

Pass Criteria	Execute and pass the scanning component test case if no deviation in Expected result in ALM with verification evidence.
Fail Criteria	Execute and fail the scanning component test case if there is deviation in expected result in ALM with verification evidence.

2.3. Required test environment

Name	Type	Release / Version	Remark	Mams number	Due date
m_prep_rx_v m	script	Baselined in Git	NA	DEP115855/00	NA
m_imn_chann el_test	script	Baselined in Git	NA	DEP113320/00	NA
m_imn_presca n	script	Baselined in Git	NA	DEP113322/01	NA
m_imn_diff_v m	script	Baselined in Git	NA	DEP113316/00	NA
Methods m_imn_metal. pl	script	Baselined in Git	NA	DEP115831/01	NA
PDF Compare (PDFCMP)	script	Baselined in Git	NA	DEP113369/04	NA
TestFramework	script	Baselined in Git	NA	DEP141702 /00	NA
m_amn_test_ environment.p l	script	Baselined in Git	NA	DEP117147/00	NA

<i>CTS_ENCASE_log_parser.pl</i>	<i>script</i>	Baselined in Git	NA	DEP119773/00	NA
<i>CTS_MDME_intlv_test_parser.pl</i>	<i>script</i>	Baselined in Git	NA	DEP120307/01	NA
<i>Methods m_imn_asl_ts_pCASL</i>	<i>script</i>	Baselined in Git	NA	DEP125876/00	NA
<i>Methods m_imn_asl_ts_star</i>	<i>script</i>	Baselined in Git	NA	DEP125877/00	NA
<i>Methods m_imn_prof_log</i>	<i>script</i>	Baselined in Git	NA	DEP126117/01	NA
<i>Doxygen</i>	<i>script</i>	Baselined in Git	NA	DEP121850/01	NA
<i>ScanAcqState</i>	<i>script</i>	Baselined in Git	NA	DEP149752/01 (VLD) DEP149984/00 (TAF)	NA
<i>googletest\googlemock</i>	<i>framework</i>	1.7	NA	DEP162686, TAF GoogleTest/Google Mock Any DEP162684 - VLD GoogleTest/Google Mock Any	NA

3. Test Specification

3.1. Test Cases

3.1.1. Essentials

3.1.1.1. SWCMP.Scan.HAZ.MI.30.Retro.RespPeak.tc1

Test ID	62795
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Reddy, Srikanth (310031195) on 2025-07-22 12:10:12 Meaning: QA Approved - (QA) Signed by: Koonen, Jos (nly95806) on 2025-07-21 13:18:32 Meaning: SME Approved - (Business) Signed by: Gorkum, Robbert van (320189046) on 2025-07-21 09:36:35 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the UI parameters for resp. peak coverage and resp. peak factor required by hazard requirement SYS.Mit.HAZ-MI.30 (99580).

Requirements Coverage:

Req ID	Requirement Tag	Description				
99641	SWCMP.Scanning.HAZ.MI.30.Retro.RespPeak	Name	Range	Visible when	Description	System / Option restrictions
		Resp peak coverage	Default, user defined	respiratory comp = retrospective	Specifies the type of resp phase coverage used to control	N/A

Req ID	Requirement Tag	Description				
					the amount of extreme motion included in maxima and minima peaks.	
		Resp peak factor	0.1 - 1.0	resp phase coverage = user defined	Specifies the value of the user defined respiratory peak coverage for determining maxima and minima peaks, excluding outliers.	N/A

Test steps:

Step	Description	Expected Result
Prep 1	Open the testframework and configure the system as a WA15 (1.5T Ingenia) system using "set test environment for methods". Confirm that the protocol database is set to tedb.db Create a new patient Add a 4DMR-RT protocol to the scan list from the folder: Hospital/203 Motion/RespirationCompensation/RetroRespBinned_T1 Set anatomy and anatomic region to abdomen. Start exam. Acknowledge any popups	N/A (preparation)
Veri 1	Open the protocol for editing. On the motion tab, confirm that "Resp. peak coverage" parameter is present. Set "Resp. peak coverage" to "default".	"Resp. peak coverage" parameter is present. With "Resp. peak coverage" set to "default" confirm that "Resp. peak factor" is not present. Objective evidence: screenshot showing the Motion tab - "Resp. peak coverage" parameter is present

Step	Description	Expected Result
		and "Resp. peak factor" is not present.
Veri 2	Set "Resp. peak coverage" to "user defined" Check range of "Resp. peak factor" Set "Resp. peak factor" to 0.0. Set "Resp. peak factor" to 1.0.	"Resp. peak coverage" parameter is present With "Resp. peak coverage" set to "user defined" confirm that "Resp. peak factor" is present "Resp. peak factor" has a range of 0.0-1.0. "Resp. peak factor" can be changed to 0.0. "Resp. peak factor" can be changed to 1.0. Objective evidence: screenshot showing the Motion tab - "Resp. peak coverage" parameter is present and "Resp. peak factor" can be set to 1.0.

3.1.2. ME 3DVIEW

3.1.2.1. SWCMP.Scan.3DVIEW.Conflicts.tc1.v2

Test ID	25236
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-06-20 11:40:46 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2016-06-20 11:05:27 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2016-06-20 10:56:46 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Check the occurrence of parameter combination conflicts

Requirements Coverage:

Req ID	Requirement Tag	Description
56979	SWCMP.Scan.3DVIEW.Conflicts.v2	A conflict is generated if 3D VIEW is used in combination with any of the following parameters: - Echoes > 1 - Shot mode = single-shot - Stacks > 1 and Stacks as packages = no - Chunks > 2 and chunk scan order = default

Req ID	Requirement Tag	Description
		A conflict is generated if a Reference tissue is selected that is inappropriate for the anatomy as defined by the main parameter 3D VIEW. The 3D VIEW parameter forces other parameter settings (through conflicts): - Brain T1 and Brain BB use Scan technique SE exclusively, and anti-DRIVE - Brain T2 uses Scan technique SE exclusively - Brain FLAIR uses Scan technique IR exclusively, and T2prep - Brain IR uses Scan technique IR dual - Any 3D VIEW with refocusing flip angle below 90 deg. and refocusing control level advanced requires startup echoes ≥ 4, Any 3D VIEW with refocusing flip angle below 90 deg. and refocusing control level simple requires startup echoes ≥ 1 The 3D VIEW parameter forces other parameter settings (through conflicts) and conflict resolution guidance is provided to the user: - Nerve SPAIR uses Scan technique set to SE - Nerve SPAIR uses Fat suppression set to SPAIR - Nerve STIR uses Scan technique set to IR - Nerve STIR uses Fat suppression set to no

Test steps:

Step	Description	Expected Result								
Step 1	- Use m_amn_set_test_environment.pl and select configuration wa30t - Load scan BrainViewT2 from Head_CNS/TSE Check conflict generation for the combination of "3D VIEW" = "Brain T2" with: - Echoes > 1 - Shot mode = single-shot - Stacks > 1 and Stacks as packages = no - or Chunks > 2 and chunk scan order = default	Conflicts are generated when 3D VIEW" = "Brain T2" in combination with: - Echoes > 1 - Shot mode = single-shot - Stacks > 1 and Stacks as packages = no (NB: no conflict for 'Stacks' > 1 and 'Stacks as packages' = yes) - or Chunks > 2 and chunk scan order = default (NB: no conflict for 'chunks' = 2, and no conflict for 'chunks' >2 and 'chunk scan order' = 'LR' or 'RL').								
Step 2	Check the (range) conflicts that are generated for inappropriate combinations of Reference tissue values and 3D VIEW anatomy choices. An appropriate Reference tissue is important, because the shape of a refocusing flip angle sweep may depend on it in the case of Refocusing control set to tissue specific.	The table below lists all allowed combinations of 3D VIEW values with Reference Tissue values. <table><tr><td>Brain</td><td>Grey matter, White matter, CSF, UserDefined</td></tr><tr><td>Spine/Nerve</td><td>Grey matter, White matter, CSF, Skeletal muscle, Verteb. Marrow, Disk, UserDefined</td></tr><tr><td>MSK</td><td>Skeletal muscle, Bone marrow, Fat, UserDefined</td></tr><tr><td>Female/General</td><td>Skeletal muscle,</td></tr></table>	Brain	Grey matter, White matter, CSF, UserDefined	Spine/Nerve	Grey matter, White matter, CSF, Skeletal muscle, Verteb. Marrow, Disk, UserDefined	MSK	Skeletal muscle, Bone marrow, Fat, UserDefined	Female/General	Skeletal muscle,
Brain	Grey matter, White matter, CSF, UserDefined									
Spine/Nerve	Grey matter, White matter, CSF, Skeletal muscle, Verteb. Marrow, Disk, UserDefined									
MSK	Skeletal muscle, Bone marrow, Fat, UserDefined									
Female/General	Skeletal muscle,									

Step	Description	Expected Result	
		<i>Pelvis, Prostate</i>	<i>Fat, Liver, Pancreas, Kidney, Spleen, Female pelvis, Male pelvis, UserDefined</i>
		<i>Breast</i>	<i>Skeletal muscle, Fat, Breast, UserDefined</i>
		Conflicts are given for all other combinations of 3D VIEW and Reference Tissue.	
Step 3	Reload scan BrainViewT2 from Head_CNS/TSE. Switch 3D VIEW to Brain T1. Solve conflicts on 'DRIVE' by setting its value to 'anti'. Decrease 'startup echoes' to '1' and check for conflicts. Then switch to Brain T2. Check for conflicts on 'startup echoes'.	'3D VIEW' = 'Brain T1' allows startup echoes = 1 and higher. - A conflict is generated for the combination of '3D VIEW' = 'Brain T2' and 'startup echoes' < 4.	
Step 4	- Switch '3D VIEW' to 'Brain FLAIR'. - Switch 'Scan technique' to all possible values and try to resolve conflicts.	- A conflict is generated on '3D VIEW' = 'Brain FLAIR' and 'Scan technique' is set to 'MIX' or 'SE'. 3D VIEW = FLAIR is disabled when 'Scan technique' is set to 'FFE'. - A conflict is generated on '3DVIEW' = 'Brain FLAIR' in combination with 'Scan technique' = 'IR' and 'T2 prep' = 'no'. - No conflict on Scan technique = IR and T2 prep = 'yes'.	
Step 5	- Switch '3D VIEW' to 'Brain DIR'. - Switch 'Scan technique' to all possible values and try to resolve and check for conflicts.	- A conflict is generated on '3DVIEW' in combination with 'Scan technique' = 'SE' or 'MIX'. - The '3D VIEW' parameter is disabled when 'Scan Technique' is set to 'FFE'. - A conflict is generated on '3D VIEW' is 'Brain DIR' in combination with 'Scan technique' = 'IR' and 'IR dual' = 'no'. - No conflict is generated when '3D VIEW' is 'Brain DIR' is combined with 'Scan technique' = 'IR' and 'IR dual' = 'yes'.	
Step 6	- Switch 3D VIEW to Brain T1. - Set 'Scan technique' to 'SE' and check for conflicts on the parameter 'DRIVE'.	A conflict is generated when: '3D VIEW' = 'Brain T1' - 'Scan technique' = 'SE' and 'DRIVE' = 'yes' or 'no'. The conflict is resolved when 'DRIVE' is set to 'anti'.	
Step 7	Switch 3D VIEW to Nerve SPAIR.	A conflict is generated on '3D VIEW' = 'Nerve	

Step	Description	Expected Result
	Switch 'Scan technique' to all possible values and try to resolve conflicts.	SPAIR' and 'Scan technique' = 'IR' or 'MIX'. - The '3D VIEW' parameter is disabled when 'Scan technique' is set to 'FFE'. - A conflict is generated when: '3D VIEW' = 'Nerve SPAIR', 'Scan technique' = 'SE' and 'Fat suppression' = 'no, SPIR' or 'ProSet'. - The conflict is resolved when 'Fat suppression' is set to 'SPAIR'.
Step 8	- Switch 3D VIEW to Nerve STIR. - Switch 'Scan technique' to all possible values and try to resolve conflicts.	- A conflict is generated on '3D VIEW' = 'Nerve STIR' in combination with 'Scan technique' = 'SE' or 'MIX'. - The parameter '3D VIEW' is disabled when 'Scan technique' is set to 'FFE'. - A conflict is generated when: '3D VIEW' = 'Nerve STIR', 'Scan technique' = 'IR' and 'Fat suppression' = 'SPIR', 'SPAIR' or 'ProSet'. - The conflict is resolved when 'Fat suppression' is set to 'no'.

3.1.2.2. SWCMP.Scan.3DVIEW.FlipAngleSweep.tc1.v1

Test ID	22336
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-04 13:28:47 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: IJperen, Gert van (nly95968) on 2015-02-04 10:07:59 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 09:55:45 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

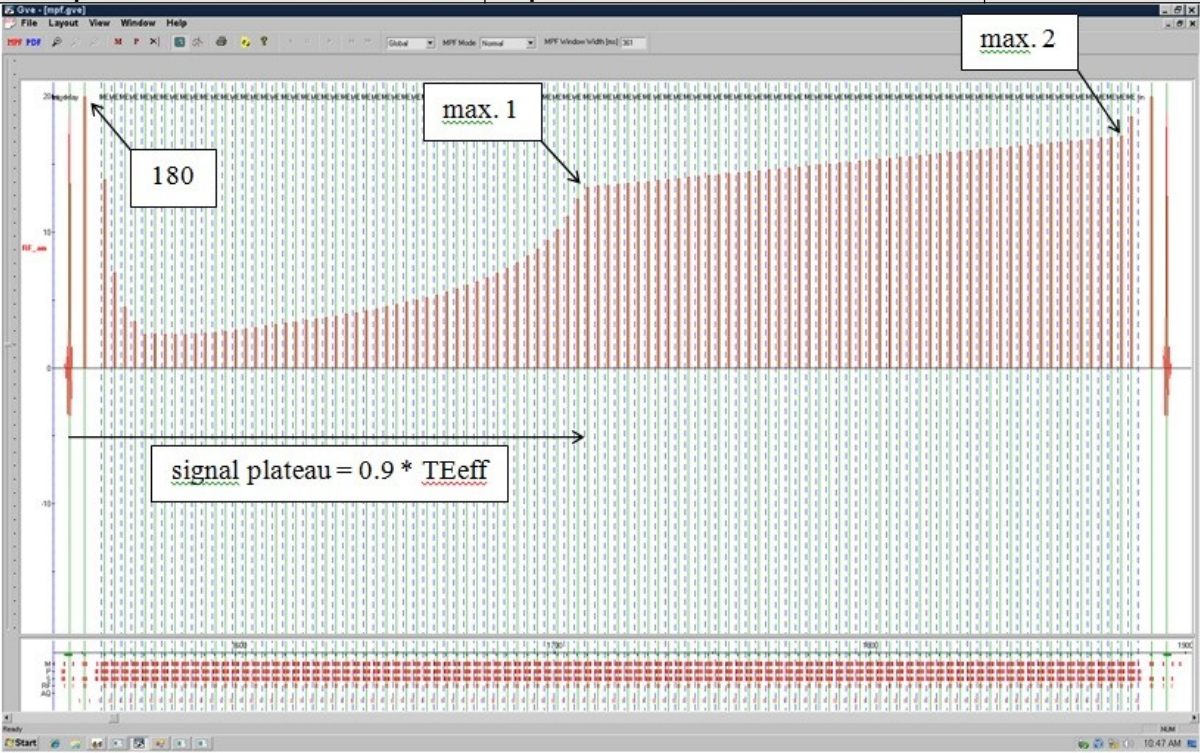
Flip angle sweep for refocusing pulses, enabling a more even distribution of signal strength over the full duration of the TSE shot.

Requirements Coverage:

Req ID	Requirement Tag	Description
21546	SWCMP.Scan.3DVIEW.FlipAngleSweep	3D VIEW uses variable refocusing angles across the TSE shot, tailored for each possible setting of the main parameter 3D VIEW (application + contrast).

Test steps:

Step	Description	Expected Result	Attachments
Step 1	- Use m_amn_set_test_environment.pl and select configuration wa30t - Load scan BrainViewT2 from Head_CNS/TSE - Modify the parameter 3D VIEW from Brain T2 to TEST - Note the values of plateau and TEeff from the INFO page - Run the scan, and visualize a TSE shot in GVE mpf mode, as shown in the attached image	- Verify that the train of refocusing pulses shows a flip angle sweep, starting from 180 deg (the first block pulse), going down to some minimum value in 5 (or more) steps, then slowly increasing towards a first value max. 1, and then increasing linearly towards a second value max. 2. - Estimate the values of max. 1 and max. 2 (by comparing with the first 180 deg), and verify that they are 120 deg and 155 deg respectively. These values correspond to the values of Control Scan parameters 3D VIEW max1 refoc. angle and max2 refoc. angle respectively (last array element (64), for 3D VIEW set to value TEST). Modify these value(s) and verify that the flip angle sweep is adapted accordingly. - Verify that the time from the 90 deg excitation to the max. 1 refocusing pulse is equal to the value of plateau as seen on the INFO page, and is equal to $0.9 * TE_{eff}$. The factor 0.9 corresponds to Control Scan parameter 3D VIEW plateau TE ratio (last array element (64), for 3D VIEW set to value TEST). Modify the value and verify that the flip angle sweep is adapted accordingly.	

Step	Description	Expected Result	Attachments
			

3.1.2.3. SWCMP.Scan.3DVIEW.FreeTurboFactor.tc1.v1

Test ID	22339
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-04 13:28:53 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: IJperen, Gert van (nly95968) on 2015-02-04 10:08:01 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 09:55:47 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:
Any TSE factor value is possible, completely free of discretization or rounding effects, and free of any coupling to acquired P or S matrix size.

Requirements Coverage:

Req ID	Requirement Tag	Description
21549	SWCMP.Scan.3DVIEW.FreeTurboFactor	3D VIEW avoids any discretization effects in the choice of TSE factor. Any coupling between acquired k-space matrix size in Y- or Z-direction and TSE factor is eliminated.

Test steps:

Step	Description	Expected Result
Step 1	- Use m_amn_set_test_environment.pl and select configuration wa30t - Load scan BrainViewT2 from Head_CNS/TSE	- Verify that you can modify the value of the parameter TSE factor from the starting value 100 to values 101, 102, 103, ... and that the value of ACQ voxel MPS (mm) as displayed on the INFO page does not change (it stays at 1.00/1.00/1.00). This is unlike conventional (non-3D VIEW) 3D TSE, where internal discretisations take place, that affect the precise value of ACQ voxel MPS (mm) for a given required voxel size. - Verify that you can modify the value of the parameter TSE factor to values higher than the value of ACQ matrix M x P as displayed on the INFO page (in this case 256). This is unlike conventional (non-3D VIEW) 3D TSE, where TSE factor greater than ACQ matrix P is impossible.
Step 2	Modify turbo direction to Z	- Verify that you can modify the value of the parameter TSE factor from the starting value 100 to values 101, 102, 103, ... This is unlike conventional (non-3D VIEW) 3D TSE, where internal discretisations take place, that generate conflicts between the parameters TSE factor and slices. - Verify that you can modify the value of the parameter TSE factor to values higher than the value of slices. This is unlike conventional (non-3D VIEW) 3D TSE, where TSE factor greater than slices is impossible.

3.1.2.4. SWCMP.Scan.3DVIEW.Nonselective.tc1.v2

Test ID	52729
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:21:41 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:52:36 Meaning: SME Approved - (Technical)

	Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:22 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the use of non-slab selective refocusing pulses

Requirements Coverage:

Req ID	Requirement Tag	Description
21548	SWCMP.Scan.3DVIEW.Nonselective	3D VIEW uses the shortest possible echo spacings across the TSE shot, enabled by non-selective block-shaped refocusing pulses.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScan3dViewNonselective in the results file is zero.

3.1.2.5. SWCMP.Scan.3DVIEW.Par.tc1

Test ID	41291
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:41 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 10:28:27 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:14:47 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of 3D VIEW parameters.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and	The reported number of failed tests related to

Step	Description	Expected Result
	inspect the results file.	SwCmpScan3DViewPar in the results file is zero.

3.1.2.6. SWCMP.Scan.3DVIEW.PartialNSA.tc1.v1

Test ID	22340
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-04 13:28:59 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: IJperen, Gert van (nly95968) on 2015-02-04 10:08:04 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 09:55:51 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

NSA = 2 scans with the 2nd NSA executed partially, limited to central phase encoding steps

Requirements Coverage:

Req ID	Requirement Tag	Description
21550	SWCMP.Scan.3DVIEW.PartialNSA	3D VIEW allows the 2nd NSA in a NSA = 2 scan to be executed partially.

Test steps:

Step	Description	Expected Result
Step 1	- Use m_amn_set_test_environment.pl and select configuration wa30t - Load scan BrainViewT2 from Head_CNS/TSE - Set NSA to 2	Verify that a parameter partial NSA (no, yes) appears (only with 2, not with > 2)
Step 2	Switch between no and yes	Verify that this corresponds to a proportional reduction in scan time.
Step 3	The amount of reduction is controlled by the Control Scan parameter partial 2NSA fraction. Modify the value.	Verify that the scan time reduction effect changes proportionally.

3.1.2.7. SWCMP.Scan.3DVIEW.tc1.v1

Test ID	22334
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Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-04 13:29:06 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: IJperen, Gert van (nly95968) on 2015-02-04 10:08:09 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 09:55:54 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Coverage, isotropic resolution, Scan time

Requirements Coverage:

Req ID	Requirement Tag	Description
45392	SWCMP.Scan.3DVIEW.v1	The technique provides 3D TSE imaging with specific optimisations (.FlipAngleSweep, .Ultrashort, .Nonselective) to enable whole organ coverage with (near) isotropic resolution in clinically acceptable scan time, resulting in whole brain 1 mm ³ coverage in 5:00 min scan time, or equivalent for other applications.

Test steps:

Step	Description	Expected Result
Step 1	- Use m_amn_set_test_environment.pl and select configuration wa30t - Load scan BrainViewT2 from Head_CNS/TSE - Modify parameters such as to reach 1*1*1 mm³ resolution, 200 mm coverage, TR = 2000 msec, using SENSE factor 3.0	- Verify that such a scan has a predicted scantime below 5:00 min scan time - Verify that the scan runs successfully This fulfills the claim "isotropic 1 mm³ whole brain coverage in 5:00 min scan time", even for long TR like 2000 msec, as required for T2 contrast.

3.1.2.8. SWCMP.Scan.3DVIEW.Ultrashort.tc1.v2

Test ID	55188
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2022-07-28 10:07:52 Meaning: QA Approved - (QA)

	Signed by: Sharma, Nishant (320059006) on 2022-07-28 07:37:20 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2022-07-27 10:38:14 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Check the use of the shortest possible echo spacing, enabled by non-equidistant first echo time.

Requirements Coverage:

Req ID	Requirement Tag	Description
21547	SWCMP.Scan.3DVIEW.Ultrashor t	3D VIEW uses the shortest possible echo spacings across the TSE shot, enabled by a slightly larger first (non-equidistant) echo time.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScan3DViewUltrashort in the results file is zero.

3.1.3. ME APT

3.1.3.1. SWCMP.Scan.APT.Design.tc1.v1

Test ID	55206
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thamanna Chandrashekarappa, Deepak Shekhar (320098697) on 2022-07-28 10:12:51 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2022-07-28 07:51:59 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2022-07-27 14:44:09 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The test is interactive using application software installation on a system where the scanner hardware is simulated

Requirements Coverage:

Req ID	Requirement Tag	Description
64372	SWCMP.Scan.APT.SatPulse	APT shall make use of a saturation pulse that alternates playout between amplifiers. Alternating pulse between two amplifiers is used to overcome the rf amplifier duty_cycle constraints.
64373	SWCMP.Scan.APT.ImageType	Scanning shall use a separate image type for communicating APT Weighted images to recon
64379	SWCMP.Scan.APT.Sequence	For APT the TSE-Dixon sequence shall be used for data acquisition.
64380	SWCMP.Scan.APT.AcqVoxelSize	The minimum voxel size shall be limited to 1.5 x 1.5 x 4 mm in the APT imaging sequence to keep acceptable SNR levels.
64384	SWCMP.Scan.APT.FatSup	Fat suppression (SPIR) shall be supported with APT
64860	SWCMP.Scan.APT.SatPulseDur	Duration of saturation pulse shall be editable in the UI. The default shall be 2s.
64861	SWCMP.Scan.APT.PulseStrength	The B1rms of the saturation pulse shall be editable in the UI. The default shall be 2uT.
68841	SWCMP.Scan.APT.Restrictions	The following scan-protocol parameters shall be restricted for APT 1) SENSE acceleration in phase direction is restricted to the range 1 - 1.6 2) SENSE acceleration in slice direction is restricted to 1 3) Gradient mode is restricted to 'Regular' 4) SofTone mode is restricted to 'ComforTone' 5) Fat Suppression is restricted to SPIR

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanAptDesign in the results file is zero.

3.1.3.2. SWCMP.Scan.APT.SatPulseShimming.tc1

Test ID	38971
Revision	1
Product Model	All
Gradient System	
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-04-29 16:17:02 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Verburg, Frank (nly95979) on 2019-04-29 15:47:34 Meaning: SME

	Approved - (Business)
	Signed by: Hey, Silke (300170272) on 2019-04-26 17:07:10 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test covers the separation of RF shimming settings for imaging and APT saturation pulses, by inspecting the log-file for B1 shim requests for each of the two modes.

Requirements Coverage:

Req ID	Requirement Tag	Description
72339	SWCMP.Scan.APT.SatPulseRF Shimming	For APT, the system shall employ RF shimming for the saturation pulses separately from the imaging RF pulses.

Test steps:

Step	Description	Expected Result
Prep 1	1. Use the test framework to set configuration as 3T Ingenia 2. Set the simulation mode to 'Acq at host, Recon at Reconstructor' 3. Enable PrescanOptimizer logging trace by opening bsmglog.dat and adding the following line under the [default] header: PrescanOptimizer.*.trace = system 4. Start a new logfile. 5. Open application SW and create a new examination.	Not applicable, prep only
Step 2	1. Select a protocol with RF shimming and APT: [Hospital DB]\Head\Brain\APT\APT_10slices. 2. Open the protocol for editing. 3. On the Geometry tab, disable SENSE.	The protocol should load and validate without conflicts
Step 3	Execute scan	Scan completes without issue
Step 4	• Inspect logfile using the following query: <i>(process = 'oc-scan-engine' and text contains 'Response is available for request') or text contains 'm_Name'</i> • Attach the logfile as objective evidence.	Logfile shows repeated logging of responses for pairs of B1shimrequests. There are requests for imaging transmit/receive and requests for alterate mode transmit/receive.
cleanup	Remove the log trace from bsmglog.dat .	N.a. cleanup only.

3.1.3.3. SWCMP.Scan.APT.UI.tc1.v2

Test ID	51125
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-06-11 09:04:28 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Mariappan, Yogesh kannan (310134237) on 2021-06-11 08:12:35 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:17 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability and verification of APT UI parameters.

Manual Test cases covered :

SWCMP.Scan.APT.UI.tc1

Requirements Coverage:

Req ID	Requirement Tag	Description
64371	SWCMP.Scan.APT.UI	APT shall be enabled by the parameter "APT" in the Contrast tab

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanAptUI in the results file is zero.

3.1.4. ME ASL**3.1.4.1. SWCMP.Scan.ASL.Par.Multiphase.tc1**

Test ID	26474
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-15 13:50:15 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-

	14 11:02:08 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-14 08:53:52 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the multiphase UI parameters for STAR and STAR Angio are as specified and are not available for pCASL.

Requirements Coverage:

Req ID	Requirement Tag	Description										
57577	SWCMP.Scan.ASL.Par.Multiphase	The following UI parameters will appear on the Dyn/Ang tab for STAR or STAR Angio: <table><tr><th><i>parameter name</i></th><th><i>description and allowed values.</i></th></tr><tr><td>phases</td><td>Number of phases acquired within the cycle duration.</td></tr><tr><td>duration (ms)</td><td>Cycle duration: Repetition time between consecutive labels. (only available when phases>1).</td></tr><tr><td>phase interval</td><td>Time between consecutive phases. (only available when phases>1). range: shortest, user defined, or longest shortest = minimal shot duration longest = acquisition duration / phases</td></tr><tr><td>(ms)</td><td>User defined choice of the phase interval (only available when phases>1 and phase interval = 'user defined').</td></tr></table>	<i>parameter name</i>	<i>description and allowed values.</i>	phases	Number of phases acquired within the cycle duration.	duration (ms)	Cycle duration: Repetition time between consecutive labels. (only available when phases>1).	phase interval	Time between consecutive phases. (only available when phases>1). range: shortest, user defined, or longest shortest = minimal shot duration longest = acquisition duration / phases	(ms)	User defined choice of the phase interval (only available when phases>1 and phase interval = 'user defined').
<i>parameter name</i>	<i>description and allowed values.</i>											
phases	Number of phases acquired within the cycle duration.											
duration (ms)	Cycle duration: Repetition time between consecutive labels. (only available when phases>1).											
phase interval	Time between consecutive phases. (only available when phases>1). range: shortest, user defined, or longest shortest = minimal shot duration longest = acquisition duration / phases											
(ms)	User defined choice of the phase interval (only available when phases>1 and phase interval = 'user defined').											

Test steps:

Step	Description	Expected Result
Preparation	Switch off Option 010, set the scan protocol database to TEDB. Start the Examcards.	N.A.
Step 1	Drag the protocol 'Head CNS / FLL / STAR_TS' to the scan list and open it in the parameter editor. Set the reference tissue to 'Grey matter'. Go to the 'DYN/Ang' Tab.	The parameters 'phases' is visible.
Step 2	Change the value of 'phases' to '5'.	The parameters 'duration' and 'phase interval' are visible. The available values for 'phase interval' are 'shortest', 'longest' and 'user defined'.
Step 3	Change the value of 'phase interval' to 'user defined'	The parameter '(ms)' is visible.
Step 4	Drag the protocol 'Head CNS / FLL / STAR_ANGIO' to the scan list and open it in the parameter editor. Go to the 'DYN/Ang' Tab.	The parameters 'phases', 'duration', 'phase interval' and '(ms)' are visible. The available values for 'phase interval' are 'shortest', 'longest' and 'user defined'.
Step 5	Drag the protocol 'Head CNS / FLL / pCASL_TS' to the scan list and open it in the parameter editor. Go to the 'DYN/Ang' Tab.	The parameters 'phases', 'duration', 'phase interval' and '(ms)' are not visible.

3.1.4.2. SWCMP.Scan.ASL.Par.SourceImagesChoice.tc1

Test ID	26475
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-15 13:50:18 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-14 11:02:11 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-14 08:53:55 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the source image selection parameter is available and offers the specified values.

Requirements Coverage:

Req ID	Requirement Tag	Description
56671	SWCMP.Scan.ASL.SourceImage sChoice	ASL shall provide a choice for saving the source images: <i>parameter name</i> <i>description and allowed values.</i>
		ASL source images range: None, Last Control, All None = No source images Last Control = images from the last phase and dynamic of the control images All = All source images (all label and control and all phases and all dynamics).

Test steps:

Step	Description	Expected Result
Preparation	Switch off Option 010, set the scan protocol database to TEDB. Start the Examcards.	N.A.
Step 1	Drag the protocol 'Head CNS / FLL / STAR_TS' to the scan list and open it in the parameter editor. Set the reference tissue to 'Grey matter'. Go to the 'Postproc' Tab.	The parameters 'ASL source images' is visible. The available values are 'None', 'Last Control' and 'All'.
Step 2	Drag the protocol 'Head CNS / FLL / STAR_ANGIO' to the scan list and open it in the parameter editor. Go to the 'Postproc' Tab.	The parameters 'ASL source images' is visible. The available values are 'None', 'Last Control' and 'All'.
Step 3	Drag the protocol 'Head CNS / FLL / pCASL_TS' to the scan list and open it in the parameter editor. Go to the 'Postproc' Tab.	The parameters 'ASL source images' is visible. The available values are 'None', 'Last Control' and 'All'.

3.1.4.3. SWCMP.Scan.ASL.Par.tc1.v1

Test ID	26217
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-04 13:11:35 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Engelen, Walter van (nly95948) on 2016-02-29 14:46:30 Meaning: SME Approved - (Business) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-02-26 15:51:37 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies if all required ASL techniques are available at the UI.

Test steps:

Step	Description	Expected Result
Preparation	Switch off Option 010, set the scan protocol database to TEDB and start examcards. Open the 'HEAD_CNS\FLL\STAR_TS' in the parameter editor.	N.A.
Step 1	Go to the "dyn/ang" tab and select the "Arterial Spin labeling" parameter.	"STAR", "pCASL" and "STAR Angio" are available.

3.1.4.4. SWCMP.Scan.ASL.Patent.tc1

Test ID	28399
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-06-07 12:23:30 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2016-06-07 10:33:16 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-06-06 17:38:34 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies that certain combinations of ASL, 3D GRASE, profile orders and NSA's are not allowed.

It uses a protocol with pCASL and 3D GRASE as a starting point.

Requirements Coverage:

Req ID	Requirement Tag	Description
60952	SWCMP.Scan.ASL.Patent	The following scan parameter combination is not allowed in combination with any blood labeling scan (ASL): • Scan mode = 3D • Fast imaging mode = GRASE • profile order = anything other than low-high • NSA > 1

Test steps:

Step	Description	Expected Result
Preparation	Make sure Option 010 is enabled, set the scan protocol database to TEDB and start examcards. Open the 'HEAD_CNS\FLL\3D_ASL' in the parameter editor.	N.A.
Step 1	Go to the "Contrast" tab and set the "profile order" parameter to "low_high". Go to the "Motion" tab and set the "NSA" parameter to "2".	The protocol is not in conflict.
Step 2	Go to the "Contrast" tab and set the "profile order" parameter to "linear".	The protocol is in conflict.
Step 3	Set the "profile order" parameter back to "low_high". Set the "profile order" parameter back to "asymmetric".	The protocol is in conflict.
Step 4	Set the "profile order" parameter back to "low_high". Go to the "Dyn/Ang" tab and set the "Arterial Spin labeling" parameter to "STAR".	The protocol is not in conflict.
Step 5	Go to the "Contrast" tab and set the "profile order" parameter to "linear".	The protocol is in conflict.
Step 6	Set the "profile order" parameter back to "low_high". Set the "profile order" parameter back to "asymmetric".	The protocol is in conflict.
Step 7	Set the "profile order" parameter back to "low_high". Go to the "Dyn/Ang" tab and set the "Arterial Spin labeling" parameter to "TIMESLIP".	The protocol is not in conflict.
Step 8	Go to the "Contrast" tab and set the "profile order" parameter to "linear".	The protocol is in conflict.
Step 9	Set the "profile order" parameter back to "low_high". Set the "profile order" parameter back to "asymmetric".	The protocol is in conflict.

3.1.4.5. SWCMP.Scan.ASL.pCASL.3DReadout

Test ID	25219
Revision	1
Product Model	Any
Gradient System	Any

Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-04 13:11:37 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-04 13:06:04 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2016-03-02 09:45:55 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies that the pCASL can be run with 3D multi-shot segmented grase.

Requirements Coverage:

Req ID	Requirement Tag	Description
56076	SWCMP.Scan.ASL.pCASL.3DReadout	The system shall support the acquisition of images acquired with pCASL flow labeling in combination with a segmented 3D GraSE readout.

Test steps:

Step	Description	Expected Result
preparation	Switch off Option 010, set the scan protocol database to TEDB and start examcards. Create a new exam.	N.A.
Step 1	Select the scan 'Head CNS / FLL / 3D_pCASL_qASL'. Ensure that the 'scan mode' is '3D', 'Fast Imaging Mode' is 'GraSE', 'shot mode' is 'multishot' and 'Arterial Spin labeling' is 'pCASL'. Run the scan.	The scan shall be successfully performed without unsolvable conflicts.

3.1.4.6. SWCMP.Scan.ASL.pCASL.Control.tc1

Test ID	26301
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-07 16:46:05 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-07 16:27:56 Meaning: SME Approved - (Business)::Bulk Transitions
	Signed by: Graaf, Ruud de (nly95749) on 2016-03-07 15:08:39 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the pCASL control pulse train is the same as the pCASL label pulse train except for the 180 degree phase increment on every second pulse.

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125876).

Requirements Coverage:

Req ID	Requirement Tag	Description
58928	SWCMP.Scan.ASL.pCASL.Contr ol.v1	pCASL must apply as a control module the same train of RF pulses applied at the labeling module at the labeling plane, but the phase increment must be 180 between RF pulses so that the spins experience no net inversion.

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_pCASL.pl - m_imn_asl_ts_pCASL.acq to the patch directory. Run the script 'm_imn_asl_ts_pCASL.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Compare the files in 'G:\log' named m_imn_asl_ts_pCASL_<configuration>_train1.dat and m_imn_asl_ts_pCASL_<configuration>_train2.dat for all available configurations. These are the object attributes for the first and the second labelling train of the first scan.	The attributes for the pulses are the same with the exception of the rf_phase_step1 of the RF::pCASL_1_ objects. The rf_phase_factor1 * rf_phase_step1 for the RF::pCASL_1_ objects should differ by 180 when comparing the two RF pulse trains.

3.1.4.7. SWCMP.Scan.ASL.pCASL.Cycle.tc1

Test ID	26216
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-07 16:46:11 Meaning: QA

	Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-07 16:28:01 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-07 15:08:42 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the pCASL scan technique uses cycles. It verifies that on consecutive cycles, the pCASL pulse train switches between CONTROL and LABEL type and that labels are send to reconstruction to indicate the type of RF pulse train.

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125876).

Requirements Coverage:

Req ID	Requirement Tag	Description
57241	SWCMP.Scan.ASL.Cycle.v1	The ASL acquisition scheme shall consist of cycles, which contain a labelling step and one or more acquisition steps. The labelling step contains a control or a label pulse. The system will switch between control and label on consecutive cycles.
58922	SWCMP.Scan.ASL.Reconstructi on_Label.v1	MR-Methods shall provide labels for the reconstruction software in order to distinguish between label and control images (Requirements - Figure 5).

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_pCASL.pl - m_imn_asl_ts_pCASL.acq to the patch directory. Run the script 'm_imn_asl_ts_pCASL.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Inspect the files 'G:\log' named m_imn_asl_ts_pCASL_<configuration>_cycle.dat for all available configurations.	Each scan consists of multiple cycles. Each cycle contains a pCASL train and an AQ. There are two types of pCASL pulse trains: - A train with pulses which all have the same relative phase (=LABEL) • A train where the relative phase differs by 180 on each consecutive pulse (=CONTROL). On consecutive cycles the pCASL pulse train switches between the LABEL and the CONTROL

Step	Description	Expected Result
		type.
Step 2	Check the Reconstruction_label values for all AQ's.	The CONTROL type pulse trains are trailed by an AQ with the reconstruction label 0. The LABEL type pulse trains are trailed by an AQ with the reconstruction label 1.

3.1.4.8. SWCMP.Scan.ASL.pCASL.Label.tc1

Test ID	26299
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-22 10:03:55 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-22 09:39:41 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Polders, Daniel Louis (310171444) on 2016-03-22 08:42:05 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the pCASL label pulse train is as specified.

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125876).

Requirements Coverage:

Req ID	Requirement Tag	Description
58927	SWCMP.Scan.ASL.pCASL.Label.v1	pCASL must apply as a labeling module a train of excitation pulses at the labeling plane to label the blood through a process which mimics flow-driven adiabatic inversion (Requirements - Figure 3). The ratio between the selection gradient and the gradient average shall not be smaller than 10 ($G/G_{av} \geq 10$) and the ratio between pCASL interval and RF pulse duration shall not be bigger than 2.60 ($t / < 2.60$). The AFP shall also be fulfilled for a mean blood velocity v equal to 40 cm/s ($1 / T2_{blood} < G_{av} * v / B1_{av} < * B1_{av}$).

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging:	N.A.

Step	Description	Expected Result
	- m_imn_asl_ts_pCASL.pl - m_imn_asl_ts_pCASL.acq to the patch directory. Run the script 'm_imn_asl_ts_pCASL.pl' from the command line. Wait for the script to finish.	
Step 1	Inspect the files in 'G:\log' named m_imn_asl_ts_pCASL_<configuration>_cycle.dat for all available configurations.	Each pCASL train consists out of multiple pulses
Step 2	Inspect the files in 'G:\log' named m_imn_asl_ts_pCASL_<configuration>.dat for all available configurations. Find the first GR::pCASL_0_ and GR::pCASL_1_ objects in the log files. The selection gradient strength (G) is equal to the value of gr_str of the GR::pCASL_0_ object. Calculate the gradient average (Gav) by adding ((gr_dur - gr_slope)*gr_str*gr_repetitions) of GR::pCASL_0_ and GR::pCASL_1_ and dividing this by sq_dur of SQ::pCASL. Determine the pCASL interval by subtracting the rf_time of RF::pCASL_0_ from the rf_time of RF::pCASL_1_. Determine the pulse duration by inspecting the rf_dur of RF::pCASL_0_. Determine the B1 average (B1avg) by dividing rf_b1*rf_teff of RF::pCASL_0_ through the pCASL interval.	G/Gav > 10. pCASL interval / pulse duration < 2.60. $10 < Gav * 1000 * 0.40 / B1avg < 267.522 * B1avg$

3.1.4.9. SWCMP.Scan.ASL.pCASL.Par.tc1.v1

Test ID	25218
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Gorissen, Ben (nly14691) on 2016-08-11 15:19:10 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-08-11 11:34:33 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-08-11 09:03:32 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the parameters for pCASL are visible.

Test steps:

Step	Description	Expected Result
Preparation	Switch off Option 010, set the scan protocol database to TEDB and start examcards. Open the 'HEAD_CNS\FLL\pCASL_TS' protocol in the parameter editor.	N.A.
Step 1	Go to the Contrast tab. Ensure that "Scan mode" is set to "MS". Go to the DYN/Ang Tab. Ensure that "Arterial Spin Labeling" is set to "pCASL". Change the "label duration" to "1800".	The parameters "label distance (mm)" and "post label delay (ms)" are visible. The parameter "normalized" is not visible.
Step 2	Set the label distance to the minimal value using the decrement button. Go to the Geometry tab and increase the FOV FH with a value between 10 and 15mm.	A conflict is raised on the 'label distance (mm)' and on 'slices'.
Step 3	Increase the 'label distance' by clicking on it.	The conflict is resolved. The value is changed to the previous value with half of the increase of the FOV added ($\pm 0.5\text{mm}$).
Step 4	Set the value of the 'label distance' to '1000'	The parameter value is set to 300mm ($\pm 0.5\text{mm}$).
Step 5	Set the value of the 'post label delay' to '300'	The parameter value is set to 800ms ($\pm 0.5\text{ms}$).
Step 6	Change the 'TR' to shortest on the 'Contrast' tab. Set the value of the post label delay to '3000' on the 'Dyn/Ang' tab.	The parameter value should be set to 2500ms ($\pm 0.5\text{ms}$).
Step 7	Open the 'HEAD_CNS\FLL\3D_pCASL_qASL' protocol in the parameter editor. Go to the Contrast tab. Ensure that "Scan mode" is set to "3D". Go to the Dyn/Ang Tab.	The parameters "label distance (mm)", "post label delay (ms)" and "normalized" are visible. The allowed values for "normalized" are 'yes' and 'no'.

3.1.4.10. SWCMP.Scan.ASL.pCASL.Par.tc2.v2

Test ID	52730
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:22:22 Meaning: QA Approved - (QA)::Bulk

	Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-16 09:57:02 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:22 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the sequence timing for the pCASL label delay is correct.

Requirements Coverage:

Req ID	Requirement Tag	Description				
87632	SWCMP.Scan.Validator.ASL.Par.v1	Parameter name	Range	Visible when	Description	System / Option restrictions
		Arterial Spin Labeling	no FAIR STAR pCASL Time-SLIP STAR Angio	Always	Enables user to select one of the Arterial Spin Labeling techniques.	Application specific restrictions are imposed by the following options: M_OPT.A WOPT_ARTERIAL_SPIN_LABEL (option bit 35) enables STAR ASL technique M_OPT.A WOPT_PCASL (option bit 78) enables pCASL ASL technique

Req ID	Requirement Tag	Description					
						M_OPT.A WOPT_OPTION10 (option bit 501) enables FAIR and Time-SLIP ASL techniques M_OPT.A WOPT_CINEMA (option bit 109) enables STAR Angio ASL technique	
		label type	If pCASL or STAR Angio - parallel slab - free slab If Time-SLIP - parallel slab - free slab - pencil beam	Arterial Spin Labeling = pCASL, STAR Angio or Time-SLIP	The label type defines the volume used to label.	Label type "free slab" will be available with pCASL only under the availability of the option: M_OPT.A WOPT_OPTION10 (option bit 501)	
		Label Offc. AP(P = +mm)	-220 - 220	label type = free slab or pencil beam	Determines the position of the label volume as the	N.A	

Req ID	Requirement Tag	Description				
					distance to a central plane.	
		Label Offc. RL(L = +mm)	-220 - 220	label type = free slab or pencil beam	Determines the position of the label volume as the distance to a central plane.	N.A
		Label Offc. FH(H = +mm)	-220 - 220	label type = free slab or pencil beam	Determines the position of the label volume as the distance to a central plane.	N.A
		Label Ang. AP(deg)	-180 - 180	label type = free slab or pencil beam	Determines the angulation of the current volume around the AP axis	N.A
		Label Ang. RL(deg)	-180 - 180	label type = free slab or pencil beam	Determines the angulation of the current volume around the AP axis	N.A

Req ID	Requirement Tag	Description				
		Label Ang. FH(deg)	-180 - 180	label type = free slab or pencil beam	Determines the angulation of the current volume around the FH axis	N.A
		orientation	transverse sagittal coronal	label type = free slab	Determines the orientation of the labeling slab.	N.A.
		diameter (mm)	5 - 300	label type = pencil beam	Controls the diameter of the labeling beam.	N.A.
		beam direction	AP RL FH	label type = pencil beam	Direction of the long axis of the pencil beam	N.A.
		label thickness (mm)	1 - 300	Arterial Spin Labeling = STAR , STAR Angio, FAIR or Time-SLIP	Thickness of the labeling slab.	N.A.
		label gap (mm)	0 - 200	Arterial Spin Labeling = STAR, STAR Angio or Time-SLIP	Gap between the imaging slab and the labeling	N.A.

Req ID	Requirement Tag	Description				
				^ label type = parallel slab	slab.	
		label distance (mm)	min value - 300	Arterial Spin Labeling = pCASL	Distance between Label and Imaging volume. The min. value is calculated from the image volume size to prevent saturation of the first slices.	N.A.
		post label delay (ms)	The actual range depends on other paramete rs.	Arterial Spin Labeling = pCASL	Time between the end of the continuou s labeling and the beginning of the readout.	N.A.
		label delay (ms)	The actual range depends on other paramete rs	Arterial Spin Labeling = STAR, STAR Angio or Time-SLIP	Time between the single labeling pulse and the beginning of the readout.	N.A.
		label location	F, H, R, L, A or P dependin g on the	Arterial Spin Labeling = pCASL or	Determin es the position of the	R,L, A,P label locations and

Req ID	Requirement Tag	Description				
			slice orientation (Transverse, Sagittal or Coronal)	Time-SLIP Û Label type = parallel slab	label with respect to the imaging stack.	Sagittal and Coronal slice orientations will require M_OPT.A WOPT_OPTION10 (option bit 501)
		label duration (ms)	0 - 5000	Arterial Spin Labeling = pCASL	Duration of the labeling RF pulse.	The parameter will be available when M_OPT.A WOPT_OPTION10 (option bit 501)
		vascular crushing	no, yes	Arterial Spin Labeling = pCASL	Reduce the vascular artifacts by dephasing the signal from label present at the time of imaging.	The parameter will be available when M_OPT.A WOPT_OPTION10 (option bit 501)
		backgr. suppression	no, user defined, auto 2 pulse, auto 4 pulse	Arterial Spin Labeling = pCASL	Suppress signals from background tissue.	For selection of background suppression other than "auto 2 pulse", software

Req ID	Requirement Tag	Description				
						option M_OPT.A WOPT_OPTION10 (option bit 501) is required.
		backgr. suppr. pulses (ms)	0 - 10000	backgr. suppression = user defined	Timing of excitation of background pulses.	N.A.
		normalized	no, yes	Arterial Spin Labeling = pCASL	Enables normalization of the perfusion images with a proton density image.	The parameter will be available when M_OPT.A WOPT_PCASL_NORMALIZED (option bit 117) is enabled for Scan mode = 3D For Scan mode = 2d or MS additionally software option M_OPT.A WOPT_OPTION10 (option bit 501) is required.
		phases	1-50	All Arterial Spin Labeling	Number of phases acquired within	If ASL = pCASL then the parameter

Req ID	Requirement Tag	Description				
				techniques	the cycle duration.	r will be available only when M_OPT.A WOPT_OPTION10 (option bit 501) is enabled.
		duration (ms)	0-10000	phases > 1	Cycle duration: Repetition time between consecutive labels.	N.A.
		phase interval	shortest user defined longest	phases > 1	Time between consecutive phases.	N.A
		(ms)	The actual range depends on other parameters	phase interval = user defined	User defined choice of the phase interval.	N.A
		pulse type	wow_ir hypsec	Arterial Spin Labeling = FAIR	Type of the inversion pulse.	N.A
		adjust FAIR ref	no, yes	Arterial Spin Labeling = FAIR ^ Dynamic study = individual	This parameter defines whether the dynamic scans are performed in combination with	N.A

Req ID	Requirement Tag	Description				
					adjusted widths of the inverted regions in the reference images for FAIR ASL.	
		adjust value	0.1 - 100	adjust FAIR ref = yes	It is used to decrease the inversion width of the reference images as a function of the dynamic scans.	N.A
		mask	no, yes	Arterial Spin Labeling = Time-SLIP	Enables acquisition of mask image.	N.A
		inversion scheme	nonselect - select - nonselect selective only	Arterial Spin Labeling = Time-SLIP	Define the order of the two inversion pulses.	N.A
		backgr. suppr. delay (ms)	longest user defined	Inversion scheme = select - nonselect	Delay between the second inversion pulse and image acquisition.	N.A

Req ID	Requirement Tag	Description
		In addition software option M_OPT.AWOPT_OPTION10 shall allow the following combinations: 1. Scan technique != FFE EPI ^ ASL = pCASL 2. Scan mode = 2D ^ ASL = pCASL 3. Scan technique != GRASE ^ scan mode = 3D ^ ASL = pCASL 4. Fat suppression = SPAIR ^ ASL = pCASL 5. Scan technique = SE EPI ^ ASL = STAR 6. Scan technique = FFE TFE ^ ASL = STAR 7. Scan technique = FFE TFE ^ ASL = STAR Angio

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanAsIPcaslPar in the results file is zero.

3.1.4.11. SWCMP.Scan.ASL.pCASL.Presat.tc1

Test ID	26298
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-07 16:46:18 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-07 16:28:08 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-07 15:08:47 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the pCASL scan technique contains a WET presat module and that this WET presat module is as specified .

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125876).

Requirements Coverage:

Req ID	Requirement Tag	Description
58930	SWCMP.Scan.ASL.Presat.v1	ASL must have a WET pre saturation module (Requirements - Figure 2), applied at the imaging plane before labeling/control module. The interval (t) between the RF pulses and the flip angle of each one of them () must be

Req ID	Requirement Tag	Description
		optimized according to field strength for brain perfusion. RF pulses shall be selective (selective gradient is different from 0) and: for 1.5T, t = 10.0 ms and 1 = 106.8, 2 = 86.8, 3 = 76.0, 4 = 153.6 degrees, and for 3.0T, t = 8.0 ms and 1 = 89.17, 2 = 98.37, 3 = 82.31, 4 = 156.77 degrees.

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_pCASL.pl - m_imn_asl_ts_pCASL.acq to the patch directory. Run the script 'm_imn_asl_ts_pCASL.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Inspect the files in 'G:\log' named m_imn_asl_ts_pCASL_<configuration>_cycle.dat for all available configurations.	Each pCASL train is preceded by a WET pulse.
Step 2	Inspect the files in 'G:\log' named m_imn_asl_ts_pCASL_<configuration>.dat for all available configurations. Find the first RF::wet object in the log files. Calculate the interval t between the pulses by adding the values of rf_dur and rf_pause. Inspect the value of the rf_angle. Find the first GR::presat_s_0_ object in the log files. Inspect the gradient strength (gr_str).	The interval is as specified for the field strength. The angle is as specified for the field strength. The gradient strength is different from 0.

3.1.4.12. SWCMP.Scan.ASL.pCASL.Shim.tc1

Test ID	26302
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-07 16:46:21 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-07 16:28:11 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-07 15:08:50 Meaning: SME Approved - (Technical)::Bulk

	Transitions
Approval Status	Approved

Description:

This test verifies that the AS prep phase is repeated when the labeling plane location is changed. It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125876).

The script executes 2 scans. The second scan has a different labeling location compared to first scan.

Requirements Coverage:

Req ID	Requirement Tag	Description
58929	SWCMP.Scan.ASL.pCASL.Shim.v1	Shimming must always be performed in the labeling plane and preparation phase must be repeated if the labeling plane changes position. The AS prep phase shall be repeated if the labeling plane location is changed.

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_pCASL.pl - m_imn_asl_ts_pCASL.acq to the patch directory. Run the script 'm_imn_asl_ts_pCASL.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Inspect the files 'G:\log' named m_imn_asl_ts_pCASL_<configuration>_cycle.dat for all available configurations.	Just before the start of the second scan (search 'Scan starts') a message is given that 'AS' will be performed.

3.1.4.13. SWCMP.Scan.ASL.pCASL.SPD.tc1.v1

Test ID	32566
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 16:57:24 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Prins, Wim (nly95928) on 2017-03-23 16:21:13 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-22 14:43:33 Meaning: SME Approved - (Technical)

Approval Status	Approved
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Description:

The Spin Proton Density (SPD) image is acquired as the first dynamic when 'normalized' is set to yes. The SPD image doesn't contain any labeling or control pulses and no background suppression in the imaging volume.

This test verifies that the image of the first dynamic has a considerable larger signal than in the other dynamics when 'normalized' is set to yes.

Requirements Coverage:

Req ID	Requirement Tag	Description
56077	SWCMP.Scan.ASL.pCASL.SPD	The system shall support the acquisition of SPD images in which the pCASL labeling pulses and background suppression pulses are disabled.

Test steps:

Step	Description	Expected Result
Preparation	Switch off Option 010, set the scan protocol database to TEDB and start the MR application. Create a new exam. Connect a head coil and put in a 3l bottle phantom, which is filled with the liquid described below. Select the scan 'Head CNS / FLL / Survey' and run it. The phantom liquid shall consist of demi water with the following additives (per 1000ml demi water): + 770mg CuSO4 5H2O + 2000mg NaCl + 1ml arquad (1% solution) + 0.15ml H2SO4 (0.1N solution) Note: The 3l bottle in the standard 1.5T phantom set should match the phantom liquid specification.	N.A. (preparation)
Step 1	Select the scan 'Head CNS / FLL / 3D pCASL qASL' and plan it on the phantom. Set 'ASL source images' to 'All' on the 'Postproc' tab. Run the scan. Measure the signal in the middle slice for each dynamic in the second series.	The signal of the 1st dynamic is at least 3 times higher than the signal of the subsequent dynamics.
Step 2	Select the scan 'Head CNS / FLL / 3D pCASL qASL' and plan it on the phantom. Set 'normalized' to 'no' on the 'Dyn/Ang' tab. Set 'ASL source images' to 'All' on the 'Postproc' tab. Run the scan. Measure the signal in the middle slice for each dynamic in the second series.	The signal of the 1st dynamic is less than 2 times higher than the signal of the subsequent dynamics.

3.1.4.14. SWCMP.Scan.ASL.STAR.Control.tc1.v2

Test ID	52731
Revision	1
Product Model	Any

Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:22:24 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:59:59 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:22 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the control module for STAR and STAR Angio scan techniques are as specified .

Requirements Coverage:

Req ID	Requirement Tag	Description
58924	SWCMP.Scan.ASL.STAR.Contro l.v1	STAR must apply as a control module a unique adiabatic inversion pulse with frequency sweep reversing halfway at the labeling plane so that the spins experience no net inversion (Requirements - Figure 4b). The adiabatic condition for the inversion pulse shall be fulfilled ($B1 > BW / \gamma$).

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanAslStarControl in the results file is zero.

3.1.4.15. SWCMP.Scan.ASL.STAR.Cycle.tc1

Test ID	26303
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-10 15:33:49 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-10 11:59:53 Meaning: SME Approved - (Business)::Bulk Transitions

	Signed by: Graaf, Ruud de (nly95749) on 2016-03-09 09:13:50 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the STAR and STAR angio scan technique uses cycles. It verifies that on consecutive cycles, the system switches between CONTROL and LABEL type and that labels are sent to reconstruction to indicate the type of RF pulse.

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125877).

Requirements Coverage:

Req ID	Requirement Tag	Description
57241	SWCMP.Scan.ASL.Cycle.v1	The ASL acquisition scheme shall consist of cycles, which contain a labelling step and one or more acquisition steps. The labelling step contains a control or a label pulse. The system will switch between control and label on consecutive cycles.
58922	SWCMP.Scan.ASL.Reconstruction_Label.v1	MR-Methods shall provide labels for the reconstruction software in order to distinguish between label and control images (Requirements - Figure 5).

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_star.pl - m_imn_asl_ts_star.acq to the patch directory. Run the script 'm_imn_asl_ts_star.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Inspect the files 'G:\log' named m_imn_asl_ts_star_<configuration>_cycle.dat for all available configurations.	Each scan consists of multiple cycles. Each cycle contains a RF::star pulse and one or more AQ's. The system switches between the RF::star_0_ and the RF::star_1_ pulse on consecutive cycles.
Step 2	Check the Reconstruction_label values for all AQ's.	The RF::star_0_ pulses are trailed by AQ's with the reconstruction label 0. The RF::star_1_ pulses are trailed by AQ's with the reconstruction label 1.

3.1.4.16. SWCMP.Scan.ASL.STAR.Label.tc1.v2

Test ID	52732
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:22:26 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:00:51 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:22 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the labeling module for STAR and STAR Angio scan techniques.

Requirements Coverage:

Req ID	Requirement Tag	Description
58923	SWCMP.Scan.ASL.STAR.Label.v1	STAR must apply as a labeling module a single adiabatic inversion pulse at the labeling plane (Requirements - Figure 4a). The adiabatic condition for the inversion pulse shall be fulfilled ($B1 > BW / \gamma$).
57017	SWCMP.Scan.ASL.STARAngio.LabelTechnique	STAR Angio shall use the STAR labelling technique.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanAslStarLabel in the results file is zero.

3.1.4.17. SWCMP.Scan.ASL.STAR.Par.tc1

Test ID	26238
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-07 16:46:25 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Engelen, Walter van (nly95948) on 2016-03-07 09:52:08 Meaning: SME Approved - (Business) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-04 13:06:08 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the UI parameters for STAR and STAR Angio are as specified.

Test steps:

Step	Description	Expected Result
Preparation	Switch off Option 010, set the scan protocol database to TEDB. Enable the scan simulation. Start the MR application. Create an examination. Drag the protocol 'Head CNS / FLL / SURVEY' into the scanlist and start the scan. Wait until the scan is completed.	N.A.
Step 1	Drag the protocol 'Head CNS / FLL / STAR_TS' to the scan list and open it in the parameter editor. Go to the DYN/Ang Tab.	The parameters "label thickness", "label delay" and "label gap" are visible.
Step 2	Drag the protocol 'Head CNS / FLL / STAR_Angio' to the scan list and open it in the parameter editor. Go to the 'DYN/Ang' Tab.	The parameters "label type", "label thickness" and "label delay" are visible.
Step 3	Set "label type" to "free slab".	The parameter "orientation" is visible. The label offcenter and angulation parameters on the 'Offc/Ang' tab are visible.
Step 4	Change the orientation and angulation of the slab in Planscan.	The values of the label offcenter and angulation parameters on the 'Offc/Ang' tab are changed.
Step 5	Set "label type" to "parallel slab"	The parameter "label gap" on the 'Dyn/Ang' tab is visible. The parameter "orientation" on the 'Dyn/Ang' tab is not visible. The label offcenter and angulation parameters on the 'Offc/Ang' tab are not visible.

3.1.4.18. SWCMP.Scan.ASL.STAR.Postsat.tc1

Test ID	26305
Revision	1
Product Model	Any
Gradient System	Any

Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-10 15:33:55 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-10 11:59:57 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-09 09:13:55 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the STAR and STAR Angio scan techniques contain a post saturation module and that this post saturation module is as specified .

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125877).

Requirements Coverage:

Req ID	Requirement Tag	Description
58925	SWCMP.Scan.ASL.STAR.Postsat.v1	STAR must have a post saturation module, composed of a single 90 degree saturation pulse, applied at the imaging plane after the labeling or control module. The RF pulses shall be selective (selective gradient is different from 0).

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_star.pl - m_imn_asl_ts_star.acq to the patch directory. Run the script 'm_imn_asl_ts_star.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Inspect the files in 'G:\log' named m_imn_asl_ts_star_<configuration>_cycle.dat for all available configurations.	Each RF::star pulse is succeeded by a Postsat module.
Step 2	Inspect the files in 'G:\log' named m_imn_asl_ts_star_<configuration>.dat for all available configurations. Find the first RF::presat_4_ object in the log files. Inspect the value of the rf_angle. Find the first GR::presat_s_1_ object in the log files. Inspect the gradient strength (gr_str).	The rf_angle is 90. The gradient strength is different from 0.

3.1.4.19. SWCMP.Scan.ASL.STAR.Presat.tc1

Test ID	26304
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-10 15:33:57 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-03-10 11:59:59 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-09 09:13:57 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the STAR and STAR Angio scan techniques contain a WET presat module and that this WET presat module is as specified .

It uses a test script to automate scanning on multiple configurations and filtering of the system log files (VLD DEP125877).

Requirements Coverage:

Req ID	Requirement Tag	Description
58930	SWCMP.Scan.ASL.Presat.v1	ASL must have a WET pre saturation module (Requirements - Figure 2), applied at the imaging plane before labeling/control module. The interval (t) between the RF pulses and the flip angle of each one of them () must be optimized according to field strength for brain perfusion. RF pulses shall be selective (selective gradient is different from 0) and: for 1.5T, t = 10.0 ms and 1 = 106.8, 2 = 86.8, 3 = 76.0, 4 = 153.6 degrees, and for 3.0T, t = 8.0 ms and 1 = 89.17, 2 = 98.37, 3 = 82.31, 4 = 156.77 degrees.

Test steps:

Step	Description	Expected Result
Preparation	Copy the following files from the tst directory in the clearcase BB imaging: - m_imn_asl_ts_star.pl - m_imn_asl_ts_star.acq to the patch directory. Run the script 'm_imn_asl_ts_star.pl' from the command line. Wait for the script to finish.	N.A.
Step 1	Inspect the files in 'G:\log' named	Each RF::star pulse is preceded by a WET pulse.

Step	Description	Expected Result
	m_imn_asl_ts_star_<configuration>_cycle.dat for all available configurations.	
Step 2	Inspect the files in 'G:\log' named m_imn_asl_ts_star_<configuration>.dat for all available configurations. Find the first RF::wet object in the log files. Calculate the interval between the pulses by adding the values of rf_dur and rf_pause. Inspect the value of the rf_angle. Find the first GR::presat_s_0_ object in the log files. Inspect the gradient strength (gr_str).	The interval is as specified for the field strength. The angle is as specified for the field strength. The gradient strength is different from 0.

3.1.4.20. SWCMP.Scan.ASL.STARAngio.Combinations.tc1

Test ID	26259
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-04 13:11:42 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2016-03-01 08:28:27 Meaning: SME Approved - (Business) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-02-29 16:21:19 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test case verifies that STAR Angio can only combined with the specified acquisition parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description
56669	SWCMP.Scan.ASL.STARAngio.Combinations	STAR Angio shall require the following acquisition parameters: • Scan mode = 3D • Scan technique = FFE • Acquisition Mode = Cartesian • Fast imaging mode = TFEPI • Angio/Contrast enh. = no • Quantitative Flow = no • Images = M A conflict shall occur if these parameters are not set correctly in

Req ID	Requirement Tag	Description
		combination with the STAR Angio.

Test steps:

Step	Description	Expected Result
Step 1	Switch off Option 010, set the scan protocol database to TEDB and start examcards. Open the 'HEAD_CNS\FLL\STAR_Angio' protocol in the parameter editor.	The protocol should not be in conflict.
Step 2	Change the value of the 'Scan mode' parameter. Try to resolve conflicts (without changing 'Scan mode' and 'Arterial Spin labeling') until you have conflicts on 'Arterial Spin labeling' and 'Scan mode' (more is allowed). Repeat this for each value of 'Scan mode'	The conflicts can never be resolved if the 'Scan mode' is different from the specified value.
Step 3	Repeat the procedure described in step 1 for the parameter 'Scan technique'	The conflicts can never be resolved if the 'Scan technique' is different from the specified value.
Step 4	Repeat the procedure described in step 1 for the parameter 'Acquisition mode'	The conflicts can never be resolved if the 'Acquisition mode' is different from the specified value.
Step 5	Repeat the procedure described in step 1 for the parameter 'Fast imaging mode'	The conflicts can never be resolved if the 'Fast imaging mode' is different from the specified value.
Step 6	Repeat the procedure described in step 1 for the parameter 'Angio/Contrast enh.'	The conflicts can never be resolved if the 'Angio/Contrast enh.' is different from the specified value.
Step 7	Repeat the procedure described in step 1 for the parameter 'Quantitative Flow'	The conflicts can never be resolved if the 'Quantitative Flow' is different from the specified value.
Step 8	Repeat the procedure described in step 1 for the parameter 'Images'	The conflicts can never be resolved if the 'Images' is different from the specified value.

3.1.4.21. SWCMP.Scan.ASL.STARAngio.UserGuidance.tc1

Test ID	26258
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-29 11:58:56 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-29 11:42:44 Meaning: SME Approved - (Technical)::Bulk Transitions

	Signed by: Koonen, Jos (nly95806) on 2016-03-25 13:30:32 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test case verifies that user guidance is given for STAR Angio in case of conflicts arising due to limitations by the gradient duty cycle model.

Requirements Coverage:

Req ID	Requirement Tag	Description
57019	SWCMP.Scan.ASL.STARAngio. UserGuidance	For STAR Angio the system shall provide user guidance to resolve conflicts arising due to limitations by the gradient duty cycle model.

Test steps:

Step	Description	Expected Result
Step 1	Change the configuration of the VM to Ingenia 1.5T with a 781 gradient amplifier. Switch off Option 010, set the scan protocol database to TEDB and start examcards. Set 'gr duty cycle model' in the scan control parameters to 'no'. Set 'recon memory model' in the scan control parameters to 'no'. Open the 'HEAD_CNS\FLL\STAR_Angio' protocol in the parameter editor. Go to the 'Dyn\Angio' tab. Change the 'phase interval' to 'shortest', the 'duration' to '3000' and increase the value of 'phases' to the highest possible value that does not generate a conflict. Accept the changes.	The protocol should not be in conflict.
Step 2	Set 'gr duty cycle model' in the scan control parameters to 'yes'. Open the protocol in the editor.	The protocol should be in conflict. User guidance to resolve the conflict should be shown in the 'Assistance' tab.

3.1.5. ME B0Shim

3.1.5.1. SWCMP.Scan.B0Shim.B0Map.pars.v1.tc1

Test ID	41151
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:45 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 10:30:18 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:15:40 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of B0Map parameters.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB0ShimB0MapPars in the results file is zero.

3.1.5.2. SWCMP.Scan.B0Shim.pars.tc1

Test ID	41288
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:43 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 10:30:58 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:16:42 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of B0Shim parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description
79539	SWCMP.Scan.B0Shim.pars.v1	A table with more than 5 columns is not supported by this version of the DocGen macro. For requirements look into the requirements document or

Req ID	Requirement Tag	Description
		ALM for the actual requirement.
98439	SWCMP.Scan.B0Shim.pars.v2	A table with more than 5 columns is not supported by this version of the DocGen macro. For requirements look into the requirements document or ALM for the actual requirement.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB0ShimPars in the results file is zero.

3.1.5.3. SWCMP.Scan.B0Shim.tc1.v1

Test ID	46248
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Krishnakumar, Ashwin (ing02466) on 2020-05-15 09:14:00 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2020-05-11 14:53:18 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2020-05-06 13:27:09 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

B0 shimming preparation phase execution. For reproducibility reason this test case requires to be verified on a 3T system configuration equipped with Higher Order Shimming (HOS) hardware.

Requirements Coverage:

Req ID	Requirement Tag	Description
21069	SWCMP.Scan.B0Shim.Auto	FC MR-Methods shall support an automatic B0-homogeneity improvement correction including Volume shim. The scan protocol shall contain parameters that specifies which correction technique is to be used.
21070	SWCMP.Scan.B0Shim.HOS	MR-Methods controls the higher order shimming of the B0 field.

Test steps:

Step	Description	Expected Result
Step 1	- With command: <i>tf</i>, start the testframework, and select Ingenia 3T configuration. - Start the ExamCards Test Environment. Select any convenient scanning protocol.	- Log lines from the PDF and MPF code should be visible in the log file. - Execution of the first scan should produce a full execution of all preparation phases, except

Step	Description	Expected Result
	- Set <i>System/System Tuning.../Tuned Hardware/HOS Calibration Status</i> = PASS - Set <i>QuickSurvey</i> = No, on the PostProc tab - Set the <i>Shim</i> method selection to "PB volume" on the Contrast tab , - Set <i>Preparation phases</i> to "auto" on the PostProc tab, - Perform the scan. - Check the log file. The HS preparation phase should execute and log lines from both PDF and MPF code should be visible. - Repeat the same scan. - Repeat the same scan with the <i>Shim</i> method selection set to "volume". - Check the log file. The AS preparation phase should execute and log lines from both the PDF and MPF should be visible.	the AS preparation phase, and the following log lines should be observed: PDF: MPUPR: HS is requested MPUPR: HS will be performed, no appropriate previous result found. MPF: HS: SHIMS: X = ... HS: SHIMS: Y = ... HS: SHIMS: Z = ... HS: SHIMS: Z2 = ... HS: SHIMS: ZX = ... Simulation values will appear in log. HS: SHIMS: ZY = ... HS: SHIMS: X2-Y2 = ... HS: SHIMS: 2XY = ... PDF: MPUPR: HS is requested MPUPR: HS will be skipped, results taken from scan 0 MPF: // nothing PDF: MPUPR: AS will be performed, no appropriate previous result found MPUPR: HS will not be performed MPF: AS: SHIMS -- AS: SHIMS -- AS: SHIMS -- AS: SHIMS -- AS: SHIMS -- AS: SHIMS -- Verify that: - HS preparation phase execution will occur for the first test scan. - HS preparation phase execution will be skipped in the second test scan. - AS preparation phase execution will occur for the third test scan.

3.1.5.4. SWCMP.Scan.B0Shim.tc2.v1

Test ID	46253
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Krishnakumar, Ashwin (ing02466) on 2020-05-15 09:14:38 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2020-05-11 14:53:40 Meaning: SME Approved - (Technical)

	Signed by: Graaf, Ruud de (nly95749) on 2020-05-06 13:27:26 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

B0 shimming preparation phase execution for spectroscopy. For reproducibility reason this test case requires to be verified on a 3T system configuration equipped with Higher Order Shimming (HOS) hardware.

Requirements Coverage:

Req ID	Requirement Tag	Description
21069	SWCMP.Scan.B0Shim.Auto	FC MR-Methods shall support an automatic B0-homogeneity improvement correction including Volume shim. The scan protocol shall contain parameters that specifies which correction technique is to be used.
21070	SWCMP.Scan.B0Shim.HOS	MR-Methods controls the higher order shimming of the B0 field.

Test steps:

Step	Description	Expected Result
Step 1	- With command: <i>tf</i>, start the testframework, and select Ingenia 3T configuration. - Set <i>System/System Tuning.../Tuned Hardware/HOS Calibration Status</i> = PASS - Start the application and select a new patient - Run first just an imaging scan survey. - Create a new logfile with <i>tf</i> - Under "Phantom Studies" select "Spectro" and select the "WS_exc" protocol. - In the CoilState simulator, connect dSteam Interface, and Connect "TR Head" coi to TFTINT.ConTRC. - Disengage posterior coil by using the <i>pstest</i> command or disengaging it in the ExamCard Properties menu option - Set the "Shim" to "PB volume". - Adjust the scan parameters, setting "PB order" to "first". - Start the scan (first time). - Adjust the scan parameters, setting "PB order" to "second". - Start the scan (second time). - Adjust the scan parameters, setting "Shim" to "no". - Start the scan (third time). - Check the log file output.	<i>The higher order shim preparation phase should execute a first order shim determination for the first scan and the following log lines should appear (select text contains 'MPUPR: HS': or text contains 'HS: SHIMS:'</i> MPUPR: HS is requested MPUPR: HS will be performed, no appropriate previous result found. HS: SHIMS: X = ... HS: SHIMS: Y = ... Simulation values will appear in the log. HS: SHIMS: Z = ... <i>The second order shim forces that the prep phase is performed.</i> MPUPR: HS is requested MPUPR: HS will be performed, no appropriate previous result found. HS: SHIMS: X = ... HS: SHIMS: Y = ... HS: SHIMS: Z = ... HS: SHIMS: Z2 = ... HS: SHIMS: ZX = ... Simulation values will appear in log. HS: SHIMS: ZY = ... HS: SHIMS: X2-Y2 = ... HS: SHIMS: 2XY = ... <i>The higher order shim preparation phase should not be executed for the third scan. Hence, MPF shouldn't produce any HS logging.</i> MPUPR: HS will not be performed <i>Pass/Fail Criteria: The HS preparation phase should execute with the logging like above.</i>

3.1.5.5. SWCMP.Scan.B0Shim_B1Cal.tc1

Test ID	9970
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-04 13:27:48 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 09:54:38 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Jurrissen, Michel (nly95853) on 2015-02-03 11:54:58 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Purpose: to verify if B0 and B1 image based shimming is operational.

Configuration: WA 3T TX

Note: This test also uses the calculator for B0 and B1 shimming. The calculators itself is implemented in a software unit/building block with a building block test to verify the calculators results.

Requirements Coverage:

Req ID	Requirement Tag	Description
21072	SWCMP.Scan.B0Shim.B0Map	Subsystem Scanning must make it possible to measure within 70 seconds a B0 map that can be used for image based shimming.
21071	SWCMP.Scan.B0Shim.ImageBased	Image based B0 shimming with user selected ROI, or automatically determined through SmartExam, shall be supported for 3.0T and 1.5T.
21073	SWCMP.Scan.B0Shim.Calculators	FC MR-Methods shall implement the calculator that calculates the correction gradients, based upon a B0 map (known as image based B0 shimming).

Test steps:

Step	Description	Expected Result
Step 1	Install test examcards Open Application SW Select patient Select examcard int_3t_tx_wa	Examcards opens without conflicts
Step 2	Delete all scans after 3D10_FFE	n.a.

Step	Description	Expected Result
Step 3	Start examcard	First scan runs
Step 4	Plan B0 map and 3D10_FFE	Scans planned
Step 5	Start scan	Scan runs towards completion.

3.1.6. ME B1CAL

3.1.6.1. SWCMP.Scan.B1CAL.CardiacTX.Shimming.PrepPhases.tc1.v2

Test ID	25733
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-01 11:29:56 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-01 08:55:38 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-01-29 09:07:05 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

B1 calibration scans and RF shim performance
Reuse of F0 AS preparation phases

Purpose:

If the preparation phases of a scan with RF shimming are set to *auto* , the F0 and AS preparation phase results of a previous scan may only be re-used if RF shimming was used during those preparation phases.

Requirements Coverage:

Req ID	Requirement Tag	Description
16819	SWCMP.Scan.B1CAL.CardiacTX.Shimming.PrepPhases	If the preparation phases of a scan with RF shimming are set to 'auto', the preparation phase results of a previous scan may only be re-used if RF shimming was used during those preparation phases. This requirement is valid for the following preparation phases: the F0 and B0 (AS)

Test steps:

Step	Description	Expected Result
Prep 1	Reuse of F0 AS preparation phases: Preparations - To define the test environment in VMware use the script: C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select a 3 Tesla system. - Copy from the Clearcase SW archive the Examcard: "<code>system\scanning\mrm\methods\imaging\tst\m_imn_b1cal_ts.ExamCard</code>" To the test machine in the folder: "G:\importpp\ Double click on the Examcard. The Examcard will be imported in the Examcards folder "Other\Inbox" - In the trace definition file "<code>C:\templbstrace.dat</code>" enable the trace channels <code>mmiffe debug *</code> <code>mmirfe debug *</code> <code>maobjlog mpf *</code> In a VMware Virtual test machine the trace definition file is located on the G drive: "<code>G:\site\lbstrace.dat</code>" - In the MRBoot Configuration Manager start the; - Application Software (For Test machines) - Examcards Scan Environment. (For Virtual Test machines) - Select a new patient. From the Examcards folder "Other\Inbox" drag the Examcard "m_imn_b1cal_ts" in to the scanlist. - Use the "Logging Application" tool to create a new current log file - Run the Survey scan from the scanlist. Now plan all the other scans in the scanlist, then run all the scans.	NA (preparation)
Step 1	Open the log file: G:\Log\devlogcurrent.log with the logbrowser program and search for the keywords: 'm_Name', 'M2D_TFEtra', 'F0 will be', 'AS will be'.	Verify that F0 and AS are repeated ('F0/AS will be performed ...') for the first three scans (no RF shim, RF shim adaptive, RF shim volume) but that they are reused ('F0/AS will be skipped ...') by the following RF shimmed scans.

3.1.6.2. SWCMP.Scan.B1CAL.DA.NonSelective.tc1.v2

Test ID	25754
Revision	1
Product Model	Any
Gradient System	Any

Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-01 11:30:01 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-01 08:55:43 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-01-29 09:07:10 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Inspection of dualAngle B1 calibration sequence

dualAngle TSE scans to not allow for a slice selective RF excitation pulse.

Requirements Coverage:

Req ID	Requirement Tag	Description
45526	SWCMP.Scan.B1CAL.DA.v1	A B1 mapping/calibration method based on the Double Angle technique shall be implemented.
45530	SWCMP.Scan.B1CAL.DA.NonSelective.v1	A B1 mapping/calibration method based on the TSE <i>dualAngle</i> technique without slice-selective excitation shall be implemented.

Test steps:

Step	Description	Expected Result
Prep 1	Preparations - To define the test environment in VMware use the script: C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select a 3 Tesla system. - Start <i>ExamCards Environment</i>. - Enter <i>New Patient</i>. Start the <i>Coilstate Simulator UI</i> and connect the <i>Anterior</i> coil to <i>SBM.DCC0</i>.	NA (Preparation)
Step 1	Open Control Parameter Editor and set the parameters MT B1 calibration scan UI to full on tab Scan. Note: The scan control parameter needs to be set in order to have access to all examination parameters of the B1 calibration protocols. Select PreScans\Testscans\B1_cal_DA_TSE protocol from the TEDB Hospital tab	Verify that the protocol validates without conflict.
Step 2	Set the parameter B1 field map selective to yes.	Verify that a conflict is raised on: Scan mode technique, B1 field map and B1 field map selective. Note: For the dualAngle B1 map technique with TSE read-out slice selection is performed using the refocusing pulses.

Step	Description	Expected Result
Step 3	Set the parameter B1 field map selective back to no.	Verify that the protocol is validated without conflicts.
Step 4	Open Control Parameter Editor and set the parameters MT B1 calibration scan UI to default on tab Scan.	Control parameter correctly set back.

3.1.6.3. SWCMP.Scan.B1CAL.DA.PhaseInsensitive.tc1.v3

Test ID	52717
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:13:06 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:37:36 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:23 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test performs Inspection of dualAngle B1 calibration sequence.

For dualAngle TSE scans the gradient scheme needs to be adapted for a phase-insensitive TSE read-out.

Requirements Coverage:

Req ID	Requirement Tag	Description
16804	SWCMP.Scan.B1CAL.DA.Phase Insensitive	For the TSE <i>dualAngle</i> technique without slice-selective excitation a phase-insensitive read-out scheme shall be implemented to ensure correct B1 values.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB1CalDaPhaseInsensitive in the results file is zero.

3.1.6.4. SWCMP.Scan.B1CAL.DA.Slice_prof_corr.tc2.v4

Test ID	52715
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:13:08 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:37:37 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:23 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies that DualAngle FFE scans to allow for a slice selective RF excitation pulse. Inspection of dualAngle B1 calibration sequence

Requirements Coverage:

Req ID	Requirement Tag	Description
45529	SWCMP.Scan.B1CAL.DA.Selective	A FFE based B1 mapping/calibration method based on the <i>dualAngle</i> technique with slice-selective excitation shall be implemented.
16805	SWCMP.Scan.B1CAL.DA.SliceProfCorr	In case a slice-selective excitation pulse is applied during the B1 calibration scan, MR-Methods shall instruct reconstruction to perform a pulse specific slice profile correction.
74597	SWCMP.Scan.B1CAL.AutoInsert.CardiacTX.Method.v1	For Cardiac acquisitions, the system shall automatically insert a cardiac triggered B1 calibration scan using a FFE EPI dualAngle technique with slice-selective excitation.
16807	SWCMP.Scan.B1CAL.CardiacTX.Pulse	Cardiac B1 calibration scans shall use the SG_150_100_167 pulse

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB1CalDualAngleSliceProfCorr in the results file is zero.

3.1.6.5. SWCMP.Scan.B1CAL.DA.tc1.v2

Test ID	25759
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-01 11:30:09 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-01 08:55:54 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-01-29 09:07:20 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Inspection of dualAngle B1 calibration sequence (saturation prepulse)

The dualAngle method is based on a saturation prepulse.

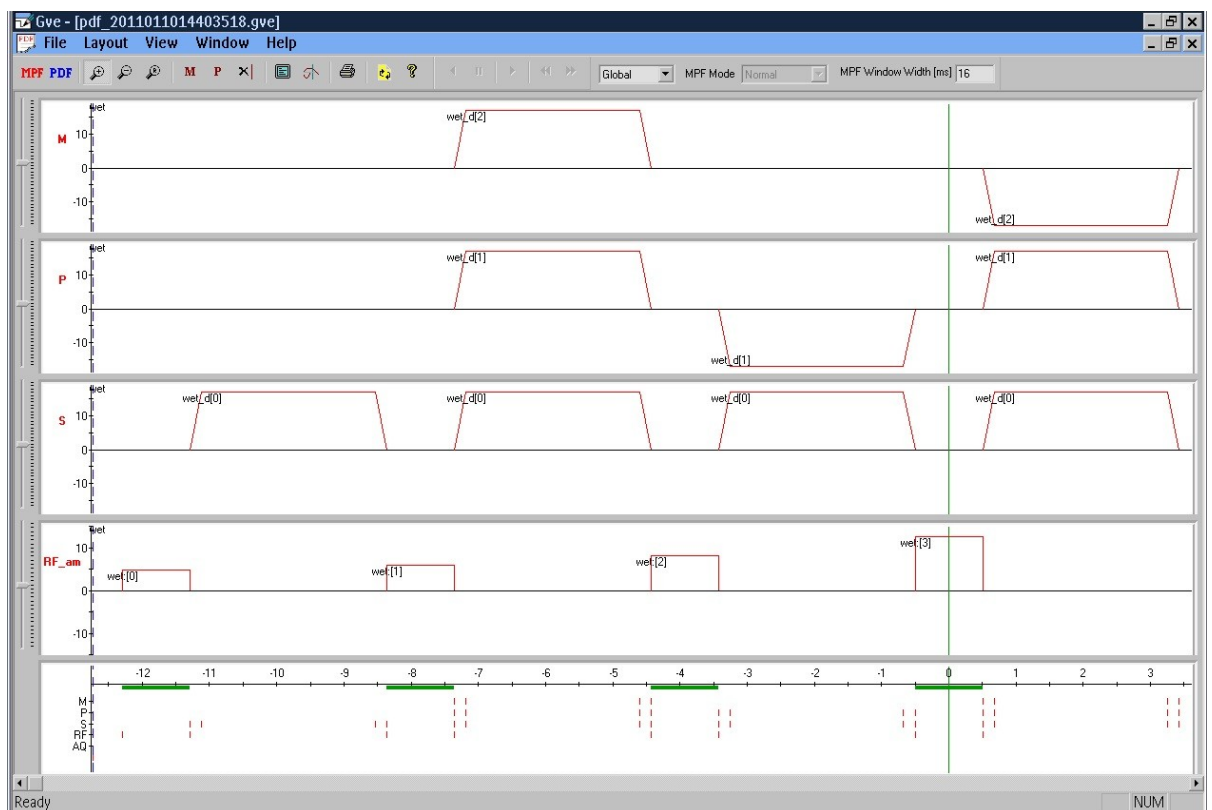
Requirements Coverage:

Req ID	Requirement Tag	Description
45526	SWCMP.Scan.B1CAL.DA.v1	A B1 mapping/calibration method based on the Double Angle technique shall be implemented.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Preparations - To define the test environment in VMware use the script: C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select a 3 Tesla system. - Start <i>ExamCards Environment</i>. - Enter <i>New Patient</i>. - Start the <i>Coilstate Simulator UI</i> and connect the Anterior coil to <i>SBM.DCC0</i>.	N.A. (Preparation)	None
Step 1	Make sure the trace channel "GVE * *" is enabled in C:\temp\bstrace.dat In a VMware Virtual test machine the trace definition file is located on the G drive: "G:\site\bstrace.dat" Select protocol PreScans\Testscans\B1_cal_DA_TSE	Verify that the protocol validates without conflict.	None

Step	Description	Expected Result	Attachments
	from the TEDB Hospital tab.		
Step 2	Start GVE. Load the PDF from the file pdf.gve and select the PDF sequences wet and base. Use the Probe tool to measure the amplitudes of the WET pulse consisting of 4 block pulses.	- Verify that the ratio of the block pulses to be 1.0 / 1.28 / 1.75 / 2.68 (Refer fig Attached). Note: The ratio of the WET block pulse is controlled via the scan control parameter WET saturation pulse angles which are set by default to 72.0 / 92.0 / 126.0 / 193.0. - Verify timing and orientation of the WET gradients. (Refer to fig. attached) - Verify that the time delay between the reference point of the WET pulse and the excitation pulse of the base sequence to be 650ms using the Measure tool. Note: The saturation delay is controlled via the examination parameter B1 field map saturation delay on the tab postproc. Only visible if the scan control parameter MT B1 calibration scan UI is set to full.	



3.1.6.6. SWCMP.Scan.B1CAL.DA.tc1.v3

Test ID	52716
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:13:10 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:37:39 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:23 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test performs inspection of dualAngle B1 calibration sequence (saturation prepulse)

Requirements Coverage:

Req ID	Requirement Tag	Description
45526	SWCMP.Scan.B1CAL.DA.v1	A B1 mapping/calibration method based on the Double Angle technique shall be implemented.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB1CalDa in the results file is zero.

3.1.6.7. SWCMP.Scan.B1CAL.DREAM.tc1.v4

Test ID	52718
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:13:12 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:37:41 Meaning: SME Approved -

	(Technical)::Bulk Transitions
	Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:23 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test inspects the validity of the DREAM B1 mapping sequence.

Requirements Coverage:

Req ID	Requirement Tag	Description
45527	SWCMP.Scan.B1CAL.DREAM	A B1 mapping/calibration method based on the DREAM technique shall be implemented.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB1CalDream in the results file is zero.

3.1.6.8. SWCMP.Scan.B1CAL.DTR.tc1.v3

Test ID	52714
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharappa, Deepak Shekhar (320098697) on 2021-12-16 14:13:14 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:37:43 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:24 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies that for dualTR scans an additional TR delay is required and slice selection and gradient spoiling is adapted.

Requirements Coverage:

Req ID	Requirement Tag	Description
45480	SWCMP.Scan.B1CAL.DTR.v1	A B1 mapping/calibration method based on the Actual Flip Angle technique shall be implemented.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanB1CalDtr in the results file is zero.

3.1.6.9. SWCMP.Scan.B1CAL.GEN.Acquisition.tc1.v2

Test ID	25763
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-01 11:30:18 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-01 08:56:02 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-01-29 09:07:31 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

B1 calibration scans and RF shim performance
Transmit channel switching

Purpose:

Transmit channels are switched in run time for B1 Calibration scans

This test can only be run in a simulated environment, not on a test bay.

Make sure that the trace channels and object logging as described in the preparations are properly set.

Requirements Coverage:

Req ID	Requirement Tag	Description
16797	SWCMP.Scan.B1CAL.GEN.Acquisition	A B1 calibration scan shall acquire in a single scan the B1 Maps of all different transmit channels.

Test steps:

Step	Description	Expected Result
Prep 1	Preparations - To define the test environment in VMware use the script: C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select a 3 Tesla system. - Open in Notepad the trace file: C:\temp\bstrace.dat In a VMware Virtual test machine the trace definition file is located on the G drive: "G:\site\bstrace.dat" - Add the trace channels mmiffe debug * mmirfe debug * MAOBJLOG MPF * - Save the modifications - Start ExamCards Environment. - Enter <i>New Patient</i>.	N.A. (Preparation)
Step 1	From the TEDB Hospital tab. run the protocols: PreScans\Testscans\B1_cal_DTR, PreScans\Testscans\B1_cal_DA_TSE, PreScans\PreScans\B1_calibration, PreScans\PreScans\B1_calibration_CT In the logfile search for the text strings: 'iffe_dynamic_scan' and 'perform_ph' and 'bitmask'	Verify that the bitmask_mux attribute of the RF`ex pulse has the value 1 in the first dynamic and the value 2 in the second dynamic scan. The current dynamic number can be identified for the FFE based B1 Calibration scans by the logged value 'iffe_dynamic_scan' and for the TSE based scan by the current phase number 'perform_ph 1' (dynamic scan 0), perform_ph 0' (dynamic scan 1)

3.1.6.10. SWCMP.Scan.B1CAL.GEN.PO_OVERRULES.tc1.v3

Test ID	39098
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-04-29 16:17:06 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Verburg, Frank (nly95979) on 2019-04-29 15:49:27 Meaning: SME Approved - (Business) Signed by: Hey, Silke (300170272) on 2019-04-26 17:11:42 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

B1 calibration scans and RF shim performance

Purpose:

In order to avoid overtipping problems in brain applications the drive scales provided by the B1 shim calculator are overwritten by the drive scales found by the power optimization.

Test Strategy

This test checks the mechanism used to select the drive scale values from PO prep-phase rather than from the B1 shim calculator, in case of brain applications.

Requirements Coverage:

Req ID	Requirement Tag	Description
16815	SWCMP.Scan.B1CAL.GEN.PO_Overrules	For head scans, the drive scales determined by the B1 calibration scan are overruled by the drive scales determined by the preparation phase for power optimization.

Test steps:

Step	Description	Expected Result
Prep 1	1. In Notepad, open the trace file: C:\temp\bstrace.dat (On a Virtual test machine: "G:\site\bstrace.dat") 2. Add the trace channels - mmiffe debug * - mmirfe debug * - MAOBJLOG MPF * and save the modifications 3. Use the testframework to configure the VM as a 3 Tesla system. 4. Select the TEDB protocol database 5. Start ExamCards Environment. 6. Enter New Patient.	N.A. (Preparation)
Step 1	1. Create a new log file using the Logging Application tool. 2. Run the B1_calibration protocol from Hospital >PreScans->PreScans folder. 3. Insert protocol '2D_FE_Posterior' from Hospital > PreScans > Testscans folder. 4. On the contrast tab, change RF Shims from 'fixed' to 'adaptive', and Shim to 'auto'. 5. On the coils tab, unselect Smart Select, selecting the Q-body coil as receive coil. 6. Run the scan.	Verify that the logfile contains the following line for the scan '2D_FE_Posterior': PO: (after PO) using drive scale from B1 shim tool
Step 2	1. In the coil simulator attach the 'dS Base' coil 2. Copy the same scan, re-enable Smart Select to use the 'dS Base' coil as a receive coil. 3. Run the scan	Verify that the log file contains the line: PO: PO result used for shimmed scan, overrules drive scale from RF shim calculator Note: This line indicates that the drives scales provided by the B1 shim calculator are overwritten by the drive scales found by the power optimization.
Step 3	Run the scan again	Verify again that the log file contains a new line: PO: PO result used for shimmed scan, overrules drive scale from RF shim calculator Note: In this case the PO preparation phase was skipped. Nevertheless, drives scales provided by the B1 shim calculator are overwritten by the drive scales found by an earlier power optimization.
Wrap Up	Attach the logfile as objective evidence.	N.A. (Wrap Up)

3.1.6.11. SWCMP.Scan.B1CAL.GEN.ReceiveRFShim.tc1.v2

Test ID	25765
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-01 11:30:22 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-01 08:56:05 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-01-29 09:07:37 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

B1 calibration scans and RF shim performance
Receive RF shim for QBC scans

Purpose:

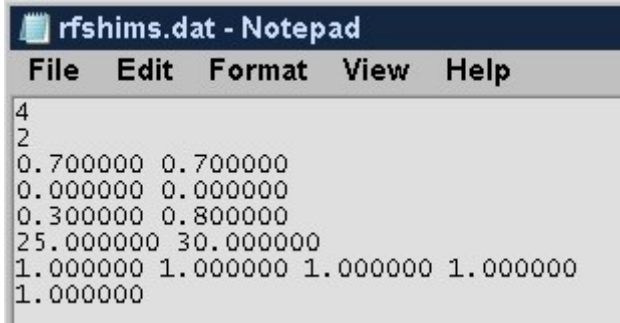
If the QBC is used for receive for an RF shimmed scan reconstruction is instructed the receive RF shim values will be derived from the B1 Calibration scans

Requirements Coverage:

Req ID	Requirement Tag	Description
16813	SWCMP.Scan.B1CAL.GEN.ReceiveRFShim	When images are reconstructed using signals from QBC elements and the RF shimming methods is not fixed, MR-Methods shall instruct reconstruction to use the receive RF shim values derived from the B1 calibration scan.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Preparations - Open in Notepad the trace file: <code>C:\temp\bstrace.dat</code> In a VMware Virtual test machine the trace definition file is located on the G drive: "G:\site\bstrace.dat" - Add the trace channels <code>mmirfe debug *</code> <code>mmirfe debug *</code> <code>MAOBJLOG MPF *</code> - Save the modifications - To define the test environment in	N.A. (Preparation)	None

Step	Description	Expected Result	Attachments
	VMware use the script: C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select a 3 Tesla system. 1. Start ExamCards Environment. 2. Enter New Patient. 3. <i>Take care that the pateint reference postion has been selected before scanning. E.g. by running a scan before the mentioned scan.</i>		
Step 1	Modify the content of the file G:\site\rfshims.dat as shown in Figure Attached. Load the protocol M2D_TFEtra_no again and modify the value RF shims to shimtool and Verify that Body coil is explicitly selected as receive coil.	Verify in the Sequence Development Viewer that the following RC parameters are filled as indicated: - o RC_phase_per_channel_arr: 5.76, 5.85 - o RC_scale_per_channel_arr: 0.8, 0.3 The first set of values is only true if the body coil receive phase are both 0.0. This is always the case on a ROF. On a test system those phase have to be added in the following way: phase[0] = $-\pi \cdot (30 + QBody_receive_phase[0]) / 180$ phase[1] = $-\pi \cdot (25 + QBody_receive_phase[1]) / 180$ If the resulting phases are negative, 2pi has to be added.	
 The screenshot shows a Notepad window titled 'rfshims.dat - Notepad'. The menu bar includes File, Edit, Format, View, and Help. The text content of the file is as follows: <pre>4 2 0.700000 0.700000 0.000000 0.000000 0.300000 0.800000 25.000000 30.000000 1.000000 1.000000 1.000000 1.000000 1.000000</pre>			

3.1.6.12. SWCMP.Scan.B1CAL.Pars.tc1.v3

Test ID	41378
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-06-18 14:38:19 Meaning: QA Approved - (QA)

	Signed by: Engelen, Walter van (nly95948) on 2019-06-18 12:43:35 Meaning: SME Approved - (Business)
	Signed by: Verburg, Frank (nly95979) on 2019-06-18 10:33:01 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

For B1 calibration scans only a limited number of parameters is displayed in the *ExamCard UI*.

Requirements Coverage:

Req ID	Requirement Tag	Description
74597	SWCMP.Scan.B1CAL.AutoInsert .CardiacTX.Method.v1	For Cardiac acquisitions, the system shall automatically insert a cardiac triggered B1 calibration scan using a FFE EPI dualAngle technique with slice-selective excitation.

Test steps:

Step	Description	Expected Result
Prep 1	Preparations 1. To define the test environment in VMware use the script: C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select a 3 Tesla system. 1. Start <i>ExamCards Environment</i> . 2. Enter <i>New Patient</i> .	N.A. (Preparation)
Step 1	From the TEDB Hospital tab select: • PreScans\PreScans\B1_Calibration • PreScans\PreScans\B1_Calibration_CT Select the protocols for modification. Ensure Respiratory compensation is set to "breath hold".	Verify that only the following protocol parameters are displayed For B1_Calibration : Slice Offc. (AP, RL, FH) and Respiratory compensation and Breath hold mode are displayed on the tabs: Geometry to offc/ang. For B1_Calibration_CT : Device, Slice Offc. (AP, RL, FH), Respiratory compensation and Breath hold mode are displayed on the tabs: Geometry to offc/ang.
Step 2	Press Cancel to leave the parameter editing mode and open the Control Parameter Editor. Make sure option 47 is enabled under Service Application >>Configuration >> Software Licenses >> Enable/Diable Option Switches.	Verify that the scan control parameter MT B1 calibration scan UI is displayed on the tab Scan.
Step 3	Set MT B1 calibration scan UI to full, press Apply and leave the Control Parameter Editor. Select again the protocol B1_Calibration and B1_Calibration_CT for modification.	Verify that the parameters described in the requirements SWCMP.Scan.B1CAL.Pars.v1, SWCMP.Scan.B1CAL.CardiacTX.Pars.v1 and SWCMP.Scan.B1CAL.GEN.RFShim.Pars are visible for the different B1 Calibration protocols. Verify that the range of the parameter RF shims is limited to fixed for these B1 Calibration scans.

3.1.6.13. SWCMP.Scan.B1CAL.tc1.v3

Test ID	58122
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:53:47 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:16:25 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11- 15 11:08:30 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

B1 calibration scans and RF shim performance
Acquisition of B1 calibration scans

Purpose:

Acquisition of B1 calibration scans are based on the dualTR and the DualAngle TSE method.

Requirements Coverage:

Req ID	Requirement Tag	Description
16797	SWCMP.Scan.B1CAL.GEN.Acquisition	A B1 calibration scan shall acquire in a single scan the B1 Maps of all different transmit channels.
16814	SWCMP.Scan.B1CAL.GEN.Shim Box	For RF shimming based on a user-defined volume the same shim box shall be used as for B0 shimming.
16817	SWCMP.Scan.B1CAL.CardiacTX .Shimming	RF shim=Volume will be used for Cardiac.
45528	SWCMP.Scan.B1CAL.DA.Breath hold.v1	Methods provides B1 calibration/mapping methods that allow data collection within a single breath-hold.
74597	SWCMP.Scan.B1CAL.AutoInsert .CardiacTX.Method.v1	For Cardiac acquisitions, the system shall automatically insert a cardiac triggered B1 calibration scan using a FFE EPI dualAngle technique with slice-selective excitation.
74606	SWCMP.Scan.B1CAL.AutoInsert .Method.v1	For all applications except for Cardiac, Methods provides a fixed B1 calibration scan based on the DREAM technique that will automatically be inserted by the system.

Test steps:

Step	Description	Expected Result
Prep 1	1. Configure a VM with Ingenia CX 3.0T configuration 1. Start workflow. 2. Enter New Patient. 3. Import ExamCard <i>m_imn_b1cal_ts</i> and add it to the Inbox. Note: The ExamCard <i>m_imn_b1cal_ts</i> can be found in <code>\system\mrm\methods\imaging\tst\</code>	N.A. (Preparation)
Step 1	Run the Survey and plan the following scans	• Verify that only one shim box used for B0 and B1 shimming is visible. • Verify on the contrast page that for the protocol M2D_TFEtra_volL the examination parameter <i>multitransmit</i> is set to yes and that the parameter <i>RF Shims</i> is displayed allowing to select between the various RF shimming options.
Step 2	Set SENSE to yes in M2D_TFEtra_volL and enable breath hold. Press Start Scan to run the remaining scans.	• Verify that all scans were successfully acquired. • Verify that for the M2D_TFEtra_volL scan ask for a breath hold.

3.1.6.14. SWCMP.Scan.B1CAL.Workflow.v4

Test ID	39099
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-04-29 16:17:08 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Verburg, Frank (nly95979) on 2019-04-29 15:49:45 Meaning: SME Approved - (Business) Signed by: Hey, Silke (300170272) on 2019-04-26 17:06:34 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

B1 calibration scans and RF shim performance
B1 Calibration workflow

Purpose

The latest acquired B1 Calibration scan should be used for RF shim calculations. Therefore, after re-acquiring a new B1 calibration scan, a new calculation job should be executed (using the new data).

Test strategy

This test checks this mechanism by repeating the inserted prescan and checking the logfile for shim calculation job using the new prescan.

Requirements Coverage:

Req ID	Requirement Tag	Description
16816	SWCMP.Scan.B1CAL.Workflow	The most recent made B1Map shall be used for calculating the RF shim settings. This has also as consequence that the RF shim settings must be recalculated for already prepared but not executed scan protocols as soon as a B1 map is made.

Test steps:

Step	Description	Expected Result
Prep 1	1. Use the testframework to configure the VM as a 3 Tesla system. 2. Select the TEDB 3. Enable the listing of automatic inserted prescans in the scan list by changing ShowPreScans to 'true' in the file 'C:\Program Files (x86)\PMS\Mipnet91\examcards_win_cs.exe.config' 4. Start simulation mode (Acq on Host, Recon on Host). 5. Start a new logfile. 6. Start the full application software. 7. Start a new examination 8. Add the protocol Head CNS \ EPI \ P_SEEPI_MS to the scanlist. 9. Select the protocol for editing, and set the following values: a. On the Contrast tab, set RF Shims to 'adaptive'. b. On the Postproc tab, set Quick Survey to 'no'. c. On the Coils tab, select Smart Select.	N.A. (Preparation)
Step 1	Run the protocol.	A B1_calibration scan is inserted automatically. All inserted scans complete successfully.
Step 2	• Press the 'Repeat Prescans' button. • Create a copy of the protocol • start the scan	A new B1_calibration prescan is inserted automatically. The scan runs to completion.
Step 3	• Open the devlogcurrent.log file. • Query for: <i>text contains '(ShimCalculator): GetImageParamRec'</i> <i>or text contains 'Using RF shims' or text contains 'RepeatPrescans' or text contains 'OnInvalidatePrescans'</i> • Attach the devlogcurrent.log file to this step as objective evidence.	1. The logging shows the series used as input for the shim calculator, and the RF shim values used by scan engine. 2. The logging shows the 'OnRepeatPrescans clicked' message, as well as logging from examcards 'OnInvalidatePrescans started: Clear examcard PrescanResponses and prescanOptimizer'. 3. Logging shows that the new B1 calibration scan series is used by the shim calculator after Repeat Prescans has been pressed.
Post	Restore C:\Program Files (x86)\PMS\Mipnet91\examcards_win_cs.exe.config to the original value.	N.A (wrap-up)

3.1.7. ME Cardiac

3.1.7.1. SWCMP.Scan.Cardiac.ArrhythmiaRejection.tc1.v2

Test ID	36174
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-11-03 13:54:30 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2017-11-03 12:46:15 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-11-03 11:08:45 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

All RR-intervals within the accept window are accepted.

Requirements Coverage:

Req ID	Requirement Tag	Description
10057	SWCMP.Scan.Cardiac.ArrhythmiaRejection	The arrhythmia reject technique can be enabled by the operator to skip acquisitions for suspected RR-intervals during prospective cardiac- and gated sweep-synchronization. Suspected RR-intervals deviate too much from the average duration.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Add "MMICARD DEBUG *" to bstrace.dat. Run m_amn_set_test_environment.pl command on VMWare and set the system type to 1.5T Ingenia. Enter simulation mode for scannersim and reconsim. Connect a dS head coil in coil simulator to DCC1	N. A. (preparation)	None
Step 1	Execute the attached batchfile <teima_card_sync.acq> with the Batch Interpreter.	The batch shall successfully complete after execution.	 teima_card_sync.zip
Step 2	Search for ICARD->total_rejections for	ICARD->total_rejections = 0.	None

Step	Description	Expected Result	Attachments
	arrh1 in the devlog file		
Step 3	Search for ICARD->total_rejections and ICARD->ave_card_freq_bpm for the first G_2D1_arrh70 scan in the devlog file	ICARD->total_rejections = 0 and ICARD->ave_card_freq_bpm = 65 +/- 2.	None
Step 4	Search for ICARD->total_rejections and ICARD->ave_card_freq_bpm for the second G_2D1_arrh70 scan in the devlog file	ICARD->total_rejections = 10 +/- 2 and ICARD->ave_card_freq_bpm = 59 +/- 2.	None
Step 5	Search for ICARD->total_rejections and ICARD->ave_card_freq_bpm for the third G_2D1_arrh70 scan in the devlog file	ICARD->total_rejections = 23 +/- 2 and ICARD->ave_card_freq_bpm = 67 +/- 2.	None
Step 6	Search for ICARD->total_rejections and ICARD->ave_card_freq_bpm for the fourth G_2D1_arrh70 scan in the devlog file. Search for "Arrhythmia rejection disabled after many sequential rejections" for this scan in the devlog file.	ICARD->total_rejections = 16 +/- 2 and ICARD->ave_card_freq_bpm = 79 +/- 2. Expected logging shall be observed in the devlog file.	None
Step 7	Search for ICARD->total_rejections and ICARD->ave_card_freq_bpm for the fifth G_2D1_arrh70 scan in the devlog file. Search for "Arrhythmia rejection disabled after many sequential rejections" for this scan in the devlog file.	ICARD->total_rejections = 19 +/- 2 and ICARD->ave_card_freq_bpm = 80 +/- 2. Expected logging shall be observed in the devlog file.	None
Step 8	Search for "Arrhythmia occurred" in the devlog file and inspect a pair of lines enclosing it, as shown in the example below. MMICARD(debug): Next Kernel: delay (12), hph: 0, sweep: 5 MMICARD(debug): CARDIAC: GAS: Arrhythmia occurred! Remeasure one cardiac cycle. MMICARD(debug): Next Kernel: measure (12), hph: 0, sweep: 4	Expected logging shall be observed in the devlog file. The figure between brackets shall vary, but shall be the same for each pair. The sweep number shall vary as well. With the exception of the first and last sweep, the sweep number shall be one higher in the first line than in the last line of each pair.	None
Step 9	Search for "Arrhythmia detected" in the devlog file and inspect the lines above and below this line, as shown in the example below: MMICARD(debug): Before: par_meas 0 seg 0 p_inner 20 p_outer 0 serial_meas 0 MMICARD(debug): Arrhythmia detected, count 10 MMICARD(debug): After : par_meas 0 seg 0 p_inner 19 p_outer 0 serial_meas 0 Remove 'mmicard debug *' from bstrace.dat file.	Expected logging shall be observed in the devlog file. If p_inner in the first line is non-zero, p_inner in the second line will be one less and all other figures will be the same.	None

Step	Description	Expected Result	Attachments
	Attach the devlog file as objective evidence		

3.1.7.2. SWCMP.Scan.Cardiac.GatedSweep.tc1

Test ID	7060
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Verify the appearance and range of the cardiac gated sweep synchronization parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description
10055	SWCMP.Scan.Cardiac.GatedSweep	It shall be possible to define cardiac gated sweep synchronized scan protocols that allow adjustment of the start-time and acquisition-duration of a specific heart phase.

Test steps:

Step	Description	Expected Result												
Step 1	Select "Head CNS / CARD / G_2D1_1hp" in the scan protocol editor.	Verify that the parameters are present on the motion tab with the specified ranges: <table><tr><th>Parameter name</th><th>Range</th></tr><tr><td>Cardiac Synchronization</td><td>no, trigger, gate, retrospective</td></tr><tr><td>device</td><td>ECG, PPU, Internal, External</td></tr><tr><td>R-R window (%)</td><td>0-100</td></tr><tr><td>Gate delay</td><td>shortest, user defined</td></tr><tr><td>(ms) Present if "Gate delay" has value</td><td>x-y</td></tr></table>	Parameter name	Range	Cardiac Synchronization	no, trigger, gate, retrospective	device	ECG, PPU, Internal, External	R-R window (%)	0-100	Gate delay	shortest, user defined	(ms) Present if "Gate delay" has value	x-y
Parameter name	Range													
Cardiac Synchronization	no, trigger, gate, retrospective													
device	ECG, PPU, Internal, External													
R-R window (%)	0-100													
Gate delay	shortest, user defined													
(ms) Present if "Gate delay" has value	x-y													

Step	Description	Expected Result	
		"user defined". "x-y" depends on other parameters.	
		width	shortest, user defined, longest
		(ms) Present if "width" has value User defined. "x-y" depends on other parameters.	x-y

3.1.7.3. SWCMP.Scan.Cardiac.GatedSweep.tc2

Test ID	7067
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test the division of profiles over the gates when manipulating the gated width. Check the contents of the info page.

Preparation

- Enable trace logging for the "debug" channel of MMICARD. Make sure that the other trace channels in MMICARD are disabled.
- Restart the *scanner* process using *permproc*
- Select TEDB using the Preset Procedures Database Selection tool. This is the protocol database with test protocols.
- Start ExamCards.

Make sure the common tests in Test description Common tests have been performed with the gated sweep test protocol.

Requirements Coverage:

Req ID	Requirement Tag	Description
10055	SWCMP.Scan.Cardiac.GatedSweep	It shall be possible to define cardiac gated sweep synchronized scan protocols that allow adjustment of the start-time and acquisition-duration of a specific heart phase.

Test steps:

Step	Description	Expected Result
Step 1	Select test protocol "Head CNS / CARD / G_2D1_epi".	Verify that the value of the user defined gate width is 550ms in the protocol parameter while it is 700 ms on the info page.
Step 2	Change the gate width protocol parameter into 600ms and leave the parameter field.	Verify that the gate width protocol parameter automatically is adapted to 700ms and the info page still displays a gate width of 700 ms.
Step 3	Try out different gate widths between 50 and 900 ms and find out what values are allowed.	- Verify that the following widths are accepted: 50, 100, 150, 200, 250, 350 and 700 ms. - Verify that no conflicts occurred.
Step 4	Gate delay: - Verify the gate delay to be 100 ms both in the protocol parameter and on the info page. - Modify the gate delay and make sure the protocol does not get into conflict. Check if the info page is updated accordingly.	The gate delay protocol parameter and the info page always show the same value.
Step 5	Shortest gate delay: Set the gate delay to "shortest" and check the info page.	The user defined gate delay disappears from the protocol parameters. The info page shows the gate delay to be 41 ms. (16ms + 0.5 TR).

3.1.7.4. SWCMP.Scan.Cardiac.GatedSweep.tc3.v2

Test ID	52719
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:14:02 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:43:14 Meaning: SME Approved - (Technical)::Bulk Transitions

	Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:24 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that a Cardiac Gated Sweep scan is modelled correctly in the SOR objects.

Requirements Coverage:

Req ID	Requirement Tag	Description
10055	SWCMP.Scan.Cardiac.GatedSweep	It shall be possible to define cardiac gated sweep synchronized scan protocols that allow adjustment of the start-time and acquisition-duration of a specific heart phase.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanCardiacGatedSweep in the results file is zero.

3.1.7.5. SWCMP.Scan.Cardiac.GatedSweep.tc6

Test ID	7071
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Actual synchronization of the gating to the (simulated) ECG signal is tested. The requirement on minimal gate delay is tested.

Preparation

- Enable trace logging for the "debug" channel of MMICARD. Make sure that the other trace channels in MMICARD are disabled.
- Restart the *scanner* process using *permproc*
- Select TEDB using the Preset Procedures Database Selection tool. This is the protocol database with test protocols.
- Start ExamCards.
- Select TEDB as the current protocol database.
- Start Scan Software.
- Enable physiology simulation with one RR-interval of 1000ms (=60bpm).

Requirements Coverage:

Req ID	Requirement Tag	Description
10055	SWCMP.Scan.Cardiac.GatedSweep	It shall be possible to define cardiac gated sweep synchronized scan protocols that allow adjustment of the start-time and acquisition-duration of a specific heart phase.

Test steps:

Step	Description	Expected Result
Step 1	Select test protocol "Head CNS / CARD / G_2D1_sync" (user defined gate delay of 19ms) and start the scan.	Two markers in the physiology display appear just after the R-tops in the physiology display. Actually ten markers would be expected, but due to performance considerations not every marker is displayed. The scan is completed normally.

3.1.7.6. SWCMP.Scan.Cardiac.GatedSweep.tc7

Test ID	7072
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

IQ on a phantom is tested. All aspects of gated sweep are tested here, except the correct recording of different heart phase, since the phantom is stationary.

Preparation

- Enable trace logging for the "debug" channel of MMICARD. Make sure that the other trace channels in MMICARD are disabled.
- Restart the *scanner* process using *permproc*
- Select TEDB using the Preset Procedures Database Selection tool. This is the protocol database with test protocols.
- Start ExamCards.
- Select TEDB as the current protocol database.
- Start Scan Software.
- Enable physiology simulation with one RR-interval of 1000ms (=60bpm).

Requirements Coverage:

Req ID	Requirement Tag	Description
10055	SWCMP.Scan.Cardiac.GatedSweep	It shall be possible to define cardiac gated sweep synchronized scan protocols that allow adjustment of the start-time and acquisition-duration of a specific heart phase.

Test steps:

Step	Description	Expected Result
Step 1	- Position the head phantom in the head-coil - Select test protocol "Head CNS / BFFE / a6_M2D_GAS" and start the scan. - Wait for the scan to complete and load the resulting images in the viewing context - Play the images as a movie.	No flash effects should be visible.

3.1.7.7. SWCMP.Scan.Cardiac.prospective.perfusion.tc1

Test ID	26120
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-04-13 13:26:51 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Visser, Kees (nly22601) on 2016-04-13 11:59:32 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2016-04-13 11:55:33 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test checks the specific timing of a cardiac perfusion scan.

Requirements Coverage:

Req ID	Requirement Tag	Description
94223	SWCMP.Scan.Cardiac.Prospective.Perfusion.v1	Cardiac perfusion scans need to anticipate for increasing heart rates. This situation is recognized by the following conditions: - Prospective cardiac triggering - Trigger delay unequal to longest - Arrhythmia rejection = no - no trigger period in beats = 1 - Fast imaging mode TFE - Multiple slice 1 heart phase - Interleaved slices = yes - no multi-beat block dynamics In this situation, the RR interval percentage is respected while planning the sequence, i.e. it determines the allowed timing interval to place

Req ID	Requirement Tag	Description
		sequences and in practice it determines the longest allowed trigger delay. But during scanning, it does not complete the interval with delay sequences and it will recognize R-peaks. This functionality allows conflict free modification of cardiac frequency, as long as the timing for the defined slice acquisitions within the RR interval do not exceed the no-trigger period. The picture below illustrates three situations for an RR-interval of 1000 ms (60 BPM) and an first RR-interval percentage of 10% leading to a so-called "no-trigger period" of 900 ms: The first two pictures illustrate the non-cardiac perfusion case where only RR-intervals longer than 900 ms are accepted. Even though the three slices are acquired withing say 700 ms and the scanner only executes a delay sequence. In the last picture, the delay sequence is removed and the scanner starts accepting R-peaks as soon as the slices are acquired. - The maximum accepted heart rate cannot be more than twice the entered heart rate - The heart rate mentioned on the info page will in this case have an additional value between brackets. This value represents the maximum heart rate

Test steps:

Step	Description	Expected Result
Step 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system - Add protocol "Head CNS TFE i3_ct_ms_tfe" to the execution list - Make sure the entered heart rate is 60 BPM - Check the info page	Info page indicates the entered heart rate (60) and the maximum allowed heart rate between brackets should be between 90 and 110.
Step 2	Use Sequence Development Viewer to inspect the SOR::cycle	The duration of SOR::cycle should be between 600 and 700 ms. The delay sequences SQ::FFE_rep_delay and SQ::end_rep_delay should either be zero or non-existent
Step 3	- Change the trigger delay on the Motion page from "shortest" to "longest" - Inspect the info page - Add the logfile to this step	The maximum accepted heart rate (between brackets) should decrease from 99 to 67

3.1.7.8. SWCMP.Scan.Cardiac.Prospective.tc1

Test ID	7059
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]

Approval Status	Approved
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Description:

Verify the appearance and range of the cardiac prospective synchronization parameters.

Test steps:

Step	Description	Expected Result	
Step 1	Select "Head CNS / CARD / T_2D3_5hp" in the scan protocol editor.	Verify that the parameters are present on the motion tab with the specified ranges:	
		Parameter name	Range
		Cardiac Synchronization	no, trigger, gate, retrospective
		device	ECG, PPU, Internal, External
		R-R window (%)	0-100
		Number of heart phases	single phase, user defined, maximum
		defined phases <i>"x-y" depends on other parameters.</i>	x-y
		phase interval	shortest, user defined, longest
		(ms) Present if "phase interval" has value "user defined". <i>"x-y" depends on other parameters.</i>	x-y
		arrhythmia rejection	no, yes
		trigger delay	shortest, user defined, mid

Step	Description	Expected Result	
			diastole, end diastole, user def. table, longest
		adaptive	no, yes

3.1.7.9. SWCMP.Scan.Cardiac.Retrospective.tc1.v2

Test ID	37855
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-05-14 09:41:55 Meaning: QA Approved - (QA) Signed by: Mariappan, Yogesh kannan (310134237) on 2018-05-14 08:30:01 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2018-05-14 08:22:53 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This is to test the range of the cardiac retrospective synchronization parameters and the two ways of specifying the heart phases.

Test steps:

Step	Description	Expected Result
Prep 1	Set the system configuration by executing m_amn_set_test_environment.pl Select 1.5T Ingenia. Start a new exam and provide the mandatory parameters. Load the protocol "Head CNS / CARD / R_2D7_16hp".	N/A
Step 1	In Motion tab: Check the range for the following parameters:	

Step	Description	Expected Result
	a. Cardiac Synchronization. b. Device. c. R-R window (%). d. arrhythmia rejection.	a. Range for Cardiac Synchronization shall include "no, trigger, gate, retrospective". b. Range for device shall include "ECG, PPU, Internal" c. Range for R-R window shall be 0 to 100 %. d. Range for arrhythmia rejection shall include "no, yes".
Step 2	In Motion tab: a. Check for the range of: Recon heart phases b. Set the value of Recon heart phases to 40.	a. Range for Recon heart phases shall be 1 to 256. b. Recon heart phases shall be set to 40 without any conflict.
Step 3	In Contrast tab: Set the Fast imaging mode to 'TFE' and the shot mode to 'default'. In Motion tab: Check the range for the below mentioned parameters a. number of heart phases. b. ACQ heart phases (%)	a. Range for number of heart phases shall include "User defined, maximum". b. Range for ACQ heart phases (%) shall be 50-100.
Step 4	Set the number of heart phases to 'Maximum', and check the range for ACQ phase interval.	Range for the ACQ phase interval shall be 0-500.
Step 5	Ensure that the Heart Rate (beats/min) is set to 60 in the Exam card property editor. In Motion tab: Ensure the parameter 'Heart rate > 250 bmp' is set to 'no'. a. Set the value of ACQ phase interval to 20 and check the value of REC heart phases in info page. b. Set the value of ACQ phase interval to 40 and check the value of REC heart phases in info page.	a. There shall be no conflicts and the value of the REC heart phases shall be 50. b. There shall be no conflicts and the value of the REC heart phases shall be 25.

3.1.8. ME Coil

3.1.8.1. SWCMP.Scan.Coil.Auto.Acquisition.tc1.v1

Test ID	33233
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Buunen, Bas (nly95831) on 2017-04-18 15:03:47 Meaning: QA Approved - (QA)

	Signed by: Visser, Kees (nly22601) on 2017-04-18 11:43:46 Meaning: SME Approved - (Business)
	Signed by: Graaf, Ruud de (nly95749) on 2017-04-18 11:23:38 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test checks the translation of receive channel specification in a protocol into channels specified for reconstruction and for acquisition. The test is automated and runs with biperformer on VM-Ware with at least the configuration "systemP" installed or a WS/STM or RTM.

Test Description

Tests are run for two simulated configurations (WA_3T, 3T_TX) in a VMWARE environment. No real system is involved. For each configuration, the contents of the FE object and RC parameters are checked.

The test configurations are:

1. WA_3T: Ingenia, with the anterior coil additional connected and the posterior coil not in holdback.
2. 3T_TX: 3T Achieva TX (dual transmit) 16 channels with SENSE_NEURO_VASC and SENSE_SPINE_15 connected

Tests for WA_3T:

1. The coil Survey Scan
2. Protocol "fstn2D", DYN, "Head CNS". This results in a body coil scan
3. Batch file modification of the used coil in Anterior with a limited set of channels selected.
4. Batch file modification of the used coil in Posterior with a limited set of channels selected

Tests for 3T_TX:

1. The coil Survey Scan
2. Protocol "fstn2D", DYN, "Head CNS". This results in a body coil scan
3. Batch file modification of the clinical mode coils and the channels selected. Spine3 (All Channels NV, only first 8 channels for Spine)

TestWare

Running the tests is automated by a perl script that uses the biperformer batch interpreter in batch mode
Perl_script: m_imn_channel_test.pl.

The perl script configures the system, starts the batches and compares the result. Note that for comparing the result the logfile will be stripped.

The batch files are:

1.
 - a. m_imn_channel_test.acq: implements the tests for WA_3
 - b. m_imn_ach_channel_test.acq: implements the tests for 3T_TX

Reference files used for comparing the results:

WA_3T

m_imn_channel_test_1_fe.ref : fe object test_case 1
 m_imn_channel_test_2_fe.ref : fe object test_case 2
 m_imn_channel_test_3_fe.ref : fe object test_case 3
 m_imn_channel_test_4_fe.ref : fe object test_case 4

m_imn_channel_test_5.ref : RC Parameters test_case WA

3T_TX

m_imn_ach_channel_test_1_fe.ref : fe object test_case 1 achieve

m_imn_ach_channel_test_2_fe.ref : fe object test_case 2 achieve

m_imn_ach_channel_test_3_fe.ref : fe object test_case 3 achieve

m_imn_ach_channel_test_5.ref : RC Parameters test_case achieve

Test Procedure:

The test is automated by using the batch interpreter. Here is described how to run the test on a VMWARE machine.

Requirements Coverage:

Req ID	Requirement Tag	Description
18971	SWCMP.Scan.Coil.Auto.Acquisiti on	For coil elements that are not selected, no data are acquired during the scan. Coils for which no coil elements are selected, are detuned during data acquisition.

Test steps:

Step	Description	Expected Result
Step 1	Install VMWARE machine configuration CDAS "atlsystemp".	None
Step 2	Login gyrotest	None
Step 3	Run "m_imn_channel_test.pl" in command prompt	None
Step 4	Wait until completed (about 7 minutes:). Attach the generated ref files as objective evidence	Check command prompt window. Test is succeeded when it shows that all has succeeded. ChannelTest m_imn_channel_test_1_fe.ref: Succeeded ChannelTest m_imn_channel_test_2_fe.ref: Succeeded ChannelTest m_imn_channel_test_3_fe.ref: Succeeded ChannelTest m_imn_channel_test_4_fe.ref: Succeeded ChannelTest m_imn_channel_test_5.ref: Succeeded ChannelTest m_imn_ach_channel_test_1_fe.ref: Succeeded ChannelTest m_imn_ach_channel_test_2_fe.ref: Succeeded ChannelTest m_imn_ach_channel_test_3_fe.ref: Succeeded ChannelTest m_imn_ach_channel_test_4_fe.ref: Succeeded ChannelTest m_imn_ach_channel_test_5.ref: Succeeded. When not successful action has to be taken:. When the reference files are outdated. Archive the versions created in G:\Log as the actual reference files in clearcase and please explain in a V&V why this update has taken place. In all other cases a PR should be submitted.

3.1.8.2. SWCMP.Scan.Coil.Auto.HeadCoilDe-selection.tc1

Test ID	27016
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-04-15 15:02:20 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-04-06 16:56:55 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Rozijn, Tom (nly95848) on 2016-04-06 14:41:23 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test for Automatic deselection of the head coil based on positioning.

Requirements Coverage:

Req ID	Requirement Tag	Description
59666	SWCMP.Scan.Coil.Auto.HeadCoilDe-selection	When the system detects that a scan is planned further than 50 cm away from the head table position, the system will exclude the head clinical mode of the connected head coil from being selected.

Test steps:

Step	Description	Expected Result
Prep	Run the C:\nmr\gyroscan\m_amn_set_test_environment.pl script and configure the system as any IRF system. Connect the dS Head coil. Start the graphical Patsup simulation. Put the table in about 100mm, and push the lightvisor button. Enter a new patient and make sure the patient position is 'head first'. Select the F_2D1 protocol from Hospital\Head CNS\FE.	N.A. Preparation.
Step 1	Plan the scan about 600 mm into the feet direction. Note that on the info page the reported limiting SAR is 'local torso SAR'. Start the scan. Inspect devlogcurrent.log with the query: text contains 'ACS RESULTS'. Here is shown which channels are selected. Archive the logfile by making a printscreen or	Only channels from the Posterior coil are selected.

Step	Description	Expected Result
	saving the query result as a text file, and attach it to this step.	

3.1.8.3. SWCMP.Scan.Coil.Auto.HeadCoilSelection.tc1

Test ID	26932
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-04-15 15:02:22 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-04-06 16:56:58 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Rozijn, Tom (nly95848) on 2016-04-06 14:41:25 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test for Automatic selection of the head coil based on positioning.

Requirements Coverage:

Req ID	Requirement Tag	Description
59541	SWCMP.Scan.Coil.Auto.HeadCoilSelection	When the system detects that a scan is planned at the head reference table position, the system automatically selects the head clinical mode of the connected head coil(s). No other clinical mode and no elements of other coils will be selected. This mechanism is active for the patient position 'head first'.

Test steps:

Step	Description	Expected Result
Prep	Run the C:\nmr\gyroscan\m_amn_set_test_environment.pl script and configure the system as any IRF system. Connect the dS Head coil. Start the graphical Patsup simulation. Put the table in about 100mm, and push the lightvisor button. Enter a new patient and make sure the patient position is 'head first'. Select the F_2D1 protocol from Hospital\Head CNS\FE.	N.A. Preparation.

Step	Description	Expected Result
Step 1	Enter a FH off-center of 200 mm. Make sure that on the info page the reported limiting SAR is 'Head SAR'. Start the scan. Inspect devlogcurrent.log with the query: text contains 'ACS RESULTS'. Here is shown which channels are selected. Archive the logfile by making a printscreen or saving the query result as a text file, and attach it to this step.	Only channels from the Head coil are selected.

3.1.8.4. SWCMP.Scan.Coil.Auto.PPDE.tc1

Test ID	8963
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Coil UI for PPDE

Requirements Coverage:

Req ID	Requirement Tag	Description
18962	SWCMP.Scan.Coil.Auto.PPDE	In PPDE the scan protocols can be set to auto mode. In auto mode the scan protocol is validated with the proposed coil parameters. In PPDE the proposed coil parameters are visible to enable coil-dependent protocol development for scans with automatic coil selection.
18963	SWCMP.Scan.Coil.Auto.PPDE.Replace	Coil replacement mechanism is disabled in PPDE.

Test steps:

Step	Description	Expected Result
Step 1	In the SDE environment start the PPDE tool. Select a scan protocol, e.g. Head\Brain\T1_weighted\T1W_SE. The protocol can be selected from a protocol snapshot or from a ClearCase view. In the latter case it is advisable to temporarily check out the protocol used.	- Verify that the scan coil selection can be set to 'auto' mode. - Verify that in PPDE the coil parameters are visible and the PDF is validated with these coils. - Verify that a set of coils can be selected that depends on the system configuration.

3.1.8.5. SWCMP.Scan.Coil.Auto.Q-Body.tc1

Test ID	8966
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Warning message for use of QBC

Requirements Coverage:

Req ID	Requirement Tag	Description
18969	SWCMP.Scan.Coil.Auto.Q-Body	If automatic coil selection selects only coils that are always connected to the system, while other (disconnected) coils are specified in the protocol, a warning will be displayed to the user. Rationale: User may have forgotten to connect a coil, or a coil connector is loose. Using Automatic coil selection, the risk is that such a mistake goes unnoticed until long after the examination, resulting in poor image quality.

Test steps:

Step	Description	Expected Result
Step 1	- Use a simulator with a 3T Achieva configuration, make sure that all coils are enabled. (Same as in Preparation section under User Defined Coil Selection) - Start the ExamCards Scan Environment - Make sure no coils are connected in the coil simulator. - Import the Examcard 'te_ts_coil_selection_SH8.ExamCard' from the software archive, testbase building block. - Make a new exam - Drag the examcard te_ts_coil_selection_SH8.ExamCard in. - Start scan	a warning message pops up: Warning: SmartSelect selected the QBC, but the protocol indicates another coil. Did you connect all the coils you wish to use?

3.1.8.6. SWCMP.Scan.Coil.Auto.ReconMemory.tc1.v1

Test ID	30088
Revision	2
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on

	2017-03-23 16:57:26 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 15:48:47 Meaning: SME Approved - (Technical) Signed by: Visser, Kees (nly22601) on 2017-03-22 15:25:43 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Coil restrictions

Requirements Coverage:

Req ID	Requirement Tag	Description
64103	SWCMP.Scan.Coil.Auto.ReconMemory.v1	Recon memory calculations are performed with the default clinical mode. If there is not enough memory due to the huge number of channels, then recon compression will be advised via conflict handling.
18964	SWCMP.Scan.Coil.Auto.SENSE	If no SENSE-approved coils are connected, then all SENSE scans will result in a conflict on automatic coil selection and SENSE.
18965	SWCMP.Scan.Coil.Auto.SENSE.Factor	SENSE factors higher than the number of selected coil elements are just allowed, but a message is displayed as a warning to the user.

Test steps:

Step	Description	Expected Result
Prep 1	By using m_amn_set_test_environment.pl configure the system as a WA configuration (tedb + all coils enabled). Use simulation mode Acq on host, no recon, which uses reconstructor configuration settings independent of actual hardware. Set the control scan parameter 'Coil UI selection mode' to 'Full Manual'.	N.A. (preparation)
Prep 2	Select scan protocol from tedb Head CNS\PIIP_FEEPI_MS.	N.A. (preparation)
Step 1	Using the CoilState Simulator UI, connect: - dS Anterior coil (on SBM.DCC0) - dS Base coil (on SBM.DCC1) - dS Head coil (on NVC_BASE.CONNBASE)	No parameter conflict should be displayed.
Prep 3	Switch off Smart select.	N.A. (preparation)
Prep 4	Select all channels of the posterior, anterior, ds Base and ds Head coils. Remove the Flex S coil (by pressing the X button).	N.A. (preparation)
Step 2	Make it a scan producing a lot of raw data. Set number of slices to 20. Set ACQ voxel size AP (mm) to the smallest possible value.	A parameter conflict should be displayed on several parameters, including Recon compression. A message should be displayed to indicate that the total amount of data in this scan is too large.
Step 3	Set Recon Compression to Default.	No parameter conflict should be displayed. Objective evidence: Central log file.

Step	Description	Expected Result

3.1.8.7. SWCMP.Scan.Coil.Auto.SmartFoldoverSup.tc1

Test ID	8967
Revision	1
Product Model	Ingenia;Marlin
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Smart fold-over suppression

This test can also be run on an Achieva dStream system.

Requirements Coverage:

Req ID	Requirement Tag	Description
18970	SWCMP.Scan.Coil.Auto.SmartFoldoverSup	If a scan protocol defines foldover suppression by oversampling and the Posterior and/or Anterior coil is selected by automatic coil selection, then extra foldover suppression with SENSE is applied to prevent these foldover artifacts. This will be done fully automatic, invisible to the user, and will not cost any extra scan time. This extra foldover suppression is only done: - when the PC and/or AC are automatically selected, so only in auto coil selection mode. - when the foldover direction is FH or RL. When the foldover direction is AP, no foldover artifacts are expected and performing extra SENSE is not applied to prevent any IQ degradation. - when SENSE is enabled Selection of large coverage coils like the Posterior and/or Anterior coil, may result in foldover artifacts. For example when using the anterior coil for scanning a knee, foldover may occur of the other knee. To prevent these artifacts, SENSE is used to unfold these artifacts. By scanning a larger FOV with a larger SENSE factor, the scan time remains the same, and foldover suppression is guaranteed.

Test steps:

Step	Description	Expected Result
Step 1	On a scanner select a scan protocol for scanning, for example Head\Brain\T2-weighted\T2W TSE. Use the Posterior coil only, i.e. do not connect other coils such as NVC-base. Place a row of phantoms on the patient table distributed over at least 500 mm in the FH direction. Choose scan orientation 'sagital', with phase encoding	The scan runs correctly, the resulting images do not show fold-over artifacts or SENSE artifacts due to intrinsic fold-over.

Step	Description	Expected Result
	direction FH. Enable SENSE, SENSE factor with minimal value of 1.6. (Actual SENSE factor may be set to a higher value by the SW). Enable fold-over suppression by oversampling, adjust the H and F oversampling regions to resolve a possible parameter conflict.	

3.1.8.8. SWCMP.Scan.Coil.Auto.tc1.v1

Test ID	22192
Revision	1
Product Model	Ingenia;Marlin
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-04 13:31:36 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 09:52:05 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Jurrissen, Michel (nly95853) on 2015-02-04 09:29:04 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test can also be run on an Achieva dStream system.

Requirements Coverage:

Req ID	Requirement Tag	Description
18957	SWCMP.Scan.Coil.Auto.Conflicts	All conflicts caused by coil parameters should be suppressed until the actual automatic coil selection has been performed. If the conflicts are not resolved by automatic coil selection, the scan will not start and a conflict is set. The user can now solve the conflict, because the pre-defined coil parameters are replaced by the result of automatic coil selection. After solving the conflict the automatic coil selection is performed again, because the user might have changed the planning.
18955	SWCMP.Scan.Coil.Auto.StartScan	In smart select mode, the coils are automatically selected after starting the scan. So there is no feedback during planning the scan protocol. After starting the scan, it is validated in examcards with the actual coil choice.
21021	SWCMP.Scan.Methods.ExamProperties	The validation of a scan protocol shall include exam properties. Exam properties consists of context information such as patient position, patient orientation, table position, connected coils and information required for defining and approving a set of scans with inherent strict inter-scan timing relationships.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	This test does not only cover the correct insertion of coil selection jobs but also the correct insertion of prescans (coil survey scans and SENSE reference scans). This test must be performed on an OTM with an iRF table including a Posterior coil. Connect as many iRF coils as available. Preferably connect the NVC-base, NVC-head/neck and 2 Anterior coils, so in total maximal 60 coil elements can be selected. Use multiple phantoms to simulate a 'complete' patient, for example a bottle in the NVC coil and a body phantom between the posterior and anterior coil. Set the light visor at the centre of the NVC coil. Optional: to enable the display of prescans and coil selection jobs, set the key ShowPreScan to true in configuration file C:\Program Files (x86)\PMS\Mipnet42\examcards_win_cs.exe.config, and restart examcards. Use the mrmethode\testbase\res\te_irf_workflow.examcard from the SW archive.	The coil protocol parameters are hidden: not visible on the geometry page.	None
Step 2	Run the complete examcard.	- All scans, prescans and coil selection jobs should complete successfully. - Before auto coil selection, the coils are replaced automatically by the connected coils. Changes in scan timing etc. with automatic coil selection are avoided. No parameter conflicts occur. - Scans in the 'head' region use the NVC-base and NVC-Head/neck. After the scan, the 'coils' tab shows used coils and coil elements. - Scans in the 'abdomen' region use the Anterior and Posterior coils. After the scan, the 'coils' tab shows used coils and coil elements. Refer Attached Fig1	

Step	Description	Expected Result	Attachments
	1,1 ✓ Survey 2,1 ✓ CoilSurveyScan 3,1 ✓ ACS cor1 ✓ CoilSel - ACS 4,1 ✓ SenseRefScan 5,1 ✓ ACS_clear cor1 ✓ CoilSel - ACS_clear 6,1 ✓ ACS_clear_FCC cor1 ✓ CoilSel - ACS_clear_FCC 7,1 ✓ ACS_sense2_... cor1 ✓ CoilSel - ACS_sense2_FCC 8,1 ✓ 2st cor2st ✓ CoilSel - 2st 9,1 ✓ 2st_move_table cor2st ✓ CoilSel - 2st_move_table ✓ CoilSel - 2st_move_table 10,1 ✓ 2st_restricted cor2st ✓ CoilSel - 2st_restricted ✓ CoilSel - 2st_restricted 11,1 ✓ CoilSurveyScan 12,1 ✓ ACS corA1 [A] ✓ CoilSel - ACS 13,1 ✓ CoilSurveyScan 14,1 ✓ SenseRefScan 12,2 ✓ ACS_clear_FCC corA2 [A] ✓ CoilSel - ACS_clear_FCC 12,3 ✓ ACS_sense2_... corA3 [A] ✓ CoilSel - ACS_sense2_FCC 15,1 ✓ ACS corA1 [B] ✓ CoilSel - ACS 15,2 ✓ ACS_clear_FCC corA2 [B] ✓ CoilSel - ACS_clear_FCC 15,3 ✓ ACS_sense2_... corA3 [B] ✓ CoilSel - ACS_sense2_FCC		
Step 3	Repeat this test with Posterior coil in holdback.	all scans, prescans and coil selection jobs should complete successfully, this time without Posterior coil.	None

3.1.8.9. SWCMP.Scan.Coil.Auto.TransmitCoil.tc1.v1

Test ID	30101
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.

Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-02 16:58:03 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2017-03-02 16:24:08 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-11-09 09:18:58 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The purpose of this test is to show that if a transmit-receive coil is connected, that this will always be selected automatically.

(If the QBC coil would have been used in such a case, the other coil would have been broken)

Requirements Coverage:

Req ID	Requirement Tag	Description
21021	SWCMP.Scan.Methods.ExamProperties	The validation of a scan protocol shall include exam properties. Exam properties consists of context information such as patient position, patient orientation, table position, connected coils and information required for defining and approving a set of scans with inherent strict inter-scan timing relationships.
18968	SWCMP.Scan.Coil.Auto.TransmitCoil	If a transmit coil is connected, the automatic coil selection will automatically select all coil elements of this transmit coil. Rationale: connected transmit coils prohibit the use of the Q-body.

Test steps:

Step	Description	Expected Result
Step 1	Use a (full) examcards simulator environment configured as a 3T Ingenia system. Use simulation mode 'Acq on host, no recon'. In a command prompt start the ptest.exe program. From the main menu select 9 (Test control service), then 1 (Connect to test control service), 3 (Override Posterior coil state) and finally 3 to simulate a posterior coil in hold-back position. Create a new examination. Choose a scan protocol, e.g. Head\Brains\T2_weighted\T2W_TSE. Click 'Accept' to leave the protocol editor. Now connect a transmit-receive head coil. Press start scan When the scan is ready load the executed protocol again in the editor. Inspect on the coil tab which coil was used.	The T/R Head coil was used

3.1.8.10. SWCMP.Scan.Coil.ClinicalModes.AutoCoilCombinations.tc1

Test ID	13175
Revision	1
Product Model	Achieva;Marlin
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Manual coil selection HST coil.

Tests can be excuted on a STM, ROF or VMWARE

Requirements Coverage:

Req ID	Requirement Tag	Description
27619	SWCMP.Scan.Coil.ClinicalModes.AutoCoilCombinations	For automatic coil selection the CoilSurveyScan scans all clinical modes for coil combinations as long as they are not explicitly excluded. The CoilSurvey scan will only scan 1H coils.

Test steps:

Step	Description	Expected Result
Step 1	Configure the system as a T30 Achieva with 16 receivers and enable all coils.	None.
Step 2	Start the acquisition simulator and the coil simulator.	None.
Step 3	Connect the SENSE_NEURO_VASC and the SENSE_SPINE_15 coil.	None.
Step 4	Start the application soft ware and create a new examination.	None.
Step 5	Drag a CoilSurveyScan (from the PreScans\PreScans folder) to the execution list.	Verify on the coil tab that the CoilSurvey scan consists of 6 stack (one for each clinical mode). Verify that appart from the Q-Body coi, for the SENSE-NV-16 and SENSE-Spine-15 coil combination the following clinical modes are scanned: • SENSE-NV-16 • SENSE-Spine-15 (clinical mode name is not visible) • SENSE-Spine-15.A - SENSE-NV-15.HNPC • SENSE-Spine-15.ABC - SENSE-NV-15.NPC • SENSE-Spine-15.ABCD - SENSE-NV-15.PC Verify that the TopOff clinical modes (clinical modes containing SENSE-NV-16.TO-**) are not scanned.

3.1.8.11. SWCMP.Scan.Coil.ClinicalModes.AutoDefault.tc1

Test ID	8977
Revision	1
Product Model	Ingenia;Marlin
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Head coil selection de-selection

This test can also be run on an Achieva dStream system.

Requirements Coverage:

Req ID	Requirement Tag	Description
21021	SWCMP.Scan.Methods.ExamProperties	The validation of a scan protocol shall include exam properties. Exam properties consists of context information such as patient position, patient orientation, table position, connected coils and information required for defining and approving a set of scans with inherent strict inter-scan timing relationships.
64108	SWCMP.Scan.Coil.ClinicalModes,AutoDefault.v1	In SmartSelect mode, during planning and before coils have been selected automatically, the validation is done using the 'default clinical mode'. Default clinical mode is determined using connected coils, table position and stack information. Rationale: This mechanism ensures that head coils are not used in body table positions, body coils are not used in head table positions and that the correct SAR is used.

Test steps:

Step	Description	Expected Result
Step 1	This test must be performed on a scanner with an iRF patient table. Depending on FH offcentre, automatic coil selection starts from either NVC base/head/neck coils or the anterior/posterior coils. Set the control recon parameter 'Auto coil: ch. sel. SNR loss (%)' to 0.0. With this setting automatic coil selection returns all input coils. Again use the te_irf_workflow examcard. For the geometry corA3 change the FH offcentre to -600 mm. On the Examcard page engage the posterior coil. To bring the posterior into the magnet, move the patient table out of the magnet and back in. Click 'start-scan' to run the te_irf_workflow examcard.	- Scans planned at FH offcentre > -50 mm shall use the full NVC-base, NVC-head/neck coils, but not the posterior and anterior coils. Check this on the 'coils' tab of the completed scans. - Scans with FH offcentre < -500 mm shall use the posterior and anterior coils but not the NVC-base, NVC-head/neck coils. Check this on the 'coils' tab of the completed scans.

3.1.8.12. SWCMP.Scan.Coil.ClinicalModes.AutoSingleCoils.tc1

Test ID	13169
Revision	1
Product Model	Multiva;Omniva
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

CoilSurvey scan HST coil

Tests can be excuted on a STM, ROF or VMWARE.

Tests only need to be perfomred on one of the mentioned configurations (Multiva or Omniva).

Requirements Coverage:

Req ID	Requirement Tag	Description
27620	SWCMP.Scan.Coil.ClinicalModes.AutoSingleCoils	For automatic coil selection the CoilSurveyScan scans the clinical modes (coil parts) for he single coils as long as they are no subset of an other clinical mode (in terms of channels and coil function) or as long as they are not explicitly excluded.

Test steps:

Step	Description	Expected Result
Step 1	Configure the system as a T15 Multiva with 16 receivers and enable all coils.	None.
Step 2	Start the acquisition simulator and the coil simulator.	None.
Step 3	Connect the HST_SC_AH_PH and the HST_AC coil.	None.
Step 4	Start the application soft ware and create a new examination.	None.
Step 5	Drag a CoilSurveyScan (from the PreScans\PreScans folder) to the execution list.	Verify on the coil tab that the CoilSurvey scan consists of 7 stack (one for each clinical mode). Verify that appart from the Q-Body coi, for the HST coil combination the following clinical modes are scanned: - Head - NVA - Cervical Spine - Cardiac - Abdomen - Pelvis
Step 6	Now disconnect the HST_AC coil.	None.
Step 7	Drag a CoilSurveyScan (from the PreScans\PreScans folder) to the execution list.	Verify on the coil tab that the CoilSurveyScan consists of 7 stacks (one for each clinical mode).

Step	Description	Expected Result
		Verify that the clinical modes "Cardiac", "Abdomen" and "Pelvis" for the coil combination HST_SC_AH_PH and HST_AC are not scanned. Verify that instead, the clinical modes "Thoracic Spine", "Lumber Spine" and "Sacrum" for the single HST_SC_AH_PH are scanned. Verify that these 'single coil' clinical modes were not scanned in Step 5 when both the HST_SC_AH_PH and HST_AC were connected.

3.1.8.13. SWCMP.Scan.Coil.ClinicalModes.Exclude.tc1

Test ID	13172
Revision	1
Product Model	Multiva;Omniva
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Manual coil selection HST coil.

Tests can be executed on a STM, ROF or VMWARE.

Tests only need to be performed on one of the mentioned configurations (Multiva or Omniva).

Requirements Coverage:

Req ID	Requirement Tag	Description
27659	SWCMP.Scan.Coil.ClinicalModes.Exclude	It must be possible to exclude a single coil clinical mode from being scanned when it physically covers the same area of one of the coils in a coil combinations clinical mode although the single coil clinical mode is no subset of the combination clinical mode in terms of channels and coil function. For example when the HST_CS_AH_PH coil is connected together with the HST_AC coil, the "Thoracic Spine", "Lumber Spine" and "Sacrum" clinical modes are excluded. This exclusion is specified for the coil parts in the coil resource files.

Test steps:

Step	Description	Expected Result
Step 1	Configure the system as a T15 Multiva with 16 receivers and enable all coils.	None.
Step 2	Start the acquisition simulator and the coil simulator.	None.
Step 3	Connect the HST_SC_AH_PH and the HST_AC coil.	None.

Step	Description	Expected Result
Step 4	Start the application software and create a new examination.	None.
Step 5	Set the general control parameter "Coil UI selection mode" to 'Restricted Manual'.	None.
Step 6	Drag the Head\Brain\Survey\Survey scan to the execution list.	None.
Step 7	- Open the protocol for editing and go to the coil tab. - Make sure SmartSelect is switched off. When necessary, solve conflicts by removing the non-connected coils. - Select the HST_SC_AH_PH (HeadSpine-16) and HST_AC (Torso-16) coil.	Verify that the user can select the clinical modes "NVA", "Cervical Spine", "Cardiac", "Abdomen" and "Pelvis", which are all clinical modes for the two selected coils. Verify that the "Head" clinical mode is not selectable. This clinical mode does not contain the HST_AC coil.
Step 8	Now deselect the HST_AC (Torso-16) coil.	Verify that now the "Head" clinical mode is also selectable and that still the clinical modes "Cardiac", "Abdomen" and "Pelvis" are selectable, although the Torso-16 coil is not selected but still connected. Verify that the clinical modes "Thoracic Spine", "Lumbar Spine" and "Sacrum" are not selectable because the HST_AC coil is still connected.
Step 9	Now only select the HST_AC (Torso-16) coil.	Verify that now there are no selectable clinical modes (because there is only one applicable clinical mode for the selected coil).

3.1.8.14. SWCMP.Scan.Coil.ClinicalModes.tc1

Test ID	13179
Revision	1
Product Model	Achieva;Marlin
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Manual coil selection HST coil.

Requirements Coverage:

Req ID	Requirement Tag	Description
27618	SWCMP.Scan.Coil.ClinicalModes	Clinical modes are always derived from the connected coil set and are based on the specified clinical modes for coil combinations and the specified coil parts for the single coils.

Test steps:

Step	Description	Expected Result
Step 1	Configure the system as a T30 Achieva with 16 receivers and enable all coils.	None.
Step 2	Start the acquisition simulator and the coil simulator.	None.
Step 3	Connect the SENSE_NEURO_VASC and the SENSE_SPINE_15 coil.	None.
Step 4	Start the application software and create a new examination.	None.
Step 5	Set the general control parameter "Coil UI selection mode" to 'Restricted Manual'.	None.
Step 6	Drag the Head\Brain\Survey\Survey scan to the execution list.	None.
Step 7	- Open the protocol for editing and go to the coil tab. - Make sure SmartSelect is switched off. When necessary, solve conflicts by removing the non-connected coils. - Select the SENSE-NV-16 and SENSE-Spine-15 coil.	Verify that for this coil combination the user can select the following clinical modes: - SENSE-Spine-15.A - SENSE-NV-16.HNPC - SENSE-Spine-15.ABC - SENSE-NV-16.NPC - SENSE-Spine-15.ABCD - SENSE-NV-16.PC - SENSE-Spine-15.A - SENSE-NV-16.TO_HNPC - SENSE-Spine-15.ABC - SENSE-NV-16.TO_NPC Verify that the complete (single) SENSE-NV-16 and complete (single) SENSE-Spine-15 are not selectable. These clinical modes do not contain both coils. Verify that the TopOff (*.TO_*) clinical modes are selectable while the CoilSurveyScan did not scan them
Step 8	Now select the SENSE-Spine-15 only.	Verify that since for this coil all elements can be selected simultaneously, there are no multiple clinical modes to select from.
Step 9	Now select the SENSE-NV-16 coil only.	Verify that the selectable clinical modes are: SENSE-NV-16 and SENSE-NV-16.TO_HNPC

3.1.8.15. SWCMP.Scan.Coil.OtherCoils.tc1.v2

Test ID	32991
Revision	1
Product Model	Achieva
Gradient System	Any
Options	Multi Nuclei
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-12 14:03:03 Meaning: QA Approved - (QA)

	Signed by: Arunachalam, Sankaralingam (ing07488) on 2017-04-11 13:03:42 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-11 07:55:15 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Other coils and multinuclei

Requirements Coverage:

Req ID	Requirement Tag	Description
27652	SWCMP.Scan.Coil.OtherCoils.Scanmode	When in scanmode, when other coil is connected, the user must choose which one of the three possible coils is actually connected (= User defined coil selection).
27650	SWCMP.Scan.Coil.OtherCoils.Prepmode	In Prep-mode, user can insert any of the 3 coils in the protocol and that coil will be used for validation. The fact that 3 coils have the same hardware id is irrelevant.

Test steps:

Step	Description	Expected Result
Prep 1	Configure a VM as T30 Achieva with MN. In the service application, in Coil Administration, make sure the "Other" coils (Other-13C, Other-1H, Other-31P, Other-MN and Other-Decouple) are enabled. Create a new current log file	n/a (preparation only)
Prep 2	Start the Examcards ScanEnvironment CoilState simulator, disconnect all coils Import the examcard TS_Coil_Selection_MN.examCard (from ...\\mrmethode\\testbase\\res\\)	n/a (preparation only)
Step 1	<u>Without</u> creating a new exam (in prep mode), load the examcard in the scan list.	No conflicts validating the scans. Note: If there are conflicts, verify again that the system has a valid MN configuration and the 'Other' coils are available. Also, make sure that Examcard is imported <u>after</u> the system is properly configured
Step 2	Open the 31P_OTH in the parameter editor and go to the Coils tab.	Coil 'Other-31P' is selected.
Step 3	add the Other-MN coil from the 'All Coils' tab on the right.	The 'Other-MN' coil will be the selected coil and the

Step	Description	Expected Result
	Next, remove the Other-31P	protocol is not in conflict.
Step 4	Close the editor Create a new exam Load the same examcard again Run the Survey scan. In the Coilstate Simulator, connect the 'Other-Decouple' coil to ConnTRC, and the 'Other-13C' coil to ConnTRC_MN.	13C_OTH, 31P_OTH and MN_OTH are not in conflict.
Step 5	Open '13C_OTH' in the editor and switch to the Coils tab.	Among others, the 3 coils 'Other-13C', 'Other-31P' and 'Other-MN' are displayed. Only 'Other-13C' is selectable.
Step 6	Open '31P_OTH', and switch to the Coils tab.	Among others, the 3 coils 'Other-13C', 'Other-31P' and 'Other-MN' are displayed. Only 'Other-31P' and 'Other-MN' are selectable.
Step 7	Deselect the 'Other-31P' coil and select the 'Other-MN' coil.	The 'Other-MN' coil will be the selected coil and the protocol is not in conflict.
Admin	zip the current log file and attach it to this test run as objective evidence.	N/a (administration)

3.1.8.16. SWCMP.Scan.Coil.ReplacementNotConfiguredCoils.tc1.v1

Test ID	30384
Revision	1
Product Model	Achieva
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-11-14 18:32:55 Meaning: QA Approved - (QA) Signed by: Verburg, Frank (nly95979) on 2016-11-11 13:29:58 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2016-11-09 09:36:19 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

When a none configured coil has a suitable alternative as configured coil it must be replaced by that configured coil.

Requirements Coverage:

Req ID	Requirement Tag	Description
27830	SWCMP.Scan.Coil.Replacement NotConfiguredCoils	When a none configured coil has a suitable alternative as configured coil it must be replaced by that configured coil. With suitable is meant developed for the same purpose.. The following groups of coils are considered to serve the same purpose: • Body: Multi channel coils for covering the body like heart, abdomen etc... • Endo: Group of coils dedicated for Endo. • Microscopy. Group of coils dedicated for microscopy. • Breast. Group of coils dedicated for breast. • Knee. Group of coils dedicated for knee. • Shoulder. Group of coils dedicated for shoulder, • Wrist. Group of coils dedicated for wrist. • Small extremities. Group of coils serving small extremities. • Large extremities. Group of coils for large extremities. • Ankle. Group of coils for the ankle. • Flex. Group of flex coils.
27831	SWCMP.Scan.Coil.DefaultClinica IModeSelection	The default clinical mode will be derived from the expected connected coil set. This will be based on the following rules: • Dedicated coils are preferred above anything else. • User connected coils are preferred above always connected coils • Within user connected coils are also generally connected coils. The generally connected are less preferred. • Of the remaining coil combinations the clinical mode is selected that uses the greatest amount of channels and when there are more modes with the same amount of channels the one which comes first in the Alphabetical order Generally connected means coils the user can leave connected on the system, without the intension to use it. Examples are the HST coil for multiva/omniva and the posterior and NVC_Base for Ingernia.

Test steps:

Step	Description	Expected Result
Step 1	Check in the file MPUCOIL_adapt_protocol.cpp, in the function ucoil_replace_non_configured_coils(), if coilreplacement for none configured coils of the different groups are implemented.	Coil replacement for these groups must exist: * head * sense_body_type * endo * mcroscopy * breast * knee * shoulder * wrist * small_extremities * ankle * large_extremities * flex
Step 2	Check that coils are not replaced. * Install VMWARE machine * Run m_amn_set_test_environment.pl and select Ingernia 3.0T * Select the scanprotocols database * Start examcards environment * Do NOT select a patient * Select abdomen\angio\Survey\Survey * Verify on coil tab the coils displayed	Coils displayed are: * ds_Anterior * ds_Posterior * Q_body
Step 3	Ingernia protocol converted to achieva. * On an installed vmware machine run m_amn_set_test_environment.pl and choose achieva 1.5T * Do NOT select a patient	Coils displayed are: * SenseXL2A-Torso * SenseXL2P-Torso * Q_Body

Step	Description	Expected Result
	* Copy the SWCMP_Scan_Coil_ReplacementNotConfigured.ExamCard from the archive (mrmetho ds/tst) and import it. * Select this examcard * Verify of one protocol the coils on the coil tab	

3.1.8.17. SWCMP.Scan.Coil.TwoConnectorCoils.tc1.v1

Test ID	58139
Revision	1
Product Model	IngCX;Marlin
Gradient System	Any
Options	
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:53:09 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:16:04 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11-15 11:07:33 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

32 channel head coil with one connector connected.

Requirements Coverage:

Req ID	Requirement Tag	Description
27798	SWCMP.Scan.Coil.TwoConnectorCoils	For coils with two connectors it is not allowed to scan when only one of the connectors is connected to the system.

Test steps:

Step	Description	Expected Result
Step 1	Configure the system as a Ingenia CX 3.0T.	None.
Step 2	Start ScannerSim and the Coil Simulator. Start the application SW.	None.
Step 3	Connect the ds head1 32 ch coil.	None.
Step 4	Drag an arbitrary imaging protocol (Head CNS/ TSE/ F_basic_TSE) from the protocol database to the execution list and open it for editing.	Verify that the protocol is in conflict. Check that a conflict is set on SmartSelect, regardless whether SmartSelect is switched on or off. Check that the following mesaages is shown: " with the HEAD1_32_DCI coil, associated

Step	Description	Expected Result
		HEAD2_32_DCI coil is not conected '."

3.1.8.18. SWCMP.Scan.Coil.UnifConv.tc1.v1

Test ID	30405
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-11-14 18:32:38 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2016-11-14 14:29:56 Meaning: SME Approved - (Technical) Signed by: Verburg, Frank (nly95979) on 2016-11-11 13:28:19 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Coil and uniformity conversions

Use a scanner or simulator with a 3T configuration.

Get examcards 'te_r41_to_optimus' and 'te_r3_to_optimus' from the SW archive (mrmethode\testbase). Insert these examcards in the execution list, which will immediately trigger a Smart Select and uniformity conversion. Check if protocol parameters 'Smart Select' (coils tab) and 'Uniformity' (geometry tab) are correctly converted by verifying the expected output of the table below.

Requirements Coverage:

Req ID	Requirement Tag	Description
18993	SWCMP.Scan.Coil.Conversion	During loading examcards in the execution list, protocols will automatically have Automatic Coil Selection. One exception, if a non-smartselectable or manual-selectable coil is used in the protocol then automatic coil selection is not switched on

Test steps:

Step	Description	Expected Result
Prep 1	Use a scanner or simulator with a 3T configuration. Get examcards 'te_r41_to_optimus' and 'te_r3_to_optimus' from the SW archive (mrmethode\testbase). Insert these examcards in the execution list, which will	N.A.

Step	Description	Expected Result
	immediately trigger a Smart Select and uniformity conversion. Check if protocol parameters 'Smart Select' (coils tab) and 'Uniformity' (geometry tab) are correctly converted by verifying the expected output for each scan, after they have been loaded in the editor.	
Step 1	Load Scan R41Manual	Smart Select = Yes Uniformity = Clear
Step 2	Load scan R41Smart	Smart Select = YES Uniformity = Classic
Step 3	Load scan R3HeadClassic	Smart Select = YES Uniformity = Clear
Step 4	Load scan R3BodyTuned	Smart Select = YES Uniformity = body tuned Clear
Step 5	Load scan R3QBody	Smart Select = NO
Step 6	Load scan R3Breast	Smart Select = YES Uniformity = body tuned Clear
Step 7	Load scan R3BreastRFShim	Smart Select = YES Uniformity = Clear
Step 8	Load scan R3MSKClassic	Smart Select = YES Uniformity = Synergy
Step 9	Load scan R3MSKClear	Smart Select = YES Uniformity = Clear
Step 10	Load scan R3SV	No conflict on uniformity Smart Select = YES
Step 11	Load scan R3ktBlast	No conflict on uniformity Smart Select = YES
Step 12	Load scan R3EndoStrong	Smart Select = No

3.1.9. ME Diffusion

3.1.9.1. SWCMP.Scan.Diffusion.Par.tc1.v2

Test ID	41150
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya

	(310175887) on 2019-05-21 10:11:50 Meaning: QA Approved - (QA)
	Signed by: Engelen, Walter van (nly95948) on 2019-05-20 09:26:14 Meaning: SME Approved - (Technical)
	Signed by: Graaf, Ruud de (nly95749) on 2019-05-20 09:13:38 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the display of diffusion related parameters.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanDiffusionPars in the results file is zero.

3.1.9.2. SWCMP.Scan.Diffusion.Par.tc2.v1

Test ID	43735
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-12-05 17:59:22 Meaning: QA Approved - (QA)
	Signed by: Hey, Silke (300170272) on 2019-12-05 16:40:56 Meaning: SME Approved - (Business)
	Signed by: Engelen, Walter van (nly95948) on 2019-12-05 16:35:20 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the specifics of the input diffusion file for the selection "read from file" or "from file" for DTI diffusion mode.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	On a workstation run the script C: \\nmr\gyroscan\commands\m_amn_set_test_environment.pl Select the WA 30 (3T Ingenia) configuration. Start the Application Software Create a new exam. Create new log file. Select the following protocol under the Hospital folder Head CNS\Diffusion\A1_DT/SShEPI Set NSA to value 1.	Scan Protocol is validated without any conflicts.	None
Step 2	Copy the attached dti_vectors_input.txt file that is attached with this test step in E:\Export of the workstation. Rename the file to dti_vectors_input.txt Replace this file with the existing file if any. Set the parameter "directional resolution" to "read new file"	The "directional resolution" is set to "from file" choice without any conflicts. The range of "directional resolution" increases by one. The new choice that is added is "from file".	 dti_vectors_input.zip
Step 3	Open the input file in E:\Export. Note down the name of the file as present in the first line of the file. Observe the addition of a new parameter on the info page.	On the info page a new parameter "User defined DTI scheme" appears. The first entry of the parameter is same as the first entry in the file. The numbers in the bracket indicate the following (1) number of vectors (2) maximum b-factor in the file	None
Step 4	Copy the attached dti_vectors_input.txt file that is attached with this test step in E:\Export of the workstation. Rename the file to dti_vectors_input.txt. Replace this file with the existing file if any. Set the parameter "directional resolution" to "read new file"	The "directional resolution" is set to "from file" choice without any conflicts. On the info page, a new parameter "User defined DTI scheme" appears. The first entry of the parameter appears as "NO_NAME". The numbers in the bracket indicate the following (1) number of vectors (2) maximum b-factor in the file.	 dti_vectors_input.zip
Step 5	Test for 2048 vectors as part of the input file.	"directional resolution" will be set to "from file" and no conflicts occur. The scan is validated successfully.	

Step	Description	Expected Result	Attachments
	Replace the "dti_vectors_input.txt" file with the file attached with this step. This file consists of 2048 vectors. Set "directional resolution" = "read from file". The file selected will be the latest file. Open Sequence Development Viewer. Go to parameters tab. Go to VAL01_DIFF_all_pars. Check the value of VAL01_DIFF_nr_vectors.	VAL01_DIFF_nr_vectors on the sequence development viewer reads 2048.	 dti_vectors_input.zip
Step 6	Test for NSA Set "NSA" = 2. Run the scan.	Protocol validates and executes successfully.	None
Step 7	Test for the maximum number of directions present in an input file. Rename the attached text file to "dti_vectors_input.txt" and place it in E:\Export. This file consists of 1025 directions. Set "directional resolution" = "read from file". The file selected will be the latest file.	The value of the "directional resolution" changes to "error reading file" and a conflict appears on this parameter. The following error message appears in the message bar - "Too many directions provided."	 dti_vectors_input.zip
Step 8	Remove the last line from the input file - "dti_vectors_input.txt". Select "read from file" for "directional resolution". Open Sequence Development Viewer. Go to parameters tab. Go to VAL01_DIFF_all_pars. Check the value of VAL01_DIFF_nr_directions.	"directional resolution" will be set to "from file" and no conflicts occur. The scan is validated successfully. VAL01_DIFF_nr_directions on the sequence development viewer reads 1024.	None
Step 9	Test for the maximum number of b values present in the input file. Rename the attached text file to "dti_vectors_input.txt" and place it in E:\Export. This file consists of 1025 b-values. Set "directional resolution" = "read from file". The file selected will be the latest file.	The value of the "directional resolution" changes to "error reading file" and a conflict appears on this parameter. The following error message appears in the message bar - "Too many b-weightings provided."	 dti_vectors_input.zip
Step 10	Remove the last line from the input file - "dti_vectors_input.txt". Set "directional resolution" = "read from file"	"directional resolution" will be set to "from file" and no conflicts occur. The scan is validated successfully. VAL01_DIFF_nr_weightings on the	None

Step	Description	Expected Result	Attachments
	Open Sequence Development Viewer. Go to parameters tab. Go to VAL01_DIFF_all_pars. Check the value of VAL01_DIFF_nr_weightings.	sequence development viewer reads 1024.	
Step 11	Remove the input file from E:\Export. Set "directional resolution" = "read from file"	"directional resolution" goes to "error reading file" and a conflict appears on this parameter. The following message is seen on the message bar. "The file E:/Export/dti_vectors_input.txt could not be found".	None
Step 12	Test an incomplete file - That is remove one of the four columns from the file. The four columns indicate 1) first three columns are the directions 2) last column represents the b value for the corresponding direction. Rename the attached text file to "dti_vectors_input.txt" and place it in E:\Export. This text file has one out of the four columns missing.	"directional resolution" goes to "error reading file" and a conflict appears on this parameter. The following message is seen on the message bar. "Incomplete vector encountered in file E:/Export/dti_vectors_input.txt".	 dti_vectors_input.zip
Step 13	Test for acquisition of more than one b=0 weightings. Rename the attached text file to "dti_vectors_input.txt" and place it in E:\Export.. Set "directional resolution" = "read from file". Open Sequence Development Viewer. Go to parameters tab. Go to VAL01_DIFF_all_pars. Check the values of the array elements 1-6 of the parameter VAL01_DIFF_weighting_ix_array. Verify that the values of the array elements 1 and 4 have the equal values 0. These indicate the position of b=0 vectors of the attached file. Start Scan.	The file is read without any conflicts. The scan is validated successfully. The scan completes successfully.	 dti_vectors_input.zip
Step 14	Test the saving of the input file along with the protocol. Note the timestamp of the file dti_vectors_mps.txt in the location E:	The examcard is successfully imported. "directional resolution" = "from file" on the contrast tab.	None

Step	Description	Expected Result	Attachments
	\Export. Go to HeadCNS\Diffusion\DTI_FILE_1K_BVAL. Choose the protocol Check the option set for "directional_resolution" on the Contrast tab. Resolve any conflicts. Note the latest timestamp of the file "dti_vectors_mps.txt" in E:\Export.	E:\Export consists of the file: "dti_vectors_mps.txt". The time stamp is updated after the protocol is dragged onto the launchpad. Attach the log file as evidence.	

3.1.9.3. SWCMP.Scan.Diffusion.PhaseInsensitivity.tc1.v2

Test ID	52733
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:23:13 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:01:39 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:24 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the Phase Insensitivity Methods for TSE Diffusion, SPLICE and Alsop (by elimination).

Requirements Coverage:

Req ID	Requirement Tag	Description
63682	SWCMP.Scan.Diffusion.PhaseInsensitivity.Splice	Phase insensitive TSE diffusion imaging shall be obtained using the SPLICE technique. The SPLICE technique separates the odd and even parity signals of each echo, and sampling both signals. After the diffusion gradients, the readout is done. By prolonging the readout gradient GR:m, the different phase components are separated in two echo trains (E1 and E2). Both E1 and E2 are sampled, in 2 separate measurements, and two magnitude images are obtained, which are added to get a final image. When using the SPLICE phase insensitivity technique with a single-shot readout, the minimum number of startup echoes is 1.

Req ID	Requirement Tag	Description
67928	SWCMP.Scan.Diffusion.PhaseIn sensitivity.Alsop	Phase insensitive GraSE diffusion imaging shall be obtained by the Alsop elimination technique. After the diffusion gradients, the magnetization will be fully dephased by a gradient. The non-MG component is eliminated by a 90 degrees RF pulse. The sequence is based on the normal TSE and GraSE sequence with the following, additional objects: GR'deph_pi to dephase the magnetization after the diffusion gradients, RF'pi to flip the non-MG component along the z-axis, GR's_pi to do this slice selectively, GR'reph_pi in the base sequence to rephase the magnetization in order to get the first echo and GR'deph_pi_ME and GR'reph_pi_ME for dephasing and rephasing in the ME sequence.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanDiffusionPhaseInsensitivity in the results file is zero.

3.1.9.4. SWCMP.Scan.Diffusion.SPIR.tc1.v1

Test ID	32467
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-21 14:44:50 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-21 12:47:27 Meaning: SME Approved - (Technical) Signed by: Rozijn, Tom (nly95848) on 2017-03-21 11:35:33 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Residual eddy currents (mainly B0) can degrade SPIR performance. Verify the measures taken to counterbalance these effects:

Requirements Coverage:

Req ID	Requirement Tag	Description
12381	SWCMP.Scan.Diffusion.TSE_GRASE	MR-Methods shall support TSE and GRASE readout technique for diffusion scans as described in section Diffusion weighted TSE and GraSE. DWI based on TSE or GraSE sequences. Phase navigators are applied because of the general multi-shot character.).
12383	SWCMP.Scan.Diffusion.MSh_EPI_Nav	MR-Methods shall support Multi Shot diffusion EPI technique with Phase Navigator as described in section Specific requirements multi-shot diffusion with phase navigators.
12382	SWCMP.Scan.Diffusion.SSh_EPI	MR-Methods shall support Single Shot diffusion EPI technique without Phase Navigator as described in section Specific requirements single shot diffusion without phase navigator.
12380	SWCMP.Scan.Diffusion.Navigator_Reacquisition	MR-Methods shall support correction and/or reacquisition of data based in Phase Navigator input for Multi Shot scans as described in section Navigator based correction and reacquisition. Multi-shot diffusion sequences include a phase navigator that is used for phase correction and navigator quality assessment. In case the navigator quality is found insufficient, the acquired portion of k-space is remeasured.).
12385	SWCMP.Scan.Diffusion.Eddy_Currents	MR-Methods shall prevent problems related to eddy currents due to diffusion gradients as described in section Eddy currents

Test steps:

Step	Description	Expected Result
Prep 1	Position a 3 liter water phantom and a fat bottle in the head coil. Bottles should have its axis horizontal and approximately in the middle of the coil. This will ensure nice circular cross-sections of the bottles when scanning a transversal slice. Select protocol i1_DW/SSh_iq and modify the TE to the shortest possible value. Change the Uniformity to Classic. Enable SPIR and auto shim. Perform this scan. Perform this scan again in sequence development after having set all array elements of the parameter MP_DIFF_c00_spir_[mps]_prev_exs from the parameter group IMP_DIFF_all_pars to 0.	NA (preparation)
Step 2	Compare these two scans in compare mode	Fat suppression in the first scan is identical or better then the fat suppression in the second scan. Also verify that in the first scan no undesired water suppression is present.

3.1.9.5. SWCMP.Scan.Diffusion.tc1.v1

Test ID	32761
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-03 16:02:29 Meaning: QA Approved - (QA)

	Signed by: Graaf, Ruud de (nly95749) on 2017-04-03 15:46:19 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-04-03 15:06:49 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
12381	SWCMP.Scan.Diffusion.TSE_GR ASE	MR-Methods shall support TSE and GRASE readout technique for diffusion scans as described in section Diffusion weighted TSE and GraSE. DWI based on TSE or GraSE sequences. Phase navigators are applied because of the general multi-shot character.).
12383	SWCMP.Scan.Diffusion.MSh_EP I_Nav	MR-Methods shall support Multi Shot diffusion EPI technique with Phase Navigator as described in section Specific requirements multi-shot diffusion with phase navigators.
12382	SWCMP.Scan.Diffusion.SSh_EP I	MR-Methods shall support Single Shot diffusion EPI technique without Phase Navigator as described in section Specific requirements single shot diffusion without phase navigator.
12380	SWCMP.Scan.Diffusion.Navigator_Reacquisition	MR-Methods shall support correction and/or reacquisition of data based in Phase Navigator input for Multi Shot scans as described in section Navigator based correction and reacquisition. Multi-shot diffusion sequences include a phase navigator that is used for phase correction and navigator quality assessment. In case the navigator quality is found insufficient, the acquired portion of k-space is remeasured.).
12385	SWCMP.Scan.Diffusion.Eddy_C urrents	MR-Methods shall prevent problems related to eddy currents due to diffusion gradients as described in section Eddy currents

Test steps:

Step	Description	Expected Result						
Prep 1	- Using the service application, set the system to have 'BASIC' PPU support; make note of the previously set value to be able to reset when done with this test. - Position a 3 liter water phantom in the head coil or NVC coil. Bottle should have its axis horizontal and approximately in the middle of the coil. This will ensure a nice circular cross-section of the bottle when scanning a transverse slice. - Run with the batch interpreter the m_imn_diff_iq.acq batch file. This batch file executes two different scan types. For each scan type the variations listed in Table 4-1 are scanned. For each variation, images as well as viewable raw data are reconstructed. <table><tr><td>Single Shot EPI</td><td>All 8 scans</td></tr><tr><td>Multi Shot Epi</td><td>All 8 scans</td></tr><tr><td>TSE</td><td>Only the first</td></tr></table>	Single Shot EPI	All 8 scans	Multi Shot Epi	All 8 scans	TSE	Only the first	N.A. (preparation)
Single Shot EPI	All 8 scans							
Multi Shot Epi	All 8 scans							
TSE	Only the first							

Step	Description		Expected Result
		two scans and the first two analysis steps are required	
Step 1	Go to the viewing context (Review context) and drag scan 3,1 in the upper-left view port. Draw an ellipse (circle) (Right mouse button, measurement) that precisely encloses the cross-section of the bottle phantom as good as possible. Copy this ROI to all other images of slice 1 of scan 3,1. Increase the number of columns and row such that all images are visible.		All copied ROIs should enclose the object in the same manner as the original ROI. A deviation of 1-2 acquisition pixels is allowed, for a Omni gradient system 2-3 acquisition pixels is allowed.
Step 2	For scan 3,3 the same procedure applies: drag scan 3,3 to the upper-right view port, now draw an elliptical ROI around the peak in k-space of scan 3,3. It might be necessary to zoom in. Copy this ROI to other images of slice 1. Increase the number of columns and row such that all images are visible.		The position of ROI relative to the images is the same for all ROIs.
Step 3	Drag scan 4,1 and scan 4,3 in respectively to the lower-left view port and lower-right view port. Copy ROI (via "copy" and "paste") from the upper-left view port to the same slice in column in the lower-left view port and subsequently to all other images of the same slice. Increase the number of columns and row such that all images are visible.		The ROIs in column 1 should enclose the object similar or better than the ROIs in column 3. Also the edge of the phantom in column 1 should have similar or less blurring than column 3.
Step 4	Drag scan 5,1 and scan 5,3 in respectively to the lower-left view port and lower-right view port. Copy ROI (via "copy" and "paste") from the upper-left view port to the same slice in column in the lower-left view port and subsequently to all other images of the same slice. Increase the number of columns and row such that all images are visible.		The ROIs in column 1 should enclose the object similar or better than the ROIs in column 3. Also the edge of the phantom in column 1 should have similar or less blurring than column 3.
Step 5	Drag scan 6,1 and scan 6,3 in respectively to the lower-left view port and lower-right view port. Copy ROI (via "copy" and "paste") from the upper-left view port to the same slice in column in the lower-left view port and subsequently to all other images of the same slice. Increase the number of columns and row such that all images are visible. Use the ROI statistics (right click on ROI -> Show Details) to verify that for the different diffusion directions, but for the same slice and diffusion weighting, the average signal is similar or more uniform in upper-left viewport than in column lower-left view port.		The average signal does not vary more than 10% in column 1 over the diffusion directions.
Step 6	Copy ROI from the upper-right view port to the same slice in the lower-right view port and subsequently to all other images of the same slice.		The ROIs in upper-right view port should enclose the object similar or better than the ROIs in the lower-right view port.
Step 7	Go to the viewing context (Review context) and drag scan 7,1 in the upper-left view port. Draw an ellipse (circle) (Right mouse button, measurement) that precisely		All copied ROIs should enclose the object in the same manner as the original ROI. A deviation of 1-2 acquisition pixels is allowed, for an Omni gradient

Step	Description	Expected Result
	encloses the cross-section of the bottle phantom as good as possible. Copy this ROI to all other images of slice 1 of scan 7,1. Increase the number of columns and row such that all images are visible. It might be useful to use right mouse button -> "Toggle dimensions" to display all images of the same slice from left to right. Add a screenshot of the comparison to this step.	system 2-3 acquisition pixels is allowed.
Step 9	For scan 7,3 the same procedure applies: drag scan 7,3 to the upper-right view port, now draw an elliptical ROI around the peak in k-space of scan 7,3. It might be necessary to zoom in. Copy this ROI to other images of slice 1. Increase the number of columns and row such that all images are visible.	The position of ROI relative to the images is the same for all ROIs.
Step 10	Drag scan 8,1 and scan 8,3 in respectively to the lower-left view port and lower-right view port. Copy ROI (via "copy" and "paste") from the upper-left view port to the same slice in column in the lower-left view port and subsequently to all other images of the same slice. Increase the number of columns and row such that all images are visible. It might be useful to use right mouse button -> "Toggle dimensions" to display all images of the same slice from left to right. Add a screenshot of the comparison to this step.	The ROIs in column 1 should enclose the object similar or better than the ROIs in column 3. Also the edge of the phantom in column 1 should have similar or less blurring than column 3.
Step 11	Drag scan 9,1 and scan 9,3 in respectively to the lower-left view port and lower-right view port. Copy ROI (via "copy" and "paste") from the upper-left view port to the same slice in column in the lower-left view port and subsequently to all other images of the same slice. Increase the number of columns and row such that all images are visible. It might be useful to use right mouse button -> "Toggle dimensions" to display all images of the same slice from left to right. Add a screenshot of the comparison to this step.	The ROIs in column 1 should enclose the object similar or better than the ROIs in column 3. Also the edge of the phantom in column 1 should have similar or less blurring than column 3.
Step 12	Drag scan 10,1 and scan 10,3 in respectively to the lower-left view port and lower-right view port. Copy ROI (via "copy" and "paste") from the upper-left view port to the same slice in column in the lower-left view port and subsequently to all other images of the same slice. Increase the number of columns and row such that all images are visible. It might be useful to use right mouse button -> "Toggle dimensions" to display all images of the same slice from left to right. Use the ROI statistics (right click on ROI -> Show Details) to verify that for the different diffusion directions, but for the same slice and diffusion weighting, the average signal is similar or more uniform in upper-left viewport than in column lower-left view port. Add a screenshot of the comparison to this step.	The average signal does not vary more than 10% in column 1 over the diffusion directions.

Step	Description	Expected Result
	Use the service application to revert the PPU setting changed in the preparation step.	

3.1.9.6. SWCMP.Scan.Diffusion.tc2.v1

Test ID	33102
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-13 12:06:38 Meaning: QA Approved - (QA) Signed by: Crijns, Sjoerd (310257923) on 2017-04-12 14:36:02 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-12 14:02:07 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the behavior for oblique images for diffusion scans.

Requirements Coverage:

Req ID	Requirement Tag	Description
12384	SWCMP.Scan.Diffusion.Display_DB	MR-Methods shall provide information to correct label the acquired images with respect to diffusion direction as described in section Gyroview UI with correct gray scale deepness as described in section Software interfaces
12382	SWCMP.Scan.Diffusion.SSh_EP I	MR-Methods shall support Single Shot diffusion EPI technique without Phase Navigator as described in section Specific requirements single shot diffusion without phase navigator.
12380	SWCMP.Scan.Diffusion.Navigator_Reacquisition	MR-Methods shall support correction and/or reacquisition of data based in Phase Navigator input for Multi Shot scans as described in section Navigator based correction and reacquisition. Multi-shot diffusion sequences include a phase navigator that is used for phase correction and navigator quality assessment. In case the navigator quality is found insufficient, the acquired portion of k-space is remeasured.).

Test steps:

Step	Description	Expected Result
Step 1	Copy a DWI protocol from the clinical presets into the execution list. Modify it to use gradient overplus. Set the nr of b-factors to 2.	Scan runs without problems.

Step	Description	Expected Result
	Scan this protocol.	
Step 2	Load the images in the viewing context.	Only "I" images have been generated.
Step 3	Copy the previously scanned protocol and modify it to make a DTI scan with directional resolution: medium. Scan this protocol using 'start scan'.	Scan runs without problems.
Step 4	Load the images in the viewing context. Attach Screenshot as objective evidence.	Per slice, 15 "O" images and the "I" images are displayed.
Step 5	Switch the CRC parameter ICRC_diff_o_images_in_db (UI name = Diffusion O images in DB) to yes. Do not store the modified value.	None.
Step 6	Repeat the DWI scan from step 1 and load the images in the viewing context.	Both "I" and "O" images have been generated.

3.1.10. ME Dixon

3

3.1.10.1. SWCMP.Scan.Dixon.Combinations.tc1.v3

Test ID	25690
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-02-02 11:21:38 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Huang, Alan (310109142) on 2016-02-02 10:39:36 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2016-02-02 09:44:42 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result
Step 1	1. Disable all clinical science keys	Check for the following conflicts:

Step	Description	Expected Result
	2. Enable the mDIXON and mDixon MRA software options. 3. Load TEDB. 4. Load: 'Head CNS -> FE -> i1_mffe_Dixon_w'	1. Conflict on Fast Imaging mode = EPI, TFEPI. 2. Conflict on Scan mode = 2D, MS, M2D. 3. Conflict on more than 2 echoes.
Step 2	1. Increase stacks to 2. 2. Change stacks as packages to yes. 3. Change CE profile order per stack to yes. 4. Change one of the profile order for stack A or B to CENTRA.	Check that no conflict occurs on profile order for stack A or B.

3.1.10.2. SWCMP.Scan.Dixon.Combinations.tc2.v3

Test ID	32555
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:37 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:10:54 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-03-22 09:27:33 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Multi acquisition TSE Dixon using VMware

Test steps:

Step	Description	Expected Result
Step 1	Start VMware and load the TEIMADB.IDX test database. Make sure that all CSK SW options are disabled. Create a new logfile. Load: 'Head CNS -> TSE -> i1_SE_MS Dixon_wfio This MS protocol has "mDIXON" (on the contrast page) switched to "yes". Attach the logfile as objective evidence.	Check the Dixon output images on the postprocessing page. Check that all selections (water, IP, OP, and fat) can be made. Check for the following conflicts: • Only SE / TSE allowed. Conflict on GraSE, EPI fast imaging modes. • Conflict on 3D. • Conflict on more than 1 echo. • Set "partial echo" on the contrast page to yes. Conflict on partial echo. • Enable any cardiac synchronization method. Check for a conflict. • Enable any respiratory compensation method. Check for a conflict. • Enable any Navigator compensation method. Check for a conflict.

Step	Description	Expected Result
		Enable magnetization preparation pulses like MTC, flow labeling pulses, SPIR pulses and STIR. Check for a conflict

3.1.10.3. SWCMP.Scan.Dixon.finef0.tc1

Test ID	25119
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-08-27 15:08:00 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2015-08-27 12:59:44 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2015-08-27 11:40:21 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
56369	SWCMP.Scan.Dixon.finef0.v1	Dixon shall use fine f0 determination.

Test steps:

Step	Description	Expected Result
Preparation	Use m_amn_set_test_environment.pl and select configuration Ingenia 1.5T. Clear the logfile by opening a command window and typing 'p_logfile_nu createnewlogfile'. Start application software and create a new examination. Load: 'Head CNS -> FE -> i1_mffe_Dixon_w'	N.A. (preparation)
Step 1	In the scan protocol parameter editor, verify on the Contrast page, Shim = 'default' and verify on the Postproc page, Preparation phases = 'full.' Then run the scan.	Verify in G:\Log\devlogcurrent.log that the following two statements appear. F0: fine f0 determination expected PREPDUR: F0: fine: took ### msec. where # represents a number.

3.1.10.4. SWCMP.Scan.Dixon.maTSE.tc1.v2

Test ID	24013
Revision	2
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Buunen, Bas (nly95831) on 2015-09-28 16:30:10 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2015-09-10 15:03:02 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-09-10 12:59:43 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Multi acquisition TSE Dixon

This test can be performed on a 1.5T or 3.0T system. The water/fat phantom can be found in the reservation system under the name: "Bottle 1000cc Mineral water/Sunflower oil ph_144" or " Bottle 3000cc Fluid 13/Sunflower oil ph_143".

Requirements Coverage:

Req ID	Requirement Tag	Description
55696	SWCMP.Scan.Dixon.maTSE.TechUnique.v2	Dixon TSE multi acquisition scanning shall be supported. The multi-acquisition approach is where two acquisitions are employed to acquire the echoes required for Dixon reconstruction. The two Dixon echoes are acquired such that phase differences between water and fat can accumulate. The combination of the two acquisitions is used to reconstruct Dixon images.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Start the PDF with the TEIMADB.IDX test database. Place the phantom in the bore. The required scans can be found in 'Head CNS -> TSE'	None	None

Step	Description	Expected Result	Attachments
Step 2	Run the i1_TSE_MS HR_Dixon_wi scan with any abdominal or head coil and turn SENSE off.	No problems should occur. o After completing scan execution measure the reconstruction time checking the logfile. Note the time the scan has been completed and reconstruction has been completed. The difference should not be larger than 45 sec. o Check that the i1_TSE_MS HR_Dixon_wi scan produces a water image with a good suppression of the fat signal (the expected result is displayed in Figure 4-1) and in-phase image showing both water and fat displayed as in Figure 3-5.	

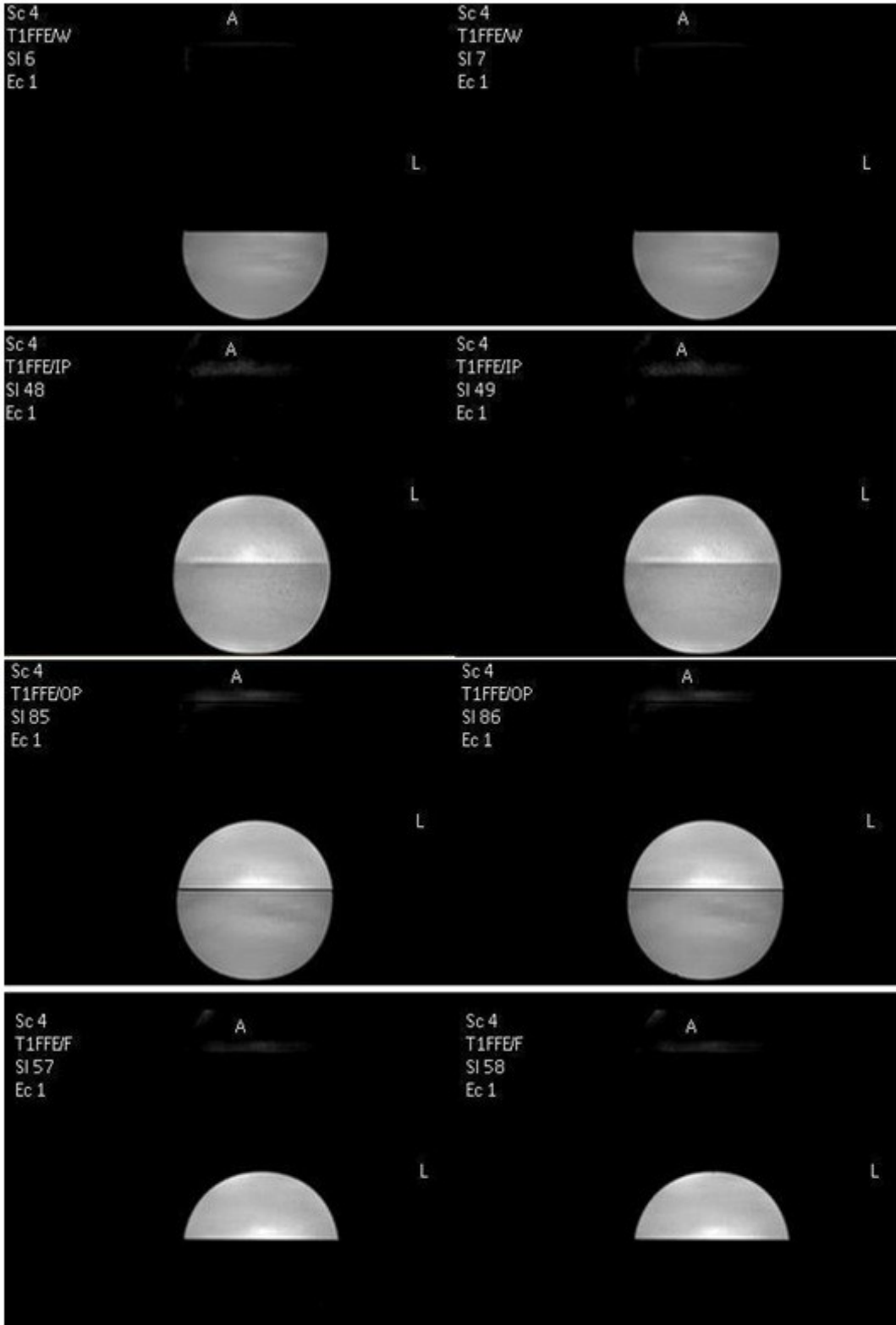
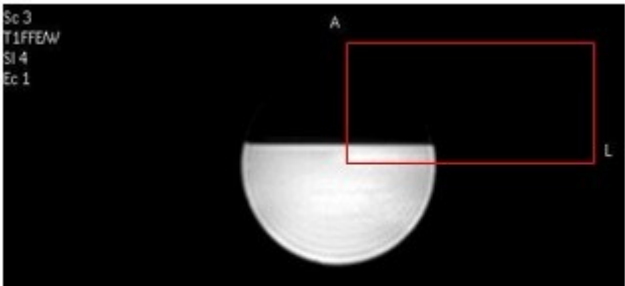
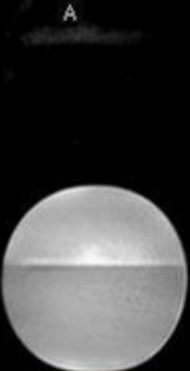
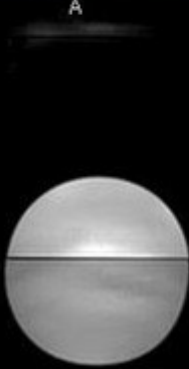
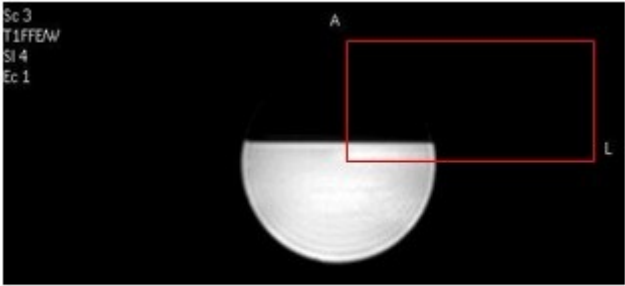
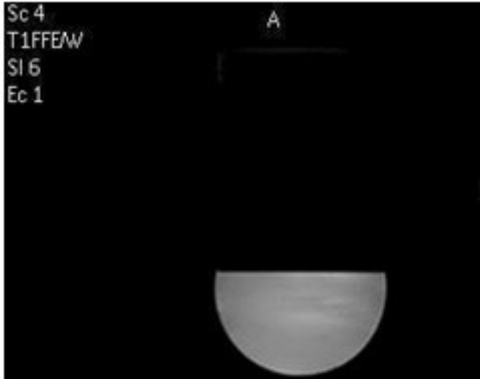
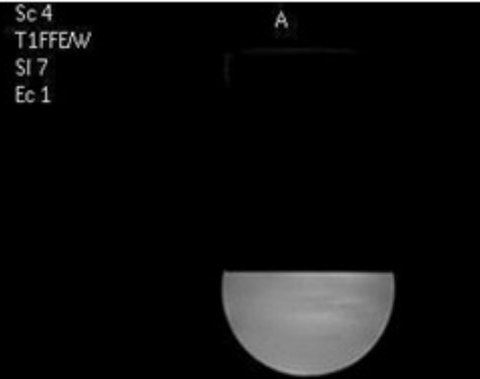
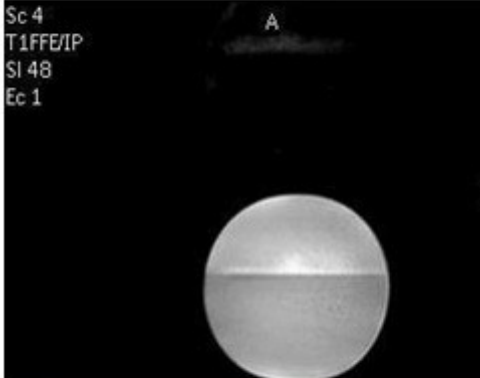
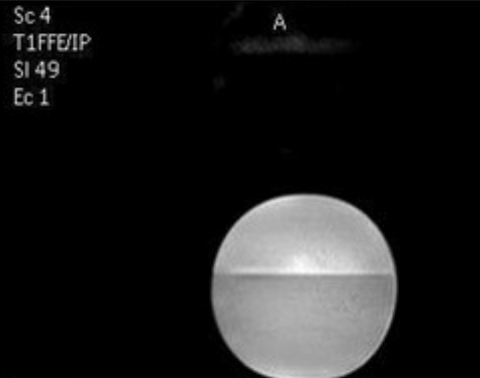
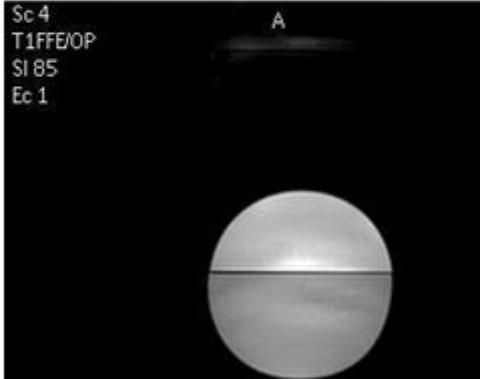
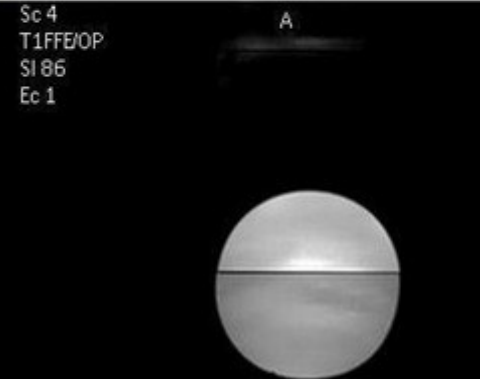
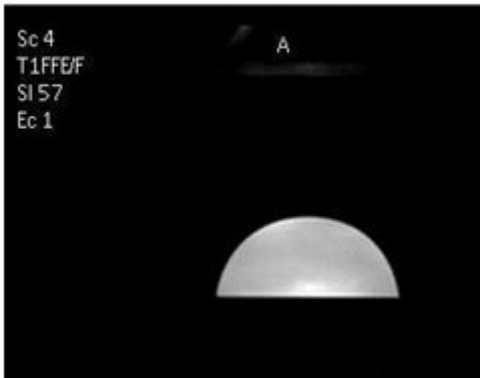
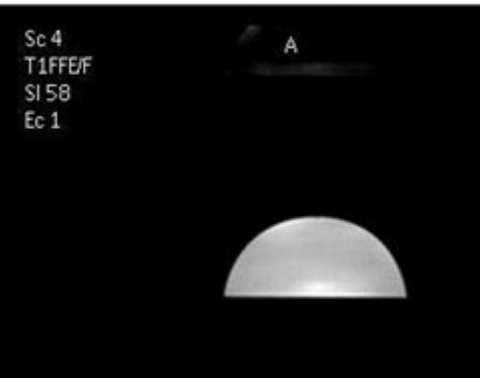
Step	Description	Expected Result	Attachments
	Sc 4 T1FFE/W SI 6 Ec 1	Sc 4 T1FFE/W SI 7 Ec 1	
	Sc 4 T1FFE/IP SI 48 Ec 1	Sc 4 T1FFE/IP SI 49 Ec 1	
	Sc 4 T1FFE/OP SI 85 Ec 1	Sc 4 T1FFE/OP SI 86 Ec 1	
	Sc 4 T1FFE/F SI 57 Ec 1	Sc 4 T1FFE/F SI 58 Ec 1	

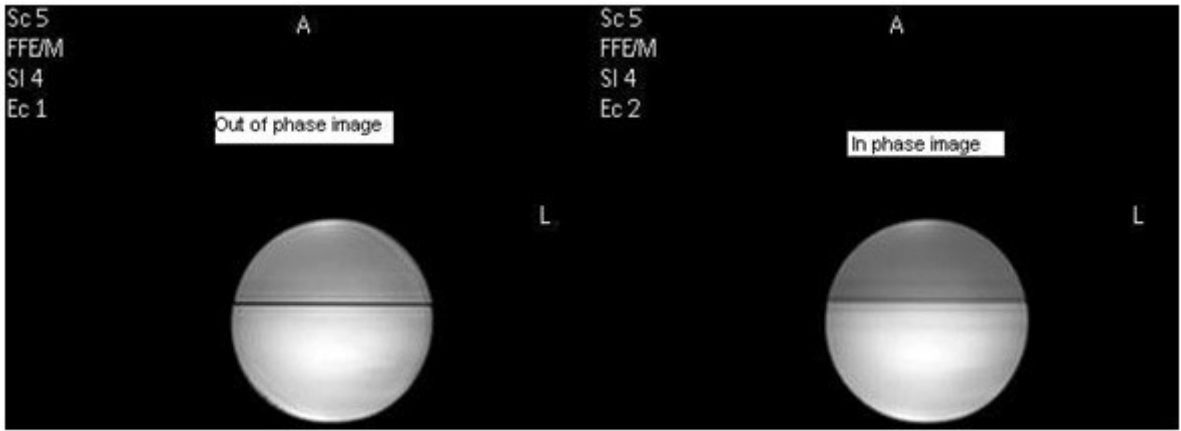
Figure 3-5: W/IP/OP/F

Step	Description	Expected Result	Attachments
	Sc 3 T1FFW SI 4 Ec 1  <i>Figure 4-1: ROI placement over fat</i>		
Step 3	Run the i1_TSE_MS Dixon_wi scan with the abdominal or head coil and turn SENSE off.	Check that i1_TSE_MS Dixon_wi scan produces water, IP images (expected result displayed in Figure 3-5).	

Step	Description	Expected Result	Attachments
	Sc 4 T1FFE/W SI 6 Ec 1		Sc 4 T1FFE/W SI 7 Ec 1
	Sc 4 T1FFE/IP SI 48 Ec 1		Sc 4 T1FFE/IP SI 49 Ec 1
	Sc 4 T1FFE/OP SI 85 Ec 1		Sc 4 T1FFE/OP SI 86 Ec 1
	Sc 4 T1FFE/F SI 57 Ec 1		Sc 4 T1FFE/F SI 58 Ec 1
<i>Figure 3-5: W/IP/OP/F</i>			
Step 4	Run the i1_TSE_IP scan (expected result displayed in) and compare the IP images	Check that the appearance of water, and the transition area of water and fat is similar	None

Step	Description	Expected Result	Attachments
	from both acquisitions.	in the images produced by both sequences. Check the scan time i1_TSE_MS Dixon_wi is comparable within 20% to i1_TSE_IP.	
Step 5	Run the i1_TSE_W scan and compare the images from the scan with the i1_TSE_MS Dixon_wi water images. Place a ROI over the fat segment as shown in Figure 4-1	Check the average signal value from both scans. The average signal of the i1_TSE_MS Dixon_wi water images should be equal or lower than the respective i1_TSE_W images.	
 <i>Figure 4-1: ROI placement over fat</i>			
Step 6	Run the SE_MS Dixon_wfio scan with the abdominal or head coil and turn SENSE off.	Check that SE_MS Dixon_wfio scan produces water, IP, OP, fat images (expected result displayed in Figure 3-5).	

Step	Description	Expected Result	Attachments
	Sc 4 T1FFE/W SI 6 Ec 1		
	Sc 4 T1FFE/IP SI 48 Ec 1		
	Sc 4 T1FFE/OP SI 85 Ec 1		
	Sc 4 T1FFE/F SI 57 Ec 1		
<i>Figure 3-5: W/IP/OP/F</i>			

Step	Description	Expected Result	Attachments
Step 7	Compare the IP scans with the d_TSE_IP scan (expected result displayed in Figure 3-6).	Check that the appearance of water, and the transition area of water and fat is similar in the images produced by both sequences.	
 <i>Figure 3-6: Black line in OP image; no black line in IP image</i>			

3.1.10.5. SWCMP.Scan.Dixon.maTSE.tc2

Test ID	24917
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-12-09 09:41:00 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Polders, Daniel Louis (310171444) on 2015-12-08 11:09:25 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-12-07 15:31:04 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test case is designed to check that the inputs for the profile order regression test have been set correctly. Although this test can be performed on any configuration, the Ingenia 1.5T configuration has been chosen for reproducibility of results.

The first two steps test the calculation of the ord_shot_offset attribute for odd turbo factor with an odd number of shots and even turbo factor with an odd number of shots for Dixon-TSE scans.

The last two steps test the calculation of the ord_shot_offset attribute for even turbo factor with an even number of shots and odd turbo factor with an even number of shots for normal TSE scans.

Requirements Coverage:

Req ID	Requirement Tag	Description
55697	SWCMP.Scan.Dixon.maTSE.Tec hnique.Motion.Averaging	Motion averaging with multi acquisition Dixon TSE scanning shall be enabled. Dixon-TSE motion averaging can be implemented by decoupling shot orders of Dixon acquisitions as illustrated for in-phase (IP) and out-phase (OP) acquisition example.

Test steps:

Step	Description	Expected Result
Setup	Use m_amn_set_test_environment.pl and select configuration Ingenia 1.5T. Start application software and create a new examination Load: 'Head CNS-> TSE -> i1_TSE_MS_Dixon_wi'	N.A. (preparation)
Step 1	Double click on the protocol to open it for editing, then right-click and open Sequence Development Viewer. Under the Objects tab, go to the ORD object.	Check that the attribute ord_shot_offset is 11.
Step 2	Close the Sequence Development Viewer. Double click on the protocol i1_TSE_MS_Dixon_wi, and change the TSE factor to 12. Right-click and open the Sequence Development Viewer. Under the Objects tab, go to the ORD object.	Check that the attribute ord_shot_offset is 21.
Step 3	Close the Sequence Development Viewer. Load: 'Head CNS -> TSE -> F_basic_TSE'. Double click on the protocol to open it for editing, right click on the protocol, and open Sequence Development Viewer. Under the Objects tab, go to the ORD object.	Check that the attribute ord_shot_offset is 0.
Step 4	Close the Sequence Development Viewer. Double click on the protocol F_basic_TSE, and change the TSE factor to 9. Right-click and open the Sequence Development Viewer. Under the Objects tab, go to the ORD object.	Check that the attribute ord_shot_offset is 0.

3.1.10.6. SWCMP.Scan.Dixon.meFFE.technique.EchoTimeRestrictions.tc1.v1

Test ID	55232
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2022-07-28 10:13:40 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2022-07-28 07:55:10 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2022-07-27 10:42:49 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test is designed to test echo time restrictions for 1.5T and 3T introduced for Dixon-FFE.

There are three mechanisms being tested here:

- 1) Assuming that the echo times is set to shortest/shortest, if increasing the first echo time will result in an allowable combination, then the first echo time should automatically increase.
- 2) Assuming that the echo times is set to shortest/shortest, if increasing the first echo time will not result in an allowable combination, then conflict guidance should guide the user to the highest resolution that results in allowable echo times.
- 3) If either the first or second echo time falls into an out-phase condition, this should not be allowed.

Requirements Coverage:

Req ID	Requirement Tag	Description
55803	SWCMP.Scan.Dixon.meFFE.Technique.EchoTimeRestrictions.T15.v2	For 1.5T, mDIXON-FFE shall support TE0 and ES shown by the green region.
55804	SWCMP.Scan.Dixon.meFFE.Technique.EchoTimeRestrictions.T30.v2	For 3.0T, mDIXON-FFE shall support TE0 and ES shown by the green region.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanDixonmeFFEtechniqueEchoTimeRestrictions in the results file is zero.

3.1.10.7. SWCMP.Scan.Dixon.meFFE.technique.tc1.v2

Test ID	24015
Revision	3
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Buunen, Bas (nly95831) on 2015-09-28 16:30:26 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2015-09-10 15:03:05 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-09-10 12:59:47 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Interactive testbay test

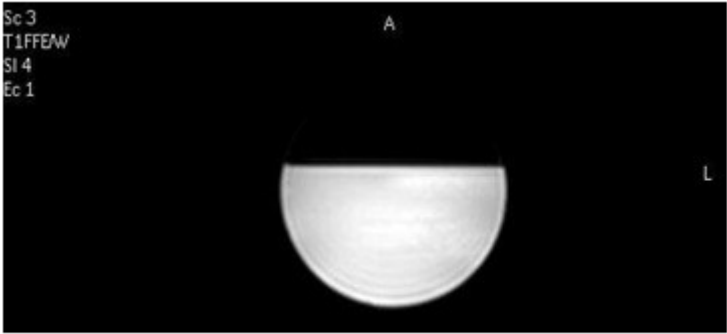
This test can be performed on a 1.5T or 3.0T system. The water/fat phantom can be found in the reservation system under the name: "3l. Bottle with 1.5 l. L11 and 1.5 l. Mineral oil_ph_096".

Requirements Coverage:

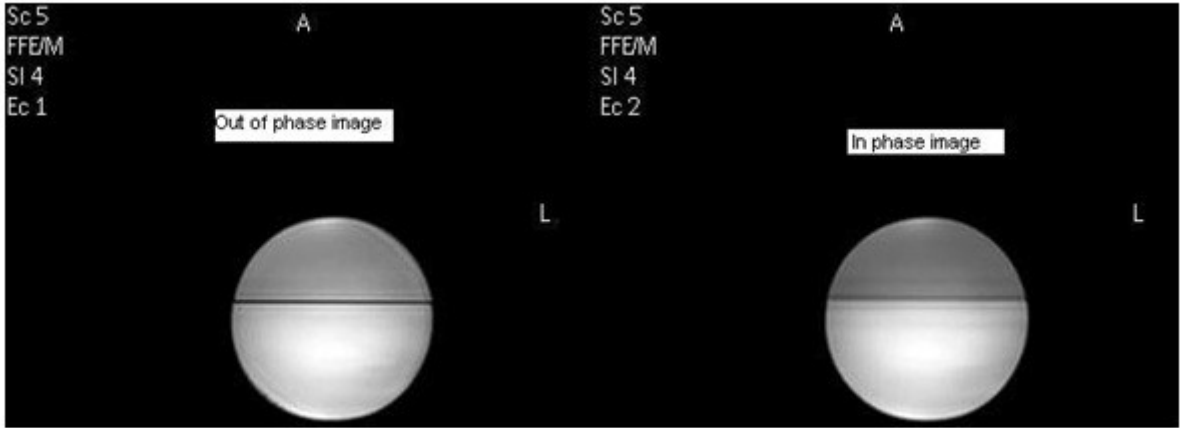
Req ID	Requirement Tag	Description
55690	SWCMP.Scan.Dixon.meFFE.technique.v3	Dixon shall make use of a standard multi-echo FFE and TFE and take extra measures for phase correction.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Start the PDF with the tedb test database. Place the phantom in the bore. The required scans can be found in 'Head CNS -> FE'	None	None
Step 2	Plan the i1_mffe_Dixon_w scan on the centre of the phantom, select any abdominal or head coil and turn SENSE off. Run it, no problems should occur.	Verify that the i1_mffe_Dixon_w Dixon scan produces a water image with a good suppression of the fat signal (the expected result is displayed in Figure 3-4).	

Step	Description	Expected Result	Attachments
	Sc 3 T1FFEAW SI 4 Ec 1  <i>Figure 3-4: Water-only</i>		
Step 3	Plan the i1_mffe_Dixon_wfio scan on the centre of the phantom, select any abdominal or head coil and turn SENSE off. Run it.	Verify that i1_mffe_Dixon_wfio Dixon scan produces water, IP, OP, fat images (expected result displayed in Figure 3-5).	

Step	Description	Expected Result	Attachments
	Sc 4 T1FFE/W SI 6 Ec 1	Sc 4 T1FFE/W SI 7 Ec 1	
	Sc 4 T1FFE/IP SI 48 Ec 1	Sc 4 T1FFE/IP SI 49 Ec 1	
	Sc 4 T1FFE/OP SI 85 Ec 1	Sc 4 T1FFE/OP SI 86 Ec 1	
	Sc 4 T1FFE/F SI 57 Ec 1	Sc 4 T1FFE/F SI 58 Ec 1	
<i>Figure 3-5: W/IP/OP/F</i>			

Step	Description	Expected Result	Attachments
Step 4	Run the d_FFE_OP_IP scan (expected result displayed in Figure 3-6) and compare the IP and OP images from both acquisitions.	Verify that the appearance of water and fat, and the transition area of water and fat is similar in the images produced by both sequences (a dark line should be visible at the transition between water and fat in out-of-phase images but not in in-phase images).	
 <i>Figure 3-6: Black line in OP image; no black line in IP image</i>			

3.1.10.8. SWCMP.Scan.Dixon.meTSE.tc1

Test ID	22342
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-05 14:12:39 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2015-02-05 09:55:27 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-04 15:07:02 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
66057	SWCMP.Scan.Dixon.meTSE.Tec hnique.v3	Dixon TSE multi-echo scanning shall be supported with intra-shot interleaved echoes. Figure 1. Sequence diagram for mDIXON-TSE multi-echo with intra-shot interleaved echoes. Multi-echo Dixon TSE consists of an intra-shot interleaved multi-echo TSE sequence with the acquisition of shifted echoes every other TSE refocusing pulse. In the example of figure 1, the top row shows a normal TSE sequence, and the bottom shows a multi-echo (number of echoes = 2) intra-shot interleaved mDIXON-TSE sequence.

Test steps:

Step	Description	Expected Result	Attachments
Setup	- On FSF page, enable all commercial software options. Disable all clinical science keys. - Use m_amn_set_test_environment.pl and select configuration wa30t - Open G:\Site\bstrace.dat and add the following line: GVE * * - Start application software. - Load: 'Pelvis -> PROPELLER -> sT2W_TSE_MVXD and if necessary, resolve any conflicts. - Run the scan to create mpf.gve	N. A. (preparation)	None
Step 1	Open GVE, load mpf.gve: Verify that non-shifted echoes and shifted echoes are acquired one after another with the same profile by measuring the positive and negative areas of phase encode gradients.	Positive area and negative area of phase encode gradients for non-shifted and shifted echoes are equal.	 nonshifted and shifted echo same p

3.1.10.9. SWCMP.Scan.Dixon.Par.tc1

Test ID	41066
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-06-11 11:42:45 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2019-06-11 10:58:18 Meaning: SME Approved - (Business) Signed by: Mariappan, Yogesh kannan (310134237) on 2019-06-11 10:14:52 Meaning: SME Approved - (Technical)

Approval Status	Approved
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Description:**Test steps:**

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanDixonPar in the results file is zero.

3.1.10.10. SWCMP.Scan.Dixon.Quant.Parameters.tc1

Test ID	8654
Revision	2
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-04-09 14:54:45 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-04-08 10:34:34 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: IJperen, Gert van (nly95968) on 2015-04-07 14:08:08 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Start VMware and select the tedb test database.

Requirements Coverage:

Req ID	Requirement Tag	Description
55692	SWCMP.Scan.Dixon.Quant.Tech nique	Dixon-Quant shall be enabled with a 3D mDIXON FFE acquisition sequence using 6 echoes, which allows for reconstruction of Fat Fraction and/or T2*/R2* images in the liver.

Test steps:

Step	Description	Expected Result
Step 1	Load: 'Head CNS -> FE -> i1_mffe_DixonQuant_ff_t2star_r2star'	Verify the range of the parameter Dixon image type for the parameter combinations: water, fat, IP, OP, FF, T2*, R2* selectable

3.1.10.11. SWCMP.Scan.Dixon.Quant.Restrictions.tc1.v2

Test ID	32892
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-05 17:43:05 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-04-05 17:05:39 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on 2017-04-05 15:24:37 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test case is designed to check conflicts placed on the Dixon-Quant sequence.

Step 1 checks for required conflicts on UI parameters.

Step 2 checks in the code that the proper coils are enabled for Dixon-Quant.

Step 3 checks that when a Dixon-Quant approved coil is selected, the scan runs until completion.

Step 4 checks that when a non-Dixon-Quant approved coil is selected, the protocol is in conflict.

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl and select the configuration 'Ingenia 3.0T'. - Use the service application to enable "dS Anterior" and "dS Base" coils - Make sure that all CSKs are switched off. - Ensure the use of TEDB. - Start application software and create a new examination. - Start ExamCardScanEnvironment - Start the graphical patsup simulator; set the light visor and move the table in. - Start the coil state simulator and attach the ds Anterior coil - start a new log file	n/a (preparation)
Step 1	Load protocol: 'Head CNS -> FE -> i1_mffe_DixonQuant_ff_t2star_r2star'	The protocol has the following conflicts: 1. Flip angle fixed to 3 degrees (3T) 2. Echo spacing longer than 1.3ms 3. First echo time longer than 1.5ms 4. Number of echoes smaller than 6 5. Conflict on halfscan and partial echo
Step 2	Open "ClearCase_view_of_stream_under_test\system\dc\coils\coilconfiguration\src\". Open a command prompt and type "findstr "DixonQuantApproved" *.res > DixonQuantApproved.txt.	Verify that 'DixonQuantApproved' appears in the 'flags' field for the following six coils: • mpgcoil_anterior • mpgcoil_sense_xl_torso • mpgcoil_hst_ac • mpgcoil_hst_ac_8 • mpgcoil_sense_xl_torso_ant_2

Step	Description	Expected Result
Step 3	- Exit examcards - Use Scan Protocol Database Selection to select scanprotocols.pdb. - Start Application software and enter a new patient. - Start the Simulation Mode (Acq on host, Recon on host). - Open Coilstate Simulator UI and plug-in the dS Anterior coil. - Go to the following directory Abdomen->Liver->Specials and pull over the mDIXON-Quant_BH protocol to the execution list. - Click on Start Scan.	mpgcoil sense xl torso post 2 Verify that the scan runs until completion.
Step 4	- in the Coilstate Simulator UI - o unplug the ds Anterior coil - o Plug in the dS Base coil - Go to the following directory Abdomen->Liver->Specials and pull over the mDIXON-Quant_BH protocol to the execution list. Attach the log file to this step as objective evidence.	Verify that there is a conflict on the mDIXON parameter.

3.1.11. ME Dynamics

3.1.11.1. SWCMP.Scan.Dynamics.4DTHRIVE.tc1

Test ID	25581
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-01 10:47:04 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2016-03-01 10:18:57 Meaning: SME Approved - (Technical) Signed by: Huang, Alan (310109142) on 2016-02-29 13:55:26 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test has been designed around the Scanning TestFramework on a VMWare machine. The script mrTest.rb is used for automatic execution of a 4DTHRIVE protocol, uses profile logging as an input and

generates a text file (report) containing the number profiles per dynamic and the number of duplicate profiles. The tester inspects the report for verification of the test.

Requirements Coverage:

Req ID	Requirement Tag	Description
56678	SWCMP.Scan.Dynamics.4DTHRIVE	A 4DTHRIVE scan protocol (Eur J Radiol 74(3):529-535,2010 Jun) shall enable a T1 weighted 3D TFE dynamic series which acquires all central kykz-profiles (keyhole) for each dynamic. The number of central kykz-profiles shall be specified by the user. The acquisition of the complete set of peripheral kykz-profiles shall be performed during the reference scan.

Test steps:

Step	Description	Expected Result
Step 1	a) Install a ROF. b) start G:\methodstestframework\src\mrTest.rb by typing: ruby mrTest.rb c) Select option: [2] Component Tests d) Select Suboption [0] Dynamics e) Select Suboption [2] Run 4DTHRIVE component test	Verify that the scans are executed by BatchInterpreter without conflict or abort.
Step 2	After the test is finished open G:\Log\methodstestframework\Dynamics\#stream\#date-time\4dthrive_detailed_report.txt	Verify in "4dthrive_detailed_report.txt" that: a) The report contains the correct configuration and current date/time. b) For all dynamic scans, the measured number of profiles is equal to the expected number of profiles. c) All dynamic scans except for the last (reference) have the same number of profiles. d) The last dynamic scan has more profiles than the other dynamic scans.
Step 3	In G:\Log\methodstestframework\Dynamics\#stream\#date-time\4dthrive_detailed_report.txt, verify that there are no duplicate profiles.	The file "4dthrive_detailed_report.txt" reports that: a) The number of double profiles in the Keyhole is 0 b) The number of double profiles in the Reference scan is 0
WrapUp	Attach 4dthrive_detailed_report.txt as supporting evidence.	N.A. (WrapUp)

3.1.11.2. SWCMP.Scan.Dynamics.4DTRAK.tc1

Test ID	25523
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-01 10:47:07 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2016-03-01 10:21:32 Meaning: SME Approved - (Technical) Signed by: Huang, Alan (310109142) on 2016-02-29 13:55:28 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test has been designed around the Methods TestFramework on a VMWare machine. The script mrTest.rb is used for automatic execution of a 4DTRAK protocol, uses profile logging as an input and generates a text file (report) containing the number profiles per dynamic and the number of duplicate profiles. The tester inspects the report for verification of the test.

Requirements Coverage:

Req ID	Requirement Tag	Description
55909	SWCMP.Scan.Dynamics.4DTRAK	A 4DTRAK scan protocol (JMRI 27:1455-1460,2008) shall enable a 3D FFE dynamic series which acquires all central kykz-profiles (keyhole) for each dynamic. The number of central kykz-profiles shall be specified by the user. The acquisition of the complete set of peripheral kykz-profiles shall be performed during the reference scan.

Test steps:

Step	Description	Expected Result
Step 1	a) Install a ROF. b) start G:\methodtestframework\src\mrTest.rb by typing: ruby mrTest.rb c) Select option: [2] Component Tests d) Select Suboption [0] Dynamics e) Select Suboption [0] Run 4DTRAK component test	Verify that the scans are executed by BatchInterpreter without conflict or abort.
Step 2	After the test is finished open G:	Verify in "4dtrak_detailed_report.txt" that:

Step	Description	Expected Result
	\\Log\methodstestframework\Dynamics\#stream\#date-time\4dtrak_detailed_report.txt	a) The report contains the correct configuration and current date/time. b) For all dynamic scans, the measured number of profiles is equal to the expected number of profiles. c) All dynamic scans except for the last have the same number of profiles. d) The last dynamic scan has more profiles than the other dynamic scans.
Step 3	InG:\Log\methodstestframework\Dynamics\#stream\#date-time\4dtrak_detailed_report.txt, verify that there are no duplicate profiles.	The file "4dtrak_detailed_report.txt" reports that: a) The number of double profiles in the Keyhole is 0 b) The number of double profiles in the Reference scan is 0
WrapUp	Attach 4dtrak_detailed_report.txt as supporting evidence.	N.A. (WrapUp)

3.1.11.3. SWCMP.Scan.Dynamics.4DTRAK_XD.breathold.tc1.v1

Test ID	32932
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-07 13:07:12 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2017-04-07 12:23:21 Meaning: SME Approved - (Technical) Signed by: Nabuurs, Christine (310147336) on 2017-04-07 11:46:26 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test is a manual test using the MR application on a VMWare machine on any configuration.

The purpose of this test is to verify the conflicts on the combination of 4DTRAK_XD with breatholds and split dynamics for protocols with and without a mask.

Requirements Coverage:

Req ID	Requirement Tag	Description
56679	SWCMP.Scan.Dynamics.4DTRAK_XD.breathhold	For 4DTRAK_XD, profiles shall only be shared within one breathhold or within one scan identity of a split dynamic scan.

Test steps:

Step	Description	Expected Result
Prep 1	a) Start VMware and run: C:\nmr\gyroscan\commands\ m_amn_set_test_environment.pl. Select any configuration. b) Load the following protocol: Hospital/Head CNS/DYN/ i1_3D_FFE_dyn_centra_keyh_viewsh.	N.A. (preparation).
Step 1	Go to "Motion" and set the parameter "Respiratory compensation" to "breath hold". Start the scan.	The scan runs without conflicts. A breath hold is performed per pre scan. Only one breath hold is performed for the scan itself.
Step 2	Load the following protocol: Hospital/Head CNS/DYN/ i1_3D_FFE_dyn_centra_keyh_viewsh Right mouse click on the examcard. Select "Group/Split" and "Split". Select "Group/Split" and "Edit Group". Choose 1 dynamic scan for the first split (in this case the mask) and 5 for the second split. Ensure that the delay mode is set to shortest for both splits. Run the scan.	The scan runs without conflicts or aborts. The second split runs through all 5 dynamic scans without prompting for manual starts.
Step 3	Set the number of dynamics of the first group to 2. Exit the Edit mode. Check for conflicts.	Conflicts appear on the split scans and the info line shows "Viewsharing. First split may only contain the mask. Only one split is allowed."
Step 4	a) Set the number of dynamics for the first group back to 1. b) Set the delay mode of the second Group to "Manual". Check for conflicts. c) Set the delay mode of the second Group to "Manual start". Check for conflicts. d) Run the scan.	a) The conflict disappears. b) Conflicts appear on - the split scans - dyn scan range - viewsharing c) The conflict disappears. d) The scan runs without conflicts or aborts. For the second split scan, manual start is required.
Step 5	Set the delay modes back to "Shortest". Split the lower Group. Check for conflicts.	Conflicts appear on - the split scans - dyn scan range - viewsharing The info line shows "Viewsharing. First split may only contain the mask. Only one split is allowed."
Step 6	Load the following protocol: Hospital/Head CNS/DYN/ i1_3D_FFE_dyn_centra_keyh_viewsh. Set "mask" to "no". Select "Group/Split" and "Split".	Conflicts appear on: - the split scans - viewsharing - mask The infobar shows the following message: "Split dynamics with viewsharing require a mask to

Step	Description	Expected Result
		be acquired".
WrapUp	Make a printscreen of the result of step 6 and attach this to the test report for evidence.	N.A. (WrapUp)

3.1.11.4. SWCMP.Scan.Dynamics.4DTRAK_XD.Restrictions.tc1

Test ID	25530
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-12-22 13:50:04 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Beck, Gabrielle (nly95705) on 2015-12-22 13:21:54 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-12-22 10:59:54 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test is a manual test using the MR application on a VMWare machine on any configuration.

The purpose of this test is to check the limitation on the minimal nr of dynamics for a 4DTRAK_XD scan with a given peripheral density (%):

1. with mask
2. without mask.

Requirements Coverage:

Req ID	Requirement Tag	Description
55889	SWCMP.Scan.Dynamics.4DTRAK_XD.Restrictions	A 4DTRAK_XD scan protocol shall only be executed when: - the viewsharing factor is lower than or equals the total nr of dynamics in the case of using no mask. - the viewsharing factor is lower than or equals the total nr of dynamics -1 in the case of using a mask

Test steps:

Step	Description	Expected Result
Prep 1	a) Start VMware and run: C:\nmr\gyroscan\commands\	N.A. (preparation).

Step	Description	Expected Result
	m_amn_set_test_environment.pl. Select any configuration. b) Load the following protocol: Hospital/Head CNS/DYN/ i1_3D_FFE_dyn_centra_keyh_viewsh. Note: For each separate step return to the settings of this original protocol.	
Step 1	Set "dyn scans" to all possible values <6.	It is not possible to set "dyn scans" to values <6 when "peripheral density" = 20 and "mask" = yes.
Step 2	Repeat step 1 for "mask" = no.	It is not possible to set "dyn scans" to values <5 when "peripheral density" = 20 and "mask" = no.
WrapUp	Make a printscreen of the result of step 2 and attach it to the test report for evidence.	N.A. (WrapUp)

3.1.11.5. SWCMP.Scan.Dynamics.4DTRAK_XD.tc1

Test ID	25524
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-01 10:47:09 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2016-03-01 10:23:56 Meaning: SME Approved - (Technical) Signed by: Huang, Alan (310109142) on 2016-02-29 13:55:30 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test has been designed around the Methods TestFramework on a VMWare machine. The script mrTest.rb is used for automatic execution of the test. This script runs a 4DTRAK_XD protocol, uses profile logging as an input and generates a text file (report) containing the number profiles per dynamic and the number of duplicate profiles. The tester inspects the report for verification of the test.

Requirements Coverage:

Req ID	Requirement Tag	Description
55886	SWCMP.Scan.Dynamics.4DTRAK_XD	A 4DTRAK_XD scan protocol (MRM 61:1242-1248, 2009) shall enable a 3D FFE dynamic series which acquires all central kykz-profiles (keyhole) and only a part of all peripheral (non-keyhole) kykz-profiles for each

Req ID	Requirement Tag	Description
		dynamic. The acquisition of one complete set of peripheral profiles shall be equally distributed over a number of sequential dynamics defined by the user (viewsharing factor = 100/peripheral density %).

Test steps:

Step	Description	Expected Result
Step 1	a) Install a ROF. b) start G:\methodstestframework\src\mrTest.rb by typing: ruby mrTest.rb c) Select option: [2] Component Tests d) Select Suboption [0] Dynamics e) Select Suboption [1] Run 4DTRAK-XD component test	Verify that the scans are executed by BatchInterpreter without conflict or abort.
Step 2	After the test is finished open G: \Log\methodstestframework\Dynamics\#stream\#date-time\ 4dtrak_xd_detailed_report.txt	Verify in "4dtrak_xd_detailed_report.txt" that: a) The report contains the correct configuration and current date/time. b) The measured number of profiles is equal to the expected number of profiles. c) All dynamic scans have the same number of profiles.
Step 3	In G: \Log\methodstestframework\Dynamics\#stream\#date-time\ 4dtrak_xd_detailed_report.txt, verify that there are no duplicate profiles.	The file "4dtrak_xd_detailed_report.txt" reports that: a) The number of double profiles in the Keyhole is 0 b) The number of double profiles in the Periphery is 0
WrapUp	Attach 4dtrak_xd_detailed_report.txt as supporting evidence.	N.A. (WrapUp)

3.1.11.6. SWCMP.Scan.Dynamics.Keyhole.mask.tc1

Test ID	25521
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-01 10:47:11 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Engelen, Walter van (nly95948) on 2016-03-01 10:24:42 Meaning: SME Approved - (Technical) Signed by: Huang, Alan (310109142) on 2016-02-29 13:55:32 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test has been designed around the Methods TestFramework on a VMWare machine. The script mrTest.rb is used for automatic execution of a 4DTRAK and a 4DTRAK-XD scan, uses profile logging as an input and generates a text file (report) containing the number profiles per dynamic and number of duplicate profiles. The tester inspects the text file for verification of the test.

Requirements Coverage:

Req ID	Requirement Tag	Description
55919	SWCMP.Scan.Dynamics.Keyhole.mask	A 4DTRAK or 4DTRAK_XD scan protocol shall enable a 3D FFE dynamic series using Keyhole which acquires a complete set of central and peripheral ky(kz)-profiles during the first dynamic.

Test steps:

Step	Description	Expected Result
Step 1	a) Install a ROF. b) start G:\methodtestframework\src\mrTest.rb by typing: <pre>ruby mrTest.rb</pre> c) Select option: [2] Component Tests d) Select Suboption [0] Dynamics e) Select Suboption [3] Run Keyhole mask component test	Verify that the scans are executed by BatchInterpreter without conflict or abort.
Step 2	After the test is finished open G:\Log\methodtestframework\Dynamics\#stream\#date-time\mask_detailed_report.txt and inspect the results for test case 4dtrak_mask	Verify in the file "mask_detailed_report.txt" below "4dtrak_mask" that: a) The report contains the correct configuration and current date/time b) For all dynamic scans, the measured number of profiles is equal to the expected number of profiles. c) All dynamic scans except for the first (mask) and the last (reference) have the same number of profiles. d) The first and the last dynamic scan have more profiles than the other dynamic scans. e) The number of duplicate profiles in the Keyhole is 0

Step	Description	Expected Result
		f) The number of duplicate profiles in the Mask is 0 f) The number of duplicate profiles in the Reference Scan is 0
Step 3	In G: \\Log\methodtestframework\Dynamics\#stream\#date-time\mask_detailed_report.txt inspect the results for test case 4dtrak_xd_mask	Verify in the file "mask_detailed_report.txt" below "4dtrak_xd_mask" that: a) For all dynamic scans, the measured number of profiles is equal to the expected number of profiles. b) All dynamic scans except for the first (mask) have the same number of profiles. c) The first dynamic scan has more profiles than the other dynamic scans. d) The number of duplicate profiles in the Keyhole is 0 e) The number of duplicate profiles in the Mask is 0 f) The number of duplicate profiles in the Periphery is 0
WrapUp	Attach mask_detailed_report.txt as supporting evidence.	N.A. (WrapUp)

3.1.11.7. SWCMP.Scan.Dynamics.Keyhole.Par.tc1

Test ID	41293
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:49 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:00:49 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:28:15 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of dynamic keyhole parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description			
94023	SWCMP.Scan.Dynamics.Keyhole.Par.v1	Name	Range	Visible when	Description
		Keyhole	yes/no	Dynamic study = individual	no: all profiles are acquired each dynamic yes: only the the central profiles are acquired each dynamic (except for the mask and reference)
		central size (%)	15 - 100 1 - 100 1 - 95	Keyhole = yes, CE profile order = not CENTRA Keyhole = yes, CE profile order = CENTRA, viewsharing = no Keyhole = yes, CE profile order = CENTRA, viewsharing = yes	specifies the acquired procentual number of profiles (ky for 2D and kykz for 3D) each dynamic with respect to the total number of profiles needed to reconstruct a complete dynamic image.
		viewsharing	yes/no	Fast Imaging Mode = none, Keyhole = yes,	Allows for 4DTRAK_XD

Req ID	Requirement Tag	Description			
				CE profile order = CENTRA	
		peripheral density %	5.0 - 50.0	viewsharing = yes	specifies the percentage of periperhal (non-keyhole) kykz-profiles that are sampled each dynamic (excluding the mask). Value is rounded to the closest value of 100/viewshar ing_factor.
		sharing direction	central, forward, backward	viewsharing = yes	specifies the direction in which the peripheral kspace of a dynamic scan is shared to: central: to reconstruct a given dynamic scan the peripheral kspace from the temporally closest dynamic scans is used forward: the peripheral kspace is shared forward in time towards

Req ID	Requirement Tag	Description			
					later dynamic scans backward: the peripheral kspace is shared backward in time towards earlier dynamic scans
		measurements	1-NSA	CE profile order = not CENTRA	NSA >1, the number of averages per keyhole scan that can be selected separately from the reference scan. This parameter can only be set to values >1 up to the number of averages when NSA >1.
		mask	yes/no	Keyhole = yes, CE profile order = CENTRA, reference scan = last	no: the system shall acquire during the first dynamic the number of profiles as specified by 4DTRAK (only central profiles) or 4DTRAK_XD (central and

Req ID	Requirement Tag	Description			
					part of peripheral profiles. yes: the system shall acquire during the first dynamic a complete set of k-space profiles (all central and peripheral profiles).
		reference scan	first/last last user defined	Keyhole = yes, CE profile order = not CENTRA Keyhole = yes, CE profile order = CENTRA, Fast Imaging Mode = none Fast Imaging Mode = TFE, Keyhole = yes, CE profile order = CENTRA	first: the reference scan (central + all peripheral profiles) is acquired during the first dynamic. If a mask is yes, the mask will be used as reference. last: the reference scan (central + all peripheral profiles) is acquired during the last dynamic. The peripheral profiles of the reference scan will be used for reconstruction of all other dynamics.

Req ID	Requirement Tag	Description			
					user defined: for each dynamic scan, the dynamic scan number is indicated which will be used as a reference to provide the peripheral k-space data for reconstructio n. As a result, multiple reference scans can be acquired.
		dyn : ref	NA	reference scan = user defined	This parameter takes string input and allows values in the range "1 - number of dynamic scans" Defines for each dynamic scan which other dynamic scan will be used as a reference. Internal logic forces the closest available reference scan to be chosen.

Req ID	Requirement Tag	Description			
		compression (%)	[0 -100]	Keyhole = yes, reference scan = user defined	The compression percentage is the percentage of the number of profiles in the shared positive and negative k-space regions with respect to the complete Keyhole acquisition part

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanDynamicsKeyholePars in the results file is zero.

3.1.12. ME ENCASE

3.1.12.1. SWCMP.Scan.ENCASE.tc1.v2

Test ID	55233
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2022-07-28 10:14:35 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2022-07-28 07:58:50

	Meaning: SME Approved - (Technical)
	Signed by: Graaf, Ruud de (nly95749) on 2022-07-27 14:50:46 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
65934	SWCMP.Scan.ENCASE.v1	Subsystem scanning shall provide a parameter to enable ENCASE. The parameter shall reside on the Geometry tab. The parameter name: ENCASE enable with possible values = YES / NO The ENCASE scan technique shall be compatible with: • Stacks = 1 • Orientation = Coronal • Foldover suppression = NO • TONE pulse = NO • Respiratory compensation = breath hold or no
42902	SWCMP.Scan.ENCASE.SequenceTypes	ENCASE functionality shall be enabled with 3D FFE and TFE scans, including b-FFE and b-TFE.
42903	SWCMP.Scan.ENCASE.Orientation	When ENCASE functionality is enabled then: • Only orthogonal coronal imaging shall be allowed. • Angulation will be limited to a small value of +/- 5 degrees.
42904	SWCMP.Scan.ENCASE.Breathhold	Protocols using ENCASE shall support a breath hold per stack acquisition strategy. Typical per-stack breath hold shall not exceed 20 seconds respecting the requirements for coverage and resolution.
42907	SWCMP.Scan.ENCASE.RL.FOV	The ENCASE functionality shall limit the RL FOV range to maximal 450mm.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCmpScanEncase in the results file is zero.

3.1.12.2. SWCMP.Scan.ENCASE.tc2.v2

Test ID	52720
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:15:19 Meaning: QA Approved - (QA)

	Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:43:16 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:25 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test checks several features of the ENCASE implementation.

Requirements Coverage:

Req ID	Requirement Tag	Description
42908	SWCMP.Scan.ENCASE.Sagittal Excitation	When ENCASE is enabled, RF selection is performed on the preparation direction. In the default clinical use of ENCASE, the RF selection and preparation occur in the RL direction.
42909	SWCMP.Scan.ENCASE.Minimu mPhasePulse	The RF pulse used for the ENCASE acquisition shall be encase_4.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanEncase in the results file is zero.

3.1.13. ME EPI

3.1.13.1. SWCMP.Scan.EPI.DynPhNav.Pars.tc1.v1

Test ID	32723
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-29 16:25:59 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-03-29 16:06:29 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on

	2017-03-29 13:27:46 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

EPI and GraSE are used through a few dedicated parameters. The status of these parameters (visibility, range) must be checked interactively.

These tests may be executed in a Test Bay environment, or using the Examcard simulator of a simulated Workstation installed in a virtual machine on a SDE PC.

Both must have protocol database "tedb.dat" installed and selected. A suitable starting protocol is located in the folder \Head CNS\EPI\, e.g. P_SEEPI_2D.

Requirements Coverage:

Req ID	Requirement Tag	Description												
32277	SWCMP.Scan.EPI.DynPhNav	MR-Methods shall acquire a dynamic EPI phase navigator to reduce the ghost level as a result of changing gradient delays. The Dynamic EPI phase navigator sequence will be acquired at the beginning of each dynamic scan and use no phase encoding, a reduced flip angle between 2 and 20 degrees and weak slice selection (see Figure below). The Dynamic EPI phase navigator is compatible with dynamic F0 stabilization, prospective motion correction, multi-shot EPI and multi package imaging. Conflicts will be given in combination with non-EPI scans and Cardiac or Respiratory synchronization.												
34541	SWCMP.Scan.EPI.DynPhNav.Parameters	The Dynamic EPI phase navigator is available as the sub-Option "enhanced" of the already existing Dynamic parameter "Dynamic stabilization". If enhanced is selected, both the dynamic EPI phase navigator as well as the "original" dynamic stabilization is applied. <table><tr><td>param:</td><td>EX_DYN_drift_comp_enable</td></tr><tr><td>type:</td><td>enum</td></tr><tr><td>range:</td><td>no, regular, enhanced</td></tr><tr><td>default:</td><td>no</td></tr><tr><td>enable:</td><td>option enhanced available for dynamic FFE EPI scans at 1.5 T</td></tr><tr><td>help text:</td><td>see text file in software archive</td></tr></table>	param:	EX_DYN_drift_comp_enable	type:	enum	range:	no, regular, enhanced	default:	no	enable:	option enhanced available for dynamic FFE EPI scans at 1.5 T	help text:	see text file in software archive
param:	EX_DYN_drift_comp_enable													
type:	enum													
range:	no, regular, enhanced													
default:	no													
enable:	option enhanced available for dynamic FFE EPI scans at 1.5 T													
help text:	see text file in software archive													

Test steps:

Step	Description	Expected Result
Step 1	Run this test on a VMWare or STM/RTM configured as WA30 1. Select tedb. 2. Open the Application software	N/A (preparation)

Step	Description	Expected Result
	3. Load the scan protocol "Head CNS, EPI, FEEPI_dyn_phnav". 4. Make sure that this protocol validates without conflict. 5. Create a new logfile	
Step 2	Parameter availability: On the dynamic page, verify the availability of the parameter "Dynamic stabilization" with allowed values "no", "regular" and "enhanced". Ensure that "Dynamic stabilization" is set to 'enhanced'.	The parameter "Dynamic stabilization" with allowed values "no", "regular" and "enhanced". Dynamic stabilization is ensured to be set to 'enhanced' and no conflict occurs.
Step 3	EPI multi-shot: Set the parameter "EPI shot mode = multi-shot".	No conflict occurs
Step 4	Combination with prospective motion correction: Set the parameter "prospect. motion corr. = yes""	No conflict occurs
Step 5	Multiple packages: Set the parameter "Minimum number of packages > 1"	No conflict occurs
Step 6	Set the parameter "prospect. motion corr." = "no" Select "Fast Imaging mode" to values unequal to "EPI"	A conflict appears on the parameters "Dynamic stabilization", "Fast Imaging mode" and "Scan technique"
Step 7	Select "Cardiac synchronization unequal no "	The parameters "Dynamic stabilization" and "Cardiac synchronization" are in conflict.
Step 8	Set the parameters "TR" = "user defined" and "Slice scan order" = "FH" . Select "Respiratory compensation" to values unequal to "no".	The parameters "Dynamic stabilization" and "Respiratory compensation" are in conflict for "Respiratory compensation" set to values "trigger", "gate" and "breath hold". When "Respiratory compensation" is set to "PEAR", parameters "EPI" and "Respiratory compensation" are in conflict. Attach the log file as objective evidence.

3.1.13.2. SWCMP.Scan.EPI.DynPhNav.tc1

Test ID	14552
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Tests for EPI and GraSE are included in teima_total (see VS: FC MR-Methods chapter 3) to be executed on a Test Bay,

and pdfcompare (see VS: FC MR-Methods chapter 4) to be executed on a workstation.

Both teima_total and pdfcompare follow the same list of test cases (protocols). All protocols are located in the folder \Head CNS\EPI\, are for completeness listed in the table below.

Protocol name	Description
P_FEEPI_dyn	FFE multi-shot EPI with dynamic study
P_SEEPI_2D	SE EPI in Scan mode 2D
P_SEEPI_MS	SE EPI in Scan mode Multi Slice
P_SEEPI_HS	SE EPI using Halfscan
F_SEEPI_HS	SE EPI using Halfscan
F_SEEPI_angl	SE EPI using scan plane angulations
F_SEEPI_fos	SE EPI using Foldover suppression by oversampling
F_SEEPI_M2D	SE EPI in Scan mode M2D
F_IREEPI	EPI with Scan technique IR
F_MIXEPI	EPI with Scan technique MIX
F_MSEEPI	SE EPI using Modified SE
F_MSEEPI_M2D	SE EPI using Modified SE in Scan mode M2D
F_2SEEPI_HS	SE EPI with 2 echoes, including Halfscan for 1st echo
F_2SEEPI	SE EPI with 2 echoes
P_FEEPI_2D	FFE EPI in Scan mode 2D
P_FEEPI_MS	FFE EPI in Scan mode Multi Slice
P_FEEPI_HS	FFE EPI using Halfscan
F_FEEPI_HS	FFE EPI using Halfscan
F_FEEPI_angl	FFE EPI using scan plane angulations
F_FEEPI_fos	FFE EPI using Foldover suppression by oversampling

F_FEEPI_M2D	FFE EPI in Scan mode M2D
P_GraSE_MS	GraSE in Scan mode MS
P_GraSE_7x5	GraSE (TSE factor 7, EPI factor 5)
P_GraSE_angl	GraSE using scan plane angulations
F_GraSE_dual	GraSE with 2 echoes
presto_4080i	FFE shifted echo (TE > TR), 3D, EPI with EPI factor 3
epi-mobi	FFE EPI using Multi Station (moving table)
GraSE_lohi	GraSE with profile order low-high
GraSE_velcom	GraSE with flow compensation
FEEPI_2dpc	FFE EPI 2D phase correction acquisition/reconstruction
FEEPI_3dmzy	FFE EPI Scan mode 3D, kz blipped EPI
FEEPI_angio	FFE EPI in combination with PC Angio mode
FEEPI_QF	FFE EPI in combination with Quantitative flow
i6_SE_equi	SE EPI single-shot, temporal slice spacing equidistant
i6_SE_min	SE EPI single-shot, temporal slice spacing minimum
i6_FFE_equi	FFE EPI single-shot, temporal slice spacing equidistant
i6_FFE_min	FFE EPI single-shot, temporal slice spacing minimum
a6_ssh_255	SE EPI multi-shot, EPI factor 255

Requirements Coverage:

Req ID	Requirement Tag	Description
32277	SWCMP.Scan.EPI.DynPhNav	MR-Methods shall acquire a dynamic EPI phase navigator to reduce the ghost level as a result of changing gradient delays. The Dynamic EPI phase navigator sequence will be acquired at the beginning of each

Req ID	Requirement Tag	Description
		dynamic scan and use no phase encoding, a reduced flip angle between 2 and 20 degrees and weak slice selection (see Figure below). The Dynamic EPI phase navigator is compatible with dynamic F0 stabilization, prospective motion correction, multi-shot EPI and multi package imaging. Conflicts will be given in combination with non-EPI scans and Cardiac or Respiratory synchronization.

Test steps:

Step	Description	Expected Result
Step 1	Run this test on a VmWare or STM/RTM (with system type set to "simulator") configured as WA30. 1. Select tedb. 2. Copy the scripts m_imn_epi_dyn_ph_nav_ts.acq and m_imn_epi_dyn_ph_nav_ts.pl from yourview\system\mrmmethods\imaging\tst to G:\patch.	N/A (preparation)
Step 2	Execute the perl script in a command window. Check the results logfile in G:\log\m_imn_epi_dyn_ph_nav_ts_comp_date.dat after the end of the test.	For all executed scans the following results are shown: - the measured and expected flip angle are equal within 1% accuracy and are smaller than 10 degrees - the measured and expected phase encoding gradient area is 0 - the measured and expected excitation gradient ratio is < 0.5 - the measured and expected gradient area ratio within 1% accuracy is -0.390 The final result of the test as indicated in the logfile is PASSED

3.1.13.3. SWCMP.Scan.EPI.Par.tc1

Test ID	41061
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:51 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:01:33 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:28:51 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of EPI parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description			
75165	SWCMP.Scan.EPI.Par				
		Name	Range	Visible when	Description
		Epi factor	3,5,...,255 (odd numbers only)	EPI or TFEPI or GraSE	This parameter determines the number of ky profiles collected per excitation and in such a way specifies the acceleration factor in multishot EPI and GRASE sequences with respect to non-EPI sequences.
		epi direction	[Y,Z]	3D and (EPI or TFEPI or GraSE)	3D EPI readout with "epi direction" set to "Z", blips the phase encodings in the Z direction

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanEpiPar in the results file is zero.

3.1.13.4. SWCMP.Scan.EPI.phasecorr.tc1

Test ID	7676
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

These tests may be executed in a Test Bay environment, having protocol database "tedb.dat" installed and selected. Suitable protocols are located in the folder \Head CNS\EPI\.

Requirements Coverage:

Req ID	Requirement Tag	Description
17431	SWCMP.Scan.EPI.phasecorr	MR-Methods shall provide image domain phase correction methods to reduce the influence of gradient delays as described in sections Single sign phase correction and Dual sign phase correction

Test steps:

Step	Description	Expected Result
Step 1	Select protocol P_SEEPI_MS and make a first scan, using the head coil and the head phantom. Locate the Reconstruction Control parameter "EPI 1d phase correction", and modify its value from "Non-linear" to "No". This effectively disables the phase correction mechanism. Make a second scan, by repeating the protocol of the first scan. Return to the Reconstruction Control parameter "EPI 1d phase correction", and modify its value back to the normal value "Non-linear".	Verify that the ghost artifact intensity in the first scan, i.e. the one with a phase correction mechanism enabled, is lower than that in the second scan.

3.1.14. ME Fatsat

3.1.14.1. SWCMP.Scan.Fatsat.LIPO.tc1

Test ID	14531
Revision	1
Product Model	Any
Gradient System	Any

Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description								
33928	SWCMP.Scan.Fatsat.LIPO.Pars	Fat suppression method by gradient reversal offresonant suppression shall be enabled by a parameter called "Grad Rev Fat suppression". Gradient Reversal Fatsat shall be applicable in combination with SPIR or SPAIR. Protocol parameters: <table><tr><th>Name</th><th>Range</th><th>Visible when</th><th>Description</th></tr><tr><td>Grad Rev Fat suppression</td><td>yes, no</td><td>Parameter "Diffusion mode" = DWI or DTI</td><td>fat suppression by gradient reversal</td></tr></table>	Name	Range	Visible when	Description	Grad Rev Fat suppression	yes, no	Parameter "Diffusion mode" = DWI or DTI	fat suppression by gradient reversal
Name	Range	Visible when	Description							
Grad Rev Fat suppression	yes, no	Parameter "Diffusion mode" = DWI or DTI	fat suppression by gradient reversal							
27960	SWCMP.Scan.Fatsat.LIPO	It shall be possible to apply Fat suppression method by gradient reversal offresonant suppression for spin echo DWI/DTI sequences at 3T (and higher). Gradient reversal fat suppression method is based on gradient reversal of the selection gradient for the refocusing pulse(s) in a SE or TSE sequence [1, 2]. The gradient reversal leads to a localization shift of off-resonant species in the opposite direction with respect to spin excitation. The consequence is that off-resonant signal such as fat is suppressed, since the excited spins will not experience the refocusing pulse(s). References: [1] Park HW, Kim DJ, Cho ZH. Gradient reversal technique and its applications to chemical-shift-related NMR imaging. Magn Reson Med 1987;4(6):526-536. [2] Gomori JM, Holland GA, Grossman RI, Geftter WB, Lenkinski RE. Fat suppression by section-select gradient reversal on spin-echo MR imaging. Work in progress. Radiology 1988;168(2):493-495.								

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Configure RTM as WA30. Disable the CSK option05. Startup application SW. Select protocol a1_dw_lipo in subanatomy Head CNS/SPIR in the protocol editor. (If this protocol does not exist yet in TEDB, use the attached ExamCard by dragging it in the execution list.).	n.a. (preparation)	 a1_DW\$47LIPO.zip

Step	Description	Expected Result	Attachments
	Make sure that the protocol has no conflicts.		
Step 2	On the contrast page select parameter <i>Grad Rev Fat Suppression</i> .	The parameter Grad Rev Fat Suppression is visible with range <i>no, yes</i> .	None
Step 3	1. Set parameter <i>Grad Rev Fat Suppression</i> value to <i>yes</i> . 2. Set parameter <i>fat suppression</i> to value <i>SPIR</i> . 3. Start this scan.	Grad Rev Fat Suppression and SPIR can be combined. The scan completes successfully.	None
Step 4	1. Set parameter <i>Grad Rev Fat Suppression</i> value to <i>yes</i> . 2. Set parameter <i>fat suppression</i> to value <i>SPAIR</i> . 3. Start this scan.	Grad Rev Fat Suppression and SPAIR can be combined. The scan completes successfully.	None
Step 5	Switch <i>diffusion mode</i> to "no". Resolve the conflict that arises on the parameter "Grad Rev Fat suppression", by changing its value to "no" (the only allowed value). After this, observe whether "Grad Rev Fat suppression" is visible on contrast page.	"Grad Rev Fat Suppression" parameter is no longer visible.	None

3.1.14.2. SWCMP.Scan.Fatsat.Pars.tc1

Test ID	9881
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware
2. Select the tedb database
3. Start *ExamCards Environment*.

Description: available options for fat/water suppression

Preparation: Load scan SPAIR_MS_TFE from Pelvis/autoSPAIR

Requirements Coverage:

Req ID	Requirement Tag	Description
23579	SWCMP.Scan.Fatsat.ProSet	A Spatial and Spectral Selective binomial Excitation Pulse to selectively excite water or fat is implemented. See chapter ProSet.
23578	SWCMP.Scan.Fatsat.SPAIR	Presaturation of the fat signal using adiabatic inversion. See chapter SPAIR.
23577	SWCMP.Scan.Fatsat.SPIR	Presaturation of the fat signal using a prepulse preceeding the imaging sequence. See chapter SPIR.

Test steps:

Step	Description	Expected Result
Step 1	Go to the contrast page and check the available options for Fat suppression.	- Verify that three different options are available: no, SPIR, SPAIR, and ProSet. - Verify that the same range is available for the parameter Water suppression

3.1.14.3. SWCMP.Scan.Fatsat.ProSet.tc1

Test ID	9888
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use Vmware
2. Select tedb
3. Select a WA30 configuration
4. Start ExamCards

Description: Verify the correct implementation of the ProSet prepulse

Preparation:

- Load scan 2dssp_B1 from Head CNS/2DSSP
- Open the Sequence Development Viewer and create a pdf snapshot

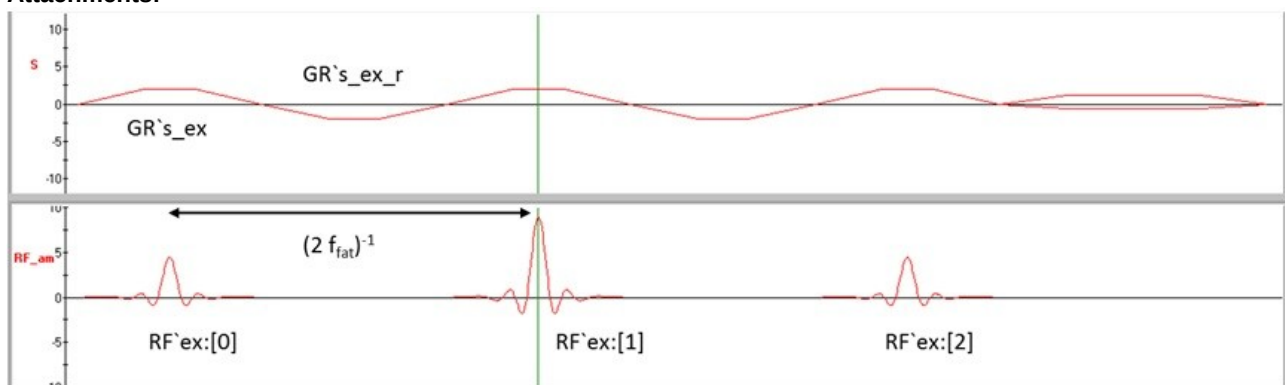
Requirements Coverage:

Req ID	Requirement Tag	Description
23579	SWCMP.Scan.Fatsat.ProSet	A Spatial and Spectral Selective binomial Excitation Pulse to selectively excite water or fat is implemented. See chapter ProSet.

Test steps:

Step	Description	Expected Result
Step 1	Verify the correct implementation of the ProSet sequence	RF`ex consists of 3 pulses (RF`ex:[0], RF`ex:[1], RF`ex:[2]) Refer Fig Attached
Step 2	Verify the correct implementation of the ProSet RF amplitudes	The ratio of the RF amplitudes is 1-2-1 Refer Fig Attached
Step 3	Verify the correct implementation of the distance of the ProSet RF pulses	The distance between the pulses corresponds to 1.15ms ($1/(2 \cdot f_{fat})$ with $f_{fat}=434\text{Hz}$ at 3T) Refer Fig Attached
Step 4	Verify the correct implementation of the distance of the ProSet selection gradients	The selection gradients are symmetrical i.e. the positive gradient lobes (GR`s_ex) have the same gradient area as the negative gradient lobes (GR`s_ex_r). They only have opposite sign. Refer Fig Attached

Attachments:



3.1.14.4. SWCMP.Scan.Fatsat.SPAIR.ID.auto.availability.tc1

Test ID	9887
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware

2. Select the tedb database
3. Start *ExamCards Environment*.

Description: Certain parameter combinations are not allowed with autoSPAIR

Preparation: Load scan SPAIR_MS_TFE from Pelvis/autoSPAIR

Requirements Coverage:

Req ID	Requirement Tag	Description
23596	SWCMP.Scan.Fatsat.SPAIR.ID.auto.availability	autoSPAIR will be available in combination with: • Respiratory, navigator triggering/tracking and breath-holding • Rest slabs • Diffusion weighted imaging using EPI read-out It is not available in combination with: • FFE sequences without T1 enhanced contrast • TFE prepulses • Cardiac triggering

Test steps:

Step	Description	Expected Result
Step 1	Verify conflicts in combination with autoSPAIR	Conflicts occur for - Cardiac synchronization not no (motion page) - Contrast enhancement not T1 - TFE prepulse not no

3.1.14.5. SWCMP.Scan.Fatsat.SPAIR.ID.auto.default.tc1

Test ID	9885
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware
2. Select the tedb database
3. Start *ExamCards Environment*.

Description: Behavior of inversion delay parameter for autoSPAIR

Preparation: Load scans SPAIR_MS_TFE from Pelvis/autoSPAIR

Requirements Coverage:

Req ID	Requirement Tag	Description
23595	SWCMP.Scan.Fatsat.SPAIR.ID.uto.default	When switching from SPAIR inversion delay auto to user defined, the optimal inversion delay is taken as default value.

Test steps:

Step	Description	Expected Result
Step 1	Verify the default value for the user defined SPAIR delay.	When switching from inversion delay auto to user defined the optimal inversion delay is taken as default value.

3.1.14.6. SWCMP.Scan.Fatsat.SPAIR.ID.auto.default.tc1.v2

Test ID	55341
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2022-08-05 06:19:04 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2022-08-04 15:13:59 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2022-08-04 11:47:53 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Check behavior of inversion delay parameter for autoSPAIR

Requirements Coverage:

Req ID	Requirement Tag	Description
23595	SWCMP.Scan.Fatsat.SPAIR.ID.uto.default	When switching from SPAIR inversion delay auto to user defined, the optimal inversion delay is taken as default value.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanFatSatSpairIDAutoDefault in the results file is zero.

3.1.14.7. SWCMP.Scan.Fatsat.SPAIR.ID.auto.tc1

Test ID	9886
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware
2. Select the tedb database
3. Start *ExamCards Environment*.

Description: A conflict is given if the automatically calculated inversion delay is too short.

Preparation: Load scans SPAIR_MS_TFE and SPAIR_eTHRIVE from Pelvis/autoSPAIR

Requirements Coverage:

Req ID	Requirement Tag	Description
23592	SWCMP.Scan.Fatsat.SPAIR.ID.automato	If SPAIR inversion delay auto is chosen the SPAIR inversion delay will be adapted automatically based on the current sequence timing (see chapter automatic SPAIR inversion delay calculation). A conflict is generated if the optimal SPAIR delay is too short.

Test steps:

Step	Description	Expected Result
Step 1	In scan SPAIR_MS_TFE go to the contrast page and modify the parameter profile order to linear	Conflicts occur on the inversion delay, the TFE shot interval, and the profile order.
Step 2	In scan SPAIR_MS_TFE modify the shot mode to single shot	A conflict appears on the inversion delay, profile order and shot mode and not any more on the shot interval (not visible)
Step 3	Load scan SPAIR_eTHRIVE from Pelvis/autoSPAIR	No conflict occurs.
Step 4	In scan SPAIR_eTHRIVE modify the parameter TFE shot	Conflicts occur on the inversion delay, the shot

Step	Description	Expected Result
	interval from shortest to user defined using a value of 170ms.	mode, the TFE shot interval, the profile order, and the halfscan factor z.

3.1.14.8. SWCMP.Scan.Fatsat.SPAIR.ID.tc1.v2

Test ID	52721
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:16:05 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Hegde, Shashank (310195826) on 2021-12-16 12:43:17 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:25 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the correct implementation of the SPAIR inversion delay for user defined and shortest

Requirements Coverage:

Req ID	Requirement Tag	Description
23590	SWCMP.Scan.Fatsat.SPAIR.ID. UserDefined	If SPAIR inversion delay user defined is chosen the distance between the center of the SPAIR pulse and the center of the excitation pulse will be chosen as specified by the user.
23591	SWCMP.Scan.Fatsat.SPAIR.ID.s hortest	If SPAIR inversion delay shortest is chosen the minimally possible inversion delay is implemented placing the SPAIR pulse directly in front of the imaging sequence.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanFatsatSpairId in the results file is zero.

3.1.14.9. SWCMP.Scan.Fatsat.SPAIR.Info.tc1

Test ID	9884
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware
2. Select the tedb database
3. Start *ExamCards Environment*.

Description: Display of info parameters for SPAIR

Preparation: Load scans SPAIR_MS_TFE from Pelvis/autoSPAIR

Test steps:

Step	Description	Expected Result
Step 1	Verify the display of SPAIR info parameters	For inversion delay equal to shortest or user defined, the optimal SPAIR inversion delay SPAIR inv. Delay auto. and the SPAIR TR are displayed. For inversion delay equal to auto those parameters are not displayed

3.1.14.10. SWCMP.Scan.Fatsat.SPAIR.Pars.tc1

Test ID	9883
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware
2. Select the tedb database
3. Start *ExamCards Environment*.

Description: available options for SPAIR inversion delay

Preparation: Load scans SPAIR_MS_TFE and SPAIR_MS_TSE from Pelvis/autoSPAIR

Test steps:

Step	Description	Expected Result
Step 1	Verify the presence and range of the parameter inversion delay for SPAIR fat suppression	For SPAIR fat suppression the parameter inversion delay is present and has the range: o FFE: user defined, shortest, auto. o SE: user defined, auto
Step 2	Verify the presence and range of the parameter inversion delay for SPAIR water suppression	For SPAIR water suppression the parameter inversion delay is present and only a user defined value can be specified for both SE and FFE.
Step 3	Verify the presence and range of the parameter frequency offset for SPAIR fat/water suppression	For SPAIR fat/water suppression the parameter frequency offset is present and has the range: Default, user defined
Step 4	Verify the presence and range of the parameter power for SPAIR fat suppression only	For SPAIR fat suppression the parameter power for SPAIR fat suppression is present and has the range o 1,2

3.1.14.11. SWCMP.Scan.Fatsat.SPIR.ID.tc1.v2

Test ID	52722
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:16:08 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:39:58 Meaning: SME Approved - (Technical)

	Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:25 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the implementation of the SPIR inversion delay

Requirements Coverage:

Req ID	Requirement Tag	Description
23586	SWCMP.Scan.Fatsat.SPIR.ID	The SPIR delay time (the time duration from the center of the prepulse to the center of the excitation pulse) is made as short as possible by placing the SPIR pulse directly in front of the imaging sequence. See chapter <i>SPIR inversion delay</i> .

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanFatsatSpirId in the results file is zero.

3.1.14.12. SWCMP.Scan.Fatsat.SPIR.Pars.tc1

Test ID	9882
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use VMware
2. Select the tedb database
3. Start *ExamCards Environment*.

Description: available options for SPIR fat/water suppression

Preparation: Load scan SPAIR_MS_TFE from Pelvis/autoSPAIR

Test steps:

Step	Description	Expected Result
Step 1	Go to the contrast page and select SPIR fat suppression	- Verify the presence and range of the parameter strength: weak, medium, strong. - Verify the presence and range for the parameter frequency offset: default, user defined.
Step 2	Select SPIR water suppression	Verify the range for the parameter frequency offset: default, user defined.

3.1.14.13. SWCMP.Scan.Fatsat.SPIR.PulseType.tc1

Test ID	9890
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Preparations

1. Use Vmware
2. Select tedb
3. Select a WA30 configuration
4. Start ExamCards

Description: Verify the correct implementation of the SPIR pulse shape

Preparation:

Load scan SPIR_base from Head CNS/SPIR
Open the Sequence Development Viewer

Requirements Coverage:

Req ID	Requirement Tag	Description
23585	SWCMP.Scan.Fatsat.SPIR.Pulse Type	The SPIR pulse shape OPT_EX_2 is used. See chapter <i>SPIR pulse shape</i> .

Test steps:

Step	Description	Expected Result
Step 1	Verify the correct pulse shape of the SPIR RF pulse	RF`spir:shape = opt_ex_2

3.1.14.14. SWCMP.Scan.Fatsat.SPIR.Strength.tc1

Test ID	9891
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:*Preparations*

1. Use Vmware
2. Select tedb
3. Select a WA30 configuration
4. Start ExamCards

Description: Verify the correct implementation of the SPIR pulse angle and strength

Preparation: Load scan SPIR_base from Head CNS/SPIR

Requirements Coverage:

Req ID	Requirement Tag	Description
23581	SWCMP.Scan.Fatsat.SPIR.Strength.weak	The SPIR flip angle is adapted such that a fat suppression level as defined in the scan control parameter SPIR strength[0] is achieved.
23582	SWCMP.Scan.Fatsat.SPIR.strength.medium	The SPIR flip angle is adapted such that a fat suppression level as defined in the scan control parameter SPIR strength[1] is achieved.
23583	SWCMP.Scan.Fatsat.SPIR.strength.strong	The SPIR flip angle is adapted such that a fat suppression level as defined in the scan control parameter SPIR strength[2] is achieved.
23584	SWCMP.Scan.Fatsat.SPIR.angle	The proper inversion pulse angle is calculated based on the sequence repetition time in order to achieve zero fat magnetization at the center of the excitation pulse. See chapter <i>SPIR flip angle</i> .

Test steps:

Step	Description	Expected Result
Step 1	- Change the parameter TE from user defined to shortest - Modify SPIR strength from strong to medium and weak - Open the Sequence Development Viewer	$RF_spir:angle (medium) = CSC_spir_strength[1] * RF_spir:angle (strong)$ $RF_spir:angle (weak) = CSC_spir_strength[0] * RF_spir:angle (strong)$

3.1.14.15. SWCMP.Scan.Fatsat.tc1.v1

Test ID	32735
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-07 15:18:00 Meaning: QA Approved - (QA) Signed by: Mariappan, Yogesh kannan (310134237) on 2017-04-07 13:51:59 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2017-04-06 08:38:06 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Based on the known T1 of fat and a theoretical description of the magnetization behavior for different sequence types, the required inversion delay for SPAIR fat suppression is calculated in the pdf validation phase. In order to verify the intended implementation of this functionality, it is sufficient to check the outcome of the pdf validation for a number of possible sequence types. For this purpose, a batch script has been designed which performs this verification for 1.5T Achieva and 3.0T Ingenia configurations. Since the underlying theoretical model depends on various sequence parameters the following sequence combinations are tested:

- Scan technique: SE (TSE, EPI, GraSE), FFE (TFE, EPI,TFEPI)
- Scan mode: MS, 2D/3D/M2D
- Profile order: low high, linear, asymmetric (TSE), reversed linear (TFE,TFEPI)
- SENSE
- REST slabs
- DWI
- Suppression levels: weak, medium, strong
- Respiratory/navigator triggering

In addition the angle step adaptation for SPIR fat suppression is tested in case of respiratory and navigator triggering.

Description:

Automatic calculation of the SPAIR inversion delay for the following test cases on 1.5T Achieva, and 3.0T Ingenia configurations

- *Spin echo sequences:*
 - *Test case 1: Scan mode & varying TRspair*
 - 1.1 SPAIR_MS_TSE_TR2000
 - 1.2 SPAIR_MS_TSE_TR3000
 - 1.3 SPAIR_MS_TSE_TR4000
 - 1.4 SPAIR_2D_TSE_TR1000

- 1.5 SPAIR_2D_TSE_TR2000
 - 1.6 SPAIR_2D_TSE_TR3000
- Test case 2: Scan mode & varying suppression levels
 - 2.1 SPAIR_MS_TSE
 - 2.2 SPAIR_MS_TSE_med
 - 2.3 SPAIR_MS_TSE_weak
 - 2.4 SPAIR_2D_TSE
 - 2.5 SPAIR_2D_TSE_med
 - 2.6 SPAIR_2D_TSE_weak
- Test case 3: Fast imaging techniques
 - 3.1 SPAIR_2D_TSE
 - 3.2 SPAIR_2D_SE
 - 3.3 SPAIR_2D_SEEPI
 - 3.4 SPAIR_3D_GraSE
- Test case 4: SENSE, REST slabs, and DWI
 - 4.1 SPAIR_MS_TSE_SENSE
 - 4.2 SPAIR_MS_TSE_REST
 - 4.3 SPAIR_2D_TSE_SENSE
 - 4.4 SPAIR_2D_TSE_REST
 - 4.5 SPAIR_MS_SEEPI_str_DWI
- Test case 5: Profile order
 - 5.1 SPAIR_MS_TSE_linear
 - 5.2 SPAIR_MS_TSE_low_high
 - 5.3 SPAIR_MS_TSE_asym
 - 5.4 SPAIR_2D_TSE_linear
 - 5.5 SPAIR_2D_TSE_low_high
 - 5.6 SPAIR_2D_TSE_asym
- Gradient echo sequences
 - Test case 6: Scan mode & varying TRspair
 - 6.1 SPAIR_MS_TFE_TR7.8
 - 6.2 SPAIR_MS_TFE_TR10.0
 - 6.3 SPAIR_MS_TFE_TR15.0
 - 6.4 SPAIR_3D_TFE_TR5.8
 - 6.5 SPAIR_3D_TFE_TR10.0
 - 6.6 SPAIR_3D_TFE_TR15.0
 - 6.7 SPAIR_eTHRIVE
 - Test case 7: Scan mode & varying suppression levels
 - 7.1 SPAIR_MS_TFE_TR7.8
 - 7.2 SPAIR_MS_TFE_med_TR7.8
 - 7.3 SPAIR_MS_TFE_weak_TR7.8
 - 7.4 SPAIR_3D_TFE_TR5.8
 - 7.5 SPAIR_3D_TFE_med_TR5.8
 - 7.6 SPAIR_3D_TFE_weak_TR5.8
 - Test case 8: Fast imaging techniques
 - 8.1 SPAIR_3D_TFE_TRshortest
 - 8.2 3D_FFE_TRshortest
 - 8.3 3D_FFEPI_TRshortest
 - 8.4 SPAIR_3D_TFEPI_TRshortest
 - Test case 9: SENSE, REST slabs

- 9.1 SPAIR_MS_TFE_SENSE
- 9.2 SPAIR_MS_TFE_REST
- 9.3 SPAIR_3D_TFE_SENSE
- 9.4 SPAIR_3D_TFE_REST
- o Test case 10: Profile order
 - 10.1 SPAIR_MS_TFE_low_high
 - 10.2 SPAIR_MS_TFE_linear
 - 10.3 SPAIR_MS_TFE_rev_lin
 - 10.4 SPAIR_3D_TFE_low_high
 - 10.5 SPAIR_3D_TFE_linear
 - 10.6 SPAIR_3D_TFE_rev_lin
- Navigator and respiratory triggering
 - o Test case 11: navigator and respiratory triggering
 - 11.1 SPAIR_rnav
 - 11.2 SPAIR_RT
 - o Test case 12: navigator and respiratory triggering (SPIR)
 - 12.1 SPIR_rnav
 - 12.2 SPIR_RT
 - 12.3 SPIR_DWI_rnav
 - 12.4 SPIR_DWI_RT

VLD for test script m_imn_spir_ts.pl is archived in Agile (DEP133729)

Requirements Coverage:

Req ID	Requirement Tag	Description
23592	SWCMP.Scan.Fatsat.SPAIR.ID.a uto	If SPAIR inversion delay auto is chosen the SPAIR inversion delay will be adapted automatically based on the current sequence timing (see chapter automatic SPAIR inversion delay calculation). A conflict is generated if the optimal SPAIR delay is too short.
23593	SWCMP.Scan.Fatsat.SPAIR.ID.a uto.trig	The SPAIR inversion delay is adapted in runtime in order to correct for steady state loss of the fat magnetization during the trigger delay. See section <i>SPAIR inversion delay step for respiratory triggering</i> .
23594	SWCMP.Scan.Fatsat.SPAIR.ID.a uto.ethrive	For 2D/3D TFE sequences in combination with automatic inversion delay and TFE shot interval shortest, the TFE shot interval is adapted to allow the optimal inversion delay.
23587	SWCMP.Scan.Fatsat.SPIR.trig	The SPIR pulse angle is adapted in runtime in order to correct for steady state loss of the fat magnetization during the trigger delay. See chapters <i>SPIR angle step for respiratory triggering</i> and <i>SPIR angle step for navigator triggering</i> .
23596	SWCMP.Scan.Fatsat.SPAIR.ID.a uto.availability	autoSPAIR will be available in combination with: • Respiratory, navigator triggering/tracking and breath-holding • Rest slabs • Diffusion weighted imaging using EPI read-out It is not available in combination with: • FFE sequences without T1 enhanced contrast • TFE prepulses • Cardiac triggering

Test steps:

Step	Description	Expected Result
Prep 1	Select tedb database on VMWare. Copy the following files:	N.A. (Preparation)

Step	Description	Expected Result
	<i>m_imn_spir_ts.pl</i> <i>m_imn_spir_ts.acq</i> to the <i>patch</i> directory. The files can be found in ClearCase under <i>your_views\system\lrm\methods\imaging\tst</i>	
Step 1	Run the script <i>m_imn_spir_ts.pl</i>. This script will automatically change the configuration to different configurations (1.5T Achieva and 3.0T Ingenia) and run the protocols using the batch interpreter. Check that the following files <i>m_imn_spir_ts_*.dat</i>, <i>m_imn_spir_ts_*.log</i> and <i>m_imn_spir_ts_comp_*.log</i> are generated in <i>G:\log</i> for the two configurations. The file <i>m_imn_spir_ts_comp_*.log</i> contains the values of the different SPIR or SPAIR parameters measured for different protocols against the values calculated using the theoretically derived autoSPAIR equations. Attach the <i>m_imn_spir_ts_*.log</i>, <i>m_imn_spir_ts_*.dat</i>, and <i>m_imn_spir_ts_comp_*.dat</i> files as objective evidence.	The values of the measured parameters in the file <i>m_imn_spir_ts_comp_*.log</i> shall be within 1% accuracy of the calculated values in the same file.

3.1.15. ME FFE

3.1.15.1. SWCMP.Scan.FFE.CE_No.tc1

Test ID	7232
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test instruction

1. Drag the protocol to the execution list.
2. Start the scan protocol editor
3. Check the availability of the Contrast enhancement parameter and it's range.

Requirements Coverage:

Req ID	Requirement Tag	Description
17203	SWCMP.Scan.FFE.CE_No	MR-Methods shall provide Gradient Field Echo (FFE) technique as described in section 4.1 with the related Protocol Parameters described in section 4.2.

Test steps:

Step	Description	Expected Result
Step 1	Load scan protocol \Head CNS\FE\F2D1 in the protocol editor and check presence of parameters and their ranges	The specified Contrast enhancement parameter is present The values can be entered within the ranges specified in requirement FFECENo Contrast enhancement Range: no, T1, T2, balanced

3.1.15.2. SWCMP.Scan.FFE.FlowComp.tc1

Test ID	7235
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test instruction

1. Drag the protocol to the execution list.
2. Start the scan protocol editor
3. Check the availability of the Flow compensation parameter and it's range.

Requirements Coverage:

Req ID	Requirement Tag	Description
17209	SWCMP.Scan.FFE.FlowComp	MR-Methods shall provide Flow Compensation FFE technique, a variant of FFE technique which reduces flow artifacts and is used in Inflow imaging, as described in section 10.1 with related Protocols Parameters described in section 10.2

Test steps:

Step	Description	Expected Result
Step 1	Load scan protocol \Head CNS\FE\F2D9 in the protocol editor and check presence of parameters and their ranges	The specified Flow compensation parameter is present The values can be entered within the ranges specified in requirement FFEFlowComp Flow compensation Range: no, yes

3.1.15.3. SWCMP.Scan.FFE.Inflow.tc1

Test ID	7234
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test instruction

1. Drag the protocol to the execution list.
2. Start the scan protocol editor
3. Check the availability of the Angio/ Contrast enh parameter and it's range.

Requirements Coverage:

Req ID	Requirement Tag	Description
17208	SWCMP.Scan.FFE.Inflow	MR-Methods shall provide FFE technique which supports Inflow imaging, as described in section 9.1 with related Protocols Parameters described in section 9.2.

Test steps:

Step	Description	Expected Result
Step 1	Load scan protocol \User defined 1\2344P\IM2D Inflow in the protocol editor and check presence of parameters and their ranges	The specified Angio/ Contrast enh parameter is present The values can be entered within the ranges specified in requirement FFEInflow Angio / Contrast enh Range: no, inflow, phase contrast, contrast enh slice overlap Range: no, yes

3.1.15.4. SWCMP.Scan.FFE.ShiftedEcho.tc1

Test ID	7233
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test instruction

1. Drag the protocol to the execution list.
2. Start the scan protocol editor
3. Check the availability of the Shifted echo parameter and it's range.

Requirements Coverage:

Req ID	Requirement Tag	Description
17207	SWCMP.Scan.FFE.ShiftedEcho	MR-Methods shall support a variant of FFE technique in which the echo time can be longer than the repetition time, as described in section 8.1 with the Protocol Parameters described in section 8.2.

Test steps:

Test steps:										
Step	Description	Expected Result								
Step 1	Load scan protocol \Head CNS\EPI\presto4080i in the protocol editor and check presence of parameters and their ranges	The specified Shifted echo parameter is present The values can be entered within the ranges specified in requirement FFShiftedEcho <table><tr><td>TE>TR shift</td><td>Range: 1-3</td></tr><tr><td>M add factor</td><td>Range: -10..0 – 10.00</td></tr><tr><td>P add factor</td><td>Range: -10..0 – 10.00</td></tr><tr><td>S add factor</td><td>Range: -10..0 – 10.00</td></tr></table>	TE>TR shift	Range: 1-3	M add factor	Range: -10..0 – 10.00	P add factor	Range: -10..0 – 10.00	S add factor	Range: -10..0 – 10.00
TE>TR shift	Range: 1-3									
M add factor	Range: -10..0 – 10.00									
P add factor	Range: -10..0 – 10.00									
S add factor	Range: -10..0 – 10.00									

3.1.15.5. SWCMP.Scan.FFE.T2star.tc1

Test ID	13613
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description								
28057	SWCMP.Scan.FFE.T2star	T2* Maps and R2* Maps shall be available for Multi-Echo Gradient echo acquisitions, including FFE and TFE, both 2D (2D, M2D, MS) and 3D. MLE will be the default calculation method for R2*/T2* maps. RLSQ method will be optionally available for T2/T2* (in Reconstruction Control Parameter "T2(*), R2(*)calculation method"). (Note: it's not possible to request for R2/R2* images with RLSQ) Protocol parameters: <table><tr><th>Name</th><th>Range</th><th>Visible When</th><th>Description</th></tr><tr><td> </td><td> </td><td> </td><td> </td></tr></table>	Name	Range	Visible When	Description				
Name	Range	Visible When	Description							

Req ID	Requirement Tag	Description				
		Calculated images	The range is extended for FFE/TFE with [T2*, R2*]	Always	Reconstructs T2* and/or R2* maps	
		T2* clipvalue (ms)	0.00 - 2047.00	Calculated images equals (includes) T2*	The "T2* Clipvalue (ms)" parameter specifies the maximum calculated T2* value (ms). All pixels with a T2* larger than the clip value (including pixels with an apparently negative T2*) are set to this clip value.	
		R2* clipvalue (1/s)	1.00 - 10000.00	Calculated images equals (includes) R2*	The "R2* clip value (1/s)" parameter specifies the maximum calculated R2* value (1/s). All pixels with an R2* larger than the clip value, are set to this clip value. (a large R2* corresponds to a short T2*).	

Req ID	Requirement Tag	Description
		Note: The clipvalues have impact on the display ("window & level") of the images, but also on the discretization of the T2*/R2* images: for a 12 bit image, only 4096 discrete values are possible.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Configure RTM as 3T. Startup application SW. Select protocols i1_2dt2star and i1_3dt2star in subanatomy Head CNS/FE in the protocol editor. (If this protocol does not exist yet in TEDB, use the attached ExamCard by dragging it in the execution list.) Make sure that the protocol has no conflicts	n.a. (preparation)	 FFET2star.zip
Step 2	Make sure that the parameter "Calculated images" has R2* and T2* enabled.	Parameter "Calculated images" shall contain both R2* and T2*. No conflicts.	None
Step 3	Start the 2D scan.	Scan completes successfully.	None
Step 4	Start the 3D scan.	Scan completes successfully.	None

3.1.16. ME k-space_disturb

3.1.16.1. SWCMP.Scan.k-space_distrib.Halfscan.tc1

Test ID	32412
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-11 09:44:29 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-04-11 07:44:12 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-04-10 17:04:21

	Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The purpose of this test is to verify Halfscan functionality.

This test is a manual test using the MR Application on a VMWare machine on a 1.5T Ingenia configuration for reproducibility. The tester verifies the correct setting for the ENC attributes "min_encoding_numbers[]", "max_encoding_numbers[]" and "profiles[]".

Requirements Coverage:

Req ID	Requirement Tag	Description
65704	SWCMP.Scan.k-space_distrib.Halfscan	The scan protocol shall support the following Halfscan k-space manipulation techniques: - Partial acquisition of k-space in Y-direction by omitting part of the ky encoding steps at one side of k-space, defined by the partial matrix factor. - Partial acquisition of k-space in Z-direction by omitting part of the kz encoding steps at one side of k-space, defined by the partial partition factor. - Combination of the above.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	a) On a VMWare, run C:\nmr\gyroscan\m_amn_set_test_environment.pl and select the WA15 configuration. b) Create a new logfile. c) Start the Application software and go to the database folder Hospital\Head CNS\FE. Drag the following protocol into the execution list: F_3D_10 Go to the Geometry page and change the following values: FOV: FH = 100, RL = 100, AP = 50 ACQ voxel size: FH = 1, RL = 1, AP = 1 Slice oversampling: user defined oversample factor: 1	N.A. (preparation).	None
Step 2	Run the scan and open devlogcurrent and use the following filter query: text icontains ': min_encoding_numbers' or text icontains ': max_encoding_numbers' or text icontains 'max_profiles' or text icontains '= "F_3D10" ' Check the logfile for correct min and max encoding numbers.	The logfile contains: <pre>m_Name ="F_3D10" 01 01 00: min_encoding_numbers : -99 -50 -25 0 01 01 00: max_encoding_numbers : 100 49 24 0</pre>	None
Step 3	Copy the previous scan and change the values of the following parameters on the contrast page:	The logfile contains: <pre>m_Name ="F_3D10" 01 01 00: min_encoding_numbers :</pre>	None

Step	Description	Expected Result	Attachments
	Halfscan = yes factor Y = 0.6 Run the scan and open devlogcurrent and use the same filter query as used in step 2 to check the min and max encoding numbers.	-99 -10 -25 0 01 01 00: max_encoding_numbers : 100 49 24 0	
Step 4	Copy the previous scan and change halfscan factor Y to 0.8. Run the scan and open devlogcurrent and use the same filter query as used in step 2 to check the min and max encoding numbers.	The logfile contains: m_Name ="F_3D10" 01 01 00: min_encoding_numbers : -99 -30 -25 0 01 01 00: max_encoding_numbers : 100 49 24 0	None
Step 5	Copy the previous scan and change halfscan factor Y to 1.0 and halfscan factor Z to 0.75. . Run the scan and open devlogcurrent and use the same filter query as used in step 2 to check the min and max encoding numbers.	The logfile contains: m_Name ="F_3D10" 01 01 00: min_encoding_numbers : -99 -50 -13 0 01 01 00: max_encoding_numbers : 100 49 24 0	None
Step 6	Copy the previous scan and change halfscan factor Y to 0.6 (factor Z remains 0.75). Run the scan and open devlogcurrent and use the same filter query as used in step 2 to check the min and max encoding numbers. Export the part of the logfile only containing the output of the filter query. (in devlogcurrent.log click in the upper window: File-->Export View). Add the generated file to this test step as objective evidence for this entire testcase. (see attached example).	The logfile contains: m_Name ="F_3D10" 01 01 00: min_encoding_numbers : -99 -10 -13 0 01 01 00: max_encoding_numbers : 100 49 24 0	 exported_log.zip

3.1.16.2. SWCMP.Scan.k-space_distrib.IrregularSamplingPattern.tc1

Test ID	32705
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-28 12:44:55 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Graaf, Ruud de (nly95749) on 2017-03-28 12:02:05 Meaning: SME Approved - (Technical)
	Signed by: IJperen, Gert van (nly95968) on 2017-03-28 11:53:54 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

The sampling pattern generated by a Poisson disk algorithm is characterised by the following properties:

- It has a prescribed total number of samples (N), where N depends on the matrix size and the required acceleration factor
- Sampling points are randomly assigned on a grid of discrete k_y, k_z positions
- Sampling points should not "cluster", and there should be no large unsampled areas ("holes")
- So we can observe a minimum distance between any two sampling points (a), and
- we can observe areas with radius (b) where extra samples could have been added without violating that minimum distance (a)

The algorithm finds the largest possible value for (a) for a given N.

Ideally, the "hole size" (b) would then be equal to the minimum distance (a).

In practice, the ratio b/a will be somewhat larger than one, due to the stochastic nature of the process, and the fact that on the grid only certain discrete values for the distances a or b are possible.

Also, low acceleration factors lead to almost fully sampled patterns, where the whole "clustering vs. hole" concept loses its meaning.

A realistic upper limit for the observed value of the ratio b/a , in sufficiently undersampled regime, is 1.300.

A realistic upper limit for the sampling density where a meaningful ratio b/a can be estimated, is 0.1.

The prescribed sampling density (sdf) is usually variable, meaning it is a function of the radial distance to the k-space origin.

Observed sampling densities can only be locally averaged values, so we need to average over concentric rings,

with radial distance in a narrow range $[r_{\min}, r_{\max}]$. Within each narrow ring, the observed sdf is then supposed to be constant.

The above mentioned distance (a) and (b) will depend on the variable sdf, and the ratio b/a must be evaluated per individual ring.

Test steps:

Step	Description	Expected Result
Prep 1	• On a VMWARE (or ROF) run the C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl script. Select a WA15 configuration (1.5T Ingenia). • Add the trace channel "MAOBJENC cs_shutter_analysis *" to the bstrace.dat file, which is located at G:\site on VMWARE (or C:\temp on a ROF).	N.A. Preparation.

Step	Description	Expected Result
	- Start the 'ExamCards Scan Environment' on VWMARE (or 'Application Software' on a ROF. And create a new exam. - Start a new logfile.	
Step 1	- From the Hospital folder select the following protocol: Head CNS\CSSENSE\3DFFE_CS2. - Underneath the CS-SENSE parameter, change reduction factor to the maximum allowed value of 32.	The protocol validates without any issue.
Step 2	- Open the file G:\temp\analysis_file.dat, e.g. in Notepad. - The table lists sampling density (sdf) as a function of radial distance to origin (ring number 'f'). - Check that the sdf value decreases for increasing distance (ring number 'f').	For each ring number 'f', the value of sdf is lower than the value for the previous ring.
Step 3	For each grid point in a ring the minimum distance to any measured point is calculated. The maximum value found within a ring for these minimum values is used to characterize holes. The minimum value found for these minimum values is used to characterize cluttering. - The table in the G:\temp\analysis_file.dat file also lists the ratio of these found maximum and minimum values as a function of radial distance to origin (ring number "f"). - Check that this ratio is below 1.300 for a ring with sampling density (sdf) below 0.1 - Check that the overall worst case ratio for any ring with sdf below 0.1 is below 1.300	Expected result: - The ratio is below 1.300 for a ring with sampling density (sdf) below 0.1. "The file contains a last line stating "Worst case ratio is xxx" where xxx is smaller or equal 1.300"
Wrap up	- Attach the file G:\temp\analysis_file.dat and the G:\log\devcurrent.log files as objective evidence to the test case. - Remove the G:\temp\analysis_file.dat. - Remove the trace channel "MAOBJENC cs_shutter_analysis *" from the bstrace.dat file.	N.A. (Wrap up).

3.1.17. ME MBSENSE

3.1.17.1. SWCMP.Scan.MBSENSE.Pars.tc1.v1

Test ID	40642
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:53 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:03:46 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:43:45 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description			
74857	SWCMP.Scan.MBSENSE.Par	Name	Range	Visible when	Description
		MB SENSE	yes, no	(SingleShot (SE-EPI or FFE-EPI)) and (MS or M2D)	enables or disables MB SENSE technique
		MB Factor	Coil/ technique dependent, as specified below.	MB SENSE	determines the number of slices that are going to be simultaneously excited and acquired.
		MB shift	default , user defined	MB SENSE	The MB-shift is an integer value that describes the $((1/n)*FOV)$ shift that is applied for MB imaging.

Req ID	Requirement Tag	Description			
		((1/n)*FOV)	1 - 6	user defined MB Shift	Shift to be applied for MB imaging.
		MB Factor range:			
		coil	scan technique	minimum	maximum
		32ch head coil	SE-EPI	2	4
		32ch head coil	FE-EPI	2	8
		ds Head coil dS Head/Neck coil	SE-EPI	2	2
		ds Head coil dS Head/Neck coil	FE-EPI	2	2

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanMbSensePars in the results file is zero.

3.1.17.2. SWCMP.Scan.MBSENSE.RF.GR.tc1

Test ID	26140
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-06-10 09:40:17 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Graaf, Ruud de (nly95749) on 2016-06-10 08:05:22 Meaning: SME Approved - (Technical)::Bulk Transitions
	Signed by: Polders, Daniel Louis (310171444) on 2016-06-09 11:10:46 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test must be performed on a SDE VMWare with the complete software application installed. The scanner configuration must be set to 3.0T wide aperture.

The tester must be familiar with the "ExamCards Scan Environment".

The RF pulses used in the sequence will be sampled, a FFT over the RF waveform will be performed and used as an approximation of the slice profile. The distance between the excited slices must be equal to the theoretical value (within 10% error margin). The RF waveforms and the slice profiles can be found under G: \Temp with the names "mb_waveform_<rf function>_<rf number>.xls" and/or "mb_slice_prof_<rf function>_<rf number>.xls". Within the file mb_ts_rf_id_<rf number>.txt in the same directory it is possible to find the results of the component test specification.

Requirements Coverage:

Req ID	Requirement Tag	Description
45434	SWCMP.Scan.MBSENSE.RF.SliceDistance	If "MB Factor" scan protocol parameter is different than 1, the distance between the simultaneously excited slices (Multiband_dist) shall be: $\text{Multiband_dist} = (\text{inter_slice_distance}) * (\text{total_number_of_slices}) / (\text{MB Factor}), \text{ where}$ inter_slice_distance is the distance between the center of one slice and the center of the adjacent slice; total_number_of_slices is the total number of slices in the selected stack.
57081	SWCMP.Scan.MBSENSE.Phase Cycling.GR.Blip_S	The area of the blip gradient (area_blip_s) shall be: $\text{area_blip_s} = 1.0 / (\text{Multiband_shift} * \text{Gamma} * \text{Multiband_dist}) \text{ where}$ Multiband_shift is the fraction of the Field of View (FOV) which determines the spatial shift introduced in the phase direction of slices that are going to be simultaneously excited and acquired. For example, Multiband_shift 2 means FOV / 2 shift of the adjacent slice in the phase direction; Gamma is the gyromagnetic ratio of the proton; Multiband_dist is the the distance between the simultaneously excited slices. Multiband_dist is define as: $\text{Multiband_dist} = (\text{inter_slice_distance}) * (\text{total_number_of_slices}) / (\text{MB Factor}), \text{ where}$

Req ID	Requirement Tag	Description
		inter_slice_distance is the distance between the center of one slice and the center of the adjacent slice; total_number_of_slices is the total number of slices in the selected stack.
57089	SWCMP.Scan.MBSENSE.Phase Cycling.GR.B_Blip_S	The area of the balancing blip gradient (area_b_blip_s) shall be: $\text{area_b_blip_s} = (\text{Multiband_shift} - 1) / (\text{Multiband_shift} * \text{Gamma} * \text{Multiband_dist})$ where Multiband_shift is the fraction of the Field of View (FOV) which determines the spatial shift introduced in the phase direction of slices that are going to be simultaneously excited and acquired. For example, Multiband_shift 2 means FOV / 2 shift of the adjacent slice in the phase direction; Gamma is the gyromagnetic ratio of the proton; Multiband_dist is the the distance between the simultaneously excited slices. Multiband_dist is define as: $\text{Multiband_dist} = (\text{inter_slice_distance}) * (\text{total_number_of_slices}) / (\text{MB Factor}),$ where inter_slice_distance is the distance between the center of one slice and the center of the adjacent slice; total_number_of_slices is the total number of slices in the selected stack.
45433	SWCMP.Scan.MBSENSE.RF	The RF pulse used by the MultiBand SENSE technique shall excite simultaneously the number of slices specified by "MB Factor" scan protocol parameter.
57091	SWCMP.Scan.MBSENSE.Phase Cycling.GR.Shift	Multiband_shift shall be equal 2, if "MB Factor" is lower or equal to 4. If "MB Factor" is higher than 4, Multiband_shift shall be equal to: $\text{Multiband_shift} = (\text{MB Factor}) / 2.$ If Multiband_shift is fractional, it should be rounded up.
45435	SWCMP.Scan.MBSENSE.Phase Cycling.GR	MBSENSE technique shall use balancing blip gradient (b_blip_s) and blip gradient (blip_s) in the slice direction.

Test steps:

Step	Description	Expected Result
Step 1 - Preparation	On a SDE VMWare, make sure that the "Development" software option switch is disabled (that can be checked in Field Service Application -> Software Options -> Activate/Deactivate Optionswitches).	Preparation only
Step 2 - Preparation	Install scanprotocols (Start -> All Programs -> MR Development -> Configuration -> Scan protocol database selection), enable the mb_cts trace channel (in G:\Site, open the file bstrace.dat and add the following line: "mpumb mb_cts * ") and start "ExamCards Scan Environment" in "MR Boot Configuration Manager".	Preparation only
Step 3	Load the protocol Head\Brain\BOLD_imaging\FE_EPI. If the protocol is in conflict with the message "The total amount of data in this scan is too large", open "Edit Scan	The parameter "MB SENSE" is visible with range: • yes • no

Step	Description	Expected Result
	Control Parameters..." and in the "Scan" tab, set "recon memory model" to "no". Go to the geometry page verify the visibility and range of the parameter "MB SENSE".	
Step 4	Start the coil simulator (Start -> All Programs -> MR Development -> Scan Simulations -> Coilstate Simulator UI) and connect a head coil. Start the table simulator (Start -> All Programs -> MR Development -> Scan Simulations -> Start Graphical Patsup Simulation), set the "Light Visor" in the upper part of the table (where the head coil would be placed) and drag the table to the isocenter. Go to the "Offc/Ang" tab in the ExamCard and change the "FH (H =+ mm)" parameter in order to set up a head scan. That can be confirmed with the parameter "Head SAR" visible in the "Info"page.	The parameter "Head SAR" is visible in the "Info"page.
Step 5	Select "MB SENSE" equal to "yes" and go through all the range of "MB Factor". After every new input of "MB Factor", open the file(s) "mb_ts_rf_id<rf number>.txt" and inspect the file.	• "Input : Multiband_factor" must be equal to "Measured: Multiband_factor". • "Input : Multiband_dist" must be equal to "Measured: Multiband_dist" within 10% error margin. • balancing blip gradient sequence id must be equal to the base sequence id. • blip gradient sequence id must be equal to the xbase sequence id. • balancing blip gradient direction id and blip gradient direction id must be equal to the slice direction id. • The balancing blip gradient area must be equal to $(\text{Multiband_shift} - 1) / (\text{Multiband_shift} * \text{Gamma} * \text{Multiband_dist})$. • The blip gradient area must be equal to $1.0 / (\text{Multiband_shift} * \text{Gamma} * \text{Multiband_dist})$. • If Multiband_factor is lower or equal to 4, Multiband_shift must be equal 2. Otherwise, Multiband Shift must be equal to $\text{Multiband_factor} / 2$ rounded up if fractional.
Step 6	Repeat step 3, 4 and 5 loading the protocol Head\Brain\Diffusion\DWI in step 2 instead of Head\Brain\BOLD_imaging\FE_EPI.	Verify expected results accordingly.

3.1.18. ME MDME

3.1.18.1. SWCMP.Scan.MDME.NoiseClippingFactor.tc1

Test ID	24174
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-04-13 09:00:14 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Graaf, Ruud de (nly95749) on 2015-04-13 07:47:29 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Devaraj, Ajit (usd43945) on 2015-04-10 17:47:52 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Added the following line to g:\site\bstrace.dat mpaladm check_sp310 * Start the ExamCards Scan Environment. Unzip and Import the attached examcard <u>TS MDME with Phantom.examcard</u> Create new patient and drag the examcard into the execution list. The execution list contains the following scan protocols: Survey MDME GraSE MDME TSE	N.A (Preparation)	 TS MDME with Phantom.zip
Step 1	Run the scan protocol 'Survey.' Open the scan protocol MDME GraSE in the protocol editor. Press 'Accept' to accept the validation. Both MDME GraSE and MDME TSE change from scan protocol state 'needs planning' to 'ready to run' indicating that the scan protocol validated successfully. Press 'Start Scan' to run the protocols in the execution list.	All scans complete successfully.	None
Step 2	Copy the file g:\log\devlogcurrent.log to a folder on an SDE. Unzipped the attached file in the same folder location. In a command window, navigator to the same folder location. Run the command R:	For the MDME scans, the value of the first element of RC_noise_clip_factors = 0.001 (to three digits).	 filter_query_MDME_noise_clip.zip

Step	Description	Expected Result	Attachments
	\\MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe devlogcurrent.log @filter_query_MDME_noise_clip.txt > MDME_filter_extract.log. The parsed file will contain entries in the format of: RC_noise_clip_factors = Performance Logging [ScanExec] 0 value [Scan name]= Indicating the first element of the noise_clip_factors array follow by the scan that was run. Example: RC_noise_clip_factors = 0.001000000005, Performance Logging [ScanExec] 0 value [Scan name]=MDME TSE		

3.1.19. ME Methods

3.1.19.1. SWCMP.Scan.Methods.3DSliceInterpolation.tc1

Test ID	25527
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-12-21 12:01:41 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2015-12-14 13:20:18 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Huang, Alan (310109142) on 2015-12-14 11:51:28 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Flexible 3D slice interpolation

Introduction

3D slice interpolation can be achieved by choosing a higher acquisition voxel size in the slice selection direction than reconstruction voxel size.

Test description

The test requires a VMware, RTM or STM, with TEDB test protocol database installed.

The test consists of a number of manual tests varying protocol parameters in order to verify the behavior of the UI.

Requirements Coverage:

Req ID	Requirement Tag	Description
56420	SWCMP.Scan.Methods.3DSliceInterpolation	The 3D slice interpolation is defined as the ratio between the acquisition and reconstruction voxel size in the slice selection direction. Manipulation of the voxel size in slice selection direction shall show the following behavior: - when changing the acquisition voxel size in slice selection direction, the reconstruction voxel size will change accordingly in order to maintain the same slice interpolation factor - when changing the reconstruction voxel size in the slice selection direction ds_rec. The acquisition voxel size ds_acq remains unchanged which will lead to a modified slice interpolation factor $SIP = ds_acq/ds_rec$

Test steps:

Step	Description	Expected Result
Preparation	On a VMware select an Ingenia 3T configuration and TEDB database and open the ExamCards Application. Select the protocol Head CNS\TSE\F_3D_sl_128 and open it for edit.	N.A. (Preparation)
Step 1	On the geometry page, change the parameter 'ACQ voxel size FH' to 2mm and check the value of 'Recon voxel size FH'.	The parameter 'Recon voxel size FH' is automatically changed to 2mm as well so that 'ACQ voxel size FH' = 'Recon voxel size FH'.
Step 2	On the Info page, check the values of the parameters 'ACQ voxel MPS (mm)', 'REC voxel MPS (mm)' and 'Act. slice gap'.	The Info parameter 'ACQ voxel MPS (mm)' in slice direction is equal to 'ACQ voxel size FH' and 'REC voxel MPS (mm)' in slice direction is equal to 'Recon voxel size FH'. The Info parameter 'Act. slice gap' is 0.
Step 3	On the geometry page, change the parameter 'Recon voxel size FH' to 1mm.	The parameter 'ACQ voxel size FH' remains unchanged so that 'ACQ voxel size FH' = 2 * 'Recon voxel size FH'.
Step 4	On the Info page, check the values of the parameters 'ACQ voxel MPS (mm)', 'REC voxel MPS (mm)' and 'Act. slice gap'.	The Info parameter 'ACQ voxel MPS (mm)' in slice direction is equal to 'ACQ voxel size FH'. The Info parameter 'REC voxel MPS (mm)' in slice direction is equal to 'Recon voxel size FH'. The Info parameter 'Act. slice gap' is 'Recon voxel size FH' - 'ACQ voxel size FH'.
Step 5	On the Summary page, check the value of the last value for the parameter 'Voxel'.	The displayed value is equal to the parameter 'ACQ voxel size FH' on the geometry page.
Step 6	On the geometry page, change the parameter 'Recon voxel size FH' to 1.4mm.	The parameter 'ACQ voxel size FH' remains unchanged.

Step	Description	Expected Result
Step 7	On the Info page, check the values of the parameters 'ACQ voxel MPS (mm)' , 'REC voxel MPS (mm)' and 'Act. slice gap'.	The Info parameter 'ACQ voxel MPS (mm)' in slice direction is equal to 'ACQ voxel size FH'. The Info parameter 'REC voxel MPS (mm)' in slice direction is equal to 'Recon voxel size FH'. The Info parameter 'Act. slice gap' is 'Recon voxel size FH' - 'ACQ voxel size FH'.

3.1.19.2. SWCMP.Scan.Methods.64Kimages.tc1

Test ID	13519
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
28116	SWCMP.Scan.Methods.64Kimages	It shall be possible to acquire at least 65536 images.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Run this test on a RTM configured with a valid product configuration. 1. Select tedb 1. Load scan protocol tedb / Head CNS / COMB/64000_imgs in the protocol editor (If this protocol does not exist yet in TEDB, use the attached ExamCard by dragging it in the execution list.) 1. Make sure that the protocol is conflict free	N.A. (preparation)	 FE_EPI_64K.zip
Step 2	Start the scan	The scan should complete successfully	None

3.1.19.3. SWCMP.Scan.Methods.Breathhold.tc1.v1

Test ID	32579
Revision	1
Product Model	Achieva dStream;Ingenia S
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-29 16:06:20 Meaning: QA Approved - (QA) Signed by: Geraedts, Tom (nly22893) on 2017-03-29 15:47:58 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-29 15:25:05 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
42762	SWCMP.Scan.Methods.BreathholdStart	Component ME Methods shall signal the event EventBreathhold when a breath-hold is about to start before acquiring data of a breath-hold period.
53201	SWCMP.Scan.Methods.BreatheAgain	Component ME Methods shall signal the event EventBreatheAgain when a breathhold period is finished.
54498	SWCMP.Scan.Methods.ReportingRespTriggerDetected	When the measurement waits for cancel from the host and processing respiratory input is requested, the respiratory input shall be processed and when a respiratory trigger is detected it shall be reported via the event EventRespTriggerDetected to the host with a latency lesser than 200ms.

Test steps:

Step	Description	Expected Result
Step 1	Run gyroscan\commands\m_amn_set_test_environment.pl command on a Recon Workstation and set the system type to 1.5T Ingenia. Confirm that on the PSC Service Application the Miscellaneous Box under the Scanner Hardware configuration is set to AIBo. Confirm that Autovoice is enabled at the system level setting. Create new a logfile Load the scan protocol Head_CNS/RESP/B_3D7 in the protocol editor. Resolve conflicts (if any). Start the scan.	N.A.(preparation)
Step 2	Open the devlogcurrent.log file. Search for the following entries at the beginning of each breathhold in the devlogcurrent.log file: "DasProxyImpl Notify(EventBreathhold)" and "ScanSetPerformer::HandleEvent(EventRespTriggerDetected)"	The following entries are logged at the beginning of each breathhold in the devlogcurrent.log file: "DasProxyImpl Notify(EventBreathhold)" and "ScanSetPerformer::HandleEvent(EventRespTriggerDetected)"
Step 3	Open the devlogcurrent.log file. Search for the first instance of "CancelWaitingFor" in the devlogcurrent.log	The difference between the two time stamps (TimeStamp1 and TimeStamp2) should be less

Step	Description	Expected Result
	file and observe its time stamp (say TimeStamp1). Search for the immediate preceding instance of "ScanSetPerformer::HandleEvent(EventRespTriggerDetected)" or "ScanSetPerformer::HandleEvent(EventRespTriggerDetected) skipped." in the devlogcurrent.log file and observe its time stamp (say TimeStamp2).	than 200ms.
Step 4	Open the devlogcurrent.log file. Search for the following entries at the end of each breathhold in the devlogcurrent.log file: "DasProxyImpl Notify(EventBreatheAgain)" Attach the logfile as objective evidence	The following entries are logged at the end of each breathhold in the devlogcurrent.log file: "DasProxyImpl Notify(EventBreatheAgain)"

3.1.19.4. SWCMP.Scan.Methods.ComforTone.tc1.v4

Test ID	58148
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:52:29 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:15:41 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11-15 11:07:00 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Requirements Coverage:						
Req ID	Requirement Tag	Description				
81742	SWCMP.Scan.Methods.Constraints.ComforTone.v3	Subsystem Scanning shall implement ComforTone mode that will restrict the predicted acoustical noise level that is exposed to the patient. This mode shall be available for FFE, SE, TSE, TFE, FFE-EPI, SE-EPI and DWI-EPI. The predicted acoustical noise level restriction varies per system configuration: <table><tr><td>System configuration</td><td>Maximum allowed acoustic noise (dBA)</td></tr><tr><td>1.5T Wide bore</td><td>99</td></tr></table>	System configuration	Maximum allowed acoustic noise (dBA)	1.5T Wide bore	99
System configuration	Maximum allowed acoustic noise (dBA)					
1.5T Wide bore	99					

Req ID	Requirement Tag	Description	
		1.5T Small bore	103
		3.0T Elition	99
		3.0T Small bore	105
		3.0T Ingenia	107

Test steps:

Step	Description	Expected Result												
Step 1	Use m_amn_set_test_environment.pl to set the simulator software for a T15 Ingenia. Enable scanner simulation, recon on the host or the reconstructor (if available). Drag the attached ComforTone Examcard into the scanner execution list. For each of the scan protocols open the parameter editor. On the info page inspect the SPL value. For each protocol calculate the acoutical noise level as displayed SPL + 98 dB.	Verify that for each protocol FFE, SE, TSE, TFE, FFE-EPI, SE-EPI, DWI-EPI, coil survey, SENSE reference scan, all using Softone mode = ComforTone, the calculated acoustical noise level is equal to or lower than 99 dB.												
Step 2	Run the scans in the execution list. When the scans are completed open the log file using Logbrowser. Query the log file with the filter "text contains 'm_Name' or text contains 'Sound Pressure Level' "	Verify that he scans are performed succesfully. Verify that all scans including any inserted coil survey or scan SENSE refence scan have a Predicted Sound Pressure Level equal to 1 dB or lower, corresponding to an acoustical noise level of 99 dB or lower.												
Step 3	Open the resource file mpicscdb.i targeted on the test system and inspect the parameter values of parameter CSC_restricted_spl_limits located at the 2nd position in the array. Write down these acutal values.	For each system configuration: Add 98dBA to the value located at the 2nd position in the array and verify that the sum does not exceed the specified restricted acoustic noise (dBA). <table> <tr> <th>System configuration</th><th>System configuration translation</th><th>Restricted acoustic noise (dBA)</th></tr> <tr> <td>1.5T Wide bore</td><td><1_5T && WA_CYL></td><td><value> + 98 <= 99</td></tr> <tr> <td>1.5T Small bore</td><td><1_5T && !WA_CYL></td><td><value> + 98 <= 103</td></tr> <tr> <td>3.0T Elition</td><td><3_0T && WA_CYL && GCT_WA2 ></td><td><value> + 98 <= 99</td></tr> </table>	System configuration	System configuration translation	Restricted acoustic noise (dBA)	1.5T Wide bore	<1_5T && WA_CYL>	<value> + 98 <= 99	1.5T Small bore	<1_5T && !WA_CYL>	<value> + 98 <= 103	3.0T Elition	<3_0T && WA_CYL && GCT_WA2 >	<value> + 98 <= 99
System configuration	System configuration translation	Restricted acoustic noise (dBA)												
1.5T Wide bore	<1_5T && WA_CYL>	<value> + 98 <= 99												
1.5T Small bore	<1_5T && !WA_CYL>	<value> + 98 <= 103												
3.0T Elition	<3_0T && WA_CYL && GCT_WA2 >	<value> + 98 <= 99												

Step	Description	Expected Result		
		3.0T Ingenia	<3_OT && WA_CYL && ! GCT_WA2 >	<value> + 98 <= 107
		3.0T Ingenia CX	<3_OT && ! WA_CYL>	<value> + 98 <= 105

Attachments:

ComforTone.zip

3.1.19.5. SWCMP.Scan.Methods.Conflict.Automatic.3D_multi_chunk.tc1

Test ID	13170
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Automatic conflict resolving 3D multichunk

Introduction

When changing the number of slices in a 3D multi-chunk scan to arbitrary values, conflicts are automatically resolved by rounding the number of slices upwards to satisfy the applicable "multiple" criterion. At the same time, a maximum FOV of 400 mm is enforced.

Test description

The test requires an RTM or STM, with TEDB test protocol database installed. The test consists of a number of interactive test steps involving protocol modification and Planscan.

Analysis

Observe the parameter behaviour as described under 'expected results', both on the Summary Tab and the legacy Geometry and Conflicts (Info) Tab.

Requirements Coverage:

Req ID	Requirement Tag	Description
27649	SWCMP.Scan.Methods.Conflict. Automatic.3D_multi_chunk	At an update of the number slices in a 3D multi-chunk scan, the number of slices is adapted to a multiple of the number of chunks. Two cases exist, where the number of chunks itself may not or may be adapted respectively: - In case of a fixed (user defined) number of chunks, the number of slices per chunk is increased/decreased. The chunk thickness is changed accordingly. Possible values for number of slices are multiples of the number of chunks. - In case of a fixed (user defined) chunk thickness, the number of chunks is changed accordingly. Possible values for number of slices are multiples of the number of slices per chunk, as defined by the user defined chunk thickness and the slice thickness. Possible values for number of slices are also multiples of the number of chunks (which in this case is <i>not</i> user defined).

Test steps:

Step	Description	Expected Result
Step 1	Make a suitable Survey scan, e.g. select F_2D1 from TEDB, Head CNS, FE Change FOV to 500 mm and change Orientation into Sagittal.	A completed scan appears as image in the Planscan windows, ready to visualize the planning of a subsequent Transverse scan.
Step 2	Select F_3D9 from TEDB, Head CNS, FE, and open for planning. Press and hold the shift key, and move the mouse to an edge slice, until the # sign appears in Planscan. Perform a series of asymmetric FOV changes (increases) by dragging the mouse, on alternate sides of the stack of slices.	The thickness of the stack of slices is increased, but never beyond 400 mm. On the Summary Tab this number is indicated as FOV in FH direction.
Step 3	Go to Geometry Tab, and locate parameter Multi-chunk. Change value from no to yes. Change value of the parameter chunks from 2 to 3. Use the asymmetric FOV Planscan way to change the the number of slices.	The number of slices is rounded upwards to a multiple of 3 (= the user defined number of chunks). The toggle step of the parameter slices is set to 3. The maximum number of slices is 198, corresponding to FOV 396 mm.
Step 4	Go to Geometry Tab, and locate parameter Multi-chunk. Change value from yes to by thickness. Change value of the parameter chunk thickness to 10 mm. Use the asymmetric FOV Planscan way to change the the number of slices.	The number of slices is rounded upwards to a multiple of 5 (= the user defined chunk thickness 10 mm divided by the slice thickness 2 mm). The toggle step of the parameter slices is set to 5. The maximum number of slices is now 100, corresponding to FOV 200 mm. This because the maximum number of chunks is 20.
Step 5	Change value of the parameter chunk thickness to 30 mm. Use the asymmetric FOV Planscan way to change the the number of slices.	The number of slices is rounded upwards to a multiple of 15 (= the user defined chunk thickness 30 mm divided by the slice thickness 2 mm). The toggle step of the parameter slices is set to 15. The maximum number of slices is now 195, corresponding to FOV 390 mm. In that case the number of chunks is 13.

3.1.19.6. SWCMP.Scan.Methods.Conflict.Automatic.Geo.Offcenters.tc1

Test ID	11915
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test Cases .Conflict.Automatic.Geo.Offcenters

Requirements Coverage:

Req ID	Requirement Tag	Description
21025	SWCMP.Scan.Methods.Conflict.Automatic.Geo.Offcenters	Stacks are not allowed to be planned anywhere. Therefore, the stack offcenters are controlled to be within specified ranges. These ranges depend, amongst others, on the table position, coils used, system type, use of prescans, the offcenters of other stacks of the same scan and patient orientation.

Test steps:

Step	Description	Expected Result
GEO_O FFC_A	Single stack scan is moved beyond the anterior planning limitations. The offcenter is moved back to the anterior limit.	Anterior offcenter equals the maximum allowed offcenter of 160mm.
GEO_O FFC_P	Single stack scan is moved beyond the posterior planning limitations. The offcenter is moved back to the posterior limit.	Posterior offcenter equals the maximum allowed offcenter of 160mm.
GEO_O FFC_F	Single stack scan is moved beyond the feet planning limitations. The offcenter is moved back to the feet limit.	Feet offcenter equals the maximum allowed offcenter.
GEO_O FFC_H	Single stack scan is moved beyond the head planning limitations. The offcenter is moved back to the head limit.	Head offcenter equals the maximum allowed offcenter.
GEO_O FFC_L	Single stack scan is moved beyond the left planning limitations. The offcenter is moved back to the left.	Left offcenter equals the maximum allowed offcenter of 220mm.
GEO_O FFC_R	Single stack scan is moved beyond the right planning limitations. The offcenter is moved back to the right limit.	Right offcenter equals the maximum allowed offcenter of 220mm.
GEO_O FFC_M Stack_H	Multi-stack (3 stacks) scan without table movement. One stack is moved in head direction such that a table stroke is required to scan all stacks.	The other 2 stacks are moved with the moved stack in head direction such that the scan is not in conflict and can be performed.
GEO_O FFC_M Stack_F	Multi-stack (3 stacks) scan without table movement. One stack is moved in feet direction such that a table stroke is required to scan all stacks.	The other 2 stacks are moved with the moved stack in feet direction such that the scan is not in conflict and can be performed.
GEO_O FFC_M	Multi-stack (3) scan without table movement. The scan needs a prescan. One stack is moved in feed direction	The other 2 stacks are moved with the moved stack in feed direction such that the scan is not in

Step	Description	Expected Result
Stack_ prescan	such that a table stroke is required to fit all stacks within the prescan.	conflict and can be performed.

3.1.19.7. SWCMP.Scan.Methods.Conflict.Automatic.tc1

Test ID	9981
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Automatic conflict resolving test

Introduction

Instead of setting conflicts, the pdf can automatically update EX parameters in order to resolve the conflict. No user assistance is used. This test covers the automatic adjustment of EX parameters.

Test description

The runs several batches and performs logfile analysis to check whether conflicts were resolved correctly. The specific test cases are described in the companion documents. For each companion document a batch is started, e.g. for TFE m_imn_auto_conflict_tfe.acq.

For the test, an RTM is required. A script is used to automatically connect the correct coils, enable trace channels, run the batches and perform the test analysis.

Analysis

In the batches, examination parameters are set to bring the selected protocol into conflict. The conflicts are automatically to be resolved. Trace channel logging will provide the information how the conflict resolution is performed. The tests scripts select the required log statements from the logfile. Next, analysis is performed on these log statements whether the conflict was resolved and whether this was done in the correct way.

The flow diagram is as follows:

Start test script

First subscript selected, e.g. m_imn_auto_conflict_tfe

set used coils

set trace channels

start new logfile

perform batch

select required trace logging

analyse logging

results of the analysis is displayed and put into result logfile

reset trace channels

Second subscript selected

Not yet available, because currently the test script only contains only one automated subscript.

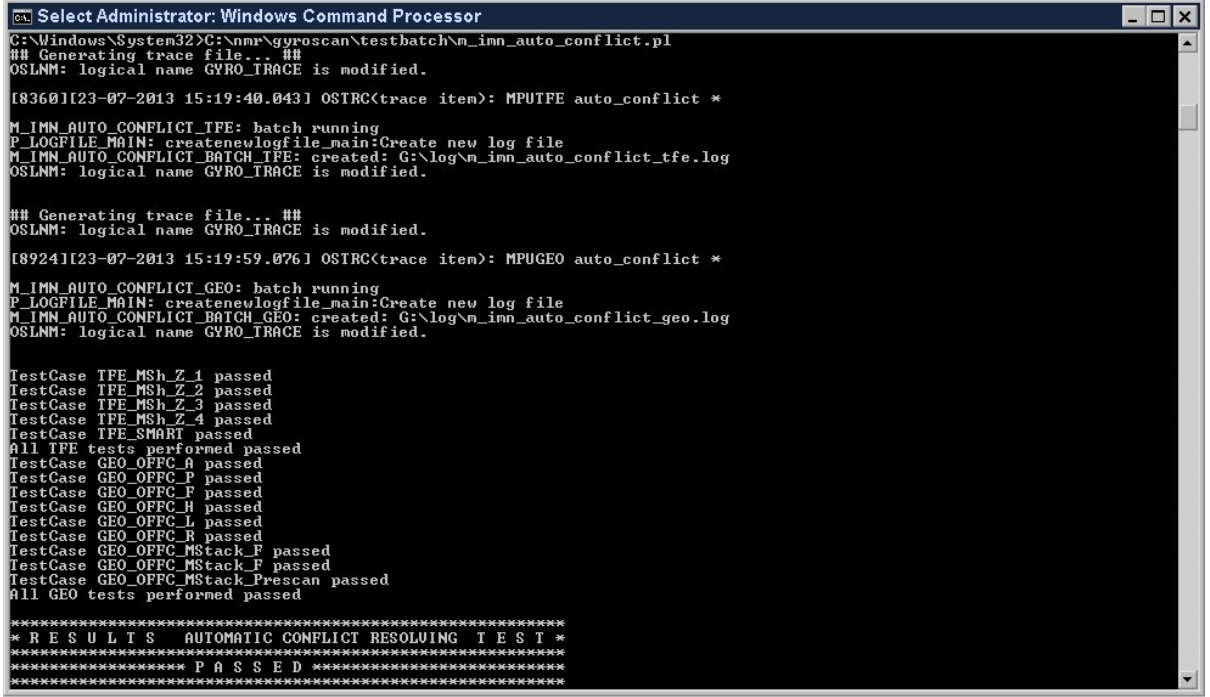
Last subscript selected

Report total test result

Requirements Coverage:

Req ID	Requirement Tag	Description
17768	SWCMP.Scan.TFE.MSh_3D	At an update of the number of acquired slices, for Multi-shot TFE with turbodirection Z conflicts shall automatically be resolved by: - Increasing the number of shots in case the nr_shots >= sqrt(TFE_factor) - Increasing the TFE factor in case in case the nr_shots < sqrt(TFE_factor).
21024	SWCMP.Scan.Methods.Conflict. Automatic	Conflicts shall be automatically resolved based on the last change of the user, when no user guidance is needed.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	1. Configure the RTM as T30 Achieva. 2. Enable all coils. 3. Select TEDB as protocol database. 4. Start the simulator (ACQ on host, recon on reconstructor) 5. Start a command prompt. 6. From the command prompt, start the script: c:\nmr\gyroscan\testbatch\m_imn_auto_conflict.pl. This will run the test.	Verify that after the testscript has run in the command prompt, it is indicated that all tests are passed (see attached screenshot). Note: when all tests are passed, this is also reflected in the total result status (2nd last line in the screenshot). This can also be verified by inspection of the logfile generated during the test: g:\log\m_imn_auto_conflict_results.log.	
 <pre> C:\Windows\System32\cmd.exe /c c:\nmr\gyroscan\testbatch\m_imn_auto_conflict.pl ## Generating trace file... ## OSLNM: logical name GYRO_TRACE is modified. [83601]23-07-2013 15:19:40.0431 OSTRC(trace item): MPUTFE auto_conflict * M_IMN_AUTO_CONFLICT_TFE: batch running P_LOGFILE MAIN: createnewlogfile_main:Create new log file M_IMN_AUTO_CONFLICT_BATCH_TFE: created: G:\log\m_imn_auto_conflict_tfe.log OSLNM: logical name GYRO_TRACE is modified. ## Generating trace file... ## OSLNM: logical name GYRO_TRACE is modified. [89241]23-07-2013 15:19:59.0761 OSTRC(trace item): MPUGEO auto_conflict * M_IMN_AUTO_CONFLICT_GEO: batch running P_LOGFILE MAIN: createnewlogfile_main:Create new log file M_IMN_AUTO_CONFLICT_BATCH_GEO: created: G:\log\m_imn_auto_conflict_geo.log OSLNM: logical name GYRO_TRACE is modified. TestCase TFE_MSh_Z_1 passed TestCase TFE_MSh_Z_2 passed TestCase TFE_MSh_Z_3 passed TestCase TFE_MSh_Z_4 passed TestCase TFE_SMART passed All TFE tests performed passed TestCase GEO_OFFC_A passed TestCase GEO_OFFC_P passed TestCase GEO_OFFC_F passed TestCase GEO_OFFC_H passed TestCase GEO_OFFC_L passed TestCase GEO_OFFC_R passed TestCase GEO_OFFC_MStack_F passed TestCase GEO_OFFC_MStack_F passed TestCase GEO_OFFC_MStack_Prescan passed All GEO tests performed passed ***** * R E S U L T S A U T O M A T I C C O N F L I C T R E S O L V I N G T E S T * ***** ***** P A S S E D ***** ***** </pre>			

3.1.19.8. SWCMP.Scan.Methods.Conflict.UserGuidance.tc1.v1

Test ID	58147
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:51:46 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:15:16 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11- 15 11:06:20 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

User guidance Conflict Solving

Introduction

When a conflict is set, for certain cases the pdf can generated suggestions on how to solve the conflict. These suggestions are prevalidated after which the user can click on the suggestion to resolve the conflict.

Test description

The test requires an VM/RTM with TEDB test protocol database installed. The test consists of a number of interactive test steps involving protocol modification.

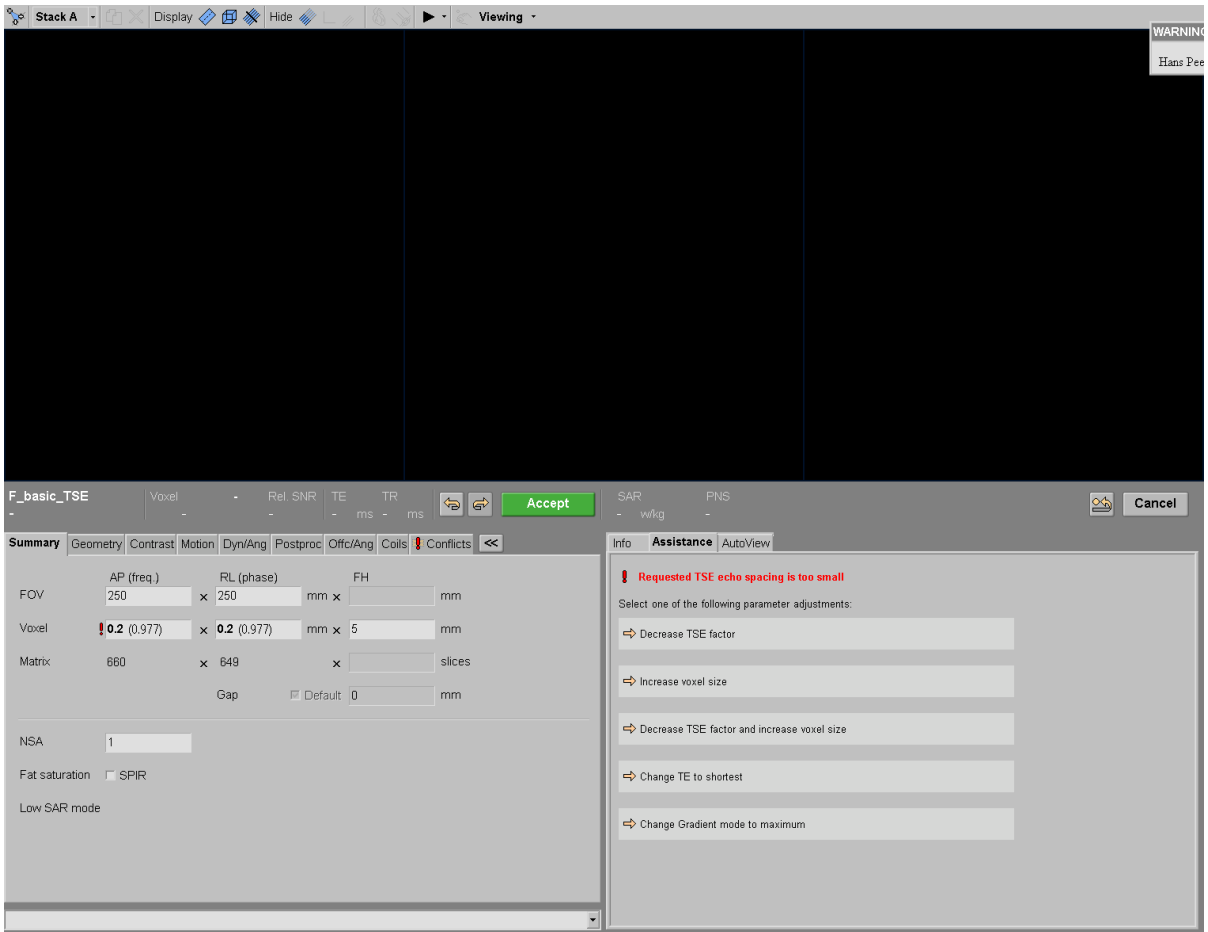
Analysis

Observe the parameter and conflict behavior when the conflict is invoked and resolved. Only valid suggestions should be shown.

Requirements Coverage:

Req ID	Requirement Tag	Description
21026	SWCMP.Scan.Methods.Conflict. UserGuidance	For a number of cases, described in this document and companion documents, the validate function shall return, in case of a conflict, suggestions to solve the conflict. The suggestions consists of: 1. Instruction for the UI telling what has to be displayed. 2. Scan Protocol Parameter modifications that must be applied when a certain suggestion is selected

Test steps:

Step	Description	Expected Result	Attachments
Step 1	1. Configure a VM as an Ingenia CX 3.0T (787 amplifiers) 2. Select TEDB as protocol database 3. Start Examcards 4. Make a patient with 50 kg weight. 5. Add the following protocol to the scanlist: Head_CNS\TSE\F_basic_TSE 6. Edit the protocol by modifying the voxel size in read direction from 0.98 to 0.2 mm. 7. Check that the system provides guidance to resolve the TSE timing conflict. (See figure below) 8. Choose one option by clicking on it and check that the conflict disappears. 9. Click the undo button and check that the same guidance appears. 10. Choose another option and check that the conflict disappears.	• Verify that Guidance is provided • Verify that the conflict is resolved for both provided options • Verify that the conflict reappears after undo.	
			

3.1.19.9. SWCMP.Scan.Methods.Constraints.FOV.NarrowBore1.tc1

Test ID	25249
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2015-10-15 17:54:39 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-10-15 17:37:50 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-10-15 17:36:19 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
57206	SWCMP.Scan.Methods.Constraints.FOV.NarrowBore1	The maximum allowed FOV shall be limited by the hardware capabilities and available options. For the configuration(s) specified in supported Product Model/Field Strength combinations given in the details of this requirement the system shall set the maximum allowed FOV as follows: The upper limit of FOV in AP direction shall be at least 520 mm The upper limit of FOV in LR direction shall be at least 520 mm The upper limit of FOV in FH direction shall be at least 520 mm

Test steps:

Step	Description	Expected Result
Step 1	Install appropriate SW in test environment	N/A. preparatory step.
Step 2	Set system configuration to the supported configuration that is specified by the requirement.	N/A. preparatory step.
Step 3	Start Application software or ExamCards test environment and create a new exam.	N/A. preparatory step.
Step 4	Select a protocol from the Philips database, open the protocol in editor and go to the Geometry tab of the protocol editor.	N/A. preparatory step.
Step 5	Hover the mouse pointer over the value of parameter FOV AP (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 520 as specified in the requirement.
Step 6	Hover the mouse pointer over the value of parameter FOV RL (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective	The maximum value of the range shown in the tooltip is at least 520 as specified in the requirement.

Step	Description	Expected Result
	evidence.	
Step 7	Hover the mouse pointer over the value of parameter FOV FH (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 520 as specified in the requirement.

3.1.19.10. SWCMP.Scan.Methods.Constraints.FOV.NarrowBore2.tc1

Test ID	25250
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2015-10-15 17:54:41 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-10-15 17:37:51 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-10-15 17:36:21 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
57207	SWCMP.Scan.Methods.Constraints.FOV.NarrowBore2	The maximum allowed FOV shall be limited by the hardware capabilities and available options. For the configuration(s) specified in supported Product Model/Field Strength combinations given in the details of this requirement the system shall set the maximum allowed FOV as follows: The upper limit of FOV in AP direction shall be at least 490 mm The upper limit of FOV in LR direction shall be at least 490 mm The upper limit of FOV in FH direction shall be at least 490 mm

Test steps:

Step	Description	Expected Result
Step 1	Install appropriate SW in test environment	N/A. preparatory step.
Step 2	Set system configuration to the supported configuration that is specified by the requirement.	N/A. preparatory step.
Step 3	Start Application software or ExamCards test environment and create a new exam.	N/A. preparatory step.

Step	Description	Expected Result
Step 4	Select a protocol from the Philips database, open the protocol in editor and go to the Geometry tab of the protocol editor.	N/A. preparatory step.
Step 5	Hover the mouse pointer over the value of parameter FOV AP (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 500 as specified in the requirement.
Step 6	Hover the mouse pointer over the value of parameter FOV RL (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 500 as specified in the requirement.
Step 7	Hover the mouse pointer over the value of parameter FOV FH (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 500 as specified in the requirement.

3.1.19.11. SWCMP.Scan.Methods.Constraints.FOV.Offcenter.tc1

Test ID	25252
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2015-10-15 17:54:42 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-10-15 17:37:53 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-10-15 17:36:22 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
57209	SWCMP.Scan.Methods.Constraints.FOV.Offcenter	The maximum allowed Offcenter shall be limited by the hardware capabilities and available options. For the configuration(s) specified in supported Product Model/Field Strength combinations given in the details of this requirement the system shall set Offcenter range as follows: The lower limit of the AP Offcenter shall be -160 mm (± 10 mm) and the upper limit of AP Offcenter shall be +160 mm (± 10 mm). The lower limit of the LR Offcenter shall be -220 mm (± 10 mm) and the

Req ID	Requirement Tag	Description
		upper limit of LR Offcenter shall be +220 mm (± 10 mm).

Test steps:

Step	Description	Expected Result
Step 1	Install appropriate SW in test environment	N/A. preparatory step.
Step 2	Set system configuration to the supported configuration that is specified by the requirement.	N/A. preparatory step.
Step 3	Start Application software or ExamCards test environment and create a new exam.	N/A. preparatory step.
Step 4	Select a protocol from the Philips database, open the protocol in editor and go to the Offc/Ang tab of the protocol editor.	N/A. preparatory step.
Step 5	Hover the mouse pointer over the value of parameter Stack Offc. AP (P =+mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The lower limit of the AP Offcenter is -160 mm (± 10 mm) and the upper limit of AP Offcenter is +160 mm (± 10 mm) as specified in the requirement.
Step 6	Hover the mouse pointer over the value of parameter Stack Offc. RL (L =+mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The lower limit of the RL Offcenter is -220 mm (± 10 mm) and the upper limit of RL Offcenter is +220 mm (± 10 mm) as specified in the requirement.

3.1.19.12. SWCMP.Scan.Methods.Constraints.FOV.WideBore1.tc1

Test ID	25251
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2015-10-15 17:54:46 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-10-15 17:37:55 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-10-15 17:36:24 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
57208	SWCMP.Scan.Methods.Constraints.FOV.WideBore1	The maximum allowed FOV shall be limited by the hardware capabilities and available options. For the configuration(s) specified in supported Product Model/Field Strength combinations given in the details of this requirement the system shall set the maximum allowed FOV as follows: The upper limit of FOV in AP direction shall be at least 540 mm The upper limit of FOV in LR direction shall be at least 540 mm The upper limit of FOV in FH direction shall be at least 470 mm

Test steps:

Step	Description	Expected Result
Step 1	Install appropriate SW in test environment	N/A. preparatory step.
Step 2	Set system configuration to the supported configuration that is specified by the requirement.	N/A. preparatory step.
Step 3	Start Application software or ExamCards test environment and create a new exam.	N/A. preparatory step.
Step 4	Select a protocol from the Philips database, open the protocol in editor and go to the Geometry tab of the protocol editor.	N/A. preparatory step.
Step 5	Hover the mouse pointer over the value of parameter FOV AP (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 540 as specified in the requirement.
Step 6	Hover the mouse pointer over the value of parameter FOV RL (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 540 as specified in the requirement.
Step 7	Hover the mouse pointer over the value of parameter FOV FH (mm) to make the tool tip visible. Make a screenshot while tooltip is visible and attach to step result as objective evidence.	The maximum value of the range shown in the tooltip is at least 470 as specified in the requirement.

3.1.19.13. SWCMP.Scan.Methods.CryoCooler.tc1.v1

Test ID	32300
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-22 12:17:03 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2017-03-22 10:32:53 Meaning: SME Approved - (Business)

	Signed by: Graaf, Ruud de (nly95749) on 2017-03-20 11:12:28 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Code review of Switching off the CryoCooler

Requirements Coverage:

Req ID	Requirement Tag	Description
21042	SWCMP.Scan.Methods.CryoCooler	The cryo-cooler should not be switched off at all. However, due to IQ deterioration as result of induced B0 fluctuations, the cryo-cooler may be switched off for the following scans conditions: FFE-SSH-EPI with ESP > 40ms FFE-TFE with: TE > 5 && TR > 40ms FFE-MSH-EPI with TE > 5 && TR > 40

Test steps:

Step	Description	Expected Result
Step 1	Open file methpdf\src\mpicscdb.res	Verify that CSC_GEN_cryo_shutoff_mode is set to "MGUCSC_CRYO_SEQ_DEPENDENT2" for all supported configurations.
Step 2	Open file methpdf\src\mpudef_comb.cpp	Verify that the value of MPF->switch_cryo_off is set to 'No' for all supported configurations, given the default value for CSC_GEN_cryo_shutoff as obtained from the previous step, with the following exceptions. MPF->switch_cryo_off is set to 'Yes' • for FFE-SSH-EPI with ESP > 40ms • for FFE or TFE with TE >= 5 && TR >= 40ms • for FFE-MSEPI with TE >= 5 && TR >= 40ms

3.1.19.14. SWCMP.Scan.Methods.Dynamics.Geo-updates.tc1

Test ID	9971
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

BOLD and Prospective Motion Correction

This test can be performed on any system running the complete application software.

Purpose

This test is used to check the methods implementation of BOLD and Prospective Motion Correction. It tests:

- - o the interface of geometry updates
 - o effects on sequence timing
- - o

Preparations:

Use a ROF/RTM configured as WA_3T.

Or VMWARE: Memory 4Gbytes Processors 4 recsim configured to use 3Gbytes and WA_3T and G:

\systemstartup\s_sys*.ini renamed.

Startup the complete application.

Requirements Coverage:

Req ID	Requirement Tag	Description
21086	SWCMP.Scan.Methods.Dynamic s.Geo-updates	The scan protocol shall accept and process geometry updates for dynamic scans during scanning on behalf of Prospective Motion Correction.

Test steps:

Step	Description	Expected Result
Step 1	Open Application SW. Plug in the NVC_BASE and NVC_HEAD coils. Import the ExamCard m_imn_bold_test.examcard stored in mrmetho\stst. Perform all scans. These scans have settings that test the ranges of BOLD scanning.	Scans succeed.
Step 2	Make a copy of the scan FFE_EPI. Edit the protocol to switch off prospective motion correction (In tab Dyn/Ang, set 'prospect. motion corr.' to 'no').	No conflicts occur and the timing (TE/TR) of the scan does not change (info tab).
Step 3	Make a copy of the scan T1W_PRESTO. Edit the protocol: set 'Dynamic study' to 'individual'; 'dyn scans' to 30; 'prospect. motion corr.' to 'yes'.	Conflict occurs with 'Reconstructon mode' 'immediate'.
Step 4	Change 'Reconstruction mode' from 'immediate' to 'real time'. Press accept.	Conflict disappears.
Step 5	Press Start Scan	All scans complete.

3.1.19.15. SWCMP.Scan.Methods.ExtendTableOffsetLimit.tc1

Test ID	15905
Revision	1
Product Model	Ingenia;Marlin
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic

	signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

The location of the script is \system\coils\spt\sptdatabase\res\

Requirements Coverage:

Req ID	Requirement Tag	Description
35547	SWCMP.Scan.Methods.ExtendTableOffsetLimit	- It shall be possible to extend the maximum allowed table offset up to 220mm via a control scan parameter setting in the batch interpreter. - This control parameter shall not be visible to the operator in the control scan parameter editor.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Configure system for MR-RT in FSF. (Open Scanner Hardware Configuration in FSF and select laser positioning system as DORADOnova)	N.A. (preparation)	 DORADONova.zip
Step 2	Open Geometric_Fidelity_QA (sp_rt_wa00gc_geometric_fidelity_qa.acq) in notepad and locate line containing text \$ scan 1,0,"geometric_fidelity_m200_c",wait Remove lines between the above line and the line containing text \$ startup_init	Script shall be editable in notepad and it shall be possible to remove contents from it.	None
Step 3	Start Application Software. Open Geometric_Fidelity_QA (sp_rt_wa00gc_geometric_fidelity_qa.acq) script from System Performance Tool under SPT UI. Verify that this script will set the control parameter CSC_BI_remove_table_offset_limit = YES and will also move the table to -200mm.	The script shall contain the commands: \$ modify_n UCSC_all_pars CSC_BI_remove_table_offset_limit = YES; and \$ move -200	 SPT.zip
Step 4	Run the script.	The script shall complete the scan successfully.	None
Step 5	Open the generated MIP scan in ImageView, Increase Image Information Annotation to Maximum and check the FH offset	The displayed FH offset shall be -200.	 CtrlParamYes.zip
Step 6	1. Open the script Geometric_Fidelity_QA (sp_rt_wa00gc_geometric_fidelity_qa.acq) and edit scan parameter CSC_BI_remove_table_offset_limit = NO	The script shall complete the scan successfully.	None

Step	Description	Expected Result	Attachments
Step 7	2. Run the modified script Open the generated MIP scan in ImageView, Increase Image Information Annotation to Maximum and check the FH offset	The displayed FH offset shall not be -200 but -160.	 CtrlParamNo.zip

3.1.19.16. SWCMP.Scan.Methods.NSA.PartialNSA.Pars.tc1.v2

Test ID	32420
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-22 16:58:49 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-03-22 16:47:47 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on 2017-03-22 09:23:21 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result
Prep	Change the configuration of the RTM to a WA30 (Ingenia 3.0T). Disable the development key. Load the TEDB. Start a new logfile. Load: 'Head CNS -> TSE -> F_TSE_basic' Change: NSA to 1.5.	Preparation successfull.
Step 1	Verify that the following settings are not allowed in combination with a Partial NSA. The protocol will be automatically modified. • Scan mode = 3D • Fast Imaging mode = GraSE and EPI • kt=BLAST = YES. • Acquisition mode = radial, spiral or MultiVane. • Halfscan = Yes or default. Attach the devlogcurrent.log file to this step as onbjective evidence.	When changing the parameter settings to these settings, the protocol will be automatically modified to disable the partial NSA, or a conflict is given.

3.1.19.17. SWCMP.Scan.Methods.NSA.PartialNSA.Pars.tc2

Test ID	15413
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result
Step 1	Change the configuration of the RTM to a WA30 (Ingenia 3.0T). Disable the development key. Load the TEDB. Load: 'Head CNS -> TSE -> F_TSE_basic' Change: NSA, succesively from 1.1 to 10.	Check for the following behavior of NSA parameter: 1. NSA parameter should increment in minimum 0.1 increases from 1.1 until 5.0. Beyond 5.0, NSA should increment in 1.0 increases.

3.1.19.18. SWCMP.Scan.Methods.ProtocolPars.LocScanOrder.tc1.v1

Test ID	30556
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-11-25 10:34:27 Meaning: QA Approved - (QA) Signed by: Crijns, Sjoerd (310257923) on 2016-11-21 11:55:53 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2016-11-21 11:35:27 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
56160	SWCMP.Scan.Methods.Protocol Pars.LocScanOrder	The scan protocol shall be able to specify the order in which scan volumes must be scanned. With "scan volumes" is meant: - Slices (2D volumes) in a multi-slice scan, or - Chunks (3D volumes) in a 3D scan with multi-chunk = yes in the scan protocol.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Run this test on an RTM configured as Ingenia 3.0T system. - Set default slice numbering directions (System -> Image Display Settings, Unlock to customize settings (password "Philips"), Reset, Save). - Add following line of text to the file "C:\Temp\bstrace.dat" to enable trace logging needed for this test: MP* ts_loc_scan_order * - Create a new examination. - Import attached ExamCards "slice and chunk scan order ascend.ExamCard" and "slice and chunk scan order descend.ExamCard"	Preparations shall be completed without errors.	 slice and chunk scan order ascend.z  slice and chunk scan order descend.
Veri 1	- Start a new log file. - Load ExamCard "slice and chunk scan order ascend.ExamCard". - Delete the loaded ExamCard from the Scanlist. - Search log file devlogcurrent.log for "LOC_SCAN_ORDER".	The search result in log file devlogcurrent.log shall contain the same lines of text as in attachment "slice and chunk scan order ascend.txt". Objective evidence: Snapshot of search result in log file devlogcurrent.log.	 slice and chunk scan order ascend.z
Veri 2	- Start a new log file. - Load ExamCard "slice and chunk scan order descend.ExamCard". - Delete the loaded ExamCard from the Scanlist. - Search log file devlogcurrent.log for "LOC_SCAN_ORDER".	The search result in log file devlogcurrent.log shall contain the same lines of text as in attachment "slice and chunk scan order descend.txt". Objective evidence: Snapshot of search result in log file devlogcurrent.log.	 slice and chunk scan order descend.
Prep 2	Set reversed slice numbering directions for all orientations (System -> Image Display Settings, Unlock to customize settings (password "Philips"), set R -> L for sagittal, P -> A for Coronal, H -> F for Transverse, Save).	Preparations shall be completed without errors.	None
Veri 3	- Start a new log file. - Load ExamCard "slice and chunk scan order ascend.ExamCard". - Delete the loaded ExamCard from the Scanlist. - Search log file devlogcurrent.log for "LOC_SCAN_ORDER".	The search result in log file devlogcurrent.log shall contain the same lines of text as in attachment "slice and chunk scan order ascend.txt" in step Veri 1. Objective evidence: Snapshot of search result in log file devlogcurrent.log.	None
Veri 4	- Start a new log file. - Load ExamCard "slice and chunk scan order descend.ExamCard".	The search result in log file devlogcurrent.log shall contain the same	None

Step	Description	Expected Result	Attachments
	order descend.ExamCard". - Delete the loaded ExamCard from the Scanlist. - Search log file devlogcurrent.log for "LOC_SCAN_ORDER".	lines of text as in attachment "slice and chunk scan order descend.txt" in step Veri 2. Objective evidence: Snapshot of search result in log file devlogcurrent.log.	
Post 1	- Set default slice numbering directions (System -> Image Display Settings, Unlock to customize settings (password "Philips"), Reset, Save). - Remove the line of text that was added to the file "C:\Temp\bstrace.dat" in step Prep 1.	Cleanup actions shall be completed without errors.	None

3.1.19.19. SWCMP.Scan.Methods.RESPSensor.NoBreathingDetected.tc1.v1

Test ID	39087
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2018-11-30 07:28:19 Meaning: QA Approved - (QA) Signed by: Mariappan, Yogesh kannan (310134237) on 2018-11-30 04:25:31 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2018-11-29 12:13:41 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test case invokes the test script from the "Testframework". It is automated to

1. Execute an acq file on Batch Interpreter.
2. Extract the relevant log lines from devlogcurrent.log into the file LOGFILE_EXTRACT.log.
3. Compare the resulting LOGFILE_EXTRACT.log with the reference file respSensor-nobreathing.ref for resulting differences
4. Generate a report that contains the differences found. In case of no differences, the report mentions that there are no differences found.

Requirements Coverage:

Req ID	Requirement Tag	Description
72891	SWCMP.Scan.Methods.RESPSensor.NoBreathingDetected.v1	The measurement shall notify the message "Scanning without respiratory control (no signal during last 20 seconds)" when respiratory breathing cannot be detected for 20 seconds for the respiratory compensation techniques - Trigger and Gating.

Req ID	Requirement Tag	Description
		Note: The measurement continues for the respiratory compensation techniques - PEAR and Gating. The Triggering technique halts the measurement till 20s after which the scan continues. For technique Breath-hold the sensor is not required for the measurement.

Test steps:

Step	Description	Expected Result
Step 1	- On an test machine (Workstation / RTM) open a command tool. - Give command: tf - Go to "Testing: Component" - Choose "No breathing detected"	The test script executes successfully and the output files are produced on G: <code>\\log\testframework\respSensor_noBreathing</code> <code>\\<STREAM>\\<SWID>\\<DATETIME></code>
Step 2	In the location of the output file, open "<code>respSensor_noBreathing_detailed_report.txt</code>" Attach the file "<code>respSensor_noBreathing_detailed_report.txt</code>" as objective evidence.	No differences are reported in the output file. The file contains the message " <i>No differences found</i> ".

3.1.19.20. SWCMP.Scan.Methods.RESPSensor.NotConnected.tc2.v1

Test ID	28635
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2018-11-30 07:29:05 Meaning: QA Approved - (QA) Signed by: Mariappan, Yogesh kannan (310134237) on 2018-11-30 04:28:49 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2018-11-29 12:15:21 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

When Physiology simulation is active, it simulates the condition that respiratory sensor is connected. When it is switched off this corresponds to no respiratory sensor connected.

This test case invokes the test script from the "*Testframework*". It is automated to

1. Execute an acq file on Batch Interpreter.
2. Extract the relevant log lines from devlogcurrent.log into the file *LOGFILE_EXTRACT.log*

3. Compare the resulting *LOGFILE_EXTRACT.log* with the reference file *respSensor-notConnected.ref* for resulting differences
4. Generate a report that contains the differences found. In case of no differences, the report mentions that there are no differences found.

Requirements Coverage:

Req ID	Requirement Tag	Description
59502	SWCMP.Scan.Methods.RESPSensor.NotConnected	The measurement shall aborts and notify the message "No respiratory sensor detected" , when no respiratory sensor is connected and one of the following respiratory compensation techniques is used: - Triggering, - PEAR - Gating. Note: for technique Breath-hold the sensor is not required for the measurement.

Test steps:

Step	Description	Expected Result
Step1	• On a test machine (Workstation \ RTM) open a command tool. • Give command: tf • Go to "Testing: Component" • Choose "Resp sensor not connected"	The script runs successfully and the output files are produced in G:\log\testframework\respSensor_notConnected\ <STREAM>\<SWID>\<DATETIME>
Step 2	In the location of the output files, open <i>respSensor_notConnected_detailed_report.txt</i> Attach <i>respSensor_notConnected_detailed_report.txt</i> as objective evidence.	No diiferences are reported in the output file. The file contains the message "No differences found"

3.1.19.21. SWCMP.Scan.Methods.ScanInfo.TableMoveSpeed.tc1.v2

Test ID	32573
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 16:57:29 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 15:56:33 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on

	2017-03-22 16:09:12 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Table movement is used within examcards. MR-Methods has to derive from a scan step if high tabletop movement speed must be used. This test is integrated in the examcard m_imn_ce.examcard which specifically perform a basic contrast enhanced study, consisting of one scanset containing.

1. CE_LowerLegl first dynamic
2. CE_UpperLeg first dynamic
3. BolusTRack
4. CE_UpperLeg second dynamic
5. CE_LowerLeg second dynamic.

The software is instrumented with logging that will create TableMovement requests

The logfile is filtered on these requests and compared against a reviewed/ expected reference file.

When both are equal the test is succeeded.

Requirements Coverage:

Req ID	Requirement Tag	Description
25480	SWCMP.Scan.Methods.ScanInfo .TableMoveSpeed	An interface shall be provided to specify the requested Table Movement Speed for a scan step. The tabletop movement speed may only be high (>180mm/s) for a resumed dynamic and a Contrast Enhanced scan (BolusTrack).

Test steps:

Step	Description	Expected Result
Prep	Use the m_amn_set_test_environment.pl script to configure the system as a WA30 Ingenia DDAS system.	Preparation successfull.
Step 1	Open a command prompt. Run C:\nmr\gyroscan\testbatch\m_amn_table.pl. If the test is passed, in the command prompt: "Table Test: Succeeded" will be shown, otherwise it will show "Table Test: Failed". Add the table test reference file 'm_amn_table.ref' which can be found in G:\log to this step as objective evidence.	The test completes, and the command prompt shows: "Table Test: Succeeded".

3.1.19.22. SWCMP.Scan.Methods.SDM.tc1.v2

Test ID	52734
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharaappa, Deepak Shekhar (320098697) on 2021-12-16 14:25:04 Meaning: QA Approved - (QA)

	Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:02:29 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:25 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies Sequence Development Mode (SDM).

Requirements Coverage:

Req ID	Requirement Tag	Description
21038	SWCMP.Scan.Methods.SDM	An interface shall be provided that allows an operator with design choice knowledge to view the generated measurement and recon descriptions and to make (minor) modifications. This is called Sequence Development Mode (SDM).

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanMethodsSdm in the results file is zero.

3.1.19.23. SWCMP.Scan.Methods.ShimApproved.tc1.v1

Test ID	58145
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:50:54 Meaning: QA Approved - (QA)
	Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:14:50 Meaning: SME Approved - (Technical)
	Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11- 15 11:05:50 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verify that only RF shimming-approved coils are used with RF shimming

Test with a single coil connected

This test will verify that a RF shimming can only be applied if all the coils used in the protocol and / or present in the scanner are RF shimming-approved. This test can be performed in the VM (Ingenia CX 3.0T configuration).

Requirements Coverage:

Req ID	Requirement Tag	Description
21096	SWCMP.Scan.Methods.ShimApproved	Shimming may only be used when no coils are connected by the user that are not shim approved

Test steps:

Step	Description	Expected Result	Attachments
Step 1	1. Configure a VM with Ingenia CX 3.0T configuration. 2. Select the TEDB database 3. Use the service application to make sure that all the coils are enabled 4. Start the application software 5. Create a new patient 6. Select the following protocol: "Hospital/ Pelvis/Multitransmit/T2W_SPAIR_shim med" and drag it into the execution list. 7. Simulate the presence of Anterior coils. 8. Modify the protocol in order to work in "Multi-transmit" mode. And modify it to smart select. 9. Ensure RF shimming is set to Adaptive.	Verify that no conflict appears.	None
Step 2	Open System tuning and set RF shimming approved to no for the anterior coil. (Refer Fig1 Attached)	Verify that a conflict appears (see Fig2 attached).	 Fig1_RF_shim.zip  Fig2_Conflict.zip

3.1.19.24. SWCMP.Scan.Methods.TunedHWPars.tc1.v1

Test ID	32251
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-16 11:43:56 Meaning: QA Approved - (QA) Signed by: Polders, Daniel Louis (310171444) on 2017-03-16 10:01:03 Meaning: SME Approved - (Business)

	Signed by: Graaf, Ruud de (nly95749) on 2017-03-16 09:42:37 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Verify that the tuned HW parameters are used.

Requirements Coverage:

Req ID	Requirement Tag	Description
21046	SWCMP.Scan.Methods.TunedHWPars	MR-Methods shall use the tuned. Results of system adjustments.

Test steps:

Step	Description	Expected Result
Step 1	Code review: 1. Open mpghw.cpp in a editor 2. Verify that tuned hardware parameters are read after the fixed HW parameters in method MPGHW_init_arch.	Tuned hardware parameters are read after fixed HW parameters.

3.1.19.25. SWCMP.Scan.Methods.Unific.tc1

Test ID	35602
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-09-21 08:44:12 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2017-09-11 16:24:23 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-09-08 15:08:29 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test case checks the parameters for the uniformity correction filter, which depend on the body part scanned.

Test steps:

Step	Description	Expected Result
Prep 1	1. Start VM and run p_apco_simconfig.pl DDAS -a in a command window. 2. Select the configuration WA30. 3. Select Start ==>MR-DEV Configurations==>Scan Protocol Database Selection and select tedb 4. Select Start==>MR-DEV Scan Simulations==>Scan Simulations==> Simulation mode (Acq on Host, Recon on Host). 5. Start software in the MR Boot Configuration Manager 6. Create a new logfile using p_logfile_nu createnewlogfile 7. Select Start==>MR-DEV Scan Simulations==>Coilstate Simulator UI 8. Select Start==>MR-DEV Scan Simulations==>Start Graphical Patsup Simulation	N.A. Prep steps.
Step 2	1. Plug in dS Head into SBM.DCC0 in the device simulator. 2. Move the patient table to position the light visor for the head position and press light visor.	Confirm that info tab shows 'Head SAR'
Step 3	1. Open a protocol Hospital/Head CNS/FE/F_2D1 2. On PostProc tab, change Uniformity correction to yes.	Observe no conflicts
Step 4	Open Sequence Development Viewer, on the parameters tab examine the following parameters: VAL01__PROC_unific_filter VAL03__PROC_unific_mode RC_unific_correction_strength RC_unific_h_reg_factor RC_unific_regularity RC_unific_biasfield_max	The value of the parameters are as follows: VAL01__PROC_unific_filter = yes VAL03__PROC_unific_mode = MPUPROC_UNIFIC_BRAIN RC_unific_correction_strength = 17 RC_unific_h_reg_factor = 9 RC_unific_regularity = 1.0 RC_unific_biasfield_max = 1.65
Step 5	1. Plug in dS Anterior coil into SBM.DCC0 in the device simulator. 2. Move the patient table to position the light visor for the torso position and press light visor.	Confirm that info tab shows 'Local Torso SAR'.
Step 6	Use the same sequence as above, F_2D1 with the uniformity correction set to Yes.	No conflicts observed on the protocol.
Step 7	In the Sequence Development Viewer, on the parameters tab examine the following parameters: VAL01__PROC_unific_filter VAL03__PROC_unific_mode RC_unific_correction_strength RC_unific_h_reg_factor RC_unific_regularity RC_unific_biasfield_max	The value of the parameters are as follows: VAL01__PROC_unific_filter = yes VAL03__PROC_unific_mode = MPUPROC_UNIFIC_BODY RC_unific_correction_strength = 17 RC_unific_h_reg_factor = 9 RC_unific_regularity = 0.75 RC_unific_biasfield_max = 1.8
Step 8	In Postproc tab set the reconstruction mode to 'on the fly'.	Conflict should be observed between reconstruction mode and uniformity correction.

3.1.19.26. SWCMP.Scan.ProtocolPars.tc1.v1

Test ID	47841
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.

Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-08-27 10:17:57 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2020-08-27 09:52:24 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2020-08-26 15:50:58 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Purpose

Verify the UI behavior and interfaces of scan protocol pars. The test is performed on VMWARE.

Requirements Coverage:

Req ID	Requirement Tag	Description
21009	SWCMP.Scan.Methods.Protocol Pars.Stacks	The scan protocol shall be able to specify more than one scan volume to be scanned. This enables optimizing strategies like interleaved scanning of different scan volumes, so while spins in one volume are relaxing the spins in another scan volume can be scanned.
21056	SWCMP.Scan.Methods.IR.Pars	The scan protocol shall contain a parameter that specifies the inversion delay with a range of (10ms – 5 sec).
21054	SWCMP.Scan.Methods.SE.Pars	The scan protocol shall contain parameters to specify the following properties of SE: 1. Number of echoes for 2D/3D (1-8) and for MS (1, 2) 2. Min. TE of train of echoes (N*10 ms) 3. TE 1st echo (3 – 50 ms) 4. TE 2nd echo (50 - 200 ms) 5. TR for SE (100ms – 100 sec) and for IR (250ms -50sec) 6. Image types modulus, real, imaginary, phase
21013	SWCMP.Scan.Methods.Constraints.HW	Hardware capabilities shall be derived from the configuration information and MR pulse sequences shall be defined that do not exceed these hardware capabilities.
21014	SWCMP.Scan.Methods.Constraints.HW.GradientSystem	MR pulse sequences are derived that do not exceed the gradient capabilities. The gradient capabilities are expressed in: amplifier limits, max strength, max slew rate and gradient duty cycle.
21015	SWCMP.Scan.Methods.Constraints.AcousticNoise	MR pulse sequences shall be optimized in order to meet the acoustic noise exposure constraints as identified by the protocol.
26842	SWCMP.Scan.Methods.Constraints.PNS	For every sequence, the Methods software shall evaluate PNS against established limits for the system configuration, and optimize/adapt the sequence in order to prevent exceeding the threshold of intolerable PNS.
21022	SWCMP.Scan.Methods.Conflict.Check	The scan protocol shall be validated. The following items will be checked: 1. the MR pulse sequence execution timing 2. the available software options 3. the available / connected hardware

Req ID	Requirement Tag	Description
		- incompatible features - Exposure limits
21023	SWCMP.Scan.Methods.Conflict.Pars	An interface is provided to pass the set of conflicting scan protocol parameters to the UI. (the conflict can be solved by modifying scan protocol parameters in this set).
21026	SWCMP.Scan.Methods.Conflict.UserGuidance	For a number of cases, described in this document and companion documents, the validate function shall return, in case of a conflict, suggestions to solve the conflict. The suggestions consists of: - Instruction for the UI telling what has to be displayed. - Scan Protocol Parameter modifications that must be applied when a certain suggestion is selected
21027	SWCMP.Scan.Methods.Enabled Pars	An interface shall be provided that specifies the set of protocol parameters that is modifiable.
21028	SWCMP.Scan.Methods.ParameterRange	An interface shall be provided that to specify the range of a scan protocol parameter.
26843	SWCMP.Scan.Methods.ScanInfo.SED	SED represents the RF energy deposition a patient is exposed to during a scan and is given in kJ/kg. Predicted value of SED is calculated as the product of the predicted value of the whole body SAR and the predicted scan duration. For dynamic scans the value is calculated as the product of duration of one dynamic step, SAR of one dynamic step and number of total dynamics.
26844	SWCMP.Scan.Methods.ScanInfo.PNS	For every sequence, the Methods software shall evaluate PNS against established limits for the system configuration and display on the info page in % w.r.t. PNS threshold.
26845	SWCMP.Scan.Methods.ScanInfo.maxB1rms	Predicted value of maximum B1rms shall be calculated by evaluating B1rms over any 10 s interval of the scan during sequence definition.
21053	SWCMP.Scan.Methods.SE	A scan protocol shall be able to perform the SE scanning technique.
21055	SWCMP.Scan.Methods.IR	Inversion Recovery shall be possible for the following scan techniques: SE, TSE and EPI
21076	SWCMP.Scan.Methods.InteractiveF0.protocol	The scan protocol shall contain a parameter that enables manual F0 selection based on the measured F0 response.
21077	SWCMP.Scan.Methods.InteractiveF0.Execution	An interface shall be provided to instruct the scanner to measure the F0 response and wait for the manual selection of the F0.

Test steps:

Step	Description	Expected Result
Prep1	Preparation: - Install the RTM (or STM) machine - Login as gyrotest. Open command window and start "p_apco_simconfig.pl -a" - Select the configuration "WA30". - Select tedb by using Start==>MR Development==>Configuration==> "Scan Protocol Database Selection". - Select Start==>MR_Development==>Configuration==> "Install Test Examcards".	N.A. (preparation)

Step	Description	Expected Result
	6. Remove/rename the file G: \systemstartup\s_sys_SystemStartupProperties.ini 7. Restart the machine and login as gyrotest 8. Select Start==>MR_Development==>Scan Simulations==> "Simulation mode (Acq on Host, Recon on Host)". 9. Start "Application Software" in the MR Boot Configuration Manager	
Veri 1	Select examcard "Hospital->test->int_3t_tx " Perform Exam and select survey and Advanced parameter editor	Verify that the patient weight parameter is displayed (prep mode).
Veri 2	Select 'New Patient , Enter mandatory patient details select test examcard "int_3t_tx_wa". and Perform Exam	No conflicts in the examcard.
Veri 3	select survey and open geometry tab verify that you see the geometry parameters FOV FH/AP, stacks slices etc...	Geometry parameters are available. Stacks : 3
Veri 4	Open contrast tab Verify range scan mode (2D,3D,MS,M2D) Verify range technique (SE, IR, MIX, FFE)	Verify that the contrast parameters scan mode and scan technique are displayed and that their ranges are as expected.
Veri 5	Close and select protocol <<3D10_FFE>> TR is set to a 'shortest'. Verify effect of changing SAR mode (parameter is on the contrast page). Initially, it is set to 'high'. 1. Set SAR mode to 'moderate'. Verify on info page that scan dur increased and SAR < ~75% [73-75] 2. Set SAR mode to 'low'. Verify on info page that scan dur increased and SAR < ~50% [48-50] 3. Set SAR mode to 'ultra low'. Verify on info page that scan dur increased and SAR < ~25% [23-25] 4. Set SAR mode to 'user defined' and set the level to 65%. Verify on info page that scan dur is decreased (relative to 3.) and the SAR < ~65% [63-65]	1. SAR < [73-75] 2. SAR < [48-50] 3. SAR < [23-25] 4. SAR < [63-65]
Veri 6	Close and select protocol <<3D10_FFE>> Set TR to 'shortest' and NSA to 50. (Set Multitransmit to YES, if a Non-WA TX system configuration is chosen) Make sure that no coils are connected in Coilstate Simulator. (For an iRF system posterior coil is required to be available) Verify the effect of changing B1 mode <parameter on contrast page> 1. Set B1 Mode to 'user defined' and select the maximum B1 (13.5) 2. Select maximum B1 (8.0) 3. Select maximum B1 (4.0)	1. Local Torso SAR < 100%. Note the values of Scan time, Whole Body SAR and SED. The value of SED must be approximately* equal to Scan time (in seconds) x Whole body SAR / 1000.0. 2. Local Torso SAR < 100%. The value of Whole Body SAR remains the same and Scan time increases. The value of SED must be approximately* equal to Scan time (in seconds) x Whole body SAR / 1000.0. 3. Local Torso SAR < 52%. The value of Whole Body SAR decreases by about 50% and Scan time increases. The value of SED must be approximately* equal to Scan time (in seconds) x Whole body SAR / 1000.0. *Due to limited precision of displayed values of SAR and SED, the displayed value of SED can deviate up to 10% from the calculated value.
Veri 7	Close and select protocol <<2DSI_SENSE>>	The maximum B1+rms should be displayed in InfoPage.

Step	Description	Expected Result
	Check that maximum B1+rms is displayed in InfoPage.	
Veri 8	Close and select protocol <<3D10_FFE>> Set TR to 50 ms. Verify SoftTone (Silent) mode <parameter on contrast page> 1. Verify SoftTone mode = 'no' and verify Sound Pressure Level 2. Change SoftTone mode = 'yes' and verify Sound Pressure Level 3. Change SoftTone mode = 'user defined' factor 3.0 and verify Sound PressureLevel	Within 0.5 dB, values should be equal to 1. Note the Sound Pressure Level. 2. Sound Pressure Level is decreased. 3. Sound Pressure Level is decreased. (SPL may have a negative value).
Veri 9	Press close > and Press Start Scan Select all not executed scans in Exam Overview <ctrl left mouse button>> and press keyboard <delete>.	Survey is completed without abort. Exam Overview shall be displayed with one executed scan.
Prep 2	Browse in protocol database to select protocol "Head CNS->DIFFUSION->a6_DW/bperf2" with drag and drop. Clear geometry name <<Geo1>> Remove tick mark 'Smart Select' (Q-body is selected)	N.a.
Veri 10	Start Advanced parameter editor and press <<contrast>> Verify PNS mode <parameter on the contrast page> 1. Verify PNS mode = moderate and verify info page PNS level 2. Change PNS mode = high and verify info page PNS level 3. Change PNS mode = low and verify info page PNS level	Within 5%, percentage values should be equal to 1. PNS/Level 80% / normal 2. PNS/Level 89% / 1st level 3. PNS/Level 60% / normal
Veri 11	Select << coils >> tab. Tick mark 'Smart Select'.	Posterior coil is now selected. No conflicts are shown.
Veri 12	Select <<conflicts>> tab.	No conflicts shown; Info pars (e.g. Rel. SNR) are visible.
Veri 13	Select <<postproc>> tab. Change "Interactive F0" to "select". Press <<proceed>> and press <Start Scan>>. UI should display:	The window with graph interactive F0 determination is displayed.
Veri 14	Press <<proceed>> in datamonitoring window.	Scan finishes
Veri 15	Purpose Verify SE parameters and IR parameters 1. Select protocol Head CNS/F-3D-SCANS/ F-3D-SE. 2. Select <contrast> tab and change nr echoes to 2. 3. Verify info page (i.e., <<Conflicts>> tab). 3. Change Multi (ms) 10.0, a conflict occurs. Try with key <<- >> to decrease the value. The value should not decrease. (There are conflicts) 4. Change scan mode in MS: Conflict disappears and info page shows info parameters. 5. Try to change 'TE second' by entering 49. 6. Enter 201 7. Change scan technique to IR (conflict occurs). Go to the conflict page and remove conflict with "Fold over suppression" by selecting no. 8. Change IR delay to 9 ms. 9. Change IR delay to 5001 ms	Verify that 1. No Conflicts displayed 2. Act TR: 171. Act TE: 30/60 3. Min selectable Multi (Ms) is 10ms. 4. Act TE 10/100 5. Value is set to 50 6. Value is set to 200 7. IR delay is displayed with value 400 8. Conflict and IR delay set to 10 9. No conflict; IR delay set to 5000; scan duration more than 50 minutes.

3.1.20. ME Mobi

3.1.20.1. SWCMP.Scan.Mobi.Shutter.tc1

Test ID	7391
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

MobiTrak/MobiFlex: Modified reconstruction shutter

Verify the large reconstruction shutter for MobiTrak scans

For **cylindrical** systems, a modified shape of the reconstruction shutter has been implemented to enlarge the viewable area of MobiTrak or MobiFlex images. This 'large shutter' is the cross-section of an increased size ellipsoidal shutter and a box shaped shutter to cut off parts of the images which correspond to inhomogeneous regions at large FH offcentre positions. The combined shape is an ellipsoid with two flat faces at the F and H sides. The 'large shutter' is used for all multi-stack scans with 'table movement per stack' set to 'yes' or GeoLink'ed scans in an ExamCard.

Requirements Coverage:

Req ID	Requirement Tag	Description
17571	SWCMP.Scan.Mobi.Shutter	MobiTrak scans have an adapted image shutter to minimize loss of information in regions between the stations (stacks), as specified in chapter Modified reconstruction shutter for MobiTrak and MobiFlex.
21021	SWCMP.Scan.Methods.ExamProperties	The validation of a scan protocol shall include exam properties. Exam properties consists of context information such as patient position, patient orientation, table position, connected coils and information required for defining and approving a set of scans with inherent strict inter-scan timing relationships.

Test steps:

Step	Description	Expected Result
Step 1	Select 'mobi 3d' with 'stack alignment' FH and table movement. Modify to the largest possible field of view and coronal slice orientation. With current stack B, introduce an AP angulation of 10 degrees. Adapt the (AP) offcentres to the position of your phantom but avoid range conflicts for the offcentres. Perform the scan.	Verify that the reconstruction shutter as described above has been applied to all images. Edges of the shutter should be visible in the images, through the phantom or in the background noise.

Step	Description	Expected Result
Step 2	With current stack B, introduce an RL angulation of 20 degrees and again perform the scan.	Verify that the straight boundary of the shutter now has a position which shifts per slice within each stack.

3.1.20.2. SWCMP.Scan.Mobi.Shutter.tc2

Test ID	7392
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

MobiTrak/MobiFlex: Modified reconstruction shutter

Verify the large reconstruction shutter for MobiFlex scans

For **cylindrical** systems, a modified shape of the reconstruction shutter has been implemented to enlarge the viewable area of MobiTrak or MobiFlex images. This 'large shutter' is the cross-section of an increased size ellipsoidal shutter and a box shaped shutter to cut off parts of the images which correspond to inhomogeneous regions at large FH offcentre positions. The combined shape is an ellipsoid with two flat faces at the F and H sides. The 'large shutter' is used for all multi-stack scans with 'table movement per stack' set to 'yes' or GeoLink'ed scans in an ExamCard.

Requirements Coverage:

Req ID	Requirement Tag	Description
17571	SWCMP.Scan.Mobi.Shutter	MobiTrak scans have an adapted image shutter to minimize loss of information in regions between the stations (stacks), as specified in chapter Modified reconstruction shutter for MobiTrak and MobiFlex.
21021	SWCMP.Scan.Methods.ExamProperties	The validation of a scan protocol shall include exam properties. Exam properties consists of context information such as patient position, patient orientation, table position, connected coils and information required for defining and approving a set of scans with inherent strict inter-scan timing relationships.

Test steps:

Step	Description	Expected Result
Step 1	Plan a 3-scan FFE ExamCard with scan mode 3D. This scan protocol is available as 'mobi 3d' with the number of stacks changed to 1. Link the 3 scans via a 'GeoLink'. Check the reconstruction parameters in Sequence development mode (parameters -> RC).	`RC_shutter_type = RC_ELLIPSOID_AND_BOX `RC_shutter_radiuses [xyz] = `CRC_large_shutter_radiuses [ xyz ] `RC_shutter_box_sizes [xyz] = `CRC_large_shutter_box_sizes [xyz]

Step	Description	Expected Result
Step 2	Optional: Modify to the largest possible field of view and coronal slice orientation. Use Scan Alignment for the 3 scans in the FH direction. Introduce an AP angulation of 10 degrees in the 2nd scan. Adapt the (AP) offcentres to the position of your phantom but avoid range conflicts for the offcentres. Perform the scan. Verify that the reconstruction shutter as described above has been applied to all images. Edges of the shutter should be visible in the images, through the phantom or in the background noise.	
Step 3	Optional: In the 2nd scan introduce an RL angulation of 20 degrees and again perform the scan. Verify that the straight boundary of the shutter now has a position which shifts per slice within each scan.	

3.1.20.3. SWCMP.Scan.Mobi.TableTopExtender.tc1.v2

Test ID	58144
Revision	1
Product Model	IngCX
Gradient System	Any
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-11-10 15:16:46 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11-10 11:54:52 Meaning: SME Approved - (Business) Signed by: Oorschot, Joep van (310224063) on 2023-11-08 14:27:42 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The FH FOV is dependent on license option "Extended FOV".

Requirements Coverage:

Req ID	Requirement Tag	Description
17572	SWCMP.Scan.Mobi.TableTopExtender	MrMethods shall support scanning with an extended patient table stroke using a table top extender. Functionality and limitations are specified in chapter Intera/Achieva extended table stroke using a Table Top Extender.

Test steps:

Step	Description	Expected Result
Prep 1	- Select Ingenia CX 3.0T configuration from test framework. - Create a new log file. - Take the following protocol from the TEDB protocol database: "HEAD CNS \ MobiTrak \ i1_TTE".	N.A. (preparation)
Step 1	- Make sure that license key "Extended FOV" is switched on (Service Application/Configuration/Software License/Extended FOV) - Note the value of the scan control parameter "Mobi geo. corr. max. FH size". - Note the maximum of the range of the FH FOV parameter. - Note the maximum of the range of the RL FOV parameter.	The scan control parameter and FH FOV are the same.
Step 2	- Switch off license key "Extended FOV" (Service Application/Configuration/Software License/Extended FOV). - Note the value of value of the scan control parameter "Mobi maximum FH size". - Note the maximum of the range of the FH FOV parameter. - Note the maximum of the range of the RL FOV parameter. - Attach the log file to this step as objective evidence.	The scan control parameter and FH FOV are the same. The range of the RL FOV is unaffected (maximum 530 mm) compared to value found in previous step.

3.1.20.4. SWCMP.Scan.Mobi.Trak.tc1.v1

Test ID	32717
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-29 12:02:05 Meaning: QA Approved - (QA) Signed by: Crijns, Sjoerd (310257923) on 2017-03-29 10:03:47 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-29 09:54:28 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Check patient table positions derived from stack offcentres

If the operator does not interrupt table movement, the patient table should move to the correct optimal positions, determined by the stack offcentres.

Table movement during a scan (MobiTrak)

Requirements Coverage:

Req ID	Requirement Tag	Description
17562	SWCMP.Scan.Mobi.Trak	MrMethods shall support multi station scanning within a single scan. This means instructing the tabletop to move during the scan and correct for the positioning inaccuracy of tabletop in the scan location. See chapter MobiTrak: Table movement during a scan for more details.

Test steps:

Step	Description	Expected Result
Prep	On a TB, start with the TEDB protocol database and disable all CSK SW options. Place 3 bottles or phantoms which can be distinguished easily in MR images on the patient table with roughly 200 mm spacings. Place the phantoms close to the scanner side of the patient table. This will ensure that table movement can bring each phantom to the centre of the magnet. Not satisfying this condition may result in a scan abort. To minimize motion of the phantoms after table movement, make sure that the bottles are stabilized, e.g. using sand bags. Use the light visor to define the patient reference point centred on the middle of the three phantoms. Create a new logfile.	N/A (Preparation)
Step 1	Perform a survey scan with patient position head-first. A suitable protocol can be found in TEDB, anatomy Head/CNS, sub-anatomy MobiTrak, scan name 'survey'.	The scan completes correctly.
Step 2	Plan a 3-stack FFE scan, scan mode M2D, FH offcentres -200, 0, +200 and table movement per stack, patient position head first. This protocol is available as 'mobi m2d' in TEDB. Perform this scan. You will be prompted for table movement once before the scan and twice during the scan to bring each subsequent stack (A,B,C) to centre of the system. Move the patient table as requested. Observe that the table moves to bring stacks A, B and C to the isocentre of the magnet. When the scan is completed, check the images. In the Database Browser (Start/MR Development/Configuration/Database Browser), use the Blobviewer to inspect the 'preparation parameters'.	Verify that the scan was performed with the expected 3 patient table positions -200, 0, +200, with less than 6 mm deviations. Check the images to verify that the expected phantom is visible in each of the images. Note: If the table happens to be already at the correct position within 30 mm there will be no table movement for the first stack.
Step 3	Select the survey scan again and modify it to 'Patient position = feet first'. Press Start Scan to perform it.	The scan completes correctly.
Step 4	Select the mobi m2d protocol, change it to 'Patient position = feet first' and perform the scan again. This time, you should be prompted to move the patient table twice during the scan.	Verify that this time the table positions were close to MINUS the FH offcentres to for stacks A, B and C respectively (e.g. 200, 0, -200). The images shall show the expected phantoms.

Step	Description	Expected Result
Step 5	Modify the scan protocol to a SE protocol, 'Patient position = head first', scan mode MS, stacks as packages = yes, and perform the scan again. Attach the logfile as objective evidence.	You should be prompted to move the table twice during the scan. After the scan, verify via blob viewer, 'preparation parameters' that the table positions were close to the FH offcentres for stacks A,B,C respectively (e.g. -200, 0, 200). Again verify the correct contents of the images.

3.1.20.5. SWCMP.Scan.Mobi.Trak.tc2.v1

Test ID	32461
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-21 14:44:49 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:39:48 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-21 12:19:41 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Check the allowed table position deviations

For interrupted or ignored table movements, the allowed deviations from the optimal table positions should be according to specification. For MobiTrak or MobiFlex scans and SENSE prescans the allowed deviation is always the minimum (6 mm, see RS: MobiTrak/MobiFlex Methods). The same is true for all dynamic scans except the first of a scan. For scans with a large FOV the allowed table position deviation is 6 mm. With smaller FOV the allowed deviation is $(0.7 * \text{max_fh_size} - \text{fh_fov})/2$ where max_fh_size comes from the control scan parameter 'Mobi maximum FH size'. The value may be further reduced by the COIL attribute 'max_allowed_table_offset' of the applied receive coils.

Note: which stack is measured first depends on the parameter 'Stack scan order'.

Table movement during a scan (MobiTrak)

These tests involve the correctness of scan geometry and stored geometrical information with table movement during a scan.

Requirements Coverage:

Req ID	Requirement Tag	Description
17562	SWCMP.Scan.Mobi.Trak	MrMethods shall support multi station scanning within a single scan. This means instructing the tabletop to move during the scan and correct for the positioning

Req ID	Requirement Tag	Description
		inaccuracy of tabletop in the scan location. See chapter MobiTrak: Table movement during a scan for more details.

Test steps:

Step	Description	Expected Result
Prep 1	- On a test bay, load the TEDB protocol database and make sure that CSK SW options are disabled. - Place 3 bottles or phantoms which can be distinguished easily in MR images on the patient table with roughly 200 mm spacings. Place the phantoms close to the scanner side of the patient table. This will ensure that table movement can bring each phantom to the centre of the magnet. Not satisfying this condition may result in a scan abort. To minimize motion of the phantoms after table movement, make sure that the bottles are stabilized, e.g. using sand bags. - Use the light visor to define the patient reference point centred on the middle of the three phantoms. - Load protocol Head CNS/MobiTrak/mobi m2d a 3-stack FFE MobiTrak protocol. Use the QBC as the only receive coil. Modify the protocol to a dynamic study with two manually started dynamic scans. - Create a new logfile	N/A (preparation)
Step 1	Perform this scan. Interrupt table movement or, after table movement, move the table from its optimal position by e.g. 50 mm. Press proceed or start-scan to start the scan.	The system prompts for table movement because the allowed deviation is only 6 mm.
Step 2	When prompted for table movement of the second and third stacks, again try to wrongly position the table.	For any position deviation which exceeds 6 mm prompts for table movement appear.
Step 3	Repeat the attempts to wrongly position the table for stacks A, B and C of the second dynamic scan. Again the maximum allowed deviation is 6 mm for all stacks.	The prompt to move the table re-appears as long as the table is not within 6 mm from the actual position used in the first dynamic scan.
Step 4	- Load protocol Head CNS/MobiTrak/mobi m2d. - Modify the protocol to 'table movement per stack' to 'no' and Stack alignment 'no'. - Make the offcentre values equal to zero for all 3 stacks. - On the motion page set 'Respiratory compensation' to 'breath-hold' with 3 slices per breath-hold and breathhold guidance to manual. - Change 'slice scan order' to 'ascend'. - Modify the protocol to a dynamic study with two manually started dynamic scans. - Start this scan. - When the breath-hold prompt appears, move the table from its optimal position by e.g. 60 mm. - Press proceed or start-scan to start the scan.	The system prompts for table movement
Step 5	Move the table by 40 mm in the correct direction (resulting in a table deviation of less than 25mm).	The scan runs without a prompt for table movement.
Step 6	- Start the scan once more and move the patient table to the optimal position and perform the scan. - Load the images of this scan and of the previous scan in the viewer and subtract them.	The deviation of the two images is less than 1mm.
Step 7	Load the scan of step 1 and step 4 in the viewing environment. For both scans, subtract the images of the first dynamic scan from the second dynamic scan. Attach the logfile as objective evidence.	For both scans, the deviation of the two images is less than 1mm.

3.1.20.6. SWCMP.Scan.Mobi.Trak.tc4

Test ID	7366
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Check table movement not allowed by a receive coil

Check that table movement per stack is forbidden by a conflict if (one of) the receive coil(s) does not allow table movement.

Table movement during a scan (MobiTrak)

These tests involve the correctness of scan geometry and stored geometrical information with table movement during a scan. Some of the subsections below contain hazard aspects, see TS: Preparation Phases.

- To perform the tests for patient table movement, use a standard TB and start with the TEDB protocol database and required commercial options. The test protocols for MobiTrak are found in the anatomy Head/CNS, sub-anatomy MobiTrak.
 - Place 3 bottles or phantoms which can be distinguished easily in MR images on the patient table with roughly 200 mm spacings. Place the phantoms close to the scanner side of the patient table. This will ensure that table movement can bring each phantom to the centre of the magnet. Not satisfying this condition may result in a scan abort. To minimize motion of the phantoms after table movement, make sure that the bottles are stabilized, e.g. using sand bags.
 - Use the light visor to define the patient reference point centred on the middle of the three phantoms.
- For scans with movement of the patient table, there are several testable items, as described below.

Note: The tests of Check patient table positions derived from stack offcentres should be performed for the applicable version(s) of the patient support, i.e MT patient support, Fast Trak Lateral, etc.

Requirements Coverage:

Req ID	Requirement Tag	Description
17562	SWCMP.Scan.Mobi.Trak	MrMethods shall support multi station scanning within a single scan. This means instructing the tabletop to move during the scan and correct for the positioning inaccuracy of tabletop in the scan location. See chapter MobiTrak: Table movement during a scan for more details.

Test steps:

Step	Description	Expected Result
Step 1	For an available coil, for example the NVC-base coil, set the attribute "Table movement" to no. (Execution List right mouse menu -> System Tuning -> Tab Coil -> Select dS Base, scroll down to 'Table movement', set parameter 'no'.) Select the scan protocol 'mobi m2d'. Connect the NVC base coil. There must be a parameter conflict for 'Move table per stack = yes'. If there are conflicts for other parameters, e.g. offcentres, first resolve these until the conflict for 'Move table per stack' appears.	Check that the expected parameter conflict occurs.

3.1.20.7. SWCMP.Scan.Mobi.Trak.tc5.v1

Test ID	32471
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-22 16:41:54 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-03-22 15:58:32 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-03-21 17:01:29 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Use the simulator to test the positioning of parallel REST slabs

Requirements Coverage:

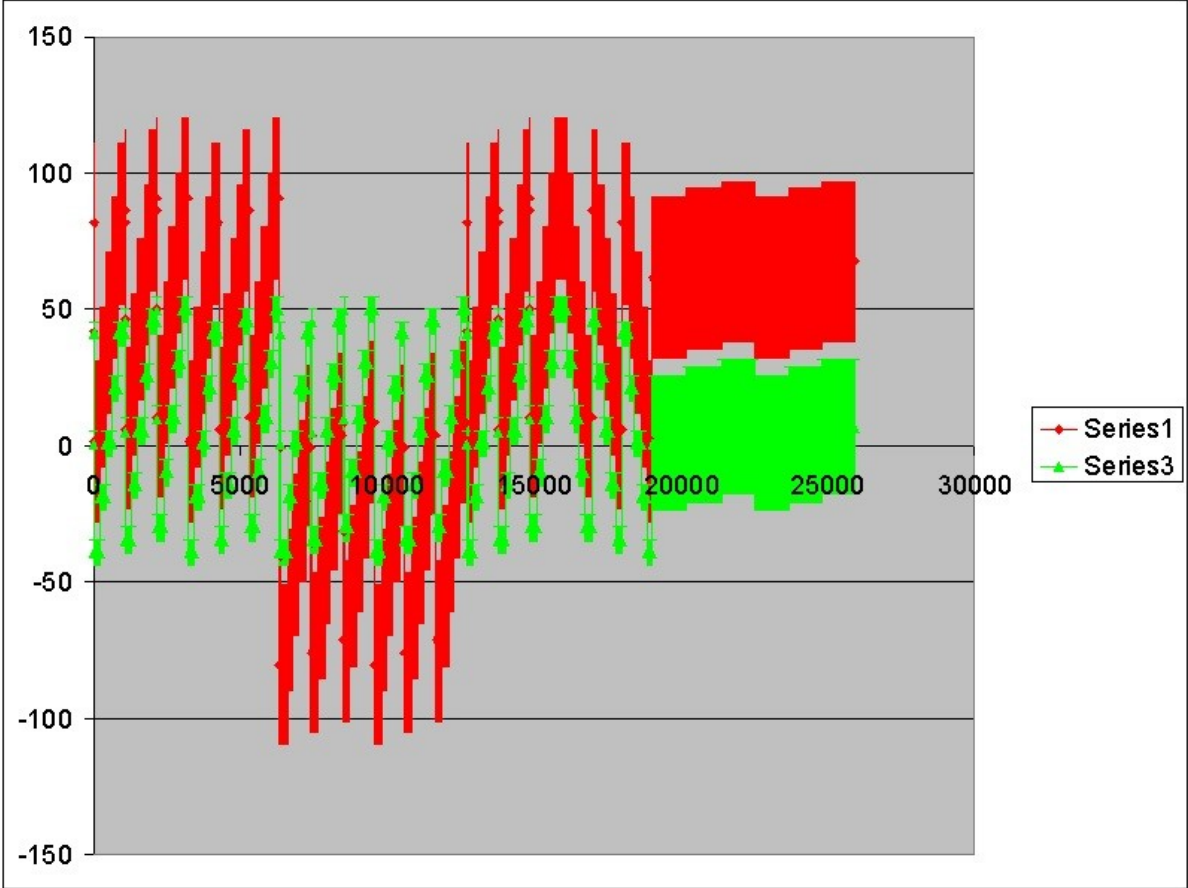
Req ID	Requirement Tag	Description
17562	SWCMP.Scan.Mobi.Trak	MrMethods shall support multi station scanning within a single scan. This means instructing the tabletop to move during the scan and correct for the positioning inaccuracy of tabletop in the scan location. See chapter MobiTrak: Table movement during a scan for more details.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	On a VMware install or select TEDB as your preset procedure database. Make sure that all CSK SW options are switched off. Start ExamCards. Verify that the files objlog_XXX.cf are present. They are created when you first run a scan on the VMware. In order to enable object logging, copy the PERL script m_imn_mobi_oblog.pl	N/A (preparation)	None

Step	Description	Expected Result	Attachments
	(available in mrmethods\imaging\tst) to G:\Temp and let it run using m_imn_mobi_oblog.pl enable. Create a new logfile.		

Step	Description	Expected Result	Attachments
Step 1	- Select the protocol 'Head CNS\MobiTrak\MobiM2D REST' with 3 transverse stacks and 2 dynamic scans. Reduce the scan resolution to 64 and scan percentage to 25%. In order to be able to do that, change the scan control parameter 'resolution definition' to 'classic'. Start the scan and enable object logging as described above. Run the scan (press PROCEED several times). - Modify the REST slab position to 'feet'. Again, start the scan, check the object logging attributes and selected objects, and press PROCEED several times to complete the scan. - Change back to REST slab position 'head'. On the dynamic scan page, change 'stack yo-yo order' to 'yes' and perform the scan with object logging. - Change back to 'yo-yo stack order' equal 'no', modify the scan protocol to a 3D scan and select 'stacks as packages' equal to 'yes'. Again perform the scan with object logging. Result analysis: - Copy the PERL script m_imn_mobi_rest.pl in mrmethods\imaging\tst to the log directory and execute it. Running the PERL script: perl m_imn_mobi_rest.pl devlogcurrentxxx.log > mobi.dat - The output consists of 5 columns: an incrementing excitation pulse counter, the position of the REST slab, half the thickness of the REST slab, the position of the slice or 3D volume and half it's thickness. - A graphical representation of this information can be obtained by EXCEL. Paste the data from mobi.dat into the indicated columns of sheet1 of m_imn_mobi_rest.xlsx in mrmethods\imaging\tst. Compare the result to the reference plot on the left side of sheet 1. Disable object logging by running: m_imn_mobi_oblog.pl disable. Attach the logfile as objective evidence. Explanation: The MobiM2D REST protocol has a 10 degrees angulation. Without angulation, corresponding slices and slabs in each stack are expected at the same distance from the centre of the magnet because the (simulated) table movement brings each stack to the centre of the magnet. With the	The plot shows the same behavior of slice and REST slab positions as displayed in the reference plot on the left side of the excel sheet. Verify that the REST slab is at the expected position with the expected thickness (REST slab thickness) and the expected distance (REST gap) from the slice. In the part corresponding to the second scan, the position of the REST slab has changed from being above the stack to being below the stack (foot vs head position of the REST slab). The part corresponding to the third scan gives similar results, with a reversed stack order in the second dynamic scan. In the part corresponding to the 3D scan the volume and REST slab are always at the same positions with respect to the magnet centre within 10mm accuracy. The gap between REST slab and 3D volume should have the expected constant value (REST slab gap).	

Step	Description	Expected Result	Attachments
	10 degrees angulation in the scan protocol, some differences arise between the stacks. For the 3D scan, the (simulated) table movement will take care to remove most of the offsets with respect to the magnet iso-centre.		
			

3.1.20.8. SWCMP.Scan.Mobi.TrakAlign.tc1

Test ID	7385
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

MobiTrak: Stack alignment

Supine, Head First, Stack alignment FH, LR, AP

'Stack alignment' has been introduced to automatically make the stacks of a multi-stack scan parallel to each other and to arrange them along a line to mimic one 'superstack'. The alignment axis can be in-plane along the M or P axis or perpendicular to the slices, i.e. in the S direction. In fact the user interface allows to chose a 'patient' alignment axis AP, LR or FH and stacks are aligned along either M, P, or S, depending on which of these is closest to this patient alignment axis.

Requirements Coverage:

Req ID	Requirement Tag	Description
17568	SWCMP.Scan.Mobi.TrakAlign	For MobiTrak scans, MrMethods shall support automatic alignment of stack orientations and automatic adjustment of stack offcentres to a specified stack overlap. Specified in chapter Stack alignment for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	- Select the protocol 'survey', reset geometry and perform the scan. Load the images in the planscan environment. - Plan a 3-stack 3D scan (transverse, 5 slices per stack) and the same patient position and orientation as the previous scan. A suitable protocol is 'mobi 3d'. Set the current stack to 'B' and then change the orientation to transverse. With stack alignment 'FH' and stack overlap equal to zero, the slice orientation of all stacks should now be transverse. The stacks should be aligned along the FH axis with continuously increasing slice numbers and constant spacings of slices. This looks like 3 stacks glued together in a single 'superstack' of 15 slices. Change the angulations of stack B and observe that the stacks continue to be aligned: angulations of stack B are copied to stacks A and C and the offcentres of A and C are adjusted to keep the stacks in one line. - Set the current stack to B, switch stack alignment to 'LR' and set stack overlap to 100 mm. Slices should now be aligned along the LR axis. The overlap between subsequent stacks should be 100 mm. Within each stack, slice numbers increase in the FH direction (transverse slice orientation). Stacks A-C are placed along the LR axis from L to R, with slice numbers increasing from stack A to stack C. Note: the placement of stacks is from L to R, which is the opposite to the direction of increasing offcentres (R to L). This inconsistency is related to our convention that slice numbers increase from L to R and not from R to L. - Now switch stack alignment to 'AP'. Slices should now be aligned along the AP axis. The overlap between subsequent stacks should be 100 mm. Within each stack, slice numbers increase in the FH direction (transverse slice orientation). Stacks A-C are placed along the AP axis from A to P, with slice numbers increasing from stack A to stack C. - Set stack alignment back to FH with stack overlap	For all cases of this section scan graphics behaves as indicated.

Step	Description	Expected Result
	equal to zero. Change scan mode to 'MS'. Now you should see 15 equally spaced slices with slice centre to centre intervals of 15 mm (slices thickness + gap). Modify the stack overlap to -15 mm: an additional gap between stacks of 15 mm. Now the slice to slice intervals should be 15 mm within each stack and 30 m between stacks.	

3.1.20.9. SWCMP.Scan.Mobi.TrakAlign.tc2

Test ID	7386
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

MobiTrak: Stack alignment

Perform the previous tests for a prone patient position

'Stack alignment' has been introduced to automatically make the stacks of a multi-stack scan parallel to each other and to arrange them along a line to mimic one 'superstack'. The alignment axis can be in-plane along the M or P axis or perpendicular to the slices, i.e. in the S direction. In fact the user interface allows to chose a 'patient' alignment axis AP, LR or FH and stacks are aligned along either M, P, or S, depending on which of these is closest to this patient alignment axis.

Requirements Coverage:

Req ID	Requirement Tag	Description
17568	SWCMP.Scan.Mobi.TrakAlign	For MobiTrak scans, MrMethods shall support automatic alignment of stack orientations and automatic adjustment of stack offcentres to a specified stack overlap. Specified in chapter Stack alignment for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	- Make a large FOV survey scan, patient position head first, orientation prone and load the images in the planscan environment. - Select 'mobi 3d', set the patient orientation to 'prone' and repeat the previous tests.	None

3.1.20.10. SWCMP.Scan.Mobi.TrakDynamics.tc1

Test ID	7383
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Dynamic scan features for MobiTrak

Reconstruction mode immediate vs. immediate first dynamic.

In dynamic studies with a manually started second dynamic scan, two types of immediate reconstruction are available. Mode 'immediate' will reconstruct all dynamic scans including the first after scan completion. Mode 'immediate 1st dyn' will reconstruct the first dynamic scan before prompting to start the second dynamic scan.

Requirements Coverage:

Req ID	Requirement Tag	Description
17567	SWCMP.Scan.Mobi.TrakDynamics	MrMethods shall support special dynamic scan functionality for MobiTrak multi-station pre and post contrast scanning, specified in chapter Dynamic scan issues for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	Select scan protocol 'mobi 3d'. Set 'dynamic study' to 'yes', 'dyn scans' to 2 and 'dyn scan times' to 'manual'. Set 'Reconstruction mode' to 'immediate'. Perform the scan and move the patient table as requested by the system. Activate autoview.	Verify that the reconstruction of the first dynamic scan is performed after completion of the scan, i.e. after completing dynamic scans 1 and 2.
Step 2	Now change 'Reconstruction mode' to 'immed. 1st dyn' and perform the scan again. This time, verify that the first dynamic scan is reconstructed before the prompt to start the second dynamic scan appears. Modify the scan protocol from FFE to SE, scan mode from 3D to MS. Eliminate the conflict by using 'foldover suppression' of 125mm on both sides. Use a larger voxel size to reduce the scan duration.	Verify that the 'Reconstruction mode' choices 'immediate' and 'immed 1st dyn' behave as indicated above.

3.1.20.11. SWCMP.Scan.Mobi.TrakDynamics.tc2

Test ID	7384
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Dynamic scan features for MobiTrak

Dynamic scan yo-yo order

For dynamic studies with multiple stacks, the stack order in subsequent dynamic scans may be always the same or it may alternate in subsequent dynamic scans. This latter behaviour is obtained by setting 'dynamic scan yo-yo order' to 'yes'. This choice will be useful in combination with 'table movement per stack' to reduce the number of table movements required.

Requirements Coverage:

Req ID	Requirement Tag	Description
17567	SWCMP.Scan.Mobi.TrakDynamics	MrMethods shall support special dynamic scan functionality for MobiTrak multi-station pre and post contrast scanning, specified in chapter Dynamic scan issues for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	Select protocol 'mobi m2d', and make it a dynamic study with 3 manually started dynamic scans. Set 'dynamic scan yo-yo order' to 'yes'. Start the scan. During the first dynamic scan, there are table movements for each subsequent stack. For the second dynamic scan, stack 'C' is measured first. No table movement because the table was positioned for stack 'C' already. Subsequently, table movements for stacks 'B' and 'A'. For the third dynamic scan, the stack order is reversed again. No table movement for stack 'A' which is first. Subsequently, table movements for stacks 'B' and 'C'.	Scan runs correctly with stack order / table movements as indicated.
Step 2	Repeat this test with 'dynamic scan yo-yo order' to 'no'. In this case, the stack order is ABC for both dynamic scans and the table has to be moved back to the stack A position before the start of the second dynamic scan. The images should be similar to those of the previous scan.	Scan runs correctly with stack order / table movements as indicated.
Step 3	Modify the scan protocol from FFE to SE and scan mode from M2D to MS with 'stacks as packages' set to 'yes'. Repeat the previous tests (yo-yo no and yes) and verify that again the stack order alternates in subsequent dynamic scans.	Scan runs correctly with stack order / table movements as indicated.

3.1.20.12. SWCMP.Scan.Mobi.TrakMIP.tc1

Test ID	7390
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

MobiTrak: Multi-stack orthogonal MIP

Check Multi-stack MIP

Multi-stack scans with orthogonal MIP enabled on the processing page should produce 3 orthogonal MIP images for each input stack.

Requirements Coverage:

Req ID	Requirement Tag	Description
17570	SWCMP.Scan.Mobi.TrakMIP	For MobiTrak scans MrMethods shall support immediate orthogonal MIP images for each station (stack), as specified in chapter Multi-stack orthogonal MIP for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	- Select the protocol 'mobi-m2d' and enable orthogonal MIP. There will be a conflict between MIP and the number of slices for stack A. The minimum number of slices for MIP is 5. Increase the number to 5. Now, there will a conflict on the current stack. Switch to stacks B and C and also change the numbers of slices to 5. This should solve the conflict. Also check that stack type and a negative slice gap equal to -slice thickness lead to a conflict. Perform the scan and check the MIP images. - Select 'mobi 3d'. Enable orthogonal MIP. After completing the scan, a MIP entry should be found in the patient database. This MIP entry should contain 3 orthogonal MIP images per stack and dynamic scan of the corresponding scan entry. Check the image information: scan orientation, offcentres, angulations. - Change the scan protocol to 10 slices and 2 chunks. Perform the scan. Similar MIP images for all stacks should be present.	User interface behavior and produced images are as indicated.

3.1.20.13. SWCMP.Scan.Mobi.TrakProfOrder.tc1.v1

Test ID	32718
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-29 12:03:14 Meaning: QA Approved - (QA) Signed by: Crijns, Sjoerd (310257923) on 2017-03-29 10:05:29 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-29 09:59:10 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

MobiTrak: profile order per stack

Check profile order per stack

For multi-stack scans with 'Angio mode' equal to 'contrast enhanced', this parameter allows to use different profile orders (low-high, high-low, linear, rev-linear) per stack.

Requirements Coverage:

Req ID	Requirement Tag	Description
17569	SWCMP.Scan.Mobi.TrakProfOrder	For MobiTrak scans MrMethods shall support support different profile orders per station (stack), specified in chapter Profile order per stack for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	Select 'Head CNS\MobiTrak\mobi 3d', select 'angio mode' equal to 'contrast enhancement' and activate 'profile order per stack'. Select profile orders 'high-low', 'linear' and 'low-high' for stacks A, B and C respectively. Perform the scan while you are in the RF shielded examination room.	Verify from the sound of the scanner that different profile orders are used. Check normal completion of the scan and a reasonable image quality.

3.1.20.14. SWCMP.Scan.Mobi.TrakProfOrder.tc2.v1

Test ID	58108
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:49:15 Meaning: QA

	Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:13:49 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11-15 11:04:30 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

MobiTrak: profile order per stack

For multi-stack scans with 'Angio mode' equal to 'contrast enhanced', this parameter allows to use different profile orders (low-high, high-low, linear, rev-linear) per stack.

Requirements Coverage:

Req ID	Requirement Tag	Description
17569	SWCMP.Scan.Mobi.TrakProfOrder	For MobiTrak scans MrMethods shall support support different profile orders per station (stack), specified in chapter Profile order per stack for MobiTrak.

Test steps:

Step	Description	Expected Result
Step 1	- Plan and perform a Head CNS / MobiTrak/Survey scan. - Select the protocol Head CNS / MobiTrak / mobi 3d. Set angio mode to 'contrast enhanced' and 'profile order per stack' to 'yes'. - Set 'Move table per stack' to 'no'. - Disable 'Stack alignment' and plan the 3 stacks to cover roughly the same region - Adapt the protocol to a TR near 4 ms: lower scan matrix (64 or 128), larger FOV (change scan control parameter 'resolution definition' to 'classic'), partial echo, minimum WF shift. - Run the scan and verify that it completes normally without any scan aborts. - Set 'dynamic study' to 'yes' and repeat the test for 'dyn scans' equal to 10 with 'dyn scan times' equal to 'shortest'. - Set 'yo-yo stack order' to 'yes' and again perform the scan.	In all cases verify that no scan aborts or errors occur.

3.1.20.15. SWCMP.Scan.Mobi.TrakProfOrder.tc3.v2

Test ID	26641
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-29 11:58:30 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-29 11:35:21 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on 2016-03-22 19:46:33 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

MobiTrak: profile order per stack
DAS performance checks

For multi-stack scans with 'Angio mode' equal to 'contrast enhanced', this parameter allows to use different profile orders (low-high, high-low, linear, rev-linear) per stack.

Requirements Coverage:

Req ID	Requirement Tag	Description
17569	SWCMP.Scan.Mobi.TrakProfOrder	For MobiTrak scans MrMethods shall support support different profile orders per station (stack), specified in chapter Profile order per stack for MobiTrak.

Test steps:

Step	Description
Step 1	On a testsystem or RTM, create a file r_rcrepatch.ini in G:\patch. Add the line 'modify system gyro_rec_label_log SYMBOLIC'. In the command line type 'permproc stop recon' and after the scanner processes have stopped 'permproc start recon'. Start the Application software and select 'Head CNS\MobiTrak\mobi m2d'. Set the following parameter values: 'Angio / Contrast enh.' = 'contrast enh.' 'CE profile order per stack' = 'yes' - order for stack A,B,C.. 1 ' high_low' 2 'linear'

Expected Result
The profile orders per stack agree with the values in the executed scan: A: high_low B: linear C: low_high

Attachments
 m_imn_prof_log.zip

Step	Description	Expected Result	Attachments
	3 'low_high'		
	Run the scan. After the scan has finished, use the PERL script ... \\methods\\imaging\\tst\\m_imn_prof_log.pl (see also in attachment) to extract profile order information from the log file. Open the output file (G: \\log\\m_imn_prof_log.dat) in excel. Check the date, time and configuration. Make a graph of the profile orders (column 'phase enc'). Compare the orders per stack with the attached reference file (.xlsx).		

3.1.21. ME MRE

3.1.21.1. SWCMP.Scan.MRE.Breath-hold.tc1

Test ID	25292
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:34 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:26 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:13 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_environment.pl. Select 3T Ingenia as the option. Start Application Software from MR Boot Configuration Manager Unzip and import the attached examcard <u>TS MRE Run Gender F.examcard</u>	N.A. (Preparation)	 TS MRE Run Gender F.zip

Step	Description	Expected Result	Attachments
	Press 'New Exam' and set the gender of the patient to 'Female'. Prescribe other mandatory parameters. ==== Scan a survey Load Inbox-> TS MRE Run Gender F ExamCard. All scans in the queue should be ready run. If they are not then remove the geometry name to allow them to be ready to run.		
Step 1	Inspect the parameter information page for the MRE 4 SL scan. Prescribe 450x450mm FOV, Enable SENSE with acceleration set to 2 and make slice as 1	Scan time should be less than or equal to 20sec	None
Step 2	Press 'Start Scan' to run the contents of the scan queue.	All scans run to completion and reconstruct successfully	None

3.1.21.2. SWCMP.Scan.MRE.ExtTrigger.tc1

Test ID	25980
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:37 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:28 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:16 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Import the attached examcard <u>TS MRE with Phantom.examcard</u> Press 'New Exam' and load Inbox->TS MRE WITH PHANTOM ExamCard. Scan Survey Start Sequence Development Editor	NA	 TS MRE WITH PHANTOM.zip
Step 1	Scan MRE 4SL exam-card Sequence Development Editor will come-up. Continue the scan till you see MRE 4SL on the title bar. Under parameters, check for MPF_FFE_ELASTO_trig_phases and MPF_FFE_ELASTO_synch_ext_devices [Under MPF_ELASTO_all_aprs] Under Objects select SOR, Select ex, expand till you see SOR object as shown in the attachment Double click on the SY:: external_device and check for sy_pulse_bit and sy_timp_step. <i>sy_time_step in SY Object is calculated as $\frac{1}{4} * (\frac{1}{\text{Frequency}}) * 1000$. This represents the time delay corresponding to 90 deg phase. (Full period = 360 deg, $\frac{1}{4} = 90$ deg).</i> <i>Also, 1/4 comes from number of trigger phases which is 4 for MRE acquisition.</i> <i>For 1 of 4 acquisition = $0 * 90 \text{ deg} = 0$ phase</i> <i>For 2 of 4 acquisition = $1 * 90 \text{ deg} = 90$ phase</i> <i>For 3 of 4 acquisition = $2 * 90 \text{ deg} = 180$ phase</i> <i>For 4 of 4 acquisition = $3 * 90 \text{ deg} = 270$ phase</i> <i>For 60Hz frequency (default value) it should be 4.2 with a tolerance of +/- 0.1</i>	MPF_FFE_ELASTO_trig_phases = 4 MPF_FFE_ELASTO_synch_ext_devices = Yes SOR object should show SY::external_device before the RF pulse as shown in the SOR Object attachment SY object have sy_pulse_bit set to ON. sy_time_step = 4.2 with a tolerance of +/- 0.1	 SOR_Object.zip  SY_Object.zip

3.1.21.3. SWCMP.Scan.MRE.ImageTypes.tc1

Test ID	25289
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:40 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:30 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:18 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Import the attached examcard <u>TS MRE with Phantom.examcard</u> Press 'New Exam' and load Inbox->TS MRE WITH PHANTOM ExamCard. Scan Survey	N.A. (preparation)	 TS MRE WITH PHANTOM.zip
Step 2	Scan MRE 4SL exam-card Open the scanned images by double clicking on the scanned exam-card	FFE/M and MRE/P images for all the 4 phases (0, 90, 180 and 270) will be available	None

3.1.21.4. SWCMP.Scan.MRE.Par.tc1

Test ID	40986
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya

	(310175887) on 2019-05-23 10:32:13 Meaning: QA Approved - (QA)
	Signed by: Bajaj, Naveen (310105618) on 2019-05-23 09:21:52 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2019-05-23 08:53:38 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of MRE parameters.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanMrePar in the results file is zero.

3.1.21.5. SWCMP.Scan.MRE.PatExperience.tc1.V1

Test ID	32098
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-06 15:43:54 Meaning: QA Approved - (QA)
	Signed by: Hegde, Shashank (310195826) on 2017-04-06 14:46:40 Meaning: SME Approved - (Business)
	Signed by: Graaf, Ruud de (nly95749) on 2017-04-04 12:21:12 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl. Select 3T Ingenia.	N.A. (Preparation)

Step	Description	Expected Result
	Start Application Software from MR Boot Configuration Manager. Press 'New Exam' and set the gender of the patient to 'Female'. Prescribe other mandatory parameters. ==== Load scan protocol /Hospital/Abdomen/MRE/FFE_MRE_Comptibility_4sl Select the Anatomy as Abdomen and the anatomic region as Liver in the exam card property window. Resolve the clinical science key related conflict for MRE frequency from the conflicts tab and remove the geometry name if needed. Ensure that the respiratory compensation is set to "breathhold" in the motion tab. The scan should be ready to run,	
Step 1	Check the number of slices in the summary tab. Check the number of breath holds for this scan.	The number of breathholds shall be equal to the number of slices.
Step 2	Set the 'Patient experience scan' in the motion tab to 'Yes'. Check the number of breath holds for this scan. Attach a screenshot that shows the number of slices and the number of breathholds as evidence.	As the scan inserts an additional breath hold to prepare the patient for the experience, the number of breathholds shall be equal to (number of slices +1)

3.1.21.6. SWCMP.Scan.MRE.PatOri.tc1.v1

Test ID	32727
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-30 08:48:10 Meaning: QA Approved - (QA) Signed by: Hegde, Shashank (310195826) on 2017-03-30 08:21:08 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-30 07:41:00 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select an Achieva configuration. Start Application Software from MR Boot Configuraiton Manager Press 'New Exam', prescribe the mandatory parameters and start the exam. ===== Load the scan protocol /Hospital/Abdomen/MRE/FFE_MRE_Comptibility_4sl. The scan shall be ready to run. If it is not, then remove the geometry name or resolve the clinical science key related conflict to allow it to be ready to run. Select the Anatomy as Abdomen and the anatomic region as Liver in the exam card property window.	N.A. (Preparation)
Step 1	Select the Exam Card property editor. Check that the Patient Orientation to 'Head First'. Check that the protocol is in the 'ready to run' state and not in conflict.	The Patient Orientation shall be head first. The scan shall be in 'ready to run' state and no conflict shall be observed.
Step 2	Select the Exam Card property editor Set the Patient Orientation to 'Feet First'. Check that the scan is now in conflict. Navigate to the conflict tab and check the parameters in conflict.	The Patient Orientation is feet first. The scan shall be in conflict. The conflicted parameters shall be Patient body position set to 'feet first' (equivalent to Patient Orientation on the property editor) and MRE enable set to yes. Please attach a screenshot showing the conflicts between these parameters as evidence.

3.1.21.7. SWCMP.Scan.MRE.SeriesAttributes.tc1

Test ID	25291
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:47 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:36 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:24 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Unzip and import the attached examcard TS MRE ATTR.examcard Ensure option 36 is enabled Create new patient and drag the examcard into the execution list.	N.A. (Preparation)	 TS MRE ATTR.zip
Step 1	Check the exam queue contains Survey MRE 4 SL Run all scans to completion.	The exam queue contains the expected scans. Scans run to completion and reconstruct successfully.	None
Step 2	Open database browser and connect to the repository. Go to the series scanned (Patient->Study->Examination) above, right click on the MRE 4SL series and select write to file. In the pop-window, select save as to save the text file. Open the text file and search for mre	mre_frequency and mre_meg_frequency should match the Frequency parameter prescribed on UI mre_amplitude should match the Wave amplitude parameter prescribed on UI mre_meg_amplitude should match the Gradient strength parameter prescribed on UI mre_meg_direction should match the Direction parameter prescribed on UI, mre_no_of_motion_cycles should be 3 mre_no_of_phase_delays should be 4 mre_meg_pairs should be 1	None

3.1.21.8. SWCMP.Scan.MRE.Technique.tc1

Test ID	25287
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:49 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:39 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:26 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Edit c:\temp\bstrace.dat Add the following line (if it is not already present): GVE ** After adding the line, open a command window. Type the command logd. Check that the line GVE ** appears in the logd start up (see attachment). Start the GVE. (Start->MR Development->Scan Simulations->GVE) Display Object Names in the GVE. Unzip and import the attached examcard <u>TS MRE with Phantom.examcard</u> Ensure option36 is installed	N.A. (Preparation)	 logd_shows_bstrace_entry_GVE.zip  TS MRE WITH PHANTOM.zip

Step	Description	Expected Result	Attachments
	Press 'New Exam' and set the gender of the patient to 'Phantom' Load the examcard TS MRE with Phantom. Run Survey		
Step 1	Adjust the scan MRE 4 SL. The scan starts with TR = 50 ms and MRE frequency = 60 Hz. Verify that the adjustment step (up/down) available for TR is 16.67 ms. Adjust the TR to 66.67, 83.33, and 100 ms using the up/down adjustment arrows for TR. Note: some rounding at the 0.01 ms place may be observed. Set the TR to 52 ms by entering the value directly in the parameter value field. Observe that the UI sets the TR to 50 ms. Set the TR to 59 ms by entering the value directly in the parameter value field. Observe that the UI sets the TR to 66.67 ms. Set the TR to 65 ms. Observe that the UI sets the TR to 66.67 ms. Press 'Accept' to accept the changes. Run the scan	No conflicts arise during these adjustments. The adjustment step for TR is 16.67 ms. The TR after rounding has occurred matches the list of 50, 66.67, 83.33, and 100 ms. Tolerance is for TR value is +/- 0.01 ms. Setting the TR to a number that is not a multiple of 16.67 ms will result in the TR moving to the closest multiple. Setting the TR to 52 ms results in TR adjusted to 50 ms. Setting the TR to 59 ms results in TR adjusted to 66.67 ms. Setting the TR to 65 ms results in TR adjusted to 66.67 ms.	None
Step 2	Open the pdf.gve file that was generated by the previous step. In the 'Select PDF Sequence' menu select 'base' to be loaded twice. (see attached screen grab SelectPDF_load_base_x2.png) Verify the GVE plot resemble the attached screen grab baseX2_gve_plot.png.	The pdf.gve file loads and the SQ`base is displayed twice.	 SelectPDF_load_base_x2.zip  baseX2_gve_plot.zip
Prep 2	Select the GVE measure tool (the 'M' button). Position the mouse point on the first green line on the RF axis and click the left mouse button to enter measurement mode. Move the mouse to the second green while holding down the left mouse button. (see attached screen grab GVE_mouse_over_for_measurement.png).	N.A. (Preparation)	 GVE_mouse_over_for_measurement.zip

Step	Description	Expected Result	Attachments
	When position over the second green line, release the left mouse button. Upon release of the left mouse button, the 'Measurement' window appears.		
Step 3	Check in the measurement window popup that the TR is 66.67 ms to within +/- 0.3 ms.	The TR as measured with the GVE measurement tool will be 66.67 ms +/- 0.3 ms.	 GVE_measured_on_RF_axis.zip
Step 4	Adjust the scan MRE 4 SL. Adjust the TR to 83.33 ms using the up/down arrows or direct text entry. Note: some rounding at the 0.01 ms place may be observed. Check that the UI set the TR to 83.33 ms. Press 'Accept' to accept the changes. Run the scan.	No conflicts arise during this adjustment. The TR after rounding has occurred is 83.33 ms +/- 0.01 ms.	None
Step 5	Open the pdf.gve file that was generated by the previous step. In the 'Select PDF Sequence' menu select 'base' to be loaded twice.	The pdf.gve file loads and the SQ`base is displayed twice.	None
Prep 3	Select the GVE measure tool (the 'M' button). Position the mouse point on the first green line on the RF axis and click the left mouse button to enter measurement mode. Move the mouse to the second green while holding down the left mouse button. When position over the second green line, release the left mouse button. Upon release of the left mouse button, the 'Measurement' window appears.	N.A. (Preparation)	None
Step 6	Check in the measurement window popup that the TR is 83.33 ms to within +/- 0.3 ms. The inaccuracy is actually a function the limitations of the GVE and the tester's accuracy at selecting each point.	The TR as measured with the GVE measurement tool will be 83.33 ms +/- 0.3 ms.	None

Step	Description	Expected Result	Attachments
Step 7	Adjust the scan MRE 4 SL. Set the TR to 50 ms by entering the value directly in the parameter value field. Verify that the UI sets the TR to 50 ms. Adjust the MRE frequency to 90 Hz. Observe that the UI sets the TR to 55.56 ms. Adjust the TR to 44.44, 55.56, 66.67, 77.78, 88.89, and 100 ms using the up/down adjustment arrows for TR. Note: some rounding at the 0.01 ms place may be observed. Check that the adjustment step (up/down) available for TR is 11.11 ms. Adjust the TR to 40, 50, 60, 70, 80, and 90 ms entering the values directly in the parameter value field. Note: some rounding at the 0.01 ms place may be observed. Check that the UI sets the TR to the closest multiple of 11.11 ms. Set the TR to 100 ms by entering the value directly in the parameter value field. Press 'Accept' to accept the changes. Run the scan	No conflicts arise during these adjustments. The adjustment step for TR is 11.11 ms. Tolerance in TR is +/- 0.01 ms. The TR after rounding has occurred matches the list of 44.44, 55.56, 66.67, 77.78, 88.89, and 100 ms. Setting the TR to a number that is not a multiple of 11.11 ms will result in the TR moving to the closest multiple. TR entered --> UI set TR table: 40 ms --> 44.44 ms; 50 ms --> 55.56 ms; 60 ms --> 55.56 ms; 70 ms --> 66.67 ms; 80 ms --> 77.78 ms; 90 ms --> 88.89 ms	None
Step 8	Open the pdf.gve file that was generated by the previous step. In the 'Select PDF Sequence' menu select 'base' to be loaded twice.	Open the pdf.gve file that was generated by the previous step. In the 'Select PDF Sequence' menu select 'base' to be loaded twice.	None
Prep 4	Select the GVE measure tool (the 'M' button). Position the mouse point on the first green line on the RF axis and click the left mouse button to enter measurement mode. Move the mouse to the second green while holding down the left mouse button. When position over the second green line, release the left mouse button. Upon release of the left mouse button, the 'Measurement' window appears.	N.A. (Preparation)	None
Step 9	Check in the measurement window popup that the TR is 100.0 ms to within +/- 0.3 ms.	The TR as measured with the GVE measurement tool will be 100.0 ms +/- 0.3	None

Step	Description	Expected Result	Attachments
		ms.	
Prep 5	Start the Sequence Development Editor	N.A	None
Step 10	Adjust the scan MRE 4 SL Set the TR = 50ms and Frequency to 60Hz. Scan it. On Sequence Development Editor, Go to parameters tab and check for VAL12_ELASTO_ffe_cycles (under VAL12_ELASTO_ffe_all_pars)	VAL12_ELASTO_ffe_cycles should be 3	None
Step 11	Adjust the scan MRE 4 SL Set the TR = 50ms and Frequency to 100Hz. Scan it. On Sequence Development Editor, Go to parameters tab and check for VAL12_ELASTO_ffe_cycles (under VAL12_ELASTO_ffe_all_pars)	VAL12_ELASTO_ffe_cycles should be 5	None

3.1.21.9. SWCMP.Scan.MRE.UI.tc1

Test ID	25293
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-03-22 09:08:57 Meaning: QA Approved - (QA) Signed by: Nimbargi, Vijay (310198564) on 2016-03-21 11:29:38 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2016-03-16 14:00:07 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration.	N.A. (Preparation)	

Step	Description	Expected Result	Attachments
	Start Application Software from MR Boot Configuraiton Manager Ensure Option 36 is enabled. If it is not enabled then enable it and follow the instructions in field service framework to activate it. Unzip and import the attached examcard <u>TS MRE CSK.examcard</u>		 TS MRE CSK.zip
Step 1	Load Inbox-> TS MRE CSK ExamCard. Check that all scans are ready to run. If they are not then removed the geometry name to allow them to be ready to run. In particular, the scan CSK SE MRE is ready to run and not in conflict.	All scans in the queue should be ready run. No conflict occurs for the scan CSK SE MRE because the option 36 is enabled.	None
Step 2	Check for the availability and range for MRE enable parameter under Motion tab	MRE enable parameter is present and provides option yes and no	None
Step 3	After setting MRE enable to Yes, check for availability and range of values for Direction	Direction parameter is present and provides option M,P and S	None
Step 4	After setting MRE enable to Yes, check for availability and range of values for Frequency	Frequency parameter is available and has range from 1-1000	None
Step 5	After setting MRE enable to Yes, check for availability and range of values for MSG dur(T0)	MSG dur(T0) parameter is availalbe and has range from 1-100	None
Step 6	After setting MRE enable to Yes, check for availability and range of values for Gap(T0)	Gap(T0) parameter is available and has range from 1-100	None
Step 7	After setting MRE enable to Yes, check for availability and range of values for TE(T0) Change the frequency to 10 Hz and check the range of values Change the frequency back to 65 Hz and check the range of values	TE(T0) parameter is available, the echo time shall be 7 and shall have range from 7-100 This frequency change shall raise a conflict and shall modify the range to 1 - 100. The conflict shall resolve and the parameter shall have the range 7-100.	None
Step 8	After setting MRE enable to Yes, check for availability and range of values for Sign of 2nd gradient	Sign of 2nd gradient parameter is available and has auto , same sign and opposed sign as options	None

Step	Description	Expected Result	Attachments
Step 9	After setting MRE enable to Yes, check for availability and range of values for Amplitude	Amplitude parameter is available and has zero and maximum as options	None
Step 10	After setting MRE enable to Yes, check for availability and range of values for Enforce even length	Enforce even length parameter is available and has yes and no as options	None
Step 12	After setting MRE enable to Yes, check for availability and range of values for harmonic	harmonic parameter is available and has range from 1-10	None

3.1.21.10. SWCMP.Scan.MRE.UI.tc2

Test ID	25975
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:55 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:43 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:30 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Import the attached examcard <u>TS MRE with Phantom.examcard</u> Press 'New Exam' and load Inbox->TS MRE WITH PHANTOM ExamCard. Ensure option 36 is enabled For MRE protocol check for below parameters	N.A. (Preparation)	 TS MRE WITH PHANTOM.zip

Step	Description	Expected Result	Attachments
Step 1	Check for availability and range of MRE enable parameter under motion tab	MRE enable parameter is available and provides Yes and no as options	None
Step 2	After setting MRE enable to Yes, check for availability and range of values for Wave amplitude	Wave amplitude is available should provide low, moderate, high and user_defined as options	None
Step 3	After setting MRE enable to Yes, Select user_defined for Wave amplitude and check for availability and range of values for Percentage	Percentage is available and range is 0 to 100%	None
Step 4	After setting MRE enable to Yes, check for availability and range of values for Frequency	Frequency is available and range is 40 to 200Hz	None
Step 5	After setting MRE enable to Yes, check for availability and range of values for Direction	Direction is available and options are FH, AP, RL	None
Step 6	After setting MRE enable to Yes, check for availability and range of values for Gradient strength	Gradient strength is available and range is 0 to 100	None
Step 7	After setting MRE enable to Yes, check for availability and range of values for Patient experience scan	Patient experience scan is available and provides yes and no options	None

3.1.21.11. SWCMP.Scan.MRE.UI.tc3

Test ID	25284
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:18:58 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:44 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:32 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Edit c:\temp\bstrace.dat Add the following line (if it is not already present): GVE * * After adding the line, open a command window. Type the command logd. Check that the line GVE * * appears in the logd start up (see attachment). Start the GVE. (Start->MR Development->Scan Simulations->GVE) Display Object Names in the GVE. Unzip and import the attached examcard <u>TS MRE with Phantom.examcard</u> Ensure option36 is enabled Press 'New Exam' and set the gender of the patient to Phantom Load Inbox->TS MRE WITH PHANTOM ExamCard. Run Survey	N.A. (Preparation)	 logd_shows_bstrace_entry_GVE.zip  TS MRE WITH PHANTOM.zip
Step 1	>Adjust the scan MRE 4 SL. Check on the GEOMETRY tab that: slice orientation is coronal. fold over direction is RL. Check on the MOTION tab that: MRE direction is set to AP Set the Gradient strength to 5 [Motion->MRE enable->Gradient strength] Press 'Accept' to accept the changes. In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z.	The value of MRE direction can be changed to AP. Gradient strength can be set to 5 The MEG appears on the S and X axes as demonstrated in attached GVE screen grab. Absolute maximum value of the MEG gradients should be 5 Attach screen grab of GVE to report, include the date of the test in the filename of attachment.	 gve-meg-AP_slice_color.zip

Step	Description	Expected Result	Attachments
	Check that the motion encoding gradient (MEG) which is the combination GR`FFE_ELASTO[0] and GR`FFE_ELASTO[1] is on the expected axis the sequence frame(M,P,S) and physical frame (X,Y,Z).		
Step 2	>Adjust the scan MRE 4 SL. Check on the GEOMETRY tab that: slice orientation is coronal. fold over direction is RL. Change the MRE direction to RL. Set the Gradient strength to 10 [<i>Motion->MRE enable->Gradient strength</i>] Press 'Accept' to accept the changes. Run the scan. In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the motion encoding gradient (MEG) which is the combination GR`FFE_ELASTO[0] and GR`FFE_ELASTO[1] is on the expected axis the sequence frame(M,P,S) and physical frame (X,Y,Z).	The value of MRE direction can be changed to RL. Gradient strength can be set to 10 The MEG appears on the P and Y axes as demonstrated in attached GVE screen grab. Absolute maximum value of the MEG gradients should be 10 Attach screen grab of GVE to report, include the date of the test in the filename of attachment.	 gve-meg-LR_slice_c or.zip
Step 3	>Adjust the scan MRE 4 SL. Check on the GEOMETRY tab that: slice orientation is coronal. fold over direction is RL. Change the MRE direction to FH. Press 'Accept' to accept the changes. Run the scan In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the motion encoding gradient (MEG) which is the combination GR`FFE_ELASTO[0] and GR`FFE_ELASTO[1] is on the expected axis the sequence frame(M,P,S) and physical frame (X,Y,Z).	The value of MRE direction can be changed to FH. The MEG appears on the M and Z axes as demonstrated in attached GVE screen grab.	 gve-meg-FH_slice_c or.zip

Step	Description	Expected Result	Attachments
Step 4	>Adjust the scan MRE 4 SL. Check on the GEOMETRY tab that: fold over direction is RL. Change the slice orientation to transverse. Change the MRE direction to FH. Press 'Accept' to accept the changes. Run the scan In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the motion encoding gradient (MEG) which is the combination GR`FFE_ELASTO[0] and GR`FFE_ELASTO[1] is on the expected axis the sequence frame(M,P,S) and physical frame (X,Y,Z).	The MEG appears on the S and Z axes (as demonstrated in attached GVE screen grab for the slice orientation of transverse and MEG of FH).	 gve-meg-FH_slice_trans.zip
Step 5	>Adjust the scan MRE 4 SL. Check on the GEOMETRY tab that: fold over direction is RL. Check the slice orientation is transverse. Change the MRE direction to AP Press 'Accept' to accept the changes, and press "Start Scan". In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the MEG appears on the X axis for MEG direction of AP.	The MEG appears on the M and X axes.	None
Step 6	>Adjust the scan MRE 4 SL. Check on the GEOMETRY tab that: fold over direction is RL. Check the slice orientation is transverse. Change the MRE direction to RL. Press 'Accept' to accept the changes, and press "Start Scan". In the GVE, select Open PDF and open PDF.gve that was just created.	The MEG appears on the P and Y axes.	None

Step	Description	Expected Result	Attachments
	Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the MEG appears Y axis for MEG direction of LR.		
Step 7	>Adjust the scan MRE 4 SL. Change the slice orientation to sagittal. Change the fold over direction to AP (if not done by the system). Change the MRE direction to FH. Press 'Accept' to accept the changes. Run the scan. In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the motion encoding gradient (MEG) which is the combination GR`FFE_ELASTO[0] and GR`FFE_ELASTO[1] is on the expected axis the sequence frame(M,P,S) and physical frame (X,Y,Z).	The MEG appears on the M and Z axes (as demonstrated in attached GVE screen grab).	 gve-meg-FH_slice_sag.zip
Step 8	>Adjust the scan MRE 4 SL. Check the slice orientation is sagittal. Check the fold over direction is AP (if not done by the system). Change the MRE direction to AP. Press 'Accept' to accept the changes, and press "Start Scan". In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the MEG appears on the X axis for MEG direction of AP.	The MEG appears on the P and X axes.	None
Step 9	>Adjust the scan MRE 4 SL. Check the slice orientation is sagittal. Check the fold over direction is AP (if not done by the system).	The MEG appears on the S and Y axes.	None

Step	Description	Expected Result	Attachments
	Change the MRE direction to RL. Press 'Accept' to accept the changes, and press "Start Scan". In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. In the GVE, Change the layout to show M,P,S and X,Y,Z. Check that the MEG appears on the Y axis for MEG direction of RL.		

3.1.21.12. SWCMP.Scan.MRE.UI.tc4

Test ID	25285
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-11 11:19:01 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2016-02-10 11:46:46 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-10 11:36:35 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Set the system configuration by executing m_amn_set_test_enviroment.pl, select any configuration. Start Application Software from MR Boot Configuraiton Manager Edit c:\temp\bstrace.dat Add the following line (if it is not already present): GVE * *	N.A. (Preparation)	 logd_shows_bstrace_entry_GVE.zip  TS MRE FREQ.zip

Step	Description	Expected Result	Attachments
	After adding the line, open a command window. Type the command logd. Check that the line GVE * * appears in the logd start up (see attachment). Start the GVE. (Start->MR Development->Scan Simulations->GVE) Display Object Names in the GVE. Unzip and import the attached examcard <u>TS MRE with Phantom.examcard</u> Ensure option36 is enabled Press 'New Exam' and set the gender of the patient to Female Load Inbox->TS MRE FREQ ExamCard. Run Survey		
Step 1	>Adjust the scan MRE 4 SL. On the MOTION tab: Attempt to enter an MRE frequency < 40 Hz or > 200 Hz. Check that attempts to enter a value outside of this range shall not be accepted by the UI and the value is changed to the nearest end point value.	The entry of a value outside the range of 40.0 to 200.0 Hz will result in the value being set to the closest end point.	None
Step 2	>Adjust the scan again. Check the MRE Direction is set to AP and the slice orientation is coronal. Set the MRE frequency to 40 Hz. Press 'Accept' to accept the changes. Run the scan. In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. Check that the FFE_elasto[0] (two) and FFE_elasto[1] objects appear on the slice axis. Use the measurement tool to verify the duration of the gradient objects FFE_elasto[0] + FFE_elasto[1] + FFE_elasto[0]. <i>An example of the</i>	In the GVE, The sequence plot has the FFE_elasto[0] and FFE_elasto[1] objects on the S (selection) axis. The duration of the three gradient lobes is 25.0 ms +/- 0.1 ms.	 MEG-40Hz_in_GVE.zip

Step	Description	Expected Result	Attachments
	<i>measurement in the GVE has been attached.</i> The expected gradient duration is $1/(40 \text{ Hz}) = 25.0 \text{ ms}$.		
Step 3	>Adjust the scan again. Set the MRE frequency to 200 Hz. Press 'Accept' to accept the changes.Run the scan In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. Check that the FFE_elasto[0] (two) and FFE_elasto[1] objects appear on the slice axis. <i>An example of the measurement in the GVE has been attached.</i>	In the GVE, the sequence plot has the FFE_elasto[0] and FFE_elasto[1] objects on the S (selection) axis. The duration of the three gradient lobes is 5.0 ms +/- 0.1 ms.	 MEG-200Hz_in_GVE.zip
Step 4	>Adjust the scan again. Set the MRE frequency to 60 Hz. Press 'Accept' to accept the changes. Run the scan In the GVE, select Open PDF and open PDF.gve that was just created. Select sequence base for display. Check that the FFE_elasto[0] (two) and FFE_elasto[1] objects appear on the slice axis. <i>An example of the measurement in the GVE has been attached.</i>	In the GVE, the sequence plot has the FFE_elasto[0] and FFE_elasto[1] objects on the S (selection) axis. The duration of the three gradient lobes is 16.7 ms +/- 0.1 ms.	 MEG-16p67_in_GVE.zip

3.1.22. ME MSP

3.1.22.1. SWCMP.Scan.MSP.behavior.tc1

Test ID	7401
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved.

	This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Test basic Functionality with Manual Scan Pause and count down teller on RTM

Requirements Coverage:

Req ID	Requirement Tag	Description
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Step 1	Start Head CNS / TSE / F_singleshot	Verify that the scan starts
Step 2	Push the "Pause scan" button.	Verify that the scan is paused within 3 seconds.
Step 3	Push the F8 key.	Verify that the scan completes
Step 4	Copy Head CNS / TSE / F_singleshot scan, change the TR to 10000 ms and start the scan	Verify that the "Start scan" button is replaced by the "Pause scan" button after the preparation phase is performed.
Step 5	Push the "Pause scan" button.	1. Verify that an info message appears with a count down teller starting with 10 seconds. 2. Verify that the count down teller counts down and the scan is paused. 3. Verify that the "Pause scan" button is replaced by a "Resume scan" button. 4. Verify that a pop up window appears that allows you to proceed the scan.
Step 6	Resume the scan pushing the F8 key.	Verify that the scan resumes.
Step 7	Again push the F8 key to pause the scan.	1. Verify that an info message appears with a count down teller starting with 10 seconds. 2. Verify that the count down teller counts down and the scan is paused.
Step 8	Push F8 key to start the scan	The scan starts and completes successfully

3.1.22.2. SWCMP.Scan.MSP.behavior.tc2.v2

Test ID	32550
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-27 17:21:35 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-27 11:58:53 Meaning: SME Approved - (Technical) Signed by: Visser, Kees (nly22601) on 2017-03-24 08:12:25 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Use testbay

Requirements Coverage:

Req ID	Requirement Tag	Description
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Prep	create a new current log file	n/a (preparation)
Step 1	Start Head CNS / CARD / G_2D4_das	Verify that the patient communication device is disabled.
Step 2	Push the "Pause scan" button.	1. Verify that the scan is paused. 2. Verify that the patient communication is enabled.
Step 3	Pause and resume the scan several times and finally let the scan complete	Verify that the scan will be pause and resumed upon request and finally completes.
Step 4	As Objective Evidence, attach the log file to the test run.	n/a (administration)

Step	Description	Expected Result

3.1.22.3. SWCMP.Scan.MSP.behavior.tc3

Test ID	7403
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Use testbay

Requirements Coverage:

Req ID	Requirement Tag	Description
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Step 1	1. Start Head CNS / CARD / G_2D1_1hp 2. Open the RF door.	Verify that the scan is aborted.
Step 2	1. Again start Head CNS / CARD / G_2D1_1hp 2. Pause the scan 3. Open the RF door	Verify that the scan is not aborted.
Step 3	1. Close the RF door 2. Resume the scan	Verify that the scan completes

3.1.22.4. SWCMP.Scan.MSP.behavior.tc4

Test ID	7404
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Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Use testbay

Requirements Coverage:

Req ID	Requirement Tag	Description
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Step 1	1. Start Head CNS / CARD / G_2D1_arrh70. 2. Pause the scan. 3. Enter the MR suit and reposition the table.	Verify that the scan is aborted

3.1.22.5. SWCMP.Scan.MSP.behavior.tc5

Test ID	7405
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Use testbay

Requirements Coverage:

Req ID	Requirement Tag	Description
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Step 1	1. Start Head CNS / CARD / G_2D1_arrh70. 2. Pause the scan.	Verify that the started scan is paused.
Step 2	1. Enter the MR suit and close the RF door. Wear ear protection for this test! 2. Push the start PICU button.	Verify that the scan resumes.
Step 3	Push the PICU start button.	Verify that the scan is paused.
Step 4	1. Leave the MR suit. 2. Close the RF door. 3. Push the F8 button	The scan resumes and completes.

3.1.22.6. SWCMP.Scan.MSP.behavior.tc6

Test ID	7406
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Use testbay

Requirements Coverage:

Req ID	Requirement Tag	Description
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and

Req ID	Requirement Tag	Description
		the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Step 1	1. Start Head CNS / PROF_ORDER / i1_3D_CE_3st. 2. Pause the scan while the table is moving to the final position.	Verify that the scan is not paused.
Step 2	When, after table movement, the scan continues click the Scan Pause button When the scan has stopped, click the "Resume scan" button.	Verify that the scan pauses, resumes and finishes in the normal way.

3.1.22.7. SWCMP.Scan.MSP.IQ.tc1

Test ID	7398
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Use testbay and head phantom

Requirements Coverage:

Req ID	Requirement Tag	Description
18089	SWCMP.Scan.MSP.IQ	Using the manual scan functionality should not degrade the IQ.

Test steps:

Step	Description	Expected Result
Step 1	1. Start HeadCNS \ TSE \ F_basic_TSE 2. Pause the scan after about 30% and after about 60% scan progress. 3. After scan completion start this scan again but do not	Verify that image quality of the scan that has not been paused and resumed is equivalent to the scan that has been paused and resumed.

Step	Description	Expected Result
	pause this scan.	

3.1.22.8. SWCMP.Scan.MSP.Restrictions.tc1

Test ID	7407
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Use testbay

Requirements Coverage:

Req ID	Requirement Tag	Description
18086	SWCMP.Scan.MSP.Restrictions	The manual scan pause functionality is not available in combination with the features mention in section Restrictions
65797	SWCMP.Scan.MSP.behavior.v2	If the scan is paused the patient communication device is enabled and the scanner room can be entered. Note that moving the patient table leads to a scan abort. In case that the scan cannot be paused within the 2-3 sec requirement, a count down teller indicates the time to pause the scan. The user always can stop the scan if "pausing" the scan takes too long using the conventional "stop scan" functionality. The scan is continued by pushing the "Resume Scan" button. In case that the scan is proceeded the patient communication device is disabled, the RF door must be closed and the scan is resumed after some startup cycles.

Test steps:

Step	Description	Expected Result
Step 1	Start the int_fast_mobiflex scan	Verify that the 'Start scan' button is changed into the 'Next scan' button and that the F8 button also behaves like the 'Next scan button'. Hence Pause scan is not possible.
Step 2	Start an interactive scan	Verify that the 'Start scan' button is changed into the 'Next scan' button and that the F8 button also behaves like the 'Next scan button'. Hence Pause scan is not possible.
Step 3	Start a breathhold scan	Verify that the 'Start scan' button is changed into the 'Next scan' button and that the F8 button also

Step	Description	Expected Result
		behaves like the 'Next scan button'. Hence Pause scan is not possible.

3.1.22.9. SWCMP.Scan.MSP.UI.tc1.v2

Test ID	33110
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-13 14:51:09 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Visser, Kees (nly22601) on 2017-04-12 09:43:30 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-12 09:00:34 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Test basic Functionality without Manual Scan Pause on RTM

Requirements Coverage:

Req ID	Requirement Tag	Description
18087	SWCMP.Scan.MSP.UI	The manual scan pause UI is specified in section UI under General Requirements.

Test steps:

Step	Description	Expected Result
Step 1	Log onto the scanner. Start the Application software. Start a new patient. Disable the Manual Scan Pause application by disabling the "Scan Pause" CSC parameter in the control parameter environment (Scan Pause is No).	Verify CSC-parameter can be set to no.
Step 2	Start HeadCNS \ FE \ F_2D1 Attach logfile as objective evidence to this test step	1. Verify that the "Start scan" button is not replaced by the "Pause scan" button. 2. Verify that pushing the F8 key does not lead to a pause of the scan. 3. Verify that pushing the "stop scan" button once leads to stop of the scan (recon may continue).

3.1.22.10. SWCMP.Scan.MSP.UI.tc2.v2

Test ID	33111
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-13 14:51:10 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Visser, Kees (nly22601) on 2017-04-12 09:44:14 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-12 09:02:04 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Test basic Functionality with Manual Scan Pause on RTM

Requirements Coverage:

Req ID	Requirement Tag	Description
18087	SWCMP.Scan.MSP.UI	The manual scan pause UI is specified in section UI under General Requirements.

Test steps:

Step	Description	Expected Result
Step 1	Create new logfile. Start application software. Enable the Manual Scan Pause application by enabling the "Manual Scan Pause" CSC parameter in the control parameter environment (Manual Scan Pause is Yes)	Verify CSC-parameter can be changed to yes.
Step 2	Start Head CNS \ FE \ F_2D1	Verify that the "Start scan" button is replaced by the "Pause scan" button after the preparation phase is performed.
Step 3	Push the "Pause scan" button.	1. Verify that the scan is paused within 3 seconds. 2. Verify that the "Pause scan" button is replaced by a "Resume scan" button. 3. Verify that a pop up window appears that allows you to proceed the scan.
Step 4	Resume the scan pushing the "Resume scan" button.	Verify that the scan is resumed.
Step 5	Again push the F8 key.	1. Verify that the scan is paused within 3 seconds. 2. Verify that the "Pause scan" button is replaced by a "Resume scan" button. 3. Verify that a pop up window appears that allows you to proceed the scan.
Step 6	Push the "Proceed" from the proceed pop up window to resume scanning.	Verify that the scan is resumed.
Step 7	Push the F8 key to pause the scan.	Verify that the scan is paused.
Step 8	Again push the F8 key and wait for scan completion.	Verify that the scan is resumed.

Step	Description	Expected Result
Step 9	Again start the scan. Pause the scan. Push the "stop scan" button. Attach logfile as objective evidence to this test step.	Verify that pushing the "stop scan" button once leads to stop of the scan (reconstruction may continue).

3.1.23. ME NearMetal

3.1.23.1. SWCMP.Scan.NearMetal.Combinations.tc1.v2

Test ID	35603
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-10-03 10:43:49 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2017-10-02 16:24:12 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-10-02 16:22:06 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

UI test using Exam Cards environment on a VM, which verifies parameter combinations and conflicts for SEMAC+VAT and MARS.

Test steps:

Step	Description	Expected Result
Prep 1	- Select tedb - Create a new log file - Open Exam Cards Environment - Select Hospital/Pelvis/NearMetal/T2W_TSE_Semac_weak	N.A. Preparation steps
Step 2	In the protocol T2W_TSE_Semac_weak verify the visibility of the parameter "O-MAR" on the Geometry tab in combination with the following parameters: "Scan technique" other than "SE" or "IR" "Fast Imaging Mode" other than "None" or "TSE"	- The parameter "O-MAR" is not visible in combination with "Scan technique" other than "SE" or "IR" - The parameter "O-MAR" is not visible in combination with "Fast Imaging Mode" other than "None" or "TSE"

Step	Description	Expected Result
Step 3	Verify the range of the parameter "RF shim" with the parameter "O-MAR" other than "No"	The range of the parameter "RF shim" in combination with "O-MAR" other than "No" is "Fixed" only
Step 4	In the protocol T2W_TSE_Semac_weak verify the following parameter combinations: "MARS+VAT" or "SEMAC+VAT" in combination with "water-fat-shift" equal to "maximum" "MARS+VAT" or "SEMAC+VAT" in combination with "Uniformity Correction" on PostProc tab "SEMAC+VAT" in combination with "Halfscan" in z "SEMAC+VAT" in combination with "TSE 3D non-selective" "SEMAC+VAT" in combination with "3D view MSK T1" "SEMAC+VAT" in combination with "Scan mode" other than "3D" "SEMAC+VAT" in combination with "mDixon" other than "No" "SEMAC+VAT" in combination with "fat suppression/water suppression" equal to "ProSet" Ensure that the posterior coil is selected, e.g. select 'Smart Select'. Recon compression is only enabled with a multi channel coil. "SEMAC+VAT" in combination with "Recon compression"	- A conflict appears between "O-MAR" and "water-fat_shift" - A conflict appears between "O-MAR" and "Uniformity correction" - A conflict appears between "O-MAR" and "Halfscan" - A conflict appears between "O-MAR" and "TSE 3D non-selective" - A conflict appears between "O-MAR" and "3D view" after resolving the conflict on startup echoes - A conflict appears between "O-MAR" and "Scan mode" - A conflict appears between "O-MAR" and "mDixon" - A conflict appears between "O-MAR" and "Fat Suppression/Water Suppression" - The parameter "Recon compression" does not show the option "No".
Step 5	- Open the protocol T2w_TSE_Semac_weak for edit. - On the Geometry page, verify the displayed range of the parameter O-MAR. - Repeat the test for changing values of the Scan Control Parmeter "Enable VAT" to "No", "SEMAC only" and "MARS only". - Attach log file as objective evidence.	- Scan control Parameter "Enable VAT" set to "Both": The parameter O-MAR has the range "NO", "MARS+VAT", "SEMAC+VAT" - Scan control Parameter "Enable VAT" set to "No": The parameter O-MAR has the range "NO", "MARS", "SEMAC" - Scan control Parameter "Enable VAT" set to "SEMAC only": The parameter O-MAR has the range "NO", "MARS", "SEMAC+VAT" - Scan control Parameter "Enable VAT" set to "MARS only": The parameter O-MAR has the range "NO", "MARS+VAT", "SEMAC"

3.1.23.2. SWCMP.Scan.NearMetal.Pars.tc1

Test ID	22361
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-02-10 12:55:18 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Dingemans, Henk (nly95779) on 2015-02-09 16:05:36 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2015-02- 09 12:56:34 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:**Test steps:**

Step	Description	Expected Result
Step 1	In the protocol Pelvis\NearMetal\T2w_TSE_Semac_weak go to the geometry page verify the visibility and range of the parameter "O-MAR".	The parameter "O-MAR" is visible with range • No • MARS+VAT • SEMAC+VAT
Step 2	Select "O-MAR" equal to "SEMAM+VAT" and verify the visibility and range of the parameter "distortion correction".	The parameter "distortion correction" is visible with the range: • weak • medium • strong

3.1.23.3. SWCMP.Scan.NearMetal.Reggression.tc1

Test ID	22360
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-02-10 12:55:20 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-09 16:05:38 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2015-02- 09 12:56:36 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This regression test verifies the metal functionality using a batchinterpreter script. The script contains the following test scans which are derived from the scans in tedb/pelvis/NearMetal/.

- - o T2W_Semac_VAT: A SEMAC scan with VAT
 - o T2W_Semac_weak: A SEMAC scan with weak distortion correction
 - o T2W_Semac_med: A SEMAC scan with medium distortion correction
 - o T2W_Semac_strong: A SEMAC scan with strong distortion correction
 - o T2W_Semac_weak_7: A SEMAC scan with weak distortion correction and a frequency range of 7kHz
 - o T2W_Semac_med_12: A SEMAC scan with medium distortion correction and a frequency range of 12kHz
 - o T2W_Semac_strong_18: A SEMAC scan with strong distortion correction and a frequency range of 18kHz
 - o STIR_TSE_Semac: A STIR SEMAC scan with weak distortion correction and a frequency range of 7kHz
 - o STIR_TSE_Mars: A STIR MARS scan

Specific sequence parameters are logged using the enabled metal_test trace channel and object logging. They are filtered from the logfile and stored in the file G:\log\m_imn_metal.dat. The following test cases are verified:

- VAT gradients
- VAT acquisition frequency
- SEMAC off-resonance suppression
- B0 shim
- SEMAC RF shape
- SEMAC number of z-encodings
- SEMAC factor
- STIR bandwidth

The results of the tests including the final pass/fail statement are written in the logfile G:\log\m_imn_metal_compare.log.

Test steps:

Step	Description	Expected Result
Step 1	On a SDE VMWare, install tedb and copy m_imn_metal.pl and m_imn_metal.acq from your_view\mrmmethods\imaging\tst\ to G:\patch. Run a scan in ExamCard to ensure that the object logging files are created in G:\Temp and object logging can be enabled by the script. Run the perl script m_imn_metal.pl in the command window. This script will automatically run a set of test cases.	Preparation only
Step 2	Once the regression test is finished, open the file G:\log\m_imn_metal_compare.log.	On the top of the regression logfile verify that the correct date and time, the intended software stream and SWID and used input files are logged.
Step 3	Inspect the logfile for the test: Timing of readout and VAT gradients	Verify that the actual and expected values are equal or within an indicated precision range.
Step 4	Inspect the logfile for the test: Check AQ frequency	Verify that the actual and expected values are equal or within an indicated precision range.

Step	Description	Expected Result
Step 5	Inspect the logfile for the test: Check SEMAC ORS	Verify that the actual and expected values are equal or within an indicated precision range.
Step 6	Inspect the logfile for the test: B0 shim test	Verify that the actual and expected values are equal or within an indicated precision range.
Step 7	Inspect the logfile for the test: RF shape test	Verify that the actual and expected values are equal or within an indicated precision range.
Step 8	Inspect the logfile for the test: SEMAC sequence test	Verify that the actual and expected values are equal or within an indicated precision range.
Step 9	Inspect the logfile for the test: SEMAC factor test	Verify that the actual and expected values are equal or within an indicated precision range.
Step 10	Inspect the logfile for the test: METAL STIR bandwidth test	Verify that the actual and expected values are equal or within an indicated precision range.

3.1.24. ME NonCartesian

Attachments:



3D radial stack of
stars ExamCard.zip

3.1.24.1. SWCMP.Scan.Radial.Combinations.tc1.v2

Test ID	62375
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-06-16 13:42:50 Meaning: QA Approved - (QA) Signed by: Gorkum, Robbert van (320189046) on 2025-06-16 13:06:58 Meaning: SME Approved - (Technical) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-06-16 11:17:07 Meaning: SME Approved - (Business)

Approval Status	Approved
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Description:

The purpose of this test is to test protocol parameter combinations of 3D radial stack of stars. This test is a pdf test automation test. The test case tests functional incompatibility that results in combination conflicts. The most important functional compatibilities as specified in the requirement are tested in TEIMA.

Requirements Coverage:

Req ID	Requirement Tag	Description
98071	SWCMP.Scan.Radial.combinations.v1	The Radial functionality shall be compatible with: • mDIXON: no, yes • all slice orientations • RF select. FOS: no, yes • multi-stack with move table per stack combination • Prepulses: REST slabs, TFE prepulse, MTC, T2prep, MSDE prepulses* • Respiratory compensation: no, breath hold, retrospective • NSA ≥ 1 • Dynamic study: individual, block *For retrospective respiratory compensation, the supported prepulses are TFE Prepulse and REST slabs.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanRadialCombinations in the results file is zero.

3.1.24.2. SWCMP.Scan.Radial.Par.tc1

Test ID	41065
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:56 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:46:55 Meaning: SME Approved - (Business) Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 10:50:40 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the availability of Radial parameters.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanRadialPar in the results file is zero.

3.1.24.3. SWCMP.Scan.Radial.Technique.tc1.v2

Test ID	52735
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:26:14 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:03:26 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:26 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the 3D radial stack of stars technique.

Requirements Coverage:

Req ID	Requirement Tag	Description
55959	SWCMP.Scan.Radial.method	3D inplane radial sampling shall be available for FFE and TFE scan techniques with cartesian outplane sampling.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanRadialMethod in the results file is zero.

3.1.25. ME PrePhases

3.1.25.1. SWCMP.Scan.PrepPhases.AS.tc1

Test ID	7949
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

AS phase: Automatic Shimming

AS Volume selection and slice selection

Requirements Coverage:

Req ID	Requirement Tag	Description
64304	SWCMP.Scan.PrepPhases.AS.v1	The Automatic Shimming (AS) preparation phase improves the local magnetic field homogeneity by adjustment of linear gradients. The linear gradient errors shall be measured locally. The derived linear shim corrections are incorporated in the gradient waveforms. The duration of the coarse AS preparation phase on the DAS shall not exceed 11000 msec per volume (stack).

Test steps:

Step	Description	Expected Result
Step 1	Verify that the preparation phase indeed operates on the intended volume or slice by putting a phantom in an offcentre position in the bore (AS should succeed) and then moving the phantom to the isocentre.	Predicted outcome: autoshim echo signals should be down at least 10 (inspect using the monitoring viewer).

3.1.25.2. SWCMP.Scan.PrepPhases.AS.tc2

Test ID	7950
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this

	item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

AS phase: Automatic Shimming

AS Shim loops

Requirements Coverage:

Req ID	Requirement Tag	Description
64304	SWCMP.Scan.PrepPhases.AS.v1	The Automatic Shimming (AS) preparation phase improves the local magnetic field homogeneity by adjustment of linear gradients. The linear gradient errors shall be measured locally. The derived linear shim corrections are incorporated in the gradient waveforms. The duration of the coarse AS preparation phase on the DAS shall not exceed 11000 msec per volume (stack).

Test steps:

Step	Description	Expected Result
Step 1	Run a 3-stack SPIR scan with 'Shim' set to 'auto'.	Verify that monitoring displays the signals as described in RS: Preparation Phases in the "Automatic Shimming (AS phase): Monitoring" chapter: for each stack A, B, C, there should be 2 iterations, each of which over 3 orientations x, y, and z. (These orientations may occur in any order, e.g. z, y, x.)

3.1.25.3. SWCMP.Scan.PrepPhases.AS.tc3

Test ID	7951
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

AS phase: Automatic Shimming

AS Predictability

Requirements Coverage:

Req ID	Requirement Tag	Description
64304	SWCMP.Scan.PrepPhases.AS.v1	The Automatic Shimming (AS) preparation phase improves the local magnetic field homogeneity by adjustment of linear gradients. The linear gradient errors shall be measured locally.

Req ID	Requirement Tag	Description
		The derived linear shim corrections are incorporated in the gradient waveforms. The duration of the coarse AS preparation phase on the DAS shall not exceed 11000 msec per volume (stack).

Test steps:

Step	Description	Expected Result
Step 1	Select protocol 'tedb\Head CNS\BFFE\A6_3D_pe'. Set the FOV to 400, 'Preparation phases' to 'full', 'Shim' to 'volume', 'Scan mode' to '2D', and TE to 20 ms. Place a 3 liter bottle upright in the scanner. Change a shim component by adjusting the preset gradient offsets (add e.g. 0.1 to the value of the tuned hardware parameter: 'grad. Shims [mT/m]').	Verify that automatic shimming compensates for this.

3.1.25.4. SWCMP.Scan.PrepPhases.AS.tc4

Test ID	7952
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

AS phase: Automatic Shimming

AS Localization

Select protocol 'tedb\Head CNS\BFFE\A6_3D_pe'. Set the FOV to 400, 'Preparation phases' to 'full', 'Shim' to 'volume', 'Scan mode' to '2D', and TE to 20 ms. Place a 3 liter bottle upright in the scanner.

Requirements Coverage:

Req ID	Requirement Tag	Description
64304	SWCMP.Scan.PrepPhases.AS.v1	The Automatic Shimming (AS) preparation phase improves the local magnetic field homogeneity by adjustment of linear gradients. The linear gradient errors shall be measured locally. The derived linear shim corrections are incorporated in the gradient waveforms. The duration of the coarse AS preparation phase on the DAS shall not exceed 11000 msec per volume (stack).

Test steps:

Step	Description	Expected Result
Step 1	Position the shim volume in the bottom half of the bottle. Run the scan	Verify that this part of the bottle is indeed optimized (contains fewer bFFE artifact lines than the top part).

Step	Description	Expected Result
Step 2	Position the shim volume in the top half of the bottle. Run the scan	Verify that this part of the bottle is optimized.

3.1.25.5. SWCMP.Scan.PrepPhases.CD.tc1

Test ID	7948
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

CD phase: Channel Delay determination

Run the tests on a bottle (not the head phantom). In the "Control Parameter Editor", enable expert mode and switch on the CD preparation phase, as it is normally off. Also enable 'CD enable logging'.

Requirements Coverage:

Req ID	Requirement Tag	Description
18820	SWCMP.Scan.PrepPhases.CD	The Channel Delay preparation phase calibrates delays and phase shifts in the gradients as specified in the CD chapter

Test steps:

Step	Description	Expected Result
Step 1	Measure the delays of the 3 gradient channels and write them down.	
Step 2	Set the 'HW_grad_delay_x','HW_grad_delay_y','HW_grad_delay_z' parameters to slightly higher values, choosing different increments for each channel, e.g. 0.004, 0.003, -0.001. Avoid HW parameter settings where the difference between the fastest and the slowest channel exceeds 0.0064ms, since the system won't scan with these settings. Press 'Apply', but do not store! Run the preparation phase.	Each positive/negative increment of the HW parameter should be reflected in a corresponding positive/negative delay found by the preparation phase.
Step 3	Verify with the monitoring viewer: signals from 3 echoes must be shown for each of the 3 directions x, y, z, together with their FFTs.	
Step 4	Set the hardware parameters to the determined delays. Repeat the preparation phase	Now the delays should be zero.

Step	Description	Expected Result

3.1.25.6. SWCMP.Scan.PrepPhases.CL.exe_criteria.tc1.v2

Test ID	34909
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2017-06-22 14:28:13 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-06-21 14:26:56 Meaning: SME Approved - (Technical) Signed by: Visser, Kees (nly22601) on 2017-06-21 14:26:15 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The VM will be configured as Ingenia 3T (WA30) with a DDAS for reproducibility reason. It imports cl_prep_exe_criteria.ExamCard. It also prepares a batch file to put the posterior coil in holdback. During testing, the tester is asked to stop and restart scannersim. This simulates a reboot of the DAS after which a calibration of the control loop is necessary.

Requirements Coverage:

Req ID	Requirement Tag	Description
45449	SWCMP.Scan.PrepPhases.CL.exe_criteria	The following conditions must be satisfied in order for this preparation phase to be performed: • HW fixed parameter 'RF-Pulse control loop mode' must be set to other than 'OFF' for all applicable channels. • The control loop preparation phase must be enabled in the preparation control parameters. • The RF amplifier must be in POWERMODE_HIGH The following conditions will trigger an execution of the control loop calibration: When no previous result of the calibration is available, i.e.: • A new patient ref id is created • The patient table position/orientation changes • A different transmit coil is connected. • After a DAS boot. • Prep phase monitoring is set to 'ON'

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Use VM installed as "atlsystemp".	N/A (preparation)	
	Login and run the script		

Step	Description	Expected Result	Attachments
	m_amn_set_test_environment.pl and choose 3.0T Ingenia. Copy the the attached file CL_prep_exe_criteria.zip to the VM and unzip with right mouse button (7-zip) to E:\ Double click the extracted ExamCard, to import it.		 CL_prep_exe_criteria.zip
Prep 2	Start the Examcards Scan Environment and create a new examination. Drag the ExamCard "Other/InBox/cl_prep_exe_criteria' into the scan list.	N/A (preparation)	None
Prep 3	Start the first scan 'CL BODYT scout'. Press 'proceed' when asked to reuse the current patient position. Accept tabletop movement when asked.	N/A (preparation)	None
Prep 4	Before scan 'CL DAS reboot', the examcard is paused. Stop and restart scannersim by starting the simulation mode (Acq on Host, Recon on Host). After scannersim has been restarted, Continue the scan	N/A (preparation)	None
Prep 5	Before scan 'CL cal. off', the examcard is paused. From the "Edit Scan Control Parameters...", tab "Preparation", set "CL Calibration" to "off" using 'Apply' to save the settings (not 'Store').	N/A (preparation)	None
Prep 6	Before scan 'CL TR Head', the examcard is paused. From the "Edit Scan Control Parameters...", tab "Preparation", set "CL Calibration" to "on" using 'Apply' to save the settings (not 'Store'). Open a commandbox, go to E:\ and run: copy e:\pc_in_holdback.tacq g:\patch\ "stt.exe -b pc_in_holdback.tacq" This will put the posterior coil in holdback. Start the coilsimulator and add a Head_TR coil to the TRC connector. Continue the scan	N/A (preparation)	None

Step	Description	Expected Result	Attachments				
Prep 7	Before scan 'CL copy2 BODYT scout', the examcard is paused. Use the coilstate simulator to remove the Head_TR coil from the TRC connector Continue the scan	N/A (preparation)	None				
Prep 8	Before scan 'CL monitoring on', the examcard is paused. From the "Edit Scan Control Parameters...", tab "Preparation", set "CL Calibration" to "monitoring on" using 'Apply' to save the settings (not 'Store'). Continue the scan	N/A (preparation)	None				
Prep 9	Before scan 'CL ch1 mode OFF', the examcard is paused. Select 'system tuning...' by right clicking in the examcard environment. In the 'Fixed Hardware' tab, set RF-Pulse control loop mode' to 'OFF' for the first channel only using 'Apply' to save the settings (not 'Store'). Continue the scan	N/A (preparation)	None				
Prep 10	Before scan 'CL ch2 mode OFF', the examcard is paused. In the 'Fixed Hardware' tab, set: - RF-Pulse control loop mode' to 'AMP_AND_PH' for the first channel. - RF-Pulse control loop mode' to 'OFF' for the second channel using 'Apply' to save the settings (not 'Store'). Continue the scan	N/A (preparation)	None				
Prep 11	Before scan 'CL mode phase / ampl', the examcard is paused. In the 'Fixed Hardware' tab, set: - RF-Pulse control loop mode' to 'PHASE' for the first channel. - RF-Pulse control loop mode' to 'AMPLITUDE' for the second channel using 'Apply' to save the settings (not 'Store'). Continue the scan	N/A (preparation)	None				
Step 12	After the scan has finished: Open the logging in the LogBrowser and open G:\log\devlogcurrent.log. Use 'view' -> 'Filter query..' with the following expression to get an overview of the results: text contains 'm_Name' or text contains 'CL: performing' or text contains 'CL:	The following results should be found: <table><tr><th>Scan</th><th>CL calibration</th></tr><tr><td>"CL BODYT scout"</td><td>"CoilSurveyScan" performed for all</td></tr></table>	Scan	CL calibration	"CL BODYT scout"	"CoilSurveyScan" performed for all	None
Scan	CL calibration						
"CL BODYT scout"	"CoilSurveyScan" performed for all						

Step	Description	Expected Result		Attachments
	prep phase' or text contains 'CL: control loop' or text contains 'CL: calibration' The first scan is a simple scout scan with an in initial CoilSurvey scan. The CoilSurvey conducts a CL calibration since no CL results are available. The 'CL BODYT scout' can reuse the calibration results from the CoilSurvey scan and therefore should skip the calibration. 'CL copy1 BODYT scout' is just a copy of 'CL BODYT scout', therefore the scan can reuse the CL results and the CL phase should be skipped. 'CL table pos.change" moves the patient table. A patient position change should trigger the CL calibration. 'CL org table pos' moves the patient table back to its original position. Since the CL result at that position can be reused, the CL calibration should be skipped. 'CL DAS reboot' reboots the DAS. A DAS boot should trigger a rerun of the CL calibration 'CL cal. off' is run with CL calibration off, therefore the CL phase should be skipped. 'CL TR Head' uses a different T/R coil, therefore the CL results cannot be reused. 'CL copy2 BODYT scout' is a copy of the first scan, therefore the scan can reuse the CL results and the CL phase should be skipped. 'CL monitoring on' uses 'monitoring on' for CL phase. 'Monitor on' should trigger the CL calibration while normally the results from a previous scan can be reused. In 'CL ch1 mode OFF', the control loop is forced to 'off' for the first channel. In 'CL ch2 mode OFF', the control loop is forced to 'off' for the second channel. 'CL mode phase / ampl' changes HW tuned control loop settings. A CL calibration should be triggered for all setting other than 'OFF'.		available channels.	
		"CL BODYT scout"	prep phase is skipped, reusing results	
		"CL copy1 BODYT scout"	prep phase is skipped, reusing results	
		"CL table pos.change"	"CoilSurveyScan" performed for all available channels.	
			"CL BODYT scout" prep phase is skipped, reusing results	
		"CL org table pos."	prep phase is skipped, reusing results	
		"CL DAS reboot"	"CoilSurveyScan" performed for all available channels.	
			"CL DAS reboot" prep phase is skipped, reusing results	
		"CL cal. off"	control loop prep phase disabled	
		"CL TR Head"	performed for all available channels.	

Step	Description	Expected Result		Attachments
		"CL copy2 BODYT scout"	"CoilSurveyScan" prep phase is skipped, reusing results "CL copy2 BODYT scout" prep phase is skipped, reusing results	
		"CL monitoring on"	performed for all available channels.	
		"CL ch1 mode OFF"	performed for all available channels, except channel 1.	
		"CL ch2 mode OFF"	performed for all available channels, except for channel 2.	
		"CL mode phase / ampl"	performed for all available channels.	

3.1.25.7. SWCMP.Scan.PrepPhases.CL.post_conditions.tc1.v1

Test ID	26279
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-03-03 11:29:04 Meaning: QA Approved - (QA) Signed by: Geraedts, Tom (nly22893) on 2016-03-03 09:49:03 Meaning: SME Approved - (Business)

	Signed by: Graaf, Ruud de (nly95749) on 2016-03-03 09:20:35 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

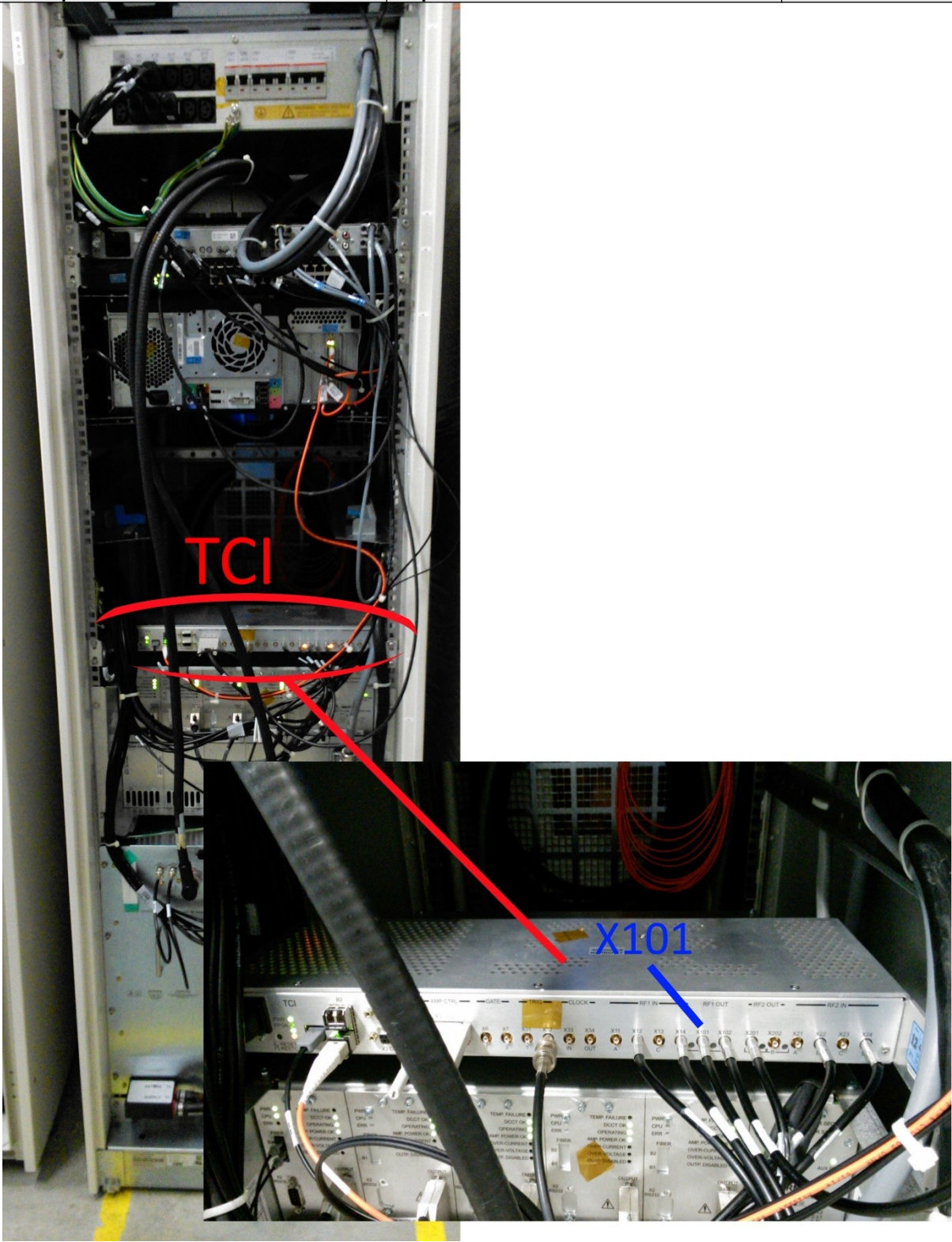
This test needs to be performed on a 3T TB with a DDAS / TCI and watercooled RF amplifier installed.

Requirements Coverage:

Req ID	Requirement Tag	Description
45450	SWCMP.Scan.PrepPhases.CL.post_conditions	If the CL calibration preparation phase detects one of the following conditions, it will log a warning message and continue the scan with the control loop disabled: 1. Too low or too high gain signal during delay measurement. 2. Too short or too long calculated delay. 3. Too low or too high gain signal during gain measurement. 4. RF amplifier in wrong mode (LOW mode) When the prep. phase is called from the STT interface: In case of one of the following conditions, a warning message will be logged and the scan continues with the control loop disabled: 1. RF amplifier in wrong mode (LOW mode). 2. Illegal transmit channel number requested.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Start the Examcard environment and create a new patient.	N/A (preparation)	None
Prep 2	Remove the cable from X101 on the TCI and leave the cable disconnected (see attachment on where to locate the cable).	N/A (preparation)	

Step	Description	Expected Result	Attachments
			
Prep 3	Copy the attached file 'CLtest.zipx' to E:\ and unzip this file (right click, select 'Extract to "CLtest"'), Double click E:\CLtest\CLtest.pl; please wait until command box closes.	N/A (preparation)	 CLtest.zip

Step	Description	Expected Result	Attachments
	This script will: - import the examcard - create a new logfile Put the 3L bottle phantom vertically in the iso centre.		
Step 4	Drag the ExamCard "Other/InBox/ cltest" into the scan list. Start the scan.	The scan aborts. The logging should mention that the gain is too low during the CL phase and that calibration failed for channel 1. The CL should be disabled for all channels. The scan should continue after the CL phase (but is aborted later on).	None
Prep 5	Reconnect the original cable from X101 on the TCI.	N/A (preparation)	None

3.1.25.8. SWCMP.Scan.PrepPhases.CL.tc1

Test ID	22202
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-05 14:11:21 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Lin, Xuefei (nly95967) on 2015-02-05 13:15:34 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-05 11:51:22 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test needs to be performed on a 3T TB with a DDAS / TCI and watercooled RF amplifier installed.

Requirements Coverage:

Req ID	Requirement Tag	Description
58573	SWCMP.Scan.PrepPhases.CL.v 2	The RF control loop calibration preparation phase will run at the beginning of each scan when no previous calibration results are available, as part of the preparation phases. It sets the parameters for the control loop (delay, gain and phase) to function properly for the current RF coil load condition.

Req ID	Requirement Tag	Description
		If the calibration parameters cannot be determined, the user will not be warned and the measurement continues with the control loop disabled for all channels. The measurement should be done at least 15 kHz off-centre compared to f0. The control loop calibration should be done before any prephase that relies on the TX gain, i.e. before PU / PO phase It shall be possible to start the CL preparation phase from System Tuning Tools (STT)
45451	SWCMP.Scan.PrepPhases.CL.logging	The delay (ns), gain and phase (deg) are calculated from the measurements during the calibration of the control loop. The calculated values will be logged in the central log file and in monitoring .dat files: - monitor_FB_loop_TX<x>_delay.dat - monitor_FB_loop_TX<x>_gain.dat - monitor_FB_loop_TX<x>_phase.dat where <x> is the number of the transmit channel, starting from 1 for the first channel. The monitoring files are used by service to perform trend analysis and e.g. detect degradation of components in the system. If the CL calibration phase succeeds, the following text will be logged: CL: performing calibration for channel <x> CL: result for channel %d: gain = %f, phase = %f, delay = %f where <x> is the number of the transmit channel, starting from 1 for the first channel. If the CL calibration cannot be determined, the following text will be logged: (1a) Gain signal too low during the delay measurement: CL: calibration for channel %d failed, low gain %f during delay calibration, check FWD power (1b) Gain signal too high during the delay measurement: CL: calibration for channel %d failed, high gain %f during delay calibration (2a) Calculated delay too short: CL: calibration for channel %d failed, short delay %f (2b) Calculated delay too long: CL: calibration for channel %d failed, long delay %f (3a) Gain signal too low during the gain measurement: CL: calibration for channel %d failed, low gain %f during gain calibration, check FWD power (3b) Gain signal too high during the gain measurement: CL: calibration for channel %d failed, high gain %f during gain calibration (4) RF amplifier in wrong mode: CL: amplifier for channel %d is in wrong mode calibration not executed (5) Illegal transmit channel number requested: CL: channel %d cannot be calibrated with provided fe object calibration not executed In case of one of the modes where the CL calibration could not be determined as indicated above, the following general warning will be logged: CL: calibration for channel %d not enabled calibration not executed
45453	SWCMP.Scan.PrepPhases.CL.performance	The calibration should not take more than 100 ms per TX channel.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Start the Application and create a new patient.	N/A (preparation)	None
Prep 2	Copy the attached file 'CLtest.zipx' to E:\ and unzip this file (right click, select 'Extract to "CLtest\"); Double click E:\CLtest\CLtest.pl; please wait until command box closes. This script will: - import the examcard - create a new logfile Put the 3L bottle phantom vertically in the iso centre.	N/A (preparation)	 CLtest.zip
Step 3	Drag the ExamCard "Other/InBox/ cltest" into the scan list. Start the scan. After the scan has finished: Start the Logging browser and open G:\log\devlogcurrent.log, check that the correct CL preparation phase messages appear in the time period of this test by using the following text in 'View' -> "Filter Query...": 'CL:' This query will show all logging produced by the CL phase.	The scan finishes successfully. For every used TX channel in the system, the logging should indicate: The calibrated values for delay, gain and phase. Timing of the CL phase (shown by the PREPDUR: logging) should indicate that the time is < the number of TX channels * 100 ms.	None
Step 4	Change the filter query in the Logging browser to: 'perform ' This query will show all performed prep phases.	All performed prepphases are shown now. The order in which the CL preparation phase occurs should be before the PU and PO phase. The timing of the prep phase should indicate this.	None
Step 5	Check that all monitoring files are created/updated with the latest scan information in the G:\monitoring directory.	The following monitoring files need to be available: - monitor_FB_loop_TX<x>_delay.dat - monitor_FB_loop_TX<x>_gain.dat - monitor_FB_loop_TX<x>_phase.dat where <x> is the TX channel For every CL adjustment, one line should be added according: <Date(YYYY-MM-DD)> <Time> <control loop delay> The same holds for the control loop gain and phase monitoring files.	None
Step 6	From the SW archive, open the file mppcl__g.c (mrmethpdf/src) and mmpcl__mxg.c (mrmethpdf/src) This step checks that the RF frequency used, is at least 15 kHz from f0.	In both files, check that the initial RF frequency of the RF object (RF`CL:freq) and that the frequency setting (RF`CL:freq_step1 x RF`CL:freq_factor1) is set to 15 kHz or higher	None

3.1.25.9. SWCMP.Scan.PrepPhases.CL.testability.tc1.v1

Test ID	25689
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-02-08 10:55:19 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-12-18 11:26:03 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-12-18 09:26:23 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

The VM will be configured as Ingenia 3T (WA30) for reproducibility reason with a DDAS. It imports cl_prep_testability.ExamCard.

Requirements Coverage:

Req ID	Requirement Tag	Description
45454	SWCMP.Scan.PrepPhases.CL.T estability	Within the simulation environment, the CL prep phase should be performed and not lead to a scan abort.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Use VM installed as "atlsystemp". Login and run the script m_amn_set_test_environment.pl and choose 3.0T Ingenia. Copy the the attached file CL_prep_testability.zipx to the VM and unzip with right mouse button. Double click the extracted ExamCard, to import it. Start the Examcards Scan Environment and create a new examination.	VM is configured as Ingenia 3T DDAS cl_prep_testability.ExamCard is imported in "Other/Inbox"	 CL_prep_testability.zip
Step 2	Drag the ExamCard "Other/InBox/ cl_prep_testability' into the scan list. Start the scan.	After the CL prep phase, at least the RCU, PU and PO prephase should be performed.	None

Step	Description	Expected Result	Attachments
	Press 'proceed' when asked to reuse current patient position. Press 'confirm and Start' when asked to allow table top movement. After the scan has finished: Start the Logging browser, check that the correct CL preparation phase messages appear in the time period of this test by using the following text in 'View' -> "Filter Query...": 'perform ' This query will show all logging produced by prep phases. This test case checks that the CL phase does not abort a scan in the simulation environment.		
Step 3	In the Logging browser, check that the correct CL preparation phase messages appear in the time period of this test by using the following text in 'View' -> "Filter Query...": 'CL:' This query will show all logging produced by the CL phase.	The logging should contain CL preparation phase logging. The CL phase should be performed, the result may be successful or error. The scan should not be aborted by the CL adjustment.	None

3.1.25.10. SWCMP.Scan.PrepPhases.F0.tc1.v2

Test ID	52268
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-09-23 14:40:26 Meaning: QA Approved - (QA) Signed by: Hegde, Shashank (310195826) on 2021-09-23 14:20:00 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-09-23 14:12:20 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

- Step 2 and Step 8 verify possible fault conditions
- Step 3 and Step 4 verify the F0 with initially an offset of a few kHz

- Step 5 verifies that the optimization will follow the centre of the defined slice package
- Step 6 verifies the retry mechanism in problematic situations
- Step 7 verifies the allowed range of the frequency shift
- Step 8 verifies
- Step 9 and Step 10 verify acquired the F0 spectrum
- Step 11 verifies the F0 phase duration

Requirements Coverage:

Req ID	Requirement Tag	Description
64125	SWCMP.Scan.PrepPhases.F0.v1	Determine the water spin resonance frequency in the centre of the stacks of slices. Check that the frequency shift is within the allowed range Do retries in problematic situations: insufficient signal strength, large frequency shift. The duration of the coarse F0 preparation phase on the DAS shall not exceed 4000 msec per volume (stack).

Test steps:

Step	Description	Expected Result
Prep 1	Start application with english UI and create an new exam Start new logfile Define a scan with • The Body Coil (QBC) selected for receive • One stack in the isocentre that - o Contains 1 slice - o Has sagittal orientation - o No angulations Place a 1 litre bottle lengthwise on the table. Write down the initial resonance frequency ('System Tuning/Tuned Hardware parameters/Resonance frequency') Perform the scan Release the existing patient	N.A. (preparation)
Step 2	Change the resonance frequency to the initial resonance frequency – 20000. Select a new patient and create a new exam Repeat the scan Release the existing patient	Verify that the F0 preparation phase aborts with the message <i>'No signal at all'.</i>
Step 3	Change the resonance frequency ('System Tuning/Tuned Hardware parameters/Resonance frequency') to the initial resonance frequency – 3000. Select a new patient and create a new exam Repeat the scan Release the existing patient	Verify that the F0 preparation phase finds a value close (< 50 Hz) to the initial resonance frequency written down in Prep 1.
Step 4	Change the resonance frequency ('System Tuning/Tuned Hardware parameters/Resonance frequency') to the initial resonance frequency + 5000. Select a new patient and create a new exam	Verify that the F0 preparation phase finds a value close (< 50 Hz) to the initial resonance frequency written down in Prep 1.

Step	Description	Expected Result
	Repeat the scan	
Step 5	Restore the initial resonance frequency value in the Tuned Hardware parameters. Copy the scan mentioned in <i>Prep 1</i> and angulate the stack such that it still intersects with the bottle. Then copy this stack and relocate the 2nd stack at an RL offcentre such that it still intersects with the bottle. Perform the scan. Inspect the logfile and use the filter <i>text icontains 'F0: Volume' or text contains 'm_Name'</i>	Verify that for the scan the F0 preparation is performed succesfully and that the F0 is measured and updated for both stacks seperately.
Step 6	Copy the scan mentioned in <i>Prep 1</i> and change the stack orientation to coronal and relocate the stack at an FH offcentre such that there is no intersection with the bottle.	Verify that the F0 is found but that the following messages appear in the logfile: F0: Warning: No signal in first volume try F0: Warning: No signal in second volume try
Step 7	Set each of the two control preparation parameters 'F0 max offset freq (Hz)' and 'F0 max offset freq (warning)' to -1Hz Repeat the scan mentioned in <i>Prep 1</i>	Verify that the following message is presented on the UI: <i>Large frequency offset! Chance of mispositioned slices!</i>
Step 8	Restore the original values of the control parameters modified in Step 7. Take the phantom out of the bore Repeat the scan mentioned in <i>Prep 1</i>	Verify that the following message is presented on the UI: <i>Cannot determine resonance frequency, no signal at all.</i>
Step 9	Place a 1 litre bottle lengthwise on the table. Enable F0 monitoring Repeat the scan mentioned in <i>Prep 1</i>	Verify that the F0 spectrum shows a single peak.
Step 10	Replace the 1 litre both by a water / fat phantom Repeat the scan mentioned in <i>Prep 1</i>	Verify that the F0 spectrum shows two peaks (a few 100Hz apart)
Step 11	Disable F0 monitoring Extract the logfile using the logbrowser filter <i>text contains 'm_Name' or text icontains 'F0:'</i> Attach the logfile extract to this step.	Verify that none of the reported <i>F0: took</i> values exceed 4000 msec per volume.

3.1.25.11. SWCMP.Scan.PrepPhases.FF.tc1.v1

Test ID	36558
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-12-20 08:52:33 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-12-20 07:27:22 Meaning: SME Approved - (Technical)

	Signed by: Oorschot, Joep van (310224063) on 2017-12-19 16:08:47 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

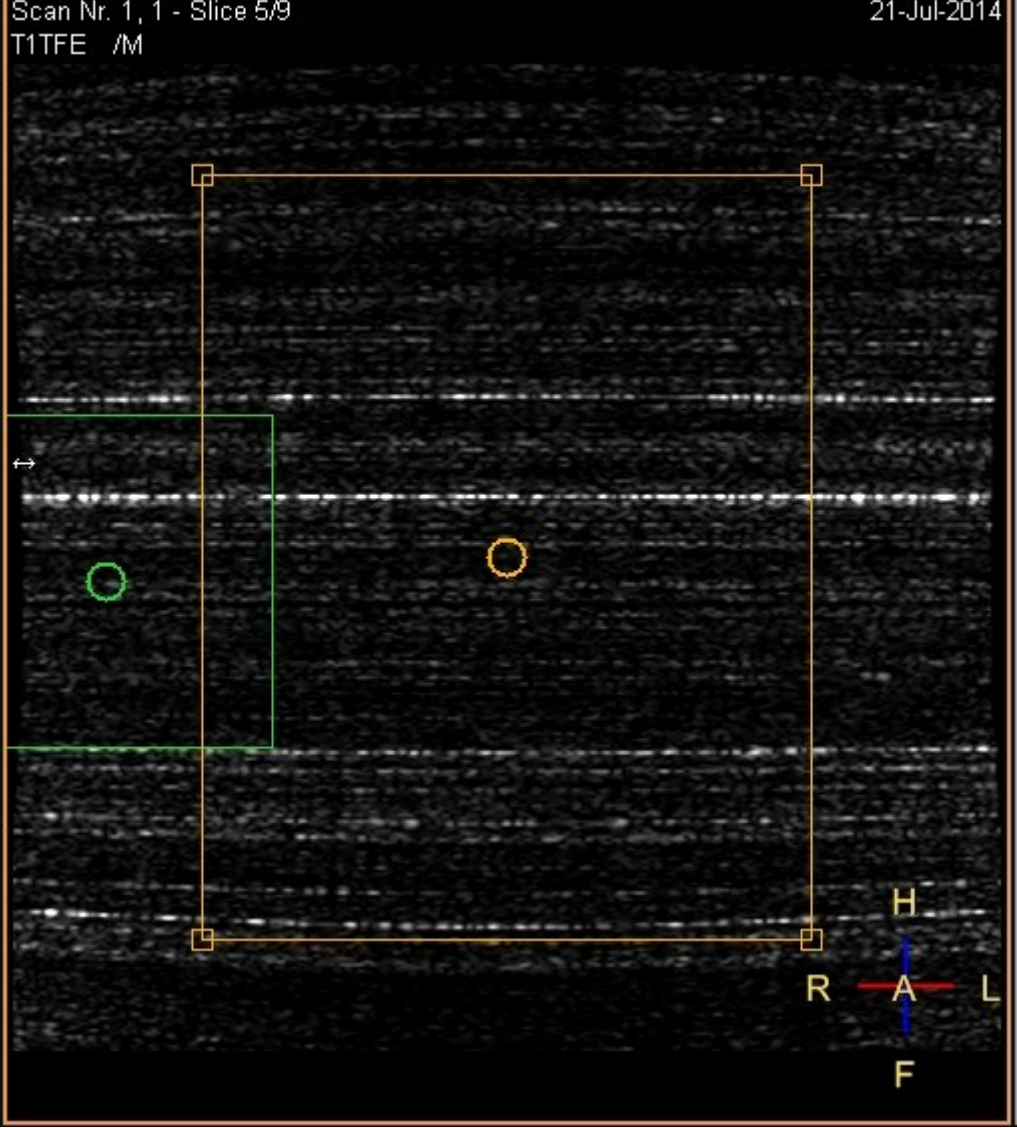
The FFtest.pl script configures VMWare as Ingenia 3T (WA30). It places simulation files on G:\patch. It imports FFtest.ExamCard. It starts 'Simulation mode (Acq on Host, Recon on Host)'. It also creates a new log file.

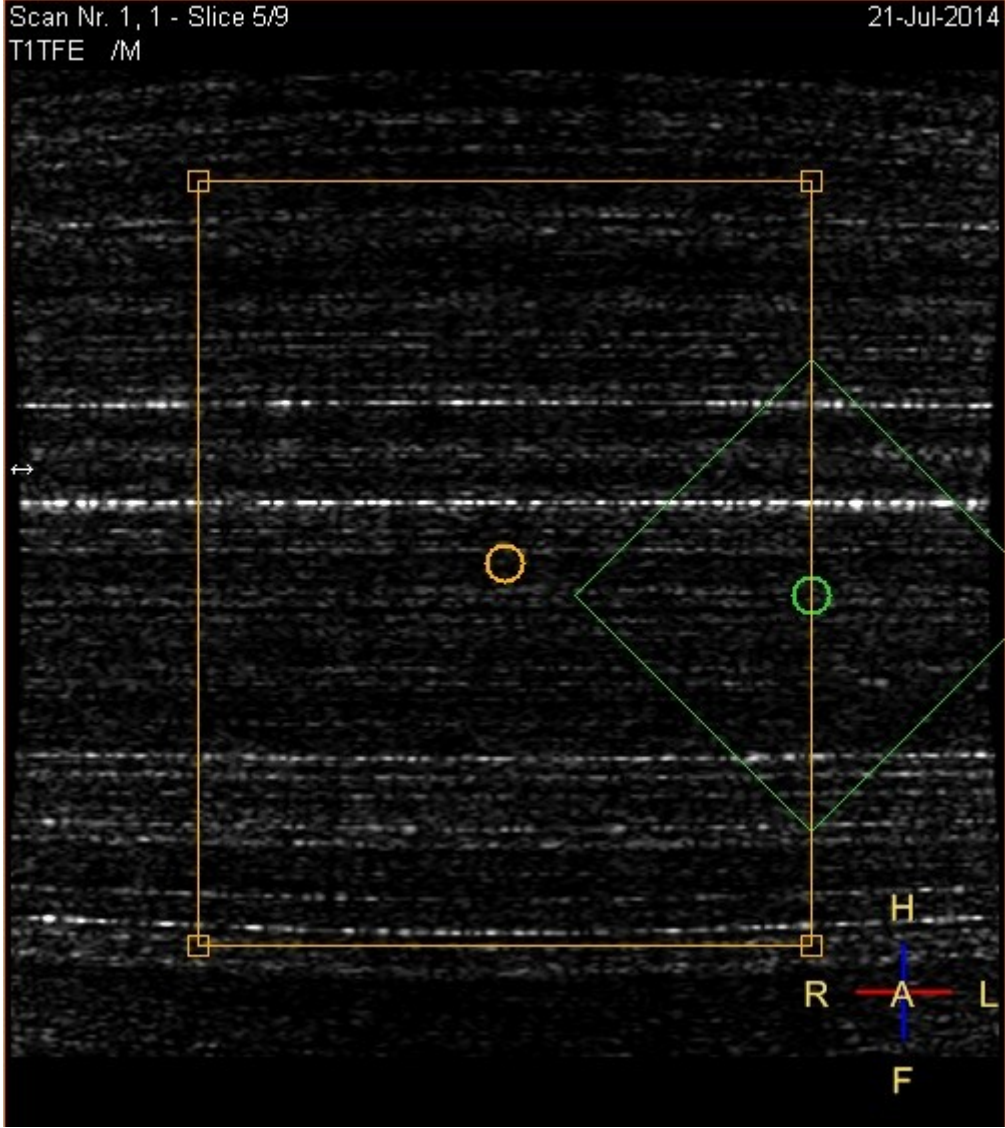
Requirements Coverage:

Req ID	Requirement Tag	Description
82679	SWCMP.Scan.PrepPhases.FF.v 1	The Fast Frequency (FF) determination preparation phase shall determine scan-to-scan magnet field drift. The drift as determined by this preparation phase shall be added to the parameter HW_resonance_freq.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Copy the attached file 'FFtest.zip' to VMware E: and unzip this file (right click, select 'Extract All...', 'Extract'). Double click E:\FFtest\FFtest.pl; please wait until command box closes, configure as an Ingenia 3T (WA30). Start the Application Software, and create a New Examination. Create a New Examination. Drag the ExamCard "Other/InBox/FFtest" into the scan list. Make sure that the System Tuning/Tuned Hardware/resonance frequency is set to 127888500 Hz. Run the Survey scan. Plan the anatomies 'SAG' and 'TRA'. Check that in the GUI the (green) shim volumes have different locations/angulations: Shim volume in SAG scans should be set to the patient right and not angulated; shim volume in TRA scans at patient left and rotated by 45 degrees.	N/A (preparation)	 FFtest.zip

Step	Description	Expected Result	Attachments
	Scan Nr. 1, 1 - Slice 5/9 T1TFE /M 21-Jul-2014 		

Step	Description	Expected Result	Attachments
	Scan Nr. 1, 1 - Slice 5/9 T1TFE /M 21-Jul-2014 		
Step 1	Run the scan T1W_SE_1 (the system will pause after the scan completes, don't continue, press 'Cancel'). Use LogBrowser to inspect the log file; use query: text contains 'Chosen resonance' or text contains 'FF: obtained base' or text contains 'FF: ok, updating' or text contains 'm_Name'	The log file shows that for the scan T1W_SE_1, the obtained base frequency was 127888416 Hz. The resonance frequency was 127888015.	None
Step 2	In the directory G:\Patch, remove the file 'simul.dat', then rename the file 'simul_2.dat' to 'simul.dat'. Run the scan T1W_FFE_2 (the system will again pause after the scan completes).	The log file shows that for the scan T1W_FFE_2, the obtained base frequency was 127888419 Hz. The resonance frequency was 127888939.	None
Step 3	In the directory G:\Patch, remove the file 'simul.dat', then rename the file 'simul_3.dat' to 'simul.dat'. Run the scan T1W_SE_3 (the system will again pause after this scan completes).	The log file shows that for the scan T1W_SE_3 the resonance frequency was 127888023, and the detected drift was 8.0 Hz	None
Step 4	In the directory G:\Patch, remove the file	The log file shows that for the scan	None

Step	Description	Expected Result	Attachments
	'simul.dat', then rename the file 'simul_4.dat' to 'simul.dat'. Run the scan T1W_FFE_4.	T1W_FFE_4 the resonance frequency was 127888955, and the detected drift was 16.0 Hz	
Step 5	Rerun the scanT1W_FFE_4.	The log file shows that for this scan the resonance frequency now was 127888955, and the detected drift was 0.0 Hz	None
Step 6	Rerun the scanT1W_SE_3.	The log file shows that for this scan the resonance frequency now was 127888034, and the detected drift was 11.0 Hz	None
Step 7	Again rerun the scanT1W_SE_3.	The log file shows that for this scan the resonance frequency now was 127888034, and the detected drift was 0.0 Hz	None
Post 1	Delete FFtest from Other/InBox. Stop the Application Software. Double click G:\Patch\FFtest-cleanup.pl. Remove G:\Patch\FFtest-cleanup.pl. Attach log file as objective evidence.	N/A (cleanup)	None

3.1.25.12. SWCMP.Scan.PrepPhases.FF.tc2

Test ID	22199
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-02-05 14:11:26 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Lin, Xuefei (nly95967) on 2015-02-05 13:15:54 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-02-05 11:51:33 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Test steps:

Step	Description	Expected Result
Step 1	Run the scan tedb / TSE / Hospital / F_MS on a 3T test bay and verify that the FF preparation phase takes less than 300 ms by searching the log file for 'FF: took'.	Log file contains line 'PREPDUR: FF: took x msec' with x < 300

3.1.25.13. SWCMP.Scan.PrepPhases.FF_MN.tc1

Test ID	30074
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-09-26 12:06:43 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2016-09-26 10:50:28 Meaning: SME Approved - (Technical) Signed by: Yildirim, Muhammed (nlv10972) on 2016-09-23 14:50:34 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies that for Multi Nuclei the FF perparation phase is not performed and that in that case the f0_max_elapsed_time is set to a lower value.

Requirements Coverage:

Req ID	Requirement Tag	Description
62165	SWCMP.Scan.PrepPhases.FF-MN	The Fast Frequency (FF) determination preparation shall be disabled for Multi Nuclei. When the FF preparation phase is disabled, the F0 preparation phase will be performed more often to compensate for F0 drift.

Test steps:

Step	Description	Expected Result
Prep 1	The test can be executed on any 3T Multi Nuclei sysetm. When executed on a simulator (ROF): make sure the system is properly configured as a 3T Multi Nuclei system. Start scan simulation: select the proper simulation mode. Start the "Coilstate Simulator UI" Start the graphical Patsup Simulator and press the "LightVisor" button. Add " MPUPR debug * " to the C:\temp\bstrace.dat file. Make the TEDB the active protocol database. Strat the application software and start a new examination.	N.A. (Preparation).

Step	Description	Expected Result
Step 2	From the TEDB select Head CNS\FE\F_2D1 protocol. On the coil tab, switch off smartselect and select the QBC. Start the scan. Copy the scan and start again.	Both scans run fine without any issue.
Step 3	Open the devlogcurrent file in the logbrowser. Apply the search query: text contains 'm_Name' or text contains 'will be' or text contains 'elapsed' or text contains 'FF: ' Verify that for both scans "MPUPR: FF will be performed" is logged. Verify that for both scans indeed the FF preparation phase is performed (follows from the 'FF: ' logging). Verify that there is logged "ff switch:1, f0_max_elapsed_time: 1800.00	For both scans the FF preparation phase is performed. FF is enabled (ff_switch: 1) The f0_max_elapsed_time = 1800.00
Step 4	Connect a MN coil, for instance the P-140. Move the table out and in to disengage the posterior coil (both on a scanner and in the Graphical Patsup simulator on a simulator) Copy the scan and set the nucleus to P31. On the Coil tab select the P-140 coil. Start the scan. Copy the scan and start again.	Both scans run without any issue.
Step 5	Open the devlogcurrent file in the logbrowser. Apply the search query: text contains 'm_Name' or text contains 'will be' or text contains 'elapsed' or text contains 'FF: ' Verify that for both scans the logging "MPUPR: FF will be performed" is NOT present. Verify that for both scans indeed the FF preparation phase is NOT performed. Verify that there is logged "ff switch:0, f0_max_elapsed_time: 240.00	For both scans the FF preparation phase is NOT performed. FF is disabled (ff_switch: 0) The f0_max_elapsed_time = 240.00
Step 6	Disconnect a MN coil. Move the table out and in to bring the posterior coil back in again (both on a scanner and in the Graphical Patsup simulator on a simulator). Copy the scan and set the nucleus to H1.	Both scans run without any issue.

Step	Description	Expected Result
	On the Coil tab select the QBC. Start the scan. Copy the scan and start again.	
Step 8	Open the devlogcurrent file in the logbrowser. Apply the search query: text contains 'm_Name' or text contains 'will be' or text contains 'elapsed' or text contains 'FF: ' Verify that for both scans "MPUPR: FF will be performed" is logged. Verify that for both scans indeed the FF preparation phase is performed (follows from the 'FF: ' logging). Verify that there is logged "ff switch:1, f0_max_elapsed_time: 1800.00	For both scans the FF preparation phase is performed. FF is enabled (ff_switch: 1) The f0_max_elapsed_time = 1800.00
Wrap up 9	Remove the " MPUPR debug * " from C:\item\bstrace.dat.	N.A. (Wrap up)

3.1.25.14. SWCMP.Scan.PrepPhases.FRC.tc1

Test ID	7955
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
18823	SWCMP.Scan.PrepPhases.FRC	The Receiver Correction preparation phase removes the intensity distortion introduced by the receiver hardware and body coil as specified in the FRC chapter

Test steps:

Step	Description	Expected Result
Step 1	Select a protocol with 'Preparation phases' to 'full' that uses the QBC for receive.	Use the monitoring viewer to verify that 'Frequency response correction' signals are acquired for all body coil channels (normally 1; on dual transmit systems, 2. These should be labeled BODY_QUAD_E1 and BODY_QUAD_E2).

Step	Description	Expected Result
Step 2	Use the monitoring viewer to determine the span of the frequency sweep (in kHz). Modify the protocol to increase (or decrease) the bandwidth per pixel, e.g. by a factor 2. Run this scan.	Verify that the sweep range has changed similarly.

3.1.25.15. SWCMP.Scan.PrepPhases.NS.tc1

Test ID	7956
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
18824	SWCMP.Scan.PrepPhases.NS	The Noise determination preparation phase determines the complex noise levels of the receiver chain as specified in the NS chapter

Test steps:

Step	Description	Expected Result
Step 1	Use a standard synergy coil in connector A or use the posterior coil.	Verify that noise is determined for all synergy channels 1. Inspect the monitoring 'NS' tab; 2. Look for 'NS:' in the logging during preparation; 3. Look for 'noise_arr' in the reconstruction logging that follows scan completion.

3.1.25.16. SWCMP.Scan.PrepPhases.PH.tc1

Test ID	7957
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]

Approval Status	Approved
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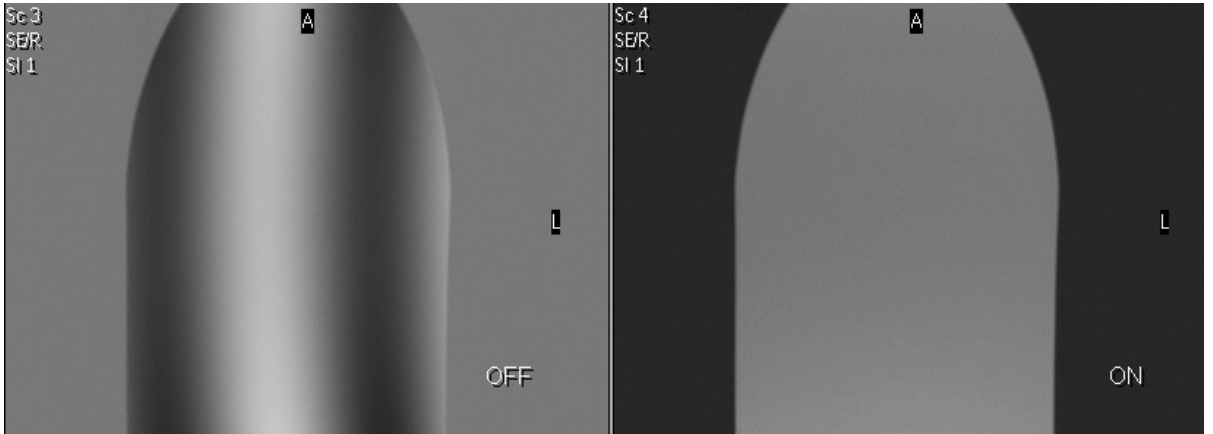
Description:

PH phase: Echo Phase determination

Requirements Coverage:

Req ID	Requirement Tag	Description
18825	SWCMP.Scan.PrepPhases.PH	The Echo Phase determination preparation phase acquires data to enable phase corrections as specified in the PH chapter

Test steps:

Step	Description	Expected Result	Attachments
Step 1	- Place a 3 liter bottle upright on the patient support and move it in. - In menu \System Tuning...\Fixed Hardware\ add 0.150 to the value of 'overall grad. delay [ms]'. Press 'Apply' (don't press 'Store'). - In menu \Control Parameter Editor...\Preparation\ set 'echo phase determination' to 'off'. Press 'Apply' (don't press 'Store'). - Select protocol \tedb\RESP\P_2D1\ and make the following modifications: - on the 'motion' tab, set 'Respiratory compensation' to 'no' - on the 'postproc' tab, set 'Images' to 'R' (i.e., real) Run this scan.	Verify that the result resembles the left of the image.	
			
Step 2	In menu \Control Parameter Editor...\Preparation\ set 'echo phase determination' to 'on'. Press 'Apply' (don't press 'Store'). Rerun the scan.	Verify that the result resembles the right of the image.	None
Step 3	To clean up: - In menu\Control Parameter Editor...\ press 'Factory Defaults...' - In menu \System Tuning...\ press 'Factory Defaults...'		None

3.1.25.17. SWCMP.Scan.PrepPhases.PMU.set_triplevel.tc1.v2

Test ID	58173
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Sastry, Shekar Lakshminarsimha (320029429) on 2023-11-16 17:50:07 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2023-11-15 11:14:25 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-11-15 11:05:05 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

PMU phase: Power Monitoring Unit

Requirements Coverage:

Req ID	Requirement Tag	Description
85265	SWCMP.Scan.PrepPhases.PMU.set_triplevel.v1	An RF amplifier's PMU trip level shall be set to at most twice the expected average power level for that amplifier, but at least as high as the PMU can reliably determine. The minimum trip level will be set to: 50W for 1H PMU in single transmit systems 25W for 1H PMU in dual transmit systems 4W for MN PMU in multi nuclei systems..

Test steps:

Step	Description	Expected Result
Step 1	Configure a VM as MR7700 3.0T + Multi Nuclei In Service application, enable all coils (including P140-Flex) Start simulator processes Start application sw Start a new log file	N.A. (preparation)
Step 2	Ensure P-140 flex coil is plugged in Device simulator. Load the protocol : Hospital > 307 Spectroscopy > MN Decoupling > 1H_Decoupling_MN_Sp Plan and Run the scan.	No conflict w.r.t protocol Scan runs successfully.
Step 3	Open the logfile with the log browser Create a filter to only visualize the scan name and the PMU trip level logging. Use the following filter expression: <i>text icontains 'scan name]' or text icontains 'PMU to be set'</i>. Attach the log file to this step as objective evidence.	Verify that for 1H PMU's: - if $P_{avg} < 13W$, then PMU shall be set to 25W - if $P_{avg} > 13W$, then PMU shall be set to at most $\text{int}(P_{avg} * 2)$ Verify that for MN PMU:

Step	Description	Expected Result
		1. a. PMU shall be set to at most $\text{int}(\text{Pavg} * 2)$ For eg. PMU: MN ch1: Current pmu limit 4 watt, Pavg 0.520 watt. PMU to be set to 4 watt. PMU: 1H DECOUPLING ch1: Current pmu limit 540 watt, Pavg 81.568 watt. PMU to be set to 130 watt. PMU: 1H DECOUPLING ch2: Current pmu limit 540 watt, Pavg 81.568 watt. PMU to be set to 130 watt.
Step 4	Copy the protocol from Step# 2. Plan the protocol Set Contrast > Proton decoupling to "No" Run the scan	No conflict w.r.t protocol Scan runs successfully.
Step 5	Open the logfile with the log browser Create a filter to only visualize the scan name and the PMU trip level logging. Use the following filter expression: <i>text icontains 'scan name]' = ' or text icontains 'PMU to be set'.</i> Attach the log file to this step as objective evidence.	Verify that for 1H PMU's: 1. a. if $\text{Pavg} < 13\text{W}$, then PMU shall be set to 25W b. if $\text{Pavg} > 13\text{W}$, then PMU shall be set to at most $\text{int}(\text{Pavg} * 2)$ Verify that for MN PMU: 1. a. PMU shall be set to at most $\text{int}(\text{Pavg} * 2)$ For eg. PMU: MN ch1: Current pmu limit 4 watt, Pavg 0.815 watt. PMU to be set to 4 watt. PMU: 1H ch1: Current pmu limit 540 watt, Pavg 0.015 watt. PMU to be set to 25 watt. PMU: 1H ch2: Current pmu limit 540 watt, Pavg 0.015 watt. PMU to be set to 25 watt.

3.1.25.18. SWCMP.Scan.PrepPhases.PMU.tc1

Test ID	7959
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

PMU phase: Power Monitoring Unit

PMU check preparation phase must NOT run for proton scans

This test is needed to assure that at 1.5T and 3.0T, the PMU check preparation phase is **NOT** executed for proton scans.

On a 1.5T or 3.0T TB, setup a proton scan experiment, and verify that the PMU check preparation phase does not run.

Test steps:

Step	Description	Expected Result
Step 1	Check that OPTION40 is turned ON.	
Step 2	Start the Application and create a new patient. Choose an imaging preset procedure and adjust it.	Verify that the scan nucleus is proton (1H).
Step 3	Start the scan	Check that the scan does not abort on the PMU check preparation phase. Also, use the Logging application to check that the following messages DO NOT appear in the time period of this test: • VP: success

3.1.25.19. SWCMP.Scan.PrepPhases.PMU.tc2.v2

Test ID	7961
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

PMU phase: Power Monitoring Unit

PMU Error detection for spectroscopy sequences

This test is needed to assure that the PMU check preparation phase detects the error condition of the CPC amplifier PMU for spectroscopy sequences at 1.5T and 3T.

On a 1.5T or 3T TB, verify that the PMU check preparation phase detects disconnection of the PMU input forward power/blanking input cable and aborts the scan for spectroscopy sequences.

Test steps:

Step	Description	Expected Result
Step 1	Check that Multi Nuclei (MN) options are turned on.	
Step 2	Start the Application and create a new patient. Choose a spectroscopy preset procedure and adjust it. Change the scan nucleus to phosphorus (P31), and adjust the scan to eliminate conflicts. Disconnect the forward power cable input at the PMU (connection label: Forward Power X1). Choose the option to start the scan	verify that the PMU check preparation aborts and the following error message is displayed: VP: Check failed. PMU forward power/blanking input cable disconnected ?
Step 3	Repeat the last step, this time disconnecting the PMU blanking input signal (connection label: Blanking input X7).	The same results should occur.

3.1.25.20. SWCMP.Scan.PrepPhases.PMU.tc2.v3

Test ID	51270
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-08-26 10:50:43 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2021-08-25 14:06:18 Meaning: SME Approved - (Business) Signed by: Gottwald, Lukas (320109596) on 2021-06-15 13:45:24 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

PMU phase: Power Monitoring Unit

Requirements Coverage:

Req ID	Requirement Tag	Description
85121	SWCMP.Scan.PrepPhases.PMU-MN.v1	The PMU check preparation phase will run at the beginning of each scan. It will check that the PMU is correctly reading power values from the amplifier. If the read values are lower than the threshold (4 W), the preparation phase will abort the scan.

Test steps:

Step	Description	Expected Result
Preparation	On a SDE VMWare, set the configuration to a DDAS system with Multinuclei capabilities. Select the tedb. Create a new patient, and connect the p-140 flex coil. Load the following protocol: Hospital/Phantom studies/SPECTRO/single .	
Step 1	Run the scan.	The scan finishes successfully.
Step 2	Change the value of 'pmu prepphase treshold MN [W]' to 25. Copy the previous scan, and start it.	Verify that the PMU check preparation aborts and the following error message is logged: VP: Check failed. PMU forward power/blanking input cable disconnected ?

3.1.25.21. SWCMP.Scan.PrepPhases.PO.tc1.v1

Test ID	31027
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-14 12:07:09 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2017-04-14 10:45:41 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-14 10:42:02 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

- Step 2, Step 3, Step 6 and Step 7 verify possible fault conditions
- Step 4 verifies the optimization algorithm following the target angle
- Step 5 verifies that the optimization will follow the centre of the defined slice package
- Step 8 verifies that the final drive scale delivered by PU phase is used as initial drive scale for PO phase
- Step 9 verifies the PO phase duration

Requirements Coverage:

Req ID	Requirement Tag	Description
64121	SWCMP.Scan.PrepPhases.PO.v1	The final drive scale derived from the PU phase shall be fine tuned by performing PO measurements.

Req ID	Requirement Tag	Description
		The PO measurements will use the centre of the defined slice package. The duration of this preparation phase on the DAS shall not exceed 2100 msec.

Test steps:

Step	Description	Expected Result
Prep 1	Start application with english UI and create an new exam Start new logfile Define a scan with the Body Coil (QBC) selected for receive. Place a 0.25 liter bottle lengthwise on the table.	N.A. (preparation)
Step 2	Set the control parameter ' <i>PO min time-SNR</i> ' to the highest possible value. Start the scan	Verify that the log file contains the following messages: <i>PO: SE time signal not above noise threshold</i> <i>PO: STE time signal not above noise threshold</i>
Step 3	Restore the value of control parameter ' <i>PO min time-SNR</i> ' to its original value. Set the control parameter ' <i>PO min freq-SNR</i> ' to the highest possible value. Start the scan again	Verify that the log file contains the following messages (<i>nnn.n</i> is some numerical value): <i>PO: SE-F0 (nnn.n) signal not above noise threshold (nnn.n)</i> <i>PO: STE-F0 (nnn.n) signal not above noise threshold (nnn.n)</i>
Step 4	Restore the value of control parameter ' <i>PO min freq-SNR</i> ' to its original value. Set the control parameter ' <i>PO target angle</i> ' (by default 90 or 135 degrees, depending on the field strength) to the other value (135 or 90). Inspect the log lines that start with ' <i>PO: .:</i> ' and verify that the last reported value matches the target angle.	Verify that the PO iterations converge to the new target angle within a margin of 10 degrees.
Step 5	Restore the value of control parameter ' <i>PO target angle</i> ' to its original value. Relocate the bottle such that it has an LR offcentre of about 150mm. Define a scan with a 1 sagittal slice through the iso centre Perform the scan	Verify that the PO phase successfully completes, as indicated in the logfile by ' <i>PO: Power optimization successfully completed</i> '.
Step 6	Restore the value of control parameter ' <i>PO target angle</i> '. Reduce the body coil attribute ' <i>Ref. scales</i> ' by half of their initial value. Set control parameter PU to OFF. Perform the scan again	Verify that the PO phase fails. The logfile shows the lines: <i>PO: No flip angle found within 10 deg from target</i> <i>PO: Power optimization failed</i>
Step 7	Restore the body coil attribute ' <i>Ref. scales</i> ' to its original value. Restore the control parameter PU switch to its original value Take the bottle manually out of the scanner Repeat the scan	Verify that the PO phase fails and that the following message appears on the UI: ' <i>Cannot determine optimum RF power</i> '.
Step 8	Extract the logfile using the logbrowser filter <i>text contains 'm_Name' or text icontains 'PO:' or text icontains 'PU: final drive scale'</i> Attach the logfile extract to this step.	Verify that the <i>final drive scale</i> value found by the PU phase is equal to the <i>initial drive scale</i> value used for the PO phase. Write down this value.
Step 9	Inspect the logfile extract of Step 8 and search for ' <i>PO: took</i> '. The value reports the actual PO phase duration.	Verify that none of the reported <i>PO: took</i> values exceed 2100 msec

3.1.25.22. SWCMP.Scan.PrepPhases.PU.tc1.v1

Test ID	31018
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-02 16:58:09 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2017-03-02 16:26:24 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Visser, Kees (nly22601) on 2017-03-02 15:20:59 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

PU phase: Pickup coil optimization

PU Drive Scale

Requirements Coverage:

Req ID	Requirement Tag	Description
64116	SWCMP.Scan.PrepPhases.PU.v1	Use the pick-up coils to determine the initial load factor of the body coil and the initial drive scale required for the PO preparation phase. The duration of this preparation phase on the DAS shall not exceed 210 msec.

Test steps:

Step	Description	Expected Result
Prep 1	Startup application and create an new exam Start new logfile Define a scan with the Body Coil (QBC) selected for receive.	N.A. (preparation)
Step 2	Perform the scan and inspect the logfile.	Write down the ' <i>pure load factor</i> ' reported by the PU phase
Step 3	Change the Pickup coil sensitivities (System Tuning/Coil/Q-Body/Pickup coil sensitivities [V/uT]) with about 10% and perform the scan. Inspect the logfile.	Write down the <i>pure 'load factor'</i> reported by the PU phase and notice 10% change in the value compared to the result of step 1.
Step 4	Change the Pickup coil sensitivities [V/uT] now by a factor 1000 and perform the scan.	The scan aborts. The logfile will show: ' <i>Not enough signal in PU phase</i> '
Step 5	Extract the PU logging from the logfile using the filter 'PU:' and check the duration 'PU: took'. Attach a screenshot of the extracted logging to this step.	Write down the duration in msec and verify that this value does not exceed 210.

3.1.25.23. SWCMP.Scan.PrepPhases.RCU.tc1.v1

Test ID	32554
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 16:57:32 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Prins, Wim (nly95928) on 2017-03-23 16:06:27 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-22 14:35:30 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

RCU phase: RF Circulator unit

This test is needed to assure that the RCU check preparation phase reports correct information about the RF circulator and its connections.

Requirements Coverage:

Req ID	Requirement Tag	Description
18826	SWCMP.Scan.PrepPhases.RCU	The RF Circulator Unit preparation phase detects RF power problems and cabling problems in the RCU hardware as specified in the RCU chapter

Test steps:

Step	Description	Expected Result
Step 1	Start the Application and create a new patient. Choose any Survey scan with body coil as transmitter from imaging preset procedure and resolve any conflict. In the "Control Parameter Editor" menu, make sure that "RCU check" is 'on'. Start the scan.	Verify that it runs successfully.
Step 2	Using the Logging Application to check that the following message appear in the time period of this test and before the PU preparation phase (the numerical values are allowed to vary for each scan). Note: The lines containing 'at RF amplifier' are not shown for water-cooled amplifier type.	RCU: perform RF circulator check RCU: Transmit on channel 1 results: RCU: forw power phase at RF amplifier[1] = -68 deg RCU: forw power of [1] at RF amplifier[1] = 17.029 mV^2 sd = 0.2 % RCU: forw power of [1] at RF circulator[1] = 14.395 mV^2 (84.5 %) sd = 0.2 % RCU: refl power of [1] at RF circulator[1] = 4.450 mV^2 (30.9 %) sd = 0.3 % RCU: vswr at RF Circulator[1] = 3.50

Step	Description	Expected Result
		RCU: RCU: forw power of [1] at RF amplifier[2] = 0.000 mV^2 sd = 63.3 % RCU: forw power of [1] at RF circulator[2] = 0.000 mV^2 (0.0 %) sd = 70.6 % RCU: refl power of [1] at RF circulator[2] = 0.034 mV^2 (0.2 %) sd = 7.2 % RCU: RCU: Transmit on channel 2 results: RCU: forw power of [2] at RF amplifier[1] = 0.000 mV^2 sd = 71.6 % RCU: forw power of [2] at RF circulator[1] = 0.000 mV^2 (0.0 %) sd = 67.4 % RCU: refl power of [2] at RF circulator[1] = 0.022 mV^2 (0.2 %) sd = 4.2 % RCU: RCU: forw power phase at RF amplifier[2] = 88 deg RCU: forw power diff phase = 156 deg RCU: forw power of [2] at RF amplifier[2] = 13.441 mV^2 sd = 0.2 % RCU: forw power of [2] at RF circulator[2] = 12.071 mV^2 (89.8 %) sd = 0.2 % RCU: refl power of [2] at RF circulator[2] = 2.604 mV^2 (21.6 %) sd = 0.3 % RCU: vswr at RF Circulator[2] = 2.73 RCU:
Step 3	Verify that after the PU phase, the following logging occurs	RCU: verifying for load factor 1.0177 RCU: success
Step 4	Verify that the following monitoring files are created or updated with the latest scan information in the directory G:\monitoring\	• monitor_Circulator_TX1-FWDcirc-FWDamp-ratio.dat (this file is not created for water-cooled amplifier type) • monitor_Circulator_TX2-FWDcirc-FWDamp-ratio.dat (this file is not created for water-cooled amplifier type) • monitor_Circulator_TX1-VSWR.dat • monitor_Circulator_TX2-VSWR.dat
Step 5	Verify that the respective formats of these files (if applicable) are like:	System Reference Number : 00083 Hospital name : BAY 8 SW Release : 4.1.2 Filename : G: \monitoring\monitor_Circulator_TX1-FWDcirc-FWDamp-ratio.dat File date : 2012-02-13 11:32:53 History length (days) : 90 Date(YYYY-MM-DD) Time FWDcirc-FWDamp-ratio [%] 2009-02-13 11:32:53 87.2 2009-02-13 11:33:35 87.6 etc.

Step	Description	Expected Result
		and System Reference Number : 00083 Hospital name : BAY 8 SW Release : 4.1.2 Filename : G: \monitoring\monitor_Circulator_TX1- VSWR.dat File date : 2012-02-13 11:32:53 History length (days) : 90 Date(YYYY-MM-DD) Time VSWR at output Circulator 2012-08-01 09:29:49 2.01 2012-08-01 09:30:18 2.01 2012-08-01 09:31:06 2.04 etc.

3.1.25.24. SWCMP.Scan.PrepPhases.RG.AttenuatorChoice.tc1

Test ID	35440
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2017-09-14 10:33:59 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2017-09-14 10:24:33 Meaning: SME Approved - (Technical) Signed by: Prins, Wim (nly95928) on 2017-09-13 18:00:34 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
67916	SWCMP.Scan.PrepPhases.RG.A tenuatorChoice	The Receive Gain (RG) optimization preparation phase determines: - The receiver attenuation to prevent ADC overflow as required for non-iRF systems. - o The aim is to adjust to maximum possible receive gain

Req ID	Requirement Tag	Description
		o If even in lowest gain there is overflow, the scan shall abort and a message shall be displayed to inform the operator that too much signal was detected during the receive optimization The actual time required for the RG phase shall be less than or equal to TR times the number of packages of the clinical scan.

Test steps:

Step	Description	Expected Result
Prep 1	1. Perform the test on a non iRF system. 2. Connect the 8 channel head coil and place a 2.0L bottle phantom inside it. 3. Position the light visor at the centre of the phantom. 4. Create new exam. 5. Create a new log file.	N/A Preparation step.
Step 2	1. In the Philips protocol database, choose the following protocol: Head/Brain/T2_weighted/T2W_TSE. 1. From the info page note down the values of a) Act.TR (ms) from the info page. b) Packages 1. Accept the protocol. 1. Press Start scan.	The scan shall complete successfully. 2a) Act TR (ms) = ____ 2b) Packages = ____
Step 3	Open devlogcurrent.log. Filter query 'RG'. Note down the time taken by the RG prep phase by seeing the time in phrase 'PREPDUR: RG: took'	The time taken by the RG prep phase shall be less than or equal to the product of 'TR (ms)' and 'Packages', the values of which were noted in Step 2a) and 2b). The measurement time taken by the RG prep phase shall not exceed the above mentioned product by more than 10%. Note: The 10% margin is provided to factor in the hardware processing time.
Step 4	In devlogcurrent.log filter query to 'text contains 'after scan: max_values:'. This will list the max register value for each channel. Attach the log file as evidence.	In the filtered query max register value does not exceed the value of 254 for any of the channels of the connected receiver coils.
Step 5	Inspect the code in 'methmpf/src/mmifre.cpp' Go to routine MMIRFE_rg_exec(.....) Repeat this for 'methmpf/src/mmiffe.cpp' and 'methmpf/src/mmqqprep.cpp'	Code is present which aborts the scan for the following condition if (overflow && gain_pos = MGOBJAQ_LOW_POS).

3.1.25.25. SWCMP.Scan.PrepPhases.RG.TSE.tc1.v1

Test ID	38348
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-07-12 13:35:11 Meaning: QA Approved - (QA) Signed by: Hegde, Shashank (310195826) on 2018-07-12 12:40:19 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2018-07-12 12:17:04 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
67915	SWCMP.Scan.PrepPhases.RG.T SE	The Receive Gain (RG) optimization preparation phase determines: The phase and gradient delay adjustment values for TSE scans. The actual time required for the RG phase shall be less than or equal to TR times the number of packages of the clinical scan.

Test steps:

Step	Description	Expected Result
Prep 1	- On an Ingenia system, connect the head and base coils and place a 2.0L bottle phantom inside it. - Position the light visor at the centre of the phantom. - Create new exam. - Create a new log file.	N.A preparation step
Step 2	- In the Philips protocol database, choose the following protocol: Head/Brain/T2_weighted/T2W_TSE. - From the info page note down the values of - Act.TR (ms) from the info page. - Packages - Accept the protocol. - Press Start scan.	The scan completes successfully.
Step 3	Open devlogcurrent.log. Filter query 'RG'. Note down the time taken by the RG prep phase by seeing the time in phrase 'PREPDUR: RG: took'	The time taken by the RG prep phase is less than or equal to the product of 'TR (ms)' and 'Packages', the values of which were noted in Step 2a) and 2b). The measurement time taken by the RG prep phase shall not exceed the above mentioned product by more than 10%. Note: The 10% margin is provided to factor in the hardware processing time.
Step 4	In the filtered query of RG in the log file, search for phrases 'RG: after interpolation A' These values indicate gradient delay and phase	The gradient delay and phase adjustment values are measured/interpolated and logged for all the slices as: 'mc' and 'rfex'.

Step	Description	Expected Result
	adjustments for a TSE scan. Attach the log file as evidence.	

3.1.25.26. SWCMP.Scan.PrepPhases.RxCalib.tc1

Test ID	7963
Revision	2
Product Model	Ingenia;Marlin
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-06-20 13:56:34 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2016-06-20 13:05:08 Meaning: SME Approved - (Technical) Signed by: Verburg, Frank (nly95979) on 2016-06-15 16:31:05 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

RX phase: Calibrations

The RX tests for the requirement RX.CALIB are automated by perl scripts. The table below describes where each perl script is to be executed, which test-cases it implements and which requirements it verifies.

Script:	Configuration	Test Cases
m_prep_rx_tb.pl	TestBay WA_3T	RX_CF_1 RX_TM_1
m_prep_rx_vm.pl	configured as WA_3T	RX_WF_1 RX_ER_1

Script:	Result to include in V&V
m_prep_rx_tb.pl	G:\Temp\m_prep_rx_tb.log
m_prep_rx_vm.pl	G:\Temp\m_prep_rx_vm.log

--	--

Besides these perl-scripts, an ExamCard, toolbox psets, and an analysis program are archived as testware in the buildingblock system/mrmethods/methodspreparation/tst.

All this is incorporated in the scan software test msi. The perl scripts (and the other parts of the testware) are targeted to "C:\nmr\gyroscan\testbatch".

m_prep_rxcf.pset	Pset to measure an exponential waveform generated on the tune coil for verifying the RX calibration.
------------------	--

m_prep_rxta.pset	Pset used for verification of the time alignment. It generates an FM sweep as RF pulse on tune coil A, which is received by all coil elements of the posterior and body coil.
rta_rxcftstdll.dll	DLL used by the toolbox pset to analyze the coarse/fine calibration tests.
rta_rxtatstdll.dll	DLL used by the toolbox pset to analyze the time adjustment tests.
m_prep_rx_test.ExamCard	ExamCard used in the tests.
m_prep_rx_tb_results.ref	Reference file for testbay tests
m_prep_rx_vm_results.ref	Reference file for vmware-based tests

The following test cases cover part of the requirement RX.CALIB.

General Information

- The tests rely on the System Test and Tuning (STT) software and dStream hardware being present.
- All tests are contained in the perl script 'm_prep_rx_vm.pl'. It contains testcases RX_WF_1, RX_ER1.

Details of test case RX_WF_1

For this test case, the perl script m_prep_rx_vm.pl does the following:

1. It starts batchinterpreter services also required for examCardscripting.
2. It starts a new logfile.
3. It connects the NVC_BASE and Anterior coils.
4. It runs the \m_prep_rx_test.ExamCard twice, by means of examCardscripting.
5. It disconnects, then reconnects the NVC_BASE and Anterior coils.
6. It again runs the \m_prep_rx_test.ExamCard.
7. It analyzes the logfile and generates a report. The expected results are:

- a. CoarseFine and TimeAlignment performed for first survey scan.
- b. Only CoarseFine performed for second survey scan.
- c. No RX Calibrations performed for the third scan.
- 8. It reports the result in the test logfile "G:\Temp\m_prep_rx_vm.log":
 - a. RF_WF_1 Succeeded/Failed

Details of test case RX_ER_1

For this test case, the perl script m_prep_rx_vm.pl does the following

1. It activates simulation errors with the specific trace channels for coarse fine errors and time alignment errors.
2. It starts a mobiflex examCard by means of examCard scripting. The expected result is:
 - a. The first scan aborts.
3. It then limits the errors to time alignment errors only by activating the MMPRX simulate_ta_errors trace channel.
4. It starts a mobiflex examCard by means of examCard scripting. The expected result is:
 - a. Time alignment fails and is performed for each tabletop position in the survey mobitrak scan.
 - b. Time alignment fails and is performed for each scan in the scanset.
5. It runs an stt toolbox scan for these cases:
 - a. No calibration required.
 - b. CF Calibration required.
6. It analyzes the logfile and reports the result in the testlog file "G:\Temp\m_prep_rx_vm.log":
 - a. RF_ER_01 Succeeded/Failed

Requirements Coverage:

Req ID	Requirement Tag	Description
18828	SWCMP.Scan.PrepPhases.RxC alib	Time alignment and the dual ADC calibration must have been performed before IRF receivers are used during MR experiments. The results of time alignment and dual ADC measurements become invalid after the receivers have been restarted. This means that the calibrations have to be redone. The dual ADC calibration must calibrate the gain difference between coarse and fine with an accuracy lower than 0.5%. This within a frequency bandwidth of 1MHz. The following assumption is considered as valid $\text{GainRatio}(\text{df}) = \text{GainRatioAtF0} + \text{FreqSlope} * \text{df}$ With df the delta between the intended frequency and f0. The dual ADC calibration must calibrate the phase difference between coarse and fine with an accuracy lower than 0.5 degrees. This within a frequency bandwidth of 1MHz. The receivers are not really linear; The FIR filter response characteristics are optimal around the centre frequency. Chosen for this frequency is the resonance frequency. Therefore it is required to recalibrate the coarse fine when the resonance frequency deviates too much from the previously optimal frequency value. A value is chosen of 10% of the typical signal bandwidth of 400kHz = 40KHz. The time alignment must measure the parameters necessary to correct timing differences between receive channels. Also called ReceiveChannelDelay. After calibration the system must satisfy the following requirements: • The timing differences between receive channels are within the

Req ID	Requirement Tag	Description
		range [-250ns, 250ns] with respect to the transmit system (25% off smallest sample interval). Please note The differences of the ReceiveChannelDelays between combined channels may not change more than 8 pico secods for 3T and 16 pico seconds for 1.5T (0.5 degree phase) in the period between measuring the sensitivity maps and doing the actual diagnostic scan. New Time alignment settings may be actualized in between scans when those scans are not related or when the time alignment can be done with a reproducibility of less than 1 degree. Time alignment itself may not introduce an additional load/situation dependable phase differences for receive channels that are also used during system calibration for other corrections. At this moment it are the receive channels of the body coil Signal received from the fine channels have more channel to channel variation than the signal received by the coarse channels. Due to temperature effect there are changes in delays of both channels. This can cause a phase differences up-to 5 degrees in 30 minutes. Therefore it is required for clinical scans where phase matters to recalibrate the delay between coarse and fine up-to 30 minutes each 3 minutes. Note: This requirement can be relaxed when a model is implemented for this phase difference that can predict the phase difference with enough accuracy (1 degree for 30 minutes). The complex sample vectors measured during the prep phases must be routed to data monitoring. This way the measurement can be displayed or monitored later on. The cluster will be identified "RX". The preparation results shall be logged in monitoring files which can be easily collected by service and used for statistical analysis. For clinical usage a scan should abort when it uses signal digitized from a receiver with incorrect coarse fine calibrated receiver. During hardware diagnostics a scan may not abort. A receiver that is not time-aligned may not cause an abortion. The system must stay silent using the assumption that only time aligned channel contributes significantly to an image. Nice To Have: A check in reconstruction that only time aligned channel contributes significantly to an image. Accompanied with an abort or warning when a not time-aligned receiver contributes to the image. Reconstruction can do so when it receives a label telling which channels are not time-aligned.

Test steps:

Step	Description	Expected Result
Step 1	In simulator environment, open a 'command box' and run p_apco_simconfig.pl -a In the menu select the WA30 configuration. Start the Field Service tool (FSF) and check that all coils are configured. If not then configure all coils in FSF, and close FSF. Then run C:\nmr\gyroscan\testbatch\m_prep_rx_vm.pl After the test has finished, verify that the expected text is displayed.	*** RX VM PrepPhase Test ** VM RX prep phase Test: Succeeded

Step	Description	Expected Result
Step 2	The test also produces the file G:\log\m_prep_rx_vm_results.ref. In case the changed results are acceptable, this file can be posted as the new reference file.	

3.1.25.27. SWCMP.Scan.PrepPhases.RxCalib.tc2

Test ID	7964
Revision	1
Product Model	Ingenia;Marlin
Gradient System	Any
Options	N.A.
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

RX phase: Calibrations

RX Test Bay Tests

The RX tests for the requirement RX.CALIB are automated by perl scripts. The table below describes where each perl script is to be executed, which test-cases it implements and which requirements it verifies.

Script:	Configuration	Test Cases
m_prep_rx_tb.pl	TestBay WA_3T	RX_CF_1 RX_TM_1
m_prep_rx_vm.pl	WMWARE configured as WA_3T	RX_WF_1 RX_ER_1

Script:	Result to include in V&V
m_prep_rx_tb.pl	G:\Temp\m_prep_rx_tb.log
m_prep_rx_vm.pl	G:\Temp\m_prep_rx_vm.log

Besides these perl-scripts, an ExamCard, toolbox psets, and an analysis program are archived as testware in the buildingblock system/mrmethods/methodspreparation/tst.

All this is incorporated in the scan software test msi. The perl scripts (and the other parts of the testware) are targeted to "C:\nmr\gyroscan\testbatch".

m_prep_rxcf.pset	Pset to measure an exponential waveform generated on the tune coil for verifying the RX calibration.
m_prep_rxta.pset	Pset used for verification of the time alignment. It generates an FM sweep as RF pulse on tune coil A, which is received by all coil elements of the posterior and body coil.
rta_rxcftestdll.dll	DLL used by the toolbox pset to analyze the coarse/fine calibration tests.
rta_rxtatestdll.dll	DLL used by the toolbox pset to analyze the time adjustment tests.
m_prep_rx_test.ExamCard	ExamCard used in the tests.
m_prep_rx_tb_results.ref	Reference file for testbay tests
m_prep_rx_vm_results.ref	Reference file for vmware-based tests

The following test cases cover part of the requirement RX.CALIB.

Either an Ingenia or an Achieva dStream system can be used for this test. (The test requires the presence of iRF hardware, in particular of a dStream Posterior coil.)

General Information

- The tests rely on the System Test and Tuning (STT) software and dStream hardware being present.
- All tests are contained in the perl script 'm_prep_rx_tb.pl'. It contains testcases RX_CF_1, RX_TM_1.

Details of test case RX_CF_1 (Coarse Fine)

For the Coarse/Fine case, the perl script performs the following tasks:

1. It runs the toolbox pset m_prep_rxcf.pset. This pset defines a measurement which receives exponential signals generated on the tune-coil-A on a receiver connected to a body coil part. This is done for two situations:
 - a. Without coarse fine calibration, for only one frequency f0.
 - b. With coarse fine calibration, for the frequencies f0, f0-500KHz, f0-250kHz, f0+125kHz, f0+485Khz.

This results in 6 STT MRD files.

1. It runs the analysis program that checks the signals in the 6 MRD files for phase continuity and amplitude continuity. Expected results:
 - a. First MRD file: a phase jump and amplitude discontinuity is observed.
 - b. Second, third, fourth, fifth and sixth MRD files: the phase is constant within certain boundaries for signals lower than 10 dBm and the ratio between the measured amplitude and expected amplitude is constant within certain boundaries.

It checks whether the monitor files are generated (.PrepPhases.RX.CALIB_09).

Details of test case RX_TM_1 Time_Alignment Working

For the Time Alignment case, the test performs the following tasks:

1. It runs the toolbox m_prep_rxta.pset. This pset defines a measurement which uses a receiver connected to a body coil part and the receivers of the posterior coil to receive an FM sweep generated on tune-coil-B. This is done for two situations (and results in 2 STT MRD files):
 1.
 - a. Without time alignment
 - b. With time alignment
1. It estimates the timing differences between the channels before time alignment and after time alignment, and whether they satisfy the requirements.
2. It checks whether the data monitor files and service monitor files are generated (.PrepPhases.RX.CALIB_09).

Requirements Coverage:

Req ID	Requirement Tag	Description
18828	SWCMP.Scan.PrepPhases.RxC alib	Time alignment and the dual ADC calibration must have been performed before IRF receivers are used during MR experiments. The results of time alignment and dual ADC measurements become invalid after the receivers have been restarted. This means that the calibrations have to be redone. The dual ADC calibration must calibrate the gain difference between coarse and fine with an accuracy lower than 0.5%. This within a frequency bandwidth of 1MHz. The following assumption is considered as valid $\text{GainRatio}(\text{df}) = \text{GainRatioAtF0} + \text{FreqSlope} * \text{df}$ With df the delta between the intended frequency and f0. The dual ADC calibration must calibrate the phase difference between coarse and fine with an accuracy lower than 0.5 degrees. This within a frequency bandwidth of 1MHz. The receivers are not really linear; The FIR filter response characteristics are optimal around the centre frequency. Chosen for this frequency is the resonance frequency. Therefore it is required to recalibrate the coarse fine when the resonance frequency deviates too much from the previously optimal frequency value. A value is chosen of 10% of the typical signal bandwidth of 400kHz = 40KHz. The time alignment must measure the parameters necessary to correct timing differences between receive channels. Also called

Req ID	Requirement Tag	Description
		ReceiveChannelDelay. After calibration the system must satisfy the following requirements: - The timing differences between receive channels are within the range [-250ns, 250ns] with respect to the transmit system (25% off smallest sample interval). Please note - The differences of the ReceiveChannelDelays between combined channels may not change more than 8 pico secods for 3T and 16 pico seconds for 1.5T (0.5 degree phase) in the period between measuring the sensitivity maps and doing the actual diagnostic scan. New Time alignment settings may be actualized in between scans when those scans are not related or when the time alignment can be done with a reproducibility of less than 1 degree. Time alignment itself may not introduce an additional load/situation dependable phase differences for receive channels that are also used during system calibration for other corrections. At this moment it are the receive channels of the body coil Signal received from the fine channels have more channel to channel variation than the signal received by the coarse channels. Due to temperature effect there are changes in delays of both channels. This can cause a phase differences up-to 5 degrees in 30 minutes. Therefore it is required for clinical scans where phase matters to recalibrate the delay between coarse and fine up-to 30 minutes each 3 minutes. Note: This requirement can be relaxed when a model is implemented for this phase difference that can predict the phase difference with enough accuracy (1 degree for 30 minutes). The complex sample vectors measured during the prep phases must be routed to data monitoring. This way the measurement can be displayed or monitored later on. The cluster will be identified "RX". The preparation results shall be logged in monitoring files which can be easily collected by service and used for statistical analysis. For clinical usage a scan should abort when it uses signal digitized from a receiver with incorrect coarse fine calibrated receiver. During hardware diagnostics a scan may not abort. A receiver that is not time-aligned may not cause an abortion. The system must stay silent using the assumption that only time aligned channel contributes significantly to an image. Nice To Have: A check in reconstruction that only time aligned channel contributes significantly to an image. Accompanied with an abort or warning when a not time-aligned receiver contributes to the image. Reconstruction can do so when it receives a label telling which channels are not time-aligned.

Test steps:

Step	Description	Expected Result
Step 1	Open a 'command box' and run the perl script m_prep_rx_tb.pl. The test proceeds autonomously and writes its result to the file G:\log\rx_test.csv. After the test, verify the contents of this file.	If the test is succesful, the file rx_test.csv contains the line RX Test: Succeeded If the test is not successful, the file rx_test.csv contains the line RX Test: Failed

Step	Description	Expected Result
Step 2	The test also produces the file G:\log\m_prep_rx_results.ref. In case acceptable changes, e.g. to the hardware, lead to the test failing, this file can be posted as the new reference file.	

3.1.25.28. SWCMP.Scan.PrepPhases.RxCalib2.tc1

Test ID	35620
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2017-09-19 14:50:42 Meaning: QA Approved - (QA) Signed by: Dingemans, Henk (nly95779) on 2017-09-19 13:25:01 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-09-19 11:06:18 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The test case is defined starting from the defect verification of MR00203612. If the fast next scan modality is used the Rx calibration phase must be skipped.

Requirements Coverage:

Req ID	Requirement Tag	Description
67950	SWCMP.Scan.PrepPhases.RxCalib2	The RX Phase drift is skipped after BolusTrak scan.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Include "MMAMN perf *" in C:\temp\bstrace.dat - Configure the RTM as a Ingenia WA15 using m_amn_set_test_environment.pl - Start the Application Software. - Start a new exam; - Import and Load the attached exam card - Create a new logfile - Run the Survey Scan	N.A. (preparation)	 SWCMP.Scan.PrepPhases.RxCalib2.tc1.z

Step	Description	Expected Result	Attachments
Step 2	- Run the first grouped scans - Extract the logfile using the attached query query.txt with the following command line: R: \\MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe "devlogcurrent.log" @query.txt > devlogcurrent_extractFast.log	Check the lines between "start preparations after scan" and "scan completed" for the second scan after m_Name = "2D_BOLUSTRAK" (the 2D_bolustrak scan) in devlogcurrent_extractFast.log No lines containing RxCSGCalib must be present	 query.zip
Step 3	- Run the second grouped scans - Extract the logfile using the same query of the previous step with the following command line: R: \\MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe "devlogcurrent.log" @query.txt > devlogcurrent_extract.log Attach the extracted logfile to this test step as objective evidence.	Check the lines between "start preparations after scan" and "scan completed" for the second scan after the last m_Name = "2D_BOLUSTRAK" (the 2D_bolustrak scan) in devlogcurrent_extract.log At least one line containing RxCSGCalib must be present	None

3.1.25.29. SWCMP.Scan.PrepPhases.RxCalibOnError.tc1.v1

Test ID	39215
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-11-30 10:50:32 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2018-11-29 16:50:29 Meaning: SME Approved - (Technical) Signed by: Visser, Kees (nly22601) on 2018-11-29 15:26:09 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test case uses simulated environment with embedded testware that can be activated to inject fault conditions.

Requirements Coverage:

Req ID	Requirement Tag	Description
72978	SWCMP.Scan.PrepPhases.RxC alibOnError.v1	- RX calibration shall notify to contact Philips service in case the calibration of an embedded receiver fails. - RX calibration shall notify to clean the connector in case the calibration of an user connected receiver fails.

Test steps:

Step	Description	Expected Result
Prep 1	- Include in bstrace.dat - o MMPRX sim_cf_phase_errors * - o MMPRX sim_errors_embedded_devices * - Use test framework to set the test configuration on a RTM on 1.5T Ingenia - Create new logfile - Select new exam - Select test protocol "Head CNS/FE/F_2D1"	N.A. (preparation)
Step 1	- Start the scan. - Acknowledge popup."Calibration of embedded failed....."	Logfile has a new logline with: oc-scan-engine ChoiceRequest::ChoiceRequest(10800, 129, 0) The scan completes after acknowledge
Step 2	- Remove the following from bstrace.dat the line "MMPRX sim_errors_embedded_devices *" - Restart the scanner simulation and wait until it is started.. - Copy previous scan - Press "Start Scan".	The scan completes
Step 3	- Copy previous scan - Connect the base-coil with the coil simulator. - Press "Start Scan" - Acknowledge popup "Calibration of Base-coil failed. Clean connector"	Logfile has a new line with: joc-scan-engine ChoiceRequest::ChoiceRequest(XPRRXCF01, 129, 0) The scan completes after acknowledge
Step 4	- Remove from bstrace.dat the line "MMPRX sim_cf_phase_errors *" - Restart the scan simulation. - Copy previous scan. :Press "Start Scan" - Attach the extracted logfile using query "text contains 'm_Name' or text contains '10800' or text contains 'XPRRXCF01'" as evidence.	The scan completes.

3.1.25.30. SWCMP.Scan.PrepPhases.TM.tc1.v1

Test ID	31017
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-02 16:58:10 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2017-03-02 16:26:25

	Meaning: SME Approved - (Business)::Bulk Transitions
	Signed by: Visser, Kees (nly22601) on 2017-03-02 15:23:01 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

TM phase:Tuning and Matching

TM Messages

Requirements Coverage:

Req ID	Requirement Tag	Description
64110	SWCMP.Scan.PrepPhases.TM.v 1	Tune all tunable coils that are selected for the subsequent preparation steps and the clinical scan and log the actual coil offset (kHz) and coil quality factor (Q). The duration of this preparation phase on the DAS shall not exceed 500 msec.

Test steps:

Step	Description	Expected Result
Step 1	1. Start application sw and create a new examination 2. Start new logfile. 3. Connect a multi channel Head coil for acquisition and position a bottle in the coil. 4. Perform a standard FFE protocol with 2 echoes (e.g. tedb/Head CNS/FE/ F_2D1; Change the number of echoes to 2) and wait for completion. 5. Inspect the logfile to check the TM preparation phase results. Each tunable coil is reported starting with 'TM: TQ' (transmit) or 'TM: RQ' (receive). a. Note that 'TM: RQ' for the body coil is not logged for all system configurations. b. Note the number of 'TM: TQ' lines differ (TX config: 4; non-TX config: 1) and will be skipped when appropriate results can be reused. c. Use the logbrowser to select the TM logging by applying the filter <i>text icontains 'TM:.'</i> or <i>text icontains 'TM will be'</i> or <i>text icontains 'TM: took '</i> and attach a screenshot of this logging extract to this step.	The logfile extract shows logline(s) starting with 'TM: TQ' that contain the coil offset (kHz) and the coil quality factor (Q).
Step 2	Inspect the screenshot created in Step 1 and look for the line 'PREPDUR: TM: took'.	Write down the duration in msec and verify that this value does not exceed 500.

3.1.26. ME PrePulses

3.1.26.1. SWCMP.Scan.PrePulses.3D-IR.Neurography

Test ID	24923
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-05-12 09:29:37 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2016-05-11 16:35:31 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on 2016-05-11 16:31:09 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test checks utilization of a specific RF pulse by inspection of the RF properties in Sequence Development Viewer. For reproducibility reasons test for 1.5T system.

Requirements Coverage:

Req ID	Requirement Tag	Description
56977	SWCMP.Scan.PrePulses.3D-IR.Neurography	For the 3DView selection "Nerve STIR", a non-selective adiabatic inversion pulse optimized for brachial plexus (HS3_4_6 or HS4_4_8) shall replace the default inversion pulse.

Test steps:

Step	Description	Expected Result
prep	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Select TEDB - Add protocol tedb\Head CNS\PrePulses\ Nerve_STIR to executionlist - Under the contrast tab, check or set the following protocol parameters: - technique = IR 3D VIEW = Nerve STIR - startup echoes = 4 - fat suppression = no	N.A. (preparation)
Step 1	Using the Sequence Development Viewer, inspect the attributes of the RF object RF::inv .	For RF::inv the attribute must be rf_shape = HS3_4_6 with pulse angle at least 1300 and bandwidth at least 1500 Hz.

3.1.26.2. SWCMP.Scan.PrePulses.BB.UI.2D.SE.tc1

Test ID	25494
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-05-25 09:51:39 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-25 08:14:53 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-05-24 15:59:34 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This is an UI test to check the availability and range of protocol parameters.

This test uses the MR Application software on a Virtual Machine, RTM, or Testbay using the m_amn_set_test_environment.pl tool to set the test environment.

Although this test is valid for all products, field strengths, and gradients, a specific MR configuration is selected for reproducibility reasons.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol 2D_SE_BB to executionlist - Under the contrast tab, set BB pulse to: yes	NA (preparation)	 CTS PrePulses_V2.zip
Step 1	Check the availability and range of the parameter delay in the UI.	The parameter delay is listed, its allowed values are default and user defined .	None
Step 2	Check the availability and range of the parameter STIR in the UI.	The parameter STIR is listed, its allowed range is no, yes .	None
Step 3	Change the BB pulse to MSDE , and inspect the availability and range of the parameter STIR in the UI.	The parameter STIR is not listed.	None
Step 4	After selecting delay: default, check if the info parameter BB inv. Delay (ms) appears.	The info parameter appears and reports a value between 10 and 2000 ms .	None

3.1.26.3. SWCMP.Scan.PrePulses.BB.UI.2D.TFE.tc1

Test ID	25498
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-05-25 09:51:44 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-25 08:14:55 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-05-24 15:59:36 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This is an UI test to check the availability and range of protocol parameters.

This test uses the MR Application software on a Virtual Machine, RTM, or Testbay using the m_amn_set_test_enviroment.pl tool to set the test environment.

Although this test is valid for all products, field strengths, and gradients, a specific MR configuration is selected for reproducibility reasons.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol M2D_TFE_tfepp_BB to scan list - Under the contrast tab, set TFE prepulse to: black blood - Set bb pulse type: default	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Check the availability and range of the parameter delay in the UI.	The parameter default is listed, its allowed range is default, user defined .	None
Step 2	Check for the availability of the STIR parameter.	The parameter STIR is not listed.	None
Step 3	Change bb pulse type from default to MSDE .	No conflict occurs.	None

3.1.26.4. SWCMP.Scan.PrePulses.BB.UI.MSDE.tc1

Test ID	25495
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-05-25 09:51:46 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-25 08:14:58 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-05-24 15:59:38 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This is an UI test to check the availability and range of protocol parameters.

This test uses the MR Application software on a Virtual Machine, RTM, or Testbay using the m_amn_set_test_environment.pl tool to set the test environment.

Although this test is valid for all products, field strengths, and gradients, a specific MR configuration is selected for reproducibility reasons.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol 3D_TFE_MSDE_OnceRefocused to executionlist	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Under the contrast tab, check the availability and range of the parameter MSDE mode in the UI.	The parameter MSDE mode is listed, its allowed range is at least [intracranial, nerve, flow ghost supp., and user defined]	None
Step 2	After selecting MSDE mode: User defined , check the availability and range of the parameter refocusing pulses	The parameter refocusing pulses is listed, its allowed range is [1, 2].	None
Step 3	After selecting MSDE mode: User defined , check the availability and range of the parameter refocusing type	The parameter refocusing type is listed, its allowed range is at least [composite, adiabatic].	None
Step 4	After selecting MSDE mode: User defined , check the availability and range of the parameter TEprep	The parameter TEprep is listed, its allowed range is at least shortest, user defined .	None

Step	Description	Expected Result	Attachments
Step 5	After selecting MSDE mode: User defined , and TEprep: user defined , check the availability and range of the parameter (ms)	The parameter (ms) is listed, its allowed range is at least 0-500 ms .	None
Step 6	After selecting MSDE mode: User defined , check the availability and range of the parameter Venc (RL, AP, FH)	The parameter Venc (RL, AP, FH) is listed, and allows the input of three float numbers, each with a range of: -30.00 to 30.00	None

3.1.26.5. SWCMP.Scan.PrePulses.BB.UI.tc1

Test ID	25491
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-05-25 09:51:48 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-25 08:15:00 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-05-24 15:59:40 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This is an UI test to check the availability and range of protocol parameters.

This test uses the MR Application software on a Virtual Machine, RTM, or Testbay using the m_amn_set_test_enviroment.pl tool to set the test environment.

Although this test is valid for all products, field strengths, and gradients, a specific MR configuration is selected for reproducibility reasons.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol 2D_SE_BB to executionlist - Add protocol M2D_TFE_tfepp_BB to executionlist	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Open the 2D_SE_BB protocol.	Check that the following items are listed under the contrast tab: BB pulse\no	None

Step	Description	Expected Result	Attachments
		- BB pulse\yes (DIR) - BB pulse\MSDE	
Step 2	Change the scan mode to M2D .	No conflict occurs.	None
Step 3	Change the scan mode to MS .	A conflict occurs on the parameters Scan Mode and BB pulse .	None
Step 4	Change the BB pulse type from yes (DIR) to MSDE .	The conflict is resolved.	None
Step 5	Change the scan mode to 3D , BB pulse type from MSDE to yes (DIR) , slice thickness to 25 , and TE to shortest .	No conflict occurs.	None
Step 6	Change the BB pulse type from yes (DIR) to MSDE	No conflict occurs.	None
Step 7	Change the scan mode to 2D .	No conflict occurs.	None
Step 8	Change the scan mode to MS .	No conflict occurs.	None
Step 9	Change the scan mode to M2D .	No conflict occurs.	None
Step 10	Open the 2D_TFE_tfepp_BB protocol.	Check that the following items are listed under the Contrast tab: - TFE Prepulse\black blood\default - TFE Prepulse\black blood\MSDE There is no item called BB pulse in the Contrast tab.	None
Step 11	Change the Water-fat shift to minimum , and change the Fast imaging mode to TFEPI .	No conflict occurs.	None
Step 12	Change the Scan mode to 2D .	No conflict occurs.	None
Step 13	Change the Fast imaging mode to TFE .	No conflict occurs.	None
Step 14	Change the Scan mode to MS .	A conflict occurs on the parameters Scan mode and bb pulse type .	None
Step 15	Change the Scan mode to 3D .	A conflict occurs on the parameters Scan mode and bb pulse type .	None
Step 16	Change the BB pulse type from default to MSDE .	The conflict is resolved.	None
Step 17	Change the scan mode to MS .	No conflict occurs.	None
Step 18	Change the scan mode to M2D .	No conflict occurs.	None
Step 19	Change the scan mode to 2D .	No conflict occurs.	None

3.1.26.6. SWCMP.Scan.PrePulses.CardiacRestGrid.UI.tc1

Test ID	25499
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Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-03-01 15:32:45 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Polders, Daniel Louis (310171444) on 2016-03-01 14:52:12 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-03-01 14:37:40 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This is an UI test to check the availability and range of protocol parameters.

This test uses the MR Application software on a Virtual Machine, RTM, or Testbay using the m_amn_set_test_enviroment.pl tool to set the test environment.

Although this test is valid for all products, field strengths, and gradients, a specific MR configuration is selected for reproducibility reasons.

Requirements Coverage:

Req ID	Requirement Tag	Description			
56767	SWCMP.Scan.PrePulses.Cardiac RestGrid.UI	The following parameters shall be listed on the contrast page:			
		Name	Range	Visible when	Description
		Motion\REST grid	-no -yes	Cardiac Synchronisation = trigger	Allow the user to select Cardiac rest grid functionality
		dimension	-lines -grid	REST grid = yes	Sets the type of grid/line
		spacing (mm)	[min value* - 30]	REST grid = yes	Sets spacing of grid lines*: the actual minimal spacing of grid lines depend on the in plane resolution.

Req ID	Requirement Tag	Description			
		tag grid angle	[-90, 90]	REST grid = yes	This parameter rotates the axis of the tagging grid or lines
		breathhold scheme [2]	[1, 20]	REST grid = yes	

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol TFE_card_grid to executionlist	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Check the availability and range of the parameter REST grid in the UI (Motion tab).	The parameter REST grid is listed, its allowed range is at least no, yes	None
Step 2	After selecting REST grid: yes , check the availability and range of the parameter dimension in the UI.	The parameter dimension is listed, its allowed range is at least lines, grid	None
Step 3	After selecting REST grid: yes , check the availability and range of the parameter spacing (mm) in the UI.	The parameter spacing (mm) is listed, its allowed range is 6.05 – 30 mm	None
Step 4	On the Geometry tab, increase the ACQ voxel size RL (mm) to 4, and AP (mm) to 4 On the Motion tab, inspect the range of the spacing (mm) parameter again.	The range of the spacing parameter (mm) is 6.05 - 30 mm .	None
Step 5	After selecting REST grid: yes , check the availability and range of the parameter breathhold scheme in the UI.	The parameter Breathhold scheme is listed, it allows for two integers to be set to 1-20	None

3.1.26.7. SWCMP.Scan.PrePulses.MSDE.Application.Cranial.tc1

Test ID	25509
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-02-15 11:32:43 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Dingemans, Henk (nly95779) on 2016-02-15 10:58:36 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Huang, Alan (310109142) on 2016-02-15 09:51:56 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test checks the correct setting of several VAL01 parameters based on the selection of a single MSDE mode. To do this, manual inspection using the Sequence Development Viewer is performed.

Requirements Coverage:

Req ID	Requirement Tag	Description
55794	SWCMP.Scan.PrePulses.MSDE.Application.Cranial	For the MSDE_mode "intracranial", MSDE shall employ the following parameters: • 2 refocusing pulses • refocusing type = composite • TE = shortest • Venc (AP/LR/FH) = 3/3/3 cm/s

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	• Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. • Import the attached examcard • Add protocol 3D_TSE_MSDE_TwiceRefocused to the scan list • Under the contrast tab, set MSDE mode to Intracranial • Validate the protocol	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Using the Sequence Development Viewer, inspect the following VAL01_MSDE parameters: VAL01_MSDE_refoc_pulses VAL01_MSDE_refoc_type VAL01_MSDE_te_enum VAL01_MSDE_venc	VAL01_MSDE_refoc_pulses = 2 VAL01_MSDE_refoc_type = composite VAL01_MSDE_te_enum = shortest VAL01_MSDE_venc = [3.00, 3.00, 3.00]	None

3.1.26.8. SWCMP.Scan.PrePulses.MSDE.Application.FlowGhostSuppression.tc1

Test ID	25510
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-02-15 11:32:45 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Dingemans, Henk (nly95779) on 2016-02-15 10:58:38 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Huang, Alan (310109142) on 2016-02-15 09:51:59 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test checks the correct setting of several VAL01 parameters based on the selection of a single MSDE mode. To do this, manual inspection using the Sequence Development Viewer is performed.

Requirements Coverage:

Req ID	Requirement Tag	Description
55795	SWCMP.Scan.PrePulses.MSDE.Application.FlowGhostSuppression	For the MSDE_mode "flow ghost supp", MSDE shall employ the following parameters: 2 refocusing pulses - refocusing type = composite - TE = shortest - Venc (AP/LR/FH) = 0.0 / 0.0 / 15.0 cm/s

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol 3D_TSE_MSDE_TwiceRefocused to the scan list - Under the contrast tab, set MSDE mode to flow ghost supp - Validate the protocol	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Using the Sequence Development Viewer, inspect the following VAL01_MSDE parameters: VAL01_MSDE_refoc_pulses VAL01_MSDE_refoc_type VAL01_MSDE_te_enum VAL01_MSDE_venc	VAL01_MSDE_refoc_pulses = 2 VAL01_MSDE_refoc_type = composite VAL01_MSDE_te_enum = shortest VAL01_MSDE_venc = [0.00, 0.00, 15.00]	None

3.1.26.9. SWCMP.Scan.PrePulses.MSDE.Application.Neurography.tc1

Test ID	25508
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-05-25 09:51:53 Meaning: QA

	Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-25 08:15:02 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2016-05-24 15:59:43 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test checks the correct setting of several VAL01 parameters based on the selection of a single MSDE mode. To do this, manual inspection using the Sequence Development Viewer is performed.

Requirements Coverage:

Req ID	Requirement Tag	Description
55793	SWCMP.Scan.PrePulses.MSDE.Application.Neurography	For the MSDE_mode "nerve", MSDE shall employ the following parameters: • 2 refocusing pulses • refocusing type = adiabatic • TE = 65 ms • Venc (AP/LR/FH) = 1/1/1 cm/s

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	• Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. • Import the attached examcard • Add protocol 3D_TSE_MSDE_TwiceRefocused to the scan list • Under the Contrast tab, set MSDE mode to nerve • Validate the protocol	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Using the Sequence Development Viewer, inspect the following VAL01_MSDE parameters: VAL01_MSDE_refoc_pulses VAL01_MSDE_refoc_type VAL01_MSDE_te_enum VAL01_MSDE_echo_time VAL01_MSDE_venc	VAL01_MSDE_refoc_pulses = 2 VAL01_MSDE_refoc_type = adiabatic VAL01_MSDE_te_enum = user defined VAL01_MSDE_echotime = 65.00 VAL01_MSDE_venc = [1.00, 1.00, 1.00]	None

3.1.26.10. SWCMP.Scan.PrePulses.MSDE.RefocusingPulses.tc1

Test ID	52738
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.

Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:18:47 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:05:46 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:26 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the setting of several RF pulses

Requirements Coverage:

Req ID	Requirement Tag	Description
55799	SWCMP.Scan.PrePulses.MSDE.RefocusingPulses	iMSDE shall support composite as well as adiabatic refocusing pulses.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesMsdeRefocusingPulses in the results file is zero.

3.1.26.11. SWCMP.Scan.PrePulses.MSDE.RF-GR_overlap.tc1.v2

Test ID	52723
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2021-12-16 14:18:49 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:41:06 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:26

	Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the absence of overlap between the durations of RF and GR objects.

Requirements Coverage:

Req ID	Requirement Tag	Description
56677	SWCMP.Scan.PrePulses.MSDE. RF-GR_overlap	MSDE shall generate a non-slice selective preparation sequence (without overlap of RF pulses with gradient pulses).

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesMsdeRfGrOverlap in the results file is zero.

3.1.26.12. SWCMP.Scan.PrePulses.MSDE.Sequence.tc1

Test ID	52724
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharaappa, Deepak Shekhar (320098697) on 2021-12-16 14:18:52 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:42:50 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:26 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the general structure of the RF pulses and gradients of the MSDE sequence

Requirements Coverage:

Req ID	Requirement Tag	Description
98171	SWCMP.Scan.PrePulses.MSDE. Sequence.v1	The MSDE sequence consists of: - Two excitation pulses (tip-down and tip-up pulses), with in between - One or two refocusing pulses, each flanked by

Req ID	Requirement Tag	Description
		3. Two gradient pulses (so 2 gradients or 4 gradients). 4. One spoiler gradient along the S-axis. Note that venc gradients are only enabled when venc > 0.0. In the case of twice refocused (improved) MSDE, this sequence is preceded by an additional gradient pair.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesMsdeSequence in the results file is zero.

3.1.26.13. SWCMP.Scan.PrePulses.MSDE.SequenceOrdering.tc1.v2

Test ID	52739
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:18:55 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:06:31 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:27 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test inspects the relative placement of the SQ_MSDE in the SOR object

Requirements Coverage:

Req ID	Requirement Tag	Description
55800	SWCMP.Scan.PrePulses.MSDE.SequenceOrdering	MSDE shall be performed directly before the sequence readout.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and	The reported number of failed tests related to

Step	Description	Expected Result
	inspect the results file.	SwCmpScanPrePulsesMsdeSequenceOrdering in the results file is zero.

3.1.26.14. SWCMP.Scan.PrePulses.MSDE.TEshortest.tc1.v2

Test ID	52725
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharaappa, Deepak Shekhar (320098697) on 2021-12-16 14:18:57 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:48:04 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:27 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies for the utilization of all available gradient strength and slew rate.

Requirements Coverage:

Req ID	Requirement Tag	Description
56601	SWCMP.Scan.PrePulses.MSDE.TEshortest	The minimum MSDE duration (TE_prep) shall be calculated automatically.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesMsdeTeShortest in the results file is zero.

3.1.26.15. SWCMP.Scan.PrePulses.MSDE.VencConversion.tc1.v2

Test ID	52726
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Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:18:59 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:47:29 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:27 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies that the user is allowed to prescribe Venc values in the patient reference frame (RL-AP-FH)

Requirements Coverage:

Req ID	Requirement Tag	Description
56676	SWCMP.Scan.Prepulses.MSDE.VencConversion	The user is allowed to prescribe Venc values in the patient reference frame (RL-AP-FH), which will be translated into the corresponding gradient 1st moment that corresponds with a 180 degree phase shift for this velocity. The first moment is derived by the following conversion from Venc (cm/s) to $m1$ (mT / ms²/m): $m1 = 0.5 * \pi / (\gamma * \text{venc}), \text{ which reduces to } m1 = 1174.33 / \text{venc}$ For a once-refocused MSDE sequence, the total $m1$ is given by: $m1_tot = G * (s+lenc) * (lenc + d1 + 2*s)$ And for a twice refocused MSDE sequence, $m1$ tot is given by: $m1_tot = G * (s+lenc) * (4*lenc + 2*d1 + 2*d2 + 8*s)$ Where G = is the plateau gradient strength, s = duration of the gradient slope, lenc the duration of the gradient plateau, d1 the duration of the refocusing pulse, d2 = the time between the second and third gradient lobe. (see figure below): The (user defined) Venc values in RL - AP - FH patient reference frame shall be converted to $m1$ (aka gradient 1st moment) values in the XYZ reference frame (i.e. along the physical gradient axes).

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesMsdeVencConversion in the results file is zero.

3.1.26.16. SWCMP.Scan.PrePulses.MTC.Par.tc1

Test ID	41063
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:28:58 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:05:12 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:45:31 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of MTC parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description			
75176	SWCMP.Scan.PrePulses.MTC.Par	Name	Range	Visible when	Description
		MTC	no on resonance off resonance off res. multi	Scan technique = SE or FFE Fast Imaging Mode = none, TSE, or TFE	Allows the acquisition of MTC acquisitions, using on-resonance, off-resonance or off-resonance multi pulses.
		nr repetitions	[1 - 100]	MTC = off res. multi	This parameter determines the number of off-resonance

Req ID	Requirement Tag	Description			
					MTC pulses applied within a shot.
		angle	[1- 1200]	MTC = off res. multi	This parameter determines the angle of each of the pulses
		duration	1 - 100]	MTC = off res. multi	This parameter determines the duration of each of the pulses
		frequency	[1 - 1000]	MTC = off res. multi	This parameter determines the frequency of each of the pulses
		MTC	[no yes]	MTC not equal to no, Dynamic study = individual and dynamic scans >1	Allows the acquisition of two dynamics, one with MTC enabled, one disabled.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesMtcPar in the results file is zero.

3.1.26.17. SWCMP.Scan.PrePulses.REPP.Par.tc1

Test ID	41062
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:29:02 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:05:15 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:46:14 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of research prepulse parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description			
78764	SWCMP.Scan.PrePulses.REPP.Par.v1	Name	Range	Visible when	Description
		Custom prepulse	[no, yes]	always	The custom prepulse is set on/off
		pulse shape	[block, sg_300_100_0,wowir, hypsec]	Custom prepulse = yes	This parameter defines the pulse shape
		elements	[1 - 16]	Custom prepulse = yes	Defines the number of elements in the composite pulse
		pulse angles	[0.00 -	Custom	This

Req ID	Requirement Tag	Description			
			1800.00]	prepulse = yes	parameter defines per element the flip angle (in degrees).
		pulse durs	[0.01 - 50.00]	Custom prepulse = yes	This parameter defines per element the duration (in [ms]).
		pulse phases	[-180.00 - 180.00]	Custom prepulse = yes	This parameter defines per element the phases (in degrees).
		pulse intervals	[0.00, 0.064 - 500.00]	Custom prepulse = yes	This parameter defines per consecutive elements the interval time (in [ms]). The interval is measured from the end of an element to the start of the next element. For a typical binomial custom pulse these intervals are set to 0.
		delay time	[0.00 - 10000.00]	Custom prepulse = yes	The time interval between the center of the

Req ID	Requirement Tag	Description			
					(composite) pulse to the center of the RF excitation pulse
		freq. offset	[-100000.00 - 100000.00]	Custom prepulse = yes	This parameter defines the transmit frequency offset (in Hz) relative to the central ¹ H Larmor frequency
		select grad in (uT/m)	[-10000.00 - 10000.00]	Custom prepulse = yes	The parameter defines the gradient strength in (uT/m)
		IR: pre SE only	[no, yes]	technique = IR and Custom prepulse = yes	The "IR: pre SE only" parameter only has effect for long IR delays and long TR. No means: custom pulses are repeated as many times as possible during a long TR, Yes means: applying the custom pulses fewer times, i.e. only just once before the excitation

Req ID	Requirement Tag	Description			
					(with 90 degrees pulse) of each slice.
		dyn custom prepulse	[no, yes]	Custom prepulse = yes and Dynamic study = individual and dyn scans>1	This "dyn custom prepulse" parameter is used as follows: No means: Custom prepulse is active in all dynamics, Yes means: Custom prepulse not active in 1st dynamic and active in all subsequent dynamics.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPrePulsesReppPar in the results file is zero.

3.1.26.18. SWCMP.Scan.PrePulses.T2Prep.UI.tc1

Test ID	25497
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-02-01 17:32:40 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk

	(nly95779) on 2016-02-01 11:16:21 Meaning: SME Approved - (Technical)::Bulk Transitions
	Signed by: Huang, Alan (310109142) on 2016-01-13 10:42:40 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This is an UI test to check the availability and range of protocol parameters.

This test uses the MR Application software on a Virtual Machine, RTM, or Testbay using the m_amn_set_test_environment.pl tool to set the test environment.

Although this test is valid for all products, field strengths, and gradients, a specific MR configuration is selected for reproducibility reasons.

Requirements Coverage:

Requirements Coverage:

Req ID	Requirement Tag	Description																
56766	SWCMP.Scan.PrePulses.T2Prep .UI	The following parameters shall be listed on the contrast page: <table><tr><th>Name</th><th>Range</th><th>Visible when</th><th>Description</th></tr><tr><td>Contrast\T2p rep</td><td>-no -yes</td><td>Fast imaging mode = not none nor EPI</td><td>Allows the user to enable the T2 preparation sequence.</td></tr><tr><td>echo time (ms)</td><td>15-500 ms</td><td>T2prep = not no</td><td>Duration of the T2prep sequence</td></tr><tr><td>refocusing pulses</td><td>1, 2, 4</td><td>T2prep = not no</td><td>Number of refocusing pulses employed.</td></tr></table>	Name	Range	Visible when	Description	Contrast\T2p rep	-no -yes	Fast imaging mode = not none nor EPI	Allows the user to enable the T2 preparation sequence.	echo time (ms)	15-500 ms	T2prep = not no	Duration of the T2prep sequence	refocusing pulses	1, 2, 4	T2prep = not no	Number of refocusing pulses employed.
Name	Range	Visible when	Description															
Contrast\T2p rep	-no -yes	Fast imaging mode = not none nor EPI	Allows the user to enable the T2 preparation sequence.															
echo time (ms)	15-500 ms	T2prep = not no	Duration of the T2prep sequence															
refocusing pulses	1, 2, 4	T2prep = not no	Number of refocusing pulses employed.															

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add protocol 3D_T2prep_FLAIR to executionlist	N.A. (preparation)	 CTS PrePulses_V2.zip
Step 1	Check the availability and range of the parameter T2prep in the UI (Contrast tab).	The parameter T2prep is listed, its allowed range is at least no, yes	None
Step 2	After selecting T2prep: yes , check the availability and range of the parameter	The parameter echo time (ms) is listed, its allowed range is 15 – 500 ms	None

Step	Description	Expected Result	Attachments
	echo time (ms) in the UI.		
Step 3	After selecting T2prep: yes , check the availability and range of the parameter refocusing pulses in the UI.	The parameter refocusing pulses is listed, its allowed range is 1,2,4	None

3.1.27. ME PreScans

3.1.27.1. SWCMP.Scan.PreScan.B0PreScan.Scheduling.EPIGeometryCorrection.tc1

Test ID	41194
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-27 11:55:57 Meaning: QA Approved - (QA) Signed by: Nabuurs, Christine (310147336) on 2019-05-20 17:23:13 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:14:12 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies that a B0 PreScan request is generated for a scan with EPI Geometry correction enabled.

Requirements Coverage:

Req ID	Requirement Tag	Description
70550	SWCMP.Scan.PreScans.B0PreScan.Scheduling	When the scanning subsystem validates a clinical scan that requires a B0 prescan, the validation output shall include a request for a B0 prescan.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPreScansB0PreScanSchedulingEPI GeometryCorrection in the results file is zero.

3.1.27.2. SWCMP.Scan.PreScans.B0PreScan.tc1.v1

Test ID	37709
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-05-17 13:40:34 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2018-05-17 13:24:24 Meaning: SME Approved - (Technical) Signed by: Rozijn, Tom (nly95848) on 2018-05-17 11:16:06 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test checks the main protocol properties of the B0 PreScan.

It checks that:

- no images are produced.
- the scan is a 'dixon' scan with a specific echo time, delta echo time and flip angle.

This test makes use of the standard application software and can be performed on a RTM.

Requirements Coverage:

Req ID	Requirement Tag	Description
70563	SWCMP.Scan.PreScans.B0PreScan.v1	When the scanning subsystem receives the request to define a B0 prescan, it will construct a prescan with fixed characteristics based on information of the clinical scan

Test steps:

Step	Description	Expected Result
Preparation	• Open the Testframework, and Configure the system as a WA30 (3.0T Ingenia) system. • Start Simulation mode. • Start the Coilsate Simulator UI. • Connect the "ds Head1 32ch" and the "ds Head2 32ch" coil. • Select the PPDB. • Start the "Application Software" and create a new patient. • Start the Graphical Patsup Simulator. Move the table	N.A. (preparation)

Step	Description	Expected Result
	200 mm in and set the Light Visor by pressing the "Light Visor" button.	
Step 1	- Load a Diffusion scan from the Head/Brain/Diffusion folder, and make sure that Multiband Sense is enabled on the geometry tab. - Observe in the scanner status area that before the clinical scan is performed, first a CoilSurveyScan, a SenseRefScan and B0_PreScan are performed. - Observe that none of the PreScans (CoilSurveyScan, a SenseRefScan and B0_PreScan) are visible in the execution list. - Observe that when going to the viewing environment, only images for the clinical scan are available.	- Before the clinical scan is performed, first a CoilSurveyScan, a SenseRefScan and B0_PreScan are performed. - The CoilSurveyScan, a SenseRefScan and B0_PreScan are not visible in the ExecutionList. - The CoilSurveyScan, SenseRefScan and B0_PreScan did not produce viewable images. The only images available are from the clinical scan.
Step 2	- Drag the B0_PreScan from the PreScans\PreScans folder into the execution list. - Change the Scan Control Parameter "B0 pre scan UI" to 'full' - Verify on the contrast page that the protocol is a 4 echo Dixon scan with a first echo time of 1.32 ms, an echo spacing of 1.14 ms and a flip angle of 10 deg. - Open the protocol and verify that selecting something else than 'yes' for the "mDIXON" parameter leads to a conflict.	- The protocol is a 4 echo Dixon scan with a first echo time of 1.32 ms, an echo spacing of 1.14 ms and a flip angle of 10 deg. - It is only possible to select 'yes' for the "mDIXON" parameter.
Step 3	Open the protocol, and set the 'SofTone mode' to Comfortone.	No conflict is given.
Step 4	- Close the Application Software. - Run the C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl script and Configure the system as a WA15 (1.5T Ingenia) system. - Drag the B0_PreScan from the PreScans\PreScans folder into the execution list. - Change the Scan Control Parameter "B0 pre scan UI" to 'full' - Verify on the contrast page that the protocol is a 4 echo Dixon scan with a first echo time of 2.64 ms, an echo spacing of 2.28 ms and a flip angle of 10 deg.	The protocol is a 4 echo Dixon scan with a first echo time of 2.64 ms, an echo spacing of 2.28 ms and a flip angle of 10 deg.

3.1.27.3. SWCMP.Scan.PreScans.Optimizer.PatientOrientation.tc1

Test ID	37674
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.

Signatures	Signed by: Moore, Liz (gbr01023) on 2018-05-17 13:42:07 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2018-05-16 14:47:15 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2018-05-16 14:12:12 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies that the patient orientation is correctly taken into account by the prescan optimizer, even when it is changed while a prescan is being acquired.

Requirements Coverage:

Req ID	Requirement Tag	Description
70550	SWCMP.Scan.PreScans.B0PreScan.Scheduling	When the scanning subsystem validates a clinical scan that requires a B0 prescan, the validation output shall include a request for a B0 prescan.

Test steps:

Step	Description	Expected Result
Prep 1	- Use the test framework to set the configuration 'Ingenia 3.0T'. - Ensure the use of TEDB - enable visibility of automatically inserted prescans: - open C:\Program Files (x86)\PMS\Mipnet91\examcards_cs.exe.config - set 'ShowPreScan' to 'true' and save the file - Start application software and create a new examination. - Start the graphical patsup simulator; set the light visor on the head position and move the table in. - Start the coil state simulator and attach the ds Head and ds Base coils - Start acquisition/recon simulation; set it to 'simulation mode (acq on host, recon on host)' - Set patient orientation to head first/supine - insert a spiral protocol in the scan list, Head/CNS > noncartesian > SE spiral dixon - start a new log file Read the next steps before executing them	n/a (preparation)
Step 1	Press 'start scan'	Three scans are inserted and started automatically: - CoilSurveyScan - SENSE reference scan - B0_prescan
Step 2	While the B0 prescan is running, change the patient	B0_prescan is successfully completed

Step	Description	Expected Result
	orientation to feet first Attach the log file to this step.	- New prescans are inserted automatically - The prescans are successfully completed - The SE spiral Dixon scan is successfully completed
Clean up	disable visibility of automatically inserted prescans: - open C:\Program Files (x86)\PMS\Mipnet91\examcards_cs.exe.config - set 'ShowPreScan' to 'false' and save the file	n/a (clean up)

3.1.27.4. SWCMP.Scan.PreScans.Optimizer.tc1.v1

Test ID	41055
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-05-02 15:20:33 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2019-05-02 10:02:21 Meaning: SME Approved - (Technical) Signed by: Visser, Kees (nly22601) on 2019-05-02 08:53:40 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Tests the prescan optimizer implementation by means of unit tests.

Requirements Coverage:

Req ID	Requirement Tag	Description
70550	SWCMP.Scan.PreScans.B0PreScan.Scheduling	When the scanning subsystem validates a clinical scan that requires a B0 prescan, the validation output shall include a request for a B0 prescan.
70575	SWCMP.Scan.PreScans.Optimizer.Logging	The prescan optimizer shall be able to dump its state into a log file.
70559	SWCMP.Scan.PreScans.Optimizer.Interface	The prescan optimizer shall implement an interface that allows a client to - notify prescan optimizer that requests for a clinical scan changed - notify prescan optimizer that a prescan changed state - clear prescan optimizer - set the list of clinical scans to be managed by the prescan optimizer - get the available prescan responses - get next action to be performed for a clinical scan - create a response based on a completed manually inserted prescan

Test steps:

Step	Description	Expected Result
Step 1	on the SDE, open a command prompt	No failures are reported

Step	Description	Expected Result
	- navigate to the software archive, baselined on the version to be tested - go to the folder that contains the prescan actions code > ... system\scanning\prescanactions\mak - Unit tests are executed as part of the build process. Build and run test executables by issuing the command 'mrmake -b .'	

3.1.27.5. SWCMP.Scan.PreScans.ScanTime.tc1.v4

Test ID	57465
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-07-13 14:14:11 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2023-07-13 13:15:47 Meaning: SME Approved - (Business) Signed by: Engelen, Walter van (nly95948) on 2023-07-13 11:34:50 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies specified scan time for coil survey, SENSE reference and B1 calibration scan.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanPreScansScanTime in the results file is zero.

3.1.27.6. SWCMP.Scan.PreScans.tc1.v6

Test ID	46104
Revision	1
Product Model	Any

Gradient System	Any
Options	N.A.
Signatures	Signed by: Krishnakumar, Ashwin (ing02466) on 2020-04-16 05:40:57 Meaning: QA Approved - (QA) Signed by: Mariappan, Yogesh kannan (310134237) on 2020-04-15 13:13:29 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-04-15 12:22:23 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This automated test covers the automatic insertion of Prescans and ShimCalculation jobs by executing a number of validations in the Batch Interpreter environment, as well as via the Examcards scripting environment. It covers the following prescan types:

- Automatic Coil Selection
- Insertion and reuse Coil Survey scan
- Insertion and reuse of Sense Reference Scan (including metal scan requesting a TSE ref scan)
- PDF generating B0 map requests and ability to create a ready to run B0map for these requests.
- Insertion and reuse of B1 map based RF shimming

Two tools are used:

- Batch interpreter
- Examcards simulator

In batch interpreter, we can make prescan in a batch, and see if a next scan reuses that prescan or not. Scan requests are extracted to create prescans which are checked on correct validation behavior.

The Examcards simulator is used to perform the following tests that cannot be performed by the batch interpreter:

- handling of table positions
- actual automatic inserting of prescans
- handling of shim calculations

Requirements Coverage:

Req ID	Requirement Tag	Description
45461	SWCMP.Scan.SENSE.coca.prescan.v1	Predefined prescan protocols (SENSE reference and coil survey scan) are required to provide reconstruction SW coil sensitivity maps for • SENSE unfolding • uniformity corrections • automatic coil selection • recon compression (FCC)
45470	SWCMP.Scan.PreScans.Results.v1	• The clinical scan shall fill its coil parameters according to the results of the Coil Selection Reconstruction job.
45471	SWCMP.Scan.PreScans.TablePosition.v1	General requirement:

Req ID	Requirement Tag	Description
		- The optimal table position of a Pre Scan is set to that of the requesting scan. The Coil Survey Scan and the SENSE Reference scan provide information about coil sensitivity profiles. Since the posterior coil moves with respect to the patient when the table is used, for the Coil Survey Scan and SENSE Reference scan some special requirements on the table position holds: - When the Coil Survey Scan or the SENSE Reference scan use the posterior coil, the Coil Survey Scan or SENSE Reference scan must be forced to the optimal table position of the requesting clinical scan. - When a clinical scan uses the posterior coil, the table must be forced to the actual table position on which the Pre Scan was made. - A scan can use a Pre Scan that was inserted by another scan. In that case, the table position on which the Pre Scan was made might deviate from the optimal table of the requesting scan. As long as the difference between the actual position and the optimal table position is smaller than the max_allowed_table_offset, this is allowed. Otherwise a new PreScan needs to be inserted. Since the information provided by the B1 calibration is independent of coil sensitivity profiles, these special posterior coil requirements do not hold for the B1 Calibratin scans.
74597	SWCMP.Scan.B1CAL.AutoInsert .CardiacTX.Method.v1	For Cardiac acquisitions, the system shall automatically insert a cardiac triggered B1 calibration scan using a FFE EPI dualAngle technique with slice-selective excitation.
74606	SWCMP.Scan.B1CAL.AutoInsert .Method.v1	For all applications except for Cardiac, Methods provides a fixed B1 calibration scan based on the DREAM technique that will automatically be inserted by the system.
70550	SWCMP.Scan.PreScans.B0PreS can.Scheduling	When the scanning subsystem validates a clinical scan that requires a B0 prescan, the validation output shall include a request for a B0 prescan.

Test steps:

Step	Description	Expected Result
Step 1	Open a command window and run 'tf' From the component level tests, select and run the prescan regression test (all tests).	Test executes succesfully.
Step 2	- Inspect the results at: 'G:\Log\testframework\PrescanRegression\[GitRepository]\[GitBranch]\[timestamp]\PrescanRegression_detailed_report.txt' - Attach this file as objective evidence	File contains the following content : The filtered logfile is identical to the reference file The filtered logfile is different from the reference file Self-Test successful The filtered logfile is identical to the reference file The filtered logfile is identical to the reference file The filtered logfile is identical to the reference file The filtered logfile is identical to the reference file The filtered logfile is identical to the reference file

3.1.27.7. SWCMP.Scan.PreScans.tc1.v7

Test ID	61759
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Reddy, Srikanth (310031195) on 2025-05-21 17:56:27 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-05-21 17:26:22 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2025-05-15 10:46:26 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This automated test covers the automatic insertion of Prescans and ShimCalculation jobs by executing a number of validations in the Batch Interpreter environment, as well as via the Examcards scripting environment. It covers the following prescan types:

- Automatic Coil Selection
- Insertion and reuse Coil Survey scan
- Insertion and reuse of Sense Reference Scan (including metal scan requesting a TSE ref scan)
- PDF generating B0 map requests and ability to create a ready to run B0map for these requests.
- Insertion and reuse of B1 map based RF shimming

Two tools are used:

- Batch interpreter
- Examcards simulator

In batch interpreter, we can make prescan in a batch, and see if a next scan reuses that prescan or not. Scan requests are extracted to create prescans which are checked on correct validation behavior.

The Examcards simulator is used to perform the following tests that cannot be performed by the batch interpreter:

- handling of table positions
- actual automatic inserting of prescans
- handling of shim calculations

Requirements Coverage:

Req ID	Requirement Tag	Description
45461	SWCMP.Scan.SENSE.coca.prescan.v1	Predefined prescan protocols (SENSE reference and coil survey scan) are required to provide reconstruction SW coil sensitivity maps for • SENSE unfolding • uniformity corrections • automatic coil selection • recon compression (FCC)

Req ID	Requirement Tag	Description
45470	SWCMP.Scan.PreScans.Results.v1	The clinical scan shall fill its coil parameters according to the results of the Coil Selection Reconstruction job.
45471	SWCMP.Scan.PreScans.TablePosition.v1	General requirement: - The optimal table position of a Pre Scan is set to that of the requesting scan. The Coil Survey Scan and the SENSE Reference scan provide information about coil sensitivity profiles. Since the posterior coil moves with respect to the patient when the table is used, for the Coil Survey Scan and SENSE Reference scan some special requirements on the table position holds: - When the Coil Survey Scan or the SENSE Reference scan use the posterior coil, the Coil Survey Scan or SENSE Reference scan must be forced to the optimal table position of the requesting clinical scan. - When a clinical scan uses the posterior coil, the table must be forced to the actual table position on which the Pre Scan was made. - A scan can use a Pre Scan that was inserted by another scan. In that case, the table position on which the Pre Scan was made might deviate from the optimal table of the requesting scan. As long as the difference between the actual position and the optimal table position is smaller than the max_allowed_table_offset, this is allowed. Otherwise a new PreScan needs to be inserted. Since the information provided by the B1 calibration is independent of coil sensitivity profiles, these special posterior coil requirements do not hold for the B1 Calibratin scans.
74597	SWCMP.Scan.B1CAL.AutoInsert.CardiacTX.Method.v1	For Cardiac acquisitions, the system shall automatically insert a cardiac triggered B1 calibration scan using a FFE EPI dualAngle technique with slice-selective excitation.
74606	SWCMP.Scan.B1CAL.AutoInsert.Method.v1	For all applications except for Cardiac, Methods provides a fixed B1 calibration scan based on the DREAM technique that will automatically be inserted by the system.
70550	SWCMP.Scan.PreScans.B0PreScan.Scheduling	When the scanning subsystem validates a clinical scan that requires a B0 prescan, the validation output shall include a request for a B0 prescan.

Test steps:

Step	Description	Expected Result
Step 1	Open a command window and run 'tf' From the component level tests, select and run the prescan regression test (all tests).	Test executes successfully.
Step 2	- Inspect the results at: 'G:\Log\testframework\PrescanRegression\[GitRepository]\[GitBranch]\[timestamp]\PrescanRegression_detailed_report.txt' - Attach this file as objective evidence	File contains the following content : The filtered logfile is identical to the reference file The filtered logfile is different from the reference file Self-Test successful The filtered logfile is identical to the reference file The filtered logfile is identical to the reference file The filtered logfile is identical to the reference file

Step	Description	Expected Result

3.1.28. ME ProfOrders

3.1.28.1. SWCMP.Scan.ProfOrders.CE.tc1

Test ID	51117
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-06-11 09:13:11 Meaning: QA Approved - (QA) Signed by: Mariappan, Yogesh kannan (310134237) on 2021-06-11 08:12:47 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2021-06-07 14:58:07 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability and verification of Profile Orders related parameters.

Manual Test cases covered :

SWCMP.Scan.ProfOrders.ce_UI.tc1

SWCMP.Scan.ProfOrders.ce_orders.tc1

Requirements Coverage:

Requirements Coverage:

Req ID	Requirement Tag	Description			
56393	SWCMP.Scan.ProfOrders.ce_UI	For a 2D/3D FFE scan the Dyn/Ang page shall show the following Parameters:			
		Name	Range	Visible when	Description
		CE Profile Order	high_low, low_high,	Angio/ Contrast enh.	high_low: high_low

Req ID	Requirement Tag	Description											
			linear, rev_linear high_low, low_high, linear, rev_linear, CENTRA high_low, low_high, linear, rev_linear, CENTRA, rev_CENTR	= contrast enh and Scan mode = MS Angio/ Contrast enh. = contrast enh Scan mode = 3D, Keyhole = yes Angio/ Contrast enh. = contrast enh Scan mode = 3D, Keyhole = no	outer profile order low_high: low high outer profile order linear: linear outer profile order rev_linear: reversed linear outer profile order CENTRA: radial hybrid ky-kz profile order with separate central and peripheral ky-kz part rev_CENTRA: reversed radial hybrid ky-kz profile order								
For a 3D TFE scan the Dyn/Ang page shall show the following Parameters: <table><tr><th>Name</th><th>Range</th><th>Visible when</th><th>Description</th></tr><tr><td>CENTRA</td><td>yes, no</td><td>Fast Imaging Mode = TFE Angio/Contra st enh. = no</td><td>no: CENTRA not used yes: radial hybrid ky-kz profile order with separate central and peripheral ky-kz part</td></tr></table>						Name	Range	Visible when	Description	CENTRA	yes, no	Fast Imaging Mode = TFE Angio/Contra st enh. = no	no: CENTRA not used yes: radial hybrid ky-kz profile order with separate central and peripheral ky-kz part
Name	Range	Visible when	Description										
CENTRA	yes, no	Fast Imaging Mode = TFE Angio/Contra st enh. = no	no: CENTRA not used yes: radial hybrid ky-kz profile order with separate central and peripheral ky-kz part										

Req ID	Requirement Tag	Description			
		CE Profile Order	high_low, low_high, linear, rev_linear	Angio/ Contrast enh. = contrast enh. Keyhole = no	high_low: high_low outer profile order low_high: low high outer profile order linear: linear outer profile order rev_linear: reversed linear outer profile order
56388	SWCMP.Scan.ProfOrders.ce_orders	Methods shall provide the following sequential CE outer profile orders for 2D FFE and 3D FFE/TFE scans. - low high (starting at the lowest k profile ending at the highest k profile) - high low (starting at the highest k profile, ending at the lowest k profile) - linear (starting highest negative profile, ending at the highest positive profile) - reversed linear (starting highest positive profile, ending at the highest negative profile)			

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanProfOrdersCE in the results file is zero.

3.1.28.2. SWCMP.Scan.ProfOrders.CeHybrid.Centra.tc1.v1

Test ID	55234
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharaappa, Deepak Shekhar (320098697) on 2022-07-28 10:15:24 Meaning: QA Approved - (QA)

	Signed by: Sharma, Nishant (320059006) on 2022-07-28 08:01:07 Meaning: SME Approved - (Technical)
	Signed by: Graaf, Ruud de (nly95749) on 2022-07-27 14:56:22 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The purpose of this test is to verify the peripheral profile order (CENTRA or rev_CENTRA) for a Contrast Enhanced FE protocol.

Requirements Coverage:

Req ID	Requirement Tag	Description
56395	SWCMP.Scan.ProfOrders.ce_hybrid.rev_centra	Methods shall provide a reversed centra hybrid profile order for 3D FFE scans, in which the profile order in ky-kz space is starting with the smallest radial distance from the kspace origin and moving towards the peripheral part.
56389	SWCMP.Scan.ProfOrders.ce_hybrid.centra	Methods shall provide a centra hybrid profile order for 3D FFE scans (Radiol 225:583-588, 2002), in which the profile order in ky-kz space is starting with the largest radial distance from the kspace origin and moving towards the central ky-kz profiles. Ky-kz space is split into a central part and a peripheral part.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanProfOrdersCeHybridCentra in the results file is zero.

3.1.28.3. SWCMP.Scan.ProfOrders.regression.tc1

Test ID	32564
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-28 15:22:02 Meaning: QA Approved - (QA)
	Signed by: Crijns, Sjoerd (310257923) on 2017-03-28 14:23:47 Meaning: SME Approved - (Business)

	Signed by: Graaf, Ruud de (nly95749) on 2017-03-28 14:17:28 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
34556	SWCMP.Scan.Methods.NSA.Smart	The scan protocol shall support serial number of signal averaging (SMART - Serial Motion Artifact Reduction Technique).
65704	SWCMP.Scan.k-space_distrib.Halfscan	The scan protocol shall support the following Halfscan k-space manipulation techniques: - Partial acquisition of k-space in Y-direction by omitting part of the ky encoding steps at one side of k-space, defined by the partial matrix factor. - Partial acquisition of k-space in Z-direction by omitting part of the kz encoding steps at one side of k-space, defined by the partial partition factor. - Combination of the above.
34557	SWCMP.Scan.Methods.NSA.PartialNSA	The scan protocol shall support Partial NSA factors.
65789	SWCMP.Scan.Methods.NSA.PartialNSA.Profiles.v2	Partial NSA shall be possible for linear, low-high, and asymmetric profile orders.
65786	SWCMP.Scan.k-space_distrib.Proforders.regression	Subsystem Scanning shall implement different acquisition profile orders for 2D and 3D scans in order to influence the image contrast.
66057	SWCMP.Scan.Dixon.meTSE.Technique.v3	Dixon TSE multi-echo scanning shall be supported with intra-shot interleaved echoes. Figure 1. Sequence diagram for mDIXON-TSE multi-echo with intra-shot interleaved echoes. Multi-echo Dixon TSE consists of an intra-shot interleaved multi-echo TSE sequence with the acquisition of shifted echoes every other TSE refocusing pulse. In the example of figure 1, the top row shows a normal TSE sequence, and the bottom shows a multi-echo (number of echoes = 2) intra-shot interleaved mDIXON-TSE sequence.

Test steps:

Step	Description	Expected Result
Prep1	a) On a ROF, start G:\testframework\src\main.rb by typing: ruby main.rb b) Select option: [2] Component Tests c) Select option: [1] Profile Orders Regression	N.A. (preparation)
Step 1	After the test is finished open G:\Log\methodstestframework\ProfileOrders\#stream\#date-time\profileOrders_detailed_report.txt Selftest: Verify that the correct date and time, configuration,	The correct date and time, configuration, software stream and SWID are indicated at the top of the file.

Step	Description	Expected Result
	software stream and SWID are indicated at the top of the file.	
Step 2	Open G:\Log\methodstestframework\ ProfileOrders\#stream\#date-time\ profileOrders_dashboard_report.txt Check the number of - tests to execute - failed tests - executed tests - skipped tests	The number of tests to execute is equal to the number of executed tests. The number of failed and skipped tests is 0.
Step 3	In G:\Log\methodstestframework\ ProfileOrders\#stream\#date-time\ profileOrders_detailed_report.txt, verify for each protocol in the report if differences have been found in the profile order with respect to the reference file.	No differences have been found in the profile order with respect to the reference file.
Wrap up	Attach the result files G:\Log\methodstestframework\ ProfileOrders\#stream\#date-time\ profileOrders_detailed_report.txt and G: \Log\methodstestframework\ ProfileOrders\#stream\#date-time\ profileOrders_dashboard_report.txt as supporting evidence.	N.A. (wrap up)

3.1.29. ME Propeller

3.1.29.1. SWCMP.Scan.Propeller.BasicTechnique.tc1

Test ID	27510
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-05-02 14:19:10 Meaning: QA Approved - (QA) Signed by: IJperen, Gert van (nly95968) on 2016-05-02 11:40:52 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-02 11:06:19 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test is designed to be performed with the application software that runs on a scanner simulator in combination with instrumented logging.

Test steps:

Step	Description	Expected Result
Prep 1	Set the system configuration by executing m_amn_set_test_environment.pl Choose any configuration and wait for the script to complete Start Application Software from MR Boot Configuration Manager Press 'New Exam' and set the gender of the patient to 'Female'. Prescribe other mandatory parameters	Not applicable
Step 1	Load the following protocol from TEDB: Hospital\Pelvis\PROPELLER\2D TSE	The protocol should load without conflicts
Step 2	On the contrast tab, check the options selected for "Acquisition Mode" and "scan mode"	The acquisition mode and the Scan mode should be set to "MultiVane" and "2D" respectively
Step 3	Change the "Scan mode" to MS	There should be no conflict when scan mode is set to MS
Step 4	Change the "Scan mode" to M2D	There should be no conflict when scan mode is set to M2D
Step 5	Check the selections for "Technique" and "Fast Imaging mode"	Technique should be set to "SE" and Fast Imaging mode should be set to "TSE"
Step 6	On the Postproc tab, check the selection of "Images "	Image type should be set to "M"
Step 7	Change the "Images" to R	No conflict should be observed when images is set to R
Step 8	Change the "Images" to I	No conflict should be observed when images is set to I
Step 9	Change the "Images" to P	No conflict should be observed when images is set to P
Step 10	Run the scan	Scan should complete successfully

3.1.29.2. SWCMP.Scan.Propeller.Pars.tc1.v2

Test ID	27649
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2016-05-02 14:19:42 Meaning: QA Approved - (QA) Signed by: IJperen, Gert van (nly95968) on 2016-05-02 11:40:56 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-05-02 11:06:23 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test is designed to be performed with the application software that runs on a scanner simulator in combination with instrumented logging.

Test steps:

Step	Description	Expected Result
Prep 1	Set the system configuration by executing m_amn_set_test_environment.pl Choose any configuration and wait for the script to complete Start Application Software from MR Boot Configuration Manager Press 'New Exam' and set the gender of the patient to 'Female'. Prescribe other mandatory parameters	Not applicable
Step 2	Load the following protocol from TEDB: Hospital\Pelvis\PROPELLER\T2W_FFE_MVXD	The protocol should load without conflicts
Step 3	On the contrast tab check the "Acquisition Mode"	The MultiVane option should be available and selected
Step 4	Check the range for "MultiVane percentage (%)"	The range should be 50.00 – 800.00
Step 5	Check the "Fast imaging mode"	Fast Imaging mode should be set to "none"

Step	Description	Expected Result
Step 6	Check value of Multivane percentage (%) and the range for "TRs per blade"	The MultiVane percentage (%) value should be set to 310 and the range of "TRs per blade should be "8-56"
Step 7	Set Multivane percentage (%) to 200 Check the range for "TRs per blade"	The range should be 8 – 40
Step 8	Set Multivane percentage (%) to 400 Check the range for "TRs per blade"	The range should be 8 – 64
Step 9	Set "Fast imaging mode" to TFE, "shots per blade" to 2, and "TFE factor" to 12 Check the range for "shots per blade"	The range should be 1 - 6
Step 10	Set "TFE factor" to 8 and check the range for "shots per blade"	The range should be 1 – 10
Step 11	Set "TFE factor" to 4 and check the range for "shots per blade"	The range should be 1 – 16
Step 12	On the motion tab check the range for "MultiVane gross motion correction"	The options should be no and yes

3.1.30. ME REST

The structure of this test specification is as follows. Each specific requirement chapter in the requirement specification has a corresponding test chapter. Within each chapter, tests that require the same test environment (PC, WS, STM, OTM) are gathered in sections. Within each section, specific tests are divided over sub sections.

3.1.30.1. SWCMP.Scan.REST.Circular.tc1

Test ID	7241
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]

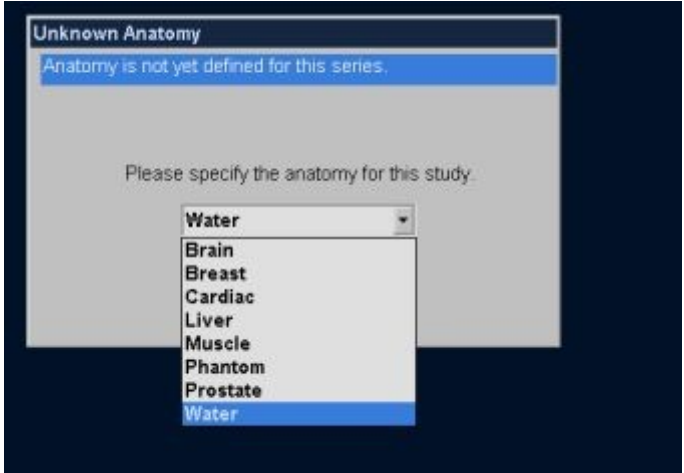
Approval Status	Approved
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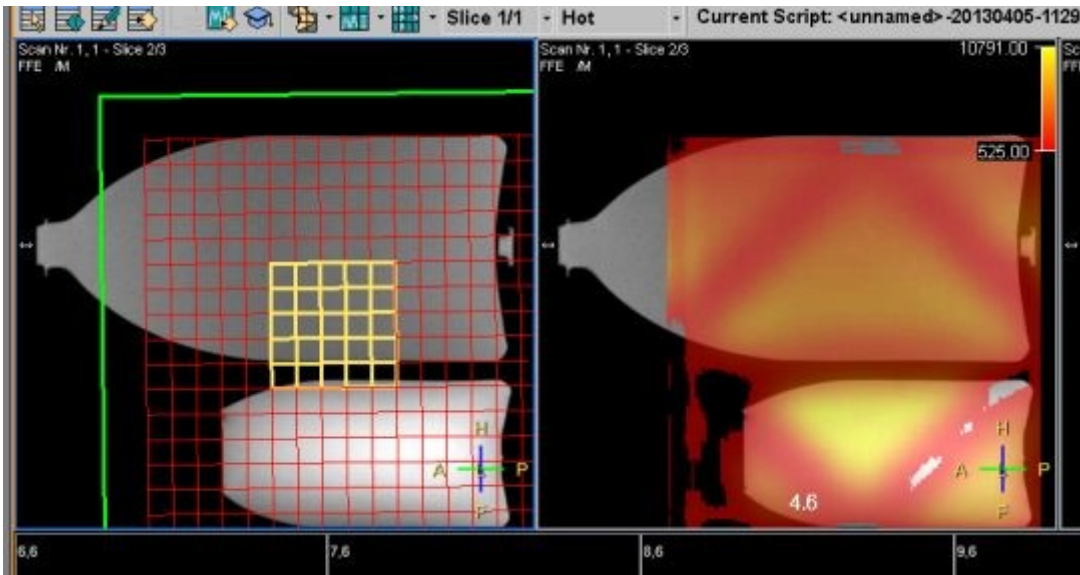
Description:

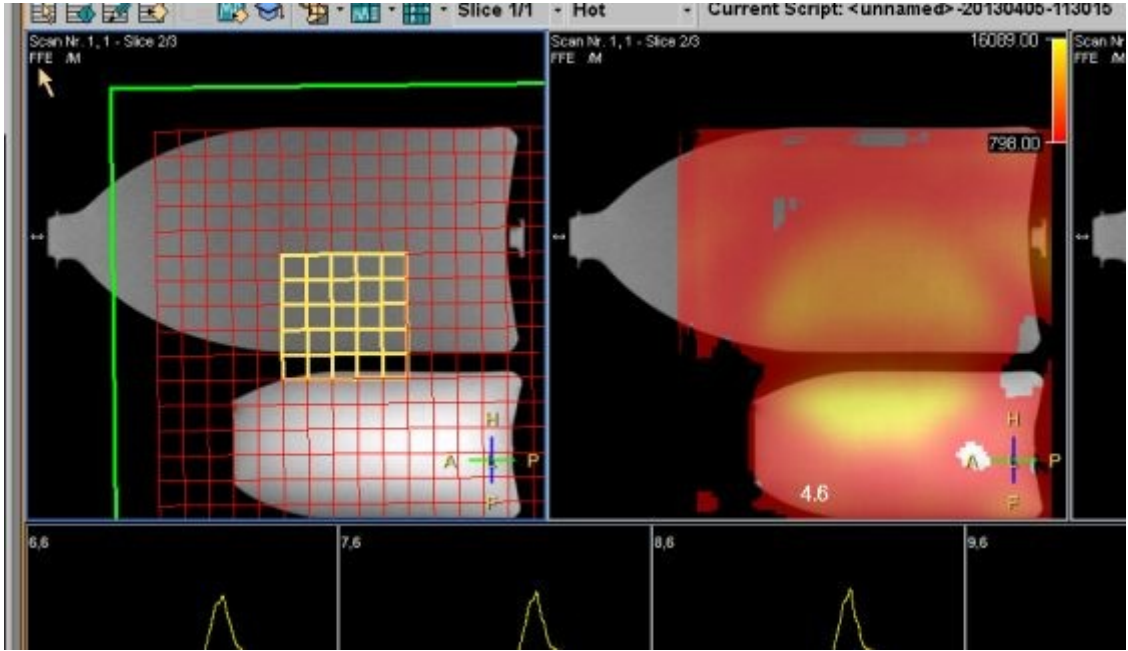
Requirements Coverage:

Req ID	Requirement Tag	Description
17395	SWCMP.Scan.REST.Circular	Circular REST slabs are supported, see section Circular REST.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Position two bottles, preferably one oil and one water upright with a small gap in feet head direction on the table.	None	None
Step 2	Use the examCard m_imn_rest_ts_circular.ExamCard found in rmethods\imaging\tst. Run the survey scan.	None	None
Step 3	Plan the geometry MyGeo so that the middle is in the gap between the two bottles. To avoid foldover artifacts, make sure the FOV fully contains the two bottles. Wait for completion 2 d csi rest 10-water	None	None
Step 4	Open the scans in SpectroView. Select 'Water' for the study anatomy (see attached figure)	None	
Step 5	Enter 'edit script', change it to the setting presented in attached figure	None	

Step	Description	Expected Result	Attachments
			
Step 6	Check the results for the 2d csi rest 4 scan	Test succeeds when the slabs are in the expected position for water (see attached figure)	
			

Step	Description	Expected Result	Attachments
Step 7	Check the results for the 2d csi rest 10 scan.	Test succeeds when the slabs are in the expected position for water (see attached figure).	
			

3.1.30.2. SWCMP.Scan.REST.tc1.v1

Test ID	32560
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:46 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:37:11 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-03-22 10:23:04 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
17392	SWCMP.Scan.REST.Parallel	Parallel REST slabs are supported, see section Parallel REST slabs.
17393	SWCMP.Scan.REST.Perpendicular	Perpendicular REST slabs are supported, see section Perpendicular REST slabs.
17394	SWCMP.Scan.REST.Free	Free REST slabs are supported, see section Free REST slabs.
17395	SWCMP.Scan.REST.Circular	Circular REST slabs are supported, see section Circular REST.

Test steps:

Step	Description	Expected Result										
Prep 1	Set configuration ingenia 3.0T using m_amn_set_test_environment.pl. Start a new log file. Enable all SW option-bits.	n/a (preparation).										
Step 1	Load protocol tedb\Hospital\Head CNS\REST\SE_MS_2PAR, Verify that the REST slab type is set to parallel. <table><tr><td>• thickness</td><td>5 - 80 mm</td></tr><tr><td>• position</td><td>feet, head</td></tr><tr><td>• gap</td><td>default, user defined</td></tr><tr><td>• power</td><td>1-4</td></tr></table>	• thickness	5 - 80 mm	• position	feet, head	• gap	default, user defined	• power	1-4	The indicated parameters can be set in the UI; the ranges are as specified.		
• thickness	5 - 80 mm											
• position	feet, head											
• gap	default, user defined											
• power	1-4											
Step 2	Set the REST slab type to perpendicular. Verify the that the following parameters can be set for REST slabs of type perpendicular, with ranges as specified in the second column. <table><tr><td>• thickness</td><td>5 - 265 mm</td></tr><tr><td>• position</td><td>anterior, posterior</td></tr><tr><td>• distance</td><td>-450 - 450 mm</td></tr><tr><td>• angulation</td><td>-90 - 90 degrees</td></tr><tr><td>• power</td><td>1-4</td></tr></table>	• thickness	5 - 265 mm	• position	anterior, posterior	• distance	-450 - 450 mm	• angulation	-90 - 90 degrees	• power	1-4	The indicated parameters can be set in the UI; the ranges are as specified.
• thickness	5 - 265 mm											
• position	anterior, posterior											
• distance	-450 - 450 mm											
• angulation	-90 - 90 degrees											
• power	1-4											

Step	Description	Expected Result										
	<table><tr><td></td><td></td></tr></table>											
Step 3	Set the REST slab type to free. Verify the that the following parameters can be set for REST slabs of type free, with ranges as specified in the second column. <table><tr><td>• orientation</td><td>transverse, sagittal, coronal</td></tr><tr><td>• thickness</td><td>5 - 80 mm</td></tr><tr><td>• power</td><td>1-4</td></tr></table>	• orientation	transverse, sagittal, coronal	• thickness	5 - 80 mm	• power	1-4	The indicated parameters can be set in the UI; the ranges are as specified.				
• orientation	transverse, sagittal, coronal											
• thickness	5 - 80 mm											
• power	1-4											
Step 4	Set the REST slab type to circular. Verify the that the following parameters can be set for REST slabs of type circular, with ranges as specified in the second column. <table><tr><td>• circular slabs</td><td>2-16</td></tr><tr><td>• thickness</td><td>5 - 265mm</td></tr><tr><td>• gap</td><td>default, user defined</td></tr><tr><td>• angulation</td><td>-90.0 – 90.0 degrees</td></tr><tr><td>• power</td><td>1-4</td></tr></table> Add a log file as objective evidence.	• circular slabs	2-16	• thickness	5 - 265mm	• gap	default, user defined	• angulation	-90.0 – 90.0 degrees	• power	1-4	The indicated parameters can be set in the UI; the ranges are as specified.
• circular slabs	2-16											
• thickness	5 - 265mm											
• gap	default, user defined											
• angulation	-90.0 – 90.0 degrees											
• power	1-4											

3.1.31. ME RNAV

3.1.31.1. SWCMP.Scan.RNAV.display.tc1.v1

Test ID	32327
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-06 10:36:26 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2017-04-06 10:18:16 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-05 16:05:26 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test uses acquisitions on a testbay to confirm that a user can show and hide the Navigator Display window at will, and that it hides automatically when the scan no longer employs navigator functionality, as described in the traced requirement.

Requirements Coverage:

Req ID	Requirement Tag	Description
9814	SWCMP.Scan.RNAV.display	The Navigator display is specified in section 6 Navigator planscan

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	- Configure the system to use TEDB. - Move the patient table out and position the body phantom transverse in the bore. - Enable Patsup simulation (Start-> MR Applications -> Service Tools -> Enable PatSup Simulation) - Perform a survey with three orientations (eg. "scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the attached figure. If not adjust the offcentres of the phantom and repeat the survey. - Start a new log file. - Enable physiology simulation (Control Scan Parameters > General > Physiology Simulation = yes, Apply, Hide)	n/a (preparation)	 planning.zip
Step 1	Select protocol "gate 100 mm" in "Head CNS/RNAV" and start the scan	Observe that the display pops up	None
Step 2	Hide the Navigator Display with the hide button	Observe that the display disappears	None

Step	Description	Expected Result	Attachments
Step 3	Recall the navigator display via the menu: 'Examination > Navigator Display...'	Observe that the Navigator Display is shown again.	None
Step 4	Modify the protocol, disable Navigator respiratory comp in the motion group and start the scan	Observe that the display disappears.	None
Step 5	Enable the Navigator Display again in via 'Examination > Navigator Display...' Attach the log file to this step as objective evidence.	The data of the previous scan must be still visible.	None

3.1.31.2. SWCMP.Scan.RNAV.display.tc2

Test ID	7002
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator display tool

Position of leading navigator

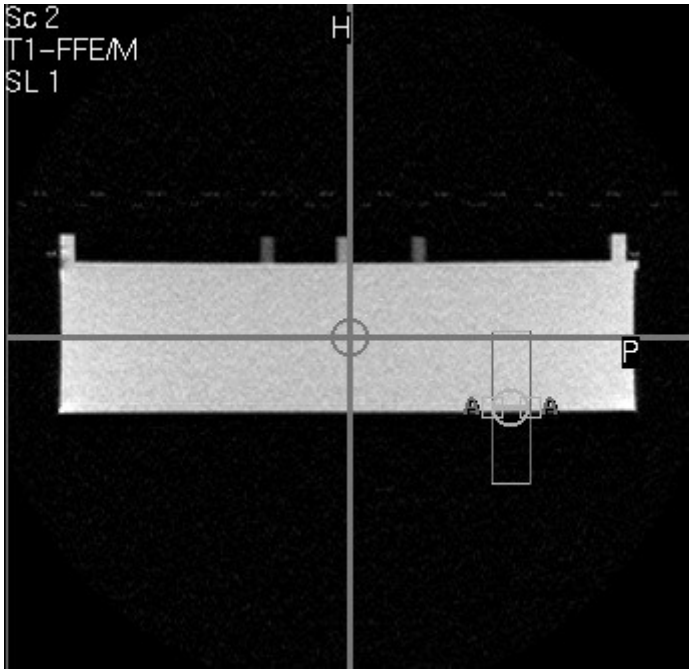
Use testbay.

- Move the patient table out and position the body phantom transverse in the bore. Switch Patsup simulation on. Patsup can be found as follows: Start-> MR Applications -> Service Tools -> Enable PatSup Simulation.
- Perform a survey with three orientations ("scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the figure below. If not adjust the offcentres of the phantom and repeat the survey.
- Enable physiology simulation

Requirements Coverage:

Req ID	Requirement Tag	Description
9814	SWCMP.Scan.RNAV.display	The Navigator display is specified in section 6 Navigator planscan

Test steps:

Step	Description	Expected Result	Attachments
Step 1	- Modify protocol "gate 100 mm" in "Head CNS/RNAV" - Position the navigator half over a phantom and half in the air like in the figure attached above, at the beginning of this section. - Perform the scan and watch the navigator display.	Watch the navigator display and see if it displays the location you entered. This is a subjective visual test, no exact measures are required. In the case of the example it means: - the top part of the navigator is white and the bottom is black because the "head" part of the navigator is in the phantom. - the green lines indicating the kernel are in the white and black part respectively. - Repeat this test for some other offcentres of the navigator kernel, e.g. with the kernel ending at the border of the phantom. - Verify that the kernel is displayed at the position as you planned it.	
			

3.1.31.3. SWCMP.Scan.RNAV.display.tc3

Test ID	7003
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator display tool

Position of trailing navigators

Use testbay.

- Move the patient table out and position the body phantom transverse in the bore. Switch Patsup simulation on. Patsup can be found as follows: Start-> MR Applications -> Service Tools -> Enable PatSup Simulation.
- Perform a survey with three orientations ("scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the figure below. If not adjust the offcentres of the phantom and repeat the survey.

Enable physiology simulation

Requirements Coverage:

Req ID	Requirement Tag	Description
9814	SWCMP.Scan.RNAV.display	The Navigator display is specified in section 6 Navigator planscan

Test steps:

Step	Description	Expected Result
Step 1	• Select protocol "gate 100 mm" in "Head CNS/RNAV" and change the navigator type into trail. • Position the navigator on a recognizable parts of the phantom. • Perform the scan and watch the navigator display.	• The navigator display shows the acquired navigators • The kernel is displayed at the position as you planned it.

3.1.31.4. SWCMP.Scan.RNAV.display.tc4

Test ID	7004
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator display tool

Position of leading and trailing navigators

Use testbay.

- Move the patient table out and position the body phantom transverse in the bore. Switch Patsup simulation on. Patsup can be found as follows: Start-> MR Applications -> Service Tools -> Enable PatSup Simulation.
- Perform a survey with three orientations ("scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the figure below. If not adjust the offcentres of the phantom and repeat the survey.

Enable physiology simulation

Requirements Coverage:

Req ID	Requirement Tag	Description
9814	SWCMP.Scan.RNAV.display	The Navigator display is specified in section 6 Navigator planscan

Test steps:

Step	Description	Expected Result
Step 1	• Modify protocol "gate 100 mm" in "Head CNS/RNAV" and change the navigator type into lead and trail. • Position the navigator on a recognizable parts of the phantom. • Perform the scan and watch the navigator display.	• The display displays two beams, a vertical scroll bar appears and you can adjust the display window. • The kernels are displayed at the position as you planned it.

3.1.31.5. SWCMP.Scan.RNAV.display.tc5.v1

Test ID	32556
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 16:38:12 Meaning: QA Approved - (QA) Signed by: Groen, Jan (nly95735) on 2017-03-23 16:25:57 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 16:20:35 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Navigator display tool

Position of two navigators

This test verifies that the navigator display is working correctly for 2 navigators.

Requirements Coverage:

Req ID	Requirement Tag	Description
9814	SWCMP.Scan.RNAV.display	The Navigator display is specified in section 6 Navigator planscan

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Prep 1	On a test bay, move the patient table out and position the body phantom transverse in the bore. Enable physiology simulation. Make sure that all CSK SW options except for Option 02 are switched off. Create a new logfile.	N/A (preparation)
Step 1	- Modify protocol "gate 100 mm" in "Head CNS/RNAV" and increase the number of navigators to two. - Position the navigators on "recognizable parts of the phantom". - Perform the scan and watch the navigator display. Attach the logfile and screenshot of the navigator display as objective evidence.	- The display displays two beams, a vertical scroll bar appears and you can adjust the display window. - The kernels are displayed at the position as you planned it.

3.1.31.6. SWCMP.Scan.RNAV.exam_align.functionality.tc1

Test ID	7025
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 3 min

Check Navigator exam alignment on 1.5T TEST BAY

Skipping mechanism for automatic preparation mode changing Navigator length, position and orientation

Test equipment, environment & configuration

- 1.5 T TEST BAY with Physiology simulation and a head phantom

Preconditions

- Take the head phantom and position it with the flat part on the table. After moving the table to the iso center switch the table movement to manual. Use the body coil, no other coils in the magnet! Perform a survey and load the images in plan scan. Position the Navigator box on the head phantom.
- Enable trace channel "mmurnav track *" and restart the scanner process

Requirements Coverage:

Req ID	Requirement Tag	Description
9818	SWCMP.Scan.RNAV.exam_align.functionality	In the automatic mode Navigator preparation shall be enforced in case of 1. Selection of a new Exam 2. Table movement 3. Navigator window selection differences 4. Gating levels are required by the scan and gating level determination has not been performed

Test steps:

Step	Description	Expected Result
Step 1	- Select "track/dyn/S" in "Head CNS/RNAV", change to a breath hold and track protocol with scale factor 1.0 and number of dynamics to 1. Duplicate the protocol. In the second protocol change the Navigator preparation mode to "auto"matic and adapt the Navigator length, position or orientation. - Start the scans	Perform the first breath hold scan. Check that starting the second breath hold scan the Navigator display and the Navigator signal from the first breath hold scan is cleaned up. After starting the second breath hold scan the Navigator signal of only the second breath hold scan is present.

3.1.31.7. SWCMP.Scan.RNAV.exam_align.functionality.tc2.v1

Test ID	32581
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:42 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:21:21 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-03-22 17:11:32 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Check Navigator exam alignment on 1.5T TEST BAY

Skipping mechanism for automatic preparation mode with the gate and track preparation phase requirement

Requirements Coverage:

Req ID	Requirement Tag	Description
9818	SWCMP.Scan.RNAV.exam_align.functionality	In the automatic mode Navigator preparation shall be enforced in case of 1. Selection of a new Exam 2. Table movement 3. Navigator window selection differences 4. Gating levels are required by the scan and gating level determination

Req ID	Requirement Tag	Description
		has not been performed

Test steps:

Step	Description	Expected Result
Prep 1	Take the head phantom and position it with the flat part on the table. After moving the table to the iso center switch the table movement to manual. Use the body coil, no other coils in the magnet! Make sure that all CSK SW options are switched off. Enable physiology simulation. Create a new logfile. Perform a survey.	N/A (preparation)
Step 1	Select "track/dyn/S" in "Head CNS/RNAV", change to a breath hold and track protocol with scale factor 1.0 and number of dynamics to 1. Plan the navigator on the head phantom. Duplicate the protocol. In the second protocol change the Navigator preparation mode to automatic and adapt to a gate and track protocol. Remove the breath hold. Make a copy of the gate and track scan, so that there are 3 scans – one initial breath hold and track scan and two gate and track scans without breath hold. Start the scans. Attach the logfile and screenshots as objective evidence.	Check that starting the second gate and track scan the Navigator display and the Navigator signal from the first breath hold scan is cleaned up. After starting the second scan the Navigator signal of only the second scan is present. Check that starting the third gate and track scan the Navigator display and signal from the previous scan stays. After starting the third scan the Navigator signal of this scan is added after a red scan separation line.

3.1.31.8. SWCMP.Scan.RNAV.exam_align.precision.tc1

Test ID	7021
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 3 min

Check Navigator exam alignment on 1.5T TEST BAY

Tracking in the measurement direction

Test equipment, environment & configuration

- 1.5 T TEST BAY with Physiology simulation and a head phantom

Preconditions

- Take the head phantom and position it with the flat part on the table. After moving the table to the iso center switch the table movement to manual. Use the body coil, no other coils in the magnet! Perform a survey and load the images in plan scan. Position the Navigator box on the head phantom.
- Enable trace channel "mmurnav track *" and restart the scanner process

Requirements Coverage:

Req ID	Requirement Tag	Description
9817	SWCMP.Scan.RNAV.examinalign .UI	The reuse of Navigator information shall be provided in case that an <i>automatic</i> Navigator preparation mode is selected
9818	SWCMP.Scan.RNAV.examinalign .functionality	In the automatic mode Navigator preparation shall be enforced in case of 1. Selection of a new Exam 2. Table movement 3. Navigator window selection differences 4. Gating levels are required by the scan and gating level determination has not been performed
9819	SWCMP.Scan.RNAV.examinalign .precision	Motion alignment of scans within an Exam shall be in the order of less or equal of 4 mm on a phantom setup

Test steps:

Step	Description	Expected Result
Step 1	• Select protocol "track/dyn/M" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section (not the grid section). Change the dynamic protocol to a breath hold and track protocol with scale factor 1.0 and number of dynamics to 1. Set the Navigator preparation mode to "auto"matic. Duplicate the protocol. • Make sure that the Slice orientation of the plane is coronal and the Foldover direction is RL. • Position the navigator kernel over the boundary of the phantom. • start the scan.	• The first breath hold scan is performed
.	• After the first breath hold scan is performed, move the mattress with the phantom about 2 cm in the feet or head direction and start the second breath hold scan.	• Verify that the Navigator display stays and a second Navigator signal is added separated by a red line. The second breath hold scan is performed.
Step 3	• Perform a subtraction of both scans.	• Both images must be at the same position (errors shall be less than 4 mm).

3.1.31.9. SWCMP.Scan.RNAV.examinalign.precision.tc2.v1

Test ID	32582
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:45 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:25:46 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-23 08:33:24 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Check Navigator exam alignment and tracking in the preparation direction.

Requirements Coverage:

Req ID	Requirement Tag	Description
9819	SWCMP.Scan.RNAV.exam_align.precision	Motion alignment of scans within an Exam shall be in the order of less or equal of 4 mm on a phantom setup

Test steps:

Step	Description	Expected Result
Prep 1	Take the head phantom and position it with the flat part on the table. After moving the table to the iso center switch the table movement to manual. Use the body coil, no other coils in the magnet! Make sure that all CSK SW options are switched off. Create a new logfile. Enable physiology simulation. Perform a survey.	N/A (preparation)
Step 1	Select protocol "track/dyn/P" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. Change the dynamic protocol to a breath hold and track protocol with scale factor 1.0 and number of dynamics to 1. Set the Navigator preparation mode to "auto". If autovoice is available, select Breathhold guidance "manual". Duplicate the protocol. Make sure that the Slice orientation of the plane is coronal and the Foldover direction is FH. Position the navigator kernel over the boundary of the phantom. Start the first scan only.	The first breath hold scan is performed.
Step 2	After the first breath hold scan is performed, move the mattress with the phantom about 2 cm in the feet or head direction and start the second breath hold scan	Check that the Navigator display is refreshed with the first breath hold and track scan. Performing the second breath hold and track scan the Navigator display stays and a second Navigator signal is added separated by a red line. The second breath hold scan is performed.

Step	Description	Expected Result
Step 3	Perform a subtraction of both scans. Attach screenshots and the logfile as objective evidence.	Both images should be at the same position (errors should be smaller than 1 mm).

3.1.31.10. SWCMP.Scan.RNAV.exam_align.precision.tc3.v1

Test ID	32583
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:44 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:25:47 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-23 08:33:01 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Check Navigator exam alignment and tracking in the selection direction.

Requirements Coverage:

Req ID	Requirement Tag	Description
9819	SWCMP.Scan.RNAV.exam_align.precision	Motion alignment of scans within an Exam shall be in the order of less or equal of 4 mm on a phantom setup

Test steps:

Step	Description	Expected Result
Prep 1	Take the head phantom and position it with the flat part on the table. After moving the table to the iso center switch the table movement to manual. Use the body coil, no other coils in the magnet! Make sure that all CSK SW options are switched off. Enable physiology simulation. Create a new logfile. Perform a survey.	N/A (preparation)
Step 1	Select protocol "track/dyn/S" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. Change the dynamic protocol to a breath hold and track	The first breath hold scan is performed.

Step	Description	Expected Result
	protocol with scale factor 1.0 and number of dynamics to 1. Set the Navigator preparation mode to "auto". If autovoice is available, select Breathhold guidance "manual". Duplicate the protocol. Make sure that the Slice orientation of the plane is transverse and position the slice anywhere in the phantom. Position the navigator kernel over the boundary of the phantom. Start the first scan.	
Step 2	After the first breath hold scan is performed, move the mattress with the phantom about 2 cm in the feet or head direction.	Check that the Navigator display is refreshed with the first breath hold and track scan. Performing the second breath hold and track scan the Navigator display stays and a second Navigator signal is added separated by a red line. The second breath hold scan is performed.
Step 3	Perform a subtraction of both scans. Attach screenshots and the logfile as objective evidence.	Both images should be at the same position (errors shall be smaller than 1 mm).

3.1.31.11. SWCMP.Scan.RNAV.exam_align.precision.tc4.v1

Test ID	32584
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:43 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:25:48 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-23 08:32:29 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Check Navigator exam alignment and tracking of breath hold and triggered scans in the selection direction.

Requirements Coverage:

Req ID	Requirement Tag	Description
9819	SWCMP.Scan.RNAV.exam_align_precision	Motion alignment of scans within an Exam shall be in the order of less or equal of 4 mm on a phantom setup

Test steps:

Step	Description	Expected Result
Prep 1	Take the head phantom and position it with the flat part on the table. After moving the table to the iso center switch the table movement to manual. Use the body coil, no other coils in the magnet! Make sure that all CSK SW options are switched off. Enable physiology simulation. Create a new logfile. Perform a survey.	N/A (preparation)
Step 1	Select protocol "track/dyn/S" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. Change the dynamic protocol to a breath hold and track protocol with scale factor 1.0 and number of dynamics to 1. Set the Navigator preparation mode to "auto". If autovoice is available, select Breathhold guidance "manual". Duplicate the protocol. Change it to a trigger and tracked scan without breath hold. Copy the breath hold and track scan as a third final scan. Make sure that the Slice orientation of the plane is transverse and position the slice anywhere in the phantom. Position the navigator kernel over the boundary of the phantom. Start the first scan.	The first breath hold scan is performed
Step 2	After the first breath hold scan and triggered scan is performed, move the table about 2 cm in or out and perform the final scan	Check that the Navigator display is refreshed with the first breath hold and track scan. Performing the second trigger and track scan Navigator signal from this scan is added to the Navigator display after a red scan separation line. The third breath hold and track scan is performed with again Navigator signal added to the Navigator display after a red scan separation line.
Step 3	Perform a subtraction of all scans Attach screenshots and the logfile as objective evidence.	All images must be at the same position (error less than 1 mm).

3.1.31.12. SWCMP.Scan.RNAV.prep_phase.tc1.v1

Test ID	32578
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:41 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:15:14 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-03-22 16:50:01 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Navigator prep phase definition and execution

Check the number of navigators performed during the navigator prep phase.

Requirements Coverage:

Req ID	Requirement Tag	Description
9815	SWCMP.Scan.RNAV.prep_phase	The Navigator preparation phase is specified in section 7 Navigator display function

Test steps:

Step	Description	Expected Result
Step 1	Make sure all CSK SW options are disabled except for Option 02. Create a new logfile. Change the control parameter that controls the navigator prep phase duration CPR_RNAV_phase_dur to 20 sec (Preparation control parameters -> enable expert mode->Nav. Preparation Phase) and choose a heart rate (HR) of 60. Select protocol Head CNS\RNAV\prep//d. In sequence development viewer check the group MPF_RNAV for the value of MPF_RNAV_prep_repetitions.	MPF_RNAV_prep_repetitions has the value 20 (CPR_RNAV_phase_dur / (60/HR))
Step 2	Select the same protocol and choose a heart rate of 30 and check the value of MPF_RNAV_prep_repetitions	MPF_RNAV_prep_repetitions has the value 10 (CPR_RNAV_phase_dur / (60/HR))
Step 3	Select the same protocol and disable now cardiac synchronization and check the value of the TFE shot interval on the info page and MP_RNAV_prep_repetitions in the sequence development viewer. Attach the logfile as objective evidence.	MPF_RNAV_prep_repetitions has the value (CPR_RNAV_phase_dur / (TFE shot interval)) Example: CPR_RNAV_phase_dur = 20s TFE shot interval = 134ms MPF_RNAV_prep_repetitions = 149

3.1.31.13. SWCMP.Scan.RNAV.prep_phase.tc2.v1

Test ID	32732
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-31 10:15:37 Meaning: QA Approved - (QA) Signed by: Oorschot, Joep van (310224063) on 2017-03-31 08:37:17 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-31 08:34:16 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Check peak at edge in prep phase. This test verifies that the software correctly detects if the object has moved outside the range of the navigator.

Requirements Coverage:

Req ID	Requirement Tag	Description
9815	SWCMP.Scan.RNAV.prep_phase	The Navigator preparation phase is specified in section 7 Navigator display function

Test steps:

Step	Description	Expected Result
Prep	On a TB, position the head phantom on the table, best on a mattress which makes it easier to move later. Make sure that all CSK SW options are switched off. Switch off kernel update by enabling the trace channel KERNELUPD_OFF in MMURNAV and restart the scanner process. Create a new logfile. Enable physiology simulation. Plan protocol "track/dyn/M" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. Enable manual start and reduce the number of dynamics to 1. Change the TFE factor to 10.	N/A (preparation)
Step 1	Start the scan in the scanner room. Wear ear protection. After the prep phase, start the dynamic scan manually. Once the scan is started move the phantom far out (>	The PR parameter PR_RNAV_pos_array must indicated values of 10000000. Logging must mention "RNAV: Peak at edge,

Step	Description	Expected Result
	50mm). Then move it back to its original position so the scan accepts the data and finishes. The navigator window will show an area without navigator signal because the phantom had moved outside the range of the navigator. Check the logfile for RNAV logging. Check the preparation parameters in the preparation blob via the database browser (Start -> MR Development -> Configuration -> Database browser). In the database browser, scroll down several layers to the series corresponding to the scan. Via a right-mouse click, the parameter blob with the PR parameters can be viewed. Attach the logfile as objective evidence.	position (incl shift!) several times.

3.1.31.14. SWCMP.Scan.RNAV.prep_phase.tc3

Test ID	7010
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Navigator prep phase on a scanner
Execute scan without prep phase

Test equipment, environment & configuration

- 1.5T or 3T scanner.

Required skill

- methods SW developer

Preconditions

- For all tests a protocol is provided in anatomy Head CNS, subanatomy RNAV in TEDB.IDX
- Enable trace channel "mmirnav testdata *" and restart the scanner process

Enable physiology simulation.

Requirements Coverage:

Req ID	Requirement Tag	Description
9815	SWCMP.Scan.RNAV.prep_phase	The Navigator preparation phase is specified in section 7 Navigator display function

Test steps:

Step	Description	Expected Result
Step 1	- Switch of the control preparation parameter for the RNAV prep phase. - Run scan prep//d	- It should be possible to execute the scan without the prep phase. - The gating level will then be set to position = 0 and the reference kernel will be the first nav during the scan. - No aborts shall occur. Gating level must be centered around the first measured position.

3.1.31.15. SWCMP.Scan.RNAV.prep_phase.tc4.v1

Test ID	32580
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 11:43:39 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 11:16:40 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-03-22 17:00:56 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Navigator prep phase on a scanner

Test prep phase in case of a breathhold scan

Requirements Coverage:

Req ID	Requirement Tag	Description
9815	SWCMP.Scan.RNAV.prep_phase	The Navigator preparation phase is specified in section 7 Navigator display function

Test steps:

Step	Description	Expected Result
Step 1	Enable physiology simulation. Create a new logfile. Open protocol "Head CNS\RNAV\breathhold" and run it. For a breathhold scan, the navigator preparation phase is skipped. Attach the logfile as objective evidence.	Verify that there is no navigator preparation phase: only one navigator at the start of each breathhold.

3.1.31.16. SWCMP.Scan.RNAV.prep_phase.tc5

Test ID	7013
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator prep phase on a scanner
Check correlation low error

Test equipment, environment & configuration

- 1.5T or 3T scanner.

Required skill

- methods SW developer

Preconditions

- For all tests a protocol is provided in anatomy Head CNS, subanatomy RNAV in TEDB.IDX
- Enable trace channel "mmirnav testdata *" and restart the scanner process

Enable physiology simulation.

Requirements Coverage:

Req ID	Requirement Tag	Description
9815	SWCMP.Scan.RNAV.prep_phase	The Navigator preparation phase is specified in section 7 Navigator display function

Test steps:

Step	Description	Expected Result
Step 1	• Select protocol prep//d. • Position the navigator somewhere in the air, far outside the phantom. • Run the scan.	• Verify that an abort occurs with an appropriate warning about navigator quality.

3.1.31.17. SWCMP.Scan.RNAV.scan.gate.tc1

Test ID	7015
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic

	signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator position measurement

Navigator gating

Test equipment, environment & configuration

- OTM with Physiology simulation and a head phantom

Preconditions

- Take the head phantom. The allow easy shifting of the phantom, put a small mattress on the patient table and position the phantom with the flat part on the mattress. Use the body coil. Perform a survey and load the images in plan scan .

Requirements Coverage:

Req ID	Requirement Tag	Description
9812	SWCMP.Scan.RNAV.scan.gate	During a Navigator scan the Navigator position is used to gate the scanner as specified in section 11 Navigator gating and tracking for CAI and 12 Navigator gating and tracking in general

Test steps:

Step	Description	Expected Result
Step 1	• Select scan protocol "gate5 mm" from "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. This protocol is a TFE with two dynamic scans. It's a gating protocol with a gate width of 5 mm. • Make sure that the Slice orientation of the plane is coronal and the Foldover direction is RL. • Position the navigator kernel over the boundary of the phantom. • Start the scan.	• The scan starts with the preparation phase and the first dynamic scan is performed.
Step 2	• After the first dynamic scan, move the mattress with the phantom about 2 cm in the feet or head direction.	• For the second dynamic scan, the shift exceeds the gate width. Data for the second dynamic scan shall be rejected.

3.1.31.18. SWCMP.Scan.RNAV.scan.pos_determ.tc1

Test ID	6998
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic

	signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 10 min

Navigator scanning

Size of the navigator (te_2drf_size)

In this test the scan control parameter *RNAV 2DRF test mode* is set to *yes*. This enables the geometry parameters for the navigator for display and copies the navigator excitation pulse to the imaging sequence.

Test equipment, environment & configuration

- This test has to be done 1.5T and 3.0T

Required skill

- Methods SW developer

Preconditions

- Move the patient table out and position the body phantom transverse in the bore. Switch Patsup simulation on. Patsup can be found as follows: Start-> MR Applications -> Service Tools -> Enable PatSup Simulation.
- Perform a survey with three orientations ("scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the figure below. If not adjust the offcentres of the phantom and repeat the survey.
- Enable physiology simulation

Requirements Coverage:

Req ID	Requirement Tag	Description
9811	SWCMP.Scan.RNAV.scan.pos_d eterm	During a Navigator scan the Navigator position is determined as specified in 10 Navigator execution and position determination
9810	SWCMP.Scan.RNAV.UI.Plansca n	The user interface and associated planscan behavior is specified in section 6 Navigator planscan.

Test steps:

Test steps:

Step	Description	Expected Result				
Step 1	• Enable control scan parameter 2DRF test mode • Change control scan parameter RNAV diameter (mm) to 15 mm • Select protocol "2drf test" in "Head CNS/RNAV" • Set gradient mode to 'maximum'. • Choose a position of the navigator within the phantom • perform the scan and load the image in plan scan • Measure the size of the navigator "full width half maximum" and fill in Table 4-1 • repeat this for a diameter of 25, 40 and 50 mm	• The images show the cross-section of a pencil beam, i.e of a cylinder. • Make sure that the size of the navigator in plan scan is equal to the control parameter value. • The diameter of the cylinder in the image must be within 20% of the expected diameter <i>Table 4-1 Size of the navigator</i> <table><tr><th>RNAV diameter (mm)</th><th>Measured</th></tr><tr><td>15</td><td></td></tr></table>	RNAV diameter (mm)	Measured	15	
RNAV diameter (mm)	Measured					
15						

Step	Description	Expected Result		
	When finished, change control scan parameter <i>RNAV diameter (mm)</i> to back to 25 mm.			
		25		
		40		
		50		

3.1.31.19. SWCMP.Scan.RNAV.scan.pos_determ.tc2

Test ID	6999
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 10 min

Navigator scanning

Size of the navigator (te_2drf_size)

In this test the scan control parameter *RNAV 2DRF test mode* is set to *yes*. This enables the geometry parameters for the navigator for display and copies the navigator excitation pulse to the imaging sequence.

Test equipment, environment & configuration

- This test has to be done 1.5T and 3.0T

Required skill

- Methods SW developper

Preconditions

- Move the patient table out and position the body phantom transverse in the bore. Switch Patsup simulation on. Patsup can be found as follows: Start-> MR Applications -> Service Tools -> Enable PatSup Simulation.
- Perform a survey with three orientations ("scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the figure below. If not adjust the offcentres of the phantom and repeat the survey.
- Enable physiology simulation

Requirements Coverage:

Req ID	Requirement Tag	Description
9811	SWCMP.Scan.RNAV.scan.pos_determin	During a Navigator scan the Navigator position is determined as specified in 10 Navigator execution and position determination
9810	SWCMP.Scan.RNAV.UI.Planscan	The user interface and associated planscan behavior is specified in section 6 Navigator planscan.

Test steps:

Step	Description	Expected Result																								
Step 1	• Enable control scan parameter 2DRF test mode • Select protocol "2drf test" in "Head CNS/RNAV" • Set gradient mode to 'maximum'. • Position the navigator in the transverse image with Navigator direction FH, choose an offcentre position of 100 mm in both AP and LR. • Perform the scan and load the image in plan scan, this enables you with the help of plan scan to measure the position of the navigator. Just change the offcentres until the green navigator overlays the disk and read the offcentres from the user interface. • Repeat this for the other offcentre positions shown in Table 4-2 When finished, change control scan parameter <i>RNAV 2DRF test mode</i> to no again	• Images are produced showing a cylinder on off-centre positions • Fill in Table 4-2 and check if the disk excited by the navigator and the planscan navigator are within 5 percent of each other <i>Table 4-2 Positioning of the navigator</i> <table><tr><td colspan="2">offcentre AP</td><td colspan="2">offcentre RL</td></tr><tr><td>desired</td><td>measured</td><td>desired</td><td>measured</td></tr><tr><td>100</td><td></td><td>100</td><td></td></tr><tr><td>100</td><td></td><td>-100</td><td></td></tr><tr><td>-100</td><td></td><td>100</td><td></td></tr><tr><td>-100</td><td></td><td>-100</td><td></td></tr></table>	offcentre AP		offcentre RL		desired	measured	desired	measured	100		100		100		-100		-100		100		-100		-100	
offcentre AP		offcentre RL																								
desired	measured	desired	measured																							
100		100																								
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-100		100																								
-100		-100																								

3.1.31.20. SWCMP.Scan.RNAV.scan.pos_determin.tc3

Test ID	7000
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 10 min

Navigator scanning

Size of the navigator (te_2drf_size)

In this test the scan control parameter *RNAV 2DRF test mode* is set to *yes*. This enables the geometry parameters for the navigator for display and copies the navigator excitation pulse to the imaging sequence.

Test equipment, environment & configuration

- This test has to be done 1.5T and 3.0T

Required skill

- Methods SW developer

Preconditions

- Move the patient table out and position the body phantom transverse in the bore. Switch Patsup simulation on. Patsup can be found as follows: Start-> MR Applications -> Service Tools -> Enable PatSup Simulation.
- Perform a survey with three orientations ("scout" in "Head CNS/RNAV") and make sure that the survey images look like the images shown in the figure below. If not adjust the offcentres of the phantom and repeat the survey.
- Enable physiology simulation

Requirements Coverage:

Req ID	Requirement Tag	Description
9811	SWCMP.Scan.RNAV.scan.pos_d eterm	During a Navigator scan the Navigator position is determined as specified in 10 Navigator execution and position determination
9810	SWCMP.Scan.RNAV.UI.Plansca n	The user interface and associated planscan behavior is specified in section 6 Navigator planscan.

Test steps:

Test steps:

Step	Description	Expected Result								
Step 1	• Modify protocol "2drf test" in "Head CNS/RNAV" • Change gradient mode to maximum. Position the navigator in the transverse image with <i>Navigator direction FH</i>, choose an offcentre position of 0 mm in both <i>AP</i> and <i>LR</i>. • Perform the scan and load the image in plan scan • Measure the size of the navigator "full width half maximum" and fill in Table 4-3 • Disable the F0 preparation phase. • Change the resonance frequency in the tuned hardware parameters with +50 Hz. • repeat this for off resonances of 100 Hz	• The diameter of the cyclinder increases due to off-resonance. • For the given frequency offsets the diameter shall be equal to the specified diameter within 50 percent <i>Table 4-3 Size of the navigator during off resonance</i> <table><tr><td>F0</td><td>measured diameter</td></tr><tr><td>on resonance</td><td></td></tr><tr><td>+50 Hz</td><td></td></tr><tr><td>+100 Hz</td><td></td></tr></table>	F0	measured diameter	on resonance		+50 Hz		+100 Hz	
F0	measured diameter									
on resonance										
+50 Hz										
+100 Hz										

3.1.31.21. SWCMP.Scan.RNAV.scan.track.tc1

Test ID	7014
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator position measurement

Offset measurement

Test equipment, environment & configuration

- OTM with Physiology simulation and a head phantom

Preconditions

- Take the head phantom. The allow easy shifting of the phantom, put a small mattress on the patient table and position the phantom with the flat part on the mattress. Use the body coil. Perform a survey and load the images in plan scan .

Requirements Coverage:

Req ID	Requirement Tag	Description
9813	SWCMP.Scan.RNAV.scan.track	During a Navigator scan the Navigator position is used to track the scan position as specified in section 11 Navigator gating and tracking for CAI and 12 Navigator gating and tracking in general

Test steps:

Step	Description	Expected Result
Step 1	• Select protocol "gate100 mm" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. This protocol is a TFE with two dynamic scans. It's a gating protocol with a gate width of 100 mm, so all profiles will be accepted. • Make sure that the Slice orientation of the plane is coronal and the Foldover direction is RL. • Position the navigator kernel over the boundary of the phantom. • Start the scan.	• The scan starts with the preparation phase and the first dynamic scan is performed.
Step 2	• After the first dynamic scan, move the mattress with the phantom about 2 cm in the feet or head direction.	• Two images are produced with a displacement between each other in the measurement direction. • Derived position are stored in <code>`PR_RNAV_pos_array`</code>.
Step 3	• Load the image in the viewing environment and	• The measured shifts of the images and that of

Step	Description	Expected Result								
	measure the displacements manually, for example in the subtracted image Read the control preparation parameters `PR_RNAV_pos_array` and fill in Table 4-4. PR parameters can be found viewed via the database browser (Start -> MR Development -> Configuration -> Database browser). In the database browser, scroll down several layers to the series corresponding to the scan. Via a right-mouse click, the parameter blob with the PR parameters can be viewed.	the navigator shall be equal within 1 mm. <i>Table 4-4 Offset measured</i> <table><tr><td>dynamic scan</td><td>`PR_RNAV_pos_array`</td></tr><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td>measured</td><td></td></tr></table>	dynamic scan	`PR_RNAV_pos_array`	1		2		measured	
dynamic scan	`PR_RNAV_pos_array`									
1										
2										
measured										

3.1.31.22. SWCMP.Scan.RNAV.scan.track.tc2

Test ID	7016
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator position measurement

Tracking in the measurement direction

Test equipment, environment & configuration

- OTM with Physiology simulation and a head phantom

Preconditions

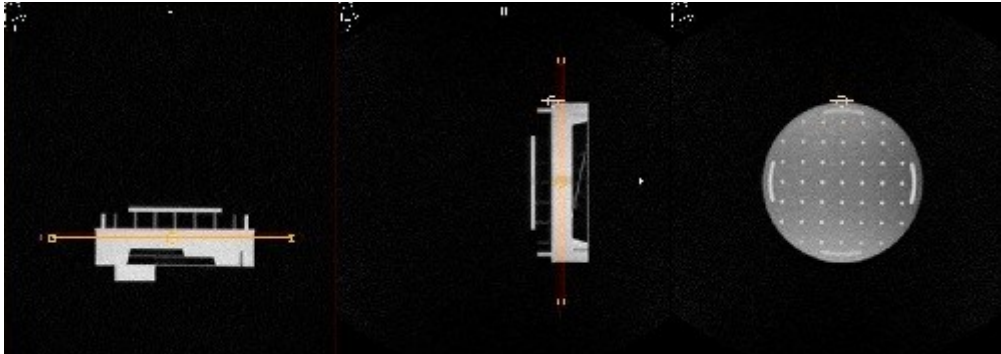
- Take the head phantom. The allow easy shifting of the phantom, put a small mattress on the patient table and position the phantom with the flat part on the mattress. Use the body coil. Perform a survey and load the images in plan scan .

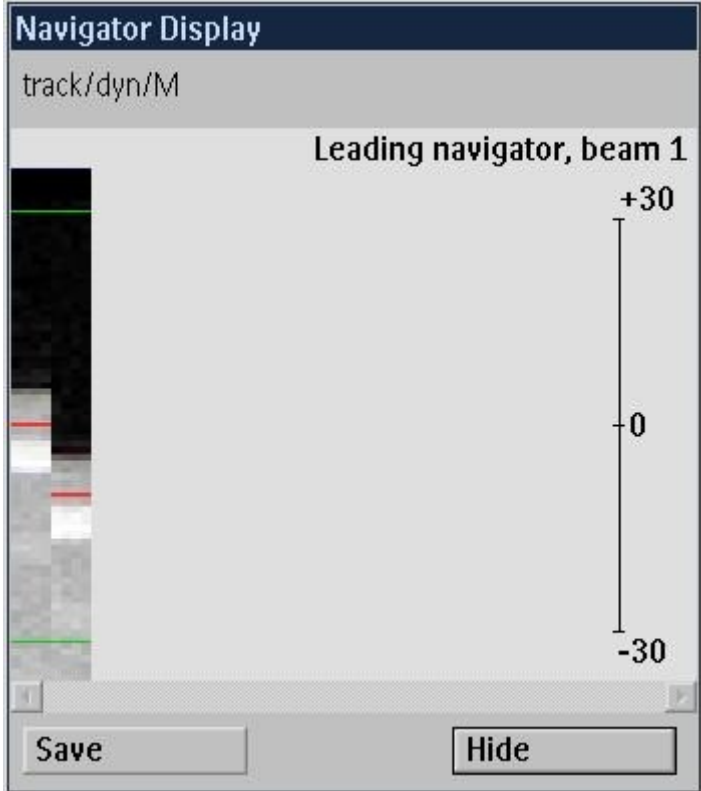
Requirements Coverage:

Req ID	Requirement Tag	Description
9813	SWCMP.Scan.RNAV.scan.track	During a Navigator scan the Navigator position is used to track the scan

Req ID	Requirement Tag	Description
		position as specified in section 11 Navigator gating and tracking for CAI and 12 Navigator gating and tracking in general

Test steps:

Step	Description	Expected Result	Attachments
Step 1	- Modify protocol "track/dyn/M" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section (i.e. The navigator planscan box should be placed such that the maximum displacement is observed when moving the phantom, see the example on the right in the image below). This protocol is a dynamic TFE with two dynamic scans with manual start before each scan. - Make sure that the Slice orientation of the plane is coronal and the Foldover direction is RL. - Make sure that the scan protocol parameter scale factor from the navigator settings is set at "1.0". - Position the navigator kernel over the boundary of the phantom, see the following image as example: Refer Attached Fig. 4.3. - Start the scan	The scan starts with the preparation phase and the first dynamic scan is performed.	
			
Step 2	- After the first dynamic scan, move the mattress with the phantom about 2 cm in the feet or head direction. - The Navigator Display should clearly indicate the movement of the phantom in the FH direction. The displacement of the phantom should fall between the boundaries of the navigator (see green lines in the image below):	Two images are produced	

Step	Description	Expected Result	Attachments
			
Step 3	Perform a subtraction of both images	Both images shall have the same phantom position (errors should be smaller than 1 mm).	None

3.1.31.23. SWCMP.Scan.RNAV.scan.track.tc3

Test ID	7018
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator position measurement

Tracking in the preparation direction

Test equipment, environment & configuration

- OTM with Physiology simulation and a head phantom

Preconditions

- Take the head phantom. The allow easy shifting of the phantom, put a small mattress on the patient table and position the phantom with the flat part on the mattress. Use the body coil. Perform a survey and load the images in plan scan .

Requirements Coverage:

Req ID	Requirement Tag	Description
9813	SWCMP.Scan.RNAV.scan.track	During a Navigator scan the Navigator position is used to track the scan position as specified in section 11 Navigator gating and tracking for CAI and 12 Navigator gating and tracking in general

Test steps:

Step	Description	Expected Result
Step 1	• Modify protocol "track/dyn/P" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. This protocol is a dynamic TFE with two dynamic scans. • Make sure that the Slice orientation of the plane is coronal and the Foldover direction is FH. • Make sure that the scan protocol parameter scale factor from the navigator settings is set at "1.0". • Position the navigator kernel over the boundary of the phantom. • Start the scan.	• The scan starts with the preparation phase and the first dynamic scan is performed.
Step 2	• After the first dynamic scan, move the mattress with the phantom about 2 cm in the feet or head direction.	• Two images are produced.
Step 3	• Perform a subtraction of both images.	• Both images shall have the same phantom position (errors must be less than 1 mm).

3.1.31.24. SWCMP.Scan.RNAV.scan.track.tc4

Test ID	7019
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Time needed: 5 min

Navigator position measurement

Tracking in the selection direction

Test equipment, environment & configuration

- OTM with Physiology simulation and a head phantom

Preconditions

- Take the head phantom. The allow easy shifting of the phantom, put a small mattress on the patient table and position the phantom with the flat part on the mattress. Use the body coil. Perform a survey and load the images in plan scan .

Requirements Coverage:

Req ID	Requirement Tag	Description
9813	SWCMP.Scan.RNAV.scan.track	During a Navigator scan the Navigator position is used to track the scan position as specified in section 11 Navigator gating and tracking for CAI and 12 Navigator gating and tracking in general

Test steps:

Step	Description	Expected Result
Step 1	• Modify protocol "track/dyn/S" in "Head CNS/RNAV" such that the slice is positioned in the spatial linearity section. This protocol is a dynamic TFE with two dynamic scans. • Make sure that the Slice orientation of the plane is transverse and position the slice anywhere in the phantom. • Make sure that the scan protocol parameter scale factor from the navigator settings is set at "1.0". • Position the navigator kernel over the boundary of the phantom. • Start the scan.	• The scan starts with the preparation phase and the first dynamic scan is performed.
Step 2	• After the first dynamic scan, move the mattress with the phantom about 2 cm in the feet or head direction.	• Two images are produced.
Step 3	• Perform a subtraction of both images.	• Both images shall have the same phantom position (errors less than 1 mm).

3.1.31.25. SWCMP.Scan.RNAV.scan.trig.tc1

Test ID	7020
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Check execution of Navigator triggering using a motion velocity regulation approach on VMware

Test equipment, environment & configuration

- SDE PC with VMware

Required skill

- methods SW developer

Preconditions

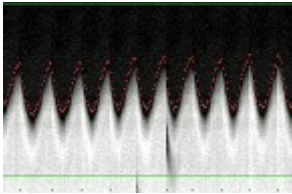
- For all tests a protocol is provided in anatomy Head CNS, subanatomy RNAV in TEDB.IDX
- Enable trace channel "mmurnav trig *" and restart the scanner process to allow easier debugging in case of problems

Time needed: 10 minutes

Requirements Coverage:

Req ID	Requirement Tag	Description
9816	SWCMP.Scan.RNAV.scan.trig	Motion velocity based triggering shall be possible. Data acquisition shall be triggered at every end expiration state. The number of missed trigger points with respect to the total number of trigger points shall be less than 20 percent

Test steps:

Step	Description	Expected Result	Attachments
Step 1	• Check that the RNAV trigger on velocity control scan parameter is set to yes. • Select protocol "M2D TFE", disable cardiac triggering and set navigator respiratory comp to triggering. • Run the scan	• The preparation phase is skipped and the scan is immediately started with triggering. The green acquisition points on the bottom of the Navigator display should be executed when the breathing signal is in the end expiration uppermost position. See attached figure.	
			

3.1.31.26. SWCMP.Scan.RNAV.WeightedGating.Par.tc1

Test ID	41060
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:29:04 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan

	(310134237) on 2019-05-21 05:06:53 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:47:46 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description				
97886	SWCMP.Scan.Validator.Respiration.Compensation.Par.v4	Name	Range	Visible when	Description	System / Option restrictions
		Respiratory compensation	no, trigger, breath hold, PEAR, gate, retrospective	always	Respiratory compensation techniques for breathing motion.	RetroResp Binned (AWOPT_MR_4D_MR_RT) enables retrospective respiratory compensation in combination with 3D Radial and CS-SENSE.
		Respiratory soft gating	no, yes	Scan type = Imaging ù Fast imaging mode = TFE ù Shot mode = MSH ù Dynamic study == contrast phases	Enables respiratory soft gating technique for 4D free breathing	N/A

Req ID	Requirement Tag	Description				
		Motion detection type	external, intrinsic	Respiratory soft gating = yes or respiratory comp = retrospective with Development SW option	Allows to specify the respiratory detection device. External is the respiratory signal detected from the VitalEye camera or respiratory belt, intrinsic is a k-space navigator acquired before each shot	Development (AWOPT_DEVELOPMENT) enables the parameter for retrospective respiratory binning
		Oversample factor	1.025-2.000	Respiratory soft gating = yes	Allows to specify the oversampling factor for respiratory soft gating	N/A
		Oversample factor	1.5-15.0 (for 1 resp phase) 2.8-15.0 (for 2-10 resp phases)	respiratory comp = retrospective	Allows to specify the oversampling factor for retrospective respiratory binning	N/A
		Navigator respiratory comp	no, gate, track, gate and track,	Scan type = Imaging U	Enables navigator respiratory	This parameter is only

Req ID	Requirement Tag	Description				
			trigger, trigger and track	Dynamic study != contrast phases	y compensa tion technique .	enabled when the SW option BodyRespi ratoryNavi gators (AWOPT_ BODYRES PNAVS) Body Resp navigators is available. RespNavE xtended (AWOPT_ OPTION02) and CardiacTri ggeredNa vigators (AWOPT_ COR_ART _IMA) enables the combinati on of this paramete r with cardiac synchroni zation. RespNavE xtended (AWOPT_ OPTION02) enables the combinati on of gate, track, gate and track for non-triggered

Req ID	Requirement Tag	Description				
						and non-breathhold scans.
		Preparation	full, auto	Navigator respiratory comp != no	Specifies the preparation mode	N/A
		Beams	1-3	Navigator respiratory comp != no	Specifies the number of respiratory navigator beams.	RespNavExtended (AWOPT_OPTION02) extends the range of 2 and 3
		Direction	FH,AP,LR	Navigator respiratory comp != no	Specifies the direction of the long axis of the navigator	RespNavExtended (AWOPT_OPTION02) extends the use of direction AP, LR
		Scale factor	-10.0 - 10	Navigator respiratory comp = Track or gate and track or trigger and track	Specifies the scale factor for slice tracking correction factors	N/A
		Weighted gating	No, Yes	Navigator respiratory comp = gate or gate and track or trigger or trigger and track Û Scan	Enables the Weighted gating option that allows user specified central	KING (AWOPT_OPTION14) enables the use of this parameter

Req ID	Requirement Tag	Description				
				mode =3D	and outer k-space profiles	
		Central gate (mm)	0.0 – 100.0	weighted gating = yes	Specifies the central gate width, the gate for the central part of k-space.	N/A
		Central dur (%)	0.0 – 100.0	weighted gating = yes	Specifies the duration of the acquisition of the central k-space	N/A
		Outer gate (mm)	0.0 – 100.0	weighted gating = yes	Specifies the outer gate, the gate for the outer part of k-space	N/A
		Gating window	0.0 – 100.0	weighted gating = No	specifies the gating window of the respiratory cycle in which data will be accepted for reconstruction	N/A
		Gating level drift	No, Yes	Navigator respirator	Enables the	N/A

Req ID	Requirement Tag	Description				
				y comp = gate or gate and track or trigger or trigger and track	update of the actual gating level every time data is acquired	
		gating level drift (mm/min)	0 - 10	Gating level drift = Yes	Specifies the gating level drift when gating level drift is set	N/A
		max slices per respiration	1 - 256	Navigator respiratory comp = gate or track or gate and track ÷ Scan mode = MS ÷ Fast Imaging mode = EPI or TFE-EPI	Specifies the maximum number of slices measured interleaved in one respiration cycle	N/A
		Type	Lead, Trail, lead and trail	Navigator respiratory comp = gate or gate and track	Specifies the type of the navigator respiratory compensation	N/A
		minimum TR (ms)	0 - 10000	Navigator respiratory comp = trigger or trigger and track	Specifies the minimum TR	N/A

Req ID	Requirement Tag	Description				
		trigger delay	Auto, user defined	Navigator respiratory comp = trigger or trigger and track	Specifies the trigger delay	N/A
		trigger delay (ms)	0 - 10000	trigger delay = User defined	Specifies the delay time in ms for user defined trigger delay	N/A
		Number of resp. phases	1, 2, 4, 6, 8, 10	respiratory comp = retrospective	Specifies the number of resp phases used for retrospective respiratory binning	N/A
Application specific restrictions are imposed by the following options:						
RadialImagingExtended (AWOPT_OPTION33) enables Respiratory triggering in combination with Radial.						

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanRnavWeightedGatingPar in the results file is zero.

3.1.32. ME SENSE

3.1.32.1. SWCMP.Scan.SENSE.availability.tc1.v1

Test ID	32765
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Revision	2
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-12 08:47:21 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-04-12 08:07:43 Meaning: SME Approved - (Technical) Signed by: Crijns, Sjoerd (310257923) on 2017-04-11 18:50:33 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Interactive tests using Examcard simulator

Requirements Coverage:

Req ID	Requirement Tag	Description
65997	SWCMP.Scan.SENSE.availability.v1	In case of SENSE, the following conflicts shall be implemented: - For multi-stack scans with only P reduction, the P direction in terms of 'patient axis' shall be equal for all stacks to prevent from getting too different noise enhancement (due to the differing geometry factors). - For multi-stack scans with only S reduction, the S direction in terms of 'patient axis' shall be equal for all stacks to prevent from getting too different noise enhancement (due to the differing geometry factors). - For multi-stack scans with both P and S reduction, P and S shall not cover more than 2 directions in terms of 'patient axis'. - For radial Stacks, the radial axis shall be along the SENSE reduction axis (= fold over axis). - A conflict shall be set on the combination of SENSE with phase navigator (Multi-shot diffusion and "fMRI echo stabilisation"1) is not allowed. - In case of P-EPI combined with S-SENSE, a conflict shall be set when $Sh_p * S_p < 4$. where "Sh_p" is the "number of shots in p-direction", i.e., the number of scan segments required to obtain the required k-space coverage in kp-direction. S_p is the SENSE-factor in p-direction. In the presence of the SENSE prototype SW key (see 2.7.2), a warning will be presented to the user with text "S-SENSE and high EPI factors: EPI ghosts may originate from distant slices". This condition will not be checked for combined turbo methods, viz. TFEPI and GraSE. In case of SENSE, the following warnings shall be implemented:

Req ID	Requirement Tag	Description
		- When SENSE is applied in the S-direction in combination with a small user defined slice oversampling factor, the following warning shall be displayed: "3D SENSE slice foldover warning: Increase slice oversample factor". - For SENSE scans with high EPI factors, the following warning shall be displayed: "SENSE with high EPI factors may cause additional ghosting".

Test steps:

Step	Description	Expected Result
Prep 1.	On a VMWARE, run the C:\nmr\gyroscan\commands\m_amn_set_test_environment.pl script and Configure the system as a WA30 (3.0T Ingenia) system. Make sure all CSK's are disabled. Start the Application Software and create a new patient. Start a new log file.	N.A. Preparation only.
Step 1	From the TEDB select the Head CNS\F-3D-SCANS\F-3D-FFE scan. Make it a 2 stacks scan. Make the first stack a transverse scan with fold-over direction RL. Make the second stack a transverse stack with fold-over direction AP. Enable SENSE and set the 'P reduction' factor to 2. Check that there is conflict on the 'P reduction', the 'current stack' and 'fold-over direction'	There is conflict on the 'P reduction', the 'current stack' and 'fold-over direction'
Step 2	Set the 'P reduction' factor to 1 and the 'S reduction' factor to 2. Make stack A a coronal scan. Check that there is conflict on the 'S reduction', the 'current stack' and 'Slice orientation'.	There is conflict on the 'S reduction', the 'current stack' and 'Slice orientation'.
Step 3	Set both the 'P reduction' factor and 'S reduction' factor to 2. Make stack A a transverse stack with fold-over direction 'AP' Make stack B a coronal stack with fold-over direction 'FH' Check that no conflict occurs.	There is no conflict.
Step 4	Now for stack B change the fold-over direction to 'RL'	There is conflict on the 'P reduction', 'S reduction',

Step	Description	Expected Result
	Check that there is conflict on the 'P reduction', 'S reduction', 'current stack', 'Slice orientation' and 'fold-over direction'.	'current stack', 'Slice orientation' and 'fold-over direction'.
Step 5	Make the scan a single stack scan. Make it a transverse stack with fold-over direction 'RL'. Set the 'P reduction' factor to 2 and the 'S reduction' factor to 1. Change the scan mode to 'MS' and set the 'stack type' to 'radial'. Check that when the 'radial axis' is unequal to the 'fold-over direction', a conflict is set on 'P reduction', 'Stack type' and 'fold-over direction'.	When the 'radial axis' is unequal to the 'fold-over direction', a conflict is set on 'P reduction', 'Stack type' and 'fold-over direction'.
Step 6	From the TEDB select the Head CNS\DIFFUSION\1a1_DWI\MshEPI scan. Check that when SENSE is not enabled and 'Uniformity' is set to 'CLEAR' the scan is in conflict on 'Uniformity', 'shot mode' and 'Diffusion mode'. Enable 'SENSE'. Check that a conflict is set on 'SENSE', 'shot mode' and 'Diffusion mode'.	There is a conflict et on 'SENSE', 'shot mode' and 'Diffusion mode'.
Step 7	From the TEDB select the Head CNS\EPI\IP_FFEPI_2D. Set "Scan mode" to '3D'. Enable SENSE and set 'fMRI echo stabilisation' to 'yes'. Check that a conflict is set on 'SENSE' and 'fMRI echo stabilisation'.	There is a conflict is set on 'SENSE' and 'fMRI echo stabilisation'.
Step 8	From the TEDB select the Head CNS\F-3D-SCANS\F-3D-FFE scan. Make it a single shot EPI scan. Enable SENSE. Set the 'P reduction' to 2 and set the 'S reduction' to 2. Check that there is a conflict set. Check that the conflict disappears when the 'P reduction' is set to 4 or higher.	The conflict disappears when the 'P reduction' is set to 4 or higher.
Step 9	Select the CNS\F-3D-SCANS\F-3D-FFE scan. Enable SENSE. Set the 'S reduction' to 2.	The message "3D SENSE slice fold-over warning: Increase slice oversample factor" is displayed.

Step	Description	Expected Result
	Set 'Slice oversampling' to 'user defined' and set the 'oversample factor' to 1.0. Check that the following message is displayed: "3D SENSE slice fold-over warning: Increase slice oversample factor".	
Step 10	Select the Head CNS\EPI\P_SEEPI_2D scan. Make it a single shot EPI scan. Remove conflicts by setting the Echo Time (TE) to 'shortest'. Enable SENSE and set the 'P reduction' to 2. Check that the displayed message changes to "SENSE with high EPI factors may cause additional ghosting".	The displayed message changes to "SENSE with high EPI factors may cause additional ghosting".
Wrap up.	Add the logfile to the test run as objective evidence.	N.A. Wrap up.

3.1.32.2. SWCMP.Scan.SENSE.coca.ui.tc1.v3

Test ID	41366
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2019-05-27 16:43:33 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2019-05-27 15:00:47 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2019-05-27 14:55:50 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:
automated test for COCA UI behaviour

Requirements Coverage:

Req ID	Requirement Tag	Description
54854	SWCMP.Scan.SENSE.coca.ui.v2	Modifiability of prescans should be restricted to the parameters that belong to the EX_coca_default_pars reference group. These are: EX_GEO->coil_selection_mode EX_GEO->cur_stack_ap_offcentre EX_GEO->cur_stack_lr_offcentre

Req ID	Requirement Tag	Description
		EX_GEO->cur_stack_fh_offcentre EX_GEO->patient_position EX_GEO->patient_orientation EX_ACQ->RF_shim EX_ACQ->multi_transmit EX_RESP->synch EX_RESP->hold EX_RESP->serial_bh EX_ACQ->measurements EX_ACQ->manual_start EX_GEO->cur_stack_clinical_mode EX_GEO->cur_stack_coil_1 EX_GEO->cur_stack_coil_1_chan EX_GEO->cur_stack_coil_1_conn EX_GEO->cur_stack_xclcoil_1 EX_GEO->cur_stack_xclcoil_1_chan EX_GEO->cur_stack_xclcoil_1_conn EX_GEO->cur_stack_coil_2 EX_GEO->cur_stack_coil_2_chan EX_GEO->cur_stack_coil_2_conn EX_GEO->cur_stack_xclcoil_2 EX_GEO->cur_stack_xclcoil_2_chan EX_GEO->cur_stack_xclcoil_2_conn EX_GEO->cur_stack_coil_3 EX_GEO->cur_stack_coil_3_chan EX_GEO->cur_stack_coil_3_conn EX_GEO->cur_stack_xclcoil_3 EX_GEO->cur_stack_xclcoil_3_chan EX_GEO->cur_stack_xclcoil_3_conn EX_GEO->cur_stack_coil_4 EX_GEO->cur_stack_coil_4_chan EX_GEO->cur_stack_coil_4_conn EX_GEO->cur_stack_xclcoil_4 EX_GEO->cur_stack_xclcoil_4_chan EX_GEO->cur_stack_xclcoil_4_conn EX_GEO->cur_stack_coil_5 EX_GEO->cur_stack_coil_5_chan EX_GEO->cur_stack_coil_5_conn EX_GEO->cur_stack_xclcoil_5 EX_GEO->cur_stack_xclcoil_5_chan EX_GEO->cur_stack_xclcoil_5_conn EX_GEO->cur_stack_coil_6 EX_GEO->cur_stack_coil_6_chan EX_GEO->cur_stack_coil_6_conn EX_GEO->cur_stack_xclcoil_6 EX_GEO->cur_stack_xclcoil_6_chan EX_GEO->cur_stack_xclcoil_6_conn EX_GEO->cur_stack_coil_7 EX_GEO->cur_stack_coil_7_chan EX_GEO->cur_stack_coil_7_conn EX_GEO->cur_stack_xclcoil_7 EX_GEO->cur_stack_xclcoil_7_chan EX_GEO->cur_stack_xclcoil_7_conn EX_GEO->cur_stack_coil_8 EX_GEO->cur_stack_coil_8_chan EX_GEO->cur_stack_coil_8_conn EX_GEO->cur_stack_xclcoil_8 EX_GEO->cur_stack_xclcoil_8_chan EX_GEO->cur_stack_xclcoil_8_conn EX_GEO->cur_stack_coil_9 EX_GEO->cur_stack_coil_9_chan EX_GEO->cur_stack_coil_9_conn EX_GEO->cur_stack_xclcoil_9

Req ID	Requirement Tag	Description
		EX_GEO->cur_stack_xclcoil_9_chan EX_GEO->cur_stack_xclcoil_9_conn EX_GEO->cur_stack_coil_10 EX_GEO->cur_stack_coil_10_chan EX_GEO->cur_stack_coil_10_conn EX_GEO->cur_stack_xclcoil_10 EX_GEO->cur_stack_xclcoil_10_chan EX_GEO->cur_stack_xclcoil_10_conn

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to <i>SwCmpScanSenseCocaUI</i> in the results file is zero.

3.1.32.3. SWCMP.Scan.SENSE.iRF.3D.tc1.v1

Test ID	32771
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-04 13:36:26 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-04-04 12:25:56 Meaning: SME Approved - (Technical) Signed by: Oorschot, Joep van (310224063) on 2017-04-04 11:26:43 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description
28146	SWCMP.Scan.SENSE.iRF.3D	For SENSE in slice direction methods sw shall instruct reconstruction to apply regularization based on a simulated slice profile.

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to configure an Ingenia 3T system - Use the service application to enable all coils - Select the TEDB as scan protocol database - Enable scan simulation and recon on host	n/a (preparation)

Step	Description	Expected Result
	- Connect the anterior coil using the coil simulator - Create a new log file - Start Examcards - Enter a new patient - Select scan Head_CNS > Syncra > bsfepi - Set Scan mode to 3D - Set SENSE S reduction to 2.0 - Start the scan.	
Step 1	Check the logfile for the reconstruction parameter 'sense_profile_regu'. Attach the logfile to this step as objective evidence.	The value of the sense_profile_regu parameter equals 1.

3.1.32.4. SWCMP.Scan.SENSE.iRF.flame.tc1

Test ID	13509
Revision	1
Product Model	Achieva dStream;Ingenia;Marlin
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Perform the test on a ROF system configured as 3T Ingenia.
Use the TEDB as scan protocol database.

Requirements Coverage:

Req ID	Requirement Tag	Description
28145	SWCMP.Scan.SENSE.iRF.flame	The system shall apply the SENSE flame artifact reduction algorithm for coronal TSE scans in the body region. The methods software shall instruct reconstruction to apply the so-called localized MAF algorithm for coronal orientation with LR phase encoding, a covered FOV larger than 500 mm and the use of body coils in combination with SENSE.

Test steps:

Step	Description	Expected Result
Step 1	- Configure the ROF as an Ingenia 3T system - Enable all coils - Select the TEDB as scan protocol database - Enable scan simulation and recon on reconstructor. - Connect the anterior coil. - Start Examcards - Enter a new patient	'sense optimization method' equals 2 'sense enable b0 csm extrapl' equals 1 'sense localized maf factor' > 0.0 'csm stab div value' > 0.0

Step	Description	Expected Result
	- Select the protocol: HEAD_CNS\RESP\B_MS25. This protocol contains respiratory compensation breath hold. - Set the echo time to shortest - Enable SENSE - Set SENSE factor to 4. - Set slice orientation to coronal. - Set the FH and LR FOV to 550. - Set scan technique to SE - Set Fast Imaging mode to TSE - Set shot mode to single shot - Set partial echo to no - Set Echoes to 1. - Start the scan. - Check the logfile for the following reconstruction parameters for this scan: 'sense optimization method' should be 2 'sense enable b0 csm extrapol' should be 1 'sense localized maf factor' should be larger than 0.0 'csm stab div value' should be larger than 0.0	

Attachments:



flamesense.zip

3.1.32.5. SWCMP.Scan.SENSE.iRF.MAF.tc1

Test ID	13510
Revision	1
Product Model	Achieva dStream;Ingenia;Marlin
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Perform the test on a ROF system configured as 3T Ingenia.
Use the TEDB as scan protocol database.

Requirements Coverage:

Req ID	Requirement Tag	Description
28144	SWCMP.Scan.SENSE.iRF.MAF	On iRF systems MAF SENSE is enabled for: transverse and sagittal body imaging with phase encoding direction AP in combination with respiratory compensation.

Req ID	Requirement Tag	Description
		breast imaging in any orientation and for any phase encoding direction.

Test steps:

Step	Description	Expected Result
Step 1	- Configure the ROF as an Ingenia 3T system - Enable all coils - Select the TEDB as scan protocol database - Enable scan simulation and recon on reconstructor. - Connect the anterior coil. - Start Examcards - Enter a new patient - Select the protocol: HEAD_CNS\RESP\B_MS25. This protocol contains respiratory compensation breath hold. - Set the echo time to shortest - Enable SENSE - Set SENSE factor to 2. - Start the scan. - After the scan is finished, check the reconstruction parameter 'sense optimization method' in the logfile. This should indicate 2 for the given scan.	sense optimization method has value 2.
Step 2	Disconnect the Anterior coil and connect a breast coil. Set the posterior coil in 'hold back'. To do this, open an command box and execute 'pctest'. Select the following menu entries: 9, 1, 3, 3. Do not leave the menu and do not close the command box. Select an arbitrary protocol from the protocol database, enable SENSE with SENSE factor 2. Start the scan. After the scan is finished, check the reconstruction parameter 'sense optimization method' in the logfile. This should indicate 2 for the given scan.	sense optimization method has value 2.

Attachments:



mafsense.zip

3.1.33. ME SWIp

3.1.33.1. SWCMP.Scan.SWlp.Autoview.tc1

Test ID	8077
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Display and handling of scan UI parameters
SWlp autoview

Description: Autoview setting for SWlp

Preparations

Select the tedb database

Start *ExamCards Environment*.

Load scan SWlp_ffe_se from Head_CNS/SWlp and open the sequence development viewer.

Requirements Coverage:

Req ID	Requirement Tag	Description
18797	SWCMP.Scan.SWlp.Autoview	Autoview image type will be set to the first selected output image type following the order SW-M, SW-P.

Test steps:

Step	Description	Expected Result
Step 1	Verify the value of the RC parameter RC_functional_av_output_images.	The RC parameter RC_functional_av_output_images has the value 1024

3.1.33.2. SWCMP.Scan.SWlp.Conflicts.tc1.v1

Test ID	32553
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-30 14:28:03 Meaning: QA Approved - (QA)

	Signed by: Koonen, Jos (nly95806) on 2017-03-30 10:50:32 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-28 13:38:33 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Display and handling of scan UI parameters
Padre conflicts

Description: Conflict setting for Padre

Requirements Coverage:

Req ID	Requirement Tag	Description
18796	SWCMP.Scan.SWIp.Conflicts	SWIp is only available for FFE sequences.If SWIp is enabled, a parameter change leads to a conflict if one of the following conditions is not met: • Scan mode: 3D • Slice orientation: transverse • contrast enhancement: T1 • Flow compensation : yes • Acquisition mode: Cartesian • Number of stacks: 1 • Recon matrix ≥ 128 • fast imaging: no, EPI • multi echo without flyback Furthermore, SWIp is not compatible with: Respiratory or cardiac synchronization, Dixon, half scan, partial echo, multi-nuclei imaging, MTC, B0/B1 maps, fat/water suppression, Angio/Contrast enh., quantitative flow, fMRI echo stabilization, arterial spin labeling, dynamic study, elastography, diffusion, MIP/MPR

Test steps:

Step	Description	Expected Result
Prep 1	1. Run m_amn_set_test_environment.pl and select any system configuration. 2. Start Scanner application. 3. Create new logfile. 4. Load scan SWIp_ffe_se from Head_CNS/SWIp. 5. If conflicts occur then resolve the conflict. In specific case that "total amount of data too large" conflict occurs reduce the scan resolution.	N.A. (preparation)
Step 2	Verify that SWIp is only available for the following scan protocol parameter settings: • Imaging sequence: FFE • Scan mode: 3D • Slice orientation: transverse • contrast enhancement: T1 • Flow compensation : yes • Acquisition mode: Cartesian • Number of stacks: 1 • Recon matrix ≥ 128 • fast imaging: no, EPI • multi echo without flyback	No conflicts occur on these scan protocol parameter settings

Step	Description	Expected Result
Step 3	Verify that for the conflict free SWIp protocol created in step 2 the following parameter changes lead to a conflict: • Respiratory • cardiac synchronization • Dixon • half scan • partial echo • MTC • B0/B1 maps • fat/water suppression • Angio/Contrast enh. • quantitative flow • shifted echo • fMRI echo stabilization • arterial spin labeling • dynamic study • elastography • diffusion • MIP/MPR Attach the logfile as objective evidence.	Conflicts occur for each of the conditions specified.

3.1.33.3. SWCMP.Scan.SWIp.UI.tc1

Test ID	8075
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Display and handling of scan UI parameters
SWIp UI

Description: UI appearance of SWIp

Preparations

Select the tedb database
Start *ExamCards Environment*.
Load scan SWIp_ffe_se from Head_CNS/SWIp.

Requirements Coverage:

Req ID	Requirement Tag	Description									
18793	SWCMP.Scan.SWlp.UI	The postproc tab will show the following parameters: <table> <tr> <th>name</th><th>range</th><th>description</th></tr> <tr> <td>SWlp</td><td>No, yes</td><td>Switch for reconstruction</td></tr> <tr> <td>Images</td><td>SW-M, SW-P</td><td>Reconstructed image types</td></tr> </table>	name	range	description	SWlp	No, yes	Switch for reconstruction	Images	SW-M, SW-P	Reconstructed image types
name	range	description									
SWlp	No, yes	Switch for reconstruction									
Images	SW-M, SW-P	Reconstructed image types									
18794	SWCMP.Scan.SWlp.images.SW_M	If the reconstructed image type SW-M is selected, an image will be created which represents the magnitude image weighted with a phase mask									
18795	SWCMP.Scan.SWlp.images.SW_P	If the reconstructed image type SW-P is selected, an image will be created which represents the phase mask									

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Verify the range of the SWlp parameter on the postproc page.	The available options for SWlp are: no, yes.	None
Step 2	Verify available image types for SWlp on the postproc page	The parameter <i>images</i> allows you to select <i>SWI-M</i> and <i>SW-P</i> instead of the standard <i>M</i> , <i>P</i> , <i>I</i> , <i>R</i> (see attached image)	 Padre_image_type_selection.zip

3.1.34. ME T1m

3.1.34.1. SWCMP.Scan.T1m.CardiacPresetAcqSchemes.tc1

Test ID	23554
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-06-11 15:42:40 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-06-11 15:38:05 Meaning: SME Approved - (Business)::Bulk Transitions

	Signed by: Cull, Tom (310129706) on 2015-06-10 21:15:24 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The purpose of this test is to that the preset acq. schemes are generated as expected.

Requirements Coverage:

Req ID	Requirement Tag	Description
56528	SWCMP.Scan.T1m.CardiacAcquisition	For T1 mapping = Cardiac Native, Cardiac Enhanced, or Cardiac User-Defined Cardiac synchronization device = trigger = ECG Number of heart phases = single phase trigger delay = end-diastole (typically) adaptive = no Respiratory compensation = breath hold user defined breath hold = no
43365	SWCMP.Scan.T1m.Sequence.bTFE	When T1 mapping is active for contrast enhancement = balanced (b-TFE) the following balanced parameters shall be available: TFE start-up echoes (number) = user defined = 10 b-TFE startup mode = half alpha b-TFE startup mode = lin. sweep up [with "b-tfe binom. Prepulse" = no] partial echo = yes partial echo factor = as desired in allowed range ultrashort = yes
43366	SWCMP.Scan.T1m.PresetAcqSchemes	Two preset MOLLI acquisition schemes shall be available: T1 Mapping = Cardiac Native --> 5s-(3s)-3s T1 Mapping = Cardiac Enhanced --> 4s-(1s)-3s-(1s)-2s

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Configure the test system as a WA15T (m_amn_set_test_environment.pl and select the WA15T option) Unzip and import the attached examcard <u>CTS T1M Cardiac N-E.ExcamCard</u> . Add the following line to the G:\Site\bstrace.dat file: MPALDM check_sp310 * Start the application. Open the Scan Control Parameter Editor. On the General tab, set Physiology	N.A. (preparation)	 physSimInfo.zip  CTS_T1M_Cardiac_N-E.zip

Step	Description	Expected Result	Attachments
	simulation to yes. By default, the No of RR intervals = 8. Set the first 8 RR intervals (ms) to 1000. Press 'Apply' to apply the required changes. Start a new set of log files. (Logging application -> Configuration -> Create new current logfile.		
Step 1	Start a New Exam and load the CTS T1M Cardiac N-E ExamCard into the scan queue. Run the SURVEY scan.	The expected scans appear in the scan queue and are not in conflict. Survey runs to completion.	None
Step 2	For the CardiacNative and CardiacEnhanced scans, check that the listed parameters match the expected values. On the Contrast tab: TFE prepulse slice selection delay (ms) PSIR T1-Mapping	Parameters match these expected values: On the Contrast tab: TFE prepulse = invert slice selection = no delay = user defined (ms) = 350 PSIR = no T1 Mapping = cardiac native (for cardiac native scan) T1 Mapping = cardiac enhanced (for cardiac enhanced scan)	None
Step 3	Run the CardiacNative scan and allow it to complete reconstruction.	Scan runs to completion and reconstructs.	None
Step 4	Run the CardiacEnhanced scan and allow it to complete reconstruction.	Scan runs to completion and reconstructs.	None
Step 5	Copy the file G:\Site\devlogcurrent.log to a folder on an SDE or other computer that has access to R:\MRtools\dev_support. Unzip the attached <u>filter_query_T1map_num_invd.zip</u> content to the same folder. (The extracted file will be <u>filter_query_T1map_num_invd.txt</u>) From the same folder, run the following command: R: MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe devlogcurrent.log @filter_query_T1map_num_invd.txt > CTS_cardiac_preset.log Compare the outfile	The output file generated this step matches the attached (zipped) reference file <u>CTS_cardiac_preset.ref</u>.	 CTS_cardiac_preset.zip  filter_query_T1map_num_invd.zip

Step	Description	Expected Result	Attachments
	CTS_cardiac_preset.log with the attached CTS_cardiac_preset.ref.		

3.1.34.2. SWCMP.Scan.T1m.CardiacUserDefinedAcqSchemes.tc1

Test ID	23555
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-06-11 15:42:42 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-06-11 15:38:07 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Cull, Tom (310129706) on 2015-06-10 21:16:00 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The purpose of this test is to verify the functionality of the cardiac user defined acq. mode.

Requirements Coverage:

Req ID	Requirement Tag	Description
56528	SWCMP.Scan.T1m.CardiacAcquisition	For T1 mapping = Cardiac Native, Cardiac Enhanced, or Cardiac User-Defined Cardiac synchronization = trigger device = ECG Number of heart phases = single phase trigger delay = end-diastole (typically) adaptive = no Respiratory compensation = breath hold user defined breath hold = no
43365	SWCMP.Scan.T1m.Sequence.b TFE	When T1 mapping is active for contrast enhancement = balanced (b-TFE) the following balanced parameters shall be available: TFE start-up echoes = user defined (number) = 10 b-TFE startup mode = half alpha b-TFE startup mode = lin. sweep up [with "b-tfe binom. Prepulse" = no] partial echo = yes partial echo factor = as desired in allowed range ultrashort = yes

Req ID	Requirement Tag	Description
43367	SWCMP.Scan.T1m.UserDefined AcqSchemes	When "T1 Mapping" = Cardiac User-Defined or MSK, the user shall be allowed to define the MOLLI scheme, The user defined MOLLI acquisition schemes will be specified via the additional parameters. The UI shall provide the user an array (Acq. intervals) to define the acquisition intervals. The user shall define an array of N elements that all greater than zero. The user shall define the N+1th element of Acq. intervals as zero to indicate the termination of the array. Acq. intervals: 1: [integer value > 0] 2: [integer value > 0] ... N: [integer value > 0] N+1: [integer values = 0] Acq. Interval Type : seconds or beats (only with "T1 Mapping" = Cardiac User-Defined) Pause interval : [integer value > 0]

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Configure the test system as a WA15T (m_amn_set_test_environment.pl and select the WA15T option) Unzip and import the attached examcard <u>CTS T1M Cardiac UDef.ExcamCard</u>. Add the following line to the G:\Site\bstrace.dat file: MPALDM check_sp310 * Start the application. Open the Scan Control Parameter Editor. On the General tab, set Physiology simulation to yes. By default, the No of RR intervals = 8. Set the first 8 RR intervals (ms) to 1000. Press 'Apply' to apply the required changes. Start a new set of log files. (Logging application -> Configuration -> Create new current logfile.	N.A. (preparation)	 physSimInfo.zip  CTS_T1M_Cardiac_UDef.zip
Step 1	Start a New Exam and load the CTS T1M Cardiac UDef ExamCard into the scan queue.	The expected scans appear in the scan queue and are not in conflict. Survey runs to completion.	None

Step	Description	Expected Result	Attachments
	Run the SURVEY scan		
Step 2	For the scan CardiacUserDef check that the listed parameters match the expected values. On the Contrast tab: TFE prepulse slice selection delay (ms) PSIR T1-Mapping Acq. interval type Acq. intervals Pause interval type Pause interval	Parameters match these expected values: On the Contrast tab: TFE prepulse = invert slice selection = no delay = user defined (ms) = 350 PSIR = no T1-Mapping = cardiac user defined Acq. interval type = seconds Acq. intervals [1] = 4 [2] = 3 [3] = 2 [4] = 0 Pause interval type = seconds Pause interval = 1	None
Step 3	Run the CardiacUserDef scan and allow it to complete reconstruction.	Scan runs to completion and reconstructs.	None
Step 4	Copy the file <u>G:\Site\devlogcurrent.log</u> to a folder on an SDE or other computer that has access to R:\MRtools\dev_support. Unzip the attached <u>filter_query_T1map_num_invd.zip</u> content to the same folder. (The extracted file will be <u>filter_query_T1map_num_invd.txt</u>) From the same folder, run the following command: R: \MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe devlogcurrent.log @filter_query_T1map_num_invd.txt > CTS_cardiac_undef.log Compare the outfile CTS_cardiac_undef.log with the attached CTS_cardiac_preset.ref.	The output file generated this step matches the attached (zipped) reference file <u>CTS_cardiac_undef.ref</u>.	 CTS_cardiac_undef.zip  filter_query_T1map_num_invd.zip

3.1.34.3. SWCMP.Scan.T1m.MSKUserDefinedAcqSchemes.tc1.v1

Test ID	32320
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-20 14:21:13 Meaning: QA

	Approved - (QA)
	Signed by: Oorschot, Joep van (310224063) on 2017-03-20 12:36:10 Meaning: SME Approved - (Business)
Approval Status	Signed by: Graaf, Ruud de (nly95749) on 2017-03-20 11:07:19 Meaning: SME Approved - (Technical)
	Approved

Description:

The purpose of this test is to verify the functionality of the MSK acq. mode.

Requirements Coverage:

Req ID	Requirement Tag	Description
56529	SWCMP.Scan.T1m.MSKAcquisition	For T1 mapping = MSK Cardiac synchronization = trigger device = internal defined phases = 1 Cycle duration (ms) = 1000 (typically) Respiratory compensation = no/breath hold user defined breath hold = no (if using breath hold)
43365	SWCMP.Scan.T1m.Sequence.b TFE	When T1 mapping is active for contrast enhancement = balanced (b-TFE) the following balanced parameters shall be available: TFE start-up echoes = user defined (number) = 10 b-TFE startup mode = half alpha b-TFE startup mode = lin. sweep up [with "b-tfe binom. Prepulse" = no] partial echo = yes partial echo factor = as desired in allowed range ultrashort = yes
43367	SWCMP.Scan.T1m.UserDefined AcqSchemes	When "T1 Mapping" = Cardiac User-Defined or MSK, the user shall be allowed to define the MOLLI scheme, The user defined MOLLI acquisition schemes will be specified via the additional parameters. The UI shall provide the user an array (Acq. intervals) to define the acquisition intervals. The user shall define an array of N elements that all greater than zero. The user shall define the N+1th element of Acq. intervals as zero to indicate the termination of the array. Acq. intervals: 1: [integer value > 0] 2: [integer value > 0] ... N: [integer value > 0] N+1: [integer values = 0]

Req ID	Requirement Tag	Description
		Acq. Interval Type : seconds or beats (only with "T1 Mapping" = Cardiac User-Defined) Pause interval : [integer value > 0]

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	Configure the test system as a WA15T (m_amn_set_test_environment.pl and select the WA15T option) Unzip and import the attached examcard <u>CTS T1M MSK.ExcamCard</u>. Add the following line to the G:\Site\bstrace.dat file: MPALDM check_sp310 * Start the application. Open the Scan Control Parameter Editor. On the General tab, set Physiology simulation to yes. By default, the No of RR intervals = 8. Set the first 8 RR intervals (ms) to 1000. Press 'Apply' to apply the required changes. Start a new set of log files. (Logging application -> Configuration -> Create new current logfile.	n/a (preparation)	 physSimInfo.zip  CTS_T1M_MSK_examcard.zip
Step 1	Start a New Exam and load the CTS T1M MSK ExamCard into the scan queue. Run the SURVEY scan	The expected scans appear in the scan queue and are not in conflict. Survey runs to completion.	None
Step 2	For the MSK scan check that the listed parameters match the expected values. On the Contrast tab: TFE prepulse slice selection delay (ms) PSIR T1-Mapping Acq. intervals Pause interval On the Motion tab: Cardiac synchronization device Cycle Duration (ms)	Parameters match these expected values: On the Contrast tab: TFE prepulse = invert slice selection = no delay = user defined (ms) = 350 PSIR = no T1-Mapping = MSK Acq. intervals [1] = 5 [2] = 3 [3] = 0 Pause interval = 3 On the Motion tab: Cardiac synchronization = Triggered device = Internal	None

Step	Description	Expected Result	Attachments
		Cycle Duration (ms) = 1000	
Step 3	Run the MSK scan and allow it to complete reconstruction.	Scan runs to completion and reconstructs.	None
Step 4	Copy the file <u>G:\Site\devlogcurrent.log</u> to a folder on an SDE or other computer that has access to R:\MRtools\dev_support. Unzip the attached <u>filter_query_T1map_num_invd.zip</u> content to the same folder. (The extracted file will be <u>filter_query_T1map_num_invd.txt</u>) From the same folder, run the following command: R: \MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe devlogcurrent.log @filter_query_T1map_num_invd.txt > CTS_T1map_MSK.log Compare the outfile CTS_cardiac_undef.log with the attached CTS_T1map_MSK.ref.	Verify in the output file generated by this step that - the SURVEY scan has 0 inversion delays - the MSK scan has 8 inversion delays using the attached reference file <u>CTS_T1map_MSK.ref</u> for guidance. Attach the output file to this step.	 filter_query_T1map_num_invd.zip  CTS_T1map_MSK_ref.zip

3.1.35. ME TFE

3.1.35.1. SWCMP.Scan.TFE.Physiology.tc1.v2

Test ID	33101
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-12 10:16:55 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2017-04-12 09:54:55 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-04-12 09:47:29 Meaning: SME

	Approved - (Technical)
Approval Status	Approved

Description:

This test is run on an Ingenia 1.5T for purposes of reproducibility. It can be run on any other configuration as well.

The minimum inversion time delay is dependent on where the k0 profile is acquired. Sequence parameters such as profile order, halfscan, single-/multi-shot, and turbo directions can affect this.

This test is designed to check that the minimum inversion time delay for the TFE prepulse and the trigger delay are calculated according to different sequence parameters that can influence where the k0 profile is acquired.

Requirements Coverage:

Req ID	Requirement Tag	Description
17766	SWCMP.Scan.TFE.Physiology	TFE segmentation in shots is very suitable to for cardiac scanning and to handle respiratory motion, specified in sections TFE physiology.

Test steps:

Step	Description	Expected Result
Preparation	Use m_amn_set_test_environment.pl and select configuration Ingenia 1.5T. Clear the logfile by opening a command window and typing 'p_logfile_nu createnewlogfile'. Enable trace channel 'MPUTFEPP TFE_testcase *' in C:\site\bstrace.dat. Start the application software and create a new examination. Load Head CNS/TFE/i1_ct. Under the Contrast tab, check that the turbo direction is Y and the profile order is linear. Under the Motion tab, check that the trigger delay is shortest.	N.A. (prep only)
Step 1	Under Contrast tab, set TFE prepulse to invert and click Accept. Open the log file (G:\Log\devlogcurrent.log) and run a search for 'TFE_testcase'	The value of Min. TI delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 2	Under the Contrast tab, set halfscan to yes, set the halfscan factor in the Y direction to 0.625, and set the halfscan factor in the Z direction to 1.0. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. TI delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 3	Under the Contrast tab, change the profile order to low_high and set halfscan to no. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. TI delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 4	Under the Contrast tab, change the turbo direction to Z and profile order to linear. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. TI delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.

Step	Description	Expected Result
Step 5	Under the Contrast tab, set halfscan to yes, set the halfscan factor in the Y direction to 0.625, set the halfscan factor in the Z direction to 1.0. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 6	Under the Contrast tab, set the halfscan factor in the Z direction to 0.8 and click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 7	Under the Contrast tab, change the profile order to low_high and set halfscan to no. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 8	Under the Contrast tab, change the Scan mode to MS and profile order to linear. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 9	Under the Contrast tab, set halfscan to yes and set the halfscan factor to 0.625. Click Accept Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 10	Under the Contrast tab, change the profile order to low_high and set halfscan to no. Click Accept Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 11	Load Head CNS/TFE/i1_ct again. Under the Contrast tab, check that the turbo direction is Y and the profile order is linear. Under the Motion tab, check that the trigger delay is shortest. Under the Contrast tab, change shot mode from multi-shot to single-shot and set TFE prepulse to invert. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 12	Under the Contrast tab, set halfscan to yes, set the halfscan factor in the Y direction to 0.625, and set the halfscan factor in the Z direction to 1.0. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 13	Under the Contrast tab, change the profile order to low_high and set halfscan to no. Click Accept Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 14	Under the Contrast tab, change the turbo direction to Z and click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 15	Under the Contrast tab, change the profile order to linear and click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 16	Under the Contrast tab, set halfscan to yes, set the	The value of Min. Tl delay equals to the sum of

Step	Description	Expected Result
	halfscan factor in the Y direction to 1.0, and set the halfscan factor in the Z direction to 0.625. Click Accept. Open the log file and run a search for 'TFE_testcase'	SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 17	Under the Contrast tab, change the profile order to low_high and set halfscan to no. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 18	Under the Contrast tab, change the Scan mode to MS and profile order to linear. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 19	Under the Contrast tab, set halfscan to yes and set the halfscan factor to 0.625. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 20	Under the Contrast tab, change the profile order to low_high and set halfscan to no. Click Accept. Open the log file and run a search for 'TFE_testcase'	The value of Min. Tl delay equals to the sum of SQ::tfepp->dur2 + SOR::shot->ref +/- 0.1.
Step 21	Load Head CNS/TFE/i1_ct again. Under the Contrast tab, check that the turbo direction is Y and the profile order is linear. Under the Motion tab, check that the trigger delay is shortest. Under the Geometry tab, set REST slabs to 1. Click Accept. Open the Sequence Development Viewer and check the value of SQ::rest->dur.	The trigger delay on the INFO page increases with SQ::rest->dur.
Step 22	Under the Contrast tab, set Fat suppression to SPIR and set the profile order to low_high. Click Accept. Open the Sequence Development Viewer and check the value of SQ::spir->dur.	The trigger delay on the INFO page increases with SQ::spir->dur.
Step 23	Load Head CNS/TFE/i1_ct again. Under the Contrast tab, check that the turbo direction is Y and the profile order is linear. Under the Motion tab, check that the trigger delay is shortest. Under the Contrast tab, set TFE prepulse to invert. Open the Sequence Development Viewer and check the value of SQ:tfepp -> dur (SQ_tfepp_dur_no_rest) Under the Geometry tab, set REST slabs to 1. Click Accept. Open the Sequence Development Viewer and check the value of SQ::rest->dur and SQ:tfepp->dur (SQ_tfepp_dur_with_rest).	The trigger delay on the INFO page increases with SQ_tfepp_dur_with_rest - SQ_tfepp_dur_no_rest + SQ::rest->dur.
Step 24	Under the Contrast tab, set Fat suppression to SPIR and set the profile order to low_high. Click Accept. Open the Sequence Development Viewer and check the value of SQ::spir->dur.	The trigger delay on the INFO page increases with SQ::spir->dur.

3.1.35.2. SWCMP.Scan.TFE.Physiology.tc2

Test ID	7423
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Cardiac triggering, multi phase (scanner)

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object: ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
---------------	--------------------

linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

```

SQ`tfep:dur2          11.4546
SQ`base:ref           2.7158
TFE factor             13
startup profiles       3
pmf                    1.0
repetition time        4.0

```

It's a linear profile order so the number of preceding profiles is $(13-1)*(1.0-0.5)/1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of $ky=0$ (or $kz=0$ in case of 3D turbo direction Z).

As an example, the trigger delay in PP **CT** is calculated from:

```

SQ`base:ref           0.8498
TFE factor             13
startup profiles       3
pmf                    1.0
repetition time        4.2

```

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

Pass criteria: all scans can be defined, behave as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17766	SWCMP.Scan.TFE.Physiology	TFE segmentation in shots is very suitable to for cardiac scanning and to handle respiratory motion, specified in sections TFE physiology.

Test steps:

Step	Description	Expected Result
Step 1	Start with PP i1_ct, change to user defined number of heart phases and increase the number of phases to the maximum, make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 2	Increase the TFE factor and verify that the maximum number of phases decreases, scan again with the new maximum of phases.	Pass/fail: The maximum number of phases decreases. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 3	Change the startup echoes to user defined 0 and verify that the maximum number of phases increases.	Pass/fail: The maximum number of phases increases.
Step 4	Change the number of phases from user defined to maximum, verify that it is the same.	Pass/fail: The maximum number of phases has not changed.
Step 5	Change to turbo direction Z, solve if necessary the conflict on TFE factor, change the number of phases to the maximum in user defined and make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 6	Decrease the TFE factor and increase the number of phases to the new maximum, make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 7	Switch to low-high profile order, make scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 8	Decrease the TFE factor and select more phases, make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 9	Select maximum number of heart phases and verify that it is the same.	Pass/fail: The maximum number of heart phases has not changed.
Step 10	Change to turbo direction Y, select user defined number of	Pass/fail:

Step	Description	Expected Result
	heart phases again and with the maximum number of phases and switch to shot mode default, this shall not lead to conflicts, the range of the number of phases must increase now, make a scan.	The range of the number of phases increases. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 11	Switch to scan mode MS and verify that only single phase measurements are possible.	Pass/fail: Only single phase measurements are possible.
Step 12	Switch to 3D, solve some conflicts, increase the number of phases and make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 13	Switch to 2D, take an average TFE factor and scan with the maximum number of phases.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 14	Switch to maximum number of heart phases and verify that it is the same.	Pass/fail: Number of Heart phases has not changed compared to previous scan. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar

3.1.35.3. SWCMP.Scan.TFE.Physiology.tc3

Test ID	7424
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Cardiac triggered, multi slice (PT6000)

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object: ENC`ima:[0];partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep:dur2	11.4546
SQ`base:ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0

repetition time 4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5)/1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of ky=0 (or kz=0 in case of 3D turbo direction Z).

As an example, the trigger delay in PP CT is calculated from:

SQ`base:ref	0.8498
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

Enable the option CARDPERFUSION with FIGMAN.

Pass criteria: all scans can be defined as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17766	SWCMP.Scan.TFE.Physiology	TFE segmentation in shots is very suitable to for cardiac scanning and to handle respiratory motion, specified in sections TFE physiology.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Start with PP i3_ct_ms_tfe, this is a MS TFE protocol with 6 slices	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar	None
Step 2	Look at the info page and verify that it makes sense:	Pass/fail: TFE acq duration <= TFE shot duration	None

Step	Description	Expected Result	Attachments
		2. From TFE shot duration, number of slices, trigger delay and no trigger period it follows how many beats it takes to complete one package, so this must be in agreement with the shot interval. The shot interval is 2 beats, so there are 3 slices per heart beat.	
Step 3	Make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar	None
Step 4	Increase the cardiac frequency to e.g. 120 bpm and look again at the info page. Make a scan (change the physiology simulation parameters!)	Pass/fail: The shot duration has not changed but the other parameters have. There is no only space for 2 slices per heart beat and the shot interval is 3 beats. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar	None
Step 5	Increase the cardiac frequency to e.g. 250 bpm and look again at the info page.	Pass/fail: The shot duration has not changed but the other parameters have. There is only space for 1slice per heart beat and the shot interval is 6 beats. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar	None
Step 6	Go back to the original preset procedure and 60 bpm and use the maximum turbo factor (i.e. 2 shots). Change the shot interval to respectively 1, 2, 3, 4, 5, 6 and 7 beats and check if the behaviour is like visualize in the attached table. So for example with a shot interval of 4 beats, you will here three beats with the collection of 2 slices each and a fourth silent beat. Refer Attached	Pass/fail: The behaviour for beats is like visualized in the table attached table. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar	 TFE_Beats.xls

Step	Description	Expected Result	Attachments
Step 7	Change now in the original preset procedure the number of slices from 6 to 5 and make a scan. Verify that you hear three slices in the first beat and two in the second beat. Increase the number of dynamic scans to say 10. Change the physiology simulation parameters such that there will be some interval shorter than the no trigger period. Make a scan and verify that the shorter periods are really skipped.	Pass/fail: You hear the acquisition of three slices in the first beat and two in the second beat. The shorter periods are really skipped - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar	None

3.1.35.4. SWCMP.Scan.TFE.Segmentation.tc1

Test ID	7410
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

3D with turbo direction Y (scanner)

In scan mode 3D with turbo directionY, the ky-profiles are divided in a number of shots. If the number of ky-profiles is not a multiple of the tfe factor, the number of ky-profiles is rounded downwards (see RS: TFE). The number of ky-profiles equals the pre_round_nr_profiles[P_ORI] attribute of the ENC object.

Pass criteria: all scans can be defined as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17762	SWCMP.Scan.TFE.Segmentation	TFE is a segmented gradient echo technique, as specified in sections TFE technique and parameters to control segmentation and contrast and Protocol parameter user interface. Segmentation in shots offers possibilities to manipulate image contrast and to handle physiological motion in an efficient way.

Test steps:

Step	Description	Expected Result
Step 1	Start with the default PP i1_3D in subanatomy TFE, change the turbo direction to Y and start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 2	Look at the range of the TFE factor, it should be: 1-64. Make a scan with TFE factor 1 and TFE factor 64, compare the result, the slices should look the same for each scan.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 3	Verify that single shot is allowed in combination with 3D.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 4	Change the matrix size to 256 and verify the increased range of the TFE factor to 1-256.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 5	Play around with scan percentage, rectangular FOV, half scan, foldover suppression (2 and 3 NSA), matrix size, SENSE and confirm a correct response of the range of the TFE factor.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 6	Change to scan technique TFE-EPI with turbo direction Y, solve the conflicts and verify that the range of the TFE factor is now number of ky-profiles divided by the EPI factor, so increasing the EPI factor leads to a decrease in the range setting of the TFE factor. For example: matrix size is 256, EPI factor is 9, range of the TFE factor is 1-28.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar

Step	Description	Expected Result
Step 7	Repeat Step 5	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 8	Change the EPI direction to Z and note that the range of the TFE factor is not influenced anymore by the EPI factor.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar

3.1.35.5. SWCMP.Scan.TFE.Segmentation.tc2.v2

Test ID	32745
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-04 09:14:05 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-04-03 13:45:08 Meaning: SME Approved - (Technical) Signed by: Verburg, Frank (nly95979) on 2017-04-03 13:19:53 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

3D with turbo direction Z (scanner)

In scan mode 3D with turbo direction Z, the kz-profiles are divided in a number of shots. So in case of 10 slices with an oversample factor of 1.28 there are 13 kz-profiles. This is difficult to divide so it is allowed to oversample with 1 or 2 slices more. This ensures that the shot length is the same for each shot.

Pass criteria: all scans can be defined as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17762	SWCMP.Scan.TFE.Segmentation	TFE is a segmented gradient echo technique, as specified in sections TFE technique and parameters to control segmentation and contrast and Protocol parameter user interface. Segmentation in shots offers possibilities to manipulate image contrast and to handle physiological motion in an efficient way.

Test steps:

Step	Description	Expected Result
Prep	Create a new logfile	N.A.
Step 1	Start with the default protocol i1_3D in subanatomy TFE, change the turbo direction to Z and start the scan	No error during protocol validation No scan abort
Step 2	Change the number of slices to 10 and look at the range of the turbo factor. Make a scan with all allowed TFE factors and compare the results.	The range of the turbo factor is 1,2,3,5,7,13 The slices of the scans with TFE factors 1,2,3,5,7,13 look the same
Step 3	Set Recon voxel size AP = 1 (meaning the reconstructed slice thickness becomes 1, which enables slice interpolation), set the number of slices to 10 and check the range of the TFE factor.	The range of the TFE factor is 1,2,3,4,7
Step 4	Set Recon voxel size AP = 2 (disabling slice interpolation) and the number of slices to 10 and define 2 chunks. The oversample factor is now 1.9. Check the range of the TFE factor.	The range of the TFE factor is 1,2,3,5,9
Step 5	Set Recon voxel size AP = 1 (enabling slice interpolation) and change the number of slices to 20 the range should remain the same but will be modified due to rounding. Check the range of the TFE factor.	The range of the TFE factor is 1,2,3,4,5,6,10
Step 6	Return to the original protocol i1_3D in subanatomy TFE with turbo direction Z, change to scan technique TFE-EPI with epi direction Y, solve the conflicts and verify the range of the TFE factor with varying EPI factors.	The range of the TFE factor is not influenced by the EPI factor.
Step 7	Change the EPI direction to Z. Resolve conflicts (if there are any), and note that the range of the TFE factor is now determined by the EPI factor: nr kz-profiles divided by the EPI factor has to be a multiple of the TFE factor. For each EPI factor, try all TFE factors which are in the range. As objective evidence attach the (zipped) logfile.	No new conflicts should occur for any combination of EPI factor and TFE factor.

3.1.35.6. SWCMP.Scan.TFE.Segmentation.tc3.v2

Test ID	32758
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-06 16:42:33 Meaning: QA

	Approved - (QA)
	Signed by: Graaf, Ruud de (nly95749) on 2017-04-06 16:20:11 Meaning: SME Approved - (Technical)
	Signed by: Crijns, Sjoerd (310257923) on 2017-04-06 14:20:23 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

3D with turbo direction radial (scanner)

In scan mode 3D with turbo direction radial the total number of profiles is divided in a number of shots and rounded downwards. In general the total number of profiles is large and the turbo factor is small so no big errors are expected in rounding.

Pass criteria: all scans can be defined as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17762	SWCMP.Scan.TFE.Segmentation	TFE is a segmented gradient echo technique, as specified in sections TFE technique and parameters to control segmentation and contrast and Protocol parameter user interface. Segmentation in shots offers possibilities to manipulate image contrast and to handle physiological motion in an efficient way.

Test steps:

Step	Description	Expected Result
Step 1	Make sure option TFE_Extended (CSK option 4) is enabled. Start with the default protocol Head CNS/TFE/ i1_3D, make sure that the turbo direction is radial and start the scan.	No error during protocol validation No scan abort
Step 2	Look at the range of TFE factor, it should be: 2-3207. Make a scan with TFE factor 2 and 3207, compare the result, the slices should look the same for each scan.	The range of the TFE factor is 2-3207 The slices look the same for the scan with TFE factor 2 and 3207
Step 3	Set TFE factor to 2. Set the reconstructed slice thickness to 1 (enabling slice interpolation) and the number of slices to 50 and verify that the number of shots is decreased to 797, make a scan with TFE factor 16.	The number of shots is decreased to 797 The scan validates and runs without abort
Step 4	Increase the matrix size to 512, increase the number of slices to the maximum (perhaps until the conflict "total amount of data too large"), leave the TFE factor to 16 and make a scan, starting the scan may take a while (up to a minute!).	The scan runs without abort

Step	Description	Expected Result
Step 5	Return to the original protocol and define 2 chunks, make a scan.	No scan validation errors The scan runs without abort
Step 6	Set the reconstructed slice thickness to 1 and the number of slices to 50 (with 2 chunks) and make a scan.	No scan validation errors The scan runs without abort
Step 7	Enable half scan and make a scan.	No scan validation errors The scan runs without abort
Step 8	Verify a conflict on 3D with turbo direction radial and TFE-EPI.	Turbo direction Radial with TFE-EPI gives a conflict.
Step 9	Change the first entry of the control scan parameter FFE 3D oversample factor from 1.28 to 1.0, enable 1dft test mode (also a control scan parameter) and load protocol Head CNS/TFE/i1_3D_64x64. Perform a scan and look at the MPR images.	The last image shows a ky-kz plane and should look like a dart board, i.e. the elliptical k-space shutter is nicely visible.

3.1.35.7. SWCMP.Scan.TFE.Segmentation.tc4

Test ID	7408
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

TFE factor and TFE shots

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom

- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object: ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep:dur2	11.4546
SQ`base:ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5) / 1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of $ky=0$ (or $kz=0$ in case of 3D turbo direction Z).

As an example, the trigger delay in PP **CT** is calculated from:

SQ`base:ref	0.8498
TFE factor	13
startup profiles	3

pmf 1.0
repetition time 4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

In this test the segmentation of profiles into TFE-shots will be checked. Starting with a default preset procedure, the TFE factor will be in- and decreased. Each step involves a validation of the protocol and a start scan. The protocol has a manual start in order to be really sure when the preparation phase have ended and the scan really starts.

Pass criteria: all scans can be defined as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17762	SWCMP.Scan.TFE.Segmentation	TFE is a segmented gradient echo technique, as specified in sections TFE technique and parameters to control segmentation and contrast and Protocol parameter user interface. Segmentation in shots offers possibilities to manipulate image contrast and to handle physiological motion in an efficient way.

Test steps:

Step	Description	Expected Result
Step 1	Start with the default PP i1_segm in subanatomy TFE, start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 2	Change the shot mode to single shot, note that the info page does not tell you the number of shots (since it is 1) but tells you what the tfe factor is (equals the matrix size = 256) and start scan. Since this is a single shot scan there is only one shot audible	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 3	Change the shot mode to multi shot again, change the number of slices to 1, and verify that the maximum tfe factor equals 128 which is the factor that leads to a 2 shot	The pass/fail criteria are: no deviations from below the mentioned expected output:

Step	Description	Expected Result
	scan, verify the number of shots on the info page, start the scan and listen. With a two shot scan, there is one startup shot so there are three shots audible!	- no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 4	Change the tfe factor to 127 and verify on the info page that the actual scan percentage decreases to $100 \cdot 127 / 128 = 99.2\%$, start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 5	Lower the tfe factor to $(256/3 \approx) 86$ and verify the number of shot is still 2 with an actual scan percentage of about 2/3, lower the tfe factor to 85 and note that suddenly the number of shots is 3, start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 6	Change the scan matrix size to the maximum and observe that the single shot mode is possible, not in conflict. Start the scan.	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 7	Return to multi shot mode (remaining maximum scan matrix, full scan) and change the tfe factor to 1 and start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 8	Change the tfe factor to 5 and start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar

Step	Description	Expected Result
Step 9	Change the tfe factor to 128 and start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: • no error during validation • no scan abort (for example due to buffer underrun) • no reconstruction aborts • regular update of the scan progress bar
Step 10	Change the tfe factor to 256 and start the scan	The pass/fail criteria are: no deviations from below the mentioned expected output: • no error during validation • no scan abort (for example due to buffer underrun) • no reconstruction aborts • regular update of the scan progress bar

3.1.35.8. SWCMP.Scan.TFE.Segmentation.tc5

Test ID	7414
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Display profile orders (Examcards-simulator)

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object: ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep:dur2	11.4546
SQ`base:ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5) / 1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of $ky=0$ (or $kz=0$ in case of 3D turbo direction Z).

As an example, the trigger delay in PP **CT** is calculated from:

SQ`base:ref	0.8498
TFE factor	13
startup profiles	3
pmf	1.0

repetition time 4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

Pass criteria: all scans can be defined as indicated and run without scan aborts. The displayed profile orders agree qualitatively with the statements given above.

Requirements Coverage:

Req ID	Requirement Tag	Description
17762	SWCMP.Scan.TFE.Segmentation	TFE is a segmented gradient echo technique, as specified in sections TFE technique and parameters to control segmentation and contrast and Protocol parameter user interface. Segmentation in shots offers possibilities to manipulate image contrast and to handle physiological motion in an efficient way.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	After starting the Examcards simulator, enable hex logging by setting the logical GYRO_REC_LABEL_LOG HEX. To do that, type in a command box: oslnm_nu modify system GYRO_REC_LABEL_LOG HEX Create a new log file. To do that, type in a command box: p_logfile_nu createnewlogfile	None	None
Step 2	Start with the default PP i1_3D in subanatomy TFE, change the 3D turbo direction to radial run the scan. Analyze the log file with the script m_imn_hexlab.pl found in clearcase in the folder mrmmethods\imaging\tst. To do so, type in a command box: First, to make sure the output folder exists: mkdir c:\nmr\site\log\ then, to analyze the log file m_imn_hexlab.pl <path_to_log_file> The output can be found in the c:\nmr\site\log\hex_lab.dat	Output file c:\nmr\site\log\hex_lab.dat contains 2 columns with numbers.	None
Step 3	display the profile order in mathcad using attached mathcad script file Fill in the TFE factor, make sure mathcad uses the the correct number of columns (2) in the input files.	The shots should be measured like the hands of a clock, starting at 9 o'clock.	

Step	Description	Expected Result	Attachments
			 display 2d turbo profile order.zip
Step 4	Enable half scan and, using above steps, display the profile order again,	The arms are now curved and the half scan effect is clearly visible, the "Inner circle" i.e. the profiles that were measured as first in a shot is still a circle, the shape is not influenced by the half scan effect.	None
Step 5	Change the scan matrix of the PP to 512 with 400 mm FOV, reduce the number of slices to 20, make the slice thickness equal to 10 mm and, using above steps, display the profile order again.	The shots should look like as much as possible straight lines.	None

3.1.35.9. SWCMP.Scan.TFE.ShotTiming.tc1.v2

Test ID	52728
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:19:58 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 14:51:12 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:28 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the parameters needed to specify shot timing by changing user interface parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description
17765	SWCMP.Scan.TFE.ShotTiming	Additional parameters are needed to specify shot timing, specified in sections TFE Shot Timing and Protocol parameter user interface.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and	The reported number of failed tests related to

Step	Description	Expected Result
	inspect the results file.	SwCmpScanTfeShotTiming in the results file is zero.

3.1.35.10. SWCMP.Scan.TFE.ShotTiming.tc2

Test ID	7421
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Gradient duty cycle

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object:

ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep:dur2	11.4546
SQ`base:ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5) / 1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of $ky=0$ (or $kz=0$ in case of 3D turbo direction Z).

As an example, the trigger delay in PP **CT** is calculated from:

SQ`base:ref	0.8498
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

Duty cycle problems in TFE are normally solved by increasing the shot interval. However in case of long shots (> 250 ms) the repetition time is increased instead of the shot interval. Test both by taking a high duty cycle procedure with a shot length above and below the 250 ms.

Pass criteria: all scans can be defined as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17765	SWCMP.Scan.TFE.ShotTiming	Additional parameters are needed to specify shot timing, specified in sections TFE Shot Timing and Protocol parameter user interface.

Test steps:

Step	Description	Expected Result
Step 1	Define the PP i6_duty and the change the repetition time to 49.99 (or leave it shortest), run the scan for about 10 minutes	Pass/fail: Scan runs for 10 minutes without scan abort.
Step 2	Now try to change the repetition time to a shot length of 250 ms and observe that this is not possible, look for the shortest TR above the value you just entered and again run the scan for ten minutes.	Pass/fail: Scan runs for 10 minutes without scan abort.

3.1.35.11. SWCMP.Scan.TFE.TFE-EPI.tc1

Test ID	7426
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

MS TFE-EPI (PT6000)

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil

- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object: ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(TFE\ factor - 1) * (pmf - 0.5) / pmf$
reverse linear	$(TFE\ factor - 1) * 0.5 / pmf$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep: dur2	11.4546
SQ`base: ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5) / 1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6\ ms$.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of $ky=0$ (or $kz=0$ in case of 3D turbo direction Z).

As an example, the trigger delay in PP **CT** is calculated from:

SQ`base: ref	0.8498
TFE factor	13

startup profiles 3
pmf 1.0
repetition time 4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

This section is about the protocols used for cardiac perfusion.

It is recommended to use a system with PT6000 gradients.

Pass criteria: all scans can be defined, behave as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17767	SWCMP.Scan.TFE.TFE-EPI	The TFE functionality shall combined with EPI readout, see sections TFE-EPI and Protocol parameter user interface.

Test steps:

Step	Description	Expected Result
Step 1	Start with PP i6_tfe-epi this is a cardiac perfusion protocol which lasts 2 minutes. Six slices are measured in two beats, the first minute are with a temporal resolution of 2 beats, the second minute with a temporal resolution of 3 beats. Switch on autoview and make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 2	Switch on the saturation prepulse, shared, slice order default, and repeat the scan, compare the result. Slice 1 and 2 are measured direct after the prepulse while slice 5 and 6 are measured at the end of the RR-interval, quite late after the prepulse.	Pass/fail: The difference between the two scans should be significant for slices 1 and 2, for slice 5 and 6 much less and for slice 3 and 4 in between. - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 3	Start with the original scan and switch off dynamic scans, enable SPIR with autoshim and low-high profile order, use slice order default and make a scan.	Pass/fail: - no error during validation - no scan abort (for example due to buffer

Step	Description	Expected Result
		underrun) • no reconstruction aborts • regular update of the scan progress bar
Step 4	Look in the viewing environment on the extended info and check that the trigger delays are correct.	Pass/fail: Trigger delays are the same for slices 1 and 2, for slices 3 and 4 and for slices 5 and 6. Also trigger delay $1 < 3 < 5$.
Step 5	Try now ascending and descending slice order and check again the trigger delays.	Pass/fail: For ascending slice order, trigger delays are same for slices 1&4, slices 2&5, slices 3&6. Trigger delay $1 < 2 < 3$ For descending slice order, trigger delays are also same for slices 1&4, slices 2&5, slices 3&6. Trigger delay $1 > 2 > 3$
Step 6	Disable cardiac triggering and verify a conflict on cardiac triggering and fast imaging mode (without the TFE extensions clinical science key)	Pass/fail: When "TFE extensions clinical science key" is disabled there is a conflict on cardiac triggering and fast imaging mode.

3.1.35.12. SWCMP.Scan.TFE.TFE-EPI.tc2

Test ID	7427
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

TFE-EPI, multi phase (scanner)

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence. The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object: ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep:dur2	11.4546
SQ`base:ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5)/1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of $ky=0$ (or $kz=0$ in case of 3D turbo direction Z).

As an example, the trigger delay in PP CT is calculated from:

SQ`base:ref	0.8498
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

Pass criteria: all scans can be defined, behave as indicated and run without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17767	SWCMP.Scan.TFE.TFE-EPI	The TFE functionality shall combined with EPI readout, see sections TFE-EPI and Protocol parameter user interface.

Test steps:

Step	Description	Expected Result
Step 1	Start with PP i6_tfe-epi switch off dynamic study and increase the number of phases to the maximum, solve the conflicts by changing to scan mode 2D, single beat and make a scan	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 2	Switch to multi shot and observe that the maximum TFE factor allows more heart phases, scan again with the new maximum of phases	Pass/fail: The maximum TFE factor allows more heart phases, - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 3	Change the number of phases from user defined to maximum, verify that it is the same	Pass/fail: The number of Heart phases has not changed compared to the previous scan.
Step 4	Change to 3D mode turbo direction Y, change the number of phases to the maximum in user defined and make a	Pass/fail:

Step	Description	Expected Result
	scan	- no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 5	Increase the TFE factor and adapt the number of phases to the new maximum, make a scan	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 6	Switch to low-high profile order, make scan	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 7	Use TFE factor 1 and select more phases, make a scan	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 8	Select maximum number of heart phases and verify that it is the same	Pass/fail: The number of Heart phases has not changed compared to the previous scan.
Step 9	Select user defined again with the maximum number of phases and verify that shot mode default is not allowed.	Pass/fail: Shot mode default is not allowed.
Step 10	Switch to scan mode MS and verify that only single phase measurements are possible	Pass/fail: Only single phase measurements are possible.
Step 11	Switch to 2D, take an average TFE factor and scan with the maximum number of phases	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar
Step 12	Switch to maximum number of heart phases and verify that it is the same	Pass/fail: The number of Heart phases has not changed compared to the previous scan.

Step	Description	Expected Result

3.1.35.13. SWCMP.Scan.TFE.TFE-EPI.tc3

Test ID	7428
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	During the introduction of electronic signature the initial approval state of this entity has been set to Approved. This was based on the fact that this item was approved as part of a pre-electronic signature project [R5.1.1 / R5.1.2 / R5.1.7]
Approval Status	Approved

Description:

Image quality TFE-EPI (scanner)

Test Description Layout:

The tests in this document are written down as short as possible. This may imply that only experienced users, as defined in the "purpose" section, understand the tests. Some of the test can also be performed with the Examcards test environment. This is indicated between parentheses after the test name.

Environmental Needs:

For some tests Examcards-simulator with the graphical viewer are sufficient. For other tests you need:

- a test system, for some tests PowerTrak 6000 gradient system (or higher) is needed, this is indicated between parentheses after the test name. These tests are also applicable for Ingenia scanners.
- the cardiac synergy coil
- the head phantom
- the test preset procedures which are defined in tedb.idx

How to calculate prepulse delays:

The prepulse delay is measured from the magnetic centre of the prepulse until the magnetic centre of the excitation of the profile $ky=0$ (or $kz=0$ in case of 3D turbo direction Z):

See diagram in attachment: **tfe_sequence.bmp**

The duration from the magnetic centre of the pulse until the end of the sequence equals the *dur2* attribute of that sequence. This sequence is followed by other prepulses like SPIR and REST, a number of startup cycles and a number of profiles that precede the measurement of $ky=0$. The duration from the beginning of the sequence until the magnetic centre of the $ky=0$ profile equals the *ref* attribute of that sequence.

The number of startup cycles can be found under sequence development mode in the IMP_TFE_all_pars group: MP_tfe_startup_profs.

The number of profiles preceding the $ky=0$ depends on the TFE factor, the profile order and the use of half scan. The partial matrix factor (pmf) can be found in sequence development mode in the encoding object:

ENC`ima:[0]:partial_matrix_factors. The second entry for turbo direction Y and the third entry for turbo direction Z. The definition is as follows:

profile order	preceding profiles
linear	$(\text{TFE factor} - 1) * (\text{pmf} - 0.5) / \text{pmf}$
reverse linear	$(\text{TFE factor} - 1) * 0.5 / \text{pmf}$
low high	0

As an example, the prepulse delay in PP **Prepulses** is calculated from:

SQ`tfep:dur2	11.4546
SQ`base:ref	2.7158
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.0

It's a linear profile order so the number of preceding profiles is $(13-1) * (1.0-0.5) / 1.0 = 6$ and the total delay is: $11.4546 + 2.7158 + 9 * 8.27 = 88.6$ ms.

How to calculate trigger delays:

The trigger delay is calculated pretty similar to the calculation of the prepulse delay. The delay is now calculated from the patient-to-sequencer delay (6 ms) + minimum synchronization delay (0.5 ms) and again the delay until the excitation of ky=0 (or kz=0 in case of 3D turbo direction Z).

As an example, the trigger delay in PP **CT** is calculated from:

SQ`base:ref	0.8498
TFE factor	13
startup profiles	3
pmf	1.0
repetition time	4.2

It's a linear profile order so the number of preceding profiles is 6 and the total delay is: $6.5 + 9 * 4.2 + 0.8455 = 45.2$ ms.

How to display the profile orders:

The profile order can be displayed by performing a scan in the Examcards-simulator with hex label logging enabled. The devcurrent.log has to be filtered with e.g. Perl (see a_ibolus_hexlab.pl) and the output can then be displayed with e.g. Excel or Mathcad (see R:/Scan SW/Csp/Mathcad/display 2D turbo profile order.mcd).

TFE-EPI can be used as a fast scan technique in coronary imaging. Verify that the clinical used protocol gives good image quality.

Pass criteria: the scan can be defined as indicated and runs without scan aborts.

Requirements Coverage:

Req ID	Requirement Tag	Description
17767	SWCMP.Scan.TFE.TFE-EPI	The TFE functionality shall combined with EPI readout, see sections TFE-EPI and Protocol parameter user interface.

Test steps:

Step	Description	Expected Result
Step 1	Select PP i6_3Dtf-epi and make a scan	Pass/fail: - no error during validation - no scan abort (for example due to buffer underrun) - no reconstruction aborts - regular update of the scan progress bar

3.1.36. ME TSE

3.1.36.1. SWCMP.Scan.TSE.Par.tc1

Test ID	41059
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:29:06 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:07:58 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-20 14:49:42 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of TSE parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
77962	SWCMP.Scan.TSE.Par.v2	Name	Range	Visible when	Description	System / Option restrictions
		TSE factor	2-256	Fast Imaging mode = TSE	Number of k-space profiles acquired per excitation.	Restrict range to 3-256 when Option06 (TSE extensions) is disabled
		turbo direction	Y, Z, radial	Fast Imaging mode = TSE & Scan mode = 3D	Controls the contents of a turbo shot within subsequent ky-kz areas.	Restrict range to "Y, radial" when Option06 (TSE extensions) is disabled
		bright fat reduction	no, yes	Refocusing control = yes & angle (deg) = [30.00 – 180.00]	This parameter enables or disables TSE bright fat reduction	Parameter is not visible when Option06 (TSE extensions) is disabled
		DRIVE	no, yes, anti	Fast Imaging mode = TSE	Enable an extra refocusing pulse plus a 'reset' RF pulse at the end of a TSE or GraSE echo train.	Restrict range to "no, yes" when Option06 (TSE extensions) is disabled

Req ID	Requirement Tag	Description				
		fid reduction	default, through-plane, strong, extra strong	Fast Imaging mode = TSE	Determines the crusher area of the applied crusher gradients.	Restrict range to "default, strong, extra strong" when Option 006 (TSE extensions) is disabled
		ultrashort	no, yes, ultra	Fast Imaging mode = TSE & Scan mode = 3D & not 3D view	Enable an increased data sampling rate to reduce echo spacings.	Restrict range to "no, yes" when Option 006 (TSE extensions) is disabled
		Refocusing control	no, constant, tissue specific, T1-optimized, T2-optimized	Fast Imaging mode = TSE	This parameter controls the refocusing pulse angles in TSE sequences	Restrict the range to "no, constant" when option bit 119 (TSERefoc Control) is disabled. Restrict the range to "no, constant,T1-optimized, T2-optimized" when option bit 06 (TSE extension

Req ID	Requirement Tag	Description				
						) is disabled. Restrict the range to "no, constant" when option bit 06 (TSE extension) is disabled & Scan mode = 3D
		angle (deg)	10 -180	Refocusing control = constant	For constant refocusing control, the user shall be able to specify a constant refocusing angle.	N.A.
		signal plateau	min. angle defined, TE defined, shot defined, user defined	Refocusing control = tissue specific	A signal plateau is a time during which signal is kept constant. Depending on the value of "signal plateau", more parameters are displayed:	N.A.

Req ID	Requirement Tag	Description				
		max1 angle (deg)	30-180	signal plateau = min. angle defined	Allows to specify the max angle of the refocusing pulses	N.A.
		ratio	0.5-2	signal plateau = TE defined	Specifies the duration of Signal plateau over TE ratio	N.A.
		ratio	0.5 -1	signal plateau = shot defined	Specifies the duration of Signal plateau over shot defined	N.A.
		(ms)	10-2000	signal plateau = user defined	Specifies the duration of user defined Signal plateau	N.A.
		max2 angle (deg)	30-180	signal plateau = min. angle defined	Allows to specify the max angle of the refocusing pulses	N.A.
		min. angle (deg)	10- 180	Refocusing control = T1-optimized or T2-optimized	Allows to specify the min angle of the refocusing pulses	N.A.

Req ID	Requirement Tag	Description				
		mid. angle (deg)	30-180	Refocusing control = T1-optimized or T2-optimized	Allows to specify the mid angle of the refocusing pulses	N.A.
		max. angle (deg)	30-180	Refocusing control = T1-optimized or T2-optimized	Allows to specify the max angle of the refocusing pulses	N.A.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanTsePar in the results file is zero.

3.1.36.2. SWCMP.Scan.TSE.prof_order.tc1

Test ID	25520
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2015-12-09 09:41:35 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Polders, Daniel Louis (310171444) on 2015-12-08 11:10:18 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-12-07 15:40:52 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

The main profile order functionality is tested in the profile order regression test. Although this test can be performed on any configuration, the Ingenia 1.5T configuration has been chosen for reproducibility of results.

This test checks that the selected profile order on the UI is correctly passed to the ORD object.

Requirements Coverage:

Req ID	Requirement Tag	Description
55700	SWCMP.Scan.TSE.profile_order.v2	The following profile orders can be chosen in the product: - low-high - starting at or near the center of k-space, and then choosing values lower or higher than the starting value in an alternating fashion, ever increasing the distance from the starting value. - linear - starting at or near the lowest k-space line, and then choosing values at equal increments, working towards the highest k-space line. - asymmetric - a flexible profile order that allows an independent choice of TE and TE spacing.

Test steps:

Step	Description	Expected Result
Preparation	Use m_amn_set_test_environment.pl to select an Ingenia 1.5T configuration. Start Examcards.	N.A. (preparation)
Step 1	Drag a 2D protocol F_basic_TSE to the execution list and start the parameter editor. Right click on the protocol F_basic_TSE and open Sequence Development Viewer. Go to the ORD object.	Verify that the first element of attribute ord_inner_order is set to linear.
Step 2	Close the Sequence Development Viewer. Change profile order to low_high. Set the user-defined echo time to 40 ms. Set TR to shortest. Right click on the protocol F_basic_TSE and open the Sequence Development Viewer. Go to the ORD object.	Verify that the first element of attribute ord_inner_order is set to low_high.
Step 3	Close the Sequence Development Viewer. Change the profile order to asymmetric. Right click on the protocol F_basic_TSE and open the Sequence Development Viewer. Go to the ORD object.	Verify that the first element of attribute ord_inner_order is set to asymmetric.

3.1.36.3. SWCMP.Scan.TSE.refocusing_control.tc1

Test ID	24924
Revision	4
Product Model	Any
Gradient System	Any
Options	N.A.

Signatures	Signed by: Moore, Liz (gbr01023) on 2016-05-02 10:31:58 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2016-05-02 09:56:10 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-04-29 15:44:00 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

This test case verifies that the refocusing flip angle sweeps are calculated correctly for the different enumerations of refocusing control. Although this test can be performed on any configuration, the Ingenia 1.5T configuration has been chosen for reproducibility of results.

Steps one to six test the T1-optimized refocusing control option.

Steps seven to ten test the T2-optimized refocusing control option. For T1-optimized and T2-optimized options, tests were introduced checking the basic situation where a four angle asymptotic sweep (T2-optimized) or single angle (T1-optimized with single angle equal to $90 + \min \text{ angle}/2$) would approach the min angle, followed by a sweep up to a mid angle, and ending with a linear sweep from the mid angle to the max angle. Tests were also introduced to test situations where the mid angle would overrule the min angle. Tests were also introduced to check combinations of parameters that should lead to conflicts.

Step eleven tests the constant refocusing control option.

Step twelve tests the no refocusing control option.

Test steps:

Step	Description	Expected Result	Attachments
Preparation	Use m_amn_set_test_environment.pl to select an Ingenia 1.5T configuration. Open G:\Site\bstrace.dat, and add the following line: MPURFE refoc * Start Examcards.	N.A. (preparation)	None
Step 1	Pull over the F_basic_TSE protocol from Head CNS/TSE. Go to the Contrast page. Set the following protocol parameters: • refocusing control = T1-optimized • min. angle = 30 • mid. angle = 120 • max. angle = 160 • TSE factor = 16 • profile order = linear Click Accept. Open G:\Site\sweep.dat.	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the attached. file.	 sweep.zip
Step 2	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol.	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the	

Step	Description	Expected Result	Attachments
	Go to the Contrast page. Set the following protocol parameters: • refocusing control = T1-optimized • min. angle = 60 • mid. angle = 80 • max. angle = 110 • TSE factor = 7 • profile order = low-high • startup echoes = 2 • TE = user defined - (ms) = 40 Click Accept. Open G:\Site\sweep.dat.	attached. file.	 sweep.zip
Step 3	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: • refocusing control = T1-optimized • min. angle = 40 • mid. angle = 80 • max. angle = 110 • TSE factor = 5 • profile order = asymmetric • startup echoes = 0 • TE = user defined - (ms) = 40 • TE spacing = user defined - (ms) = 10 Click Accept. Open G:\Site\sweep.dat.	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the attached. file.	 sweep.zip
Step 4	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: • refocusing control = T1-optimized. • min. angle = 50 • mid. angle = 80 • max. angle = 110 • TSE factor = 6 • profile order = asymmetric • startup echoes = 2 • TE = user defined - (ms) = 30 • TE spacing = user defined - (ms) = 10 Click Accept. Open G:\Site\sweep.dat.	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the attached. file.	 sweep.zip
Step 5	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: • refocusing control = T1-optimized. • min. angle = 70 • mid. angle = 80 • max. angle = 110 • TSE factor = 7 • profile order = asymmetric	Check that a conflict appears on the following parameters: • startup echoes • profile order • TE - (ms) • Refocusing control	None

Step	Description	Expected Result	Attachments
	- startup echoes = 0 - TE = user-defined - (ms) = 10 - TE spacing = 10 ms Click Accept.		
Step 6	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T1-optimized. - min. angle = 50 - mid. angle = 80 - max. angle = 110 - TSE factor = 4 - profile order = low-high - startup echoes = 0 - TE = shortest Click Accept.	Check that a conflict appears on the following parameters: - startup echoes - profile order - Refocusing control	None
Step 7	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T2-optimized - min. angle = 80 - mid. angle = 100 - max. angle = 140 - TSE factor = 25 - profile order = linear - TE = shortest Click Accept. Open G:\Site\sweep.dat.	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the attached. file.	 sweep.zip
Step 8	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T2-optimized - min. angle = 30 - mid. angle = 120 - max. angle = 160 - TSE factor = 16 - profile order = asymmetric - TE = user defined - (ms) = 60 - TE spacing = user defined - (ms) = 12 Click Accept. Open G:\Site\sweep.dat.	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the attached. file.	 sweep.zip
Step 9	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T2-optimized - min. angle = 50 - mid. angle = 120 - max. angle = 160	Verify that refocusing flip angles (second column) match +/- 1.0 with those in the attached. file.	 sweep.zip

Step	Description	Expected Result	Attachments
	- TSE factor = 13 - profile order = asymmetric - TE = user defined - (ms) = 42 - TE spacing = user defined - (ms) = 14 Click Accept. Open G:\Site\sweep.dat.		
Step 10	Close G:\Site\sweep.dat. Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T2-optimized - min. angle = 30 - mid. angle = 120 - max. angle = 160 - TSE factor = 23 - profile order = asymmetric - TE = user defined - (ms) = 16 - TE spacing = user defined - (ms) = 16 Click Accept.	Check that a conflict appears on the following parameters: - startup echoes - profile order - Refocusing control - TE - (ms)	None
Step 11	Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T2-optimized - min. angle = 100 - mid. angle = 80 - max. angle = 160 - TSE factor = 16 - profile order = linear - TE = shortest Click Accept.	Check that a conflict appears on the following parameters: - Refocusing control - min. angle - mid. angle	None
Step 12	Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = T2-optimized - min. angle = 100 - mid. angle = 150 - max. angle = 140 - TSE factor = 16 - profile order = linear - TE = shortest Click Accept.	Check that a conflict appears on the following parameters: - Refocusing control - mid. angle - max. angle	None
Step 13	Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = constant - angle (deg) = 50 - TSE factor = 9 - profile order = linear Open Sequence Development Viewer and click on Parameters tab.	Verify that refocusing flip angles are: 125.0, 50.0, 50.0, 50.0, 50.0, 50.0, 50.0, 50.0, 50.0.	None

Step	Description	Expected Result	Attachments
	Double-click on VAL03_RFE. Open VAL03_RFE_refoc_angles		
Step 14	Double click on the F_basic_TSE protocol. Go to the Contrast page. Set the following protocol parameters: - refocusing control = no - TSE factor = 12 - profile order = linear Open Sequence Development Viewer and click on Parameters tab. Double-click on VAL03_RFE. Open VAL03_RFE_refoc_angles	Verify that refocusing flip angles are: 180.0, 160.0, 160.0, 160.0, 160.0, 160.0, 160.0, 160.0, 160.0, 160.0, 160.0, 160.0.	None

Attachments:



sweep.zip

3.1.36.4. SWCMP.Scan.TSE.UI.tc1.v3

Test ID	32909
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-04-11 09:45:17 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-04-10 15:56:31 Meaning: SME Approved - (Technical) Signed by: Nabuurs, Christine (310147336) on 2017-04-07 15:01:19 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Although this test can be performed on any configuration, the Ingenia 1.5T configuration has been chosen for reproducibility of results.

Requirements Coverage:

Req ID	Requirement Tag	Description			
55699	SWCMP.Scan.TSE.Info	These parameters are available on the INFO page when TSE is enabled.			
		UI name	Units	Visible when	Description

Req ID	Requirement Tag	Description			
		TSE es	(ms)	always	Specifies the echo spacing duration.
		TSE shot	(ms)	always	Specifies the shot duration.
		Full flow comp.	yes	flow compensation is set to yes	Specifies when full flow compensation is used.
55701	SWCMP.Scan.TSE.half_scan.v2	The use of half scan factors from 0.525 - 1.0 in TSE sequences shall be supported.			
55703	SWCMP.Scan.TSE.3D_non-selective	The use of a short RF pulse shape leading to shorter echo spacings shall be supported.			
55705	SWCMP.Scan.TSE.flow_compensation.v2	The TSE sequence shall support flow compensation, which can be selected from the following modes: • no - flow compensation is not enabled. • yes - velocity compensation is performed. • sensitized - increase flow sensitivity in the phase encoding direction. If flow compensation = yes is selected, the direction parameter shall be enabled and can be selected from the following: • in-plane - flow compensation is performed in the measurement direction. • through-plane - flow compensation is performed in the slice selection direction.			

Test steps:

Step	Description	Expected Result
Preparation	Use m_amn_set_test_environment.pl to select an Ingenia 1.5T configuration. In the Service Application, make sure all research options are disabled. Start the Examcards Scan Environment. Create a new current log file.	N.A. (preparation)
Step 1	Drag a MS/2D/M2D protocol Head CNS\TSE\F_MS_dual_HS to the execution list and start the parameter editor	The protocol has no conflicts.
Step 2	Check the INFO page for TSE es and TSE shot.	TSE es and TSE shot appear on the info page.
Step 3	On the Contrast page , check the range of the parameter DRIVE	the range is 'no', yes' (Note: If 'anti' appears in the range, research option 6 is still enabled. Disable the research options, restart the Examcards Scan Environment and retest)

Step	Description	Expected Result
Step 4	check the range of the parameter 'ultrashort'	The range is 'no', 'yes'
Step 5	check the range of the parameter 'fid reduction'	the range is 'default', 'strong'
Step 6	change Scan mode to '3D', check parameter 3D non selective	3D non-selective has range 'no', 'yes'
Step 7	On the motion tab, check the range of the parameter 'Flow compensation'	The range is 'no', 'yes', 'sensitized'
Step 8	Set Flow compensation to yes, and first select direction through-plane, then select direction in-plane.	Full flow comp. appears on INFO page set to yes for both in-plane and through-plane directions.
Step 9	Set Flow compensation to sensitized	Full flow comp. appears on INFO page set to no
Administration	Compress the current log file and attach it as objective evidence.	n/a (administration)

3.1.37. ME Zoom

3.1.37.1. SWCMP.Scan.Zoom.Excitation.tc1.v2

Test ID	52736
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2021-12-16 14:27:05 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:04:15 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:29 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the excitation RF pulses and accompanying gradients for a protocol using Zoom.

Requirements Coverage:

Req ID	Requirement Tag	Description
57974	SWCMP.Scan.Zoom.Excitation	Zoom imaging shall use a non-coplanar excitation pulse with an additional selection gradient in the phase encode direction.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanZoomExcitation in the results file is zero.

3.1.37.2. SWCMP.Scan.Zoom.FOV.tc1

Test ID	25362
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2016-02-05 10:54:33 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2016-02-03 13:52:00 Meaning: SME Approved - (Technical) Signed by: Nabuurs, Christine (310147336) on 2016-02-03 10:46:08 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The purpose of this test is to test the ranges for FOV in phase encoding direction for Zoom imaging. This test is a manual test using the MR application on an RTM. A specific MR configuration is selected for reproducibility reasons.

Requirements Coverage:

Req ID	Requirement Tag	Description
57997	SWCMP.Scan.Zoom.FOV	When Fold-over suppression with Zoom option is selected, the FOV in phase encoding direction shall be restricted to 25 – 120 mm.

Test steps:

Step	Description	Expected Result	Attachments
Step 1	- Run m_amn_set_test_environment.pl and select the 1.5T Ingenia system. - Import the attached examcard - Add the protocol SE_MS_DWI, go to geometry tab - Set the FOV AP (mm) = 230, FOV RL (mm) = 120	Verify that there is no conflicts, the upper allowed value for FOV RL(mm) is 120 and scan runs successfully.	 SE_MS_DWI.zip

Step	Description	Expected Result	Attachments
Step 2	- Add the protocol SE_MS_DWI, go to geometry tab - Set the FOV AP (mm) = 100, FOV RL (mm) = 25, ACQ voxel size AP(mm) as 1.0, ACQ voxel size RL(mm) = 1.0	Verify that there is no conflicts, the allowed value for FOV RL(mm) is 25 - 120 and scan runs successfully.	None

3.1.37.3. SWCMP.Scan.Zoom.REST.tc1.v2

Test ID	52737
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharappa, Deepak Shekhar (320098697) on 2021-12-16 14:27:07 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2021-12-15 15:05:00 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-12-14 15:45:30 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the REST RF pulses and accompanying gradients for a protocol using Zoom

Requirements Coverage:

Req ID	Requirement Tag	Description
57975	SWCMP.Scan.Zoom.REST	Zoom imaging shall use 6 REST slabs, comprised of 6 RF pulses and selection gradients in the phase encode direction, before the imaging sequence for outer volume suppression.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanZoomRest in the results file is zero.

3.1.38. ME QuickSurvey

3.1.38.1. SWCMP.Scan.QuickSurvey.Behavior.CoilSelection.tc1

Test ID	35943
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2018-02-01 16:47:22 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2018-02-01 16:17:32 Meaning: SME Approved - (Technical) Signed by: Arunachalam, Sankaralingam (ing07488) on 2018-02-01 14:23:35 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The test is interactive using application software installation on a system where the scanner hardware is simulated

Requirements Coverage:

Req ID	Requirement Tag	Description
68020	SWCMP.Scan.QuickSurvey.Behavior.CoilSelection	For Quick Survey scans, the preferred coils are automatically selected to skip Coil Survey prescan and Coil Selection job. The selection of coils is based on a set of predefined rules. If a proper selection cannot be made, then automatic coil selection is used as a fallback. The Coil selection for Quick Survey shall work based on the following rules: - Select a preferred clinical mode. The preferred clinical mode is selected based on the following rules: - Dedicated coil is preferred over everything else. - Not generally connected user coils are preferred over generally connected coils. - Generally connected coils are preferred over always connected coils. - Choose only coils that have a head flag if the table is planned at head_position. - Forbid coils that have a head flag if the table is positioned such that the head coil is completely outside the isocenter. - When the preferred clinical mode (selected by rule 1) has only one coil and if that coils participates as part of multiple coil survey clinical modes, then fallback to automatic coil selection.

Req ID	Requirement Tag	Description
		NOTE: This fallback implies that coil survey prescan and coil selection job is required to determine the coils. This rule will ensure that for coils with multiple coil parts (like HST coil of Prodiva configuration), the system will fall back to automatic coil selection. - When the preferred clinical mode (selected by rule 1) has only 2 coils and is made up of Anterior Posterior combination, then the choose the Anterior-Posterior combination as the coils to scan with for the QuickSurvey scan. The rest of the rules don't have to be evaluated if this rule applies. - When the preferred clinical mode (selected by rule 1) has more than one coil and if one of the coils does not have a "Head Reference Position" defined, then fallback to automatic coil selection. NOTE: This fallback implies that that coil survey prescan and coil selection job is required to determine the coils. This condition will ensure that automatic coil selection is chosen in cases where the system cannot decide multiple coils (Head and Posterior coil are selectable). - When the preferred clinical mode (selected by rule 1) has more than one coil, all the coils have a defined "Head Reference Position" and if the "Head Reference Position" of all the coils is not the same, then fallback to automatic coil selection. NOTE: This fallback implies that that coil survey prescan and coil selection job is required to determine the coils. This condition will ensure that automatic coil selection is chosen in cases where the system cannot decide multiple coils. - If rules 2 until 5 does not apply, just choose the coils selected by rule 1 as the coils to scan with for the Quick Survey scan.

Test steps:

Step	Description	Expected Result
Prep 1	Add "MPU_QUICKSURVEY * *" to the bstrace.dat file (G:\site\ in vmware or C:\temp in other configs) Set the system configuration by executing m_amn_set_test_environment.pl. Choose 3T Ingenia configuration and wait for the script to complete. Ensure that only the posterior coil is connected in the coilstate simulator Start the Application Software from MR Boot Configuration Manager. Ensure that the devlogcurrent.log is not open in the LogBrowser and Create a new log file by typing the command "p_logfile_nu createnewlogfile" in the command window. Press 'New Exam' and Prescribe other mandatory parameters	NA (Preparation).
Step 2	Load the protocol Hospital\test\int\survey and rename it to survey1. Check the coils tab for the available coils	The Posterior coil and the Q-Body coil shall be visible in the coils tab
Step 3	Set the Quick survey parameter in the Postproc tab to "yes", accept the changes and start the scan. Once the scan successfully completes, double click the competed protocol, and check the coils tab.	Only the posterior coil shall be visible in the coils tab

Step	Description	Expected Result
	Attach a screenshot as evidence.	
Step 4	Open the devlogcurrent.log file in the LogBrowser and query for " text contains 'm_Name' or text contains 'QuickSurvey' "	The next log line shall be present in the log file: quicksurvey_select_preferred_coils: Rule 1 (selection size = 1), Quick survey coil selection selected
Step 5	Connect the coil "ds Anterior" to the SMB.DCC0 socket in the device simulator. Drag and drop the previous survey protocol, rename it to survey2 and Check the coils tab for the available coils	The coils "ds Anterior", "ds posterior" and "Q-body" shall be visible in the coils tab.
Step 6	Ensure that the Quick survey parameter in the Postproc tab is set to "yes". Select "Repeat prescans" from the "Examination" menu. Accept the changes, start the scan. Once the scan successfully completes, double click the completed protocol, and check the coils tab of the completed protocol. Attach a screenshot as evidence	Only the "ds Anterior", and "ds Posterior" coils shall be displayed in the coils tab.
Step 7	Query for " text contains 'm_Name' or text contains 'QuickSurvey' " in the devlogcurrent.log file using the LogBrowser	The next log line shall be present for survey2 in the devlogcurrent.log file: quicksurvey_select_preferred_coils: Rule 3 (AC + PC combination can be selected with out fallback to ACS), Quick survey coil selection selected
Step 8	Connect the "ds Base" coil to SBM.DCC0 socket in the device simulator and the "ds HeadNeck" coil to NVC_BASE.CONNBASE socket in the device simulator. Drag and drop the previous survey protocol, rename it to survey3 and Check the coils tab for the available coils	The coils "ds Base", "ds HeadNeck", "ds Posterior" and "Q-Body" shall be visible in the coils tab
Step 9	Ensure that the Quick survey parameter in the Postproc tab is set to "yes". Select "Repeat prescans" from the "Examination" menu. Accept the changes, start the scan. Once the scan successfully completes, double click the completed protocol, and check the coils tab of the completed protocol. Attach a screenshot as evidence	Only the "ds Base", and the "ds HeadNeck" coils shall be displayed
Step 10	Query for " text contains 'm_Name' or text contains 'QuickSurvey' " in the devlogcurrent.log file using the LogBrowser.	The next log line shall be present for survey3 in the devlogcurrent.log file: quicksurvey_select_preferred_coils: Rule 1 (selection size = 1), Quick survey coil selection selected
Step 11	Open the System Tuning page, and check the "Optimal Head TableTop pos" in the coil UI for "ds Base" and the "ds HeadNeck" coils	The "Optimal Head TableTop pos" for both the coils shall have the same value
Step 12	Modify the "Optimal Head TableTop pos" for the "ds Base" coil from 192 to 200.	The next logline shall be present for survey4 in the devlogcurrent.log file:

Step	Description	Expected Result
	Drag and drop the previous survey protocol. Rename the protocol to survey4. Ensure that the Quick survey parameter in the Postproc tab is set to "yes". Select "Repeat prescans" from the "Examination" menu. Accept the changes and start the scan. Once the scan completes successfully, query for " text contains 'm_Name' or text contains 'QuickSurvey' " in the devlogcurrent.log file using the LogBrowser.	quicksurvey_select_preferred_coils: Rule 4/5, Fallback to ACS : allCoilsHaveRefPositionDefined = 1, allCoilsHaveSameRefPositionDefined = 0" shall be present for survey4 in the devlogcurrent.log file
Prep 13	Change the system configuration by executing m_amn_set_test_environment.pl and choosing 1.5T Prodiva. Start the Application Software from MR Boot Configuration Manager. Press 'New Exam' and Prescribe other mandatory parameters	NA (Preparation)
Step 14	Load the protocol Hospital\test\int\survey and rename it to survey5. Check the the available coils in the coils tab.	The coils "ds NVS HeadSpine" and "Q-Body" shall be displayed in the coils tab
Step 15	Set the Quick survey parameter in the Postproc tab to "yes". Accept the changes and start the scan. Once the scan completes successfully, query for " text contains 'm_Name' or text contains 'QuickSurvey' " in the devlogcurrent.log file using the LogBrowser.	The next log line shall be present for survey5 in the devlogcurrent.log file: quicksurvey_select_preferred_coils: Rule 2, Fallback to ACS
Step 16	Extract the relevent log statements from the devlogcurrent.log file using the following command in the SDE pc: R: \\MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe "devlogcurrent.log" "text contains 'QuickSurvey' or text contains 'm_Name'" > devlogcurrent_extract_QS.log Attach the extracted devlogcurrent_extract_QS.log file as objective evidence. Revert the changes made to the bstrace.dat file. Restart all the processes again, by typing in the command "Start_stop_procs.pl stop && start_stop_procs.pl start"	NA (Wrap up)

3.1.38.2. SWCMP.Scan.QuickSurvey.Behavior.CoilSelection.tc2

Test ID	37321
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-03-09 11:02:04 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Visser, Kees (nly22601) on 2018-03-09 07:30:27 Meaning: SME Approved - (Business)
	Signed by: Graaf, Ruud de (nly95749) on 2018-03-08 17:10:42 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The test is interactive using application software installation on a system where the scanner hardware is simulated

Requirements Coverage:

Req ID	Requirement Tag	Description
68020	SWCMP.Scan.QuickSurvey.Behavior.CoilSelection	For Quick Survey scans, the preferred coils are automatically selected to skip Coil Survey prescan and Coil Selection job. The selection of coils is based on a set of predefined rules. If a proper selection cannot be made, then automatic coil selection is used as a fallback. The Coil selection for Quick Survey shall work based on the following rules: - Select a preferred clinical mode. The preferred clinical mode is selected based on the following rules: - Dedicated coil is preferred over everything else. - Not generally connected user coils are preferred over generally connected coils. - Generally connected coils are preferred over always connected coils. - Choose only coils that have a head flag if the table is planned at head_position. - Forbid coils that have a head flag if the table is positioned such that the head coil is completely outside the isocenter. - When the preferred clinical mode (selected by rule 1) has only one coil and if that coils participates as part of multiple coil survey clinical modes, then fallback to automatic coil selection. NOTE: This fallback implies that coil survey prescan and coil selection job is required to determine the coils. This rule will ensure that for coils with multiple coil parts (like HST coil of Prodiva configuration), the system will fall back to automatic coil selection. - When the preferred clinical mode (selected by rule 1) has only 2 coils and is made up of Anterior Posterior combination, then the choose the Anterior-Posterior combination as the coils to scan with for the QuickSurvey scan. The rest of the rules don't have to be evaluated if this rule applies. - When the preferred clinical mode (selected by rule 1) has more than one coil and if one of the coils does not have a "Head Reference Position" defined, then fallback to automatic coil selection. NOTE: This fallback implies that that coil survey prescan and coil selection job is required to determine the coils. This condition will ensure that automatic coil selection is chosen in cases where the system cannot decide multiple coils (Head and Posterior coil are selectable).

Req ID	Requirement Tag	Description
		1. When the preferred clinical mode (selected by rule 1) has more than one coil, all the coils have a defined "Head Reference Position" and if the "Head Reference Position" of all the coils is not the same, then fallback to automatic coil selection. NOTE: This fallback implies that that coil survey prescan and coil selection job is required to determine the coils. This condition will ensure that automatic coil selection is chosen in cases where the system cannot decide multiple coils. 1. If rules 2 until 5 does not apply, just choose the coils selected by rule 1 as the coils to scan with for the Quick Survey scan.

Test steps:

Step	Description	Expected Result
Prep 1	Add "MPU_QUICKSURVEY * *" to the bstrace.dat file (G:\site\ in vmware or C:\temp in other configs) Set the system configuration by executing m_amn_set_test_environment.pl. Choose 3T Ingenia configuration and wait for the script to complete. Ensure that only the posterior coil is connected in the coilstate simulator Connect ds_Base and ds_Head Start the graphical patsup simulator. Start the Application Software from MR Boot Configuration Manager. Ensure that the devlogcurrent.log is not open in the LogBrowser and Create a new log file by typing the command "p_logfile_nu createnewlogfile" in the command window. Press 'New Exam' and Prescribe other mandatory parameters Load the protocol Hospital\test\int\survey	NA (Preparation).
Step 2	Use patsup simulator to define a light_visor position on the midle of the table (abdomen) and start scan	NO conflict on protocol scan starts with the chosen protocol.and scan completes
Step 3	inspect the just scanned protocol n the parameter editor the coils tab	Only the posterior coil shall be visible in the coils tab
Step 4	Copy the protocol Define a lighth-visor position on the neck (reference_position in range 1300.0 1500.0) press start_scan	After start_scan a coil_survey is started. Then the coil slection job(s). then the scan starts and completes
Step 5	inspect the just scanned protocol n the parameter editor the coils tab	the tab shows that ds_Head, ds_Base and ds_Posterior are selected.
Step 6	Extract the relevent log statements from the devlogcurrent.log file using the following command in the SDE pc:	NA (Wrap up)

Step	Description	Expected Result
	R: \MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe "devlogcurrent.log" "text contains 'QuickSurvey' or text contains 'm_Name'" > devlogcurrent_extract_QS.log Attach the extracted devlogcurrent_extract_QS.log file as objective evidence.	

3.1.38.3. SWCMP.Scan.QuickSurvey.Behavior.tc1

Test ID	35942
Revision	3
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2018-02-05 14:39:53 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2018-02-05 12:31:23 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2018-02-05 12:13:49 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The test is interactive using application software installation on a system where the scanner hardware is simulated

Requirements Coverage:

Req ID	Requirement Tag	Description
68019	SWCMP.Scan.QuickSurvey.Behavior	As part of conversion to QuickSurvey, the following protocol modification will automatically be performed by the system for a qualified QuickSurvey protocol to skip prep phases and prescans. - Required to Skip Prep Phases for a ScanProtocol qualified as QuickSurvey - Preparation mode will be modified to AUTO. - Required to skip Prescans - To skip Sense Reference prescan,

Req ID	Requirement Tag	Description
		1. a. i. An error occurred while interpreting html data

Test steps:

Step	Description	Expected Result
Prep 1	Set the system configuration by executing m_amn_set_test_environment.pl Choose 3T Ingenia configuration and wait for the script to complete. Start the Application Software from MR Boot Configuration Manager. Press 'New Exam' and Prescribe other mandatory parameters Ensure that the logcurrent.log is not open in the LogBrowser and Create a new log file by typing the command "p_logfile_nu createnewlogfile" in the command window. Ensure that the test examcards are installed. Start the Sequence Development Editor (SDE) from MR Development.	NA (Preparation)
Step 2	Load the protocol Hospital\test\int\survey and rename the protocol to survey1. Check the value of the Quick survey parameter in the Postproc tab.	The Quick survey parameter shall have the value "default"
Step 3	Modify the value of the Quick survey parameter in the Postproc tab to 'No'. Set the values of the following parameters in the UI: Geometry Tab >> Uniformity : "Synergy" Contrast Tab >> RF Shims : "adaptive" Contrast Tab >> Shim : "volume" Postproc Tab >> Preparation phases : "full"	There shall be no conflicts
Step 4	Set the Quick survey parameter in the Postproc tab to "yes" and check for the 4 EX parameters that was changed in the last step: Uniformity, RF Shims, Shim and Preparation phases	The 4 UI parameters Uniformity, RF Shims, Shim and Preparation phases shall not be visible in the UI
Step 5	Open the logcurrent.log file in the LogBrowser and query for QuickSurvey	The log line "QuickSurvey: ScanProtocol successfully converted to QuickSurvey" shall be present in the log file.
Step 6	Set the Quick survey parameter in the Postproc tab to "no" and Check for the 4 EX parameters that was changed: Uniformity, RF Shims, Shim and Preparation phases	The 4 UI parameters Uniformity, RF Shims, Shim and Preparation phases shall be visible in the UI and shall have the values set in the step 3

Step	Description	Expected Result
Step 7	Query for "QuickSurvey" in the logcurrent.log file using the LogBrowser.	The log line "QuickSurvey: ScanProtocol disqualified as Quick Survey since QuickSurvey UI parameter was set to No by user" shall be present in the log file.
Step 8	Set the Quick survey parameter in the Postproc tab to "yes" again, accept the changes and start the scan. In the Sequence Development Editor for the Survey1 scan, check for the following 4 Val01 parameters: VAL01_PROC_uniformity_app VAL01_ACQ_rf_shim_allowed VAL01_AS_shim VAL01_ACQ_preparation_mode	The Val01 parameters shall have the following values : VAL01_PROC_uniformity_app : classic VAL01_ACQ_rf_shim_allowed : no VAL01_AS_shim : no VAL01_ACQ_preparation_mode : auto
Step 9	Continue the scan	The scan shall complete successfully.
Prep10	Connect the "ds Anterior" coil to both SBM.DCC0 and SBM.DCC1 sockets in the device simulator to ensure that the coil selection shall fall back to ACS.	NA (Preparation)
Step 11	Load the protocol Hospital\test\int\survey again and rename it to survey2. Select "Repeat prescans" from the "Examination" menu. Set the Quick survey parameter in the Postproc tab to "No" and ensure that the value of the following parameter in the UI: Geometry Tab >> Uniformity is "CLEAR" Now, set the Quick survey parameter in the Postproc tab to "yes", accept the changes and start the scan. When the sequence development editor opens for the CoilSurveyScan, please continue the scan. When the Sequence Development Editor opens for the Survey2 scan, check for the following 2 Val01 parameters: VAL01_PROC_uniformity_app VAL01_PROC_coil_survey_based_clear Attach a screenshot showing the 2 Val01 parameter values.	The Val01 parameters shall have the following values : VAL01_PROC_uniformity_app: CLEAR VAL01_PROC_coil_survey_based_clear: yes
Step 12	Continue the scan.	The scan shall complete successfully.
Step 13	Extract the relevant log statements from the logcurrent.log file using the following command in the SDE pc: R: \\MRtools\dev_support\tools\LogBrowser\v1.5\p_lgap_logquery.exe "logcurrent.log" "text contains 'QuickSurvey' or text contains 'm_Name'" > logcurrent_extract_QS.log Attach the extracted logcurrent_extract_QS.log file as objective evidence.	NA (WRAP UP)

3.1.39. SC BIPerf

3.1.39.1. SWCMP.Scan.BIPerf.DelayedRecon.tc1

Test ID	32391
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:13 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:12 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:27:58 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the reconstruct command.

Requirements Coverage:

Req ID	Requirement Tag	Description
20433	SWCMP.Scan.BIPerf.DelayedRecon	BIPerformer must support execution of delayed reconstruction job. It must also support aborting the delayed reconstruction job.

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to set any configuration. - create a new log file - Start the batch interpreter and load tc_swcmp.scan.biperf.delayedrecon.tc1.acq from clearcase (mrmethode/testbase/res)	n/a (preparation)
Step 1	Execute the script. Select a raw data file to reconstruct during the reconstruction command. Perform the delayed recon.	Message: "Delayed reconstruction completed successfully." is displayed in the information bar. A dialog box pops up with the message "press stop during next delayed recon".
Step 2	Press proceed on the dialog box. Wait for the next scan to be executed. Select a raw data file to reconstruct for the next reconstruction command. Perform the delayed recon and during its execution press the stop button. Add the log file to this step as objective evidence.	The next scan is executed; upon pressing the stop button the batch interpreter stops. An error is displayed in the infobar "Batch processing terminated by the user".

3.1.39.2. SWCMP.Scan.BIPerf.EchoStabilityScan.tc1

Test ID	32417
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:14 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:13 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:28:32 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the echo stability scan.

Requirements Coverage:

Req ID	Requirement Tag	Description
20441	SWCMP.Scan.BIPerf.EchoStabilityScan	BIPerformer must support execution of PDF routines for initializing or finishing an echo stability scan for SPT.

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to set any configuration. - create a new log file - Start the batch interpreter and load tc_swcmp.scan.biperf.ehostabilityscan.tc1.acq from clearcase (mrmmethods/testbase/res)	n/a (preparation).
Step 1	Execute the script. Add the logfile to this step as objective evidence.	Message "Batch execution completed successfully" is displayed in the info bar.

3.1.39.3. SWCMP.Scan.BIPerf.ExamProperty.tc1

Test ID	32418
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:15 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:14 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:28:57 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the setting the exam properties.

Requirements Coverage:

Req ID	Requirement Tag	Description
20434	SWCMP.Scan.BIPerf.ExamProperty	BIPerformer must support the modification of the following exam properties - Table usage - Patient position - Patient orientation

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to set any configuration. - create a new log file - Start the batch interpreter and load tc_swcmp.scan.biperf.examproperty.tc1. - acq from clearcase (mrmmethods/testbase/res)	n/a (preparation)
Step 1	Execute the script. Attach the generated G:\temp\EX_pars.txt file to this step as objective evidence.	Message: "Batch execution completed successfully" is displayed in the info bar. A file 'EX_pars.txt' is saved. in G:\Temp. In this file, the following values are defined: - EX_GEO_patient_position = "AWGEOM_PATPOS_FEET_FIRST"; - EX_GEO_patient_body_position = "AWGEOM_PATPOS_FEET_FIRST";

3.1.39.4. SWCMP.Scan.BIPerf.ModifyObject.tc1

Test ID	32419
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:16 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:15 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:29:24 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the modify_object command.

Requirements Coverage:

Req ID	Requirement Tag	Description
20438	SWCMP.Scan.BIPerf.ModifyObject	BIPerformer must modify the object values in PDF memory as per the modifications specified in any of these means • saved files • object <name-value> pairs

Test steps:

Step	Description	Expected Result
Prep 1	• Use m_amn_set_test_environment.pl to set any configuration. • create a new log file • Start the batch interpreter and load tc_swcmp.scan.biperf.modifyobject.tc1.acq from clearcase (mrmmethods/testbase/res)	n/a (preparation)
Step 1	Execute the script. Two files are create in the G:\Temp directory, aq_pre.acq and aq_post.acq. Inspect the value for the attribute aq_dur of the object AQ'DC_fixed. Add a screenshot of the comparison of the two values in the two files as objective evidence.	Message: "Batch execution completed successfully" has to be displayed in the info bar. In the file aq_post.acq, the value is 200. In the file aq_pre.acq, the same attribute has a different value.

3.1.39.5. SWCMP.Scan.BIPerf.ModifyParameter.tc1

Test ID	32392
Revision	1
Product Model	Any
Gradient System	Any

Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:18 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:16 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:29:54 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the modify command.

Requirements Coverage:

Req ID	Requirement Tag	Description
20436	SWCMP.Scan.BIPerf.ModifyParameter	BIPerformer must modify the parameter values in PDF memory as per the modifications specified in any of these means • saved files • parameter <name-value> pairs

Test steps:

Step	Description	Expected Result
Prep 1	• Use m_amn_set_test_environment.pl to set any configuration. • Start the batch interpreter and load tc_swcmp.scan.biperf.modifyparameter.tc1.script1.acq from clearcase (mrmethode/testbase/res)	n/a (preparation)
Step 1	Execute the script.	Message: "Batch execution completed successfully" has to be displayed.
Step 2	Load tc_swcmp.scan.biperf.modifyparameter.tc1.script2.acq from clearcase (mrmethode/testbase/res) in the batch interpreter and execute it. Compare the two files in the G:\Temp directory: csc_default.pset and csc_modify_set_post.pset. Add a screenshot of the comparison to this step as objective evidence.	The compared files have the same content.

3.1.39.6. SWCMP.Scan.BIPerf.ModifyTrace.tc1.trace

Test ID	10869
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 15:59:27 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 14:01:24 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 08:11:56 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Verify the functionality of the trace command

Batch File \$ trace enable, mpucoil, debug, ""

\$ wait_proceed

\$ trace disable, mpucoil, debug, ""

Pass/Fail criteria Execution completes successfully and the log file has to contain the lines:

- BI> trace enable, mpucoil,debug,*
- OSTRC(trace item) added: mpucoil debug *
- BI> trace disable, mpucoil,debug,*
- OSTRC(trace item) removed: mpucoil debug *
-

Requirements Coverage:

Req ID	Requirement Tag	Description
20442	SWCMP.Scan.BIPerf.ModifyTrace	BIPerformer must support enable or disable tracing for any specified process, module and stream.

Test steps:

Step	Description	Expected Result
Step 1	Execute the batch file with content as specified in the description. «Proceed» when requested	Message: "Batch execution completed successfully" has to be displayed.

3.1.39.7. SWCMP.Scan.BIPerf.PdfDisableReadClass.tc1

Test ID	32416
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:19 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:17 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:35:59 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the disable_read_class command.

Requirements Coverage:

Req ID	Requirement Tag	Description
20440	SWCMP.Scan.BIPerf.PdfDisableReadClass	BIPerformer must support disabling of "reading the control parameter values from global memory" during validation.

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to set any configuration. - Start the batch interpreter and load tc_swcmp.scan.biperf.pdfdisablereadclass.tc1.acq from clearcase (mrmethode/testbase/res)	n/a (preparation).
Step 1	Execute the script. Verify that the two files in the G:\Temp directory: csc_pars_pre.pset and csc_pars_post.pset have the same content. Add a screenshot of the comparison of the two files to this step as objective evidence.	Message: "Batch execution completed successfully" is displayed in the infobar. The two files csc_pars_pre.pset and csc_pars_post.pset have the same content.

3.1.39.8. SWCMP.Scan.BIPerf.PdfReset.tc1.startup_init.v1

Test ID	45837
Revision	1
Product Model	Any
Gradient System	Any

Options	N.A.
Signatures	Signed by: Krishnakumar, Ashwin (ing02466) on 2020-04-09 09:26:56 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2020-03-30 07:12:28 Meaning: SME Approved - (Business) Signed by: Engelen, Walter van (nly95948) on 2020-03-26 16:24:01 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Verifies the functionality of the startup_init command

Pass/Fail criteria: Verify that the two files in the G:\Temp directory: csc_pars_pre.pset and csc_pars_post.pset has same content.

Requirements Coverage:

Req ID	Requirement Tag	Description
20439	SWCMP.Scan.BIPerf.PdfReset	BIPerformer must support resetting the PDF memory by reading the parameter classes from Global memory

Test steps:

Step	Description	Expected Result
Step 1	Copy the batch file <i>tc_swcmp.scan.biperf.pdfreset.tc1.startup_init.acq</i> from the archive (mrmethods\testbase\res) and execute it in the batchinterpreter.	Message: "Batch execution completed successfully" has to be displayed. The two files in the G:\Temp directory: csc_pars_pre.pset and csc_pars_post.pset have the same content.

3.1.39.9. SWCMP.Scan.BIPerf.PosteriorCoilPositioning.tc1.PromptPCLock (table top movements at the magnet)

Test ID	10853
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 15:59:01 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749)

	on 2015-03-12 13:59:42 Meaning: SME Approved - (Technical)::Bulk Transitions
	Signed by: Dingemans, Henk (nly95779) on 2015-03-12 08:08:43 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Verifies the functionality of the PromptPCLock command with table top movements at the magnet using the tumble switch.

Batch File: \$ PromptPCLock

Pass/Fail criteria :After this batch file execution, log file should contain: "ExamCards(CoilsObserver): Posterior coil cart is in HeldBackAtHome state."

Requirements Coverage:

Req ID	Requirement Tag	Description
20432	SWCMP.Scan.BIPerf.PosteriorCoilPositioning	BIPerformer must support positioning of Posterior Coil in any of these ways. - Lock the Posterior coil at the home position so that it will not participate in the acquisition of data. - Unlock the posterior coil from its holdback position so that it can participate in the acquisition of data During Posterior coil positioning, it must be possible to move the table top either in manual mode or in motorized mode.

Test steps:

Step	Description	Expected Result
Step 1	- Start Examcards application: Needed to verify the posterior coil's state at the end of script execution by referring the Examcards log. - Make sure the posterior coil is Free (It can be achieved by running '\$PromptPCFree' command prior to this batch file execution). - Execute the batch file with content as specified in the description. - Choose 'No' from the prompt to choose to move the table top at the magnet. - When 'Patient Support' message comes, move the table top out to hold back the posterior coil and move it back to scan position. After this click on <proceed>	The posterior coil is successfully moved to Heldback position. Message: "Table top needs to be moved in order to disengage the posterior coil for scanning."

3.1.39.10. SWCMP.Scan.BIPerf.PosteriorCoilPositioning.tc2.PromptPCLock (remote table movements)

Test ID	10854
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 15:59:03 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 13:59:47 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 08:08:51 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Verifies the functionality of the PromptPCLock command with remote table top movements.

Batch File :\$ PromptPCLock

Pass/Fail criteria: After this batch file execution, log file should contain: "ExamCards(CoilsObserver): Posterior coil cart is in HeldBackAtHome state."

Requirements Coverage:

Req ID	Requirement Tag	Description
20432	SWCMP.Scan.BIPerf.PosteriorCoilPositioning	BIPerformer must support positioning of Posterior Coil in any of these ways. - Lock the Posterior coil at the home position so that it will not participate in the acquisition of data. - Unlock the posterior coil from its holdback position so that it can participate in the acquisition of data During Posterior coil positioning, it must be possible to move the table top either in manual mode or in motorized mode.

Test steps:

Step	Description	Expected Result
Step 1	- Start Examcards application: Needed to verify the posterior coil's state at the end of script execution by referring the Examcards log. - Make sure the posterior coil is Free (It can be achieved by running '\$PromptPCFree' command prior to this batch file execution). - Execute the batch file with content as specified in the description. - Choose 'Yes' from the prompt to choose to move the table top remotely.	The table top moves remotely to hold back the posterior coil. The table top will come back to its original position leaving the posterior coil in its hold back position. Message: "Table top needs to be moved in order to disengage the posterior coil for scanning."

3.1.39.11. SWCMP.Scan.BIPerf.PosteriorCoilPositioning.tc3.PromptPCFree (table movements at the magnet)

Test ID	10855
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 15:59:04 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 13:59:55 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 08:09:02 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Verifies the functionality of the PromptPCFree command with table top movements at the magnet using the tumble switch.

Batch File: \$ PromptPCFree

Pass/Fail criteria: After this batch file execution, log file should contain any one of the following lines:

"ExamCards(CoilsObserver): Posterior coil cart is in Travelling state."

"ExamCards(CoilsObserver): Posterior coil cart is in InIsoCenter state."

Requirements Coverage:

Req ID	Requirement Tag	Description
20432	SWCMP.Scan.BIPerf.PosteriorCoilPositioning	BIPerformer must support positioning of Posterior Coil in any of these ways. - Lock the Posterior coil at the home position so that it will not participate in the acquisition of data. - Unlock the posterior coil from its holdback position so that it can participate in the acquisition of data During Posterior coil positioning, it must be possible to move the table top either in manual mode or in motorized mode.

Test steps:

Step	Description	Expected Result
Step 1	- Start Examcards application: Needed to verify the posterior coil's state at the end of script execution by referring the Examcards log. - Make sure the posterior coil is held back (It can be achieved by running '\$PromptPCLock' command prior	The posterior coil is successfully moved to scan position. Following message has to be displayed Table top needs to be moved in order to re-engage the posterior coil for scanning.

Step	Description	Expected Result
	to this batch file execution). - Execute the batch file with content as specified in the description. - Choose 'No' from the prompt to choose to move the table top at the magnet. When 'Patient Support' message comes, move the table top out completely to pick up posterior coil and move it back in to scan position. Click on <proceed>	

3.1.39.12. SWCMP.Scan.BIPerf.PosteriorCoilPositioning.tc4.PromptPCFree (remote table movements).v1

Test ID	32724
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-29 14:08:20 Meaning: QA Approved - (QA) Signed by: Crijns, Sjoerd (310257923) on 2017-03-29 13:42:14 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-03-29 13:37:26 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Verifies the functionality of the PromptPCFree command with remote table top movements.

Requirements Coverage:

Req ID	Requirement Tag	Description
20432	SWCMP.Scan.BIPerf.PosteriorCoilPositioning	BIPerformer must support positioning of Posterior Coil in any of these ways. - Lock the Posterior coil at the home position so that it will not participate in the acquisition of data. - Unlock the posterior coil from its holdback position so that it can participate in the acquisition of data During Posterior coil positioning, it must be possible to move the table top either in manual mode or in motorized mode.

Test steps:

Step	Description	Expected Result
Preparation	- Start Graphical Patsup Simulation. - Ensure the table is located at position (0.00, 0.00) by dragging it with the mouse.	N.A. (preparation only)

Step	Description	Expected Result
	- Start the examcards application. - Create a new logfile.	
Step 1	- Open the Batch Interpreter. - Run the following commands in the batch interpreter: \$ PromptPCLock \$ move 1000 \$ PromptPCFree - When prompted, choose 'Yes' to allow the table top to move remotely.	- The tabletop moves 1000 mm (+/- 6mm) into the bore - Following message is displayed: <i>"Table top needs to be moved in order to re-engage the posterior coil for scanning."</i> - The table top moves remotely to bring the posterior coil to isocenter. - The table top will come back to its original position (Current position = 1,000.00 +/- 6 mm) after taking the posterior coil.
Step 2	Inspect the logfile. Attach the log file to this step as objective evidence.	logcurrent.log contains at least one of: <i>"ExamCards(CoilsObserver): Posterior coil cart is in Travelling state."</i> or <i>"ExamCards(CoilsObserver): Posterior coil cart is in InIsoCenter state."</i> at the relevant times.

3.1.39.13. SWCMP.Scan.BIPerf.ResourceManagement.tc1.Initial resource procurement and release at the end.v2

Test ID	45459
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Krishnakumar, Ashwin (ing02466) on 2020-05-19 05:51:28 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2020-05-18 12:52:17 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2020-03-05 10:55:43 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the claiming of scanner resources prior to the first task that needs scanner resources and release of scanner resources at the end of script execution.

Pass/Fail criteria :Scans completes successfully.

Claiming Resources message has to be displayed at the time of executing "bi_body1" scan.

Resources has to be released at the end of script execution(Look for "[CLAIM_STATUS: EndClaim]" in the log).

Requirements Coverage:

Req ID	Requirement Tag	Description
20443	SWCMP.Scan.BIPerf.ResourceManagement	BIPerformer must coordinate access to scanner resources in cooperation with the system's resource claim arbiter. This includes - Procurement of resources prior to the first task that requires the resources and release of resources at the end of script execution - Failure reporting

Test steps:

Step	Description	Expected Result	Attachments
Step 1	Create a new logfile Start the batch interpreter and scannersim Load and execute the attached file <i>testBIPerf.acq</i> Inspect logfile Attach the logfile	Logfile should contain: Claiming resources... [CLAIM_STATUS: StartClaim] Scan SUCCESSFUL. Scan SUCCESSFUL. [CLAIM_STATUS: EndClaim]	 testBIPerf.zip

3.1.39.14. SWCMP.Scan.BIPerf.ResourceManagement.tc2.Wait for Resource Claim

Test ID	10871
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-04-02 15:46:45 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-04-02 15:34:27 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2015-04-02 12:02:12 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test requires at least one preset procedure in the anatomical region: BODYT- Phantom studies

This test verifies that the Batch Interpreter will wait for the resource claim if the resources are used by other scan applications like Examcards.

Requirements Coverage:

Req ID	Requirement Tag	Description
20443	SWCMP.Scan.BIPerf.ResourceManagement	BIPerformer must coordinate access to scanner resources in cooperation with the system's resource claim arbiter. This includes Procurement of resources prior to the first task that requires the resources and release of resources at the end of script execution

Req ID	Requirement Tag	Description
		Failure reporting

Test steps:

Step	Description	Expected Result
Step 1	- Start a scan in Examcards and keep the scan in a state that it does not proceed (i.e. wait for acknowledgement of table movement). - Start a scan in BI by running the batch file content as specified below: \$ coil_check disable, checkkeepromoverrides \$ define_scan 1,BODYT,"Phantom studies" \$ scan_n "BI scan"	"Claiming Resources" has to be displayed in BI. The BI scan remains waiting to get resource claim.
Step 2	In Examcards, complete the scan.	BI scan has to start automatically once the Examcards scan is completed Message: "Scan SUCCESSFUL" has to be displayed when the BI scan is completed.

3.1.39.15. SWCMP.Scan.BIPerf.ResourceManagement.tc2.Wait for Resource Claim.v1

Test ID	45972
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Krishnakumar, Ashwin (ing02466) on 2020-04-09 09:27:46 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2020-04-08 08:55:19 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2020-04-08 08:32:31 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test requires at least one preset procedure in the anatomical region: BODYT- Phantom studies

This test verifies that the Batch Interpreter will wait for the resource claim if the resources are used by other scan applications like Examcards.

Requirements Coverage:

Req ID	Requirement Tag	Description
20443	SWCMP.Scan.BIPerf.ResourceManagement	BIPerformer must coordinate access to scanner resources in cooperation with the system's resource claim arbiter. This includes Procurement of resources prior to the first task that requires the

Req ID	Requirement Tag	Description
		resources and release of resources at the end of script execution Failure reporting

Test steps:

Step	Description	Expected Result
Step 1	- Start a scan in Examcards and keep the scan in a state that it does not proceed (i.e. wait for acknowledgement of table movement). - Start a scan in BI by running the batch file tc_swcmp.scan.biperf.resourcemangement.tc2.wait_for_resource_claim.acq (from the archive in folder mrmmethods\testbase\res)	"Claiming Resources" has to be displayed in BI. The BI scan remains waiting to get resource claim.
Step 2	In Examcards, complete the scan.	BI scan has to start automatically once the Examcards scan is completed Message: "Scan SUCCESSFUL" has to be displayed when the BI scan is completed.

3.1.39.16. SWCMP.Scan.BIPerf.SaveObject.tc1.save_object

Test ID	10863
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-25 09:43:11 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-25 07:25:43 Meaning: SME Approved - (Business) Signed by: Engelen, Walter van (nly95948) on 2015-03-24 13:55:39 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Verifies the functionality of the save_object command

Batch File: \$ define_scan 1,BODYT,"Phantom studies"
\$ seq_dev_exec_pdf
\$ save_object "RF`*", "G:\Temp\save_all_rf_objects.acq"

Pass/Fail criteria: All RF objects are saved in G:\Temp\save_all_rf_objects.acq

Requirements Coverage:

Req ID	Requirement Tag	Description
20437	SWCMP.Scan.BIPerf.SaveObject	BIPerformer must support saving of the specified PDF objects into a file.

Test steps:

Step	Description	Expected Result
Step 1	Requires at least one preset procedure in the anatomical region: BODYT- Phantom studies Execute the batch file with content as specified in the description.	Message: "Batch execution completed successfully" has to be displayed.

3.1.39.17. SWCMP.Scan.BIPerf.SaveObject.tc2.print_object

Test ID	10864
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 15:59:18 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 14:00:44 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 08:10:44 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Verifies the functionality of the print_object command

Batch File: \$ define_scan 1,BODYT,"Phantom studies"

\$ seq_dev_exec_pdf

\$ print_object "RF`*", "G:\Temp\print_all_rf_objects.acq"

Pass/Fail criteria: All RF objects are saved in G:\Temp\print_all_rf_objects.acq

Requirements Coverage:

Req ID	Requirement Tag	Description
20437	SWCMP.Scan.BIPerf.SaveObject	BIPerformer must support saving of the specified PDF objects into a file.

Test steps:

Step	Description	Expected Result
Step 1	Requires at least one preset procedure in the anatomical region: BODYT- Phantom studies	Message: "Batch execution completed successfully" has to be displayed.

Step	Description	Expected Result
	Execute the batch file with content as specified in the description.	

3.1.39.18. SWCMP.Scan.BIPerf.SaveParameter.tc1.save

Test ID	10860
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-24 12:02:05 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-24 10:52:26 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2015-03-24 10:30:14 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Verifies the functionality of the save command.

Batch File: \$ save CSC_pars, "G:\Temp\CSC_pars.txt"

Pass/Fail criteria :Control Scan parameters are saved into CSC_pars.txt

Requirements Coverage:

Req ID	Requirement Tag	Description
20435	SWCMP.Scan.BIPerf.SaveParameter	BIPerformer must support saving of the specified PDF parameters into a file.

Test steps:

Step	Description	Expected Result
Step 1	Execute the batch file with content as specified in the description.	Message: "Batch execution completed successfully" has to be displayed.

3.1.39.19. SWCMP.Scan.BIPerf.Scan.tc1

Test ID	32316
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:20 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:18 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:36:22 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the execution of the scan_n and seq_dev_scan_n commands.

Requirements Coverage:

Req ID	Requirement Tag	Description
20430	SWCMP.Scan.BIPerf.Scan	BIPerformer must support the execution of a scan in the following means • In normal mode or in sequence development mode. • Must report the acquisition and reconstruction progress • Must provide the Information parameters to be displayed on the Scan Overview Window of Batch Interpreter UI • Must allow aborting of the scan

Test steps:

Step	Description	Expected Result
Prep 1	• Use m_amn_set_test_environment.pl to set any configuration. • Start a new logfile. • Start the batch interpreter and load tc_swcmp.scan.biperf.scan.tc1.script1.acq from clearcase (mrmmethods/testbase/res)	n/a (preparation).
Step 1	Execute the script.	A first scan is executed. Scan parameters are shown during execution. The progress of scan and recon are shown during execution. A message box is displayed with the text "press stop during next scan"
Step 2	Press proceed on the dialog box. Wait for the next scan to be executed and press the stop button during its execution.	The next scan is executed; upon pressing the stop button the batch interpreter stops. An error is displayed in the infobar "Batch processing terminated by the user."
Step 3	Load tc_swcmp.scan.biperf.scan.tc1.script2.acq from clearcase (mrmmethods/testbase/res) Execute the script.	A first scan is executed. Scan parameters are shown during execution. The progress of scan and recon are shown during execution.

Step	Description	Expected Result
		A message box is displayed with the text "press stop during next scan"
Step 4	Press proceed on the dialog box. Wait for the next scan to be executed and press the stop button during its execution. Add the log file to this step as evidence.	The next scan is executed; upon pressing the stop button the batch interpreter stops. An error is displayed in the infobar "Batch processing terminated by the user."

3.1.39.20. SWCMP.Scan.BIPerf.ScanProtocolValidate.tc1

Test ID	32301
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-23 13:23:22 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2017-03-23 08:46:19 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Oorschot, Joep van (310224063) on 2017-03-21 13:36:52 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verify the functionality of define_scan command and define_scan command with modify.

Requirements Coverage:

Req ID	Requirement Tag	Description
20429	SWCMP.Scan.BIPerf.ScanProtocolValidate	BIPerformer must perform the PDF validation of a scan protocol. The validation result must contain the following: - Validation status - On request basis from client: Examination and Information parameters to be displayed on the Parameter Editor of Batch Interpreter UI

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to set any configuration. - Create a new log file. - Start the batch interpreter and load tc_swcmp.scan.biperf.scanprotocolvalidate.tc1.acq	n/a (preparation).

Step	Description	Expected Result
	from clearcase (mrmmethods/testbase/res)	
Step 1	Execute the script. Add the log file to this step as objective evidence.	- Batch execution begins - The first scan is executed - Before execution of the second scan, the user is presented with a parameter editor to make modifications - Upon pressing 'proceed', batch interpreter executes the second scan - Batch execution ends with message "Batch execution completed successfully"

3.1.39.21. SWCMP.Scan.BIPerf.SingleScanID.tc1

Test ID	60888
Revision	1
Product Model	All
Gradient System	
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-01-24 13:16:40 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-01-21 10:56:09 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2025-01-17 15:25:38 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Verifies the functionality of unique scan ID in Batch Interpreter.

Requirements Coverage:

Req ID	Requirement Tag	Description
96104	SWCMP.Scan.BIPerf.SingleScan ID	BIPerformer must support the execution of scans, with each scan having a unique scan ID.

Test steps:

Step	Description	Expected Result
Prep 1	Execute the command tf and choose, system configuration->set test environment for methods->select, any system configuration.	n/a (preparation).
Step 1	Start the batch interpreter and load spectro.acq from the path(G:\testframework\tst\teima\res\batchfiles).	- Batch execution should complete successfully. - Verify the single scan id's with log lines : "AdaptCurrentProtocol() : singleScanID".

Step	Description	Expected Result
		- The values from above traces should be unique and not be repeated. - For eg. if unique scan id is 1234. Expected log lines : "AdaptCurrentProtocol() : singleScanID=1234" - The duplication of scan id is not allowed. i.e.repetations are not expected as like below. "AdaptCurrentProtocol() : singleScanID=1234" "AdaptCurrentProtocol() : singleScanID=1234"

3.1.39.22. SWCMP.Scan.BIPerf.TabletopPositioning.tc1

Test ID	32709
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2017-03-29 16:07:10 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2017-03-29 15:21:14 Meaning: SME Approved - (Technical) Signed by: Hey, Silke (300170272) on 2017-03-28 17:27:26 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Verifies the functionality of the tabletop move functionality

Requirements Coverage:

Req ID	Requirement Tag	Description
20431	SWCMP.Scan.BIPerf.TabletopPositioning	BIPerformer must support the positioning of table top in motorized mode. It must also support aborting the table top movement.

Test steps:

Step	Description	Expected Result
Prep 1	- Use m_amn_set_test_environment.pl to set any configuration. - Start a new logfile. - Start the patsup simulator - Start the batch interpreter and load SWCMP.Scan.BIPerf.TabletopPositioning.tc1.acq script from clearcase (mrmmethods/testbase/res) - Execute the script. - Follow the instruction in the dialog box: - mark current table position with the light	n/a (preparation).

Step	Description	Expected Result
	visor - o push the table in at least 10 cm • Make note of the current table position in the patsup simulator. • Press process on the dialog box.	
Step 1	Another dialog box appears with the message "Tabletop moved 10 cm IN, CHECK". Make note of the current table position in the patsup simulator.	The difference between current and previous position shows that the table moved by 10 cm +/- 1mm inward.
Step 2	Press proceed on the dialog box. The table moves in the patsup simulator and another dialog box appears with the message "Tabletop moved 20 cm OUT, CHECK". Wait for the next scan to be executed and press the stop button during its execution.	The difference between current and previous position shows that the table moved by 20 cm +/- 1mm outward.
Step 3	Press proceed on the dialog box. A dialog box appears with the message "Press <cancel> during next table move". Press proceed on the dialog box. Following the instruction, press cancel during the table move. Attach the log file to this step as objective evidence.	Batch interpreter stops with the message "Table movement stopped by the operator.".

3.1.40. SC DataMon

3.1.40.1. Test Suites

The datamonitoring deliverables: data monitoring distributor and data monitoring saver are tested to verify that the requirements are implemented properly. Test clients are created to perform certain actions and check the expected results if possible. In case that the expected results are not met or failed to complete the actions, the test client will exit with an exit code other than 0.

3.1.40.2. Automated Regression Tests

3.1.40.2.1 SWCMP.Scan.DataMon.tc_autoall

Test ID	9875
Revision	2
Product Model	Any
Gradient System	Any
Options	

Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-17 15:52:17 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-17 09:03:20 Meaning: SME Approved - (Business) Signed by: Engelen, Walter van (nly95948) on 2015-03-16 15:56:58 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Performs automated regression test by running the script listed below.

The script runs on SDE PC and builds all executables needed for testing, it first creates a target environment under the C:\nmr directory on the test machine and copies necessary data files to the target environment and produces log files in the C:\nmr\site\log directory.

Afterwards it starts datamon distributor and datamon saver executables. It then starts 'Frame Streaming test' and 'Frame Filtering test' by starting the test clients and check the exit code afterwards. After those two test finishes, it starts 'Frame Saving test' by starting the test client and check the expected results and exit code of the test client. The 'Frame Saving test' is only considered successful if the expected results are met and the exit code is 0. At last it starts the 'Performance Test' and check the expected result. The 'Performance Test' is only considered successful if the expected results are met.

At the end of the script, a message is produced to summarize the results of the test run.

- **Successful Run**

If all test suites completed successfully, the following message is shown in the command window:

All datamon monitoring tests were successful

- **Not all Tests Succeeded**

If one or more tests were not successful, the following message appears:

Some datamon monitoring tests failed

This message is followed by a list of the executables that had errors.

Environment: SDE

Test Script: do_test_datamonmonitoring.pl

Requirements Coverage:

Req ID	Requirement Tag	Description
20968	SWCMP.Scan.DataMon.Multiple Clients	Multiple consumers (viewers) may be connected to the data monitoring distributor at the same time.
20970	SWCMP.Scan.DataMon.FrameFiltering	Consumers of data monitoring get all clusters from all generators by default. A specific client can modify the list of clusters of which it wants to receive data frames.
20969	SWCMP.Scan.DataMon.FrameSaving	The frame can also be stored in persistent storage (directory specified under logical name GYRO_EXPORT) on the request of the frame generator.

Req ID	Requirement Tag	Description
		The filename in which frames are stored is based on the frame cluster name. Subsequent frames from the same cluster are appended to this file unless a new frame file is opened for this cluster. A new frame file is opened on the request of the frame generator.
20971	SWCMP.Scan.DataMon.FrameStreaming	DataMonitoring supports streaming in the sense that the objects generated by the frame generator can be serialized into a data stream and de-serialized from it according to the interface specification.
20972	SWCMP.Scan.DataMon.Performance	The latency of received frames, size 50 Kbytes, produced at 10 frames per second must not exceed 50 milliseconds with 10 consumers, where producer and consumers reside on the same host.

Test steps:

Step	Description	Expected Result
Step 1	Run the script: do_test_datamonitoring.pl to performed automatically the test suites listed below: SWCMP.Scan.DataMon.FrameStreaming.tc1 SWCMP.Scan.DataMon.FrameFiltering.tc1 SWCMP.Scan.DataMon.FrameSaving.tc1 SWCMP.Scan.DataMon.tc1	a. Script produces this statement in command window: All datamonitoring tests were successful

3.1.41. SC ExamState**3.1.41.1. Test Suites****3.1.41.1.1 SWCMP.ExamState.General.MultipleFeeders.tc1**

Test ID	8069
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 16:01:35 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 14:53:44 Meaning: SME Approved - (Technical)::Bulk Transitions

	Signed by: Dingemans, Henk (nly95779) on 2015-03-12 13:46:36 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Verify that examinationstate is configured to support multiple feeders.

Environment: SDE

Test steps:

Step	Description	Expected Result
Step 1	Open examinationstate\res\examinationstatefeed.svcf and check the value of MultiClient.	MultiClient is set to 'true'

3.1.41.2. Automated Regression Tests**3.1.41.2.1 SWCMP.ExamState.tc_auto**

Test ID	13583
Revision	3
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-06-19 10:10:39 Meaning: QA Approved - (QA)
	Signed by: Engelen, Walter van (nly95948) on 2019-06-19 08:48:20 Meaning: SME Approved - (Business)
	Signed by: Hegde, Shashank (310195826) on 2019-06-19 08:10:26 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Performs automated regression test by running the script listed below.

The script runs on SDE PC and builds all executables needed for testing, it creates a target environment under the C:\nmr directory on the test machine and copies necessary data files to the target environment and produces log files in the C:\nmr\site\log directory.

At the end of the script, a message is produced to summarize the results of the test run.

If all test suites completed successfully, the following message is shown in the command window:

All Examinationstate tests were completed successful

If a test was not successful, the test script stops immediately with error message.

Environment: SDE

Test Script: do_test_examinationstate.pl

Requirements Coverage:

Req ID	Requirement Tag	Description
19067	SWCMP.ExamState.General.RecoveryFromIllegibility	When persistent data cannot be interpreted at start-up, the examinationstate must be initialized with default initial values. This may occur in case of file corruption or when a new version of the software is installed.
19074	SWCMP.ExamState.Interaction.FullUpdate	It must be possible to update the full examinationstate; for example initially to supply default values or persistent values.
19075	SWCMP.ExamState.Interaction.FullQuery	It must be possible to query the full examinationstate.
19076	SWCMP.ExamState.Interaction.FullNotification	It must be possible to subscribe to changes to any property of the examinationstate.
19078	SWCMP.ExamState.Interaction.ClassQuery	It must be possible to query a specific class of the examinationstate.
19077	SWCMP.ExamState.Interaction.ClassNotification	It must be possible to subscribe to changes in a specific class of examinationstate.
19080	SWCMP.ExamState.Content.Examdata	Examinationstate must contain data related to the current acquisition examination.
19081	SWCMP.ExamState.Content.ExamEnvironment	Examinationstate must contain data related to the environment of the current acquisition examination.
19082	SWCMP.ExamState.Content.PatientData	Examinationstate must contain data related to the current acquisition patient.
28197	SWCMP.ExamState.Content.PatientPhysio	Examinationstate must contain physio data related to the current acquisition patient.
19083	SWCMP.ExamState.Content.PatientPositioning	Examinationstate must contain data related to handling the position of the current acquisition patient relative to the system.
19084	SWCMP.ExamState.Content.PatientCoordinates	Examinationstate must contain data related to the positional frame of reference of the patient relative to the MR system.
19085	SWCMP.ExamState.Content.PatientPrompting	Examinationstate must contain data related to notifications to the patient for specific events and conditions occurring in the course of the current acquisition examination. The patientprompting substate contains data related to autovoice notifications and state updates occurring during the course of the current acquisition examination. Updates, queries and subscriptions on this class must be supported.
19086	SWCMP.ExamState.Content.NetworkEnvironment	Examinationstate must contain data related to the network environment of the MR system.
57725	SWCMP.ExamState.Content.ScanSetList	Examinationstate must contain data related to list of scanset in current acquisition exam.

Req ID	Requirement Tag	Description
		ScanSetList must also provide information related to the number of scansets in the list. Each ScanSetData in this list must contain following information- 1. Id A unique guid identifier for the scanset 2. Duration Aggregated duration (in seconds) for all execution steps inside a scanset. 3. Progress Aggregated acquisition progress of the scanset based on the duration and acquisition progress of each executionstep in the scanset. 4. HasBreathHoldPeriod Indicates if the scanset contains any execution step with breathhold period 5. HasTableMotion Indicates if the scanset has planned tabletop movement at any point during acquisition. 6. CurrentState The current state of scanset which can be any one of the follwing - 6.1 Ready : default state of scanset 6.2 Prepare : after dispatch of scanset or during preperation (prescans) phase 6.3 Scanning : actual acquisition phase 6.4 Finished : completion of acquisition or abort/failure of scanset 7. HasPreScansOnly Indicates if the scanset contains only prescans. 8. HasInteractiveScan Indicates if the scanset contains one or more interactive scans
57729	SWCMP.ExamState.Content.CurrentScanSet	Examinationstate must contain data related to the current scanset being executed. CurrentScanSet shall be of type ScanSetData and must contain - 1. Id A unique guid identifier for the scanset 2. Duration Aggregated duration (in seconds) for all execution steps inside a scanset. 3. Progress Aggregated acquisition progress of the scanset based on the duration and acquisition progress of each executionstep in the scanset. 4. HasBreathHoldPeriod Indicates if the scanset contains any execution step with breathhold period 5. HasTableMotion Indicates if the scanset has planned tabletop movement at any point during acquisition. 6. CurrentState The current state of scanset which can be any one of the follwing - 6.1 Ready : default state of scanset 6.2 Prepare : after dispatch of scanset or during preperation (prescans) phase 6.3 Scanning : actual acquisition phase 6.4 Finished : completion of acquisition or abort/failure of scanset 7. HasPreScansOnly Indicates if the scanset contains only prescans. 8. HasInteractiveScan Indicates if the scanset contains one or more interactive scans
28198	SWCMP.ExamState.Content.UidGen	Examinationstate must support the generation of a globally unique ID, formatted according to MR rules for use in initialization of a new frame of reference.

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Prep 1	Ensure the following folder structure is available in SDE-PC system. C:\testbin	The mentioned folder structure is available in SDE-PC.
Step 1	Run the script: do_test_examinationstate.pl to performed automatically the test suites listed below: • - o SWCMP.ExamState.General.tc1 - o SWCMP.ExamState.Interaction.ClassLibrary.tc1 - o SWCMP.ExamState.tc1 - o SWCMP.ExamState.Content.UidGen.tc1	a. Script produces this statement in command window: All examinationstate tests were successful

3.1.42. SC ResClaim

3.1.42.1. Test Suites

3.1.42.2. Automated Regression Tests

3.1.42.2.1 SWCMP.Scan.ResClaim.tc_autoall

Test ID	8576
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 16:02:25 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 14:55:01 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 14:02:06 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Performs automated regression test by running the script listed below.

The script runs on SDE PC and builds all executables needed for testing, it creates a target environment under the C:\nmr directory on the test machine and copies necessary data files to the target environment and produces log files in the C:\nmr\site\log directory.

At the end of the script, a message is produced to summarize the results of the test run.

Successful Run

If all test suites completed successfully, the following message is shown in the command window:

All resourceclaim tests were successful

Not all Tests Succeeded

If one or more tests were not successful, the following message appears:

Some resourceclaim tests failed

This message is followed by a list of the executables that had errors.

Environment: SDE

Test Script: do_test_resourceclaimarbitration.pl

Requirements Coverage:

Req ID	Requirement Tag	Description
20159	SWCMP.Scan.ResClaim.ExclusiveAccess	The Resource Claim Arbitration Service must grant access to only one client at a time. Clients must not access the resources until the claim has been granted and must de-access the resources before releasing the resource claim
20160	SWCMP.Scan.ResClaim.ClaimBehavior	In a resource claim request, the client must characterize its behavior as either polite or rude. The behavior property is passed on to the current resource claim holder, who decides how to react to it. Scanengine for example will not yield resources in the middle of a running examination to a <i>polite</i> request. It would do so to a <i>rude</i> request.
20161	SWCMP.Scan.ResClaim.ClaimHandlingOrder	<i>Rude</i> claim requests are granted before <i>polite</i> requests, but a granted <i>polite</i> request must not preempted by a <i>rude</i> request. Within a Behavior class, claim requests are handled in the order of appearance.
20162	SWCMP.Scan.ResClaim.EarlyWithdrawal	A client that has requested access rights may release its claim or disconnect from the server <i>before</i> it has been granted access. The net effect of such early withdrawal must be the same as if the client had never issued the claim.
20163	SWCMP.Scan.ResClaim.RunWithoutArbitration	For testing purposes it must be possible to run scan applications even if the <i>Resource Claim Arbitration Service</i> is not running. Evidently there will be no arbitration in such a situation. The supplied proxies must give the application the impression that scanning with full access rights is granted.
20164	SWCMP.Scan.ResClaim.NotifyNewClaim	Once a client has been granted access rights, it must be informed when another client is also requesting access rights. The current claim holder has the liberty to decide how to act on such notification, but must not hold the resource claim for longer than

Req ID	Requirement Tag	Description
		necessary.
20165	SWCMP.Scan.ResClaim.LogClaimHolder	To facilitate fault analysis it must be possible to identify the current owner of the resources by log file inspection.

Test steps:

Step	Description	Expected Result
Step 1	Run the script: do_test_resourceclaimarbitration.pl to performed automatically the test suites listed below: • SWCMP.Scan.ResClaim.tc1.Asynchronous claim • SWCMP.Scan.ResClaim.tc2.Claim behavior • SWCMP.Scan.ResClaim.tc3.Synchronous claim	Script produces this statement in command window: All resourceclaim tests were successful

3.1.43. SC ScanAcqState

3.1.43.1. Test Suites

3.1.43.1.1 SWCMP.Scan.ScanAcqState.General.MultipleFeeders.tc1

Test ID	17764
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 16:03:16 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 15:07:41 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 14:55:50 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Verify that scanacquisitionstate is configured to support multiple feeders.

Environment: SDE

Test steps:

Step	Description	Expected Result
Step 1	Open scanacquisitionstate\res\scanacquisitionstatefeed.svcf and check the value of MultiClient.	MultiClient is set to 'true'

3.1.43.2. Automated Regression Tests

3.1.43.2.1 SWCMP.Scan.ScanAcqState.tc_auto

Test ID	13610
Revision	1
Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Winter, Thijs (ing02948) on 2015-03-12 16:03:24 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Dingemans, Henk (nly95779) on 2015-03-12 15:07:51 Meaning: SME Approved - (Business)::Bulk Transitions Signed by: Graaf, Ruud de (nly95749) on 2015-03-12 14:56:03 Meaning: SME Approved - (Technical)::Bulk Transitions
Approval Status	Approved

Description:

Performs automated regression test by running the script listed below.

The script runs on SDE PC and builds all the executables needed for testing, it creates a target environment under the C:\nmr directory on the test machine and copies necessary data files to the target environment and produces log files in the C:\nmr\site\log directory.

At the end of the script, a message is produced to summarize the results of the test run.

If all test suites completed successfully, the following message is shown in the command window:

All scanacquisitionstate tests were completed successfully

If a test was not successful, the test script stops immediately with error message.

Environment: SDE

Test Script: do_test_scanacquisitionstate.pl

Requirements Coverage:

Req ID	Requirement Tag	Description
23622	SWCMP.Scan.ScanAcqState.Int	It must be possible to update the full SAS; for example initially to supply

Req ID	Requirement Tag	Description
	eration.FullUpdate	default values
23623	SWCMP.Scan.ScanAcqState.Interaction.FullQuery	It must be possible to query a complete state of the SAS.
23624	SWCMP.Scan.ScanAcqState.Interaction.FullNotification	It must be possible to subscribe to changes to any property of the SAS.
23625	SWCMP.Scan.ScanAcqState.Interaction.ClassQuery	It must be possible to query a specific class of the SAS.
23626	SWCMP.Scan.ScanAcqState.Interaction.ClassNotification	It must be possible to subscribe to changes in a specific class of SAS.
23627	SWCMP.Scan.ScanAcqState.Content.ScanData	SAS must contain data related to the current acquisition scan.
23628	SWCMP.Scan.ScanAcqState.Content.ScanProgress	SAS must contain data related to the progress of the current acquisition scan.
23629	SWCMP.Scan.ScanAcqState.Content.ScanState	SAS must contain data related to the execution phase of current scan, which can be used by UI process to show/display proper controls for interacting with the scan
23630	SWCMP.Scan.ScanAcqState.Content.ScanProperties	SAS must contain data related to the properties of currently running scan
23631	SWCMP.Scan.ScanAcqState.Content.SarData	SAS must contain data related to the properties of SAR of currently running scan
23632	SWCMP.Scan.ScanAcqState.Content.PnsData	SAS must contain data related to the properties of PNS of currently running scan
57712	SWCMP.Scan.ScanAcqState.Content.BreathHoldStateData	ScanAcquisitionState should provide information of the current state of breath hold sequence for the scans with breath hold period. The current state of breathhold can be any one of the following - 1. Idle : default state of breath hold. 2. Prepare : State that represents preparation phase for breath hold. 3. Hold : State that represents Hold of breath. 4. Recover : State that represents recovery of breath after the hold.
57714	SWCMP.Scan.ScanAcqState.Content.BreathHoldDurationData	SAS must contain data related to the duration of the breathhold for current scan. BreathHoldDurationData must contain the duration of the hold in seconds.
57723	SWCMP.Scan.ScanAcqState.Content.TableStateData	SAS must contain data related to the tabletop movement state. The current state of tabletop can be any one of the following - 1. Stationary : The default state of Table when it is not moving. 2. AboutToMove : When table is planned for movement during execution of current scan 3. Moving : When the table is moving

Test steps:

Step	Description	Expected Result
Step 1	Run the script: do_test_scanacquisitionstate.pl to performed automatically the test suites listed below: o SWCMP.Scan.ScanAcqState.Interaction.ClassLibrary.tc1	Script produces this statement in command window: All scanacquisitionstate tests were completed successful

Step	Description	Expected Result
	o SWCMP.Scan.ScanAcqState.tc1	

3.1.44. ScanEngine

3.1.44.1. SWCMP.Scan.ScanEngine.MultiSeries.tc1

Test ID	61853
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Reddy, Srikanth (310031195) on 2025-05-22 20:08:32 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-05-22 16:45:47 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2025-05-21 18:40:14 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test is executed using the application software.

Requirements Coverage:

Req ID	Requirement Tag	Description
97860	SWCMP.Scan.ScanEngine.Multi Series	The Scanengine shall retrieve multiple series created by ScanValidator for a scan and send it to recon job associated with the scan as part of the ReconDescription. The ScanEngine shall notify the series ids of all the multiple series received from recon subsystem to the userworkflow subsystem once the reconstruction is complete.

Test steps:

Step	Description	Expected Result
Prep 1	Preparation 1. Set test environment a. Open TF (test framework) => select "Tooling: System Configuration" => select "Set Test environment for Methods" => Select Ingenia 3T (WA30)	N.A. (preparation)

Step	Description	Expected Result
Prep 2	Preparation 1. Start "Application Software in the MR Boot Configuration Manager 2. Create 'New Patient' and enter mandatory patient details. 3. Select examcard Hospital => ScReq Advanced => 303 ASL => pCASL. 4. Plan the scan, open "Advanced Parameters" window a. Under "Geometry" tab => set "slices" to "10" b. Under "Postproc" tab => set "ASL source images" to "ALL" and "MIP/MPR" to "MIP".	No conflicts are observed
Step 1	Press 'Start'	1. The main scan is completed. 2. The first inline step is completed for ASL 3. The second inline step is completed for MIP.

3.1.44.2. SWCMP.Scan.ScanEngine.TablePositioning.tc1.v1

Test ID	37716
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-05-03 17:35:28 Meaning: QA Approved - (QA) Signed by: Verburg, Frank (nly95979) on 2018-05-03 16:18:43 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2018-05-02 09:16:17 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test is executed using the application software on VMWARE.

Requirements Coverage:

Req ID	Requirement Tag	Description
67790	SWCMP.Scan.ScanEngine.Table Positioning	Prior to performing (or resuming) a scan, scanengine must position the tabletop on the pragmatic tabletop position assigned to that scan. or station within that scan.

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	1. Add " TablePositionsOptimizer * * " to Bstrace.dat in G:\Site on a VMWARE machine 2. Set environment to Ingenia WA15 using tf 3. Start the scanner software	N.A. (preparation)	None

Step	Description	Expected Result	Attachments
	4. Create a new examination 5. Create a new logfile 6. Load scan protocol Hospital\Head CNS\FE\2D1 7. Rename this scan to 'off000' 8. Make sure that the scan is ready to run 9. Start the scan		
Step 1	Copy the protocol and change on the offcenter to -800.0 and rename protocol in off800.and start the scan	Table is moved after confirmation Scan completes	None
Step 2	Copy the protocol and connect the dS base using the device connect simulator and start the scan.	Scan completes	None
Step 4	Extract the log-file using the attached query query.txt in a cmd window using R: \MRTTools\dev_support\tools\LogBrowser\v1 .5\p_lgap_logquery.exe devlogcurrent.log @query.txt > devlogcurrent_extract.log Attach the extracted logfile.	Verify: that the pragmatic position of the first scan is on 1000.0 and that the following scans on 1800.0	 query.zip

3.1.45. ScanValidator

3.1.45.1. SWCMP.Scan.ScanValidator.TablePositioning.tc1.v1

Test ID	37885
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2018-05-18 08:58:26 Meaning: QA Approved - (QA) Signed by: Rozijn, Tom (nly95848) on 2018-05-17 14:29:12 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2018-05-17 13:27:07 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test is executed using the application software on VMWARE.

Requirements Coverage:

Req ID	Requirement Tag	Description
67783	SWCMP.Scan.Validator.TablePositioning	The ScanValidator(pdf) shall provide as output of the validation: - the optimal table position - the minimal table position - the maximal table position - indicate it is a "FastNextScan" .

Test steps:

Step	Description	Expected Result	Attachments
Prep 1	1. use: TF in cmd window to set the configuration for methods on Ingenia 1.5T 2. Add " TablePositionsOptimizer * * " to Bstrace.dat in G:\Site on a VMWARE machine 3. verify with the coil simulator that only the body coils and posterior coil are connected. Disconnect all other coils 4. Create a <i>NewExamination</i> 5. Create new logfile 6. Load scan protocol Hospital\Head CNS\FE\F_2D1 7. Rename this <i>Scan</i> to 'off000' 8. Make sure that the scan is ready to run 9. Start the scan	N.A. (preparation)	None
Step 1	Copy the protocol and change on the offcenter to -800.0 and rename protocol in off800.and start the scan	Table is moved after confirmation Scan completes	None
Step 2	Copy the protocol and connect the dS base using the device connect simulator and start the scan..	Scan completes	None
Step 4	Use logbrowser with attached filter in query.txt Do the verification in filtered view. Export the filtered view (File-Export view) to a file. Attach the file as objective evidence.	Verify that for the first completed scan the table range is 950, 1050 mm with a marging of 5mm Verify that for the second completed scan that the range is 1750, 1850 mm with a marging of 5mm Verify that for the third completed scan that the range is 1770, 1830 mm with a marging of 5mm	 query.zip

3.1.45.2. SWCMP.Scan.Validator.1DTracking.Par.tc1

Test ID	56182
Revision	1
Product Model	All
Gradient System	

Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-01-16 11:37:54 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2023-01-12 14:38:30 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-01-11 11:48:46 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies the availability of Scan Validator 1D.Tracking parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
97595	SWCMP.Scan.Validator.1DTracking.Par.v2	Name	Range	Visible when	Description	System/Option restrictions
		1D tracking	yes, no	Scan mode technique = FFE and Nucleus = 1H	This parameter enables or disables the 1D tracking feature.	1D tracking is visible if M_OPT.Opt1DTracking is available.
		dephase factor	0-6	1D tracking = yes	Sets a predefined strength for the dephasing gradient	N/A
		monitoring	yes, no	1D tracking = yes	Enables data monitoring window for 1d tracking data	N/A

Req ID	Requirement Tag	Description				
		dephase direction	If slice orientation n = transverse then A, P, F, H If slice orientation n = sagittal then R, L, F, H If slice orientation n = coronal then R, L, A, P	1D tracking = yes	Sets the direction of the dephase gradient	N/A
		skip 1D FFT	yes, no	1D tracking = yes	Enables the skipping of the Fourier Transformation for 1D tracking in reconstruction.	N/A
Multiple stacks in orthogonal orientations shall be supported with 1D tracking.						

Test steps:

Step	Description	Expected Result
Step 1	Execute SWCMP.Scan.PDFTTestAutomation and inspect the results file	The reported number of failed tests related to SWCMPScanValidator1DTrackingPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.3. SWCMP.Scan.Validator.3DVIEW.Par.tc1

Test ID	48604
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-11-06 07:46:56 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-11-05 13:00:13 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-11-05 12:52:07 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of 3DView parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
81720	SWCMP.Scan.Validator.3DVIEW.Par	3D VIEW	Brain T1, Brain T2, Brain FLAIR, Brain DIR, no	Scan mode = 3D Û Fast imaging mode = TSE	Defines optimal internal settings for applications in combination with contrast types.	This parameter is only enabled when the SW option 3D Brain VIEW (Bit 14) is enabled.
		Brain BB,no	The range is extended with <i>Brain BB</i> when	Prostate T2 RT,no	The range is extended with <i>Prostate</i>	Spine T1, Spine T2, no

Req ID	Requirement Tag	Description				
			option 3D Black Blood (Bit 111) is enabled.		T2 RT when option 3D RT VIEW (Bit 160) is enabled.	
		The range is extended with the specified values when option 3D Spine VIEW (Bit 6) is enabled.	Nerve SPAIR, Nerve STIR, no	The range is extended with the specified values when option 3D Nerve VIEW (Bit 108) is enabled.	MSK T1, MSK T1 FS, MSK T2, MSK T2 FS, MSK PD, MSK PD FS, no	The range is extended with the specified values when option 3D MSK VIEW (Bit 9) is enabled.
		Female Pelvis T1, General Pelvis T1, Female Pelvis T2, General Pelvis T2, Prostate T2, General Pelvis T2 RT, no	The range is extended with the specified values when option 3D Body VIEW (Bit 10) is enabled.	Breast T2, no	The range is extended with the specified values when option 3D Breast VIEW (Bit 54) is enabled.	[brain6] [brain7] [brain8] [spine 3] [spine 4] [spine 5] [spine 6] [msk 7] [msk 8] [msk 9] [msk 10] [msk 11] [msk 12] [msk 13] [msk 14] [msk 15] [msk 16] [msk 17] [msk 18] [msk 19] [msk 20] [msk 21] [msk 22] [msk 23] [msk 24] [msk 25]

Req ID	Requirement Tag	Description				
						[msk 26] [msk 27] [msk 28] [msk 29] [msk 30] [msk 31] [msk 32] [body 8] [breast 1] [breast 3] [breast 4] [breast 5] [breast 6] [breast 7] no
		The range is extended with the specified values when Option 006 TSE Extensions (Bit 505) is enabled.				
84836	SWCMP.Scan.Validator.PartialAcquisition.Par.v2	Name	Range	Visible when	Description	System / Option restrictions
		Halfscan	No, Yes (EPI, TFE-EPI or GRASE) ÷ (M2D, MS or (3D ÷ YZ encoding): default	Scan type = Imaging	Enables scan acceleration through partial acquisition of k-space	N/A

Req ID	Requirement Tag	Description				
		factor	0.6 - 1.0	Scan mode = 2D or M2D or MS ÷ Halfscan = Yes	Specifies the Halfscan factor	N/A
		factor y	0.6 - 1.0	Scan mode = 3D ÷ Halfscan = Yes	Specifies the halfscan factor in the y direction	N/A
		factor z	0.6 - 1.0	Scan mode = 3D ÷ NSA = 2.0 ÷ Halfscan = Yes	Specifies the halfscan factor in the z direction	N/A
		NSA	0.0 - 32.0 Fractional values are available for: ((cartesian ÷ 3D) or radial or MotionFree ÷ (NSA <= 5) Otherwise integer values	Always	Enables the acquisition of the averages. The scan time changes proportionally with the specified averages.	N/A
		SMART	no, yes	Acquisition mode = cartesian ÷ (NSA >= 2)	SMART averaging is a repeated single measurement scan	N/A

Req ID	Requirement Tag	Description					
						and is therefore also referred to as serial averaging: an initial measurement is made for all values of the phase encoding before the second measurement is commenced. A complete image is thus acquired before proceeding to the next averaging step. Since the time between the acquisition of the data is in the order of TR (e.g. 0.5 s - 2 s) with parallel averaging and in the order	

Req ID	Requirement Tag	Description				
					of minutes with SMART averaging, the latter produces a more pronounced decrease in the intensity of artifact.	
		reversed order	no, yes	Fast Imaging mode = EPI or TFE-EPI ÷ Echoes = 1 ÷ Halfscan = Yes	Enables a reversed order k-space acquisition.	This parameter is visible when M_OPT.A WOPT_OPTION15 (option bit 496) is enabled.
		Partial NSA	no, yes	Acquisition mode = Cartesian ÷ (Scan mode == 3D) ÷ (3D free factor == yes) ÷ (NSA == 2)	Enables the acquisition of the 2nd NSA partially (central k-values only).	N/A
		fraction	1.2 - 2.0	Partial NSA = yes	Specifies the partial NSA fraction.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidator3DViewPar in the results file

Step	Description	Expected Result
		is zero.

3.1.45.4. SWCMP.Scan.Validator.Acceleration.Par.tc1

Test ID	51052
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-15 11:55:56 Meaning: QA Approved - (QA) Signed by: Beck, Gabrielle (nly95705) on 2021-06-10 15:14:32 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:22 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator acceleration parameters.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorAccelerationPar in the results file is zero.

3.1.45.5. SWCMP.Scan.Validator.AIF.Par.tc1

Test ID	61775
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-05-23 16:59:38 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-05-

	23 16:56:28 Meaning: SME Approved - (Business)
	Signed by: Bajaj, Naveen (310105618) on 2025-05-22 19:00:42 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the availability of the arterial input function (AIF) volume.

Requirements Coverage:

Req ID	Requirement Tag	Description																								
97882	SWCMP.Scan.Validator.AIF.Par	For a scan with Dual-Acquisition Perfusion, the arterial input function (AIF) shall be acquired. The AIF volume consists only of a single slice. The AIF volume has the same characteristics as the first perfusion stack: • field of view • fold-over suppression • orientation • preparation direction • slice thickness The user is only able to modify off-center and angulation of the AIF. The Offc/Ang tab shall show the following parameters: <table><tr><th>Name</th><th>Range</th><th>Visible when</th><th>Description</th></tr><tr><td>AIF Offc. AP (P= +mm)</td><td>-200 to 200</td><td>Quantitative Perfusion = Yes</td><td>N.A.</td></tr><tr><td>RL (L= +mm)</td><td>-220 to 220</td><td>Quantitative Perfusion = Yes</td><td>N.A.</td></tr><tr><td>FH (H= +mm)</td><td>-2000 to 2000</td><td>Quantitative Perfusion = Yes</td><td>N.A.</td></tr><tr><td>AIF Ang. AP (deg)</td><td>-180 to 180</td><td>Quantitative Perfusion = Yes</td><td>N.A.</td></tr><tr><td>RL (deg)</td><td>-180 to 180</td><td>Quantitative Perfusion = Yes</td><td>N.A.</td></tr></table>	Name	Range	Visible when	Description	AIF Offc. AP (P= +mm)	-200 to 200	Quantitative Perfusion = Yes	N.A.	RL (L= +mm)	-220 to 220	Quantitative Perfusion = Yes	N.A.	FH (H= +mm)	-2000 to 2000	Quantitative Perfusion = Yes	N.A.	AIF Ang. AP (deg)	-180 to 180	Quantitative Perfusion = Yes	N.A.	RL (deg)	-180 to 180	Quantitative Perfusion = Yes	N.A.
Name	Range	Visible when	Description																							
AIF Offc. AP (P= +mm)	-200 to 200	Quantitative Perfusion = Yes	N.A.																							
RL (L= +mm)	-220 to 220	Quantitative Perfusion = Yes	N.A.																							
FH (H= +mm)	-2000 to 2000	Quantitative Perfusion = Yes	N.A.																							
AIF Ang. AP (deg)	-180 to 180	Quantitative Perfusion = Yes	N.A.																							
RL (deg)	-180 to 180	Quantitative Perfusion = Yes	N.A.																							

Req ID	Requirement Tag	Description			
		FH (deg)	-180 to 180	Quantitative Perfusion = Yes	N.A.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorAIFPar in the results file is zero.

3.1.45.6. SWCMP.Scan.Validator.Angio.Par.tc1

Test ID	55995
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-07-11 14:53:13 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2023-06-15 15:42:08 Meaning: SME Approved - (Business) Signed by: Engelen, Walter van (nly95948) on 2023-06-15 09:43:43 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test specifies availability of angiography parameters and conflicts.

Requirements Coverage:

Req ID	Requirement Tag	Description
91210	SWSUB.sSTD.Scan.Motion.Angio.REACT	REACT is a combination of a: - T2prep module, implemented with 4 refocusing pulses and echo time = 50ms, followed by a - TFE prepulse, implemented as an inversion pulse with an inversion delay set to shortest, and a - mDIXON TFE readout. In case the REACT feature is enabled, the T2prep and TFEPP-modules will automatically be enabled, and their parameters fixed. The parameters will not be visible in the scan protocol editor. In case REACT is enabled, the default number of startup echoes will

Req ID	Requirement Tag	Description				
		remain the same when enabling cardiac prospective triggering. This deviates from the standard behaviour, the startup cycles for a non-REACT scan protocol is set to 0 when cardiac prospective triggering is enabled. To prevent swaps due to low SNR in case of long second echo times (TE2) (e.g. when combining with ComforTone at 1.5T), an adjustment is made to the Dixon model to create additional Dixon no-fly zones for REACT at 1.5T. This is accomplished by lowering the threshold for max T2-star deviation for REACT at 1.5T. In the Dixon model an infinite T2-star is assumed, the T2-star deviation reflects the difference from an infinite T2-star that is allowed before the voxel size needs to be increased to reduce the T2-star relaxation (decrease TE2).				
91206	SWCMP.Scan.Validator.Angio.Par	Name	Range	Visible when	Description	System/Option restrictions
		Angio / Contrast enhanced	No, inflow, phase contrast, contrast enhanced, relaxation enhanced.	Scan mode technique = FFE, and Nucleus = 1H.	This parameter enables or disables various angiography techniques.	'Relaxation enhanced' is visible if M_OPT.REACT is available.
		Quantitative flow	No, yes.	Scan mode technique = FFE, and Nucleus = 1H.	Yes: quantitative flow technique is enabled, information is obtained from the phase difference if flowing spins.	Quantitative flow is visible if M_OPT.Q flow is available.
		PC flow directions	RL, AP, FH, RL-AP, RL-FH, AP-FH, RL-AP-FH.	Angio / Contrast enhanced = phase contrast,	This parameter specifies the direction(	N/A.

Req ID	Requirement Tag	Description					
				Or Quantitative flow = yes	s) in which flow will be encoded for a phase contrast or quantitative flow acquisition.		
		Uniform velocity	No, yes.	PC flow direction = RL-AP, RL-FH, AP-FH, or RL-AP-FH.	This parameter enables or disables uniform velocity for flow measurements in the selected direction(s).	N/A.	
		PC velocity (cm/s)	1-1000.	Uniform velocity = yes	This parameter specifies the velocity in cm/s at which maximum phase difference will occur in a phase contrast angio or quantitative flow acquisition.	N/A.	

Req ID	Requirement Tag	Description				
		RL velocity (cm/s)	1-1000.	Uniform velocity = no, and PC flow direction = AP-RL, RL-FH, AP-RL-FH.	This parameter specifies the velocity in RL-direction in cm/s at which maximum phase difference will occur in a phase contrast angio or quantitative flow acquisition.	N/A.
		AP velocity (cm/s)	1-1000.	Uniform velocity = no, and PC flow direction = AP-RL, AP-FH, AP-RL-FH.	This parameter specifies the velocity in AP-direction in cm/s at which maximum phase difference will occur in a phase contrast angio or quantitative flow acquisition.	N/A.
		FH velocity (cm/s)	1-1000.	Uniform velocity = no, and PC flow direction = AP-	This parameter specifies the velocity in FH-	N/A.

Req ID	Requirement Tag	Description					
				FH, RL-FH, AP-RL-FH.	direction in cm/s at which maximum phase difference will occur in a phase contrast angio or quantitati ve flow acquisitio n.		
		The following parameter restrictions are in place for angio/contrast enh. = relaxation enhanced: • Scan mode = 3D • Technique = FFE • Acquisition mode = cartesian • Fast Imaging = TFE - o shot mode = multishot • TFE factor - o startup echoes = default - o shot interval = user defined ▪ (ms) : can only be set to values > 2000ms, default = 2000ms. ▪ (beats) : can only be set to values > 2, default = 2 beats. - o profile order = low_high - o turbo direction = radial • echoes = 2 - o partial echo = no - o shifted echo = no • mDIXON = yes - o mode = multi-echo • Custom prepulse = no					

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorAngioPar in the results file is zero.

3.1.45.7. SWCMP.Scan.Validator.ASL.Par.tc1

Test ID	48536
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-11-06 07:46:59 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-11-05 12:58:48 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-11-05 12:52:08 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of ASL parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
87632	SWCMP.Scan.Validator.ASL.Par.v1	Parameter name	Range	Visible when	Description	System / Option restrictions
		Arterial Spin Labeling	no FAIR STAR pCASL Time-SLIP STAR Angio	Always	Enables user to select one of the Arterial Spin Labeling techniques.	Application specific restrictions are imposed by the following options: M_OPT.A WOPT_AR TERIAL_SP IN_LABEL (option bit 35) enables STAR ASL technique

Req ID	Requirement Tag	Description				
						M_OPT.A WOPT_PC ASL (option bit 78) enables pCASL ASL technique M_OPT.A WOPT_OP TION10 (option bit 501) enables FAIR and Time- SLIP ASL technique s M_OPT.A WOPT_CI NEMA (option bit 109) enables STAR Angio ASL technique .
		label type	If pCASL or STAR Angio - parallel slab - free slab If Time- SLIP - parallel slab - free slab	Arterial Spin Labeling = pCASL, STAR Angio or Time-SLIP	The label type defines the volume used to label.	Label type "free slab" will be available with pCASL only under the availability of the option: M_OPT.A WOPT_OP TION10 (option bit

Req ID	Requirement Tag	Description				
			• pencil beam			501)
		Label Offc. AP(P = +mm)	-220 - 220	label type = free slab or pencil beam	Determines the position of the label volume as the distance to a central plane.	N.A
		Label Offc. RL(L = +mm)	-220 - 220	label type = free slab or pencil beam	Determines the position of the label volume as the distance to a central plane.	N.A
		Label Offc. FH(H = +mm)	-220 - 220	label type = free slab or pencil beam	Determines the position of the label volume as the distance to a central plane.	N.A
		Label Ang. AP(deg)	-180 - 180	label type = free slab or pencil beam	Determines the angulation of the current volume around the AP axis	N.A

Req ID	Requirement Tag	Description				
		Label Ang. RL(deg)	-180 - 180	label type = free slab or pencil beam	Determines the angulation of the current volume around the AP axis	N.A
		Label Ang. FH(deg)	-180 - 180	label type = free slab or pencil beam	Determines the angulation of the current volume around the FH axis	N.A
		orientation	transverse sagittal coronal	label type = free slab	Determines the orientation of the labeling slab.	N.A.
		diameter (mm)	5 - 300	label type = pencil beam	Controls the diameter of the labeling beam.	N.A.
		beam direction	AP RL FH	label type = pencil beam	Direction of the long axis of the pencil beam	N.A.
		label thickness (mm)	1 - 300	Arterial Spin Labeling = STAR , STAR Angio,	Thickness of the labeling slab.	N.A.

Req ID	Requirement Tag	Description				
				FAIR or Time-SLIP		
		label gap (mm)	0 - 200	Arterial Spin Labeling = STAR, STAR Angio or Time-SLIP ^ label type = parallel slab	Gap between the imaging slab and the labeling slab.	N.A.
		label distance (mm)	min value - 300	Arterial Spin Labeling = pCASL	Distance between Label and Imaging volume. The min. value is calculated from the image volume size to prevent saturation of the first slices.	N.A.
		post label delay (ms)	The actual range depends on other parameters.	Arterial Spin Labeling = pCASL	Time between the end of the continuous labeling and the beginning of the readout.	N.A.
		label delay (ms)	The actual range depends on other	Arterial Spin Labeling = STAR,	Time between the single labeling	N.A.

Req ID	Requirement Tag	Description				
			parameters	STAR Angio or Time-SLIP	pulse and the beginning of the readout.	
		label location	F, H, R, L, A or P depending on the slice orientation (Transverse, Sagittal or Coronal)	Arterial Spin Labeling = pCASL or Time-SLIP Label type = parallel slab	Determines the position of the label with respect to the imaging stack.	R,L, A,P label locations and Sagittal and Coronal slice orientations will require M_OPT.A WOPT_OPTION10 (option bit 501)
		label duration (ms)	0 - 5000	Arterial Spin Labeling = pCASL	Duration of the labeling RF pulse.	The parameter will be available when M_OPT.A WOPT_OPTION10 (option bit 501)
		vascular crushing	no, yes	Arterial Spin Labeling = pCASL	Reduce the vascular artifacts by dephasing the signal from label present at the time of imaging.	The parameter will be available when M_OPT.A WOPT_OPTION10 (option bit 501)
		backgr.	no, user	Arterial	Suppress	For

Req ID	Requirement Tag	Description					
		suppressi on	defined, auto 2 pulse, auto 4 pulse	Spin Labeling = pCASL	signals from backgroun d tissue.	selection of backgroun d suppressi on other than "auto 2 pulse", software option M_OPT.A WOPT_OP TION10 (option bit 501) is required.	
		backgr. suppr. pulses (ms)	0 - 10000	backgr. suppressi on = user defined	Timing of excitation of backgroun d pulses.	N.A.	
		normalize d	no, yes	Arterial Spin Labeling = pCASL	Enables normaliza tion of the perfusion images with a proton density image.	The paramete r will be available when M_OPT.A WOPT_PC ASL_NOR MALIZED (option bit 117) is enabled for Scan mode = 3D For Scan mode =2d or MS additonall y software option M_OPT.A WOPT_OP	

Req ID	Requirement Tag	Description				
						TION10 (option bit 501) is required.
		phases	1-50	All Arterial Spin Labeling techniques	Number of phases acquired within the cycle duration.	If ASL = pCASL then the parameter will be available only when M_OPT.A WOPT_OPTION10 (option bit 501) is enabled.
		duration (ms)	0-10000	phases > 1	Cycle duration: Repetition time between consecutive labels.	N.A.
		phase interval	shortest user defined longest	phases > 1	Time between consecutive phases.	N.A
		(ms)	The actual range depends on other parameters	phase interval = user defined	User defined choice of the phase interval.	N.A
		pulse type	wow_ir hypsec	Arterial Spin Labeling = FAIR	Type of the inversion pulse.	N.A
		adjust	no, yes	Arterial	This	N.A

Req ID	Requirement Tag	Description				
		FAIR ref		Spin Labeling = FAIR ^ Dynamic study = individual	parameter defines whether the dynamic scans are performed in combination with adjusted widths of the inverted regions in the reference images for FAIR ASL.	
		adjust value	0.1 - 100	adjust FAIR ref = yes	It is used to decrease the inversion width of the reference images as a function of the dynamic scans.	N.A
		mask	no, yes	Arterial Spin Labeling = Time-SLIP	Enables acquisition of mask image.	N.A
		inversion scheme	nonselect - select - nonselect selective only	Arterial Spin Labeling = Time-SLIP	Define the order of the two inversion pulses.	N.A
		backgr.	longest	Inversion	Delay	N.A

Req ID	Requirement Tag	Description					
		suppr. delay (ms)	user defined	scheme = select - nonselect	between the second inversion pulse and image acquisition.		
		In addition software option M_OPT.AWOPT_OPTION10 shall allow the following combinations: 1. Scan technique != FFE EPI ^ ASL = pCASL 2. Scan mode = 2D ^ ASL = pCASL 3. Scan technique != GRASE ^ scan mode = 3D ^ ASL = pCASL 4. Fat suppression = SPAIR ^ ASL = pCASL 5. Scan technique = SE EPI ^ ASL = STAR 6. Scan technique = FFE TFE ^ ASL = STAR 7. Scan technique = FFE TFE ^ ASL = STAR Angio					

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorAslPar in the results file is zero.

3.1.45.8. SWCMP.Scan.Validator.B0FieldMap.Par.tc1

Test ID	56535
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-02-13 11:46:36 Meaning: QA Approved - (QA) Signed by: Bhosale, Manoj (310194149) on 2023-02-09 09:33:20 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-02-08 13:04:17 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of B0FieldMap parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
91689	SWCMP.Scan.Validator.B0FieldMap.Par					
		B0 field map	no, yes, calibrate	Scan type = Imaging and Nucleus = 1H	Switch to enable the acquisition of a B0 map or B0 Calibration scan.	The range 'yes' is only enabled when ServiceLevel_2 option or M_OPT.B0_Mapping is present and in case of Acquisition mode = Radial when also option M_OPT.RadialImagingExtended is present. The range 'calibrate' is only enabled when ServiceLevel_2 option or M_OPT.B0_Extended is present and in case of Acquisition

Req ID	Requirement Tag	Description				
						n mode = Radial when also option M_OPT.Ra dialImagin gExtended is present. B0 field map becomes invisible when only value remains in range.
		field map technique	multi- acquisitio n	B0 field map = yes or calibrate	Selection of the used technique to acquire a B0 map	N.A
		delta TE (ms)	0.1-10	field map technique =multi- acquisitio n	Echo spacing	N.A
98683	SWCMP.Scan.Validator.B0Field Map.Par.v1	Name	Range	Visible when	Descriptio n	System / Option restriction s
		B0 field map	no, yes, calibrate	Scan type = Imaging and Nucleus = 1H	Switch to enable the acquisitio n of a B0 map or B0 Calibratio	The range 'yes' is only enabled when ServiceLev el_2 (AWOPT_

Req ID	Requirement Tag	Description				
					n scan.	SS_LEVEL_2) option or M_OPT.BO_Mapping (AWOPT_BO_MAPPING) is present and in case of Acquisition mode = Radial when also option M_OPT.RadialImagingExtended (AWOPT_OPTION33) is present. The range 'calibrate' is only enabled when ServiceLevel_2 (AWOPT_SS_LEVEL_2) option and in case of Acquisition mode = Radial when also option M_OPT.RadialImagingExtended (AWOPT_OPTION33) is present.

Req ID	Requirement Tag	Description				
						B0 field map becomes invisible when only value remains in range.
		field map technique	multi-acquisition	B0 field map = yes or calibrate	Selection of the used technique to acquire a B0 map	N.A
		delta TE (ms)	0.1-10	field map technique =multi-acquisition	Echo spacing	N.A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorB0FieldMapPar in the results file is zero.

3.1.45.9. SWCMP.Scan.Validator.B1FieldMap.Par.tc1

Test ID	48537
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2023-02-22 13:58:01 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Shree K, Shruthi (Partner) (320185675) on 2023-02-22 13:11:15 Meaning: SME Approved - (Technical)

	Signed by: Graaf, Ruud de (nly95749) on 2023-02-22 12:43:17 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:
This test specifies availability of Scan Validator B1Shim parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
91387	SWCMP.Scan.Validator.B1Field Map.Par.V2	Name	Range	Visible when	Descriptio n	System / Option restriction s
		B1 field map	no, yes, calibrate	Imaging	Switch to enable the acquisitio n of a B1 map or B1 Calibratio n scan.	The range 'yes' is only enabled when ServiceLev el_2 option or M_OPT.A WOPT_OP TION47(B 1Map) option bit 464) is present. The range 'calibrate' is only enabled when hardware option MULTITRA NSMIT_2 is present.
		field map technique	Dual TR, Dual Angle, DREAM	B1 field map = Yes	Selection of the used technique to acquire a B1	N.A

Req ID	Requirement Tag	Description				
					map	
		selective	no, yes	field map technique =Dual TR or Dual Angle or DREAM	Switch to enable selective excitation	N.A
		TR extension	0.0-10000.0	field map technique = Dual TR or DREAM	TR Extension of the second acquired scan segment.	N.A
		grad spoil factor	0.0-1000.0	field map technique = Dual TR	Factor with which the spoiler gradient between the scan segments is increased.	N.A
		saturation delay	0.0-10000.0	field map technique = DUAL ANGLE	Duration of the saturation delay after the saturation pulse	N.A
		RF pulse shape	SG_150_100_167, SPREDREX	field map technique = DUAL ANGLE ^ Cardiac synchronization ! = no	RF pulse choice	This parameter is only visible when M_OPT.A WOPT_OPTION47 (option bit- 464 B1Map) is

Req ID	Requirement Tag	Description				
						present
		STE first	no, yes	field map technique = DREAM	Determines what signal is acquired as first echo or second echo. STE first 'yes': the STE is acquired as the first echo, the FID as the second echo. STE first 'no': the FID is acquired as the first echo, the STE as the second echo.	N.A
		STEAM flip angle	1.0-180.0	field map technique = DREAM	The flip angle of the STEAM prepulse which determines the B1 encoding.	N.A
		RF shims	fixed, adaptive, volume, smart	MultiTransmit Systems	RF shim setting selection Fixed: The drive scales and fhsim values stored in	The range 'smart' is only available when option M_OPT.A WOPT_S MART_BR

Req ID	Requirement Tag	Description					
					the coil attributes are used. Adaptive: RF shimming is performed based on an automatically selected shim volume. Volume: RF shimming is performed based on the user-defined shim volume. Smart: RF shimming is performed based on a shim volume which was derived using a 'smart' image segmentation.	EAST (option bit 25-SmartBreast) is present.	
		When M_OPT.AWOPT_OPTION47 is enabled, the following combination shall be allowed					
		1. RF shim != fixed for Spectro1H					
		2. Volume RF shim with non-cardiac and non-breast (no breast coil					

Req ID	Requirement Tag	Description
		connected)

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorB1FieldMapPar in the results file is zero.

3.1.45.10. SWCMP.Scan.Validator.Backfolding.Compensation.Par.tc1

Test ID	51049
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-15 11:56:27 Meaning: QA Approved - (QA) Signed by: Beck, Gabrielle (nly95705) on 2021-06-10 15:29:22 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:24 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator backfolding compensation parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
84806	SWCMP.Scan.Validator.Backfolding.Compensation.Par	Name	Range	Visible when	Description	System / Option restrictions
		Fold-over suppression	no, oversampling, rest, zoom	Oversampling is visible when Acquisition mode =	Foldover suppression techniques.	N.A.

Req ID	Requirement Tag	Description					
				cartesian, spiral, radial or MotionFree Zoom is visible when Acquisition mode = cartesian ÷ (TSE or GraSE) ÷ !MS Rest is visible when Acquisition mode = cartesian			
		in-plane (mm)	0.0 - 1200	Fold-over suppression = oversampling ÷ Acquisition mode = spiral or radial	This parameter specifies the oversample area for the scan in all in-plane directions. The entire oversample area is visible as a box surrounding the imaging FOV in PlanScan. It cannot be modified	N.A.	

Req ID	Requirement Tag	Description				
					within PlanScan.	
		* (mm)	0.0 - 1200.0	Fold-over suppression = oversampling $\dot{U}$ Acquisition mode = (cartesian or MotionFree)	This parameter specifies the oversampling area in the positive phase direction if fold-over direction = AP then replace * by P if fold-over direction = RL then replace * by L if fold-over direction = FH then replace * by H	N.A.
		* (mm)	0.0 - 1200.0	Fold-over suppression = oversampling $\dot{U}$ Acquisition mode = (cartesian or MotionFree)	This parameter specifies the oversampling area in the negative phase direction if fold-	N.A.

Req ID	Requirement Tag	Description					
				e)	over direction = AP then replace * by A if fold- over direction = RL then replace * by R if fold- over direction = FH then replace * by F		
		* (mm)	0.0 - 1200.0	Fold-over suppressi on = oversampl ing ù Acquisitio n mode = MotionFre e	This paramete r specifies the oversampl ing area in the positive measure ment direction If slice orientatio n = transverse : if fold- over direction = AP then replace * by L if fold- over	N.A.	

Req ID	Requirement Tag	Description				
					direction = RL then replace * by P If slice orientatio n = coronal: if fold-over direction = RL then replace * by H if fold-over direction = FH then replace * by L If slice orientatio n = sagittal: if fold-over direction = AP then replace * by H if fold-over direction = FH then replace * by P	
		* (mm)	0.0 - 1200.0	Fold-over suppressi on = oversampl	This paramete r specifies the	N.A.

Req ID	Requirement Tag	Description					
					ing $\dot{U}$ Acquisitio n mode = MotionFre e	oversampl ing area in the negative (n) Measure ment(M) direction if slice orientatio n = transverse : if fold- over direction = AP then replace * by R if fold- over direction = RL then replace * by A If slice orientatio n = coronal: if fold- over direction = RL then replace * by F if fold- over direction = FH then replace * by R If slice	

Req ID	Requirement Tag	Description				
					orientation = sagittal: if fold-over direction = AP then replace * by F if fold-over direction = FH then replace * by A	
Application specific restrictions are imposed by the following options: - M_OPT.Zoom (option bit 106) enables Scan mode = MS in combination with Zoom - M_OPT.Cardiac_Zoom (option bit 148) enables scan mode != MS in combination with Zoom						

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorBackfoldingCompensationPar in the results file is zero.

3.1.45.11. SWCMP.Scan.Validator.BBPulse.Par.tc1

Test ID	57108
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-07-11 14:43:07 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2023-06-21 08:30:08 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749)

	on 2023-06-21 07:44:32 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Black Blood (BB) parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
92365	SWCMP.Scan.Validator.BBPulse.Par	BB pulse	Always: No, MSDE If technique = SE: yes (DIR)	technique = SE or IR	Enables the use of Black blood prepulse.	yes(DIR) is available when M_OPT. BlackBlood d is enabled. MSDE is available when the M_OPT.M SDE is enabled. If scan mode = MS: BB pulse is available when M_OPT.BI ackBloodE xtended is enabled. If scan mode = 2D, M2D and 3D: BB pulse is available when M_OPT.BI ackBlood is

Req ID	Requirement Tag	Description				
						enabled.
		Slab margin (%)	[10, 100]	BB pulse = yes (DIR) and scan mode = MS	Percentage of the volume thickness that is added	N/A
		delay	default, user defined	BB pulse = yes (DIR)	Specifies the inversion time for black blood imaging. This is the time between the double inversion pulse and the start of the sequence.	N/A
		(ms)	[1, 1500]	Delay = user defined	Defines length of user defined delay.	N/A
		STIR	no, yes	BB pulse = yes (DIR)	Enables the short time inversion recovery (STIR) pulse in combination with black blood imaging.	N/A
		delay (ms)	[1, 1500]	STIR = yes	Delay of	N/A

Req ID	Requirement Tag	Description				
					STIR pulse	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorBBPulsePar in the results file is zero.

3.1.45.12. SWCMP.Scan.Validator.CalculatedImages.Par.tc1

Test ID	56536
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-02-13 11:43:34 Meaning: QA Approved - (QA) Signed by: Bhosale, Manoj (310194149) on 2023-02-09 09:34:31 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-02-08 12:54:02 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of the Calculated images parameters and conflicts.

Requirements Coverage:

Requirement Coverage		Description				
Req ID	Requirement Tag	Name	Range	Visible when	Description	System/Option restriction
98144	SWCMP.Scan.Validator.CalculatedImages.Par.v2					
		Calculated Images	if (B1 field map = no, B0 field map = no or SENC = no): (	Scan technique = MIX, SE or FFE and SWIp = no	Defines which images must be reconstructed	The CR range value requires software option

Req ID	Requirement Tag	Description
		RHO, SUM, R2*, CR, if (T1 Mapping = cardiac native or cardiac enhanced or cardiac user defined or MSK): T1, If (Scan technique = FFE): T2* If (Scan technique = SE, IR or MIX): T2) if (B1 field map = yes or calibrate): B1, if (B0 field map = yes or calibrate): B0, if (SENC = yes): SENC Strain SENC Anatomy and MDME = no and APT = no and ((mDixon = no or (mDixon = yes or quant and mDIXON images = Source)) or respirator y compensa tion = retrospec tive) * Note: When respirator y compensa tion = retrospec tive, calculated images and DIXON images can both be shown CardiacTri ggeredIR (AWOPT_ CARDIAC_ TRIGGERE D_IR) The T2, T2* and R2* range values require software option T2starR2s tarMappin g (AWOPT_ T2STAR_R 2STAR_M APPING)

Req ID	Requirement Tag	Description				
			no if (respiratory compensation = retrospective): no, mid-P, mid-V			
		T2 clip value (ms)	0.0-2047.0	scan technique = FFE	Defines the T2 clip value.	N/A
		R2 clip value (1/s)	1.0-10000.0	Calculated Images = R2	Defines R2 clip value.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorCalculatedImagesPar in the results file is zero.

3.1.45.13. SWCMP.Scan.Validator.Cardiac.Synchronization.Par.tc1

Test ID	48539
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharappa, Deepak Shekhar (320098697) on 2020-11-06 07:47:04 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-10-30 07:32:18 Meaning: SME Approved - (Technical)

	Signed by: Engelen, Walter van (nly95948) on 2020-10-28 15:01:52 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Cardiac Synchronization parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
97407	SWCMP.Scan.Validator.Cardiac.Synchronization.Par.v4	Name	Range	Visible when	Description	System / Option restrictions
		Cardiac synchronization	no, trigger if (! Spectro): gate, retrospective	Always	Enables use of cardiac synchronization methods	The range retrospective requires software option Retrospective (AWOPT_RETRO)
		device	ECG, PPU, External If !ASL: Internal	Cardiac synchronization != no	Specifies which physiological signal will be used to synchronize the scan with the heart phase	The range ECG requires AWHWOP_T_ECG The range PPU requires AWHWOP_T_PPU The range External requires AWASW_PIT_WIRELESS_BTU & software option TriggerExtended (AWOPT_

Req ID	Requirement Tag	Description				
						OPTION31)
		R-R window (%)	0-100	Cardiac synchronization != no	Defines estimated variations in the RR interval during the scan	N.A
		Number of heart phases	single phase, user defined, maximum	Cardiac synchronization = trigger or retrospective	This parameter defines number of heart phases	N.A
		Phase interval	shortest, user defined, longest	(Cardiac synchronization = trigger → Number of heart phases = maximum) or (Cardiac synchronization = gate → heart phases >1)	Cardiac triggering: Interval between successive phases in multiphase scan. Cardiac gating: Time between successive gates in multiphase scan	N.A
		(ms)	x-y The actual values depend on other parameters	Phase interval=user defined	The user defined phase interval in ms	N.A
		Cycle duration (ms)	0 -10000	device = Internal	Defines the cycle duration between the	N.A

Req ID	Requirement Tag	Description				
					internal triggers	
		Arrhythmia rejection	no, yes	Cardiac synchronization != no ù device != Internal	Rejects and re-measures the acquired data during RR interval when it is too short or too long	N.A
		defined phases	x-y The actual values depend on other parameters	Cardiac synchronization = trigger ù Number of heart phases = user defined	Sets the heart phases	N.A
		TR (beats)	1-40	Cardiac synchronization = Trigger ù ! (TFE)	TR in heartbeats	With software option TriggerExtended (AWOPT_OPTION31) not available: restricts range to 1 - 20
		trigger delay	shortest, user defined, mid diastole, end diastole, user def. table,	Cardiac synchronization = trigger	Determines the time between R-peak and start of excitation (for single	N.A

Req ID	Requirement Tag	Description				
			longest		excitation)or center of excitation (for multiple excitation)	
		(ms)	1-1000	Trigger delay = user defined	Determines user defined trigger delay	N.A
		(beats)	1-40	Cardiac synchronization = trigger ÷ TFE ÷ shot interval = user defined	User defined TFE shot interval in heartbeats	With software option TriggerExtended (AWOPT_OPTION31) not available: restrict range to 1-20
		No trig period (beats)	1, or the TFE shot interval in beats. (so depending on the value of the "(beats)" parameter).	(beats) >=2	The "No Trigger Period" is the period available for TFE shot acquisition, including prepulse delays, and/or trigger delays.	With software option TriggerExtended (AWOPT_OPTION31) not available: restrict range to 1
		adaptive	no, yes	Cardiac synchronization =	With "adaptive" set to	This parameter is visible

Req ID	Requirement Tag	Description				
				trigger	"yes" the actual value of the trigger delay (for the first heart phase) is modified during the scan, continuously adapting to the cardiac frequency .	when software option TriggerExtended (AWOPT_OPTION31) is enabled
		heart phases	x-y The actual values depend on other parameters	Cardiac synchronization = gate	Sets the heart phases	N.A
		Gate delay	shortest, user defined	Cardiac synchronization = gate	The time from R-top to start of time slot in which data is acquired	N.A
		(ms)	x-y The actual values depend on other parameters	Gate delay = user defined	User defined gate delay	N.A
		width	shortest, user defined, longest	Cardiac synchronization = gate	Length of a time slot in which data is	N.A

Req ID	Requirement Tag	Description				
					acquired	
		(ms)	x-y The actual values depend on other parameters	width = user defined	User defined gate width	N.A
		Recon heart phases	1- 256	Cardiac synchronization = retrospective	Sets the heart phases	N.A
		ACQ heart phases (%)	50-100	Number of heart phases = maximum → TFE → shot mode = default	Fractional number of acquired k-profiles per requested heart phases	N.A
		ACQ phase interval (ms)	x-y The actual values depend on other parameters	Number of heart phases = maximum	This parameter determines the heart phase interval	N.A
		Retro stretching	Linear, non-linear	Cardiac synchronization = retrospective → device= ECG or External	All individual RR-intervals need to be stretched to the average RR interval prior to reconstruction.	N.A

Req ID	Requirement Tag	Description
		In addition software option FlowLabelExtended (AWOPT_OPTION10) shall support the following combination: 1.Cardiac synchronization = trigger Û (ASL = STAR, pCASL or Time-SLIP) In addition software option MiscCardExtended (AWOPT_OPTION19) shall support the following combination: 1.Cardiac synchronization = trigger Û device = Internal Û !ASL scans In additon software option EasyPlanning (AWOPT_MR_EASY_PLANNING) shall support the following combination 1.Cardiac synchronization = trigger Û 3D non-selective = yes In addition, if software option Essentials_1 (AWOPT_MR_ESSENTIALS_1) is available, in order to allow Cardiac synchronization = gate with Respiratory gating for 2D/M2D Cartesian TFE, software option CineFreeBreathing (AWOPT_MR_CINE_FREE_BREATHING) must be available. In addition, if software option Essentials_1 (AWOPT_MR_ESSENTIALS_1) is not available, in order to allow Cardiac synchronization = gate with Respiratory gating for 2D/M2D Cartesian TFE, software option RespGatingExtended (AWOPT_OPTION22) must be available.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorCardiacSynchronizationPar in the results file is zero.

3.1.45.14. SWCMP.Scan.Validator.CSSENSE.Par.tc1

Test ID	43498
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-23 13:14:10 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van

	(nly95948) on 2021-06-23 13:08:23 Meaning: SME Approved - (Business)
	Signed by: Chung, Hayley (310279214) on 2021-06-22 16:29:56 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies cs sense related parameters

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorCsSensePars in the results file is zero. Attach the result file as an objective evidence.

3.1.45.15. SWCMP.Scan.Validator.Diffusion.IRIS.tc1

Test ID	61667
Revision	1
Product Model	Any
Gradient System	
Options	N.A.
Signatures	Signed by: Reddy, Srikanth (310031195) on 2025-05-21 11:39:20 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2025-05-16 11:57:02 Meaning: SME Approved - (Technical) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-05- 13 23:54:08 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies that IRIS supports full grid reconstruction. Full FOV IRIS supports SENSE intrinsic foldover suppression while it is disabled for IRIS-Zoom.

Requirements Coverage:

Req ID	Requirement Tag	Description
97710	SWCMP.Scan.Validator.Diffusion .IRIS	IRIS shall support full grid reconstruction. SENSE intrinsic foldover suppression is enabled for full FOV IRIS, while

Req ID	Requirement Tag	Description
		it is disabled for IRIS-Zoom.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file	The reported number of failed tests related to SwCmpScanValidatorDiffusionIRIS in the results file is zero

3.1.45.16. SWCMP.Scan.Validator.Diffusion.Par.tc1

Test ID	51293
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-17 13:51:39 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2021-06-17 11:22:47 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-17 10:54:50 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the display of diffusion related parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
92464	SWCMP.Scan.Validator.Diffusion.Par.v3	Name	Range	Visible when	Description	Option / System Restrictions
		Diffusion mode	no, DWI, DTI	Always	This input parameter enables diffusion-weighted or diffusion tensor	

Req ID	Requirement Tag	Description					
					imaging techniques: 'DWI': 3 orthogonal diffusion directions can be selected (preparation, measurement and slice selection). 'DTI': diffusion schemes containing 6 directions or more can be selected. This is needed for FA maps computation		
		gradient expert mode	no, yes	Diffusion mode = DWI or DTI	The input 'no' invokes 'maximum duration' to calculate the diffusion gradient duration. 'gradient expert mode' enables the display		

Req ID	Requirement Tag	Description					
						of the parameter 'gradient control' that lists the various choices for gradient duration and gradient waveforms.	
		gradient control	maximum dur, minimum dur, user defined, time efficient, twice refocused	Gradient expert mode = yes		This input parameter controls the duration of the diffusion gradient lobes. In all cases the separation of the two gradient lobes is set to the maximum possible value. The diffusion gradient strength is determined such that the requested b-value is achieved. This gradient control is	

Req ID	Requirement Tag	Description					
						particularly useful for q-space imaging. 'maximum dur': The maximum possible gradient lobe duration is used for a given set of scan parameters 'minimum dur': The maximum allowed gradient strength is used combined with the largest separation of the two gradient lobes. 'user defined': The duration of the diffusion gradient lobe is specified by the user. 'time efficient': diffusion	

Req ID	Requirement Tag	Description					
					gradient waveform that differs from the gradient waveform of 'maximum dur', 'minimum dur' and 'user defined'. 'twice refocused' : diffusion gradient waveform that differs from the gradient waveform of 'maximum dur', 'minimum dur' and 'user defined'.		
		duration(ms)	1.0 - 300.0	gradient control = user defined	This input parameter specifies the duration of the diffusion gradient lobe in the unit of ms by the user.		
		gradient overplus	no, yes	Diffusion mode = DWI or	This input parameter enables		

Req ID	Requirement Tag	Description					
				DTI	increased diffusion gradient performance allowing shorter TE, leading to SNR increase. Note that only an isotropically diffusion image is produced.		
		direction	M, P, S, no	Diffusion mode = DWI and gradient overplus = no	This input parameter allows selection of the direction in which the diffusion-sensitizing gradient will be applied. The directions are given in coordinates of the MR experiment, where: M = Measurement direction P = Phase		

Req ID	Requirement Tag	Description					
					encoding direction S = Slice selection direction		
		directional resolution	low (6), medium (15), high(32), opt6, opt8, opt10, opt12, opt16, opt20, opt24, opt32, opt40, opt48, opt56, opt64, opt80, opt96, opt112, opt128, from file, read new file, error reading file	Diffusion mode = DTI	This input parameter enables a diffusion tensor imaging scheme. Low, medium and high indicates an increasing number of diffusion directions. low: 6 medium: 15 high: 32 Schemes 'opt x' indicate schemes optimized according to electromagnetic repulsion theory, where x indicates the number of directions. 'from file' and 'read new file'		

Req ID	Requirement Tag	Description					
						A user defined direction scheme can be read from file. The first time the parameter shows up as "read new file". Once the file is read successfully, it changes to "from file". The value error reading file will be set in case a problem occurs when reading the diffusion schemes from file. See below this table the detailed information on the "read new file" and "from file" functional	

Req ID	Requirement Tag	Description					
					ity.		
		nr of b-factors	1 – 32	Diffusion mode = DWI or (Diffusion mode = DTI and directional resolution is not equal to ('from file' or 'read new file'))	This input parameter determines the number of different strengths (b-factors)for the diffusion weighting in one scan.		
		b-factor order	ascending , user defined	nr of b-factors is greater than 1	This input parameter defines the different b-factor orders in a diffusion scan: 'ascending': A linear order of b-factors from b-factor zero to the maximum b-factor will be used. 'user defined': The user can specify the measured b-factors		

Req ID	Requirement Tag	Description					
						and scanning order using an array of b-factors.	
		max b-factor	0-25000	(nr of b-factors = 1) or (nr of b-factors is greater than 1 and b-factor order = ascending)		This input parameter specifies the maximum b-factor (diffusion sensitizing level) used in the scan. Based on this parameter and the number of b-factors, b-factors will be scanned in a linear increasing series from b-factor zero up to the maximum b-factor.	The maximum value depends on the gradient configuration and can be less
		b-factors	0-25000	b-factor order = user defined		This input parameter specifies a user defined array of b-factors. The 'number of b-factors'	The maximum value depends on the gradient configuration and can be less

Req ID	Requirement Tag	Description					
					parameter determines how many values of this parameter are taken into account. The scanning order matches the order in the array.		
		average high b	no, yes, user defined	Diffusion mode = DWI or (Diffusion mode = DTI and directional resolution is not equal to ('from file' or 'read new file'))	This input parameter specifies the number of averages for images with high b-factors to achieve a relative SNR gain.		
		b-factor averages	1 - 20	average high b is user defined	This input parameter specifies a user defined sequence of b-factor averages. The first element of the		

Req ID	Requirement Tag	Description					
					sequence corresponds to the number of b-factor averages that will be used for the first b-factor measured , the second element corresponds to the second b-factor measured and so on.		
		ADC correction	no, yes	Diffusion mode = DWI or DTI and Geometry correction is not equal to 'none'	The input parameter specifies whether reconstruction will correct for gradient non-linearities in the derived ADC maps		
		Diff Gr (DELTA/delta(ms))	N.A.	gradient expert mode = yes and gradient control is not equal to 'time efficient'	This output parameter displays the interval between the diffusion gradient		

Req ID	Requirement Tag	Description					
				or 'twice refocused'	lobes (DELTA) and the diffusion gradient duration (delta) in the unit of ms		
		EPI shots	N.A.	Diffusion mode = DWI or DTI and Fast imaging mode = EPI and shot mode = multishot	This output parameter specifies the number of shots for IRIS acquisition	Available when SW option IRIS_ZOOM (AWOPT_IRIS_ZOOM_IMAGING) or IRIS (AWOPT_MR_FULL_FOV_IRIS) or DiffExtended (AWOPT_OPTION09) is present	
		User defined DTI scheme	N.A.	directional resolution = 'from file' or 'read new file'	This output parameter indicates the name of the directional scheme as stored in the user defined file as the first entry. The first entry contains		

Req ID	Requirement Tag	Description				
					NO_NAME in case the name of the directional scheme is not present in the file. The second entry indicates, in brackets, respectively the number of vectors and the maximum b-value.	
		In case the directional resolution set to 'from file' or 'read new file', the allowed range of diffusion vectors, directions and b-values shall be: • maximum of 2048 diffusion vectors • maximum of 1024 diffusion directions • maximum of 1024 b-values It shall be possible to specify the acquisition of multiple b=0 encodings in the user defined file.				
		User defined file:				
		Name of the file		dti_vectors_input.txt		
		Location		E:\Export		
First line in this file		The name of the directional scheme as an optional entry				
Every other line in this file represents a vector for which four numbers are stored.		The first three numbers concern the diffusion directions along RL, AP, FH and the fourth number concerns the diffusion b-value.				
Saved file name		dti_vectors_mps.txt. This				

Req ID	Requirement Tag	Description	
			output file shall be saved when a new user defined file is read by the protocol. It contains the name of the directional scheme along with the number of vectors and the maximum b-value as the first line. The other lines contain the diffusion directions along M, P and S directions along with the diffusion b-value.
		Saved file location	E:\Export
For DWI in combination with Fast Imaging mode = EPI and Scan technique is not equal to IR and Acceleration = CSSense: • Software option CsDwiBody (AWOPT_SMARTSPEED_DWI_BODY) enables the use of respiratory compensation techniques and dedicated body coils*. *dedicated body coils = anterior type coil in combination with a posterior type coil, breast type coils, endorectal type coil. For IRIS in combination with Fast imaging mode = EPI and Acceleration = SENSE: • Software option IRIS_ZOOM (AWOPT_IRIS_ZOOM_IMAGING) - enables multishot diffusion with Fold-over suppression = zoom and P reduction = 1. • If software option Essentials_1(AWOPT_MR_ESSENTIALS_1) is available, in order to allow multishot diffusion with Fold-over suppression not equal to zoom and P reduction greater than or equal to 1, software option IRIS (AWOPT_MR_FULL_FOV_IRIS) must be available. • If software option Essentials_1(AWOPT_MR_ESSENTIALS_1) is not available, in order to allow multishot diffusion with Fold-over suppression not equal to zoom and P reduction greater than or equal to 1, software option DiffExtended (AWOPT_OPTION09) must be available. Software option DiffExtended (AWOPT_OPTION09) enables diffusion with: • Dynamics scans.			

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorDiffusionPar in the results file is zero.

3.1.45.17. SWCMP.Scan.Validator.Dixon.Par.tc1

Test ID	48545
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-11-06 07:44:00 Meaning: QA Approved - (QA) Signed by: Bhosale, Manoj (310194149) on 2020-11-05 12:57:36 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-11-05 12:52:10 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Dixon parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
84366	SWCMP.Scan.Validator.Dixon.Par.v1	mDIXON	Always: no, yes if FFE: quant	Imaging	For FFE and (T)SE, "yes" enables mDIXON. For FFE, "quant" enables Dixon-Quant.	M_OPT.A WOPT_M DIXON(option bit 62- mDixonFFE) enables FFE- mDIXON acquisition M_OPT.A WOPT_M DIXON_TSE(option bit 61- mDixonRF

Req ID	Requirement Tag	Description					
							E) enables (T)SE-mDIXON acquisition M_OPT.A WOPT_M DIXON_Q(option bit 76 - FatT2star Quant) enables quant acquisition for anterior and Torso XL Coils
		mode	multi-acquisition, multi echo	FFE ù mDIXON ! = no ù !(spiral)	Multi-acquisition uses single, time shifted echoes instead of multiple echoes		M_OPT.A WOPT_OPTION29 DixonFFEE extended (option bit 482) enables multi-acquisition
		initial startup cycle	default, user defined	mode = multi echo ù fast imaging mode = none	"user defined" mDIXON initial startup cycles for FFE		M_OPT.A WOPT_OPTION29 DixonFFEE extended (option bit 482) enables "user defined" start-up cycles

Req ID	Requirement Tag	Description				
		(number)	0-1000	initial startup cycle= user defined	user defined startup cycles	N.A
		acquisitions	2-4	mode = multi-acquisition or spiral	No. of acquisitions	M_OPT.A WOPT_OPTION41 DixonRFE extended (option bit 470) not available: Restrict the range to 3-4 for SpinEcho
		all shift	0-10	(T)SE Û mDIXON ! = no	shift to be applied to all echoes, this can result in increased SNR.	N.A
		delta TE (ms)	0.1-10	mode = multi-acquisition or spiral	delta TE in ms	N.A
		mDIXON images	Always: no, water, fat, IP, OP mDixon-FFE or T(SE): B0 source	mDIXON ! = no	mDixon image type selection	M_OPT.A WOPT_OPTION29 DixonFFExtd (option bit 482) or M_OPT.A WOPT_OPTION41 DixonRFE extended

Req ID	Requirement Tag	Description				
			mDixon-quant: fat fraction, T2*, R2*			(option bit 470) extends the range to include "B0" and "Source"
		mDIXON source images	R, I, M, P, no	mDIXON images = source	Determines mDixon image type(s) to be reconstructed from the acquired data.	N.A
In addition M_OPT.AWOPT_OPTION29 DixonFFEextended (option bit 482) shall allow the following combinations with a FFE mDIXON acquisition: 1. Echoes > 2 2. Scan mode M2D or MS 3. MultiEcho with scan parameter “flyback” set to “yes” 4. Flow compensation 5. Respiratory compensation (trigger, PEAR, gate) (Note: gate requires additional M_OPT.AWOPT_OPTION22) 6. MTC						

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorDixonPar in the results file is zero.

3.1.45.18. SWCMP.Scan.Validator.Drive.Par.tc1

Test ID	57030
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine

	(310147336) on 2025-05-27 12:57:59 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-05-27 10:29:22 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2024-07-12 09:31:06 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:
This test specifies availability of Scan Validator Post Shot pulse parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
92278	SWCMP.Scan.Validator.Drive.Par	DRIVE	no, yes, anti	Fast Imaging mode = TSE or GRaSE	This parameter determines enhancement of T1(anti-DRIVE) or T2 (DRIVE) contrast.	For CS SENSE Propeller(MotionFree) and Vista scans, anti drive is always available otherwise anti drive is only available when AWOPT_OPTION06 is enabled."

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorAcquisitionDrivePar in the

Step	Description	Expected Result
		results file is zero.

3.1.45.19. SWCMP.Scan.Validator.DualAcquisitionPerfusion.Par.tc1

Test ID	61732
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-06-16 15:42:51 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2025-06-16 13:28:33 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-06-16 11:18:53 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of Dual-Acquisition perfusion.

Requirements Coverage:

Req ID	Requirement Tag	Description								
97715	SWCMP.Scan.Validator.DualAcquisitionPerfusion.Par	For a scan with proton imaging AND dynamic study AND cardiac synchronization AND single-shot TFE AND (M2D OR MS) the Dyn/Ang tab shall show the following parameters: <table><tr><th>Name</th><th>Range</th><th>Visible when</th><th>System / Option restrictions</th></tr><tr><td>Quantitative Perfusion</td><td>Yes/No</td><td>proton imaging AND dynamic study AND cardiac synchronization AND single-shot TFE AND</td><td>Quantitative Perfusion is visible if DualAcquisitionPerfusion (AWOPT_MR_CARDIAC_Q_PERFUSION) is available.</td></tr></table>	Name	Range	Visible when	System / Option restrictions	Quantitative Perfusion	Yes/No	proton imaging AND dynamic study AND cardiac synchronization AND single-shot TFE AND	Quantitative Perfusion is visible if DualAcquisitionPerfusion (AWOPT_MR_CARDIAC_Q_PERFUSION) is available.
Name	Range	Visible when	System / Option restrictions							
Quantitative Perfusion	Yes/No	proton imaging AND dynamic study AND cardiac synchronization AND single-shot TFE AND	Quantitative Perfusion is visible if DualAcquisitionPerfusion (AWOPT_MR_CARDIAC_Q_PERFUSION) is available.							

Req ID	Requirement Tag	Description			
				(M2D OR MS)	
		AIF saturation time (ms)	20-50	Quantitative Perfusion = Yes	N.A.
		Nr. of PD-weighted dyn.	1-4	Quantitative Perfusion = Yes	N.A.
System restrictions: Dual-Acquisition Perfusion (Quantitative Perfusion) scans will be in conflict with SENSE. TFE saturation pre-pulse is a prerequisite for Dual-Acquisition Perfusion (Quantitative Perfusion) scans.					

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorDualAcquisitionPerfusionPar in the results file is zero.

3.1.45.20. SWCMP.Scan.Validator.DualAcquisitionPerfusion.ReconInterface.tc1

Test ID	62298
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-06-16 13:58:13 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2025-06-16 13:26:01 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-06-16 11:14:56 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the function of Dual-Acquisition Perfusion.

Requirements Coverage:

Req ID	Requirement Tag	Description																																													
98524	SWCMP.Scan.Validator.DualAcquisitionPerfusion.ReconInterface	The interface to Recon (RC struct) shall contain the following information:																																													
		<table><tr><th>Type</th><th>Name</th><th>Description</th></tr><tr><td>YESNO_ENUM</td><td>qPerfusionEnable</td><td>Enables Dual-Acquisition Perfusion (quantitative perfusion)</td></tr><tr><td>int</td><td>cs_sample_pattern_nr_profs_aif</td><td>Number of profiles of the AIF sampling pattern</td></tr><tr><td>float</td><td>aif_row_ap_image_ori</td><td>AIF AP row orientation</td></tr><tr><td>float</td><td>aif_row_fh_image_ori</td><td>AIF FH row orientation</td></tr><tr><td>float</td><td>aif_row_rl_image_ori</td><td>AIF RL row orientation</td></tr><tr><td>float</td><td>aif_col_ap_image_ori</td><td>AIF AP column orientation</td></tr><tr><td>float</td><td>aif_col_fh_image_ori</td><td>AIF FH column orientation</td></tr><tr><td>float</td><td>aif_col_rl_image_ori</td><td>AIF RL column orientation</td></tr><tr><td>float</td><td>aif_ap_angulation</td><td>AIF AP angulation</td></tr><tr><td>float</td><td>aif_fh_angulation</td><td>AIF FH angulation</td></tr><tr><td>float</td><td>aif_rl_angulation</td><td>AIF RL angulation</td></tr><tr><td>float</td><td>aif_ap_offcentre</td><td>AIF AP off-center</td></tr><tr><td>float</td><td>aif_fh_offcentre</td><td>AIF FH off-center</td></tr><tr><td>float</td><td>aif_rl_offcentre</td><td>AIF RL off-center</td></tr></table>	Type	Name	Description	YESNO_ENUM	qPerfusionEnable	Enables Dual-Acquisition Perfusion (quantitative perfusion)	int	cs_sample_pattern_nr_profs_aif	Number of profiles of the AIF sampling pattern	float	aif_row_ap_image_ori	AIF AP row orientation	float	aif_row_fh_image_ori	AIF FH row orientation	float	aif_row_rl_image_ori	AIF RL row orientation	float	aif_col_ap_image_ori	AIF AP column orientation	float	aif_col_fh_image_ori	AIF FH column orientation	float	aif_col_rl_image_ori	AIF RL column orientation	float	aif_ap_angulation	AIF AP angulation	float	aif_fh_angulation	AIF FH angulation	float	aif_rl_angulation	AIF RL angulation	float	aif_ap_offcentre	AIF AP off-center	float	aif_fh_offcentre	AIF FH off-center	float	aif_rl_offcentre	AIF RL off-center
		Type	Name	Description																																											
		YESNO_ENUM	qPerfusionEnable	Enables Dual-Acquisition Perfusion (quantitative perfusion)																																											
		int	cs_sample_pattern_nr_profs_aif	Number of profiles of the AIF sampling pattern																																											
		float	aif_row_ap_image_ori	AIF AP row orientation																																											
		float	aif_row_fh_image_ori	AIF FH row orientation																																											
		float	aif_row_rl_image_ori	AIF RL row orientation																																											
		float	aif_col_ap_image_ori	AIF AP column orientation																																											
		float	aif_col_fh_image_ori	AIF FH column orientation																																											
		float	aif_col_rl_image_ori	AIF RL column orientation																																											
		float	aif_ap_angulation	AIF AP angulation																																											
		float	aif_fh_angulation	AIF FH angulation																																											
		float	aif_rl_angulation	AIF RL angulation																																											
		float	aif_ap_offcentre	AIF AP off-center																																											
		float	aif_fh_offcentre	AIF FH off-center																																											
float	aif_rl_offcentre	AIF RL off-center																																													

Req ID	Requirement Tag	Description		
		float	aif_ap_oversample_delta_offcentre	AIF AP oversample delta off-center
		float	aif_fh_oversample_delta_offcentre	AIF FH oversample delta off-center
		float	aif_rl_oversample_delta_offcentre	AIF RL oversample delta off-center
		float	saturationTime	Time from saturation (reference) to the center of k-space (k0) readout echo time.
		float	saturationTimeAIF	Time from saturation (reference) to the center of k-space (k0) readout echo time, for the AIF.
		float	acquisitionTimeDelay	Time delay from saturation (reference) to the first k-space readout echo time.
		float	acquisitionTimeDelayAIF	Time delay from saturation (reference) to the first k-space readout echo time, for the AIF.
		and updating existing parameters:		
		Type	Name	Comment
		int	total_nr_profiles	Adding the AIF number of profiles.
		int	data_rate	Adding the AIF number of profiles.
		int	nr_mixes	AIF has mix = 0 and the first perfusion images stack starts

Req ID	Requirement Tag	Description		
				at mix = 1
		int	complex_av_output_images	To be set for both mixes.
		int	signal_type_sizes	Set for AIF.
		int	min_encoding_numbers	Set for AIF.
		int	max_encoding_numbers	Set for AIF.
		int	slice_orientations	Set for AIF.
		int	centre_slice_first_package	Set for AIF.
		float	oversample_factors	Set for AIF.
		float	location_matrices	Set for AIF.
		int	min_nr_profs_total[1]	Set for AIF.
		int	min_nr_profs_each_side[1]	Set for AIF.
		ARGRC_SEQTP_ENUM	sequence_types	Set for AIF.
		ARGRC_IPTRANSF_ENUM	in_plane_transformations	Set for AIF.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorDualAcquisitionPerfusionSeriesInfo in the results file is zero.

3.1.45.21. SWCMP.Scan.Validator.DualAcquisitionPerfusion.Sequence.tc1

Test ID	61866
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-06-16 15:46:46 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2025-06-16 13:30:17 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-06-16 11:19:13 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the function of Dual-Acquisition Perfusion.

Requirements Coverage:

Req ID	Requirement Tag	Description
97898	SWCMP.Scan.Validator.DualAcquisitionPerfusion.Sequence	The number of acquired profiles for AIF shall be larger or equal to 16.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorDualAcquisitionPerfusionSequence in the results file is zero.

3.1.45.22. SWCMP.Scan.Validator.DualAcquisitionPerfusion.SeriesInfo.tc1

Test ID	62295
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-06-16 13:57:52 Meaning: QA Approved - (QA)

	Signed by: Sharma, Nishant (320059006) on 2025-06-16 13:27:25 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-06-16 11:17:01 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:
This test verifies the function of Dual-Acquisition Perfusion.

Requirements Coverage:

Req ID	Requirement Tag	Description																					
98827	SWCMP.Scan.Validator.DualAcquisitionPerfusion.SeriesInfo	The series info shall contain the following information:																					
		<table><tr><th>Type</th><th>Name</th><th>Description</th></tr><tr><td>int</td><td>no_aifs</td><td>Number of AIF volumes (= 1 for Dual-Acquisition Perfusion).</td></tr><tr><td>IMLDM_AIF_STRUCTURE</td><td>series_aif_arr</td><td>Array of AIF volume information.</td></tr><tr><td>float</td><td>aif_offcentre_ap</td><td>AIF AP off-center (as part of IMLDM_AIF_STRUCTURE)</td></tr><tr><td>float</td><td>aif_offcentre_rl</td><td>AIF RL off-center (as part of IMLDM_AIF_STRUCTURE)</td></tr><tr><td>float</td><td>aif_offcentre_fh</td><td>AIF FH off-center (as part of IMLDM_AIF_STRUCTURE)</td></tr><tr><td>float</td><td>aif_angulation_ap</td><td>AIF AP angulation (as part of IMLDM_AIF_STRUCTURE)</td></tr></table>	Type	Name	Description	int	no_aifs	Number of AIF volumes (= 1 for Dual-Acquisition Perfusion).	IMLDM_AIF_STRUCTURE	series_aif_arr	Array of AIF volume information.	float	aif_offcentre_ap	AIF AP off-center (as part of IMLDM_AIF_STRUCTURE)	float	aif_offcentre_rl	AIF RL off-center (as part of IMLDM_AIF_STRUCTURE)	float	aif_offcentre_fh	AIF FH off-center (as part of IMLDM_AIF_STRUCTURE)	float	aif_angulation_ap	AIF AP angulation (as part of IMLDM_AIF_STRUCTURE)
		Type	Name	Description																			
		int	no_aifs	Number of AIF volumes (= 1 for Dual-Acquisition Perfusion).																			
		IMLDM_AIF_STRUCTURE	series_aif_arr	Array of AIF volume information.																			
		float	aif_offcentre_ap	AIF AP off-center (as part of IMLDM_AIF_STRUCTURE)																			
		float	aif_offcentre_rl	AIF RL off-center (as part of IMLDM_AIF_STRUCTURE)																			
		float	aif_offcentre_fh	AIF FH off-center (as part of IMLDM_AIF_STRUCTURE)																			
float	aif_angulation_ap	AIF AP angulation (as part of IMLDM_AIF_STRUCTURE)																					

Req ID	Requirement Tag	Description		
		float	aif_angulation_rl	AIF RL angulation (as part of IMLDM_AIF_STRUCTURE)
		float	aif_angulation_fh	AIF FH angulation (as part of IMLDM_AIF_STRUCTURE)
		float	aif_fov_ap	AIF AP FOV (as part of IMLDM_AIF_STRUCTURE)
		float	aif_fov_rl	AIF RL FOV (as part of IMLDM_AIF_STRUCTURE)
		float	aif_fov_fh	AIF FH FOV (as part of IMLDM_AIF_STRUCTURE)
		IMLDM_RL_AP_FH_ENUM	aif_view_axis	AIF view axis (as part of IMLDM_AIF_STRUCTURE)
		and updating existing parameters:		

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorDualAcquisitionPerfusionSeriesInfo in the results file is zero.

3.1.45.23. SWCMP.Scan.Validator.Echoes.Par.tc1

Test ID	62009
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-05-28 21:49:34 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-05-28 18:15:21 Meaning: SME Approved - (Business) Signed by: Pereira, Kaitan (320169354) on 2025-05-28 17:27:35 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description				
92277	SWCMP.Scan.Validator.Echoes.Par	Name	Range	Visible when	Description	System / Option restrictions
		Startup echoes	0 -64	Fast Imaging mode = TSE, GRaSE	This parameter determines a number of echoes at start of each echo train that will be discarded.	N/A
		default, user defined	Fast Imaging mode = TFE, TFEPI	This parameter determines a	N/A	(number)

Req ID	Requirement Tag	Description				
				number of echoes at start of each TFE shot that will be discarded.		
		0-100	Startup echoes = user defined	The number of echoes discarded.		Echoes
			Always	This parameter determines the number of echoes.	N/A An error occurred while interpreting html data	TSE or GraSE
		256	(TFE, EPI or TFEPI or none) and technique = FFE	2	(EPI or none) and technique = SE, IR or MIX	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorEchoesPar in the results file is zero.

3.1.45.24. SWCMP.Scan.Validator.ElasticImageRegistration.Par.tc1

Test ID	61731
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-05-23 17:19:46

	Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-05-23 17:09:48 Meaning: SME Approved - (Business) Signed by: Bajaj, Naveen (310105618) on 2025-05-22 19:02:55 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies (fast) elastic image registration (FEIR) related parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
97807	SWCMP.Scan.Validator.ElasticImageRegistration	Name	Range	Visible when	Description	System/Option restrictions
		Elastic Motion Correction	yes, no	Case 1: 2D scan + single-shot mode + dynamics (dynamic scan) Case 2: 2D scan + single-shot mode + measurements > 1.0 (NSA, not partial NSA) Case 3: 2D scan + T1 mapping	Reconstruction algorithm to perform (fast) elastic image registration [FEIR].	This parameter is only visible when "ElasticImageRegistration" (AWOPT_MR_CARD_IAC_MOTION_CORR) is present. The parameter is in conflict if the anatomy is not equal to "Heart". In that event, the user is

Req ID	Requirement Tag	Description					
							notified by a message as the anatomy is no scan protocol parameter that can be set into conflict.
		Note: (FEIR dimension is set by recon) 2D single shot with Dynamics -> FEIR on dynamics (e.g., Quantitative Perfusion) 2D single shot with measurements (NSA) > 1 -> FEIR on measurements 2D single shot with T1 mapping (MOLLI) -> FEIR on row					

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorElasticImageRegistrationPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.25. SWCMP.Scan.Validator.EPIGeometryCorrection.Par.tc1

Test ID	41203
Revision	1
Product Model	All
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-22 09:56:07 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2019-05-21 14:32:51 Meaning: SME Approved - (Business) Signed by: Nabuurs, Christine (310147336) on 2019-05-21 12:53:08 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:
This test verifies the availability of the EPI Geometry correction parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
99059	SWCMP.Scan.Validator.EPIGeometryCorrection.Par.v1	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		EPI Geometry correction	yes no	Single shot EPI scan	Yes enables the EPI geometry correction functionality No disables the EPI geometry correction functionality. The functionality can be enabled for a diffusion scan with the following condition: - At least one b=0 weighting It can be enabled for a dynamic scan with the following conditions :	Available when software option EPI_Geometry_Correction (AWOPT_EPIC_BRAIN) is present

Req ID	Requirement Tag	Description				
					- Dynamic study set to individual - Dynamic scan times set to shortest - At least one dummy dynamic	
		EPI geom cor. Output image	Normal, MPR B0 Map, MPR Water Map, Blip Down Image	The functionality is enabled during delayed recon when EPI Geometry correction is set to Yes	Selects the output image type. 'Normal' indicates EPI geometry corrected images, the other output image types are intermediate images used during the calculation of the EPI geometry corrected images.	The range {MPR B0 Map, Blip Down Image}} is only enabled when the SW option DiffExtended (AWOPT_OPTION09) is present. MPR Water Map is only enabled when the SW option DevelopmentMode (AWOPT_DEVELOPMENT) is present.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorEPIGeometryCorrectionPar in the results file is zero.

3.1.45.26. SWCMP.Scan.Validator.ExternalControl.Par.tc1

Test ID	56736
Revision	1
Product Model	All
Gradient System	Any
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-03-14 07:45:44 Meaning: QA Approved - (QA) Signed by: Shree K, Shruthi (Partner) (320185675) on 2023-03-13 13:52:31 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-03-13 13:35:40 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description				
91525	SWCMP.Scan.Validator.ExternalControl.Par	Name	Range	Visible when	Description	System/Option restrictions
		External control	no, hifu, full flexibility, MRTC	Only when network IP-address of external console is set .	Allows you to control the MR scanner from compatible external platform, such as HIFU planning console, MR-Linac	External control is visible when SW option bit 222: MRTC is available. For External control hifu and full flexibility

Req ID	Requirement Tag	Description				
					control console or research software.	the HW configuration XTC is required. For External control MRTC, the HW configuration MRTC is required.
		ignore geo updates	no, yes	External control = hifu or full flexibility or MRTC	Geometry updates from the external control can be ignored.	
96973	SWCMP.Scan.Validator.External Control.Par.v1	Name	Range	Visible when	Description	System/Option restrictions
		External control	no, yes	SW option bit 222: ExternalControlEssential is available.	Allows control of the MR Scanner by clients using the MRTC interface.	External control is visible only when SW option bit 222: ExternalControlEssential is available.
		ignore geo updates	no, yes	External control = yes	Geometry updates from the external control	External control needs to be "yes".

Req ID	Requirement Tag	Description				
					will be ignored when set to yes.	Hence is visible only when SW option bit 222 is available.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorExternalControlPar in the results file is zero.

3.1.45.27. SWCMP.Scan.Validator.Fat.SuppressionTechniques.Par.tc1

Test ID	48542
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharappa, Deepak Shekhar (320098697) on 2020-11-06 07:47:07 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-10-30 07:28:59 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-10-28 15:01:53 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Fat Suppression Techniques parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
94186	SWCMP.Scan.Validator.Fat.Supp ression.Techniques.Par.v3	Fat suppression	No, SPAIR !Spectro: SPIR, ProSet	Parameter Series != TI	This input parameter enables the choice of fat suppression method	M_OPT.A WOPT_OPTION15 (option bit 496) enables the combination of ProSet with sequences where TE > TR
		Strength	weak, medium, strong	Fat suppression = SPIR	This input parameter sets the strength of the SPIR pulse	N/A
		frequency offset	default, user defined	Fat suppression = SPIR or SPAIR	This input parameter adapts the frequency offset of the SPIR pulse	N/A
		offset (Hz)	1.5T: 40 - 180 3T: 80 - 360	Frequency offset = user defined	This input parameter allows the user to set the desired offset frequency	N/A
		power	1, 2	Fat suppression	This input parameter	N/A

Req ID	Requirement Tag	Description				
				on = SPAIR	r allows to specify the amount of spectral fat suppression	
		inversion delay	user defined, auto FFE: shortest	Fat suppression = SPAIR	This input parameter enables the selection of the SPAIR inversion delay mode	N/A
		Suppr. level	weak, medium, strong	Inversion delay = auto	This input parameter specifies the fat suppression level	N/A
		(ms)	0 - 2000 ms	inversion delay = user defined	This input parameter specifies the time between the SPAIR prepulse and the excitation pulse.	N/A
		Pulse type	11, 121, 1331	Fat suppression = ProSet	This input parameter specifies the ProSet pulse to be used.	N/A

Req ID	Requirement Tag	Description				
		Grad Rev Fat suppr	No, yes	technique != FFE	This input parameter allows fat suppression by gradient reversal	This parameter is visible only when the system is 3T.
		SPAIR TR (ms)	N/A	Inversion delay = user defined	This output parameter specifies SPAIR TR in ms	N/A
		SPAIR inv. delay act./ auto (ms)	N/A	Inversion delay = user defined	This output parameter specifies SPAIR inversion delay in ms	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorFatSuppressionTechniques Par in the results file is zero.

3.1.45.28. SWCMP.Scan.Validator.FFE.Contrast.Par.tc1

Test ID	48544
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-11-06 07:47:10

	Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-10-30 07:01:31 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-10-28 15:01:55 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator FFE Contrast parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
98076	SWCMP.Scan.Validator.FFE.Contrast.Par.v3	Contrast enhancement	Always:No, T1, T2, Balanced If respiratory compensation = retrospective: T2 Enhanced	Technique = FFE	Enables a specific Contrast enhancement technique for FFE sequences	N/A
		B-FFE offset frequency	-250 - 250	Contrast enhancement = Balanced	Specifies the offset frequency in Hz.	This parameter is enabled when the SW option MOPT.AWO

Req ID	Requirement Tag	Description					
						PT_OPTION 08 (option bit 503) is enabled for non Marlin systems always for Marlin systems	
		delta offset frequency	-250 - 250	Contrast enhancement = Balanced Dynamic study = individual	Specifies the delta offset frequency in Hz per dynamic	- This parameter is enabled when the SW option MOPT.AWOPT_OPTION08 (option bit 503) is enabled for non Marlin systems - always for Marlin systems	

Req ID	Requirement Tag	Description				
		banding reduction	no, yes	(Technique = FFE or Fast imaging mode = TFE) ÷ Contrast enhancement = Balanced ÷ NSA >= 2	Enable banding reduction	This parameter is only enabled when the SW option M_OPT.BFFE_OFFSET_AVERAGING (option bit 146) is enabled. MOPT.AW OPT_OPTION08 (option bit 503) enables the use of banding reduction with TFE

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorFFEContrastPar in the results file is zero.

3.1.45.29. SWCMP.Scan.Validator.GeometryCorrection.Par.tc1

Test ID	43147
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-12-09 12:31:47 Meaning: QA Approved - (QA)

	Signed by: Engelen, Walter van (nly95948) on 2019-12-09 09:53:10 Meaning: SME Approved - (Business)
	Signed by: Rikkila, Suvi (320024526) on 2019-11-27 12:35:52 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
82485	SWCMP.Scan.Validator.GeometryCorrection.Par.v1	Geometry correction	- Always: None - Default 2D compensation Scan mode = 3D, M2D or MS: 3D compensation	Scan type = Imaging	Operator specifies none, 2D- or 3D-compensation or default. Default instructs the system to select 3D compensation for 3D sequences and 2D compensation for 2D sequences in case the system configuration requests to apply the geometry correction.	3D compensation is only allowed for scan mode = M2D or MS when option bit 88: MR-RT is enabled or system type is Marlin.

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorGeometryCorrectionPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.30. SWCMP.Scan.Validator.Gradient.Par.tc1

Test ID	41070
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:21:59 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 11:11:27 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:13:07 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description			
74999	SWCMP.Scan.Validator.Gradient.Par	Parameter Name	Range	Visible when	Description
		Gradient mode	regular, default, user defined, full control, maximum, enhanced, enhanced-max	Always	Controls the allowed gradient amplitude and slew rate for the different gradient

Req ID	Requirement Tag	Description			
					systems
		gradient switch	standard, parallel	Gradient mode = user defined or full control	Direct control of the gradient switch
		gradient performance	regular, level 1, level 2, level 3, level 4, level 5, level 6	Gradient mode = user defined	Controls the allowed gradient amplitude and slew rate for the different selected levels
		max strength (mT/m)	0.01 - 100.0	Gradient mode = full control	Controls the maximum allowed gradient strength(The actual value will be based on the HW configuration)
		max slew rate (T/m/s)	0.01 - 1000.0	Gradient mode = full control	Controls the maximum allowed slew rate(The actual value will be based on the HW configuration)

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorGradientPar in the results file is zero.

3.1.45.31. SWCMP.Scan.Validator.GradientBidSpec.Par.tc1

Test ID	62662
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Reddy, Srikanth (310031195) on 2025-07-29 09:39:51 Meaning: QA Approved - (QA) Signed by: Koonen, Jos (nly95806) on 2025-07-28 16:41:43 Meaning: SME Approved - (Business) Signed by: Gorkum, Robbert van (320189046) on 2025-07-25 11:08:26 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the Gradient Bid Spec related parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
99320	SWCMP.Scan.Validator.GradientBidSpec.Par	Name	Range	Visible when	Description	System/Option restrictions
		BIDSPEC mode	'no', 'gradient'	Never during normal clinical system usage.	Demonstrates the gradient Bid Spec.	This parameter and functionality shall only be available when: Patient age >= 200 + Patient weight = 1 + Patient gender =

Req ID	Requirement Tag	Description				
						phantom + ServiceLevel_2 (AWOPT_SS_LEVEL_2) enabled + CGEN.enable_bidspec_mode = yes
		full control	No, Yes	BIDSPEC mode = gradient	Offers manual control of the gradient strength and slew rate to be demonstrated.	This parameter shall only be visible when: option DevelopmentMode (AWOPT_development) is enabled. When this option is not available, the parameter is not visible but is still active (the parameter cannot be modified by the user).

Req ID	Requirement Tag	Description				
		grad str post crusher	0.0 - 120.0	full control = Yes	This paramete r specifies the gradient strength demonstr ating the gradient Bid Spec.	This paramete r shall only be visible when: option 'Develop mentMod e' (AWOPT_ developm ent) is enabled. When this option is not available, the paramete r is not visible but is still active (the paramete r cannot be modified by the user).
		grad slew rate post crusher	1.0 – 500.0	full control = Yes	This paramete r specifies the gradient slew rate demonstr ating the gradient Bid Spec.	This paramete r shall only be visible when: option Developm entMode (AWOPT_ developm ent) is enabled. When this option is not available,

Req ID	Requirement Tag	Description				
						the parameter is not visible but is still active (the parameter cannot be modified by the user).
Note: The general control parameter CGEN.ena_bidspec_mode shall only be enabled in the UI when the option ServiceLevel_2 (AWOPT_SS_LEVEL_2) is enabled.						

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMP.Scan.Validator.GradientBidSpec.Par in the results file is zero.

3.1.45.32. SWCMP.Scan.Validator.InteractivePositioning.Par.tc1

Test ID	56169
Revision	1
Product Model	Any
Gradient System	
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2022-12-12 16:16:46 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2022-12-12 13:39:08 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2022-12-12 13:19:34 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Interactive positioning means that specific geometry scan parameters are changed while scanning whereas the effect is to be seen in real-time.

Requirements Coverage:

Req ID	Requirement Tag	Description				
91285	SWCMP.Scan.Validator.InteractivePositioning.Par	Name	Range	Visible when	Description	System / Option restrictions
		Interactive positioning	Yes, No	Scan type = imaging and Nucleus = 1H	Interactive positioning means that specific geometry scan parameters are changed while scanning whereas the effect is to be seen in real-time.	Available when SW option M_OPT.InteractivePlanning (option bit 219) is present
94905	SWCMP.Scan.Validator.InteractivePositioning.Par.v2	Name	Range	Visible when	Description	System / Option restrictions
		Interactive positioning	Yes, No	Scan type = imaging and Nucleus = 1H	Interactive positioning means that specific geometry scan parameters are changed while	Available when M_OPT.InteractivePlanning (enabled by option bit 219 or option bit 222) is present.

Req ID	Requirement Tag	Description				
					scanning whereas the effect is to be seen in real-time.	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorInteractivePositioningPar in the results file is zero.

3.1.45.33. SWCMP.Scan.Validator.Metal.Par.tc1

Test ID	51114
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-17 13:56:00 Meaning: QA Approved - (QA) Signed by: Chung, Hayley (310279214) on 2021-06-17 12:30:12 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-16 08:57:24 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability and verification of Metal related parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
91691	SWCMP.Scan.Validator.Metal.Par.v1	Name	Range	Visible when	Description	System / Option restrictions

Req ID	Requirement Tag	Description				
		O-MAR	no, MARS+VAT, SEMAC+VAT	Scan technique = SE or IR and Fast Imaging mode = none or TSE	O-MAR (Optimized Metal Artifact Reduction) enables metal imaging	MARS+VAT is visible: - For all product models except Marlin, when M_OPT.VAT (option bit 92) and M_OPT.PatientConditions (option bit 105) are available - For product model Marlin when M_OPT.VAT (option bit 92) is available SEMAC+VAT is visible: - For all products except Marlin,

Req ID	Requirement Tag	Description					
							when M_OP T.VAT (option bit 92) and M_OP T.PatientCondition ns (option bit 105) and (M_OPT.S EMAC (option bit 93) or M_OP T.CsSE MAC (option bit 214)) are available For product model Marlin , when M_OPT.V AT (option bit 92) and (M_OPT.S EMAC (option bit 93) or

Req ID	Requirement Tag	Description				
						M_OPT.Cs SEMAC (option bit 214)) are available
		distortion correction	weak, medium, strong	O-MAR = SEMAC+V AT	modifies the number of z-encodings (SEMAC factor) for SEMAC	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorMetalPar in the results file is zero.

3.1.45.34. SWCMP.Scan.Validator.MRE.Par.tc1

Test ID	51287
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-17 13:59:23 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2021-06-17 11:24:39 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-16 08:59:52 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of MRE parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
85108	SWCMP.Scan.Validator.MRE.Par	Name	Range	Visible when	Description	System/Option restrictions
		MRE enable	no, yes	Always	Enables/disables MRE	MRE is visible in combination with FFE when M_OPT.MRE (option bit 90) is available MRE is visible in combination with SE when M_OPT.mrElastographySeEpi (option bit 154) is available. SE-EPI in combination with CSSENSE requires M_OPT.CsDwiBody (option bit 213).
		Frequency (Hz)	40 to 200 Hz	MRE enable is yes	Sets the frequency of the acoustic wave	
		Wave	low,	MRE	Sets the	

Req ID	Requirement Tag	Description				
		amplitude	moderate, high, user defined	enable is yes	amplitude of the acoustic wave	
		Percentage (%)	0 to 100 %	Wave amplitude is user defined	User defined value of the acoustic wave amplitude in terms of percentage	
		Direction	FH, AP, RL	MRE enable is yes	Sets the spatial direction of the motion encoding gradient (patient relative orientation)	
		Gradient strength	0-Max allowed gradient	MRE enable is yes	Sets the strength of the motion encoding gradients	The max allowed gradient depends on the actual system configuration
		Patient experience scan	no, yes	MRE enable is yes	Enables patient experience scan in which an acoustic vibrations are played prior to	

Req ID	Requirement Tag	Description				
					the main scan to acclimatize the patient to the vibrations	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorMREPar in the results file is zero.

3.1.45.35. SWCMP.Scan.Validator.MSDEMode.Par.tc1

Test ID	57222
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-07-11 14:51:03 Meaning: QA Approved - (QA) Signed by: Rozijn, Tom (nly95848) on 2023-07-10 16:20:49 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2023-07-10 10:50:54 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of Scan Validator MSDE Mode parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
92480	SWCMP.Scan.Validator.MSDEMode.Par	Name	Range	Visible when	Description	System / Option restrictions

Req ID	Requirement Tag	Description				
		MSDE mode	Intracranial, nerve, flow ghost supp, user defined	BB pulse = MSDE or bb pulse type = MSDE	Motion Sensitized Driven Equilibrium (MSDE) based blood suppression. Sequence will be optimized based on chosen application area.	Nerve is available when M_OPT.Neurography is enabled.
		refocusing pulses	(1 - 2)	MSDE mode = user defined	Set the preparation sequence to employ a single or two refocusing pulses.	N/A
		refocusing type	Composite, adiabatic	MSDE mode = user defined	Set the preparation sequence to employ composite or adiabatic inversion pulses.	N/A
		TEprep	shortest, user defined	MSDE mode = user defined	Sets the duration of the total MSDE preparation sequence.	N/A

Req ID	Requirement Tag	Description				
		(ms)	(0 - 500)	TEprep = user defined	To set the duration manually to a time longer than minimal.	N/A
		venc (RL, AP, FH) [3 values]	(-30 - 30)	MSDE mode = user defined	The minimum flow rate of the spins to be suppressed is set by this parameter.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorMSDEModePar in the results file is zero.

3.1.45.36. SWCMP.Scan.Validator.MultiNuclei.ImagingBandwidth.Par.tc1

Test ID	47863
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:08 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:41:00 Meaning: SME Approved - (Technical)

	Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:23:55 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameters to set the Bandwidth for MN Imaging.

Requirements Coverage:

Req ID	Requirement Tag	Description				
81441	SWCMP.Scan.MultiNuclei.ImagingBandwidth.Par	Name	Range	Visible when	Description	System / Option restrictions
		Acquisition Bandwidth	• minimum • user defined • maximum	Scan type = Imaging ÷ Nucleus ! = 1H	This parameter allows the operator to set the bandwidth of the acquisition / readout object.	None
		Hz/pixel	35 - 10000	Acquisition Bandwidth = user defined	This parameter allows the operator to set the acquisition bandwidth in Hz per pixel	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorMultinucleiImagingBandwidthPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.37. SWCMP.Scan.Validator.MultiNuclei.Nucleus.Par.tc1

Test ID	47861
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-09-16 10:28:10 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:42:48 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:23:56 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameters to select the nucleus and offset for the metabolite(s) of interest for MN Spectroscopy and Imaging.

Requirements Coverage:

Req ID	Requirement Tag	Description				
93207	SWCMP.Scan.MultiNuclei.Nucleus.Par.v1	Name	Range	Visible when	Description	System / Option restrictions
		Nucleus	1H 19F 31P 13C 23Na 129Xe 3He	Always	This parameter allows the operator to determine	1H <u>imaging</u> is always available

Req ID	Requirement Tag	Description				
			• 7Li • 17O		the nuclear spin on which the acquisition method of interest will be applied.	• 1H <u>spectroscopy</u> is only available when SW option bit 50: <i>Spectroscopy</i> is enabled. • For 31P: <u>spectroscopy</u> is only available when option bit 51: <i>Multi Nuclei 31P Spectroscopy</i> is enabled, <u>imaging</u> is only available only when option bit 159: <i>Distributed Multi Nuclei</i> is enabled. • For 19F, 13C, 23Na and 129Xe :

Req ID	Requirement Tag	Description				
						<u>imaging and spectro</u> are available only when option bit 159: <i>Distributed Multi Nuclei</i> is enabled. - For ³He, ⁷Li, ¹⁷O: <u>spectroscopy</u> is only available when CSK option bit 72: <i>Multi Nuclei Spectro Extensions</i> is enabled. - For ³He, ⁷Li, ¹⁷O: <u>imaging</u> is only available when CSK option bit 471: <i>Multi Nuclei Imaging Extensions</i> is enabled.

Req ID	Requirement Tag	Description				
		Offset Frequency (ppm)	-400 - 400	Nucleus != 1H	This parameter allows the operator to determine the offset frequency for the RF pulses and Acquisition object with respect to the default offset of the nucleus of interest.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorMultinucleiNucleusPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.38. SWCMP.Scan.Validator.MultiNuclei.ReceiveAttenuation.Par.tc1

Test ID	47862
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:12 Meaning: QA Approved - (QA)::Bulk

	Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:51:38 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:23:58 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameters to set the Receive Attenuation for MN Spectroscopy and Imaging.

Requirements Coverage:

Req ID	Requirement Tag	Description				
79467	SWCMP.Scan.MultiNuclei.ReceiveAttenuation.Par	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		Receive atten. [dB]	8 - 38 dB	Nucleus != 1H	This parameter allows the operator to set the receive attenuation of MN receive channels manually to prevent signal clipping.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorMultinucleiReceiveAttenuationPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.39. SWCMP.Scan.Validator.MultiParametricImaging.SyntAc.Par.tc1

Test ID	56005
Revision	1
Product Model	All
Gradient System	
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-07-04 16:58:43 Meaning: QA Approved - (QA) Signed by: Rozijn, Tom (nly95848) on 2023-07-04 16:13:29 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2023-06-26 14:57:12 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability for Scan Validator Multi-Parametric Imaging SyntAc parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
94179	SWCMP.Scan.Validator.MultiParametricImaging.SyntAc.Par.v2	Name	Range	Visible when	Description	System/Option restriction
		SyntAc	no, yes	Nucleus = 1H	This parameter enables the synthetic acquisition (SyntAc) sequences. This mode acquires	The MS-MDME sequence requires software option M_OPT_MDME (AWOPT_MDME, option bit

Req ID	Requirement Tag	Description
		acquisition data which contains multiple types of contrast (T1, T2, and proton-density maps). 102). The 3D-TPMID sequence requires software option M_OPT.Opt3DSyntac (AWOPT_MR_3DSYNTAC, option bit 229). MDME = Multi-Dynamic Multi-Echo; TPMID = T2-Prepared Multi-Inversion Delay The following general parameter restrictions are in place for SyntAc: • "Scan mode" shall be restricted to "MS" and "3D", with default "MS" • The following parameter restrictions shall be in place for MDME: • "echoes" can only be set to values ">=2" • • "slices" can only be set to values ">= 15" • • "Stacks" can only have value "1" • • "Dynamic study" can only be set to "individual" • • "dyn scans" (number of dynamic scans) can only have value "4" • • "Scan technique" shall be restricted to "SE" • • "Slice scan order" values "ascending" or "descending" can only be selected, with default "ascending" • The following parameter restrictions shall be in place for TPMID: • "echoes" can only be set to "1" • • "Flip angle" shall be restricted to "1.0 – 15.0", with default: "15.0" • • TFE "profile order" shall be restricted to "low_high" • • TFE factor "shot interval" shall be restricted to "user defined" • • TFE factor shot interval " (ms)" range shall be restricted from "800.0" to "1200.0" • • "Cardiac synchronization" shall be restricted to "No" • • "Dynamic study" shall be restricted to "No" • • "Partial echo" shall be restricted to "No" • • "Halfscan" shall be restricted to "No" •

Req ID	Requirement Tag	Description
		• "Acquisition mode" shall be restricted to "Cartesian" • • "Imaging sequence" shall be restricted to "FFE" • • "Fast imaging mode" shall be restricted to "TFE" • • "Contrast enhancement" shall be restricted to "T1" • • "TE" shall be restricted to "shortest" and "in-phase" • • "TR" shall be restricted to "shortest" • • Scan mode "3D non-selective" shall be restricted to "No" • • TFE factor "3D free factor" shall be restricted to "No" • • "startup echoes" shall be restricted to "default" • • "T2prep" shall be restricted to "Yes" and shall not be visible to the user • • "TFE prepulse" shall be restricted to "Invert" and shall not be visible to the user • • TFE factor "turbo direction" shall be restricted to "radial" •

Test steps:

Step	Description	Expected Result
Step 1	Execute SWCMP.Scan.PDFTestAutomation and inspect the results file	The reported number of failed tests related to SWCMPScanValidatorMultiParametricImagingSynt AcPar in the results file is zero.

3.1.45.40. SWCMP.Scan.Validator.MultiSeries.tc1

Test ID	61807
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Reddy, Srikanth (310031195) on 2025-05-21 19:33:49 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-05-19 17:14:04 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2025-05-19 11:41:23 Meaning: SME Approved - (Technical)

Approval Status	Approved
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Description:

This test verifies generation of multiple series

Requirements Coverage:

Req ID	Requirement Tag	Description
97864	SWCMP.Scan.Validator.MultiSeries	The ScanValidator shall create and populate multiple series generated for a scan in the SeriesDescription based on the needs defined in the ScanProtocol.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorMultiSeries in the results file is zero.

3.1.45.41. SWCMP.Scan.Validator.Par.Restriction.tc1

Test ID	56191
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2022-12-20 09:14:37 Meaning: QA Approved - (QA) Signed by: Sharma, Nishant (320059006) on 2022-12-19 15:22:05 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2022-12-19 14:38:05 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies parameter restriction.

Requirements Coverage:

Req ID	Requirement Tag	Description
91335	SWCMP.Scan.Validator.Par.Restriction	Upon request, the scan validator shall make all protocol parameters invisible, except for the following parameters which shall be modifiable: - Stack offcentres - Navigator offcentres and angulations - REST offcentres and angulations - B0 shim offcentres, angulations and sizes
99086	SWCMP.Scan.Validator.Par.Rest	For a Smart Plan Survey, the Scan Validator shall make all protocol

Req ID	Requirement Tag	Description
	restriction.Smart	parameters invisible, except for the following parameters which shall be modifiable: - Stack offcentres and angulations - Coils In addition, the following parameter display restriction shall be in place for Smart Heart survey: - Cardiac synchronization shall not be modifiable. - Cardiac device shall be modifiable when applicable. - Trigger delay shall be modifiable when applicable. (ms) shall be modifiable when applicable. - Respiratory synchronization shall not be modifiable. - Breath hold mode shall be modifiable when applicable.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorParRestriction in the results file is zero. Attach the result file as an objective evidence.

3.1.45.42. SWCMP.Scan.Validator.PartialAcquisition.Par.tc1

Test ID	48541
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-11-06 07:47:13 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-10-30 07:30:22 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-10-28 15:01:56 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Partial Acquisition parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
84836	SWCMP.Scan.Validator.PartialAcquisition.Par.v2	Halfscan	No, Yes (EPI, TFE-EPI or GRASE) ÷ (M2D, MS or (3D ÷ YZ encoding)): default	Scan type = Imaging	Enables scan acceleration through partial acquisition of k-space	N/A
		factor	0.6 - 1.0	Scan mode = 2D or M2D or MS ÷ Halfscan = Yes	Specifies the Halfscan factor	N/A
		factor y	0.6 - 1.0	Scan mode = 3D ÷ Halfscan = Yes	Specifies the halfscan factor in the y direction	N/A
		factor z	0.6 - 1.0	Scan mode = 3D ÷ NSA = 2.0 ÷ Halfscan = Yes	Specifies the halfscan factor in the z direction	N/A
		NSA	0.0 - 32.0 Fractional values are available for: ((cartesian ÷ 3D) or radial	Always	Enables the acquisition of the averages. The scan time changes proportionally	N/A

Req ID	Requirement Tag	Description				
			or MotionFree) Δ (NSA ≤ 5) Otherwise integer values		nally with the specified averages.	
		SMART	no, yes	Acquisition mode = cartesian Δ (NSA ≥ 2)	SMART averaging is a repeated single measurement scan and is therefore also referred to as serial averaging: an initial measurement is made for all values of the phase encoding before the second measurement is commenced. A complete image is thus acquired before proceeding to the next averaging	N/A

Req ID	Requirement Tag	Description				
					step. Since the time between the acquisition of the data is in the order of TR (e.g. 0.5 s - 2 s) with parallel averaging and in the order of minutes with SMART averaging, the latter produces a more pronounced decrease in the intensity of artifact.	
		reversed order	no, yes	Fast Imaging mode = EPI or TFE-EPI Echoes = 1 Halfscan = Yes	Enables a reversed order k-space acquisition.	This parameter is visible when M_OPT.AWOPT_OPTION15 (option bit 496) is enabled.
		Partial NSA	no, yes	Acquisition mode = Cartesian (Scan mode ==	Enables the acquisition of the 2nd NSA	N/A

Req ID	Requirement Tag	Description				
				3D) Û (3D free factor == yes) Û (NSA == 2)	partially (central k-values only).	
		fraction	1.2 - 2.0	Partial NSA = yes	Specifies the partial NSA fraction.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorPartialAcquisitionPar in the results file is zero.

3.1.45.43. SWCMP.Scan.Validator.PartialEcho.Par.tc1

Test ID	62010
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-05-28 21:51:57 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-05-28 18:16:50 Meaning: SME Approved - (Business) Signed by: Pereira, Kaitan (320169354) on 2025-05-28 17:29:50 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description				
92279	SWCMP.Scan.Validator.PartialEcho.Par	Name	Range	Visible when	Description	System / Option restrictions

Req ID	Requirement Tag	Description				
		Partial echo	no, yes	Always	This parameter enables or disables the use of partial echo.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorPartialEchoPar in the results file is zero.

3.1.45.44. SWCMP.Scan.Validator.PostShotPulses.Par.tc1

Test ID	51048
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-15 11:57:32 Meaning: QA Approved - (QA) Signed by: Beck, Gabrielle (nly95705) on 2021-06-10 15:28:34 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:26 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Post Shot pulse parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
84830	SWCMP.Scan.Validator.PostShotPulses.Par	Name	Range	Visible when	Description	System / Option

Req ID	Requirement Tag	Description				
						restrictions
		Drive	Yes, No, Anti	Fast imaging mode = TSE or GRASE	Enable Drive or Anti-Drive	Anti Drive is available for MultiVane-CSENSE when M_OPT.Cs MultiVane Generic (option bit 210) is enabled.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorPostShotPulsesPar in the results file is zero.

3.1.45.45. SWCMP.Scan.Validator.Prepulses.MTC.Par.tc1

Test ID	56662
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2023-02-22 13:58:03 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Shree K, Shruthi (Partner) (320185675) on 2023-02-22 13:09:43 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-02-22 12:43:18 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of MTC parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
91813	SWCMP.Scan.Validator.Prepulses.MTC.Par	MTC	no on resonance off resonance off res. multi	Always	Allows the acquisition of MTC acquisitions, using on-resonance, off-resonance or off-resonance multi pulses.	The range off res. multi is only enabled when the SW option M_OPT. MagnetizationTransferExtended (option bit 26) is present.
		nr repetitions	[1 - 100]	MTC = off res. multi	This parameter determines the number of off-resonance MTC pulses applied within a shot.	N/A
		angle	[1- 1200]	MTC = off res. multi	This parameter determines the angle of each of the pulses.	N/A

Req ID	Requirement Tag	Description				
		duration	[1 - 100]	MTC = off res. multi	This parameter determines the duration of each of the pulses.	N/A
		frequency	[1 - 1000]	MTC = off res. multi	This parameter determines the frequency of each of the pulses.	N/A
		MTC	[no yes]	MTC not equal to no, Dynamic study = individual and dynamic scans >1	Concerns a separate parameter that allows the acquisition of two dynamics, one with MTC enabled, one disabled.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorPrepulsesMTCPAr in the results file is zero.

3.1.45.46. SWCMP.Scan.Validator.Prepulses.REPP.Par.tc1

Test ID	56663
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Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2023-02-22 13:58:04 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Shree K, Shruthi (Partner) (320185675) on 2023-02-22 13:10:27 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-02-22 12:43:19 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of research prepulse parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
91814	SWCMP.Scan.Validator.Prepulse s.REPP.Par					
		Custom prepulse	[no, yes]	Always	The custom prepulse is set on/off.	The range yes is only enabled when the SW option M_OPT. ResearchP repulse (option bit 26) is present.
		pulse shape	[block, sg_300_100_0, wowir, hypsec]	Custom prepulse = yes	This parameter defines the pulse shape.	The range MSDE is only enabled when the SW option M_OPT.MSDE (option

Req ID	Requirement Tag	Description				
						bit 111) is present.
		elements	[1 - 16]	Custom prepulse = yes	Defines the number of elements in the composite pulse.	N/A
		pulse angles	[0.00 - 1800.00]	Custom prepulse = yes	This parameter defines per element the flip angle (in degrees).	N/A
		pulse durs	[0.01 - 50.00]	Custom prepulse = yes	This parameter defines per element the duration (in [ms]).	N/A
		pulse phases	[-180.00 - 180.00]	Custom prepulse = yes	This parameter defines per element the phases (in degrees).	N/A
		pulse intervals	[0.00, 0.064 - 500.00]	Custom prepulse = yes	This parameter defines	N/A

Req ID	Requirement Tag	Description				
					per consecutive elements the interval time (in [ms]). The interval is measured from the end of an element to the start of the next element. For a typical binomial custom pulse these intervals are set to 0.	
		delay time	[0.00 - 10000.00]	Custom prepulse = yes	The time interval between the center of the (composite) pulse to the center of the RF excitation pulse.	N/A
		freq. offset	[-100000.00 - 100000.00]	Custom prepulse = yes	This parameter defines the transmit frequency	N/A

Req ID	Requirement Tag	Description				
					offset (in Hz) relative to the central 1H Larmor frequency .	
		select grad in (uT/m)	[-10000.0 0 - 10000.00]	Custom prepulse = yes	The parameter defines the gradient strength in (uT/m).	N/A
		IR: pre SE only	[no, yes]	Scan technique = IR, MIX and Custom prepulse = yes	The "IR: pre SE only" parameter only has effect for long IR delays and long TR. No means: custom pulses are repeated as many times as possible during a long TR, Yes means: applying the custom pulses fewer times, i.e. only just once	N/A

Req ID	Requirement Tag	Description				
					before the excitation (with 90 degrees pulse) of each slice.	
		dyn custom prepulse (uT/m)	[no, yes]	Custom prepulse = yes and Dynamic study = individual and dyn scans>1	This "dyn custom prepulse" parameter is used as follows: No means: Custom prepulse is active in all dynamics, Yes means: Custom prepulse not active in 1st dynamic and active in all subsequent dynamics.	N/A

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorPrepulsesREPPPar in the results file is zero.

3.1.45.47. SWCMP.Scan.Validator.ReconstructionMatrix.Par.tc1

Test ID	56181
Revision	1

Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-01-23 10:30:14 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2023-01-23 07:46:14 Meaning: SME Approved - (Business) Signed by: Hegde, Shashank (310195826) on 2023-01-20 08:31:39 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test case checks the conflicts on Reconstruction matrix parameter in combination with external control parameter being set to MRTC.

Requirements Coverage:

Req ID	Requirement Tag	Description				
91320	SWCMP.Scan.Validator.ReconstructionMatrix.par	Name	Range	Visible when	Description	System/Option restrictions
		Reconstruction matrix	1, 100000	Always	Allows user to specify the resolution of output image(Pixels)	M_OPT.A WOPT_2K _IMAGING (08) enables user to set value from 0 upto 2048

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorReconstructionMatrixPar in the results file is zero.

3.1.45.48. SWCMP.Scan.Validator.Respiration.Compensation.Par.tc1

Test ID	48540
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-11-06 07:47:18 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-10-30 07:24:14 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-10-28 15:02:01 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Respiration Compensation parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
97886	SWCMP.Scan.Validator.Respiration.Compensation.Par.v4	Name	Range	Visible when	Description	System / Option restrictions
		Respiratory compensation	no, trigger, breath hold, PEAR, gate, retrospective	always	Respiratory compensation techniques for breathing motion.	RetroResp Binned (AWOPT_MR_4D_MR_RT) enables retrospective respiratory compensation in combination with 3D Radial and CS-SENSE.

Req ID	Requirement Tag	Description				
		Respirator y soft gating	no, yes	Scan type = Imaging ù Fast imaging mode = TFE ù Shot mode = MSH ù Dynamic study == contrast phases	Enables respiratory soft gating technique for 4D free breathing	N/A
		Motion detection type	external, intrinsic	Respirator y soft gating = yes or respiratory comp = retrospective with Development SW option	Allows to specify the respiratory detection device. External is the respiratory signal detected from the VitalEye camera or respiratory belt, intrinsic is a k-space navigator acquired before each shot	Development (AWOPT_DEVELOPMENT) enables the parameter for retrospective respiratory binning
		Oversample factor	1.025-2.000	Respirator y soft gating = yes	Allows to specify the oversampling factor for respiratory soft	N/A

Req ID	Requirement Tag	Description				
					gating	
		Oversample factor	1.5-15.0 (for 1 resp phase) 2.8-15.0 (for 2-10 resp phases)	respiratory comp = retrospective	Allows to specify the oversampling factor for retrospective respiratory binning	N/A
		Navigator respiratory comp	no, gate, track, gate and track, trigger, trigger and track	Scan type = Imaging ù Dynamic study != contrast phases	Enables navigator respiratory compensation technique .	This parameter is only enabled when the SW option BodyRespiratoryNavigators (AWOPT_BODYRESPNAVS) Body Res navigators is available. RespNavExtended (AWOPT_OPTION02) and CardiacTriggeredNavigators (AWOPT_COR_ART_IMA) enables the combination of this parameter with cardiac

Req ID	Requirement Tag	Description				
						synchroni- zation. RespNavE xtended (AWOPT_ OPTION02) enables the combinati on of gate, track, gate and track for non- triggered and non- breathhol d scans.
		Preparatio n	full, auto	Navigator respirator y comp != no	Specifies the preparatio n mode	N/A
		Beams	1-3	Navigator respirator y comp != no	Specifies the number of respirator y navigator beams.	RespNavE xtended (AWOPT_ OPTION02) extends the range of 2 and 3
		Direction	FH,AP,LR	Navigator respirator y comp != no	Specifies the direction of the long axis of the navigator	RespNavE xtended (AWOPT_ OPTION02) extends the use of direction AP, LR
		Scale factor	-10.0 - 10	Navigator respirator y comp = Track or	Specifies the scale factor for slice	N/A

Req ID	Requirement Tag	Description				
				gate and track or trigger and track	tracking correction factors	
		Weighted gating	No, Yes	Navigator respiratory comp = gate or gate and track or trigger and track Û Scan mode =3D	Enables the Weighted gating option that allows user specified central and outer k-space profiles	KING (AWOPT_OPTION14) enables the use of this parameter
		Central gate (mm)	0.0 – 100.0	weighted gating = yes	Specifies the central gate width, the gate for the central part of k-space.	N/A
		Central dur (%)	0.0 – 100.0	weighted gating = yes	Specifies the duration of the acquisition of the central k-space	N/A
		Outer gate (mm)	0.0 – 100.0	weighted gating = yes	Specifies the outer gate, the gate for the outer part of k-space	N/A

Req ID	Requirement Tag	Description				
		Gating window	0.0 – 100.0	weighted gating = No	specifies the gating window of the respiratory cycle in which data will be accepted for reconstruction	N/A
		Gating level drift	No, Yes	Navigator respiratory comp = gate or gate and track or trigger or trigger and track	Enables the update of the actual gating level every time data is acquired	N/A
		gating level drift (mm/min)	0 - 10	Gating level drift = Yes	Specifies the gating level drift when gating level drift is set	N/A
		max slices per respiration	1 - 256	Navigator respiratory comp = gate or track or gate and track Scan mode = MS Fast Imaging mode = EPI or TFE-EPI	Specifies the maximum number of slices measured interleaved in one respiration cycle	N/A

Req ID	Requirement Tag	Description				
		Type	Lead, Trail, lead and trail	Navigator respiratory comp = gate or gate and track	Specifies the type of the navigator respiratory compensation	N/A
		minimum TR (ms)	0 - 10000	Navigator respiratory comp = trigger or trigger and track	Specifies the minimum TR	N/A
		trigger delay	Auto, user defined	Navigator respiratory comp = trigger or trigger and track	Specifies the trigger delay	N/A
		trigger delay (ms)	0 - 10000	trigger delay = User defined	Specifies the delay time in ms for user defined trigger delay	N/A
		Number of resp. phases	1, 2, 4, 6, 8, 10	respiratory comp = retrospective	Specifies the number of resp phases used for retrospective respiratory binning	N/A
Application specific restrictions are imposed by the following options:						
RadialImagingExtended (AWOPT_OPTION33) enables Respiratory triggering in combination with Radial.						

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorRespirationCompensationPa r in the results file is zero.

3.1.45.49. SWCMP.Scan.Validator.RetroRespBinned.Par.tc1

Test ID	61984
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-05-28 21:50:56 Meaning: QA Approved - (QA) Signed by: Gorkum, Robbert van (320189046) on 2025-05-28 13:32:06 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-05-27 18:14:19 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of 4DMR-RT, also known as Retrospective respiratory binning.

Requirements Coverage:

Req ID	Requirement Tag	Description
98201	SWCMP.Scan.Validator.RetroRespBinned	4DMR-RT is enabled by respiratory compensation = retrospective. 4DMR-RT = Retrospective respiratory binning in RT; RT = radiotherapy The following general parameter restrictions are in place for 4DMR-RT: • "acceleration" can only be set to "CS-SENSE" (via conflicts) • "scan mode" can only be set to "3D" (via conflicts) • "cs-factor" shall be restricted to "2.56" and shall not be visible to the user • "Stacks" can only have value "1" • "slice orientation" shall be restricted to "transverse" • "shot mode" shall be restricted to "single-shot" • "technique" shall be restricted to "FFE" • "fast imaging mode" shall be restricted to "TFE" • "contrast enhancement" shall be restricted to "no, T1, T2 Enhanced" • "acquisition mode" shall be restricted to "radial" • "TFE prepulse" shall be restricted to "no, T2 user defined"

Req ID	Requirement Tag	Description
		• "MSDE refocusing pulses" shall be restricted to "2" and shall not be visible to the user • "MSDE refocusing type" shall be restricted to "adiabatic" and shall not be visible to the user • "venc" shall be restricted to "0.0, 0.0, 0.0" and shall not be visible to the user • "fat suppression" shall be restricted to "no, SPAIR, SPIR" • "water suppression" shall be restricted to "no, SPAIR, SPIR" • "T2 Prep" shall be restricted to "no" and shall not be visible to the user • "mDIXON" shall be restricted to "no, yes" • "mDIXON acquisition type" shall be restricted to "multi-echo" • "NSA" shall be restricted to "1" and shall not be visible to the user • "Reconstruction mode" shall be restricted to "Immediate, delayed" • "Cardiac synchronization" shall be restricted to "no" and shall not be visible to the user • "Navigator respiratory comp" shall be restricted to "no" and shall not be visible to the user • "1D Tracking" shall be restricted to "no" and shall not be visible to the user • "fMRI echo stabilization" shall be restricted to "no" and shall not be visible to the user • "Motion smoothing" shall be restricted to "no" and shall not be visible to the user • "MTC" shall be restricted to "no" and shall not be visible to the user • "multi-chunk" shall be restricted to "no" and shall not be visible to the user • "external control" shall be restricted to "no" and shall not be visible to the user • "interactive positioning" shall be restricted to "no" and shall not be visible to the user • "SyntAc" shall be restricted to "no" and shall not be visible to the user • "Angio/contrast enhancement" shall be restricted to "no" and shall not be visible to the user • "Quantitative flow" shall be restricted to "no" and shall not be visible to the user • "Arterial Spin Labeling" shall be restricted to "no" and shall not be visible to the user • "SmartPlan survey" shall be restricted to "no" and shall not be visible to the user • "SmartScout" shall be restricted to "no" and shall not be visible to the user • "Viewable Raw data" shall be restricted to "no" and shall not be visible to the user • "MIP/MPR" shall have a conflict with "yes" • "Elastic Imaging Registration" shall be restricted to "no" and shall not be visible to the user • "SWIp" shall be restricted to "no" and shall not be visible to the user • "Scan type" shall be restricted to "Imaging" • "Nucleus" shall be restricted to "1H" • "Diffusion mode" shall be restricted to "no" and shall not be visible to the user • "T1 mapping" shall be restricted to "no" and shall not be visible to the user • "partial echo" shall be restricted to "no" and shall not be visible to the user • "shifted echo" shall be restricted to "no" and shall not be visible to the user • "MRE enable" shall be restricted to "no" and shall not be visible to the user • "Dynamic study" shall be restricted to "no" and shall not be visible to the user

Req ID	Requirement Tag	Description
		The following parameter restrictions shall be in place for delayed reconstruction: • "Number of resp phases" shall not be visible to the user • "oversample factor" shall not be visible to the user • "functional images" shall be visible to the user • "resp peak coverage" shall be visible to the user • "resp peak factor" shall be visible to the user The following parameter restrictions shall be in place for contrast enhancement = "T2 enhanced": • "profile order" shall be restricted to "low-high" • "TFE prepulse" shall be restricted to "black-blood" and shall not be visible to the user • "bb pulse type" shall be restricted to "MSDE" and shall not be visible to the user • "flip angle" shall be restricted to "1.0 – 15.0" • "start up profiles" shall be restricted to "user defined" and "0" and shall not be visible to the user • "MSDE TE prep" shall be restricted to "user defined" and shall not be visible to the user • "MSDE echo time" shall be restricted to "70.0"ms and shall not be visible to the user • "mDIXON" shall have a conflict with "yes" The following parameter restrictions shall be in place for "TFE prepulse" = "T2 user defined" (available only when contrast enhancement = no, T1): • "MSDE TE prep" selection shall be available • "MSDE echo time" shall be available • "mDIXON" shall have a conflict with "yes" Under the software option Development (AWOPT_DEVELOPMENT), the following are enabled: • "slice orientation" shall also enable "sagittal, coronal". • "contrast enhancement" shall also enable "T2, balanced" • "motion detection type" shall be enabled with "intrinsic, external"

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorRetroRespBinnedPar in the results file is zero.

3.1.45.50. SWCMP.Scan.Validator.ScanMode.Par.tc1

Test ID	56179
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-01-20 09:59:13 Meaning: QA Approved - (QA)

	Signed by: Hegde, Shashank (310195826) on 2023-01-20 08:30:58 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2023-01-19 11:05:54 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test case checks the conflicts on scan mode parameter in combination with external control parameter being set to MRTC.

Requirements Coverage:

Req ID	Requirement Tag	Description				
91325	SWCMP.Scan.Validator.ScanMode.Par					
		Name	Range	Visible when	Description	System / Option restrictions
		Scan Mode	Always: 2D, 3D, - MS, - M2D Only when 1D tracking = yes 1D	Scan type = Imaging	This parameter enables user to select the scan mode	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorScanModePar in the results file is zero.

3.1.45.51. SWCMP.Scan.Validator.SDSE.Par.tc1

Test ID	61373
Revision	1

Product Model	Any
Gradient System	Any
Options	
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-04-16 10:21:37 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2025-04-15 22:37:57 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-04-07 11:18:20 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description				
97269	SWCMP.Scan.Validator.SDSE.Par	Name	Range	Visible when	Description	System/Option restrictions
		EX_SDSE.enable	no, yes	Scan Control Parameter 'Enable ScanDefinition-ScanExecution UI' is Yes.	Enables the SDSE UI.	Yes is only available when M_OPT.DevelopmentMode is present.
		The default value of the Scan Control Parameter 'Enable ScanDefinition-ScanExecution UI' is No. This parameter is only visible when M_OPT.DevelopmentMode (SYS.Option.Bit 69) is present				

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSDSEPar in the results file is zero.

3.1.45.52. SWCMP.Scan.Validator.SENSE.Par.tc1

Test ID	48538
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-11-06 07:47:20 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Bhosale, Manoj (310194149) on 2020-10-30 07:33:38 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2020-10-28 15:02:04 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator SENSE parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
93525	SWCMP.Scan.Validator.SENSE.Par.v1	SENSE	'No', 'Yes'	(Scan type = Imaging ÷ (Acquisition mode = Cartesian or Spiral or Multivane) or (Acquisition mode = Radial ÷ 3D Isotropic = No)) or (Scan type	This parameter disables/enables the sense technique	SENSE technique can be used only when a 'sense_approved' coil is connected . SENSE for spectroscopic imaging is only allowed when SW

Req ID	Requirement Tag	Description				
				= Spectroscopy ù Scan mode != SV)		option bit 15: SpectroSE NSE is enabled. M_OPT.A WOPT_OPTION37 (option bit 474) enables the output of EPI R/I/P images in combination with SENSE > 1
		M reduction (**)	1.0 - 32.0	Scan type = Spectroscopy ù SENSE = Yes	This parameter specifies the acceleration factor in the measurement direction if (GeometryOrientation = transverse): (**) is replaced by (RL) if FoldoverDirection = 'AP' (AP) if FoldoverDirection =	N/A

Req ID	Requirement Tag	Description				
					'RL' if (GeometryOrientation = coronal): (**) is replaced by (FH) if FoldoverDirection = 'RL' (RL) if FoldoverDirection = 'FH' if (GeometryOrientation = sagittal): (**) is replaced by (AP) if FoldoverDirection = 'FH' (FH) if FoldoverDirection = 'AP'	
		P reduction (**)	1.0 – 32.0	SENSE = Yes ù Acquisition mode != Spiral ù Scan mode !=	This parameter specifies the acceleration factor in the phase	N/A

Req ID	Requirement Tag	Description				
				1D	direction (**) is replaced by if (Radial): (AP) if (Cartesian): (RL) if fold-over direction = 'RL' (AP) if fold-over direction = 'AP' (FH) if fold-over direction = 'FH' if (MultiVane): (AP/RL) if slice orientation = 'transverse' (RL/FH) if slice orientation = 'coronal' (AP/FH) if slice orientation = 'sagittal'	
		S reduction (**)	1.0 - 32.0	SENSE = Yes ù Scan mode = 3D ù	This parameter specifies the acceleration	Application specific restrictions are

Req ID	Requirement Tag	Description
		O-MAR ! = SEMAC+V AT) on factor in the slice direction (* *) is replaced by (FH) if Geometry Orientatio n = 'transvers e' (AP) if Geometry Orientatio n ='coronal' (RL) if Geometry Orientatio n ='sagittal' imposed by the following options: M_OPT.A WOPT_OP TION37 (option bit 474) enables this paramete r when trajectory mode is set to 'EPI' and EPI shots is less than 4/P_reduc tion

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorSENSEPar in the results file is zero.

3.1.45.53. SWCMP.Scan.Validator.SmartPlanSurvey.Par.tc1

Test ID	41071
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:22:01 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan

	(310134237) on 2019-05-20 11:12:39 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:13:10 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies availability of smart plan survey parameters

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Parameter name	Range	Visible when	Description	System/Options restriction
97671	SWCMP.Scan.Validator.SmartPlanSurvey.Par.v1	SmartPlan survey	no, brain, knee, C-spine, L-spine, shoulder, breasts, heart global, heart local, research, heart	always	This parameter indicates that this scan must be used by the SmartExam package for automatically planning smart geometries for a certain anatomy.	"brain" is visible when SW option SmartBrain (AWOPT_SMART_BRAIN) is available. "knee" is visible when SW option SmartKnee (AWOPT_SMART_KNEE) is available. "C-Spine and L-Spine" are visible when SW option SmartSpine (AWOPT_SMART_S

						PINE) is available. "shoulder" is visible when SW option SmartShoulder (AWOPT_SMART_SHOULDER) is available. "breasts" is visible when SW option SmartBreast (AWOPT_SMART_BREAST) is available. "heart global and heart local" are visible when SW Option SmartHeartExtended (AWOPT_OPTION27) is available. "heart" is visible when SW option SmartHeart (AWOPT_MR_SMA
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						RTHEART) is available.
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Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSmartPlanSurveyPar in the results file is zero.

3.1.45.54. SWCMP.Scan.Validator.SmartSpeedPreciseReconVoxel.Par.tc1

Test ID	60702
Revision	2
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2025-02-18 17:14:11 Meaning: QA Approved - (QA) Signed by: Koonen, Jos (nly95806) on 2025-01-14 13:09:36 Meaning: SME Approved - (Technical) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2025-01-14 11:27:03 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the changes to recon voxel size when using SmartSpeed Precise.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSmartSpeedPreciseReconVoxel in the results file is zero. Attach the result file as an objective evidence.

3.1.45.55. SWCMP.Scan.Validator.Spectro.1H.SpectralEditing.BASING.Par.tc1

Test ID	47852
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:14 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:27 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:23:59 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of parameters for water or fat suppression using the BASING spectral editing technique.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
78598	SWCMP.Scan.Spectro.1H.SpectralEditing.BASING.Par	BASING pulse	- no, - fat sup, - water sup, - water +fat sup	Scan type = Spectroscopy Nucleus = 1H	This parameter enables spectral editing techniques inside the base sequence: frequency selective suppression (e.g. of water	BASING pulse MEGA is only enabled when SW option bit 143: Spectral_Editing is enabled. See Req ID 78597.

Req ID	Requirement Tag	Description				
					and/or fat). Simultaneous water+fat suppression is achieved by a dual-band chemical shift selective pulse.	
		window (Hz)	15 - 7000	BASING pulse != no	This parameter determines the bandwidth of the frequency selective editing pulses	None
		flip angle (deg)	1 - 270	BASING pulse != no	This parameter determines the flipangle of the frequency selective suppression pulse.	None
		frequency offset	- default - user defined	BASING pulse != no	This parameter allows the user to set the frequency offset of the suppression	None

Req ID	Requirement Tag	Description				
					ssion pulse to the default (water 0 ppm) or fat -3.35 ppm) resonance or selecting a user defined offset.	
		offset (PPM)	-10 - 10	frequency offset = user defined	This parameter determines the frequency offset in ppm of the editing pulse with respect to either water or fat.	None
		inv. delay (ms)	5 - 200 000	BASING pulse != no	This parameter determines the delay time between the suppression pulse and the acquisition, and is the time available	None

Req ID	Requirement Tag	Description				
					for the rephasing gradient).	
		dual Basing pulses	- no - yes	BASING pulse != no	This parameter allows the user to add a second frequency selective suppression pulse to further improve frequency selective suppression.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectro1HSpectralEditingBasingPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.56. SWCMP.Scan.Validator.Spectro.1H.SpectralEditing.MEGA.Par.tc1

Test ID	47853
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-09-16 10:28:16 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant

	(320059006) on 2020-09-16 09:53:30 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:01 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of parameters for MEGA (spectral editing) functionality.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
78597	SWCMP.Scan.Spectro.1H.SpectralEditing.MEGA.Par					
		BASING/MEGA pulse	- no - fat sup, - water sup, - water +fat sup, - MEGA	Scan type = Spectroscopy Û Nucleus = 1H	This parameter enables spectral editing techniques inside the base sequence	MEGA is only allowed when SW option bit 143: Spectral_Editing: is enabled. MEGA is only allowed on 3T DDAS systems.
		window (Hz)	15 - 7000	BASING/MEGA pulse = MEGA	This parameter determines the bandwidth of the frequency selective editing pulses	None

Req ID	Requirement Tag	Description				
		MEGA ON offset	- GABA (1.89 ppm), - user defined	BASING/ MEGA pulse = MEGA	This parameter determines the frequency target of each editing RF pulse in the sequence, when the "ON" sequence is played.	None
		frequency (PPM)	-10 - 10	MEGA ON offset = user defined	This parameter determines the frequency target of each editing RF pulse in the sequence, when the "ON" sequence is played.	None
		MEGA OFF offset	- default - user defined	BASING/ MEGA pulse = MEGA	This parameter determines the frequency target of each editing RF pulse in the sequence,	None

Req ID	Requirement Tag	Description				
					when the "OFF" sequence is played.	
		frequency (PPM)	-10 - 10	MEGA OFF offset = user defined	This parameter determines the frequency target of each editing RF pulse in the sequence, when the "OFF" sequence is played.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectro1HSpectralEditingMegaPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.57. SWCMP.Scan.Validator.Spectro.1H.WaterSuppression.Par.tc1

Test ID	47854
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:18 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:32 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:02 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of WaterSuppression parameters for Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
77480	SWCMP.Scan.Spectro.1H.Water Suppression.Par	Water suppression	- no, - VAPO R, - excitation, - inversion, - MOIST, - CHESS	Scan type = Spectroscopy Nucleus = 1H	This parameter enables or disables water suppression and allows selection of one of the available suppression techniques.	VAPOR is only allowed when option bit 142: sLASER-VAPOR is enabled.
		WS window (Hz)	5 - 7000 <i>(minimal value can be restricted by time limitation)</i>	Water suppression != no	This parameter determines the bandwidth of the frequency	None

Req ID	Requirement Tag	Description				
			s within the WS technique)		selective water suppression pulses in Hz.	
		WS prescan	• auto, • user defined • no	Water suppression != no	This parameter enables or disables the prescan Automated Water Suppression Optimisation (AWSO).	None
		WS second pulse angle	0 - 750	Water suppression = excitation	This parameter allows optimisation of the second watersuppression RF pulse to become effectively 90 degrees.	None
		WS delay (ms)	5 - 200 000	Water suppression = inversion	This parameter allows optimisation of the inversion delay (time between RF inversion	None

Req ID	Requirement Tag	Description				
					and RF excitation pulse of the acquisition sequence)	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectro1HWaterSuppression Par in the results file is zero. Attach the result file as an objective evidence.

3.1.45.58. SWCMP.Scan.Validator.Spectro.Basic.Par.tc1

Test ID	47855
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-09-16 10:28:20 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:35 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:04 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of WaterSuppression parameters for Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
77607	SWCMP.Scan.Spectro.Basic.Par	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		Samples	64 - 16384	Scan Type = Spectroscopy	This parameter allows the user to define the spectral resolution by setting the nr of points for a given Spectral BW.	None
		Spectral BW (Hz)	500 - 8000	Scan Type = Spectroscopy	Applied BandWidth for the Received Signal.	None
		NSA	1 - 65 000	Always	This parameter defines the number of acquired spectra that will be averaged to enhance SNR.	None

Req ID	Requirement Tag	Description				
82104	SWCMP.Scan.Validator.Aquisition.Type.Scantype.Par	Name	Range	Visible when	Description	System / Option restrictions
		Scan type	- Imaging - Spectroscopy	Always	This parameter enables switching between features that can be used for imaging or spectroscopy.	Spectroscopy is only allowed when option bit 50: Spectro is enabled.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroBasicPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.59. SWCMP.Scan.Validator.Spectro.Encoding_SENSE_TSI.Par.tc1

Test ID	47864
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:22 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:37 Meaning: SME Approved -

	(Technical)::Bulk Transitions
	Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:05 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameter for encoding (CSI) and the related SENSE and TSI functionalities for Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
77631	SWCMP.Scan.Spectro.Encoding _SENSE_TSI.Par	Name	Range	Visible when	Description	System / Option restrictions
		Scan mode	Always: - SV - no encoding 1D CSI 2D CSI 3D CSI Extended range for Method = SE, sLASER, PRESS or STEAM: - MS - M2D	Scan type = Spectroscopy	This parameter allows : - SV (no encoding) - CSI (chemical shift imaging) by encoding in: - one direction (1D), - two directions (2D), - or three directions (3D). - MS: multiple	3D CSI is only allowed when SW option bit 11: <i>Spectro3D</i> is enabled.

Req ID	Requirement Tag	Description				
					spectroscopic imaging slices are scanned in an interleaved sequence M2D: multiple spectroscopic imaging slices are scanned one after the other.	
		SENSE	- no, - yes	Encoding Mode != SV (no encoding) ÷ Nucleus = 1H	SENSE stands for SENSitivity Encoding This parameter allows the user to reduce scan time in spectroscopic imaging (2D CSI, 3D CSI and TSI) by skipping lines in k-space while acquiring synergy data in parallel.	SENSE for spectroscopic imaging is only allowed when SW option bit 15: <i>SpectroSENSE</i> is enabled.

Req ID	Requirement Tag	Description				
					The missing information is acquired by arrays of receiver coils with known coil sensitivities.	
		SENSE reduction factor		SENSE = yes	This parameter allows the user to apply SENSE reduction in one or both in-plane phase encoding directions (M-reduction and P-reduction).	None
		Fast Imaging mode	For Encoding mode = SV (no encoding) • none _____ For Encoding mode != SV (no encoding): • none	Method = SE, STEAM, PRESS, or sLASER	This parameter allows the user to reduce scantime in case of SpinEcho based techniques by enabling turbo	None

Req ID	Requirement Tag	Description				
			• TSI		spectroscopic imaging or multi slice imaging.	
		TSI factor	2 - 64	Fast Imaging mode = TSI	This parameter allows the user to set the nr of echoes per excitation.	None
		TSI echo spacing	0.5 - 300	Fast Imaging mode = TSI	This parameter allows the user to define the delay between two echoes in ms.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroEncodingSenseTsiPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.60. SWCMP.Scan.Validator.Spectro.MethodAndVolumeSelection.Par.tc1

Test ID	47856
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.

Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:24 Meaning: QA Approved - (QA)::Bulk Transitions
	Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:40 Meaning: SME Approved - (Technical)::Bulk Transitions
	Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:06 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of Volume Selection parameters for Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
77481	SWCMP.Scan.Spectro.MethodAndVolumeSelection.Par	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		Method	For Nucleus = 1H: - sLASER - PRESS - STEAM - SE - Pulse-Acq - ISIS For Nucleus = other than 1H: - Pulse-Acq - ISIS	Scan type = Spectroscopy	This parameter determines the method used for VOI selection by defining the nr of slice selective gradients in combination with the RF excitation and potential	sLASER is only allowed when option bit 142: sLASER-VAPOR is enabled

Req ID	Requirement Tag	Description				
					refocussing pulses.	
		Volume selection	For Method = PRESS, STEAM or sLASER • volume For Method = SE: • none, • slice, • bar For Method = Pulse Acq: • none, • slice For method = ISIS: • double volume, • volume, • slice, • bar	Scan type = Spectroscopy	This parameter determines the type of Volume Of Interest (VOI) by defining the nr of directions in which selective gradients in combination with the RF pulses are applied. • none: no slice selective gradients (excitation of entire coil penetration dept), • slice: slice selective pulses in 1	None

Req ID	Requirement Tag	Description				
					direction, • bar: slice selective pulses in 2 directions, • volume: slice selective pulses in 3 directions, • double volume: slice selective pulses in 3 directions, the ISIS subtraction scheme allows for two volumes to be selected.	
		Mixing time	• minimum • user	Method = STEAM	This parameter allows	None

Req ID	Requirement Tag	Description				
			defined		the user to set the mixing time to the shortest possible or to a user specified value.	
		(ms)	1 - 10 000 Note: minimal range is restricted by sequence objects.	Mixing time = user defined	This parameter allows the user to define the mixing time for STEAM in ms.	None
		strong crushers	- no, - yes.	Method = sLASER	This parameter allows the use of stronger crushers for the sLASER sequence in areas with higher B0 inhomogeneities (at the cost of short TE).	None
		Acq. delay	- shortest - user defined	Method = Pulse - Acq or ISIS	This parameter allows the user to use the	None

Req ID	Requirement Tag	Description				
					shortest or a tailored acquisition delay.	
		(ms)	0.1 - 50 000	Acq. delay = user defined	This parameter allows the user to specify the acquisition delay in milliseconds.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroMethodAndVolumeSelectionPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.61. SWCMP.Scan.Validator.Spectro.MN.Decoupling.Par.tc1

Test ID	47859
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:26 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:42 Meaning: SME Approved - (Technical)::Bulk Transitions

	Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:38:34 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of parameters for proton Decoupling during MN Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
77429	SWCMP.Scan.Spectro.MN.Decoupling.Par	Proton Decoupling	- no - Broad band Waltz-4 - Broad band Waltz-16 - Narrow band	Scan type = Spectroscopy Nucleus = 31P or 13C	This parameter allows the user to apply 1H decoupling during MN signal acquisition. no: no decoupling applied Broadband: Targeting decoupling to wide 1H frequency range by Waltz-4 and Waltz-16 decoupling schemes. Narrow: Targeting decoupling	None

Req ID	Requirement Tag	Description				
					g to a specific 1H resonance frequency .	
		offset freq (PPM)	-25 - 25	Proton Decoupling != no	This parameter allows the user to set the 1H decoupling offset frequency (or centre frequency) to a offset with respect to the water frequency in ppm.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroMnDecouplingPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.62. SWCMP.Scan.Validator.Spectro.MN.NOE.Par.tc1

Test ID	47985
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar

	(320098697) on 2020-09-16 10:28:28 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:44 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:07 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameter to apply 1H NOE Presaturation pulses for MN Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
81979	SWCMP.Scan.Spectro.MN.NOE.Par	Name	Range	Visible when	Description	System / Option restrictions
		NOE Prepulse	- no - broad band - narrow band - transient - times hared	Scan type = Spectroscopy Nucleus = 31P or 13C	This parameter enables a pre-saturation NOE pulse-train preceding the base sequence. Aim: to selectively saturate coupled 1H spins yielding an increase in the coupled MN	None

Req ID	Requirement Tag	Description				
					spin.	
		offset freq 1H (PPM)	-25 - 25	NOE prepulse ! = no	This parameter allows the user to define the target 1H frequency offset.	None
		mix time (ms)	1 - 50 000 (min value limited by other sequence parameter values)	NOE prepulse ! = no	This parameter allows the user to define the duration of the NOE pulse train.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroMnNoePar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.63. SWCMP.Scan.Validator.Spectro.ParameterSeries.Par.tc1

Test ID	47860
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhharappa, Deepak Shekhar

	(320098697) on 2020-09-16 10:28:30 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:47 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:08 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameter to Prepulses for Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
81433	SWCMP.Scan.Spectro.ParameterSeries.Par	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		Parameter series	- no - repetition time - inversion time - flip angle ----- Extended range, only available for: Method = STEAM ----- Extended range, only	Scan mode = Spectroscopy	This parameter allows the user to acquire a series with variable flipangle, inversion time, repetition time or decoupling frequency .	None

Req ID	Requirement Tag	Description				
			available for: Proton decouplin g != no: • decou ple freq			
		nr TEs ----- nr TRs ----- nr TIs ----- nr FAs ----- nr TMs ----- nr decouplin g offsets	1 - 128	Parameter series = echo time ----- Parameter series = repetition time ----- Parameter series = inversion time ----- Parameter series = flip angle ----- Parameter series = mixing time ----- Parameter series = decouple freq	This paramete r allows the user to define the nr of steps in the paramete r series.	None
		series spacing	For Parameter series = echo time,	Scan Type = Spectrosc opy	This paramete r allows the user	None

Req ID	Requirement Tag	Description				
			repetition time or inversion time: - evenly - logarithmic, - user defined		to define linear, logarithmic or user defined steps in a parameter series.	
			----- ----- For Parameter series = flip angle, mixing time or decouple freq - evenly - user defined			
		starting TE (ms) ----- starting TR (ms) ----- starting TI (ms) ----- starting FA (deg) ----- starting TM (ms) ----- starting dec offset (PPM)	0 - 10 000 ----- -25 - 25	Parameter series = echo time ----- ----- Parameter series = repetition time ----- ----- Parameter series = inversion time ----- ----- Parameter series = flip angle	This parameter allows the user to define the first value in a parameter series. Note: minimal starting points are handled by conflicts.	None

Req ID	Requirement Tag	Description				
				----- ----- Parameter series = mixing time ----- ----- Parameter series = decouple freq		
		ending TE (ms) ----- ending TR (ms) ----- ending TI (ms) ----- ending FA (deg)	1 - 10 000	Parameter series = echo time ` series spacing = logarithmic ----- ----- Parameter series = repetition time ` series spacing = logarithmic ----- ----- Parameter series = inversion time ` series spacing = logarithmic ----- ----- Parameter series = flip angle ` series	This parameter allows the user to define the final value in an parameter series.	None

Req ID	Requirement Tag	Description				
				spacing = logarithmic		
		est. T2 (ms) ----- est. T1 (ms)	1 - 10 000	Parameter series = echo time Δ series spacing = logarithmic ----- Parameter series = (repetition time or inversion time) Δ series spacing = logarithmic	This parameter allows the user to specify a target T1 or T2 that is used to optimise the logarithmic stepsizes.	None
		inv. bw (Hz)	50 - 3000	Parameter series = inversion time Δ series spacing = logarithmic	This parameter allows the user to set the bandwidth of the inversion pulse.	None
		inv. offset (Hz)	- 5000 - 5000	Parameter series = inversion time Δ series spacing = logarithmic	This parameter allows the user to set the offset of the inversion pulse.	None
		TE spacing (ms)	0 - 5000	Parameter series =	This parameter	None

Req ID	Requirement Tag	Description					
		----- TR spacing (ms)		echo time Ù series spacing = evenly ----- ----- Parameter series = repetition time Ù series spacing = evenly ----- ----- Parameter series = inversion time Ù series spacing = evenly ----- ----- Parameter series = flip angle Ù series spacing = evenly ----- ----- Parameter series = mixing time Ù series spacing = evenly ----- ----- Parameter series = decouple offset Ù series spacing = evenly		r allows the user to specify the stepsize for a linear (evenly spaced) paramete r series.	

Req ID	Requirement Tag	Description				
		TE values (ms)	0 - 10 000	Parameter series = echo time	This parameter allows the user to define a series of 128 user defined values for TE, TR, TI, FA or decouple frequency .	None
		----- TR values (ms)		Û series spacing = user defined	----- ----- Note: minimal allowed values are defined by conflicts.	
		----- TI values (ms)		Parameter series = repetition time		
		----- FA values (deg)		Û series spacing = user defined		
		----- TI values (ms)		Parameter series = inversion time		
		----- dec offset values (PPM)	----- -25 - 25	Û series spacing = user defined		
				----- Parameter series = flip angle		
				Û series spacing = user defined		
				----- Parameter series = mixing time		
				Û series spacing = user defined	----- ----- This parameter allows the user	

Req ID	Requirement Tag	Description				
				----- Parameter series = decouple offset Û series spacing = user defined	to define a series of 128 user defined values for decouple frequency .	
		Nr dummy scans	1 - 128	Parameter series != no	This parameter allows the user to play a nr of dummy scans before each of the series steps in order to guarantee steady state.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroParameterSeriesPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.64. SWCMP.Scan.Validator.Spectro.Presaturation.Par.tc1

Test ID	47858
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna

	Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:32 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:49 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:09 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameter to apply Presaturation for Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
81786	SWCMP.Scan.Spectro.Presaturation.Par	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		Saturation prepulse	- no, - yes	Scan type = Spectroscopy	This parameter enables a pre-saturation pulse-train preceding the base sequence to selectively saturate signals.	None
		offset freq (PPM)	- 100 - 100	Saturation prepulse ! = no	This parameter allows the user to define the frequency offset of	None

Req ID	Requirement Tag	Description				
					the prepulse(s).	
		pulse angle (deg)	100 - 5000	Saturation prepulse ! = no	This parameter allows the user to set a relative RF pulse angle for pulses within the presaturation train.	None
		window (Hz)	0 - 7 000	Saturation prepulse ! = no	This parameter defines the bandwidth of the saturation pulses, and is inversely related to the duration of the total saturation RF pulse train.	None
		sideband (Hz)	100 - 10 000	Saturation prepulse ! = no	This parameter allows the user to define minimise sidebands by changing the interval	None

Req ID	Requirement Tag	Description				
					between two saturation RF pulses.	
		delay (ms)	0 - 10 000	Saturation prepulse ! = no	This parameter allows the user to define the delay between the end of the saturation pulsetrain and the start of the excitation RF pulse.	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroPresaturationPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.65. SWCMP.Scan.Validator.Spectro.SetPulseShape.Par.tc1

Test ID	47857
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2020-09-16 10:28:34 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:53:54 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:10 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameter to select specific RF pulse shapes for specific Spectroscopy methods.

Requirements Coverage:

Req ID	Requirement Tag	Description				
81425	SWCMP.Scan.Spectro.SetPulse Shape.Par	<i>Name</i>	<i>Range</i>	<i>Visible when</i>	<i>Description</i>	<i>System / Option restrictions</i>
		RF pulse set	For Method = SE, PRESS, STEAM: - normal - sharp - classic long - classic short _____ _____ For (Method = ISIS) or (Method = Pulse-Acquire ÷ Volume = None):	Method = SE, PRESS, STEAM or ISIS or Method = Pulse-Acquire ÷ Volume = None	This parameter allows the use of specific RF pulse shapes: In case of spin echo based techniques (SE, PRESS, STEAM) this parameter defines the	None

Req ID	Requirement Tag	Description					
			• normal • block • adiabatic			RF pulse shape for both the excitation and refocusing RF pulses . • For FID based techniques (Pulse Acquire and ISIS) only the excitation RF pulse shape is defined. • normal: Pulses with small bandwidth used allowing shortest TE time • sharp: Pulses with large bandwidth and sharper	

Req ID	Requirement Tag	Description				
					excitation slice profile • classic long: Pulses used are symmetric pulse. Suitable for longer TE • classic short: Pulses used are symmetric pulse. Suitable for shorter TE • block: Large BW excitation pulse used for non selective FID acquisitions • adiabatic: Used for non selective acquisitions and/ or FID imaging sequences	

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorSpectroSetPulseShapePar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.66. SWCMP.Scan.Validator.T1Mapping.Par.tc1

Test ID	56606
Revision	1
Product Model	Any
Gradient System	
Options	N.A.
Signatures	Signed by: Moore, Liz (gbr01023) on 2023-03-09 13:06:32 Meaning: QA Approved - (QA) Signed by: Bhosale, Manoj (310194149) on 2023-02-09 09:35:29 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2023-02-08 12:48:01 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies the availability of the T1 Mapping parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
91736	SWCMP.Scan.Validator.T1Mapping.Par	Name	Range	Visible when	Description	System / Option restrictions
		T1 Mapping	no, cardiac native, cardiac enhanced, cardiac user	Always	Allows user to request for T1 maps	The cardiac native and cardiac enhanced range values

Req ID	Requirement Tag	Description					
			defined, MSK			require software option M_OPT.T1 CardiacBasic (100) The cardiac user defined range value require software option M_OPT.T1 CardiacAdvanced (101) The MSK range value require software option M_OPT.T1 MSK (104)	
		acq. interval type	beats, seconds	T1 mapping = cardiac user defined	Specifies that the duration of the acquisition intervals is measured in heart beats or seconds.		
		acq. intervals	1-10	T1 mapping = cardiac user defined or MSK	Specifies the actual duration (in beats or seconds) of each acquisition		

Req ID	Requirement Tag	Description				
					n interval in the T1 mapping acquisition scheme. Maximum of 3 intervals can be specified.	
		pause interval type	beats, seconds	T1 mapping = cardiac user defined	Specifies that the pause of the acquisition intervals is measured in heart beats or seconds.	
		pause interval	1-10	T1 mapping = cardiac user defined or MSK	Specifies the actual duration (in beats or seconds) of each pause interval in the T1 mapping acquisition scheme.	

Test steps:

Step	Description	Expected Result
Step 1	Execute SWCMP.Scan.PDFTestAutomation and inspect the result file.	The reported number of failed tests related to SwCmpScanValidatorT1MappingPar in the results file is zero.

3.1.45.67. SWCMP.Scan.Validator.TFEPrepulse.Par.tc1

Test ID	57107
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2023-07-11 13:59:56 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2023-06-21 08:29:18 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2023-06-21 07:43:51 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of TFE prepulse parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
98145	SWCMP.Scan.Validator.TFEPrepulse.Par.v2	Name	Range	Visible when	Description	System/Option restrictions
		TFE prepulse	Always: No, invert, saturate, black-blood if (respiratory compensation = retrospective and contrast enhancement != T2 enhanced) :	Fast imaging mode = TFE	Adds a prepulse to the sequence.	Black-blood is available when BlackBlood (AWOPT_BLACKBLOOD) or MSDE (AWOPT_3D_BLACK_BLOOD) is enabled.

Req ID	Requirement Tag	Description				
			T2 user defined			
		pulse type	Standard, high bandwidth	TFE prepulse = invert	Changes the inversion pulse type.	high bandwidth is available when WBLGE (AWOPT_MR_WBLGE) is enabled.
		offset	-1500-1500	Pulse type = high bandwidth	Changes the center frequency of the high bandwidth pulse, used for patients with cardiac implants.	N/A.
		pulse type	Standard, composite	TFE prepulse = saturate	Changes the saturation pulse type.	N/A.
		slice selection	No, yes	TFE prepulse = saturate or invert	Make pulse slice selective.	N/A.
		shared	No, yes	(TFE prepulse = saturate or invert) and ((Scan mode = MS or Stacks >1)	Normally the prepulse is performed before each	Shared prepulse is available when CardiacPerfusion (AWOPT_

Req ID	Requirement Tag	Description				
				and slice selection = no), or (phases > 1 or Number of heart phases = maximum) VAL01_CARD->phase_control == MPUCARD_HPC_MAXIMUM) EX_CARD->hear_phase_control	shot, with shared = yes the prepulse is performed each cycle. Can be used in case there are multiple shots within one cycle.	CARDPERFUSION) is enabled.
		delay	Shortest, user defined	TFE prepulse = invert or saturate (or black-blood and bb pulse type = default)	Delay after TFE prepulse.	N/A.
		(ms)	10-5000	Delay = user defined	Time in ms.	N/A.
		PSIR	No, yes	TFE prepulse = invert	Phase-Sensitive Inversion Recovery (PSIR), retain phase information to make a	PSIR is available when CardiacTriggeredIR (AWOPT_CARDIAC_TRIGGERED_IR) is enabled.

Req ID	Requirement Tag	Description				
					distinction between negative and positive magnitude signal during inversion recovery.	
		bb pulse type	Always: MSDE if (scan mode = 3D and acquisition mode = radial and 3D isotropic = yes) , or if (scan mode = 2D or M2D): Default	TFE prepulse = black-blood	Enables the use of Black blood prepulse.	Default is only available when BlackBlood (AWOPT_BLACKBLOOD) is available. MSDE is only available when MSDE (AWOPT_3D_BLACK_BLOOD) is available.
		bb slice thickness (mm)	[10, 30]	bb pulse type = default	Determines the slice thickness of the second 180-pulse in black-blood imaging	N/A.

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorPrepulsesTFEPar in the results file is zero.

3.1.45.68. SWCMP.Scan.Validator.TimeSeries.Acceleration.Par.tc1

Test ID	41064
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:22:03 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 11:13:50 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:13:09 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies availability of Time seriesacceleration parameters

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorTimeSeriesAccelerationPar in the results file is zero.

3.1.45.69. SWCMP.Scan.Validator.TimeSeries.F0Drift.Par.tc1

Test ID	47987
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekarappa, Deepak Shekhar (320098697) on 2020-09-16 10:28:36 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:54:00 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:11 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameters to perform Time-Resolved Spectroscopy.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
81981	SWCMP.Scan.Validator.TimeSeries.F0Drift.Par	Frequency Stabilisation	- no slice (per nsa) - voi (per dynamic)	Always	This parameter allows the user to update the F0 during scanning.	None voi (per dynamic) is only available for DDAS and only allowed when option bit 142: sLASER-VAPOR is enabled.

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorTimeResolvedSpectroF0drift Par in the results file is zero. Attach the result file as an objective evidence.

3.1.45.70. SWCMP.Scan.Validator.TimeSeries.Par.tc1

Test ID	47986
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekharaappa, Deepak Shekhar (320098697) on 2020-09-16 10:28:39 Meaning: QA Approved - (QA)::Bulk Transitions Signed by: Sharma, Nishant (320059006) on 2020-09-16 09:54:02 Meaning: SME Approved - (Technical)::Bulk Transitions Signed by: Engelen, Walter van (nly95948) on 2020-09-16 09:24:13 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test verifies the availability of the parameters to perform Time-Resolved Spectroscopy.

Requirements Coverage:

Requirements Coverage:						
Req ID	Requirement Tag	Description				
87138	SWCMP.Scan.Validator.TimeSeries.Par.v2	Name	Range	Visible when	Description	System / Option restrictions
		Dynamic study	- no - individual	Always	This parameter	The parameter

Req ID	Requirement Tag	Description				
			- block - contrast phases		r allows the user to record a time resolved series of identically acquired scans subsequently.	r selection 'contrast phases' is only enabled when the SW option SYS.Option.Bit 153: 4D free breathing is available.
		dyn scans	1 - 16384	Dynamic study != no	This parameter allows the user to define the nr of subsequent dynamics in a time resolved series.	None
		dynamic scan times	- shortest - manual user defined	Dynamic study = individual	This parameter allows the user set the temporal resolution for the time series.	None
		dummy scans	0 - 30	Dynamic study = individual	This parameter allows the user to define the nr	None

Req ID	Requirement Tag	Description				
					of dummy scans, used to improve the steady state condition.	
		duration - interval	shortest, manual, < time string>	Dynamic study = block	This parameter allows the user to define the duration and interval for each block	None
		dynamic scans per phase	1-16 384	Dynamic study = contrast phases	This parameter allows the user to define the nr of dynamic scans per contrast phase. For the first contrast phase the number of dynamics is restricted to 1. Up to 64 contrast phases are available	None
		Phase duration	shortest, < time string>	Dynamic study = contrast	This parameter allows	None

Req ID	Requirement Tag	Description				
				phases	the user to define the phase duration per contrast phase. The first contrast phase is always shortest	
		Phase start	shortest, manual	Dynamic study = contrast phases	This parameter allows the user to define the phase start per contrast phase	None

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorTimeResolvedSpectroPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.71. SWCMP.Scan.Validator.TimeSeries.Timestamp.Par.tc1

Test ID	53952
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2022-05-30 10:44:44 Meaning: QA Approved - (QA)

	Signed by: Sharma, Nishant (320059006) on 2022-05-25 10:06:20 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2022-05-09 09:54:41 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test verifies the availability of the parameters to perform Timestamp.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
91693	SWCMP.Scan.Validator.TimeSeries.Timestamp.Par.v2	Image Timestamp	no, yes	Scan type = Imaging	This parameter specifies whether images are tagged with NTP synchronized acquisition timestamps.	This parameter is only available when the DAS NTP clock is enabled in the service application and option 222: MRTC is enabled.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorTimestampPar in the results file is zero. Attach the result file as an objective evidence.

3.1.45.72. SWCMP.Scan.Validator.Trajectory.Mode.Acquisition.Par.tc1

Test ID	51047
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-15 11:58:25 Meaning: QA Approved - (QA) Signed by: Beck, Gabrielle (nly95705) on 2021-06-10 15:20:07 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:28 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator Acquisition parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
94318	SWCMP.Scan.Validator.Trajectory.Mode.Acquisition.Par.v3	Acquisition mode	Cartesian, Radial, MotionFree, MultiVane, spiral	Scan type = imaging	Acquisition mode This parameter specifies the acquisition mode with respect to k-space. Possible values: Cartesian Rectilinear sampling of k-space	Cartesian is available always Radial is available when SW option M_OPT.OptRadial (option bit 107) is available. Spiral is available when SW option

Req ID	Requirement Tag	Description				
					is performed along the readout direction kx. In some fast imaging modes (EPI, TSE), several parallel lines are sampled after each excitation. Radial The kx - ky plane is sampled with a star of lines through the origin. The transformed signals are projections of the patient or phantom. Radial imaging is available for the FFE- and TFE- techniques. MultiVane The kx - ky plane is	M_OPT.Spiral_Brain (option bit 144) is available. MultiVane is available when SW option M_OPT.Propeller (option bit 42) is available MotionFree is available when SW option M_OPT.CsMultiVaneGeneric (option bit 210) is available MotionFree and MultiVane are not visible at one time. When both SW options (M_OPT.CsMultiVaneGeneric and M_OPT.Propeller) are available then:

Req ID	Requirement Tag	Description				
					sampled with (rotating) strips of parallel lines, called blades. It can be considered as a hybrid form of cartesian and radial acquisition. MultiVane provides an internal correction for motion inconsistencies during the acquisition of one (set of) image(s). Spiral The kx - ky plane is sampled around the centre k-space while continuously receding or approaching it.	MotionFree is visible by default, but will be replaced by MultiVane when (Acceleration = no or SENSE).
		MotionFree	Default,	Acquisition	Enables	N.A.

Req ID	Requirement Tag	Description				
		Free Mode	User defined	Free Mode = MotionFree ÷ !FFE	automatic calculation of propeller specific parameters when set to default. User can manually set the values when set to user defined.	
		shots per blade	1 - 16 or smaller depending on other parameters	Acquisition mode = Multivane or (MotionFree Mode = user defined) ÷ Fast imaging mode = (TSE or TFE)	Sets the number of shots in a propeller blade	N.A.
		Multivane gross motion correction	No, yes	Acquisition mode = Multivane or (MotionFree Mode = user defined)	Sets the motion correction for propeller scans	N.A.
		TRs per blade	8 - 64 or smaller, dynamically set	Acquisition mode = (Multivane	Sets the number of TRs in a propeller	N.A.

Req ID	Requirement Tag	Description				
			dependin g on other paramete rs	e or MotionFre e) ù Fast imaging mode = none	blade	
		Acq. window	default, user def	Acquisitio n mode = spiral	This paramete r specifies if a default acquisitio n window is used or a user defined	N.A.
		Acq. window (ms)	0.5 - 100 ms	Acquisitio n mode = spiral ù Acq. Window = user def	This paramete r specifies the acquisitio n window length	N.A.
		Flow crushing	weak, medium, strong	Acquisitio n mode = spiral ù Scan technique = SE	This paramete r enables the suppressi on of flow signal via strong crusher gradients with three different strength choices.	N.A.
		Spiral deblurring	Always: yes, no If Dixon: yes	Acquisitio n mode = spiral	This paramete r enables the correction of B0-	N.A.

Req ID	Requirement Tag	Description				
					induced image blurring in reconstruction.	
		UTE FID sampling	no, yes	Acquisition mode = radial $\hat{U}$ Scan mode = 3D	When enabling the UTE FID mode, sampling starts at the center of k space, i.e., already on the rising slope of the readout gradient. It can be used to achieve nominal echo times in the order of 100 μ s.	UTE FID sampling is visible when SW option M_OPT.UI trashortTE (option bit 33) is available.
		traj. delay (us)	-10 - +10	UTE FID sampling = yes	Sets the trajectory delay (in microseconds)	N.A.
		3D isotropic	no, yes	Acquisition mode = radial $\hat{U}$ Scan mode = 3D	If set to "yes", the kspace acquisition is performed as Koosh ball and the	3D isotropic is visible when SW option M_OPT.UI trashortTE (option bit 33) is

Req ID	Requirement Tag	Description
		number of slices is set to the recon resolution . If set to "no", the kspace acquisition is performed as Stack of stars. available.
		Application specific restrictions are imposed by the following options: • M_OPT. CsMultiVaneBody (option bit 211) enables the use of respiratory compensation techniques (liver), or prone patient position (breast) in combination with CSSENSE MultiVane. • Radial CSSENSE is available when SW option M_OPT.Cs3dFreebreathing (option bit 215) is available. • M_OPT.MultiVane-XD Extensions (option bit 48) enables: - o Cardiac synchronization with MultiVane or MotionFree. - o Navigator synchronization with MultiVane or MotionFree for product model non Marlin. - o Multi-echo FFE with MultiVane. Without option bit 48 multi-echo FFE is still available for MotionFree.

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorTrajectoryModeAcquisitionPar in the results file is zero.

3.1.45.73. SWCMP.Scan.Validator.TrajectoryOrderkykzPar.tc1

Test ID	41296
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 06:22:06 Meaning: QA Approved - (QA)::Bulk Transitions

	Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-20 11:14:57 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2019-05-16 15:13:11 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

Requirements Coverage:

Req ID	Requirement Tag	Description						
74669	SWCMP.Scan.Validator.Trajectory.Order.kyz.Par							
		<table><tr><th>Name</th><th>Range</th><th>Visible when</th><th>Description</th></tr><tr><td>loop order</td><td>[zy_order, yz_order]</td><td>FFE-3D</td><td>With the 3D loop order extension it will be possible to choose the order of the ky and kz loops.</td></tr></table>	Name	Range	Visible when	Description	loop order	[zy_order, yz_order]
Name	Range	Visible when	Description					
loop order	[zy_order, yz_order]	FFE-3D	With the 3D loop order extension it will be possible to choose the order of the ky and kz loops.					

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorTrajectoryOrderkyzPar in the results file is zero.

3.1.45.74. SWCMP.Scan.Validator.Turbo.Mode.TFE.Par.tc1

Test ID	41057
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 10:21:31 Meaning: QA Approved - (QA)

	Signed by: Engelen, Walter van (nly95948) on 2019-05-21 09:01:58 Meaning: SME Approved - (Business)
	Signed by: Mariappan, Yogesh kannan (310134237) on 2019-05-21 05:12:22 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test verifies availability of TFE parameters

Requirements Coverage:

Req ID	Requirement Tag	Description			
74741	SWCMP.Scan.Validator.Turbo.Mode.TFE.Par	Name	Range	Visible when	Description
		auto shot duration	[no, yes]	TFE and CardiacProspective and cardiac phases = 1 and Shot mode = default	"Automatic shot duration" specifies the shot duration for certain cardiac frequencies. For other cardiac frequencies, a linear interpolation is made
		CENTRA(spiral)	[no, yes]	TFE-3D and turbo direction = radial	With CENTRA(spiral) enabled the ky-kz space covered in a spiral-like fashion. The first shot starts at the k-space center, subsequent shots start at ever increasing distance
		motion	[no, yes]	TFE	Allows

Req ID	Requirement Tag	Description			
		smoothing			motion smoothing with TFE sequence
		3D free factor	[no, yes]	TFE-3D	With "3D free factor" selected, the turbo factor can be chosen independently of FOV and resolution

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorTurboModeTfePar in the results file is zero.

3.1.45.75. SWCMP.Scan.Validator.Uniformity.Par.tc1

Test ID	55555
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moerland, Ad (nly95742) on 2022-10-27 14:23:15 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2022-10-27 13:29:12 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2022-10-20 15:07:59 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of Uniformity correction parameters and conflicts.

Requirements Coverage:

Req ID	Requirement Tag	Description				
90482	SWCMP.Scan.Validator.Uniformity.Par	Name	Range	Visible when	Description	System/Option restrictions
		Uniformity	Always: CLEAR, Synergy, Coil Sensitivity , If (Uniformity = Body-tuned): Body-tuned, If (Acceleration = no): Classic.	Scan mode = 1D, 2D, 3D, MS, or M2D and Acceleration = no, SENSE, CS-SENSE, or CS-SENSE-AI.	Uniformity relates to the algorithm using the coil sensitivity map for uniformity correction during image reconstruction. The parameter determines the type of homogeneity correction imposed on the images. Note: Body-tuned is only available for legacy reasons.	N.A.
Coil Sensitivity is not compatible with Uniformity Correction = yes.						

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorUniformityPar in the results file is zero.

Step	Description	Expected Result

3.1.45.76. SWCMP.Scan.Validator.UniformityCorrection.Par.tc1

Test ID	55714
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Moerland, Ad (nly95742) on 2022-10-27 14:21:57 Meaning: QA Approved - (QA) Signed by: Graaf, Ruud de (nly95749) on 2022-10-27 13:29:35 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2022-10-20 15:08:32 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

This test specifies availability of Uniformity correction parameters and conflicts.

Requirements Coverage:

Req ID	Requirement Tag	Description					
90820	SWCMP.Scan.Validator.UniformityCorrection.Par	Name	Range	Visible when	Description	System/Option restrictions	
		Uniformity correction	Yes, No.	Scan type = Imaging and nucleus =H1 and scan mode technique = SE, IR or FFE and Acceleration = no, SENSE,	Parameter enables the uniform filter, which retrospectively homogenizes the intensity of the image.	N.A.	

Req ID	Requirement Tag	Description				
				CS-SENSE, or CS- SENSE-AI and TFE PSIR = no and B1 field map = no		
		Uniformity correction is not compatible with 'on the fly' reconstruction mode.				

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorUniformityCorrectionPar in the results file is zero.

3.1.45.77. SWCMP.Scan.Validator.VOI.FOV.Par.tc1

Test ID	51050
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-15 11:58:58 Meaning: QA Approved - (QA) Signed by: Beck, Gabrielle (nly95705) on 2021-06-10 15:30:14 Meaning: SME Approved - (Technical) Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:30 Meaning: SME Approved - (Business)::Bulk Transitions

Approval Status	Approved
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Description:

This test specifies availability of Scan Validator volume of interest (VOI) parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
84807	SWCMP.Scan.Validator.VOI.FOV.Par	FOV * (mm)	5.0 – 705.0 The actual range can be restricted based on other scan parameters and hardware capabilities	Always	This parameter specifies the FOV in the measurement direction if slice orientation = transverse : if fold-over direction = AP then replace * by RL if fold-over direction = RL then replace * by AP if slice orientation = coronal: if fold-over direction = FH then replace * by AP if fold-	Max values depend on system configuration

Req ID	Requirement Tag	Description				
					over direction = RL then replace * by FH if slice orientation = sagittal: if fold-over direction = FH then replace * by AP if fold-over direction = AP then replace * by FH	
		FOV * (mm)	5.0 – 705.0 The actual range can be restricted based on other scan parameters and hardware capabilities	Always	This parameter specifies the FOV in the phase direction For Cartesian, Spectro and MultiVane-CSENSE, the value of this parameter does not have to be equal to the value of the FOV in the measurement	N.A.

Req ID	Requirement Tag	Description				
					direction. The FOV is rectangular. For all other scan protocol settings, the values in the measurement- and phase-direction are equal in value. The FOV is square. if fold-over direction = RL then replace * by RL if slice orientation = coronal: if fold-over direction = FH then replace * by FH if fold-over direction = AP then replace * by AP	
		FOV * (mm)	0.0 - 560.0 The actual	Scan mode =	This parameter	N.A.

Req ID	Requirement Tag	Description
		range can be restricted based on other scan parameters and hardware capabilities M3D, MS or M2D r specifies the FOV in the slice direction if slice orientation = transverse then replace * by FH if slice orientation = coronal then replace * by AP if slice orientation = sagittal then replace * by RL

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorVOIFOVPar in the results file is zero.

3.1.45.78. SWCMP.Scan.Validator.VOI.Stacks.Par.tc1

Test ID	51051
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Nabuurs, Christine (310147336) on 2021-06-15 12:01:02 Meaning: QA Approved - (QA)

	Signed by: Beck, Gabrielle (nly95705) on 2021-06-10 15:30:42 Meaning: SME Approved - (Technical)
	Signed by: Engelen, Walter van (nly95948) on 2021-06-09 11:45:32 Meaning: SME Approved - (Business)::Bulk Transitions
Approval Status	Approved

Description:

This test specifies availability of Scan Validator stack parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description				
		Name	Range	Visible when	Description	System / Option restrictions
91284	SWCMP.Scan.Validator.VOI.Stacks.Par.v2	type	Parallel, radial	Scan type = imaging	Type of stack	N.A.
		slices	1- 1024 The actual range can be restricted based on other parameters	Scan type = imaging	Number of slices in the imaging stack	N.A.
		slice gap	Default User defined	Scan type = imaging	Spacing between slices	N.A.
		slice orientation	Transverse, Sagittal, Coronal	Scan type = imaging	Orientation of slices in the stack	N.A.
		fold-over direction	If slice orientation = transverse	Acquisition mode = cartesian	Phase encode direction of the	N.A.

Req ID	Requirement Tag	Description				
			then AP or RL If slice orientatio n = sagittal then AP or FH If slice orientatio n = coronal then RL or FH	or MotionFre e	stack	
		fat shift direction	If slice orientatio n = transverse : if fold- over direction = AP then L or R if fold- over direction = RL then A or P If slice orientatio n = coronal: if fold- over direction = FH then L or R if fold- over direction = RL then F or	Acquisitio n mode = Cartesian or MotionFre e	Fat shift direction	N.A.

Req ID	Requirement Tag	Description				
			H If slice orientation = sagittal: if fold-over direction = FH then A or P if fold-over direction = AP then F or H			
		Stack Offc. AP (P=+mm)	-200 - 200 The actual range can be restricted based on other parameters	Scan type = imaging	This off-center determine the position of the stack as the distance to the central coronal plane.	NA
		Stack Offc. RL (L=+mm)	-220 - 220 The actual range can be restricted based on other parameters	Scan type = imaging	This off-center determine the position of the stack as the distance to the central sagittal plane.	NA
		Stack Offc.	-2000 -	Scan type	This off-	NA

Req ID	Requirement Tag	Description				
		FH (H=+mm)	2000 The actual range can be restricted based on other parameters	= imaging	center determine the position of the stack as the distance to the central transverse plane	
		Ang. AP (deg)	-180.0 - 180.0	Scan type = imaging	The angulation is defined as a rotation around the central AP axis for angulation AP.	NA
		Ang. RL (deg)	-180.0 - 180.0	Scan type = imaging	The angulation is defined as a rotation around the central RL axis for angulation RL.	NA
		Ang. FH (deg)	-180.0 - 180.0	Scan type = imaging	The angulation is defined as a rotation around the central FH axis for angulation FH.	NA

Req ID	Requirement Tag	Description

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTestAutomation and inspect the results file.	The reported number of failed tests related to SWCMPScanValidatorVOIStacksPar in the results file is zero.

3.1.45.79. SWCMP.Scan.Validator.Water.SuppressionTechniques.Par.tc1

Test ID	55343
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Thammanna Chandrashekhara, Deepak Shekhar (320098697) on 2022-08-05 06:20:00 Meaning: QA Approved - (QA) Signed by: Nisti Grigoletto Borgonovi, Arthur Felipe (310046086) on 2022-08-04 15:16:33 Meaning: SME Approved - (Business) Signed by: Sharma, Nishant (320059006) on 2022-08-04 11:46:56 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

This test specifies availability of Scan Validator water Suppression Techniques parameters.

Requirements Coverage:

Req ID	Requirement Tag	Description	Name	Range	Visible when	Description	System / Option restrictions
88724	SWCMP.Scan.Validator.Water.Suppression.Techniques.Par		Water suppression	No, SPIR SPAIR, ProSet	!Spectro	This input parameter enables the choice of water	N/A

Req ID	Requirement Tag	Description				
					suppression method	
		inversion delay(ms)	0-2000	Water suppression = SPAIR	This input parameter enables the SPAIR inversion in ms	N/A
		frequency offset	default, user defined	Water suppression = SPIR or SPAIR	This input parameter adapts the frequency offset of the SPIR pulse	N/A
		offset (Hz)	1.5T: 40 - 180 3T: 80 - 360	Frequency offset = user defined	This input parameter allows the user to set the desired offset frequency	N/A
		Pulse type	11, 121, 1331	Water suppression = ProSet	This input parameter specifies the ProSet pulse to be used	N/A
		SPAIR TR (ms)	N/A	Water suppression = SPAIR	This output parameter specifies SPAIR TR in ms	N/A
		SPAIR inv. delay (ms)	N/A	Water suppression =	This output parameter	N/A

Req ID	Requirement Tag	Description				
				SPAIR	r specifies SPAIR inv delay in ms	

Test steps:

Step	Description	Expected Result
Step 1	Execute the SWCMP.Scan.PDFTTestAutomation and inspect the results file.	The reported number of failed tests related to SwCmpScanValidatorWaterSuppressionTechnique sPar in the results file is zero.

3.1.46. TablePositionsOptimizer**3.1.46.1. SWCMP.Scan.TablePositionsOptimizer.CalculatePragmaticPositions.tc1**

Test ID	35261
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2017-10-30 18:07:10 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2017-10-30 16:35:34 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-10-19 16:33:21 Meaning: SME Approved - (Technical)
Approval Status	Approved

Description:

The test of the requirements for the TabelPositionsOptimizer are embedded in the GoogleTest tablepositionsoptimizer_gtest.exe in the building block prescanactions of the scanning subsystem. This is an automated test that covers the requirements of TablePositionsOptimizer. The functionality of the TablePositionOptimizer is developped test driven. The source code specifies all the scenarios tested.

Requirements Coverage:

Req ID	Requirement Tag	Description
67786	SWCMP.Scan.TableOptimizer.C calculatePragmaticPositions	- When the scan is indicated to be "FastNextScan" e.g. bolus track: the Table Positions Optimizer shall try if the pragmatic table position of the next scan can be used. - When the scan does not indicate to be "FastNextScan": the Table Positions Optimizer shall first try to re-use a table position used of already executed scans, so that the scan ready to be performed will profit from earlier made preparations on that position. - When none of the prepared table positions can be re-used, the Table Positions Optimizer shall define an optimal table position that is suitable for the first scan to be performed and at the same time can be re-used by maximum of not executed scans waiting. The pragmatic table position will be set to the middle of the remaining range of all not executed scans that can be done on the same pragmatic table position/

Test steps:

Step	Description	Expected Result
Step 1	Build in software component PrescanActions the executable tablepositionsoptimizer_gtest.exe.	Build succeeds
Step 2	Open cmd window cmd.exe and change the directory to where the executable is located. Type in tablepositionsoptimizer_gtest.exe. Attach the test output as objective evidence	Program runs with google test output. with no FAILED test.

3.1.46.2. SWCMP.Scan.TablePositionsOptimizer.Threading.tc1

Test ID	35262
Revision	1
Product Model	Any
Gradient System	Any
Options	N.A.
Signatures	Signed by: Buunen, Bas (nly95831) on 2017-10-30 18:07:19 Meaning: QA Approved - (QA) Signed by: Arunachalam, Sankaralingam (ing07488) on 2017-10-30 16:35:52 Meaning: SME Approved - (Business) Signed by: Graaf, Ruud de (nly95749) on 2017-10-19 17:11:02 Meaning: SME Approved - (Technical)

Approval Status	Approved
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Description:

By design the threading must be ok. This is tested by review.

Requirements Coverage:

Req ID	Requirement Tag	Description
67787	SWCMP.Scan.TableOptimizer.Threading	DesignConstraint: The TablePositionOptimizer shall be threadsafe to allow it to be used in a multithreaded environment.

Test steps:

Step	Description	Expected Result
Step 1	Check that all functions of the interface TablePositionsOptimizer has as first statement a lock on a CriticalSection Attach the list with functions that were inspected and make sure that the Stream and Product Composition specified in the test run match the code base used for inspection.	Check illustrate that all functions has a lock.

3.1.47. TestAutomation**3.1.47.1. SWCMP.Scan.PDFTestAutomation.tc1**

Test ID	41202
Revision	1
Product Model	All
Gradient System	
Options	
Signatures	Signed by: Mohan Rao, Anasuya (310175887) on 2019-05-21 11:46:51 Meaning: QA Approved - (QA) Signed by: Engelen, Walter van (nly95948) on 2019-05-15 15:28:55 Meaning: SME Approved - (Technical) Signed by: Graaf, Ruud de (nly95749) on 2019-05-15 14:46:16 Meaning: SME Approved - (Business)
Approval Status	Approved

Description:

The test describes how to execute the PDF Test Automation test cases.

Requirements Coverage:

Req ID	Requirement Tag	Description
75527	SWCMP.Scan.PDFTestAutomation	ScanValidator shall support test automation to check intended scan protocol parameter behavior.

Test steps:

Step	Description	Expected Result
Prep 1	1. Open a command window. 2. Create a new branch based on the tag/label under test. 3. Checkout this new branch. 4. Navigate to pdftestautomation building block. 5. Execute the install_pdftestautomation.bat from the res folder (this will take some minutes).	N.A.
Step 1	Execute in the new command window that opened 'runAllTests.cmd >results.log' Inspect the results.log when the execution is finished. Attach the created results file to this Step.	No failed test cases are reported in the results file.

4. Test Traceability Matrix



VTM Scanning
Voxel.xlsm

5. Document History

Revision	Date	Author	Description
A	25-Jun-2025	Kaitan Pereira	Introduced for Voxel
B	31-Jul-2025	Kaitan Pereira	Updated based on Impact Analysis

